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LEED v5 REFERENCE GUIDE

BUILDING DESIGN AND CONSTRUCTION

U.S. Green Building Council
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LEED Reference Guide for Building Design and Construction
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The LEED v5 rating systems were developed by the USGBC volunteer community and formally ratified by our broader membership. The reference guide suite is the product of the dedicated efforts of USGBC and Green Business Certification Inc. (GBCI) staff and consultants, designed to support the industry's understanding and implementation of the rating systems.

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IN REMEMBRANCE

The USGBC community lost two volunteers in 2024. We honor their memory and dedication to creating a more sustainable world through green building.

Ganesh Nayak
USGBC volunteer from 2023 to 2024

As the chair of our Equity Working Group at the USGBC, Ganesh was a long-standing leader, mentor, and champion of sustainable and inclusive design. His dedication to LEED certification and sustainable architecture transformed over four million square feet of space, leaving a lasting legacy of environmental stewardship.

Stewart Comstock
USGBC volunteer from 2020 to 2024

Stewart donated his time as the vice chair of the Sustainable Sites Technical Advisory Group, bringing decades of deep expertise in stormwater management to our work. His influence helped shape some of the most critical aspects of LEED as it evolved over the years. Stewart helped lay the foundation on which many of today's green infrastructure principles and best practices are built.

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Preface

THE CASE FOR GREEN BUILDING

Green buildings are an integral part of the solution to the environmental challenges facing the planet.

The impetus behind development of the Leadership in Energy and Environmental Design (LEED) rating systems was recognition that the design, construction, and building management industry has the expertise, tools, and technology to transform buildings and make significant advances toward a sustainable planet. LEED projects throughout the world have demonstrated the benefits of taking a green design and operations approach that reduces the environmental harm of buildings and restores the balance of natural systems.

Buildings have a major role to play in sustainability through their construction, the lifetime of their operation, and patterns of development.

What we build today, how we build it, and where we build it are profoundly important.

ABOUT LEED

LEED is a concise framework for identifying and implementing practical and measurable green building, design, construction, operations, and maintenance strategies and solutions. LEED is a voluntary, market-driven, consensus-based tool that serves as a guideline and assessment mechanism.

Within the appropriate rating system, projects that meet the prerequisites and earn enough credits to achieve the certification threshold have demonstrated performance that spans the goals in an integrated way. Certification is awarded at four levels: LEED Certified, LEED Silver, LEED Gold, and LEED Platinum. These levels incentivize higher achievement and, in turn, faster progress toward the goals.

Since its launch, LEED has evolved to address new markets and building types, advances in practice and technology, and greater understanding of the environmental and human health effects of the built environment. The Foundations of LEED detail the principles of transparency, openness, and inclusiveness that guide this evolutionary process. Learn more by exploring the Foundations of LEED at usgbc.org/resources/foundations-leed.

LEED does not certify, endorse, or promote any products, services, or companies.

HISTORY OF LEED

USGBC developed LEED for New Construction v1.0 in 1998 for the commercial building industry.

Since its launch, LEED has evolved to address new markets and building types, advances in practice and technology, and greater understanding of the environmental and human health effects of the built environment. These ongoing improvements are developed by USGBC member-based volunteer technical advisory groups, committees, subcommittees, and working groups in conjunction with USGBC staff. Once development is complete, changes are reviewed by the LEED Steering Committee and the USGBC Board of Directors and are approved by the Consensus Committees acting as the membership consensus body before being submitted to USGBC members for a ratification vote. The process is based on principles of transparency, openness, and inclusiveness, as detailed in the Foundations of LEED. To learn more, please visit usgbc.org/resources/foundations-leed.

In 2007, USGBC launched Green Business Certification, Inc. (GBCI) as an independent organization to manage the LEED certification and professional credentialing process to scale the demand for LEED and ensure ongoing quality and integrity.

For more information on the history of LEED, USGBC, and GBCI, visit: usgbc.org/about/mission-vision

LEED'S GOALS

LEED seeks to optimize the use of natural resources, promote regenerative and restorative strategies, maximize the positive and minimize the negative environmental and human health consequences of the construction industry, and provide high-quality indoor environments for building occupants. LEED emphasizes integrative design, integration of existing technology, and state-of-the-art strategies to advance expertise in green building and transforms professional practice. The technical basis for LEED strikes a balance between requiring today's best practices and encouraging leadership strategies. LEED sets a challenging yet achievable set of benchmarks that define green building for interiors, buildings, and cities and communities.

LEED v5 was developed around three central areas of impact:

- Decarbonization
- Quality of life
- Ecological conservation and restoration

Every credit and prerequisite in LEED v5 links to decarbonization, quality of life, and/or ecological conservation and restoration. The rating system annotates this connection, enabling project teams to easily shape and communicate their sustainability stories.

BENEFITS OF USING LEED

LEED is designed to address environmental challenges while responding to the needs of a competitive market. LEED-certified buildings are designed to deliver many benefits, including the following:

- Lower operating costs and increased asset value
- Reduced waste sent to landfills
- Energy and water conservation
- Conserved and restored habitat for improved ecosystem function and biodiversity
- Increased market transparency and expanded access to industry-leading green products and materials
- More healthful and productive environments for occupants
- Reductions in greenhouse gas (GHG) emissions
- Qualification for tax rebates, zoning allowances, and other incentives in many cities

By participating in LEED, owners, operators, designers, and builders make a meaningful contribution to the green building industry. By documenting and tracking buildings' resource use, they contribute to a growing body of knowledge that will advance research in this rapidly evolving field. This will allow future projects to build on the successes of today's designs and bring innovations to the market.

GBCI independently verifies sustainability performance against LEED, ensuring the integrity of outcomes and strengthening market confidence in sustainability investment and innovation.

LEED CERTIFICATION PROCESS

A LEED project begins when the team determines scope, selects the appropriate rating system, confirms alignment with the minimum program requirements (MPRs), and registers the project on the Arc platform. The team designs the project to meet the requirements for all prerequisites and for the credits they have chosen to pursue in coordination with their sustainability goals. The required documentation is compiled and submitted to GBCI for review. GBCI provides feedback on the documentation received, which gives project teams the opportunity to make any needed design and construction changes, take additional steps to improve an existing project's performance, and/or provide additional clarification needed for GBCI to determine compliance. Additional rounds of review follow, as needed and allowed, to earn certification of the project. For a more in-depth understanding of the certification process, see the Guides to Certification.

Getting Started

ABOUT THIS GUIDE

This guide explains the requirements of the LEED v5 Building Design and Construction (BD+C) rating system, offering essential background information for contractors, engineers (mechanical, structural, and civil), architects, designers (interior and landscape), designers, building operators, and LEED consultants. Early attention to rating system selection, minimum program requirements, and project boundaries ensures a smoother path to certification.

This guide is the first in a series of detailed resources for project teams pursuing LEED v5. It builds on the rating system requirements, providing background on each LEED category and a detailed explanation of the technical requirements for each credit.

WHAT'S INSIDE

This guide builds on the requirements listed in the rating system to provide background on each LEED category and an in-depth explanation of the technical requirements of each credit.

This guide contains:

- LEED v5 rating system requirements
- Rating system requirements explained
- Documentation requirements
- Referenced standards
- Further explanation

As teams review each credit and prerequisite, they should carefully evaluate the options and pathways available, ensuring they align with the project's specific goals, constraints, and characteristics.

When preparing documentation to submit for review, these materials may consist of contract documents (plans, specifications, elevations, sections, construction details, etc.) calculations, reports, manufacturer product information, photographs, and/or descriptive narratives.

The USGBC glossary is an additional resource: usgbc.org/glossary/v5

CREDIT CATEGORIES

This LEED rating system organizes the prerequisites and credits into eight categories, each addressing a fundamental aspect of sustainable, high-performance designs and construction practices:

- Integrative Process, Planning, and Assessments (IP)
- Location and Transportation (LT)

- Sustainable Sites (SS)
- Water Efficiency (WE)
- Energy and Atmosphere (EA)
- Materials and Resources (MR)
- Indoor Environmental Quality (EQ)
- Project Priorities (PR)

RATING SYSTEM SELECTION

LEED offers rating systems designed for specific project types. Projects must register under the most appropriate LEED rating system and use the guide appropriate to the project's registration. In cases where the most appropriate rating system is unclear, additional guidance is provided in the USGBC Help Center: support.usgbc.org/hc/en-us. The project team may also submit a request to consult with GBCI: support.usgbc.org/hc/en-us/requests/new. The consultation should occur prior to registration to avoid the risk of GBCI determining, during the Preliminary Review, that an inappropriate rating system has been selected.

LEED FOR BUILDING DESIGN AND CONSTRUCTION (BD+C)

New Construction and Major Renovations

Applicable to whole buildings and whole additions to buildings of various use types that are either new construction or undergoing major renovations and where at least 60% of the gross floor area is complete.

Major renovations are buildings with extensive interior alteration work in addition to work on the exterior shell of the building, primary structural components, the core and peripheral MEP (mechanical, electrical, plumbing) and service systems, and/or site work. Typically, the extent and nature of the work prevent the primary function space from being used for its intended purpose while the work is in progress, and the project team must obtain a new certificate of occupancy before reoccupying the work area.

Core and Shell Development

Applicable to whole buildings of various use types that are either new construction or undergoing major renovations, and at least 40% of the gross floor area is incomplete. An incomplete building does not have its basic floor, wall, and/or ceiling finishes installed, or essential mechanical, electrical, plumbing systems (or fixtures) necessary to occupy the space for its intended use within the contracted scope of work.

LEED FOR INTERIOR DESIGN AND CONSTRUCTION (ID+C)

Commercial Interiors

Applicable to interior renovation projects of commercial spaces (that have no scope for the exterior shell of the building and/or site work). The team installs movable furnishings, fixtures, and equipment (FF&E) to support the intended regular operations of the space.

LEED FOR BUILDING OPERATIONS AND MAINTENANCE (O+M)

Existing Buildings

Applicable to existing building projects focused on operational improvement. This rating system focuses on whole buildings that have been fully operational and occupied for at least one year. Focusing on performance-driven strategies and outcomes allows buildings in use to achieve greater efficiency.

PROJECT TYPE REQUIREMENTS

The entire gross floor area of a LEED project must be certified under a single rating system (LEED BD+C: New Construction or LEED BD+C: Core and Shell; LEED ID+C: Commercial Interiors; or LEED O+M: Existing Buildings) and is subject to all prerequisites and attempted credits in that rating system, regardless of mixed construction or space usage type.

However, in some prerequisites and credits, there may be requirements identified for specific project types, such as schools, warehouse and distribution centers, healthcare, or residential.

Project type requirements must be met if 60% or more of the gross floor area of a project comprises that project type. For example, if a project is 75% residential and 25% retail, that project must follow the residential requirements in the rating system. In the LEED BD+C: Core and Shell rating system or in a LEED BD+C: New Construction and Major Renovations project with incomplete space, the use type of the incomplete space must be based on the expected future use.

PROJECTS WITH INCOMPLETE SPACE

Projects that have incomplete space (future fit-out) can still pursue certification as long as the following criteria are met:

- Incomplete space comprises no more than 40% of the gross floor area for New Construction and no less than 40% of the gross floor area for Core and Shell.
- The team counts occupancy for the incomplete space by using projected values or, if unknown, the LEED Default Occupancy.
- A letter of commitment from the owner is provided attesting that the incomplete spaces will satisfy the requirements of each prerequisite and credit achieved by the project.
- The team provides a description of the incomplete space, identifying the remaining scope, the entity responsible for the remaining scope, and the reason the building is incomplete at the time of the final application.
- Prerequisite calculations must include the completed scope only. Credit calculations must include the completed and incomplete scope, and the incomplete scope must be held equivalent to the baseline.
- Contact USGBC prior to application submission for any attempted credits in which an adaptation for the incomplete space is not clear.

PROJECT OCCUPANCY

Occupancy counts for the project that will be used consistently throughout the application are important to establish early and share with the whole project team. Whenever possible, use actual or predicted occupancies. LEED requires occupancy to be assessed and reported for two groups: regular building occupants and visitors.

REGULAR BUILDING OCCUPANTS

Regular building occupants are routine users of a project (full- and part-time) such as:

- Employees, daily volunteers, support staff (e.g., janitors)
- Residents (dormitory, apartment, condo), overnight hotel guests, medical inpatients
- K–12 students

VISITORS

Visitors are intermittent users of a project, such as:

- Retail customers
- Medical outpatients
- Volunteers who only periodically use a building (e.g., once per week)
- Higher-education classroom students

DEFAULT OCCUPANCY COUNTS

If the occupancy is unknown and cannot be reasonably projected, use one of the following resources to estimate occupancy:

- Default occupant density from *ASHRAE 62.1-2022,* Table 6-1[1]
- Table 1

TABLE 1. Default Occupancy Counts Based on Gross Floor Area, Excluding Structured Parking

	Gross Square Feet Per Occupant		Gross Square Meters Per Occupant	
	Regular	Visitors	Regular	Visitors
General office	250	0	23	0
Retail, restaurant	435	95	40	9
Retail, grocery store	550	115	51	11
Retail, general	550	130	51	12
Retail, service	600	130	56	12
Medical office	225	330	21	31
R&D or laboratory	400	0	37	0

1. "Ventilation and Acceptable Indoor Air Quality," *ASHRAE 62.1-2022*, last accessed March 21, 2025, https://www.ashrae.org/technical-resources/bookstore/standards-62-1-62-2.

TABLE 1. Default Occupancy Counts Based on Gross Floor Area, Excluding Structured Parking (*continued*)

	Gross Square Feet Per Occupant		Gross Square Meters Per Occupant	
	Regular	Visitors	Regular	Visitors
Warehouse, distribution	2,500	0	232	0
Warehouse, storage	20,000	0	1,860	0
Educational, day care	630	105	59	10
Educational, K–12	1,300	140	121	13
Educational, postsecondary	2,100	150	195	14

NOTE: For residential (e.g., apartment or condo), use the number of bedrooms in the dwelling unit plus one, multiplied by the number of such dwelling units. For a hotel or motel, use 1.5 occupants per guest room multiplied by the total number of guest rooms. Then multiply the resulting total by 60%.

AVERAGE VERSUS PEAK OCCUPANCY

In prerequisite/credit calculations, different methods may be used to calculate occupancy:

- **Daily average**. The value that represents all the regular building occupants for a typical 24-hour day of operation. If numbers vary seasonally, use occupancy numbers that are a representative daily average over the entire operating season of the building.
- **Peak total**. The value that represents the highest number of occupants expected in the project at one time (e.g., shift overlap, events). Sometimes the building code generates the value for fire safety; but if justified, a lower number can be used.

If default occupancy counts from Table 1 are used, they must be applied as the daily average and peak occupancy values.

Peak outpatients are the highest number of outpatients at a given point in a typical 24-hour period.

EQUIVALENCIES

The LEED rating system is written based on the most up-to-date and widely available standards. However, alternative standards or compliance paths may also be available to project teams. The project priority library includes a list of equivalencies and alternative compliance paths available by credit, project types, or region.

If there is no guidance available allowing an equivalency or alternative compliance path, project teams may submit a request for USGBC to determine equivalency.

LEED Platinum Requirements

To earn LEED Platinum in LEED v5, projects must complete all prerequisites, earn a minimum of 80 points, and achieve decarbonization requirements in the following credits:

LEED BD+C

- EAc1: Electrification
- EAc3: Enhanced Energy Efficiency
- EAc4: Renewable Energy
- MRc2: Reduce Embodied Carbon

LEED ID+C

- EAc1: Electrification
- EAc2: Enhanced Energy Efficiency
- EAc3: Renewable Energy
- MRc2: Reduce Embodied Carbon

LEED O+M

- EAc1: Greenhouse Gas Emissions Reduction Performance
- EAc2: Optimized Energy Performance
- EAc5: Decarbonization and Efficiency Plans

See Appendix I for the full list of LEED Platinum requirements.

Minimum Program Requirements

The MPRs are the minimum characteristics or conditions that qualify a project to pursue LEED certification. These requirements serve as the foundation for all LEED projects and define the types of buildings, spaces, and neighborhoods that the LEED rating system is designed to evaluate.

MPR 1. MUST BE IN A PERMANENT LOCATION ON EXISTING LAND

INTENT

The LEED rating system is designed to evaluate buildings, spaces, neighborhoods, communities, and cities in the context of their surroundings. A significant portion of LEED requirements depends on the project's location; therefore, it is important that LEED projects are evaluated as permanent structures. Locating projects on existing land is important to avoid artificial land masses that have the potential to displace and disrupt ecosystems.

REQUIREMENTS

All LEED projects must be constructed and operated at a permanent location on existing land. No project that is designed to move at any point in its lifetime may pursue LEED certification. This requirement applies to all land within the LEED project.

MPR 2. MUST USE REASONABLE LEED BOUNDARIES

INTENT

The LEED rating system is designed to evaluate buildings, spaces, neighborhoods, communities, cities, and all environmental impacts associated with those projects. Defining a reasonable LEED boundary ensures the project is accurately evaluated.

REQUIREMENTS

The project team must include all contiguous land that is associated with the LEED project boundary and supports its typical operations. This includes land altered as a result of construction and features used primarily by the project's occupants, such as hardscape (parking and sidewalks), septic or stormwater treatment equipment, and landscaping. The LEED boundary may not unreasonably exclude portions of the building, space, or site to give the project an advantage in complying with credit requirements. The LEED project must accurately communicate the scope of the certifying project in all promotional and descriptive materials and distinguish it from any noncertifying space.

MPR 3. MUST COMPLY WITH PROJECT SIZE REQUIREMENTS

INTENT

The LEED rating system is designed to evaluate buildings, spaces, or neighborhoods of a certain size. The LEED requirements do not accurately assess the performance of projects outside of these size requirements.

REQUIREMENTS

All LEED projects must meet the size requirements listed below:

- **LEED BD+C and LEED O+M rating systems**
  The LEED project must include a minimum of 1,000 square feet (93 square meters) of gross floor area.
- **LEED ID+C rating system**
  The LEED project must include a minimum of 250 square feet (22 square meters) of gross floor area.

Further guidance on the MPRs can be found here: https://www.usgbc.org/leed-tools/minimum-program-requirements.

Rating System Scorecards

CREDIT CATEGORY VIEW

		New Construction	Core and Shell
Integrative Process, Planning and Assessments (IP)		**1**	**7**
IPp1	Climate Resilience Assessment	Required	Required
IPp2	Human Impact Assessment	Required	Required
IPp3	Carbon Assessment	Required	Required
IPp4	Tenant Guidelines	–	Required
IPc1	Integrative Design Process	1	1
IPc2	Green Leases	–	6
Location and Transportation (LT)		**15**	**15**
LTc1	Sensitive Land Protection	1	1
LTc2	Equitable Development	2	2
LTc3	Compact and Connected Development	6	6
LTc4	Transportation Demand Management	4	4
LTc5	Electric Vehicles	2	2
Sustainable Sites (SS)		**11**	**11**
SSp1	Minimized Site Disturbance	Required	Required
SSc1	Biodiverse Habitat	2	2
SSc2	Accessible Outdoor Space	1	1
SSc3	Rainwater Management	3	3
SSc4	Enhanced Resilient Site Design	2	2
SSc5	Heat Island Reduction	2	2
SSc6	Light Pollution Reduction	1	1
Water Efficiency (WE)		**9**	**8**
WEp1	Water Metering and Reporting	Required	Required
WEp2	Minimum Water Efficiency	Required	Required
WEc1	Water Metering and Leak Detection	1	1
WEc2	Enhanced Water Efficiency	8	7

		New Construction	Core and Shell
Energy and Atmosphere (EA)		**33**	**27**
EAp1	Operational Carbon Projection and Decarbonization Plan	Required	Required
EAp2	Minimum Energy Efficiency	Required	Required
EAp3	Fundamental Commissioning	Required	Required
EAp4	Energy Metering and Reporting	Required	Required
EAp5	Fundamental Refrigerant Management	Required	Required
EAc1	Electrification	5	4
EAc2	Reduce Peak Thermal Loads	5	5
EAc3	Enhanced Energy Efficiency	10	7
EAc4	Renewable Energy	5	4
EAc5	Enhanced Commissioning	4	3
EAc6	Grid Interactive	2	2
EAc7	Enhanced Refrigerant Management	2	2
Materials and Resources (MR)		**18**	**21**
MRp1	Planning for Zero Waste Operations	Required	Required
MRp2	Quantify and Assess Embodied Carbon	Required	Required
MRc1	Building and Materials Reuse	3	5
MRc2	Reduce Embodied Carbon	6	8
MRc3	Low-Emitting Materials	2	1
MRc4	Building Product Selection and Procurement	5	5
MRc5	Construction and Demolition Waste Diversion	2	2
Indoor Environmental Quality (EQ)		**13**	**11**
EQp1	Construction Management	Required	Required
EQp2	Fundamental Air Quality	Required	Required
EQp3	No Smoking or Vehicle Idling	Required	Required
EQc1	Enhanced Air Quality	1	1
EQc2	Occupant Experience	7	7
EQc3	Accessibility and Inclusion	1	1
EQc4	Resilient Spaces	2	2
EQc5	Air Quality Testing and Monitoring	2	–
Project Priorities (PR)		**10**	**10**
PRc1	Project Priorities	9	9
PRc2	LEED AP	1	1
Total	**Possible Points**	**110**	**110**

IMPACT AREA VIEW

Decarbonization		New Construction	Core and Shell
IP Prerequisite	Carbon Assessment	Required	Required
IP Prerequisite	Tenant Guidelines	–	Required
IP Credit	Integrative Design Process	1	1
IP Credit	Green Leases	–	6
LT Credit	Compact and Connected Development	6	6
LT Credit	Transportation Demand Management	4	4
LT Credit	Electric Vehicles	2	2
SS Credit	Heat Island Reduction	2	2
WE Prerequisite	Minimum Water Efficiency	Required	Required
WE Credit	Water Metering and Leak Detection	1	1
WE Credit	Enhanced Water Efficiency	8	7
EA Prerequisite	Operational Carbon Projection and Decarbonization Plan	Required	Required
EA Prerequisite	Minimum Energy Efficiency	Required	Required
EA Prerequisite	Fundamental Commissioning	Required	Required
EA Prerequisite	Energy Metering and Reporting	Required	Required
EA Prerequisite	Fundamental Refrigerant Management	Required	Required
EA Credit	Electrification	5	4
EA Credit	Reduce Peak Thermal Loads	5	5
EA Credit	Enhanced Energy Efficiency	10	7
EA Credit	Renewable Energy	5	4
EA Credit	Enhanced Commissioning	4	3
EA Credit	Grid Interactive	2	2
EA Credit	Enhanced Refrigerant Management	2	2
MR Prerequisite	Planning for Zero Waste Operations	Required	Required
MR Prerequisite	Quantify and Assess Embodied Carbon	Required	Required
MR Credit	Building and Materials Reuse	3	5
MR Credit	Reduce Embodied Carbon	6	8
MR Credit	Building Product Selection and Procurement	5	5
MR Credit	Construction and Demolition Waste Diversion	2	2

Quality of Life		New Construction	Core and Shell
IP Prerequisite	Climate Resilience Assessment	Required	Required
IP Prerequisite	Tenant Guidelines	–	Required
IP Prerequisite	Human Impact Assessment	Required	Required
IP Credit	Integrative Design Process	1	1
IP Credit	Green Leases	–	6
LT Credit	Equitable Development	2	2
LT Credit	Compact and Connected Development	6	6
LT Credit	Transportation Demand Management	4	4
SS Credit	Accessible Outdoor Space	1	1
SS Credit	Enhanced Resilient Site Design	2	2
SS Credit	Heat Island Reduction	2	2
WE Credit	Water Metering and Leak Detection	1	1
MR Credit	Low-Emitting Materials	2	1
MR Credit	Building Product Selection and Procurement	5	5
EQ Prerequisite	Construction Management	Required	Required
EQ Prerequisite	Fundamental Air Quality	Required	Required
EQ Prerequisite	No Smoking or Vehicle Idling	Required	Required
EQ Credit	Enhanced Air Quality	1	1
EQ Credit	Occupant Experience	7	7
EQ Credit	Accessibility and Inclusion	1	1
EQ Credit	Resilient Spaces	2	2
EQ Credit	Air Quality Testing and Monitoring	2	–

Ecological Conservation and Restoration		New Construction	Core and Shell
IP Prerequisite	Tenant Guidelines	–	Required
IP Credit	Integrative Design Process	1	1
IP Credit	Green Leases	–	6
LT Credit	Sensitive Land Protection	1	1
LT Credit	Compact and Connected Development	6	6
SS Prerequisite	Minimized Site Disturbance	Required	Required
SS Credit	Biodiverse Habitat	2	2
SS Credit	Accessible Outdoor Space	1	1
SS Credit	Rainwater Management	3	3
SS Credit	Enhanced Resilient Site Design	2	2

Ecological Conservation and Restoration		New Construction	Core and Shell
SS Credit	Heat Island Reduction	2	2
SS Credit	Light Pollution Reduction	1	1
WE Prerequisite	Water Metering and Reporting	Required	Required
WE Prerequisite	Minimum Water Efficiency	Required	Required
WE Credit	Water Metering and Leak Detection	1	1
WE Credit	Enhanced Water Efficiency	8	7
MR Prerequisite	Planning for Zero Waste Operations	Required	Required
MR Credit	Building and Materials Reuse	3	5
MR Credit	Building Product Selection and Procurement	5	5
MR Credit	Construction and Demolition Waste Diversion	2	2
EQ Prerequisite	No Smoking or Vehicle Idling	Required	Required

Integrative Process, Planning, and Assessments

OVERVIEW

As an industry, the built environment has evolved to prioritize whole-building performance and impact over isolated systems. The Integrative Process, Planning, and Assessments (IP) credit category builds on this approach in LEED v5 by emphasizing the importance of early-stage interdisciplinary collaboration and holistic, iterative planning to arrive at interrelated solutions. By using systems thinking to identify synergies among building systems and components before the design phase begins, projects prioritize the value of intentionally integrated design processes.

This holistic approach promotes the proactive consideration of how each project addresses decarbonization, quality of life (including equity and resilience) and ecological conservation and restoration from the outset. The strategies within the IP category help shape projects that are environmentally responsible, resilient, and equitable and benefit both occupants and the broader community. Key aspects of this category include:

- **Timing:** Initiating analysis in the early stages of the project (pre-design)
- **Engagement:** Involving the right people
- **Interdisciplinary collaboration:** Leveraging synergies across credit categories to maximize efficiencies and inform decision-making
- **Assessment-based understanding:** Discovering insights into natural hazards, carbon emissions, and human impacts

Decarbonization

The built environment contributes significantly to global GHG emissions and accounts for 21% of total GHG emissions and 31% of global carbon emissions in 2019 for operational carbon emissions alone.[1] The IP category's 25-year carbon assessment addresses energy use, refrigerants, embodied carbon, and transportation impacts (*IPp3: Carbon Assessment*). This assessment promotes carbon literacy and equips project teams with knowledge to apply to future projects, which fosters industry-wide progress.

This assessment pulls from other sections of the rating system to support consistent and meaningful carbon reduction (*MRp2: Quantify and Assess Embodied Carbon*, *MRc2: Reduce Embodied Carbon*, *MRc4: Building Product Selection and Procurement*).

1. IPCC, "Buildings." Chap. 9 in *IPCC Sixth Assessment Report*, n.d., https://www.ipcc.ch/report/ar6/wg3/chapter/chapter-9//.

Quality of life

Environmental justice and similar movements highlight the importance of understanding a community's social context. Additionally, practitioners are placing greater emphasis on prioritizing the health, safety, and welfare of building occupants. The American Institute of Architects *Code of Ethics and Professional Conduct* outlines their view of the obligation of designers, stating, "Members should employ their professional knowledge and skill to design buildings and spaces that will enhance and facilitate human dignity and the health, safety, and welfare of the individual and the public."[2] The IP category addresses equity by requiring an understanding of the local community, workforce, and supply chain, which encourages teams to confront inequities and positively impact their communities (*IPp2: Human Impact Assessment*).

Site-level resilience is a key topic throughout LEED BD+C: New Construction and LEED BD+C: Core and Shell. Munich Reinsurance America reported that global disaster losses in 2023 totaled $250 billion and highlighted the significant financial impact that natural disasters have worldwide.[3] The assessment aims to evaluate observed, projected, and future natural hazards for climate resilience and is meant to enhance awareness, increase transparency, reduce vulnerabilities, and ensure long-term safety and sustainability (*IPp1: Climate Resilience Assessment*).

Ecological conservation and restoration

To ensure the applicable expertise is in the room at the earliest stages (pre-design), the IP category specifically highlights goals that impact ecosystems (*IPc1: Integrative Design Process*). Strategies such as restorative site design, responsible product selection, and light pollution reduction all help improve a building's impact on the local environment (Sustainable Sites credit category and *MRc4: Building Product Selection and Procurement*). This approach encourages project teams to assess site conditions holistically and incorporate strategies that enhance biodiversity, protect natural resources, and support long-term ecological resilience.

By fostering early collaboration, holistic assessments, and interdisciplinary planning, the IP category helps project teams develop the foundation for resilient, equitable, and environmentally responsible buildings. These strategies ensure that projects not only meet immediate performance goals but also contribute to a sustainable, long-term future for both people and the planet.

2. AIA, "Design for Human Dignity and the Health, Safety, and Welfare of the Public," Canon 1, ES1.5 in *Code of Ethics and Professional Conduct*, 2024, https://www.aia.org/code-ethics-professional-conduct.
3. UNDRR, "Uncounted Costs: Data Gaps Hide the True Human Impacts of Disasters in 2023." January 17, 2024, https://www.undrr.org/explainer/uncounted-costs-of-disasters-2023.

Climate Resilience Assessment

IPp1
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To promote the comprehensive assessment of observed, projected, and future natural hazards for climate resilience with an aim to enhance awareness of hazards, increase transparency of risks, reduce vulnerabilities, and ensure long-term safety and sustainability.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Climate and Natural Hazard Assessment	

Complete a climate and natural hazard assessment.

As part of the assessment, identify observed, projected, and future natural hazards that could potentially affect the project site and building function. Address site-specific natural hazards, including, but not limited to, drought, extreme heat, extreme cold, flooding, hurricanes and high winds, hail, landslides, sea level rise and storm surge, tornadoes, tsunamis, wildfires and smoke, winter storms, and other relevant hazards (specify).

Identify two priority hazards, at minimum, to address through proposed design strategies. For each priority hazard, the project team must assess and specify the following:

- Intergovernmental Panel on Climate Change (IPCC) emissions scenario used, specifying the shared socioeconomic pathways
- Projected service life of the LEED project (e.g., fiscal year 2050 or 100 years)
- Hazard level
- Hazard risk rating
- Exposure, sensitivity, adaptive capacity, vulnerability, and overall risk levels

- Potential impact on the project site and building function
- Potential impact on the project site during construction

Where possible, use the information from the assessment to inform the planning, design, and operations and maintenance of the project and describe how project-specific strategies were considered.

REQUIREMENTS EXPLAINED

Incorporating climate resilience into the design and development of new construction projects helps mitigate the impacts of climate change and enhances the long-term durability of buildings against key vulnerabilities. A climate resilience assessment supports project teams in identifying and evaluating site-specific climate hazards that could threaten the performance, safety, and longevity of a building. This process provides a framework for embedding data-driven, forward-looking strategies that align with regional climate adaptation plans, building codes, and sustainability objectives.

Addressing climate risks during the design and construction phases not only improves a building's ability to withstand extreme weather events but also supports broader goals of public safety, economic stability, cultural preservation, and community resilience. By integrating these considerations, new development projects can better adapt to changing conditions. This ensures long-term functionality, reduced life-cycle costs, and a more sustainable environment.

This prerequisite requires project teams to conduct a comprehensive climate and natural hazard assessment, identifying current and projected hazards that may impact the project site and building operations. The assessment must address site-specific hazards, such as drought, heat, flooding, hurricanes, wildfires, tornadoes, and other relevant risks. Teams must prioritize at least two hazards and analyze them using the emissions scenarios from the Intergovernmental Panel on Climate Change's Shared Socioeconomic Pathways[4] (SSPs), specifying the project's service life and assessing hazard levels, risks, vulnerability, and potential impacts on-site operations and construction. In addition to the safety and welfare of occupants, the findings should be used to inform the project's planning, design, construction, operations, and maintenance and remain in alignment with the critical need to design for asset longevity.

Conducting a climate and natural hazard assessment is required to identify key vulnerabilities and guide operational decisions to improve project resilience over its lifespan. For many teams, especially those new to risk assessments, this will be an educational and goal-setting process. In identifying risks and vulnerabilities, teams can begin to integrate project-specific resilience strategies aimed at mitigating the impacts of natural hazards and enhancing the project's adaptation capacity.

CLIMATE AND NATURAL HAZARD RISK ASSESSMENT

A climate and natural hazard assessment is a systematic process to identify, evaluate, and understand the potential risks that climate change and natural hazards pose to a specific project

4. Maialen Iturbide, et al., "Implementation of FAIR Principles in the IPCC: The WGI AR6 Atlas Repository." *Scientific Data* 9 (2022): 629, https://doi.org/10.1038/s41597-022-01739-y.

and its functionality. This assessment considers observed and projected hazards, including extreme weather events, such as droughts, floods, wildfires, tornadoes, and hurricanes, as well as long-term phenomena such as sea level rise. The analysis involves gathering data on the likelihood, severity, and timing of these hazards using the IPCC's emissions scenarios. Key elements of the assessment include evaluating the project's exposure, sensitivity, adaptive capacity, and vulnerability to these risks, as well as identifying the overall hazard risk levels.

Climate risk emerges from the interaction of hazard, exposure, and vulnerability.[5] Hazards include climate-related physical events or trends that can cause damage or loss, whereas exposure encompasses the presence of assets, services, resources, and infrastructure that may be affected. For the purposes of this credit, natural hazards include drought, extreme heat or cold, flooding, hurricanes and high winds, hailstorms, landslides, sea level rise, storm surge, tornadoes, tsunamis, wildfires and smoke, and winter storms.[6] For each identified hazard, the team must complete the climate resilience assessment template or an equivalent, documenting exposure, risk levels, and potential mitigation strategies.

Vulnerability is the tendency or predisposition to experience negative effects. Land use, public infrastructure, the burden of disease in the population, and previous exposure to hazards can increase vulnerability. A climate and natural hazard assessment evaluates the potential risks climate change and natural hazards pose to a project, helping to identify, analyze, and plan for these risks to protect the long-term safety, functionality, and resilience of infrastructure, communities, and ecological systems.

Identification of priority hazards

Based on the assessment, project teams must identify at least two priority hazards by evaluating site-specific climate conditions, historical hazard data, projected future risks, and the building's exposure, sensitivity, and adaptive capacity. The assessment should incorporate regional climate models, hazard mitigation plans, and available climate risk databases to determine the likelihood and severity of each potential hazard. Teams should also consider how local infrastructure, soil conditions, and water management systems may exacerbate or mitigate risks.

To support this evaluation, project teams can use tools such as the *FEMA National Risk Index,*[7] *NOAA Climate Explorer,*[8] or state and municipal hazard mitigation and adaptation plans to identify patterns of past and projected hazard events. Stakeholder engagement with local authorities, utilities, and community resilience groups can further inform risk prioritization. For example, if a site is in a flood-prone area with increasing extreme precipitation events, flooding may be identified as a priority hazard due to its potential to damage flooring, walls, and electrical systems. Similarly, in regions experiencing rising temperatures, extreme heat may be prioritized because of its impact on material degradation and increased cooling loads.

5. International Energy Agency, "Climate Hazard Assessment," Climate Resilience Policy Indicator, 2022, https://www.iea.org/reports/climate-resilience-policy-indicator/climate-hazard-assessment.
6. D. D. Saulnier, A. M. Dixit, A. R. Nunes, and V. Murray, "Disaster Risk Factors—Hazards, Exposure and Vulnerability," WHO Guidance on Research Methods for Health Emergency and Disaster Risk Management, accessed March 31, 2025, https://extranet.who.int/kobe_centre/sites/default/files/pdf/WHO%20Guidance_Research%20Methods_Health-EDRM_3.2.pdf.
7. FEMA, "The National Risk Index," accessed on April 2, 2025, https://hazards.fema.gov/nri/.
8. NOAA, "The Climate Explorer," accessed April 2, 2025, https://crt-climate-explorer.nemac.org/.

Once teams identify priority hazards, they must document findings in the USGBC climate resilience assessment template or submit an equivalent external assessment tool. This process allows for climate risks to be systematically analyzed and integrated into the project's planning, design, and operational strategies.

Assessing hazards

After project teams identify two priority hazards, they must evaluate the impact by specifying the IPCC emissions scenario[9] used in the assessment, which outlines possible future atmospheric GHG concentrations. Teams should define scenarios that are both acceptable and appropriate for the project's geographic location, taking into account local climate action plans to guide selection. For instance, projects aiming to align with ambitious global climate mitigation goals or for those with shorter lifespans (20–30 years) should use SSP 1–2.6 (*Low Emissions Scenario*),[10] where less severe climate impacts are expected. Conversely, projects in high-risk areas, those with longer lifespans (50+ years), or where significant climate impacts are anticipated due to limited mitigation measures or regional vulnerabilities find *SSP 5–8.5* (High Emissions Scenario) more suitable.

Next, teams must define the projected service life of the project. For LEED projects, the projected service life refers to the project's expected lifespan, which could extend to fiscal year 2050 or up to 100 years, during which the assessment of hazard risks remains applicable. Project teams must also evaluate the hazard level and assign a hazard risk rating based on the potential severity and impact. The hazard level reflects the intensity of a specific hazard event and is categorized as low, medium, or high. Project teams should report and identify the level of each potential hazard using historical data and future projections. Teams can refer to the IPCC climate projections for historical data or climate projections. The hazard risk rating typically comes from local or regional hazard mitigation plans, and it assesses the likelihood of a hazard occurring. Teams must provide a risk rating for each identified hazard.

Project teams must assess the project's susceptibility to each hazard and consider multiple factors, including exposure, sensitivity, adaptive capacity, vulnerability, and overall risk levels. Exposure refers to the degree to which the project is vulnerable to hazards, such as its proximity to water bodies or seismic zones. Sensitivity indicates how significantly these hazards may affect the project based on factors like materials, structural design, and infrastructure. Adaptive capacity is the project's ability to adapt, withstand, or recover from the impacts of hazards through resilient design, contingency planning, or technology. A project with backup generators and multiple water supply systems has a higher adaptive capacity than a project without them.[11]

Vulnerability reflects the project's overall susceptibility, considering both its sensitivity and adaptive capacity. The overall risk level is a combined assessment of hazard severity, exposure, sensitivity, adaptive capacity, and vulnerability categorized as low, medium, or high.

Project teams must evaluate the potential impacts of each hazard on the project. Project hazards can disrupt essential services, damage structures, cause long-term operational challenges, and

9. IPCC, "IPCC Working Group I: Sixth Assessment Report," accessed on April 2, 2025, https://interactive-atlas.ipcc.ch/.
10. Iturbide, "Implementation of FAIR principles in the IPCC."
11. R. Askar, L. Bragança, and H. Gervásio, "Adaptability of Buildings: A Critical Review on the Concept Evolution," *Applied Sciences* 11, no. 10 (2021): 4483, https://doi.org/10.3390/app11104483.

impact project operations. In the operations phase, the assessment must consider how these hazards affect facility management, occupant safety, and service continuity. This includes potential disruptions to building systems, energy and water supply, indoor air quality, emergency preparedness, workforce safety, and the long-term performance of structural and mechanical components. Hazards may also affect maintenance schedules, operational costs, and the ability to provide critical services during extreme weather events.

Integration into project planning and design

As part of the climate resilience assessment, use the findings from the hazard assessment to guide the project's planning, design, operations, and maintenance phases. For instance, in the planning and design phase, teams can select less vulnerable areas on-site and implement resilient site-design strategies such as elevated foundations. The team must describe how the climate resilience assessment shaped design strategies, particularly to mitigate priority hazards, while also demonstrating alignment with specific LEED credits.

For example, if flooding is identified as a priority hazard, strategies could involve raising a building's foundation, using permeable paving, or incorporating drought-resistant landscape features, which also support LEED credits such as *SSc3: Rainwater Management* and *SSc4: Enhanced Resilient Site Design*.

Another example is when operations teams optimize HVAC performance (supporting EA credits) and implement adaptive occupant comfort strategies (supporting EQ credits) if extreme heat is a concern.

Where possible, integrate resilience considerations into project planning, design, and construction to ensure the building is adaptable to changing climate conditions. Incorporating risk-informed strategies early in the process allows teams to design for long-term durability, occupant safety, and operational continuity. Identifying site-specific hazards and vulnerabilities enables the development of design solutions that mitigate natural hazards, enhance resilience, and minimize future disruptions. This approach supports building performance, occupant well-being, and life-cycle sustainability by embedding resilience measures into structural, mechanical, and site-design strategies. Through proactive planning and design, teams can better sustain functionality, occupant well-being, and asset longevity and ensure resilience measures remain effective throughout the project's lifespan. This process facilitates adaptation to evolving threats and preserves the building's functionality and safety throughout its expected lifespan, contingent on effective strategy implementation.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	The project's climate and natural hazard assessment (using the LEED v5 Climate Resilience Assessment template or equivalent)

REFERENCED STANDARDS

- Intergovernmental Panel on Climate Change, Working Group I, Sixth Assessment Report (https://interactive-atlas.ipcc.ch/)
- FEMA National Risk Index (https://hazards.fema.gov/nri/)
- NOAA Climate Explorer (https://crt-climate-explorer.nemac.org/)

FURTHER EXPLANATION

The following explanations expand on the steps outlined in the Climate Resilience Assessment template.

STEP 1: IDENTIFY PROJECT PARAMETERS

Provide the basic project information:

- Project name
- Location (city, state/province, country)
- Project type (e.g., residential, commercial, healthcare)
- Project size (e.g., gross square meters/footage)
- Number of occupants
- Time duration facility can afford disruption (in hours). This is the amount of time the facility can be offline with no power, water, heating, cooling, or ventilation. For critical facilities, such as hospitals, this may be established by jurisdictional regulations and may be no longer than a few minutes. For some projects, such as commercial office buildings where occupants can work from home if needed, this may be days or even weeks. For projects where there is no specific code requirement, client will establish the duration based on their project requirements.

STEP 2: CLIMATE CHANGE BASIS: DETERMINE THE DESIGN SERVICE LIFE FOR THE PROJECT (SEE DEFINITIONS FOR DESIGN SERVICE LIFE)

- Determining the design service life is essential to inform all aspects of a project, from siting to systems to material selection. Jurisdictional regulations may dictate the minimum design service life, especially for critical facilities such as hospitals and schools. For all other projects, the project owner makes the final determination of design service life. Owners often ask for project team expertise in deciding what timeframe to select for the design service life.
- Professional organizations, such as the American Institute of Architects (AIA) and professional engineering associations, have recommendations for design service life based on the primary systems of a building. The AIA superspreadsheet from the Committee on the Environment (COTE) recommends design service life based on the primary structural framing system selected.[12] If the project is likely to be demolished in a short timeframe, then the design service life should reflect this, and a low-embodied carbon, reusable structural frame should

12. AIA, "COTE Top Ten Award," accessed September 8, 2025, https://www.aia.org/design-excellence/awards/cote-top-ten-awards.

be selected. If the project must stand for 200 years, then concrete, steel, mass timber, or a combination of these durable structural frames should be selected.

- Example 1: Temporary meeting hall in a public park that will be disassembled in three years when a community center, currently under construction, is completed. Recommended design service life: three years.
- Example 2: Public elementary school that will also be a community heating and cooling center and emergency meeting point. Recommended design service life: 100 years.
- Additional Considerations
  - Mechanical, electrical, and plumbing systems
  - Enclosure systems (roofs and façades)
  - Environmental product declarations (EPDs): The service life (Stage B) used for life-cycle analyses in EPDs is typically 50 years. If the project has a design service life of 30 years or less and the materials have much longer service lives, consider selecting products that can and will be reused at the end of the project's design service life.
- Resources
  - AIA COTE Superspreadsheet (download is at the bottom of the page). See Tab 09 "Change," cell 33J.[13]

STEP 3: CLIMATE CHANGE BASIS: SELECT THE EMISSIONS SCENARIO FOR THE PROJECT

- The emissions scenario is the Intergovernmental Panel on Climate Change (IPCC) Shared Socioeconomic Pathway that represents anticipated future climate for the project site. Start by checking the owner's project requirements—the project owner or their insurance company may have an emissions scenario for which the project must be designed. Jurisdictions may also have a designated scenario in their climate change action, hazard mitigation, or disaster risk management plans. If there is no predetermined scenario, the project team works together to select a scenario based on their best estimate of future global greenhouse gas emissions.
- The five emissions scenarios are described as follows. Detailed information can be found in the IPCC report:
  - SSP1-1.9 = Sustainability I (very high and immediate emissions reductions—the most optimistic scenario). This scenario does not represent the global emissions outlook in 2025 and is therefore not recommended as a default. It is grayed out in the Climate Resilience Assessment template.
  - SSP1-2.6 = Sustainability II (high and immediate emissions reductions). This scenario does not represent the global emissions outlook in 2025 and is therefore not recommended as a default. It is grayed out in the Climate Resilience Assessment template.
  - SSP2-4.5 = Middle of the Road (moderate emissions reductions).
  - SSP3-7.0 = Regional Rivalry (no emissions reductions). This scenario is not reflected in several science-based climate modeling tools (e.g., Climate Mapping for Resilience and Adaptation (CMRA) assessment tool[14]), so it is grayed out in the Climate Resilience Assessment template.

13. AIA, "COTE Top Ten Award."
14. CMRA, "Climate Mapping for Resilience and Adaptation," accessed September 8, 2025, https://resilience.climate.gov.

- SSP5-8.5 - Fossil-Fueled Development (no emissions reductions; acceleration of carbon-based fuel consumption).

- Resources
  - IPCC Sixth Assessment Report[15]

STEP 4: HAZARD ASSESSMENT

- Establish the climate changes that are likely to impact the project during its design service life. Consult science-based tools to determine climate changes for the project site; for example, increased extreme heat days, increased storm intensity, and increased drought.
- There are a number of free resources available with this information at a country, regional, and state/province level. Some Canadian provinces, U.S. states, and counties have granular data to further refine results to the jurisdictional or individual property level.
- Resources:
  - U.S. Climate Explorer Toolkit: This is a U.S. governmental tool, based on the Fourth IPCC report, with instructions for converting to the Fifth IPCC report. It covers the U.S. only.[16]
  - CMRA: This is a U.S. governmental tool that covers the U.S. only.[17]
  - Building Resilience Index: This is a global tool. Granularity of the climate snapshots varies by country.[18]
  - CAL-Adapt Local Climate Change Snapshot Tool: This provides the state of California's climate change snapshots by address.[19]
  - Our Coast Our Future: This includes the state of California ocean and bay coast maps with an interactive tool to assess flooding by address.[20]
- Many jurisdictions around the world have local climate change adaptation plans, hazard mitigation plans, disaster risk management plans, or equivalents that identify the local climate-related hazards.
  - In the U.S., these often exist at the regional, state, county and municipality level. State-level hazard mitigation plans are required by federal law, are registered with the Federal Emergency Management Agency (FEMA), and are listed on the FEMA Hazard Mitigation Plan Status website.[21]
  - Local utility districts (power, water, sewer), transportation authorities, school districts, and other local governmental organizations may also have plans with this information.
  - Many of these plans have been legally adopted by local or regional authorities, in which case they will be part of project code compliance.

15. IPCC, "Sixth Assessment Report," accessed September 8, 2025, https://www.ipcc.ch/assessment-report/ar6/.
16. NOAA, "The Climate Explorer: Explore How Climate is Projected to Change in any County in the United States," accessed September 8, 2025, https://crt-climate-explorer.nemac.org.
17. CMRA, "Climate Mapping for Resilience and Adaptation."
18. International Finance Corporation, "Building Resilience Index," accessed September 8, 2025, https://www.resilienceindex.org.
19. California Energy Commission, "Cal-Adapt: Data Driven Tools for a Resilient Future," accessed September 8, 2025, https://cal-adapt.org/tools/local-climate-change-snapshot/.
20. USGS Pacific Coastal and Marine Science Center, "Our Coast, Our Future," accessed September 8, 2025, https://ourcoastourfuture.org/about/.
21. FEMA, "Hazard Mitigation Plan Status (HMPS)," accessed September 8, 2025, https://hazards-fema.maps.arcgis.com/apps/webappviewer/index.html?id=aeb0e462543b4fa69aeaf858945e1262.

- Use the most current hazard mitigation plan, disaster risk management plan, or climate change adaptation plan from the local jurisdiction to conduct an initial hazard assessment for the project.

STEP 4.1: HAZARD ASSESSMENT: CURRENT HAZARD LEVEL

- Use the most current local jurisdictional hazard mitigation plan, disaster risk management plan, climate adaptation plan or equivalent to identify all jurisdictionally recognized relevant hazards for the project site.
  - For each applicable hazard, complete the assessment by selecting the current hazard level from the drop-down menu (choose not applicable (NA), low, medium, or high) for each hazard.
  - Note: Not all hazards will apply to the project site. Provide reasoning for not addressing any identified hazard(s). The hazard list in the Climate Resilience Assessment template is not exhaustive. Insert additional hazards identified by the local plans and/or the project team.
- Additional Guidance—General
  - For jurisdictions where hazard mitigation plans are not available or for site-specific hazard identification, consult with an environmental scientist or qualified risk/compliance officer to address all relevant conditions for the project site where possible. Use the following resources to identify high hazard conditions. For projects outside the U.S., use local science-based tools to identify project-specific hazards. Those tools should provide local equivalents of the indicated information.
- Additional Guidance by Hazard
  - **Drought**
    - Identify drought risk: Determine whether the project is in an area that has experienced moderate, severe, or extreme drought conditions for more than 25% of the time in the past 30 years. For the U.S., please refer to CMRA,[22] NOAA Historical Palmer Drought Indices,[23] and National Integrated Drought Information System—Drought Portal.[24]
  - **Earthquake**
    - Identify seismic risk in the U.S.by determining whether the project is within FEMA Earthquake zones SDC C (yellow), D (orange), or E (red). For projects outside the U.S., use local science-based tools that indicate the likelihood of earthquake occurrence.
    - If seismic risk is identified in the referenced FEMA zones listed, where possible, pursue further analysis of site and surrounding structures and service to the site. Consider the secondary hazard of fire-following-earthquake where earthquakes are a high hazard. For the U.S., refer to FEMA Earthquake Hazard maps.[25]
    - For more detailed studies and international resources see the U.S. Geological Survey (USGS) Earthquake Hazards Program[26] maps and the National Climate Assessment.

22. CMRA, "Climate Mapping for Resilience and Adaptation," accessed September 8, 2025, https://resilience.climate.gov/.
23. NOAA, "Palmer Drought Severity Index," accessed September 8, 2025, https://www.ncei.noaa.gov/access/monitoring/historical-palmers/.
24. NOAA, "National Integrated Drought Information System—Drought Portal," accessed September 8, 2025, https://www.drought.gov/data-maps-tools.
25. FEMA, "National Risk Index: Earthquake," accessed September 19, 2025, https://hazards.fema.gov/nri/earthquake.
26. USGS, "Earthquake Hazards Program: Hazards," accessed September 19, 2025, https://www.usgs.gov/programs/earthquake-hazards/hazards.

  - *ASCE 7* Hazard by Location tool uses addresses to provide hazard information.[27]
- **Extreme Heat and Cold**
  - Identify if the project is located in an area subject to extreme heat events. For the U.S., please refer to NOAA Climate Information,[28] and U.S. Climate Explorer Tool.[29]
- **Flooding**
  - Identify flood risk; determine whether project is located within a 500-year floodplain. If that information is not available, use the 100-year floodplain and add three feet (one meter) to that measurement. In addition, identify if the project includes a risk, namely, any activity or element for which even a slight chance of flooding would be too great. For the U.S., please refer to CMRA,[30] FEMA Flood Map Service Center web search portal,[31] and FM Global Flood Map.[32]
- **Hurricane and High Winds**
  - Identify hurricane and high-wind risk by determining whether the project site is located within the hurricane or high-wind susceptible region. In determining the type of risk, distinguish whether the project site has hurricane risk combined with high-wind risk, wind-borne debris, and storm surge, or if it is a project site with high-wind (non-hurricane) risk alone.
  - Hurricane-prone regions are areas vulnerable to hurricanes as defined in *ASCE 7*. For *ASCE 7-10* and *ASCE 7-16*, hurricane-prone regions are locations along the Gulf of Mexico and Atlantic coasts where the wind speed for Risk Category II buildings is greater than 115 miles per hour (185 kilometers per hour). Also included are the entirety of Hawaii, Puerto Rico, the Virgin Islands, Guam, the Northern Mariana Islands, and American Samoa. For Risk Categories III and IV, see *ASCE 7-10* or *ASCE 7-16*, Chapter 26. Note that hurricane wind speeds are increasing. Always seek the most current information.
  - For the U.S., refer to *ASCE 7-10, Minimum Design Loads for Buildings and Other Structure* (Table 1.5-1); and *ASCE 7* Windspeed tool (where sites can be looked up by address).[33]
- **Landslides and Unstable Soils**
  - Identify relative landslide, subsidence, and liquefaction risk, and critical slopes risk that may indicate unstable soil. Refer to state and local data to determine whether the project is located in a high-risk landslide area and whether there is land on or uphill of the building site with a slope exceeding 15% (6.75°). For general use only, not for site-specific information, use the U.S. Landslide Inventory and Susceptibility Map maintained by the USGS.[34]

27. ASCE, “ASCE Hazard Tool,” accessed September 8, 2025, https://ascehazardtool.org.
28. NOAA, “Climate Monitoring: Latest Monitoring Information,” accessed September 8, 2025, https://www.ncei.noaa.gov/access/monitoring/products/.
29. NOAA, “The Climate Explorer.”
30. CMRA, “Climate Mapping for Resilience and Adaptation.”
31. FEMA, “Flood Map Service Center,” accessed September 8, 2025, https://msc.fema.gov/portal/home.
32. FM Global, “Flood Map,” accessed September 8, 2025, https://www.fm.com/resources/nathaz-toolkit/flood-map.
33. ASCE, “ASCE/SEI 7-10: Minimum Design Loads for Building and Other Structures,” accessed September 8, 2025, https://www.waterboards.ca.gov/waterrights/water_issues/programs/bay_delta/california_waterfix/exhibits/docs/dd_jardins/DDJ-148%20ASCE%207-10.pdf; and ASCE, “ASCE Hazard Tool.”
34. USGS, “U.S. Landslide Inventory and Susceptibility Map,” accessed September 8, 2025, https://www.usgs.gov/tools/us-landslide-inventory-and-susceptibility-map.

- **Sea Level Rise and Storm Surge**
  - Identify impacts associated with sea level rise. Consider additional storm surge impacts in connection with hurricanes and tropical storms. In the U.S., use NOAA's Sea, Lake, and Overland Surges from Hurricanes (SLOSH) model data to interpolate storm surge.[35] In the absence of SLOSH model data, use the Surging Seas Threat Map and Forecasting Tools to establish storm surge scenarios that take sea level rise into account.[36] Sea level rise and storm surge for 2025 and beyond should use the NOAA 2012 Sea Level Rise year 2080 High Scenario combined with a 1 in 100-year floodplain (extreme flood) to determine water levels for baseline planning purposes. For projects outside the U.S., use local science-based tools that indicate storm surge and sea level rise for the project emissions scenario and design service life. For the U.S., refer to CMRA, NOAA's SLOSH tool, National Hurricane Center Storm Surge Risk Maps, Surging Seas Threat Map and Forecasting Tools for Sea Level Rise + Storm Surge, and NOAA Sea Level Rise Viewer.[37]
- **Tornado**
  - Identify tornado risk by determining whether project site is located above EF3 (136 miles per hour (219 kilometers per hour) to 156 miles per hour (251 kilometers per hour)) tornado activity area.
  - For the U.S., refer to *ASCE 7* Windspeed tool, FEMA Wind Zone Map (Figure I.1, Tornado activity), and NOAA National Climatic Data Center's U.S. Tornado Climatology.[38]
- **Tsunami**
  - Identify tsunami risk by determining whether the project is within a tsunami inundation zone. For projects outside the U.S. note that international projects may use the U.S. standard or a local equivalent, whichever is more stringent. Refer to Pacific Northwest Seismic Network's Tsunami maps, the International Tsunami Events Map from NOAA, and the National Hurricane Center Storm Surge Risk Maps.[39]
- **Wildfire and Smoke**
  - Identify wildfire risk by determining whether project is located in an area that is designated high or very high wildfire hazard potential. Also consider urban fires in densely populated urban residential areas with older construction without interior sprinklers. For projects outside the U.S., use local science-based tools that indicate the likelihood of wildfire occurrence for the project emissions scenario over the design service life. For inside the U.S., refer to CMRA and USGS Federal Fire Occurrence Map Viewer.[40]

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35. NOAA, "Sea, Lake, and Overland Surges from Hurricanes: SLOSH Display Program," accessed September 8, 2025, https://vlab.noaa.gov/web/mdl/sdp.
36. Climate Central, "Surging Seas Threat Map and Forecasting Tools for Sea Level Rise + Storm Surge," accessed September 8, 2025, https://sealevel.climatecentral.org.
37. CMRA, "Climate Mapping for Resilience and Adaptation"; NOAA, "Sea, Lake, and Overland Surges from Hurricanes"; NOAA, "National Hurricane Center Storm Surge Risk Maps," accessed September 8, 2025, https://experience.arcgis.com/experience/203f772571cb48b1b8b50fdcc3272e2c/page/Category-5; Climate Central, "Surging Seas Threat Map and Forecasting Tools"; and NOAA, "Sea Level Rise Viewer," accessed September 8, 2025, https://coast.noaa.gov/digitalcoast/tools/slr.html.
38. ASCE, "ASCE Hazard Tool"; FEMA, "Section I: Understanding the Hazards," accessed September 8, 2025, https://www.fema.gov/pdf/library/ism2_s1.pdf; NOAA, "U.S. Tornadoes," accessed September 8, 2025, https://www.ncei.noaa.gov/access/monitoring/tornadoes/.
39. Pacific Northwest Seismic Network (PNSN), "Tsunami Hazard Maps," accessed September 8, 2025, https://pnsn.org/outreach/hazard-maps-and-scenarios/eq-hazard-maps/tsunami; NOAA, "Natural Hazards Viewer," accessed September 8, 2025, https://www.ncei.noaa.gov/maps/hazards/; and NOAA, "National Hurrican Center Storm Surge Risk Maps."
40. CMRA, "Climate Mapping for Resilience and Adaptation"; and USGS, "Fire Occurrence Map Viewer: Data Services Page," accessed September 8, 2025, https://firedanger.cr.usgs.gov/apps/data-services-page.

- **Winter Storm**
  - Identify if the project is located in an area subject to frequent winter storms. Assess whether winter storms could impact essential operations or access to the site. For projects outside the U.S., use local science-based tools that indicate the anticipated frequency and intensity of winter storms for the projected emissions scenario over the design service life. For inside the U.S., refer to the ASCE Hazard Tool.[41]

STEP 4.2: SERVICE LIFE HAZARD LEVEL

- Using the design service life and emissions scenario selected for the project, assess the hazard level at the worst-case point in the project lifetime.
- Example: A project with a design service life of 100 years and emissions scenario SSP3-7.0 will look at predicted flooding 100 years from project completion at levels based on no emissions reductions.
- There may be hazards that will occur during the project design service life, based on the selected emissions scenario, that are not represented in current local plans. For example, extreme heat is a hazard to most projects but is not always included in local plans.

STEP 4.3: RISK RATING

- Select the risk rating from the drop-down menu (NA, low, medium, high) for each hazard. The risk rating can be found in local jurisdictional plans or local equivalent sources.
- For identified hazards that are not represented in the local plan, the project team should determine the risk based on the selected emissions scenario and design service life.

STEP 4.4: EXPOSURE

- Exposure is the presence of the project in a place where it could be adversely affected by a hazard.
- Select the exposure level from the drop-down menu (NA, low, medium, high) for each hazard, and adjust for the project site and type. For example, flooding may be a hazard identified by the local jurisdiction's hazard mitigation plan, but if the project site is 300 feet (100 meters) above the projected future flood plain, the project itself has low exposure to that hazard. However, if the project is dependent on infrastructure that will be exposed to that hazard—for example, for a hospital that needs to remain accessible during a flood—consider raising the exposure level.
- For identified hazards that are not represented in the local plan, the project team should determine the risk based on the selected emissions scenario and design service life.

STEP 4.5: SENSITIVITY

- Sensitivity is the degree to which a system, population, or resource is or might be affected by that hazard. For example, a primary school has high sensitivity to extreme heat, while a warehouse will have low sensitivity if the contents are unaffected by thermal conditions.
- Select the sensitivity from the drop-down (NA, low, medium, high) for each hazard.

41. ASCE, "ASCE Hazard Tool."

- For identified hazards that are not represented in the local plan, the project team should determine the sensitivity level based on the project site and type.

STEP 4.6: ADAPTIVE CAPACITY

- Adaptive capacity is the ability of a person, asset, or system to adjust to a hazard, take advantage of new opportunities, or cope with change. For example, if a project is at high risk and has high exposure to extreme heat but is designed to handle extreme heat, or can be easily modified to accommodate extreme heat, it has high adaptive capacity for that hazard. If a project is at high risk and has high exposure to sea level rise but is not designed to accommodate that level of sea level rise, it has low adaptive capacity.
- Select the adaptive capacity level from the drop-down (NA, low, medium, high) for each hazard.
- For identified hazards that are not represented in the local plan, the project team should determine the adaptive capacity based on the project site, type, and design brief.

STEP 4.7: POTENTIAL IMPACT

- Potential impact is the economic, environmental, and social influence a physical phenomenon can have on a project. These are the effects on natural and human systems that result from each hazard.
- Select the potential impact level from the drop-down (NA, low, medium, high) for each hazard.
- For identified hazards that are not represented in the local plan, the project team should determine the potential impact based on the project site, type, and design brief.

STEP 4.8: VULNERABILITY

- Vulnerability expresses the degree to which physical, biological, and socioeconomic systems are susceptible to, and unable to cope with adverse impacts of natural hazards and climate change. Vulnerability encompasses exposure, sensitivity, potential impacts, and adaptive capacity.
- Take into account the expected service life of the project and identify vulnerabilities based on changes predicted to take place during that time period.
- Select the vulnerability level from the drop-down (NA, low, medium, high) for each hazard. Solicit input from the project owner to adjust the vulnerability level, if necessary.
- For identified hazards that are not represented in the local plan, the project team should determine the vulnerability based on the project risk rating, exposure, sensitivity, adaptive capacity, and potential impact.

STEP 4.9: PRIORITY HAZARD DESIGNATION

- A priority hazard is one that poses the most serious and immediate risk.
- Based on the local plan and the results of Steps 4.1 through 4.8, determine at least two priority hazards for the project. Select NA, yes, or no from the drop-down.
- Priority hazards for the project may differ from those in the local plan, depending on the results of Steps 4.1 through 4.8.

STEP 5: SOURCES

- Upload hyperlinks to the referenced local hazard mitigation, disaster risk management, and/or climate change adaptation plans. If local plans do not incorporate climate change projections for the full design service life, upload screenshots of referenced pages from science-based tools for the project site (e.g., U.S. Climate Resilience Toolkit, CAL-Adapt). If additional references were used, upload screenshots of the referenced pages.

STEP 6: DEVELOP PRIORITY HAZARDS ACTION NARRATIVE

- Provide a narrative describing potential impacts of the priority hazards to the project site and building operations during construction and after occupancy. Explain how these potential impacts informed project decisions and identify any specific design and/or operations strategies implemented in response. Include the following:
  - The specific design and/or operations strategies that will be used to mitigate and/or adapt to each priority hazard.
  - Expected outcomes (e.g., elevation above flooding, improved comfort during extreme heat).
  - Keep the narrative brief but action oriented. This summary helps connect the assessment to real project outcomes and resilient strategies.

Human Impact Assessment

IPp2
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To ensure that project development is guided by a thorough understanding of the social context of the local community, workforce, and supply chain and to address potential social inequities.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Human Impact Assessment	

HUMAN IMPACT ASSESSMENT

Complete a human impact assessment that draws on relevant information from the following four specified categories, as applicable:

- **Demographics:** may include race and ethnicity, gender, age, income, employment rate, population density, education levels, household types, and identification of nearby vulnerable populations.
- **Local infrastructure and land use:** may include adjacent transportation and pedestrian infrastructure, adjacent diverse uses, relevant local or regional sustainability goals/commitments, and applicable accessibility codes.
- **Human use and health impacts:** may include housing affordability and availability, access to social services (e.g., healthcare, education, and social support networks), community safety and local community groups, and supply chain and construction workforce protections.
- **Occupant experience:** may include an opportunity for daylight, views, and operable windows; environmental conditions of air and water; and adjacent soundscapes, lighting, and wind patterns within the context of the surrounding buildings (e.g., a microclimate, a solar scape, neighboring structures).
- **Other:** (specify).

Where possible, use the information from the assessment to inform the planning, design, and operations and maintenance of the project and describe how project-specific strategies were considered.

REQUIREMENTS EXPLAINED

This prerequisite requires that the project team comprehensively evaluate and understand the social, economic, and environmental context of the local community, workforce, and supply chain before developing the project. To support this, teams should use methods such as site analysis, community outreach, census reports, GIS mapping, and partnerships with local organizations. This will help ensure the project aligns with community needs and promotes equitable outcomes. Teams must select key characteristics to evaluate within the categories of demographics, local infrastructure and land use, human use and health impacts, and occupant experience. These analyses and findings will guide the project's planning, design, operations, and maintenance strategies.

Ultimately, this assessment balances environmental goals with the needs and aspirations of the people affected, which fosters projects that are both ecologically and socially responsible. It supports frameworks for how designers ensure the health, safety, and welfare of those they design for. This will be an educational process for teams, especially for those who have not previously conducted similar assessments. With intentional planning, teams can integrate project-specific strategies to identify potential disparities and work collectively toward creating a more inclusive and equitable community.

HUMAN IMPACT ASSESSMENT

A human impact assessment is a process in which quantitative and qualitative data for a proposed project are collected through identifying characteristics unique to the project site and its surrounding community. Projects are required to consider the sociopolitical context of the site and the cultural makeup of neighboring residents or average income rates. Additionally, projects must identify relevant infrastructure and policy such as zoning restrictions or accessibility codes. The assessment also requires taking stock of what resources may be accessible to the residents or potential end users, including transit availability or healthcare. Last, it evaluates impacts on occupant experience, such as air and water quality. Project teams must consider how these elements interact with and impact each other to ensure that they guide project development with a comprehensive understanding of its social context.

Defining community

Project teams must first establish the scope of their assessment by identifying the community. Communities have both geographic and functional definitions. Geographic communities start with the project's neighborhood, which includes the people who live and work in and near the project and interact with it by proximity. Geographic communities can also extend beyond to include towns, cities, or counties. Functional communities include all occupants, construction workers, and visitors who come to the building. These people may or may not live nearby. Teams can shape community through various affinities or commonalities, such as age, ethnicity, income level, housing status, or educational background. The community may extend to include project team members such as architects, engineers, contractors, and designers who oversee the planning, design, and construction phases. Community in the supply chain includes material suppliers, manufacturers, distributors, and the workforce involved in production and transportation. This highlights the importance of local engagement and fair labor practices.

Address the core categories of human impact

Teams must complete a thorough assessment that evaluates the potential impact of the project on people, including living conditions, health, food security, education, and access to other resources. The assessment must include data collection and analysis of core human impact categories, such as demographics, infrastructure, health, and occupant experience factors, as well as any other relevant social impacts identified, and provide a comprehensive overview of the human impacts of project development. Project teams encourage engagement with community members and other relevant groups to gather insights, understand local needs, and validate data.

DEMOGRAPHICS

The first category evaluates the local demographics of the area surrounding the project site, which is critical to understanding how the development may influence the social fabric of the surrounding community. This process involves analyzing key demographic characteristics, including factors such as race and ethnicity, gender, age distribution, income levels, employment rates, population density, education levels, and household types. Additionally, the project includes identification of nearby vulnerable populations to consider how their needs can be addressed by its development. Teams should collaborate with nonprofit organizations that work directly with the people of the community.

LOCAL INFRASTRUCTURE AND LAND USE

The second category examines the project's impact on local infrastructure and land use, as well as identifies existing infrastructure that provides an opportunity to connect to the project. Teams must assess the adjacent public transit systems, such as walkways, bike lanes, and road networks, to ensure the project integrates well with existing mobility options and promotes sustainable transportation. The evaluation requires an analysis of diverse land uses in the vicinity, such as residential, commercial, industrial, and recreational spaces to determine how the project might influence the functional balance of the area.

Review the local community's sustainability commitments, including city-wide goals to reduce GHG emissions or promote energy efficiency, so that the project can support broader efforts to create a more sustainable future. Identify and comply with relevant accessibility codes and standards and adhere to legal requirements regarding access for people with disabilities. Follow best practices for creating inclusive, barrier-free environments.

HUMAN USE AND HEALTH IMPACTS

The third category evaluates the project's impact on human use and public health and well-being. Thoroughly evaluate the community's current access to essential resources and the overall quality of life for residents. Assess whether the project will address or alleviate these challenges, particularly regarding the availability of affordable housing. Consider the community's access or proximity to social services, such as healthcare facilities, educational opportunities, and support networks.

Community safety is important for public health and well-being. Projects that incorporate public spaces, adequate lighting, and pedestrian-friendly designs can foster a sense of safety and belonging. Projects must consider the protection and working conditions for the local supply chain and construction workforce to ensure ethical practice and fair treatment. Within reason, prioritize local procurement and employment to support the local economy, provide fair wages and benefits to workers, establish safe jobsite conditions, and implement transparent labor practices to prevent exploitation and unjust treatment.

OCCUPANT EXPERIENCE

The fourth category considers the project's impact on the overall occupant experience. The goal is to thoroughly examine how the design and construction of the project can influence the health, comfort, and well-being of its occupants. Analyze key environmental factors such as the availability of natural daylight, the quality and orientation of views, the opportunity to provide operable windows for fresh air circulation, as well as air and water quality. Consider how external elements (surrounding soundscapes, the quality of artificial and natural lighting, and the impact of wind patterns on the building and adjacent structures) affect the indoor environment and the overall experience of those inhabiting the space. Projects should aim to create a positive and health-conscious environment for its occupants.

OTHER

Project teams may include any additional relevant social factors in the human impact assessment.

INTEGRATION INTO PROJECT PLANNING AND DESIGN

Project teams should use insight from the human impact assessment to inform the project's planning, design, operations, and maintenance phases. Consider how the identified social factors inform project-specific decisions, such as changes to design features, operational practices, or community engagement strategies. Implement strategies such as these to promote inclusivity, fair labor practices, and equitable access to opportunities, and support the community's economic and social well-being. This integration is a tool to drive meaningful change within the design as well as the community, thereby creating a more resilient and sustainable project outcome.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	The project's human impact assessment (using the LEED v5 Human Impact Assessment Template or equivalent).

REFERENCED STANDARDS

- None

FURTHER EXPLANATION

USING THE HUMAN IMPACT ASSESSMENT TEMPLATE AND DATA REPOSITORY

Note: The following explanations expand on the steps outlined in the Human Impact Assessment Template.

STEP 1: IDENTIFY PROJECT PARAMETERS

Provide the basic project information.

- Location (city, state/province, country)
- Census tract (if applicable)
  - This context will aid those using the HIA Data Repository to capture granular data that can be compared to larger regional statistics. Data is currently available for the U.S.
- Project use type/program (e.g., residential, commercial, healthcare)
  - This helps establish required programming needs while also presenting an opportunity to contextualize data collected.
- At what scale will the assessment be performed? (e.g., neighborhood, city, county, state)
  - This context is necessary to understand where the data is coming from and how relevant it is to the project site. State-level data will tell a much different story than comparing counties, neighborhoods, and even census tracts in the U.S.
- Notes
  - Use this space to capture any additional information or notes.

STEP 2: SELECT INDICATORS FOR EVALUATION

Review the list of indicators provided in each section of the assessment and determine which are most applicable to the project. Select at least one indicator per category that will be expanded upon through the assessment. The following indicators and topics are provided within the template and data repository (found in the Resources and Tips section of the credit library) using publicly available resources created by U.S. federal agencies, including FEMA, Centers for Disease Control and Prevention (CDC), Census Bureau, and U.S. Environmental Protection Agency, downloaded before January 21, 2025. Within the data repository are definitions of each indicator, their relationship with human impact design, as well as the source that provides these data assumptions for U.S. projects.

- Demographics: This category focuses on understanding the demographic characteristics of both the surrounding community and the people who are expected to use or occupy the project.
  - Race and ethnicity: Percentage of White, Black/African American, American Indian or Alaska Native, Hawaiian and Other Pacific Islander, Two or More Races, Hispanic or Latinx, Middle Eastern, or North African
  - Gender: Percentage of men, women, nonbinary, or other
  - Age: Percentage of children <5 years old, youth 10–24 years old, adults >65 years old

  - Income: Median household income, percentage of individuals below poverty level
  - Employment rate: Percentage of population 16 and over in labor force
  - Population density
  - Education levels: Percentage of adults ≥25 years without high school (HS)/GED degree, adults ≥25 years with HS/GED degree, adults ≥25 years bachelor or higher degree
  - Household types: Single-headed household, multigenerational household
  - Vulnerable populations: Social vulnerability index (SVI) score, environmental justice index (EJI) score, disability, and mental health
- Local Infrastructure and Land Use: This category considers the project's relationship to existing infrastructure, services, and land use patterns at the neighborhood or district scale.
  - Adjacent transportation and pedestrian infrastructure: Traffic safety, bus service, transit service, bike score, transit score, walk score
  - Adjacent diverse uses: Access to parks, diverse uses
  - Relevant sustainability goals or commitments: Local climate action policy
  - Applicable accessibility code(s): Number of Americans with Disabilities Act (ADA) violations
- Human Use and Health Impacts: This category addresses the potential effects of the project on public and environmental health, with a focus on both short- and long-term outcomes.
  - Housing affordability and availability: Housing insecurity, percent owner occupied residences
  - Availability of social services: Food insecurity
  - Community safety: Violent crime
  - Local community groups: Community group engagement, social associations
  - Supply chain and workforce protections: Supply chain, Occupational Health and Safety Administration (OSHA) construction violations, International Labour Organization (ILO) conventions, Design for Freedom, Cradle-to-Cradle Social Fairness Score, Mindful Materials Common Materials Framework Assessment of Risks
- Occupant Experience: This category focuses on understanding the conditions of the local environment and climate, including that of air, water, sound, and light.
  - Opportunity for daylight, views, and operable windows (indoor): Equitable access to daylight and views, tree canopy coverage
  - Environmental conditions of air and water: Air quality index (AQI), ground level ozone, ambient PM2.5, population density near highways, tap water quality
  - Adjacent soundscapes, lighting, and wind patterns: Noise pollution, sky view factor, wind exposure, street canyon

Consider the project type and goals, local community needs and characteristics, as well as operational or design opportunities that will help determine what would have the most impactful outcome for the specific project. If a narrative is written for each category instead, skip down to the narrative section to describe the project's relationship to the relevant indicators (to be explained in Step 3.1).

STEP 3: DOCUMENT FINDINGS IN THE ASSESSMENT TAB

After selecting the relevant indicators for the project, there are several rows to be filled out in the spreadsheet:

- Quantitative Findings: Enter relevant data or information related to this indicator. This could include statistics, observed conditions, or other key inputs like census data, local reports, or building performance measures. Focus on summarizing what the data shows before interpreting its implications.

- Data Source: List the source(s) that informed the findings (e.g., U.S. Census, municipal reports, stakeholder feedback). Indicate the scale of the data—whether it is site-specific, local, regional, or national. See the Assumptions table in the template and resources provided in the data repository.
- Data Year: Note the year the data was collected or published. If multiple years or sources are used, summarize accordingly.
- Qualitative Findings: Use this space to expand on the findings with qualitative insights, limitations, or additional context. This is where open-ended reflections can be captured before developing narratives and implementation strategies later in the process.

STEP 3.1: USING OTHER INDICATORS

If the project chooses to use a different indicator that is not in the template, please leave findings blank and skip to the category-wide narrative at the bottom or add an indicator to the “Other” category at the end, and follow the sequence in Step 3.

STEP 4: PRIORITIZATION EXERCISE

After completing the data collection portion of the assessment, flag which data points and findings are most relevant to the project and that may have the greatest impact if the design addresses this topic in some way.[42] Use these findings to highlight priority indicators and begin contextualizing an implementation strategy. After highlighting priority indicators, it is necessary to further distill the information to determine what co-benefits[43] are most impactful for the project and what implementation strategies may come out of this exercise. Like the Alignment Process,[44] this step aims to align design with climate, health, and equity goals. Compare the demographic indicators to factors such as environmental conditions, while comparing how potential strategies connect to larger goals.[45]

See the example from the Alignment Process health situation analysis (HSA) method adapted for LEED v5. By taking note of the environmental factors, comparing demographic information with vulnerability, and noting high priority and risk health conditions, design strategies can be directly correlated to health benefits. This process links Steps 4 and 5 of the assessment.

IP

STEP 5: DEVELOP IMPLEMENTATION NARRATIVE

Once completing the assessment and prioritization, develop a narrative that describes how the project will integrate lessons learned from the data collection and analysis. Consider specific strategies that can be used, why it is being chosen, who will be involved in the implementation

42. AIA, “Design for Equitable Communities—Framework for Design Excellence,” accessed September 19, 2025, https://www.aia.org/design-excellence/aia-framework-for-design-excellence/equitable-communities.
43. Adele Houghton, “‘Co-Benefits’ as a Lens Through Which COVID-19 Building Upgrades Can Advance Environmental Sustainability, Climate Mitigation and Adaptation, and Social Equity,” *Harvard Public Health Review,* 2021, 29, https://bcphr.org/29-article-houghton/.
44. Adele Houghton, “Alignment Process,” Architectural Epidemiology and Biosity, 2024, accessed September 19, 2025, https://www.alignmentprocess.org/.
45. Adele Houghton and Xiolin (Elle) Li, *The Alignment Process Playbook, Combining Neighborhood Data to Lived Experience* (Biositu and Harvard Graduate School of Design), 15–24, https://www.alignmentprocess.org/_files/ugd/ed4ae9_c10448939ba74d8ea4d8ff7bfe98e21d.pdf.

Figure 1. Health Situation Analysis—LEED v5 Example

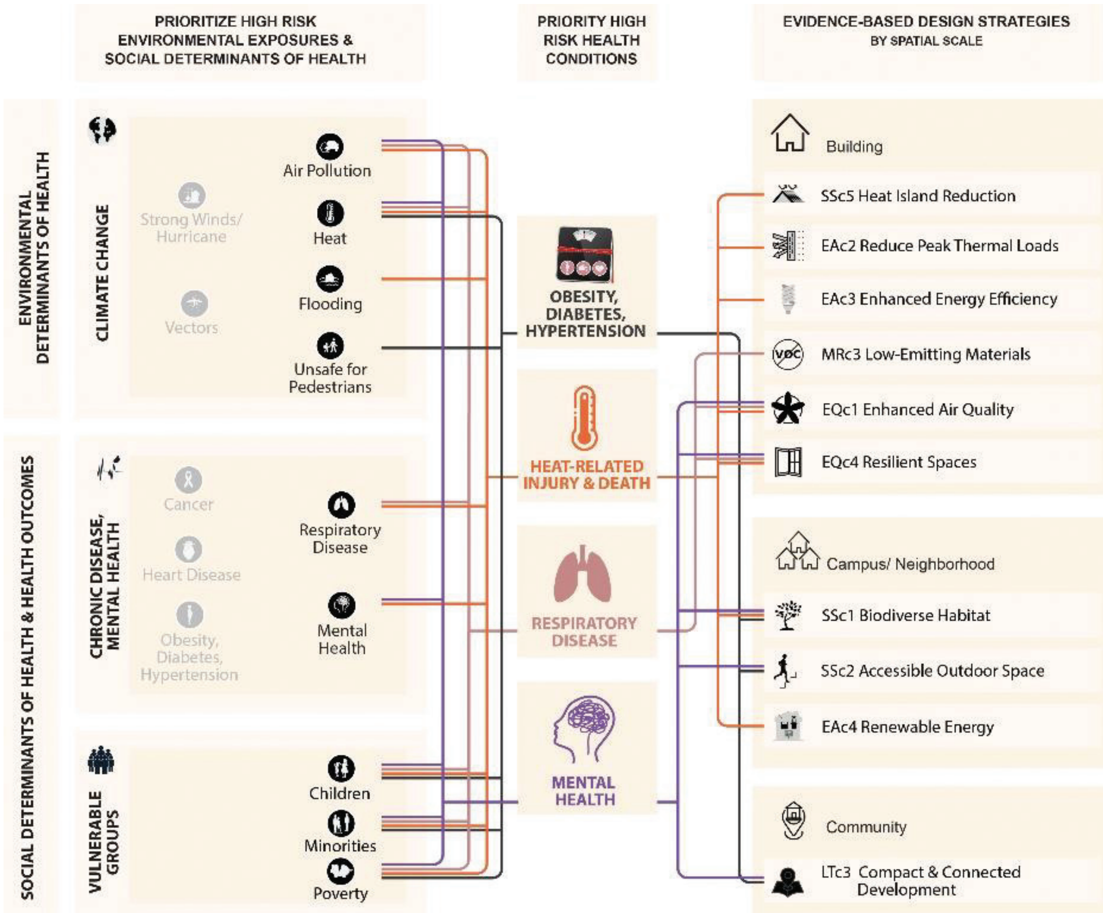

Image modified and reproduced with permission from: Houghton A and Castillo-Salgado C. *Architectural Epidemiology: Architecture as a Mechanism for Desgning a Healthier, More Sustainable, and Resilient World.* Johns Hopkins University Press: Baltimore, 2025.

of these strategies, and the expected benefits or outcomes. This is an opportunity to connect the assessment findings to actionable strategies throughout LEED credits.

SOURCES

- https://www.aia.org/design-excellence/aia-framework-for-design-excellence/equitable-communities
- https://www.alignmentprocess.org/ https://www.alignmentprocess.org/_files/ugd/ed4ae9_c10448939ba74d8ea4d8ff7bfe98e21d.pdf playbook Part 2: Combining Neighborhood Data and Lived Experience to Inform Design
- https://bcphr.org/29-article-houghton/

Carbon Assessment

IPp3
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To understand and reduce long-term direct and indirect carbon emissions, including on-site combustion, grid-supplied electricity, refrigerants, and embodied carbon.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Carbon Assessment	

CARBON ASSESSMENT

USGBC will provide the project team with a 25-year projection of the project's emissions from operations, refrigerants, and embodied carbon. The assessment will use the data from the following:

- EAp1: Operational Carbon Projection and Decarbonization Plan
- EAp5: Fundamental Refrigerant Management
- MRp2: Quantify and Assess Embodied Carbon
- LTc4: Transportation Demand Management (optional)

REQUIREMENTS EXPLAINED

This prerequisite requires the project team to conduct a 25-year carbon assessment of all emissions within the project boundary but does not require any additional data to be entered beyond what is already required by the three related prerequisites and the optional credit, if pursued.

Until recently, most projects only considered GHG emissions from operational energy use, if emissions were considered at all. However, as it has become increasingly clear that emissions from construction ("embodied carbon") and refrigerants can also be quite considerable, LEED v5 has introduced prerequisites to assess carbon emissions from all three sources as well as this unifying prerequisite, which enables project teams to compare their sources of emissions and see which are significant. This prerequisite takes the data from *EAp1: Operational Carbon Projection and Decarbonization Plan, EAp5: Fundamental Refrigerant Management, MRp2: Quantify and Assess Embodied Carbon*, as well as *LTc4: Transportation Demand Management*, if that credit has been pursued. It then provides project teams with a report and visualization showing how their different sources of emissions will compare over a 25-year time horizon. The goal is to promote more informed carbon-related decision-making.

CARBON ASSESSMENT

USGBC will develop a 25-year carbon assessment of the estimated emissions from energy use, refrigerants, embodied carbon, and for some projects, transportation from the data in the prerequisites and optional credit. It includes:

- Annual carbon emissions from each source for 25 years
- Cumulative emissions from each source each year for 25 years
- Cumulative emissions over 25 years in total and from each source and the percentage of the total from each source

Additional considerations

The information in this carbon assessment provides an overview of the various sources of carbon emissions and can help owners make informed decisions to reduce the project's emissions. Although not required for compliance, sharing the 25-year carbon projection with the owner can be beneficial. From the report, owners can extract insights on how to reduce emissions over time.

This prerequisite does not require comprehensive carbon accounting, whole-building life-cycle analyses, nor is it meant to be a substitute for more in-depth analysis. Rather, LEED v5 requires this basic cross-categorical carbon assessment to enable a broad understanding of how project emissions across sources will add up over time by using data all LEED v5 projects submitted under other prerequisites and applying reasonable assumptions.

ASSUMPTIONS BEHIND THE ASSESSMENT

Although project teams do not need to compile any calculations to complete this credit, the following section outlines the assumptions behind the USGBC-supplied carbon assessment. Project teams should conduct their own analysis in addition to this carbon assessment to produce more customized results.

Analysis period

A 25-year period was selected because it captures a significant enough time frame for projects to see the impacts of their operational carbon emissions from energy and refrigerants and from regular cycles of renovations in comparison with initial embodied carbon. It does not extend further, unlike some industry projections. This is because future uncertainties grow increasingly dominant over time (e.g., grid decarbonization and advancements in building technologies).

Operational carbon emissions

The *EAp1: Operational Carbon Projection and Decarbonization Plan* calculates the business as usual (BAU) carbon projection from operational emissions from energy over 25 years. The BAU assumes that emissions from fuel use will remain constant, and that emissions from electricity will decline by 95% over 25 years from the base year. See *EAp1: Operational Carbon Projection and Decarbonization Plan* for more information.

Refrigerant emissions

EAp5: Fundamental Refrigerant Management calculates the annual refrigerant emissions. Use a default annual leakage rate of 2% or an annual leakage rate of 1% for projects pursuing *EAc7: Enhanced Refrigerant Management,* Option 2. Limit Refrigerant Leakage.

Embodied carbon emissions

MRp2: Quantify and Assess Embodied Carbon calculates the embodied carbon in global warming potential (GWP) ($kgCO_2eq$) of the structure, enclosure, and hardscape. To calculate the embodied carbon, refer to the assumptions found in *MRp2: Quantify and Assess Embodied Carbon*.

To find the upfront embodied carbon emissions, multiply the *MRp2: Quantify and Assess Embodied Carbon* embodied carbon values by 1.5 to account for interiors and mechanical, electrical, and plumbing (MEP) products, A4, and A5 emissions. Recurring embodied carbon assumes the project undergoes renovations every 10 years. To account for the recurring embodied carbon, multiply the *MRp2: Quantify and Assess Embodied* Carbon embodied carbon values by 0.25.

These time frames are estimates based on common renovation cycles and serve illustrative purposes only.

Transportation emissions

Teams may only calculate the emissions projection from transportation when the project pursues *LTc4: Transportation Demand Management*. The projections assume no change in vehicle miles traveled (VMT) over the 25-year period, and a linear 95% decarbonization of the grid.

IP

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Completed documentation of *EAp1: Operational Carbon Projection and Decarbonization Plan*, *EAp5: Fundamental Refrigerant Management*, *MRp2: Quantify and Assess Embodied Carbon*, and if attempted, *LTc4: Transportation Demand Management*.

REFERENCED STANDARDS

- None

Tenant Guidelines

IPp4
Required
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To communicate and coordinate the sustainable design and construction features of a base building with the tenants and facilitate tenant LEED certification so that the completed space more comprehensively addresses the rating system requirements.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
Core and Shell	**N/A**
Tenant Guidelines	

TENANT GUIDELINES

Create tenant guidelines to be shared with all tenants before signing the lease, including the following content:

- A description of the sustainable design and construction features incorporated in the Core and Shell project and the project's sustainability goals and objectives, including those for tenant spaces.
- Guidance and recommendations for incorporating sustainable strategies, products, materials, and services in the tenant spaces. Refer to the attempted LEED prerequisites and credits of the base building for content requirements.
- A point of contact, from the base building team or new owner, for further coordination of base building design and construction documentation.

REQUIREMENTS EXPLAINED

This prerequisite requires that the base building project creates and distributes clear and detailed guidelines for tenants to integrate sustainability into their spaces. These guidelines serve as a road map for tenants to align their build-outs with the building's sustainability objectives, fostering collaboration between the base building team and tenants. A comprehensive set of guidelines will communicate the base building's sustainable features and provide actionable recommendations for tenants. To maximize their impact, all tenants must receive the tenant guidelines document before they sign the lease, ensuring that they consider sustainability during their design and build-out phases.

The guidelines must cover the following areas to ensure a comprehensive approach to sustainability.

Description of the sustainable design and construction features of the base building

Sustainable design and construction features should emphasize the systems and strategies that enhance energy efficiency, water conservation, materials sustainability, and indoor air quality, such as the following:

- **Energy efficiency:** LED lighting, high-efficiency HVAC systems, and renewable energy integration.
- **Water efficiency:** Low-flow fixtures, water reuse systems, and high-efficiency plumbing fixtures.
- **Sustainable materials:** Recycled-content materials, low-VOC paints, and finishes that reduce environmental impact.
- **Indoor air quality:** Advanced ventilation systems, enhanced air filtration technologies, and materials selection to minimize pollutants.

Clear sustainability goals for tenant spaces aligned with the base building's LEED certification efforts

Set specific and measurable sustainability goals for tenant spaces, ensuring alignment with the base building's LEED prerequisites and credits. These goals should help tenants

- Reduce energy and water use through submetering, efficient fixtures, and regular performance tracking.
- Enhance waste diversion by implementing robust recycling and composting programs.
- Improve indoor environmental quality through strategies such as low-emitting materials and proper ventilation design.

Providing tenants with measurable targets and examples—such as installing submeters or submitting annual sustainability performance reports—helps them support the base building's progress toward LEED certification.

Practical guidance on sustainable strategies tailored to meet the base building's sustainability targets for tenants to implement in their spaces

Offer practical, actionable recommendations for tenants to align their spaces with the base building's sustainability goals. Strategies should include

- **Water conservation:** Using low-flow fixtures and water-efficient appliances.
- **Energy efficiency:** Selecting energy-efficient lighting, appliances, and HVAC systems.
- **Material selection:** Prioritizing low-emitting and sustainable materials to improve indoor air quality.
- **Waste reduction:** Incorporating recycling programs and minimizing construction waste.

These strategies must align with the LEED prerequisites and credits pursued by the base building. Providing tailored guidance ensures tenants understand how their choices can enhance the overall sustainability of the building.

Some prerequisites and credits in the rating system highlight specific information that must be included in the tenant guidelines, at a minimum.

Aligning tenants' operations with the building's sustainability targets can play a vital role in supporting future LEED certification or recertification efforts. This alignment encourages tenants to consider sustainability in their daily operations and decision-making processes, reinforcing a culture of environmental responsibility.

Point of contact

Designate a specific point of contact, either from the base building team or the building's owner. This individual is responsible for

- Coordinating communication about the base building's sustainable design and construction documentation.
- Providing tenants with resources and support to align their spaces with the building's sustainability objectives.
- Delivering tenant training sessions, sharing regular updates, and addressing questions.

This role ensures tenants have access to the expertise and guidance needed to effectively implement the strategies outlined in the guidelines.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	The project's tenant guidelines

REFERENCED STANDARDS

- None

Integrative Design Process

IPc1
New Construction (1 point)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To support high-performance, cost-effective, and cross-functional project outcomes through an early analysis and planning of the interrelationships among systems. To provide a holistic framework for project teams to collaboratively address decarbonization, quality of life, and ecological conservation and restoration across the entire LEED rating system.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1**
Integrative Design Process	1

INTEGRATIVE DESIGN PROCESS (1 POINT)

Beginning in predesign and continuing throughout early occupancy, identify and apply opportunities to achieve synergies across disciplines and building systems through the following initiatives:

- **Integrated team:** Assemble and convene an interdisciplinary project team with diverse perspectives. Ensure the process is an equitable team effort through organized facilitation.
- **Design charette:** During predesign or early in design, conduct a charette with the owner or owner's representative and participants representing at least four key perspectives (e.g., architect, contractor, energy modeler, and community engagement representatives).
- **LEED goal setting:** Work as a team to define a set of specific and measurable project goals that address the LEED v5 impact areas of decarbonization, quality of life, and ecological conservation and restoration. Incorporate these goals into the owner's project requirements.

REQUIREMENTS EXPLAINED

This credit requires a different approach to design than the conventional, linear architectural process. Conventionally, the design and construction disciplines work separately, which leads to fragmented solutions for design and construction challenges. These solutions often create unintended consequences—some positive, but most are negative. Integrating different areas of practice helps project teams identify opportunities to significantly improve building performance and achieve synergies that yield economic, environmental, and human health benefits.

INTEGRATIVE DESIGN PROCESS

In an integrative design process (IDP), an entire team (client, designers, builders, and operators) identifies overlapping relationships, services, and redundancies among systems to identify interdependencies and benefits that would have otherwise gone unnoticed, and to increase performance and reduce costs. This approach requires project teams to have representatives from various disciplines sharing their knowledge, analyses, and ideas to inform and connect each other's work. Following this model, LEED credits become aspects of a whole rather than separate components, and the entire design and construction team can identify the interrelationships and linked benefits across multiple LEED credits.

Approaching certification using an integrative process gives the project team the greatest chance of success. The process includes three phases:

- **Discovery:** This is the most important phase of the integrative process and is an extensive expansion of pre-design. Without this phase, it can be a challenge for projects to meet environmental goals in a cost-effective way. Discovery work should take place before schematic design begins.
- **Design and construction (implementation):** This phase begins with schematic design. It resembles conventional practice but integrates all the work and collective understanding of system interactions reached during the discovery phase.
- **Occupancy, operations, and performance feedback:** This phase focuses on preparing to measure performance and creating feedback mechanisms. Assessing performance against targets is critical for informing building operations and identifying the need for any corrective actions.

To achieve economic and environmental performance, every issue and all essential voices (community, clients, designers, engineers, constructors, operators) should be brought into the project at the earliest point and before anything is designed.

Conduct this holistic process of research, analysis, and workshops in an iterative cycle that refines the design solutions. In the best scenario, teams will continue the research and workshops until the project systems are optimized, all reasonable synergies are identified, and the related strategies associated with all LEED credits are documented and implemented.

INTEGRATED TEAM

The first step involves assembling an interdisciplinary design team with the relevant and affected parties, including owners, building users, architects, engineers, contractors, and community representatives. Participants are to consider all project phases, from early design to construction and operations, to collaboratively set goals, refine strategies, and balance performance, feasibility, and costs.

During construction and procurement, contractors and builders offer insights on constructability, materials, and life-cycle impacts, and collaboration during occupancy ensures that the design intent is upheld, energy strategies are implemented effectively, and performance is monitored for continuous improvement.

DESIGN CHARETTE

The first charrette with interdisciplinary members is crucial for collective agreement on goals, priorities, and a shared project vision. Teams must proactively address major concerns early to avoid redesign delays and inefficiencies later in the project life-cycle. Leveraging tools such as energy and daylight modeling, Building Information Modeling, and life-cycle assessments (LCA) during the conceptual design phase ensures a data-driven approach to identifying conflicts and optimizing performance.

To foster engagement and collaboration, project teams must implement equitable processes by facilitating well-structured meetings, workshops, and charrettes. Resources such as the U.S. Department of Energy's *Handbook for Planning and Conducting Charrettes for High-Performance Projects*[46] provide practical checklists and agendas to guide these efforts.

LEED GOAL SETTING: DECARBONIZATION, QUALITY OF LIFE, AND ECOLOGICAL CONSERVATION AND RESTORATION GOALS

Project teams must establish measurable goals aligned with LEED v5's core impact areas: decarbonization, quality of life, and ecological conservation and restoration. It is an opportunity for project teams to further connect assessment findings to project outcomes. Experts recommend clear metrics to guide decisions, such as carbon reduction percentages, well-being outcomes, or ecosystem restoration targets.

- **Decarbonization:** Strategies include reducing operational and embodied carbon emissions. Teams can replace fossil fuel systems with renewable or electric solutions, upgrade to energy-efficient equipment, and specify low-carbon materials such as high-performance glazing and supplementary cementitious materials (SCMs) in concrete.
- **Quality of life:** A human-centered approach incorporates health, well-being, resilience, and equity into the design. Strategies such as inclusive design and biophilic design, which connect occupants to nature, reduce stress, improve air quality, and enhance cognitive performance. Selecting nontoxic and hazard-resilient materials promotes healthier indoor environments, benefits occupants' physical and mental health, and promotes long-term sustainability.
- **Ecological conservation and restoration:** Sustainable practices such as minimizing soil erosion, planting native vegetation, and integrating green infrastructure (GI) (e.g., permeable pavement, green roofs) reduce environmental impact and restore ecological functions.

46. Gail Lindsey, Joel A. Todd, Sheila J. Hayter, and Peter G. Ellis, *A Handbook for Planning and Conducting Charettes for High-Performance Projects*, U.S. Department of Energy, National Renewable Energy Laboratory, 2009, https://www.nrel.gov/docs/fy09osti/44051.pdf.

Teams must use this thorough research and analysis during the pre-design phase to inform LEED documentation requirements, including the owner's project requirements (OPR), basis of design (BOD), and construction documents. These documents are expected to clearly articulate how project goals align with integrative design principles.

Narratives should comprehensively outline strategies and analyses, such as site assessments, energy and water modeling, and LCAs, to demonstrate how the project meets sustainability objectives. Teams should include robust justifications for their design decisions to ensure clarity and accountability throughout the project life-cycle.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Evidence of the design charette date, the participants and their roles, and the name and company of the facilitator (e.g., the meeting notes for the design charettes).
			The OPR defining the goals for synergy across building systems that address decarbonization, quality of life, and ecological conservation and restoration, including how success of each goal will be measured.

REFERENCED STANDARDS

- Integrative Process (IP) ANSI Consensus National Standard Guide© 2.0 for Design and Construction of Sustainable Buildings and Communities (2012) (https://www.usgbc.org/resources/integrative-process-ip-ansi-consensus-national-standard-guide-20-design-and-construction)

FURTHER EXPLANATION

Ipc1: Integrative Design Process emphasizes three core initiatives: integrated teams, design charrette, and LEED goal setting. Together, they establish a structured, collaborative framework during the earliest project phases. This early, interdisciplinary approach is where cost savings, efficiency, and sustainability opportunities are maximized. For documentation, teams must provide the owner's project requirements (OPR), evidence of the design charette date, the participants and their roles, and the name and company of the facilitator (e.g., the meeting notes for the design charettes).The OPR defines the goals for synergy across building systems that address decarbonization, quality of life, and ecosystem conservation and restoration, including how the success of each goal will be measured.

INTEGRATED TEAM

An interdisciplinary project team should be assembled as early as pre-design, and may include representatives from architecture, key engineering disciplines (e.g., structural, mechanical, electrical, plumbing), construction and commissioning, operations/maintenance, finance, and community specialists where relevant. In contrast to a conventional linear process that brings in team members only when needed, an integrative process is inclusive from the outset, front-loading collaboration to identify synergies and prevent downstream conflicts. Engaging a broad team early on informs decisions so that the project team considers multiple facets of design; for example, decisions about building orientation or envelope will involve architects, engineers, and users, leading to a design that meets energy, comfort, and cost goals as intended. This inclusive approach not only improves technical outcomes but can also minimize incremental capital costs by allowing trade-offs (like downsizing HVAC equipment due to a better envelope) that offset high-performance features.

To get the most value from an integrated team, projects should convene a kickoff workshop led by a trained facilitator identified by the project team (either an internal team member or an external consultant experienced in integrative design). The facilitator's role is to create an inclusive environment where all voices are heard and expertise is shared openly. Effective, equitable, and inclusive facilitation techniques include rotating speaking opportunities (to avoid any one discipline dominating discussions), using anonymous input tools (e.g., digital whiteboards or polling apps that let team members contribute ideas without pressure), and setting clear ground rules at the outset. Common ground rules are "no idea is criticized," "respect everyone's time," "all ideas are welcome," and "everyone participates."[47] The facilitator should make sure the team collaboratively defines roles and responsibilities of each member (sometimes formalized in a team charter or plan) and may even create a process map of major integration points and future workshops to keep the effort organized. Such a charter or plan helps maintain accountability and buy-in across the diverse group.

DESIGN CHARRETTE

The design charrette is a pivotal, collaborative workshop that should occur during pre-design or very early schematic design—typically before finalizing the basis of design (BOD) or major project decisions—to maximize its impact on the project direction. A charrette is an intensive multidisciplinary workshop (often a full-day or even multiday session) in which stakeholders and experts gather to brainstorm and solve design challenges together. Ideally held once a clear project vision is formulated but while design options are still fluid, the charrette allows the team to explore goals and integrative solutions in real time—for example, how the building site, envelope, HVAC system, and landscape can all work in harmony. Invite a broad range of participants, with at least four distinct perspectives, to be present (for instance, the owner or owner's representative, the lead architect/designer, the general contractor or construction manager, and an energy modeler or sustainability expert). In addition, teams are encouraged to include others, such as a community engagement adviser, future facility users, and operations/maintenance staff, as these perspectives enrich the discussion with user experience, community context, and operational practicality. Engage enough charette participants to represent diverse expertise, but not so many that the workshop becomes unwieldy. If the group is large, consider break-out sessions focused on key topics (e.g., energy, water, site, wellness) with each reporting back to ensure balanced input from all stakeholders.

47. Whole Building Design Guide (WBDG), "Planning and Conducting Integrated Design (ID) Charrettes," accessed September 8, 2025, https://www.wbdg.org/resources/planning-and-conducting-integrated-design-id-charrettes#:~:text=Process%20is%20critical%20to%20successful%2C,and%20extensive%20community%20visioning%20events.

During the charrette, the facilitator (or facilitators for break-out groups) should guide participants through a structured agenda: introductions and a project overview, an exercise to establish or reaffirm the project vision, brief presentations on sustainability strategies or context (as needed for inspiration), and interactive working sessions to identify opportunities and synergies between building systems and LEED goals. One useful output of the charrette is a synergy map—a visual diagram or matrix that links proposed strategies and systems (such as the envelope, HVAC, lighting/daylighting, or water reuse) with the project's sustainability goals and LEED credits. This helps the team see where one design decision can satisfy multiple objectives or how various strategies interconnect. For example, a green roof might simultaneously contribute to thermal insulation (energy goal), stormwater management (ecosystem goal), and occupant amenity (quality of life goal). The charrette discussions should be documented in a summary report or memo that captures the key decisions made, the integrative opportunities identified (e.g., areas where disciplines found overlap or could coordinate efforts), any roadblocks or trade-offs that were revealed, and action items with responsible parties assigned. In practice, by the end of the charrette the team should have a clearer direction with unified design goals that everyone will work toward, and a set of preliminary strategies to investigate during design.[48] Importantly, the outcomes of the charrette must loop back into the project's core documents—for instance, the OPR should be updated to reflect any new goals or criteria identified, and the design team's work plan should incorporate the action items and further studies that emerged.

To demonstrate compliance with the charrette component, project teams should submit the charrette agenda and meeting invite/announcement (showing it was conducted in pre-design or early design with the required attendees), a participant list (including names, roles, and organizations to show the diversity of perspectives), and the charrette report or meeting minutes. The report should highlight the sustainability strategies discussed and any integrative synergies or major decisions coming out of the session. If a synergy map or other visuals were produced, these can be included as supporting evidence. Photographs of the workshop (e.g., flipcharts or stakeholders brainstorming) are not required but can be useful to illustrate stakeholder engagement. Be sure that the documentation clearly indicates the charrette's date (to confirm timing before or during early design) and that at least four key stakeholder categories were represented (as required by the credit). Integrating the charrette results into the OPR and BOD (discussed below) is also critical, and providing excerpts of those documents showing charrette-informed goals is an excellent way to tie it all together.

LEED GOAL SETTING

The integrative team, through exercises like the design charrette and early goal-setting workshops, must establish a set of project sustainability goals that are specific, measurable, and aligned with LEED v5's three impact areas. LEED v5 was developed around three overarching impact categories: decarbonization, quality of life, and ecological conservation and restoration. Therefore, project goals should be mapped to these themes. In practice, this means defining targets as follows:

48. G. Lindsey, J. A. Todd, and S. J. Hayter, "A Handbook for Planning and Conducting Charettes for High-Performance Projects," National Renewable Energy Laboratory (NREL), August 2003, https://docs.nrel.gov/docs/fy03osti/33425.pdf#:~:text=sustainability%20vision%20for%20the%20building,must%20consider%20how%20the%20building.

- **Decarbonization:** Requires concrete objectives for reducing carbon emissions; for example, a maximum operational energy use intensity (EUI) or greenhouse gas emissions rate, a percentage of power to be supplied by on-site renewables, and targets for lowering embodied carbon in key materials. (Example goal: Achieve ≤20 kilogram CO_2eq/m^2-year in operational carbon by Year 1 of occupancy, and source at least 30% of building electricity from on-site photovoltaics.) These goals tie into LEED energy and carbon credits and support the broader decarbonization imperative.
- **Quality of Life:** Metrics to make sure the project improves occupant well-being, comfort, and community benefits. These could include indoor environmental quality targets (e.g., maintain indoor air pollutant levels 30% below recommended limits, provide daylight and views for 90% of regularly occupied spaces, or meet certain acoustic comfort standards), commitments to conduct occupant satisfaction surveys post-occupancy, or measures to enhance social equity (such as community access to amenities or inclusive design features). (Example goal: Exceed ASHRAE 62.1 ventilation rates by 15% and achieve >80% satisfaction in post-occupancy comfort surveys, while providing a publicly accessible green space on-site.) These goals align with LEED credits in the quality of life domain (health, comfort, equity, and resilience).
- **Ecosystem Conservation and Restoration:** Objectives focused on site sustainability and environmental stewardship, like restoring a percentage of the site with native/adapted vegetation, managing stormwater to infiltrate or reuse a certain volume (beyond local requirements), reducing light pollution and habitat disturbance, or using nature-based solutions. (Example goal: Restore at least 50% of previously impervious site area to green space with native plants, capture and infiltrate the 95th percentile rain event on-site, and meet the Dark Sky criteria for all exterior lighting.) Such goals correspond to LEED Sustainable Sites, Water Efficiency, and advance the ecological conservation and restoration impact area.

Once these LEED goals are defined by the team, they must be embedded in the OPR document and tracked through design via BOD and other project plans. The OPR is the cornerstone project brief that outlines the owner's vision and performance requirements; it should explicitly list the sustainability and LEED objectives the project is striving for (with quantifiable metrics where possible). The BOD, prepared by the design team, should then describe how the design concept responds to each of those requirements—effectively linking design decisions to the stated goals and assigning responsibility for achieving them. For example, if the OPR includes "20% water use reduction beyond code," the BOD will detail the water-saving strategies (e.g., low-flow fixtures, rainwater harvesting) and identify which team member is accountable for each strategy's implementation. LEED v5 specifically asks teams to incorporate these goals into the OPR, recognizing that doing so formalizes the commitment. Documenting the goals in this way also facilitates mid-project checkpoints and ultimately the commissioning process, where the OPR/BOD alignment is reviewed. According to guidance on integrative design, outcomes of early analysis and goal setting should directly inform the OPR and BOD so that the project's aspirations are translated into actionable design criteria. Through tracking progress on each goal throughout design (e.g., via energy models, design reviews, and team meetings), the project can adjust course as needed to stay on target. This integrative feedback loop—from goal to design strategy to verification—is essential to realizing high-performance outcomes and is a hallmark of the integrative process.

To illustrate the LEED goal setting, teams should provide excerpts from the OPR and BOD that highlight the sustainability goals and their metrics—for example, a page from the OPR that lists the project's decarbonization, quality of life, and ecosystem goals (with numbers or criteria) and a corresponding page in the BOD or design report that shows how each goal is being addressed (including who is responsible). These excerpts prove that the goals were not only set but formally integrated into the design process. Include any goal-tracking tools or matrices the team used—for example, a summary table of all LEED points pursued and the person accountable or a design decision log referencing the goals. Such documentation demonstrates a clear thread from early goal setting to the project execution.

Through following these practices, project teams not only earn the Integrative Process credit, but also lay the groundwork for coordinated, high-performance outcomes throughout design, construction, and operation. Early collaboration has been shown to save time and money while improving project performance. For instance, conflicts are resolved on paper instead of in the field, and optimized solutions often mean lower life-cycle costs (avoiding expensive late design changes or oversizing of systems). Moreover, an integrative process can unlock more ambitious sustainability achievements. Studies and case projects have found that a holistic team approach can lead to higher energy efficiency and more LEED points in critical categories like Energy and Atmosphere and Indoor Environmental Quality. Equally important, it boosts stakeholder buy-in and satisfaction by creating a shared sense of purpose and ownership from the project's inception. In essence, *IPc1: Integrative Design Process* is about setting a strong foundation by getting the right people to the table early, asking the right questions, and committing to common green goals so that the project can deliver on the full promise of sustainability and human wellness that LEED v5 aspires to. With thorough documentation of this process, teams can verify that they have met the credit requirements and set their project on a course for success in the remaining LEED credits and beyond.

Green Leases

IPc2
Core and Shell (1–6 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To ensure the tenants complete the sustainable design and construction features started by the base building so that the completed space comprehensively addresses the rating system requirements.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
Core and Shell	**1–6**
Option 1. Standard Green Lease AND/OR	1–6
Option 2. Executed Standard Green Lease AND/OR	1
Option 3. Green Lease Leaders Recognition	1–6

Comply with a combination of the following options for a maximum of 6 points.

OPTION 1. STANDARD GREEN LEASE (1–6 POINTS)

Develop a standard green lease document that establishes tenant fit-out standards and incorporates tenant requirements, as described below. Commit to incorporating the green lease documentation in each future tenant lease.

For any spaces or systems intended to be fit out by the project owner, identify the standards for the future fit-out, incorporate the criteria referenced in the standard green lease document, and commit to performing the fit-out(s) in accordance with these standards.

Tenant requirements shall include standards to ensure compliance with the following prerequisites for any tenant-installed systems:

- IPp1: Climate Resilience Assessment
- IPp2: Human Impact Assessment
- IPp3: Carbon Assessment
- WEp2: Minimum Water Efficiency
- EAp2: Minimum Energy Efficiency
- EAp3: Fundamental Commissioning
- EAp4: Energy Metering and Reporting
- MRp1: Planning for Zero Waste Operations
- EQp2: Fundamental Air Quality

Points are earned by incorporating a combination of the listed best practices into the green lease (Table 1).

- Require tenant to pay for their electric and nonelectric energy and water use.
- Implement a cost recovery clause for energy-efficiency upgrades benefiting the tenant.
- Disclose tracked common area energy use, peak demand, peak thermal demand, and on-site combustion emissions to tenants.
- Disclose whole-building ENERGY STAR® or locally applicable equivalent score to tenants.
- Ensure brokers or leasing agents have energy training.
- Implement energy management best practices.
- Request annual tenant energy disclosure.
- Require energy efficiency fit-out for tenants that improves upon *EAp2: Minimum Energy Efficiency* requirement.
- Establish a tenant energy efficiency engagement and training plan.
- Meter or submeter additional tenant energy use beyond that required in *EAp4: Energy Metering and Reporting*.
- Limit on-site combustion emissions.
- Disclose tracked common area water use to tenants.
- Require water efficiency fit-out for tenants that improves upon *WEp2: Minimum Water Efficiency.*
- Meter or submeter additional tenant water use beyond that required in *WEp1: Water Metering and Reporting.*
- Implement water management best practices.
- Implement indoor air quality best practices.
- Implement thermal comfort best practices.
- Demonstrate innovation in leasing.

TABLE 1. Points for Incorporating Best Practices

Number of Additional Best Practices Incorporated into the Lease	Points
Fewer than 3	1
4–6	2
7–9	3
10–12	4
13–15	5
16 or more	6

AND/OR

OPTION 2. EXECUTED STANDARD GREEN LEASE (1 POINT)

One additional point is awarded by providing documentation of an executed tenant green lease or tenant letter of attestation meeting the criteria in Option 1.

AND/OR

OPTION 3. GREEN LEASE LEADERS RECOGNITION (1–6 POINTS)

Earn Green Lease Leaders recognition:

- At the Silver level (1 point)
- At the Gold level (3 points)
- At the Platinum level (6 points)

REQUIREMENTS EXPLAINED

This credit requires owners to develop a green lease document that establishes clear standards and best practices for tenants and their operations and is aligned with LEED goals. The green lease functions as a framework for sustainability by incorporating key clauses and operational procedures to ensure tenants and building owners actively promote sustainable practices, such as energy and water efficiency, carbon emissions reduction, indoor air quality, and waste management. A green lease ensures that both owners and tenants commit to achieving high-performance building standards. Additional recognition is available for executing the green lease or achieving certification through the Green Lease Leaders program.[49]

Pursuing green lease agreements fosters collaboration between property owners and tenants, ensuring that sustainability goals remain a priority throughout the occupancy phase. Green leases address the split incentive issue and align the costs and benefits of energy and water efficiency investments to create equitable and successful agreements for both parties.

OPTION 1. DEVELOP A STANDARD GREEN LEASE

Option 1 requires the development of a standard green lease that establishes tenant fit-out standards that incorporate tenant requirements and is aligned with the identified LEED prerequisites. For any spaces or systems intended to be fit out by the project owner, teams must identify the standards for the future fit-out for any tenant-installed systems or spaces. Tenant fit-outs must comply with the following prerequisites:

- IPp1: Climate Resilience Assessment
- IPp2: Human Impact Assessment
- IPp3: Carbon Assessment
- WEp2: Minimum Water Efficiency
- EAp2: Minimum Energy Efficiency

49. Green Lease Leaders, “Green Leasing Recognition Program,” n.d., https://greenleaseleaders.com/green-leasing/program-requirements/.

- EAp3: Fundamental Commissioning
- EAp4: Energy Metering and Reporting
- MRp1: Planning for Zero Waste Operations
- EQp2: Fundamental Air Quality

Meeting these prerequisites is a critical first step toward achieving LEED certification, because they establish minimum performance levels that must be met. It also ensures that the needs identified in the assessments are carried out through operational strategies. Once you have addressed prerequisites, teams can exceed minimum requirements by incorporating additional best practices and innovative design strategies, thereby enhancing overall building sustainability.

Incorporate green lease documentation in future leases

Owners must incorporate the green lease documentation into future tenant leases, including the criteria outlined in the standard green lease. Resources such as *the Institute for Market Transformation (IMT)* 2020 document[50] and the *BOMA International Green Lease Guide,*[51] offer detailed language examples and guidance for incorporating green clauses, operational procedures, and best practices.

Incorporate best practices into the green lease

Owners and landlords must include a tailored combination of best practices in the green lease to go beyond minimum LEED prerequisites. Best practices include the following:

- Require tenants to pay for their energy and water usage.
- Implement cost recovery clauses for energy efficiency upgrades that benefit tenants.
- Disclose tracked energy and water usage metrics (e.g., ENERGY STAR scores, peak demand).
- Ensure brokers or leasing agents receive energy training.
- Require tenant energy efficiency fit-outs that exceed LEED EA prerequisites.
- Include water and indoor air quality best practices (e.g., submetering, low-emitting materials).
- Establish tenant training and engagement programs for sustainable practices.
- Limit on-site combustion emissions and implement renewable energy strategies.
- Demonstrate innovation in leasing.

Points are awarded based on the number of best practices incorporated (1–6 points). Incorporating additional practices improves resource management and creates healthier, more sustainable tenant spaces.

OPTION 2. EXECUTED STANDARD GREEN LEASE

Teams may earn 1 point for projects that provide documentation of an executed tenant green lease or a letter of attestation meeting the criteria outlined in Option 1. Both parties must provide a copy of the green lease agreement signed by them. If the tenant has not yet finalized the green lease, they must submit a signed and dated letter as an alternative. The letter must confirm the tenant's commitment to meet the specific sustainability criteria outlined in the green lease, including

50. IMT, *Green Lease Language Examples*, 2020, imt.org/wp-content/uploads/2020/02/IMT-Green-Lease-Language-Examples-January-2020.pdf.
51. K. M. Noonan and BOMA International, et al., *Green Lease Guide*, 2018, sustainablejersey.com/fileadmin/media/Actions_and_Certification/Actions/Energy/BOMA_2018_Green_Lease_Guide.pdf.

compliance with LEED prerequisites and best practices as detailed in Option 1. Key components of the letter of attestation may include tenant information, landlord and tenant acknowledgment, commitment to sustainability requirements, specific best practices, commitment to documentation and reporting procedures, and the tenant's signature and date.

Executing a tenant green lease demonstrates a commitment to foster collaboration between landlords and tenants. This collaboration can lead to improved energy efficiency, reduced operating costs, and a more sustainable environment. The documentation serves as a tangible acknowledgment of the efforts made toward sustainability and enhances the credibility of both the landlord and the tenant. It can also elevate corporate reputations and align with broader Environmental, Social, and Governance (ESG) goals. Providing this documentation is a proactive step toward a sustainable future while securing additional recognition for the project team's initiatives.

OPTION 3. GREEN LEASE LEADERS RECOGNITION

Option 3 rewards teams that join the Green Lease Leaders Program, an initiative developed by IMT with support from the U.S. Department of Energy. Open to U.S. and international companies, including those in Canada, Europe, Australia, the United Kingdom, Japan, Costa Rica, and Mexico, this three-year program offers valuable guidance to practitioners and teams on best leasing practices to foster mutually beneficial landlord–tenant relationships. To help create more sustainable buildings and energy-efficient spaces, the program provides resources that address four key steps to achieving long-term high-performance buildings: site selection, lease negotiations, tenant fit-out, and tenant operations.

Participants benefit from established guidelines and free support while developing their green leases, as well as peer leadership recognition and substantial energy savings when leases have been implemented.[52] The program also serves as an essential avenue to demonstrate commitment to corporate ESG objectives and net-zero goals. Participating organizations enhance their reputation and contribute to broader sustainability initiatives within the community.

Green Lease Leaders awards recognition through a tiered system, which includes three levels of achievement: Silver, Gold, and Platinum.[53] The recognition criteria align with the U.S. EPA's ENERGY STAR Tenant Space recognition program.[54] The points awarded at each level correspond to the complexity and scope of the actions taken to implement green leasing practices.

- **Silver level:** Recognizes the establishment of foundational policies and business practices (e.g., a standard lease form that incorporates green lease language) that encourage reduced energy and water consumption in leased spaces. For instance, the prerequisite of the program minimum efficiency fit-out requires bests practices, such as ENERGY STAR-certified appliances and equipment; meter/submeter tenant energy; and use of only low/no volatile organic compound paints, finishes, and adhesives.

52. ENERGY STAR®, "ENERGY STAR® Tenant Space," n.d., https://www.energystar.gov/buildings/building_recognition/tenant_space_recognition.
53. Green Lease Leaders, "Program Requirements," n.d., https://www.greenleaseleaders.com/green-leasing/program-requirements/#silver.
54. ENERGY STAR®, "ENERGY STAR® Tenant Space."

- **Gold level:** Builds on silver level achievements and recognizes execution of green leases and utility-efficient tenant fit-outs. For instance, this level may require performance goal clauses within an executed lease.
- **Platinum level:** Exemplifies achievements by both the landlord and tenant to integrate environmental and social priorities into the lease and best practices. This level includes Gold requirements and additional requirements, such as documentation that verifies the implementation of the performance goal and action plan.

For all recognition levels, tenants must provide evidence that the standard lease form or corporate policy meets these two prerequisites:

- Provide sustainability contacts to landlords, and
- Require minimum efficiency standards for leased space fit-outs.[55]

Landlords must establish a standard lease form or corporate policy based on Prerequisite 1 and, depending on the property type, either Prerequisite 2a or Prerequisite 2b.

- **Prerequisite 1:** Provide sustainability contact and/or information.
- **Prerequisite 2a:** Implement cost recovery clause for energy efficiency upgrades benefiting the tenant.
- **Prerequisite 2b:** For multifamily properties, implement energy efficiency improvements during unit turns.[56]

This LEED credit option awards points based on recognition levels. The Silver level carries 1 point, because it represents a first step toward sustainability and focuses on establishing basic green leasing practices. LEED Gold carries 3 points, because it is more restrictive and acknowledges a higher level of commitment. LEED Platinum carries 6 points, because it exemplifies the highest level of achievement, where both landlords and tenants fully integrate advanced environmental and social priorities into the green lease, and it demonstrates leadership in sustainability.

Landlords and tenants must consult the Green Lease Leaders[57] website and the LEED v5 reference guides for comprehensive information on program requirements, detailed guidance on clauses, and the application process. Additionally, they must use the interactive Microsoft Excel workbook[58] or contact a program team member to help through the application process.

55. Institute for Market Transformation and U.S. Department of Energy, “Green Lease Leaders Reference Guide for Tenants,” 2021, https://www.greenleaseleaders.com/wp-content/uploads/2023/07/Tenant-Reference-Guide.pdf.
56. Institute for Market Transformation and U.S. Department of Energy, “Green Lease Leaders Reference Guide for Landlords,” 2021.
57. Green Lease Leaders, “Program Requirements,” https://www.greenleaseleaders.com/green-leasing/program-requirements/#silver.
58. IMT Comms, “Teams Application Workbook,” September 27, 2024, https://www.greenleaseleaders.com/resource/teams-application-workbook/.

DOCUMENTATION

Project Types	Options/ Paths	Paths	Documentation
All	Option 1. Standard Green Lease	All	The project's standard green lease.
			Evidence of the Core and Shell project owner's commitment to incorporating the green lease documentation in each future tenant lease (e.g., a digital signature by the Core and Shell project owner in LEED Online).
	Option 2. Executed Standard Green Lease		Executed green lease or tenant letter of commitment to performing the fit-out(s) in accordance with these standards.
	Option 3. Green Lease Leaders Recognition		Evidence of the Green Lease Leaders recognition, including level (e.g., a screenshot of the listing in the Green Lease Leaders database showing recognition level).

REFERENCED STANDARDS

- Green Lease Leaders Program (https://greenleaseleaders.com)

Location and Transportation

OVERVIEW

Location and transportation decisions play a crucial role in determining a project's long-term sustainability potential. The project's chosen location significantly influences the surrounding environment and community and affects how people access the site, who can use it, and what impact it has on local resources. By prioritizing strategies at the intersection of resource access, land-use patterns, and transportation, the Location and Transportation (LT) category guides projects toward an efficient, equitable, and low-carbon future.

Given this significance, the LT category offers the third-largest number of potential points in LEED v5. It prioritizes location-efficient sites that use existing infrastructure to promote land conservation and support compact, connected communities. Increasing urban density has manifold benefits: preserving natural habitats outside of major corridors, advancing equitable development through transportation access and community connection, and improving infrastructure efficiency.[1] Emphasizing transportation demand management (TDM) further promotes connected alternatives for mobility and equitable development. These benefits could also generate trillions of dollars in economic savings for cities before 2050.[2] One estimate suggests that a more compact approach to urban growth could reduce infrastructure capital requirements by more than US$3 trillion between 2015 and 2030.[3]

Next, the LT category recognizes electric vehicle (EV) adoption to further reduce GHG emissions and cultivate a transition to more sustainable mobility solutions.

Decarbonization

Transportation is responsible for nearly one-quarter of global energy-related carbon emissions.[4] Recognizing the enormous momentum in the transportation sector, LEED v5 introduces measures that anticipate a decarbonized future state. Strategies like transportation demand assessment, enhanced EV incentives, and support for low-carbon and micromobility alternatives, such as public transit,

1. Catlyne Haddaoui, "Cities Can Save $17 Trillion by Preventing Urban Sprawl," World Resources Institute, November 15, 2018, https://www.wri.org/insights/cities-can-save-17-trillion-preventing-urban-sprawl.
2. Haddaoui, "Cities Can Save $17 Trillion."
3. Global Commission on the Economy and Climate and New Climate Economy, "Infrastructure Investment Needs of a Low-Carbon Scenario," 2014, https://newclimateeconomy.net/sites/default/files/2023-08/Infrastructure-investment-needs-of-a-low-carbon-scenario.pdf.
4. IEA, "Energy System: Transport," https://www.iea.org/energy-system/transport.

scooters, and bikeshares, can significantly reduce associated project emissions. Public transportation on buses and trains can reduce emissions by up to two-thirds per passenger per kilometer compared to private vehicles.[5] Projects must rethink the dominance of traditional transportation approaches and encourage a fundamental mode shift away from single-occupancy vehicles (SOVs) to low-carbon alternatives (*LTc4: Transportation Demand Management, LTc5: Electric Vehicles).*

Quality of life

By promoting more equitable and healthy communities through compact and connected growth, the LT category provides pathways to affordable housing, local jobs, and sustainable transportation in the surrounding community (*LTc2: Equitable Development, LTc3: Compact and Connected Development*). This holistic approach fosters more inclusive, resilient, and economically vibrant neighborhoods. By encouraging projects to embed these principles, the category helps create communities where people can thrive.

Ecological conservation and restoration

Implementing low-carbon transportation and compact development options reduces emissions and mitigates urban sprawl, which disrupts ecosystems and natural habitats.[6] Researchers estimate that about one-third of all terrestrial species will experience habitat loss, with some species losing at least a tenth of their remaining habitat if global urbanization continues at its current rate through 2050.[7] Safeguarding sensitive lands (wetlands, prime farmland, floodplains, and steep slopes) enables projects to protect biodiversity, preserve natural carbon sinks, and bolster community resilience (*LTc1: Sensitive Land Protection, LTc3: Compact and Connected Development*).

Through these strategies, the LT category supports a transformative increase in understanding land-use choices, accelerates the adoption of EV infrastructure, fosters the transition to low-carbon transportation, and enables project teams to see the enormous potential to use location choice to support not only their own buildings but also their surrounding community.

5. Ben Welle, Anna Kustar, Thet Hein Tun, and Cristina Albuquerque, “Post-Pandemic, Public Transport Needs to Get Back on Track to Meet Global Climate Goals,” World Resources Institute, December 14, 2023, https://www.wri.org/insights/current-state-of-public-transport-climate-goals.
6. Haddaoui, “Cities Can Save $17 Trillion.”
7. William F. Laurance and Jayden Engert, “Sprawling Cities Are Rapidly Encroaching on Earth’s Biodiversity.” *Proceedings of the National Academy of Sciences* 119, no. 16 (March 2022): e2202244119, https://doi.org/10.1073/pnas.2202244119.

Sensitive Land Protection

LTc1
New Construction (1 point)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To cultivate community resilience by avoiding the development of environmentally sensitive lands that provide critical ecosystem services and reduce the environmental impact from the location of a building on a site.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1**
Option 1. Previously Developed Sites OR	1
Option 2. Previously Undeveloped Sites	1

OPTION 1. PREVIOUSLY DEVELOPED SITES (1 POINT)

Locate the development footprint on land that has been previously developed.

OR

OPTION 2. PREVIOUSLY UNDEVELOPED SITES (1 POINT)

Locate the development footprint on land that does not meet the following criteria for sensitive land:

- **Prime farmland:** Prime farmland, unique farmland, or farmland of statewide or local importance as defined by the U.S. Code of Federal Regulations, Title 7, Volume 6, Parts 400 to 699, Section 657.5 (or local equivalent for projects outside the U.S.) and identified in a state Natural Resources Conservation Service soil survey (or local equivalent for projects outside the U.S.).
- **Floodplains:** A flood hazard area shown on a legally adopted flood hazard map or otherwise legally designated by the local jurisdiction or the state. For projects in places without legally

adopted flood hazard maps or legal designations, locate on a site that is entirely outside any floodplain subject to a 1% or greater chance of flooding in any given year (100-year floodplain).

- **Notable habitat:** Land identified as habitat for one or more of the following:
  - Species listed as threatened or endangered under the U.S. Endangered Species Act or the state's endangered species act.
  - Species or ecological communities classified by NatureServe as GH (possibly extinct), G1 (critically imperiled), or G2 (imperiled).
  - Species listed as threatened or endangered species under local equivalent standards (for projects outside the U.S.) that are not covered by NatureServe data.
- **Water bodies:** Areas on or within 100 feet (30 meters) of a water body, except for minor improvements.
- **Wetlands:** Areas on or within 50 feet (15 meters) of a wetland, except for minor improvements.
- **Steep slopes:** Protect 40% of the steep slope area on the site (if such areas exist) from all development and construction activity.
  - For unstable, undeveloped slopes between 15% and 25%, protect 40% from all development.
  - For unstable, undeveloped slopes steeper than 25%, protect from all development and construction activity 60% of the steep slope area on the site.

REQUIREMENTS EXPLAINED

This credit rewards teams for considering their project's impact on the local community and ecosystems by selecting sites that minimize disruption and protect sensitive areas, such as previously developed locations. Teams pursuing this credit can still achieve points for building on locations not previously developed; however, that location must not be on prime farmland, floodplains, habitats with threatened or endangered species, waterbodies, wetlands, or steep slopes.

OPTION 1. PREVIOUSLY DEVELOPED SITES

Previously developed land is any land where infrastructure has been constructed and buildings on the site do not further disrupt sensitive land. Building on previously developed sites conserves undeveloped land from development and construction activity within the project boundary. It is essential for maintaining a diverse ecosystem and compact land development patterns, maintaining biodiversity and sustaining ecosystems that provide services such as water filtration and soil stabilization, keeping existing ecosystems intact, and preserving their ecological resilience. This supports ecosystems that are more resilient against future environmental changes and disturbances.

OPTION 2. PREVIOUSLY UNDEVELOPED SITES

If developing entirely on previously developed land is not feasible, a project must not locate the development footprint on sensitive land types, including prime farmland, flood hazard zones, imperiled species habitats, wetlands or water bodies, and their surrounding buffers or steep slopes. These types of sensitive lands have been identified as critical to biodiversity, ecosystem services, and the safety and well-being of people.

The project must be designed so that the development footprint does not encroach on sensitive areas.

Prime farmland

Prime farmland is land that has the best combination of physical and chemical characteristics for producing crops and is of major importance in meeting the needs of food and fiber. It has an adequate and dependable supply of moisture from precipitation or irrigation, a favorable temperature and growing season, acceptable acidity or alkalinity, acceptable salt and sodium content, and few to no rocks. Researchers use these lands with very rich soil for crop testing farms, and local or state regulations protect them.

Unique farmland refers to land used for producing high-value food and fiber crops such as citrus, tree nuts, olives, cranberries, as well and various fruits and vegetables. It possesses a unique combination of soil quality, growing season, moisture supply, temperature, humidity, air drainage, elevation, and aspect, all of which allow for the economical and sustainable production of high yields when properly managed.[8]

In some regions, land that does not qualify as prime or unique farmland is considered of statewide or local importance because it produces food, feed, fiber, forage, and oilseed crops. Local agencies establish the defined criteria for this type of farmland and typically include soils that closely meet the requirements for prime farmland, capable of producing high crop yields when managed using proper farming practices. In favorable conditions, some of these areas may yield crops comparable to prime farmland.[9]

Projects must avoid development on prime farmland, unique farmland, or farmland of statewide or local importance.

Floodplains

New developments must not be located on floodplains. The project team must determine the extent of flood hazard areas and identify them on a flood hazard map. Local governments, flood management agencies, and other local entities such as the Federal Emergency Management Agency (FEMA),[10] may assist teams in using GIS or geospatial data to identify the location of floodplain area.[11] Projects with the development footprint of the site must be located entirely outside any floodplain that has 1% or more chance of flooding for locations that are both addressed and not addressed by a flood hazard map. A 1% annual-chance flooding, also known as 100-year floodplain, means that there is a 1% chance that flood water will reach or surpass base flood elevation in any given year, and that structures built within this area are at a high risk of flooding. Building outside this zone reduces the likelihood of flood damage.

8. Natural Resources Conservation Service, "Soil Data Access," n.d., nrcs.usda.gov/publications/Legend%20and%20Prime%20Farmland%20-%20Query%20by%20Soil%20Survey%20Area.html.
9. Natural Resources Conservation Service, "Soil Data Access."
10. FEMA Home page, accessed March 31, 2025, https://www.fema.gov.
11. FEMA, "Flood Maps," accessed March 31, 2025, https://fema.gov/flood-maps.

Notable habitats

Notable habitats refer to areas with threatened or endangered and imperiled species. These areas are home to diverse plants, animals, and organisms with high ecological value and are vulnerable to human actions, such as forests and coastal regions. These biodiversity-rich zones provide human beings with what they eat, from the microorganisms that enrich the soil where crops grow, to the pollinators that provide fruits and nuts, and the fish that serve as the main sources of animal protein for billions of people.[12] These habitats play an essential role in maintaining ecological balance and supporting biodiversity, so projects must take action to protect them.

Threatened or endangered species

Teams must avoid locating development footprints on land that contains species listed as endangered or threatened that are listed under the U.S. Endangered Species Act or local equivalent.

Projects must also avoid land identified as a habitat for species or ecological communities classified by NatureServe.

Water bodies and wetlands

The development footprint cannot be within 100 feet of a water body or 50 feet of a wetland. The only exception is if there are minor improvements within the buffer area that allow human interaction, and the improvements do not significantly alter the existing vegetation and hydrology of the area. Avoiding development on water bodies and wetlands is essential for mitigating flooding and managing rainwater runoff, because wetlands naturally absorb excess water and reduce the impacts of heavy rainfall. Human-made water bodies or wetlands do not meet the requirements.

Projects that include water bodies and/or wetlands within the boundary must provide a map identifying the locations of any wetlands or water bodies.

Steep slopes

Projects must avoid any unstable and undeveloped steep slopes. Building on steep slopes presents risks that can affect the safety and stability of the building and pose environmental risks to the area.

All project teams must identify unstable, undeveloped steep slope area on the site. An unstable slope is susceptible to collapse or landslides, creating significant risks to human safety and infrastructure. For unstable and undeveloped slopes between 15% and 25% steepness, the project team must protect 40% of the steep slope area from all development and construction activity. For slopes steeper than 25% steepness, teams must allocate 60% of this area as protected zones where no development or construction activities will occur.

12. The Royal Society, “Why Is Biodiversity Important—with Sir David Attenborough,” YouTube video, October 11, 2021, https://youtube.com/watch?v=GIWNuzrqe7U.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Previously Developed Sites	All	Description shows the previous development of the site.
			Vicinity map or aerial images show the approximate areas of previous development on the site.
	Option 2. Previously Undeveloped Sites	All	Site map(s) showing project boundary, development footprint, any previous development, any sensitive areas, topography, and any minor improvements in required buffers.
			Narrative describing how the project team verified the criteria for prime farmland, flood hazard, notable habitat, waterbodies, wetlands, and steep slopes criteria were satisfied.
			Estimated area of sensitive land avoided.
		Steep slopes	Percentage of slopes by classification that have been developed or protected.
			Description on the protection of steep slope area according to classification or legal documents protecting slopes steeper than 15%.
			Site survey or topographic map showing steep slope areas relative to project site.

REFERENCED STANDARDS

- Natural Resources Conservation Service (NRCS) (https://www.nrcs.usda.gov)
- NatureServe (https://www.natureserve.org)
- U.S. Endangered Species Act (https://www.fws.gov/law/endangered-species-act)

FURTHER EXPLANATION

OPTION 2. PREVIOUSLY UNDEVELOPED SITES

- **International Tips**
  - International projects may find the following resources useful in accomplishing this credit:
    - The International Union for Conservation of Nature's Red List[13]
    - Global Wetlands Map by the UN Environment Programme[14]
    - The World Database on Protected Areas[15]
- **Flooding**
  - If the project area is covered in flood hazard maps, include the criteria used to delineate flood hazard area and the name of the authority that produced the maps.

13. International Union for Conservation of Nature, "Red List of Threatened Species," 2025, accessed September 19, 2025, https://www.iucnredlist.org/.
14. Global Wetland Watch, "Partners," accessed September 19, 2025, https://www.globalwetlandwatch.org/partners-and-funding/.
15. Protected Planet, "Protected Areas (WDPA)," accessed September 19, 2025, https://www.protectedplanet.net/en/thematic-areas/wdpa?tab=WDPA.

- If no flood hazard maps are available, work with an engineer, hydrologist, or other qualified professional to map the flood hazard areas subject to credit requirements; the professional should also produce a report or an executive summary of findings and supporting documents, such as site elevations or topographic maps and sections identifying the flood risk of the project site.

Steep Slopes

- Unstable slopes are any slopes that do the following:
  - Show evidence of instability, such as slumping, undercutting, mass movement, or erosion scar.
  - Are located in a landslide hazard area as mapped by local, state, or national geological surveys.
  - Are composed of unstable soil types, such as expansive clay or loose granular soils.
  - Have been identified as requiring stabilization in a civil or geotechnical engineering report.
- Suggested order of operations for calculation:
  - First, identify all slopes 15%–25% and all slopes steeper than 25%. Altogether, 40% of this area must be protected from all development and construction, meaning no development or construction can occur there.
  - Next, identify the unstable portions of slopes that are 15%–25% Forty percent of this area must be protected from all development.
  - Next, identify the unstable portions of slopes that are 25% or greater. Sixty percent of this area must be protected from all development.
  - Add the area of all unstable slopes and determine whether they meet or exceed 40% of the total area.
  - To meet the 40% area requirement, a project may need to protect both unstable and stable slopes. If they do not meet or exceed 40% of the total area, then identify stable steep slope area within the project to protect from all development and construction activity in order to meet the threshold.

Equitable Development

LTc2
New Construction (1–2 points)
Core and Shell (1–2 points)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To support the economic and social vitality of communities, provide opportunities for community members to live and work in close proximity, encourage project locations in areas with developmental challenges and promote the ecology, culture, and health of the surrounding area.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–2**
Option 1. Priority Sites	1–2
Path 1. Brownfield Remediation	2
OR	
Path 2. Historic Location	1
AND/OR	
Option 2. Housing and Jobs Proximity	1–2
Path 1. Support Local Economy	1
OR	
Path 2. Location-Efficient Affordable Housing	2
OR	
Option 3. Equitable Construction	2
Schools	**1–2**
Options 1, 2, and/or 3	1–2
OR	
Option 4. Equitable Access to Resources	2
Path 1. Public Use Spaces	1
AND/OR	
Path 2. Community Partnership	1

(continued)

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ACHIEVEMENT PATHWAYS	POINTS
Data Centers, Warehouses, and Distribution Centers	**1–2**
Options 1, 2, and/or 3 AND	1–2
Option 5. Sensitive Project Location	

OPTION 1. PRIORITY SITES (1–2 POINTS)

PATH 1. BROWNFIELD REMEDIATION (2 POINTS)

Locate the project on a brownfield where soil or groundwater contamination has been identified and where the local, state, or national authority (whichever has jurisdiction) requires its remediation. In cases of voluntary remediation by the project team, provide confirmation by the local, state, or national authority (whichever has jurisdiction), verifying that the site is a brownfield. Perform remediation to the satisfaction of the relevant authority.

OR

PATH 2. HISTORIC LOCATION (1 POINT)

Locate the project in a historic district, identified by the local government, based on a growth management plan or policy.

AND/OR

OPTION 2. HOUSING AND JOBS PROXIMITY (1–2 POINTS)

PATH 1. SUPPORT LOCAL ECONOMY (1 POINT)

Employ individuals that live within the administrative district of the project site for 15% of the construction jobs created by the LEED project.

OR

PATH 2. LOCATION-EFFICIENT AFFORDABLE HOUSING (2 POINTS)

For residential or mixed-use projects, include a proportion of new, affordable rental, and/or for-sale dwelling units priced for households earning less than the area median income (AMI). Rental units must be maintained at affordable levels for a minimum of 15 years. Existing dwelling units are exempt from requirement calculations. Meet or exceed the minimum thresholds in Table 1. Projects must meet or exceed the requirements mandated through inclusionary zoning by their local jurisdictions. Additionally, the project must achieve one of the following requirements:

- Meet the requirements of *LTc3: Compact and Connected Development, Option 2. Access to Transit*, for 2 points.
- Meet the requirements of *LTc3: Compact and Connected Development, Option 3. Walkable Location*, for 2 points.
- Locate the project in a community where the jobs-to-housing ratio exceeds 1:2 within 0.5 miles (800 meters) of walking distance.

TABLE 1. Minimum Affordable Units

Unit type	Requirements
Rental dwelling units	Rental units, at least 10% of the project's total residential floor area, priced for up to 60% AMI.
For-sale dwelling units	For-sale units, at least 10% of the project's total residential floor area, priced for up to 80% AMI.

OR

OPTION 3. EQUITABLE CONSTRUCTION (2 POINTS)

Provide access to workforce development training for construction workers through one of the following:

- **Job-related skills training:** This is achieved through on-the-job training in a Department of Labor registered apprenticeship program (or local equivalent for projects located outside the U.S.), demonstrating that 15% or more of total project construction hours were performed by participants enrolled in registered apprenticeship programs.
- **Life-skills training:** These are programs for construction workers, conducted by an organization or government entity on the construction site, covering topics such as financial literacy, debt management, first-time home buying, or entrepreneurship training, demonstrating scheduling of one course per month for the duration of construction.

SCHOOLS (1–2 POINTS)

- Meet Options 1, 2, and/or 3 (1–2 points).

OR

OPTION 4. EQUITABLE ACCESS TO RESOURCES (2 POINTS)

PATH 1. PUBLIC USE SPACES (1 POINT)

In collaboration with school authorities, ensure that at least three of the following types of spaces in the school are accessible to and available for shared use by the public:

- Auditorium
- Gymnasium
- Cafeteria
- One or more classrooms
- Playing fields and stadiums
- Joint parking

Provide access to toilets in joint-use areas after normal school hours.

AND/OR

PATH 2. COMMUNITY PARTNERSHIP (1 POINT)

In collaboration with the school authorities, contract with the community or other organizations to provide at least two types of dedicated-use spaces in the building, such as the following:

- Commercial office
- Health clinic
- Community service centers (provided by state or local offices)
- Library or media center
- Parking lot
- One or more commercial businesses

Provide access to toilets in joint-use areas after normal school hours.

DATA CENTERS, WAREHOUSES, AND DISTRIBUTION CENTERS (1–2 POINTS)

- Meet Options 1, 2, and/or 3 (1–2 points).

AND

OPTION 5. SENSITIVE PROJECT LOCATION

Locate the project building a minimum of 300 feet (90 meters) away from the property lines of the nearest sensitive receptors (e.g., residential areas, schools, day care centers, places of worship, hospitals, community centers, and public parks).

REQUIREMENTS EXPLAINED

This credit integrates equitable outcomes, community engagement, and community health as fundamental considerations for achieving equitable development outcomes. Multiple options and pathways are available to reward projects that show a measurable positive impact on the surrounding community and adopt innovative approaches to equitable outcomes in physical development.

OPTION 1. PRIORITY SITES

Many communities and governments prioritize certain redevelopment sites to address critical human equity issues. Building these sites can revitalize neighborhoods and bring social and economic benefits directly to residents, such as improved access to jobs, housing, and services. This approach helps transform underused or neglected areas into productive spaces that serve the community. Additionally, redeveloping priority sites rather than greenfield or sensitive ecological areas offers substantial environmental benefits, helping to conserve natural landscapes while supporting sustainable urban growth.

PATH 1. BROWNFIELD REMEDIATION

The U.S. Environmental Protection Agency (EPA) defines brownfields as "properties that contain or may contain a hazardous substance, pollutant, or contaminant, complicating efforts to expand, redevelop or reuse them."[16] This path promotes the redevelopment of contaminated sites, where developers remove hazardous materials from a site's soil or groundwater, thereby reducing human and wildlife exposure to environmental pollution and improving environmental health.

16. U.S. EPA, "Brownfield Overview and Definition," last updated March 27, 2025, https://19january2017snapshot.epa.gov/brownfields/brownfield-overview-and-definition_.html.

Redeveloping contaminated sites often reduces the footprint of the project's elements, with a redevelopment site using an average of 78% less land than the same project would use if it were built on a greenfield.[17]

PATH 2. HISTORIC LOCATION

This path rewards investing in historic areas, a proven strategy for maintaining and enhancing community character. Underused properties within historic districts can have a rich history that contributes to both architectural and cultural preservation when incorporated into redevelopment. The redevelopment of sites in historic districts can also reduce urban sprawl through adaptive reuse. These areas were often originally designed as walkable communities before the dominance of automobiles, with compact layouts that support pedestrian access to amenities.

OPTION 2. HOUSING AND JOBS PROXIMITY

Many metropolitan areas within the U.S. exhibited robust job growth in the favorable economic conditions that prevailed from 2010 up to the pandemic-induced recession of 2020.[18] However, not all those areas matched job growth with housing growth. Some of the nation's most successful and productive metropolitan areas failed to meet housing demand, largely because of restrictions on land use.

Projects employing individuals who reside within the administrative district help support the local economy and foster a stronger connection between the community and the project. Additionally, prioritizing the development of location-efficient affordable housing ensures that housing options are both affordable and conveniently located near essential services, public transportation, and employment opportunities. This not only reduces commuting time and environmental impact but also enhances access to resources for residents. Furthermore, projects that invest in workforce development training programs for construction workers can equip them with valuable skills that promote career growth and economic mobility.

PATH 1. SUPPORT LOCAL ECONOMY

Building projects are key drivers of economic growth and development in local communities. They generate employment, boost local economies, and enhance the overall well-being of the regions they impact. The construction industry offers jobs across a wide range of skill levels and creates opportunities within the community. In 2022 in Canada, more than 1.5 million people worked in the construction sector, which had nearly 95,000 additional jobs available.[19] Construction projects can also positively affect local businesses, such as suppliers of construction materials, who may see increased demand, sales, and revenue.

17. George W. Sherk, "Public Policies and Private Decisions Affecting the Redevelopment of Brownfields," Environmental and Energy Management Program, George Washington University, 2001, https://www.researchgate.net/publication/276292802_Public_Policies_and_Private_Decisions_Affecting_the_Redevelopment_of_Brownfields_An_Analysis_of_Critical_Factors_Relative_Weights_and_Areal_Differentials.
18. Eric Kober, "The Jobs–Housing Mismatch: What It Means for U.S. Metropolitan Areas," *Manhattan Institute*, July 7, 2021, https://manhattan.institute/article/the-jobs-housing-mismatch-what-it-means-for-u-s-metropolitan-areas.
19. Olivia Bush, "Construction Industry Statistics in Canada," *Made in CA*, updated January 3, 2025, https://madeinca.ca/construction-industry-statistics-canada/.

Project teams pursuing this path must assess and report on the number of construction jobs generated by the project as part of its contribution to local economic development. A minimum of 15% of the construction jobs created by the LEED project must employ individuals that live within the administrative district, which is defined as a division of local government such as a municipality, county, parish, or equivalent. Jobs should be counted based on full-time equivalents and can include a range of construction jobs from all phases of the project.

PATH 2. LOCATION-EFFICIENT AFFORDABLE HOUSING

Addressing the jobs-housing mismatch is essential to ensure that developers locate affordable housing near employment opportunities, thereby reducing burdens on low-income households. This mismatch occurs when housing availability is geographically disconnected from job centers and forces workers, particularly low-income individuals, to endure long commutes, increased transportation costs, and limited access to employment. Rising rents and property prices are making it harder for people to find homes they can afford, with low-income residents disproportionately affected by high utility costs and service shutoffs. Affordable housing in the right places, along with rental assistance, can reduce financial strain and support long-term stability for these communities. Also, land-use regulations are a significant barrier to affordable housing in urban communities. Zoning restrictions, density limits, and lengthy permitting processes often create challenges for builders trying to build affordable housing in high-demand areas where jobs are concentrated.

Project teams must begin by considering the location of the development and identifying the AMI for the project location. The AMI is a crucial benchmark used to determine affordability, because it represents the midpoint of household incomes in the area, with half of the households earning more and half earning less. Once teams determine AMI, they need to calculate pricing that would qualify as an affordable rental or for-sale unit based on household income levels. The square footage dedicated to these units must comprise a minimum of 10% of the project's total residential floor and meet the AMI requirements of the rating system. Teams must comply with *LTc3: Compact and Connected Development* by meeting either Option 2. Access to Transit, or Option 3. Walkable Location, to ensure that they locate projects near amenities and public transportation. Additionally, they must meet a jobs-to-housing ratio, which must exceed 1:2 within a 0.5 miles (800 meters) walking distance if pursuing the third option. A ratio exceeding 1:2 means there are more than 4.8 jobs for every four housing units within the 0.5 miles (800 meters) walking distance. A jobs-to-housing ratio of 1:2 or greater ensures jobs and housing are located together, which can shorten commuting distances and improve access to employment.

OPTION 3. EQUITABLE CONSTRUCTION

Providing workforce training is a key factor in maintaining an equitable, healthier, and supportive environment for construction workers. Training enhances the technical skills of workers and allows them to work more efficiently, produce higher-quality results, and receive progressive wage increases. Of apprentices who complete a Department of Labor registered apprenticeship program, 94% retain employment with an average annual salary of $80,000 USD.[20] Conducting training programs also makes construction workers feel valued and invested in. Research has shown that

20. Apprenticeship USA, "Explore Registered Apprenticeship," updated April 2024, https://www.apprenticeship.gov/sites/default/files/DOLIndFSApprent101-043024-508.pdf.

this can lead to higher job satisfaction, lower turnover rates, and reduced costs for recruiting and training new employees.[21]

Projects must develop and implement a strategy for providing workforce development training for construction workers that focuses on either job-related skills training in a formal apprenticeship program approved and validated by a local government agency, accredited school, labor union, or other training programs conducted by an organization or government entity.

SCHOOLS

- Refer to Options 1, 2 and/or 3.

OPTION 4. EQUITABLE ACCESS TO RESOURCES

Sharing amenity spaces in the school with the public, organizations, and businesses will bring social benefits to the local community. Shared spaces also help to reduce the need for new development, thereby preserving previously undeveloped land and avoiding the financial costs and environmental consequences of new construction. Schools that typically go unused during after-school operating hours can offer to host community programs and maximize the useful life of the building. In addition, communities may enjoy new or more convenient services.

PATH 1. EQUITABLE ACCESS TO RESOURCES

Teams must collaborate with school authorities to identify and allocate at least three types of eligible shared spaces for public use, discuss public needs to determine which spaces will be available, and obtain written confirmation specifying the selected spaces. These spaces may be in another school building within 0.25 miles (400 meters) walking distance, as long as it is part of the same campus, and they must include toilet access to shared-space users after school hours.

PATH 2. COMMUNITY PARTNERSHIP

Project teams must collaborate with school authorities to identify and allocate at least two types of eligible dedicated-use spaces within the project that will be made available to specific outside organizations and with the school authorities and the chosen outside organization(s) to determine which spaces in the project will be shared. Teams are encouraged to conduct a local community needs assessment to identify high-priority services and guide outreach to relevant organizations. Spaces must have designated toilet access to shared-space users after normal school hours.

DATA CENTERS, WAREHOUSES, AND DISTRIBUTION CENTERS

- Meet Options 1, 2, and/or 3 and Sensitive Project Location.

Data centers and warehouses situated near residential areas or other sensitive sites, such as schools and day care centers, can subject the community to issues including air and sound pollution, environmental impacts, traffic, safety, and other community disruptions. For example, data

21. Kate Rockwood, "How Learning and Development Can Attract and Retain Talent," *SHRM*, January 14, 2022, https://www.shrm.org/topics-tools/news/all-things-work/how-learning-development-can-attract-and-retain-talent.

centers produce constant noise and vibrations because they require extensive amounts of heating, ventilation, and cooling systems to maintain electronics operations.[22]

Projects must be at least 300 feet (90 meters) away from the property lines of the closest sensitive receptors. To protect the health, safety, and well-being of vulnerable populations and reduce the negative impacts of construction activities, project teams must identify and map all sensitive receptors in the surrounding area to ensure compliance and minimize disruption.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Priority Sites	Path 1. Brownfield Remediation	Documentation from authority having jurisdiction (AHJ) declaring existence of specific contamination and confirming that remediation has been completed to its satisfaction.
		Path 2. Historic Location	Map showing the project is within a historic district.
			Documentation from AHJ over historic district demonstrating the existence of the historic district (e.g., ordinance, growth management plan, government website with corresponding map, etc.).
	Option 2. Housing and Jobs Proximity	Path 1. Support Local Economy	Description of types of jobs created by the project.
			Total number of construction jobs created by the project.
			Number of local construction jobs created by the project.
			Evidence of the contributing employed individuals live within the administrative district of the project site.
		Path 2. Location-Efficient Affordable Housing	Evidence of AMI thresholds (60% and 80%) using data from a governmental entity.
			Copy of legal documentation to maintain affordable rates for at least 15 years.
			Project's total residential floor area.
			Project's total affordable residential floor area.
			Number of affordable units created (rent and sale combined).
			Confirmation showing project meets LTc1: Compact and Connected Development, Options 2 or 3, OR Documentation demonstrating project is in a community where the jobs-to-housing ratio exceeds 1.2 within a 0.5 mile (800-meter) walking distance.
	Option 3. Equitable Construction	All	Description of strategies to provide job-related skills training or life-skills training programs for construction workers and who is conducting the training.

22. Ethan R. Brush, "Noise and Vibration Considerations for Data Centers and IT Facilities," *FMJ*, December 9, 2016, https://www.acentech.com/wp-content/uploads/2021/08/Noise-and-Vibration-Considerations-for-Data-Centers-and-IT-Facilities.pdf.

(continued)

Project Types	Options	Paths	Documentation
Schools	Option 4. Equitable Access to Resources	Path 1. Public Use Spaces	Description of strategies to support accessibility to at least three school spaces and accessibility to toilets for public use.
			Floor plan and/or site plan to show where the uses occur and access to toilets.
		Path 2. Community Partnership	Description of strategies to support provision of at least two dedicated-use spaces and accessibility to toilets in the building for the community.
			Floor plan and/or site plan to show where the uses occur and access to toilets.
Data centers, warehouses, and distribution centers	All	All	Vicinity map locating the project building and sensitive receptors with property lines.

REFERENCED STANDARDS

- None

FURTHER EXPLANATION

OPTION 1. PRIORITY SITES, PATH 1. BROWNFIELD REMEDIATION

- If the project is located on a site that has already been completely assessed and remediated, the results of that assessment and remediation may be used toward achievement of this option if complete documentation is provided.
- If part of the site is found to have contamination, then the entire area within the LEED project boundary is considered a contaminated site.

OPTION 1. PRIORITY SITES, PATH 2. HISTORIC LOCATION

- A historic district is a geographically defined area that includes a concentration of historic buildings, structures, sites, or objects that together convey a unified historical, cultural, or architectural significance. These elements are typically in close physical proximity and are collectively identified as a historic district by the local government based on a growth management plan or policy.

OPTION 2. HOUSING AND JOBS PROXIMITY, PATH 1. SUPPORT LOCAL ECONOMY

- This pathway is most easily achieved when it is planned for early in the design process; for example, starting with a construction firm within the administrative district can help achieve this credit.

- Project teams should consider integrating local hiring requirements into the bidding process early on to achieve this pathway.
- Only jobs created directly from the project's activities may be counted. Secondary job creation, such as construction jobs for a café that independently opens outside of the project area in order to serve new residents and workers from the project, should not be counted.
- Documentation that may help demonstrate the hiring within the administrative district may be a construction bid requiring local hiring or contracts with subcontractors requiring local hiring. Daily construction logs may also prove useful in demonstrating the total number of employees and hours on a project.
- Projects may document job creation that arises from various phases of the project, including design, construction, and related administrative support. However, created construction jobs after LEED certification should not be included.
- It is necessary to determine both the total number of jobs created regardless of where the employed individuals reside and the number of jobs created and filled by individuals that reside within the administrative district.
- The number of jobs created by a project means the total full-time employees (FTEs) who performed work on the project.
- The simplest way to demonstrate the calculations is to add all the hours worked throughout the life of the project and divide by the total hours worked by a single full-time employee within a year (approximately 52 weeks × 40 hours/week = 2,085). This calculation results in an estimated total FTEs by pooling together all labor hours.

Example Equations:

- $\text{Total FTEs} = \dfrac{\text{Total Hours Worked}}{40\,\dfrac{\text{hours}}{\text{week}} \times 52\text{ weeks}}$
- $\text{Local FTEs} = \dfrac{\text{Total Hours Worked by Employees within Admin. District}}{40\,\dfrac{\text{hours}}{\text{week}} \times 52\text{ weeks}}$
- $\%\text{ FTEs within Admin. District} = \dfrac{\text{Local FTEs}}{\text{Total FTEs}} \times 100$

OPTION 2. HOUSING AND JOBS PROXIMITY, PATH 2. LOCATION-EFFICIENT AFFORDABLE HOUSING

- Residential floor area should be interpreted as net floor area. This includes livable floor space within a building and does not include unlivable space, such as stairwells, common spaces, mechanical rooms, walls, or columns.
- In the U.S., the area median income (AMI) is published annually by the U.S. Department of Housing and Urban Development to determine federally recognized affordability levels. The latest available AMI at the time of project registration must be used.
- For international projects, AMI may not be determined the same as in the U.S. Use the most locationally relevant and recent median income number published by a government agency.

- Inclusionary zoning is a land use policy enacted by government that requires or incentivizes developers (such as with tax abatements, parking reductions, or the right to build at higher densities) to include a certain percentage of affordable housing units in a development. Inclusionary zoning may also be referred to under other names, such as mandatory housing affordability, density bonus program, and workforce housing requirement, among others.

OPTION 4. EQUITABLE ACCESS TO RESOURCES, PATH 2. COMMUNITY PARTNERSHIP

- A dedicated-use space in a building refers to a space in a school that is always reserved for a particular use by a specific outside organization.

Compact and Connected Development

LTc3
New Construction (1–6 points)
Core and Shell (1–6 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To conserve land and ecosystem resources by encouraging development in areas with existing infrastructure. To promote livability, walkability, and transportation efficiency, including reduced vehicle distance traveled and associated emissions.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1-6**
Option 1. Surrounding Density AND/OR	1–2
Option 2. Access to Transit Path 1. Public Transit Service OR Path 2. Project-Sponsored Transit Service AND/OR	1–4 1–4 1–2
Option 3. Walkable Location	1–3
Schools	**1-6**
Options 1, 2, and/or 3 AND/OR	1–6
Option 4. Surrounding Density and Development Path 1. Surrounding Density AND/OR Path 2. Connected Site AND/OR	1–2 1–2 1–2

(*continued*)

ACHIEVEMENT PATHWAYS	POINTS
Option 5. Access to Transit or Pedestrian Access	1–4
Path 1. Access to Transit	1–4
OR	
Path 2. Pedestrian Access	1–2
Data Centers, Warehouses, and Distribution Centers	**1–6**
Options 1, 2, and/or 3	1–6
AND/OR	
Option 6. Surrounding Development and Resources	1–2
Path 1. Development and Adjacency	1–2
AND/OR	
Path 2. Transportation Resources	1–2
Healthcare	**1–6**
Options 1, 2, and/or 3	1–6
AND/OR	
Option 7. Surrounding Density	1

OPTION 1. SURROUNDING DENSITY (1–2 POINTS)

Located on a site where the surrounding existing density within 0.25 miles (400 meters) offset of the project boundary meets the values in Table 1. Use either the separate residential and nonresidential densities or the combined density values in Table 1.

TABLE 1. Points for Average Existing Density Within 0.25 Miles (400 Meters)

Combined density		Separate density			
Square Feet per Acre of Buildable Land	Square Meters per Hectare of Buildable Land	Residential Density (DU/acre)	Residential Density (DU/hectare)	Nonresidential Density (FAR)	Points
22,000	5,050	7	17.5	0.5	1
35,000	8,035	12	30	0.8	2

NOTE: DU = dwelling unit; FAR = floor-area ratio.

AND/OR

OPTION 2. ACCESS TO TRANSIT (1–4 POINTS)

PATH 1. PUBLIC TRANSIT SERVICE (1–4 POINTS)

Locate any functional entry of the project within either:

- 0.25 miles (400 meters) walking distance of existing or planned bus, streetcar, or informal transit stops.

OR

LT

- 0.5 miles (800 meters) walking distance of existing or planned bus rapid transit stops, passenger rail stations (e.g., light, heavy, or commuter rail), or commuter ferry terminals.

The transit service at these stops and stations in aggregate must meet the minimums listed in Table 2.

Both weekday and weekend trip minimums must be met.

- For each qualifying transit route, only trips in one direction are counted toward the threshold.
- If service varies by day:
  - For weekday trips, count the weekday with the lowest number of trips.
  - For weekend trips, only count the weekend day with the highest number of trips.
- If a qualifying transit route has multiple stops within the required walking distance, only trips from one stop are counted toward the threshold.
- Planned stops and stations may count if they are sited, funded, and under construction by the date of the certificate of occupancy and are complete within 24 months of that date.

TABLE 2. Minimum Daily Public Transit Service

Weekday Trips	Weekend Trips	Points
72	30	1
132	78	2
160	120	3
360	216	4

OR

PATH 2. PROJECT-SPONSORED TRANSIT SERVICE (1–2 POINTS)

Commit to providing year-round transit service (e.g., vans, shuttles, or buses) for regular occupants and visitors that meets the minimums listed in Table 3. Service must provide transportation between the project site and external destinations, such as residential areas and public transportation stations, and be guaranteed for at least three years from the project's certificate of occupancy.

Provide at least one accessible transit stop shelter within 0.25 miles (400 meters) walking distance from a functional entry of the project.

- For each qualifying transit route, total trips (inbound and outbound) are counted toward the threshold.
- If a qualifying transit route has multiple stops within the required walking distance, only trips from one stop are counted toward the threshold.

TABLE 3. Minimum Daily Project-Sponsored Transit Service

Total Daily Trips	Points
Providing shuttles	1
30	2

AND/OR

OPTION 3. WALKABLE LOCATION (1–3 POINTS)

Locate on a site that meets the location efficiency requirements in Table 4 via Walk Score® or has proximity to existing and publicly available uses within 0.5 miles (800 meters) walking distance from any functional entry.

TABLE 4. Points for Location Efficiency

Walk Score	Proximity to Uses	Points
60–69	4–7	1
70–79	8–10	2
80 or more	≥ 11	3

The following restrictions apply:

- A use may be counted as only one-use type (e.g., a retail store may be counted only once even if it sells products in several categories).
- No more than two uses in each type of use may be counted (e.g., if five restaurants are within walking distance, only two may be counted).
- The counted uses must represent at least three of the five categories.

SCHOOLS (1–6 POINTS)

- Meet Options 1, 2, and/or 3 (1–6 points).

AND/OR

OPTION 4. SURROUNDING DENSITY AND DEVELOPMENT (1–2 POINTS)

PATH 1. SURROUNDING DENSITY (1–2 POINTS)

Meet Option 1. Surrounding Density, requirements in New Construction.

AND/OR

PATH 2. CONNECTED SITE (1–2 POINTS)

Locate the project on a previously developed site that also meets one of the connected site conditions listed below:

- **Adjacent site:** At least a contiguous 25% of the project boundary must border parcels that are previously developed sites.
- **Infill site:** At least 75% of the project boundary must border parcels that are previously developed sites.

Bordering rights-of-way do not constitute previously developed land; it is the status of the property on the other side of the right-of-way that contributes to the calculation. Any part of the boundary that borders a water body is excluded from the calculation.

TABLE 5. Points for Connected Site

Type of Site	Points
Adjacent	1
Infill	2

AND/OR

OPTION 5. ACCESS TO TRANSIT OR PEDESTRIAN ACCESS (1–4 POINTS)

PATH 1. ACCESS TO TRANSIT (1–4 POINTS)

Meet Option 2. Access to Transit, listed above.

OR

PATH 2. PEDESTRIAN ACCESS (1–2 POINTS)

Locate the project with an attendance boundary where dwelling units are:

- Within 0.75 miles (1,200 meters) walking distance of a functional entry of a school building for students in eighth grade or below or ages 14 and below.
- Within 1.5 miles (2,400 meters) walking distance of a functional entry of a school building for students in ninth grade or above or ages 15 and above.

Provide pedestrian access to the site from all residential areas in the attendance boundary.

Points are awarded according to Table 6.

TABLE 6. Points for Dwelling Units Within Walking Distance

Percentage of Dwelling Units in Attendance Boundary	Points
50%	1
60%	2

DATA CENTERS, WAREHOUSES, AND DISTRIBUTION CENTERS (1–6 POINTS)

- Meet Options 1, 2, and/or 3 above (1–6 points).

AND/OR

OPTION 6. SURROUNDING DEVELOPMENT AND RESOURCES (1–2 POINTS)

PATH 1. DEVELOPMENT AND ADJACENCY (1–2 POINTS)

Locate the project on a site that meets one of the site conditions listed in Table 7.

- To qualify as an adjacent site, at least a contiguous 25% of the project boundary must border parcels that are previously developed sites.
- Bordering rights-of-way do not constitute previously developed land; it is the status of the property on the other side of the right-of-way that contributes to the calculation. Any part of the boundary that borders a water body is excluded from the calculation.

TABLE 7. Points for Development and Adjacency

Type of Site	Points
Previously developed site that was used for industrial or commercial purposes.	1
Previously developed and adjacent site with bordering parcels currently used for industrial or commercial purposes.	2

AND/OR

PATH 2. TRANSPORTATION RESOURCES (1–2 POINTS)

Locate the project on a site that has two of the following transportation resources for 1 point or all four of the following transportation resources for 2 points:

- The site is within a 10-mile (16-kilometer) driving distance of a main logistics hub.
- The site is within a one mile (1,600 meter) driving distance of an on-off ramp to a highway.
- The site is within a one mile (1,600 meter) driving distance of an access point to an active freight rail line.
- The site is served by an active freight rail spur.

A planned transportation resource must be sited, funded, and under construction by the date of the certificate of occupancy and complete within 24 months of that date.

HEALTHCARE (1–6 POINTS)

- Meet Options 1, 2, and/or 3 above.

AND/OR

OPTION 7. SURROUNDING DENSITY (1 POINT)

Locate on a site where the surrounding existing density within 0.25 miles (400 meters) offset of the project boundary meets one of the following:

- At least seven dwelling units per acre (17.5 DU per hectare) with a 0.5 floor area ratio. The counted density must be existing density, not zone density.
- At least 22,000 square feet per acre (5,050 square meters per hectare) of buildable land.

For previously developed existing rural healthcare campus sites, achieve a minimum development density of 30,000 square feet per acre (6,890 square meters per hectare).

REQUIREMENTS EXPLAINED

This credit awards project sites near essential services and in densely built locations with existing infrastructure. There are multiple paths to achieving credit compliance. Project teams should review the site location and the urban context of the project, including building density, community amenities, and public transportation routes, to optimize the options selected for the project.

OPTION 1. SURROUNDING DENSITY

Projects in a high-density area are likely to be more walkable and have greater amenities. Additionally, a denser and more compact neighborhood increases efficiency and reduces the time getting from one location to another. It provides easier access to basic services including supermarkets, pharmacies, banks, medical clinics, and offices. People can actively choose a means of travel when amenities are closely located. More importantly, high surrounding density promotes more efficient land use and achieves more sustainable growth patterns that protect natural habitats, farmland, and open spaces, ultimately leading to more resilient and livable communities.

Projects must meet minimum thresholds for residential, nonresidential, or combined density. Planners choose these minimum density requirements because they are the levels necessary to support public transit and reduce urban sprawl. Residential density is measured in dwelling units per acre. Nonresidential density is measured in floor area ratio (FAR), which is the ratio of the gross floor area to the size of the lot. FAR measures the density of the lot according to a building's gross floor area. A low FAR indicates low density, whereas a high FAR indicates higher density. The "combined density" thresholds are measured in square feet per acre (or square meters per hectare) of buildable land, and they correspond directly to the FAR for nonresidential density.

OPTION 2. ACCESS TO TRANSIT

PATH 1. PUBLIC TRANSIT SERVICE

A project located in a densely built environment with a compact and connected transportation and infrastructure network reduces environmental impact and enhances the quality of life for regular occupants. When people take public transportation, there are fewer private vehicles on the road, which reduces VMTs and associated GHG emissions and air pollution. In addition, buildings close to public transportation can significantly promote equity by providing various benefits to communities, particularly those that are underserved or disadvantaged. Accessible public transportation provides accessibility and mobility to people who do not have access to private vehicles and fosters social interaction by connecting different neighborhoods.

Projects must be located within 0.25 miles (400 meters) distance of existing lower capacity transit facilities (e.g., bus, streetcar, informal transit stops) or 0.5 miles (400 meters) of high-capacity transit facilities (e.g., bus rapid transit, rail stations). The distances from transit facilities correspond to roughly a five-minute and 10-minute walking time. Additionally, minimum transit frequency has been set to ensure a minimum viable level of service. The weekday trip thresholds in the rating system roughly correspond to leading interval times of 1–2 hours (72 trips per weekday) to 10–20 minutes (360 trips per weekday). Together, the distance and frequency work together to ensure that regular building occupants have a viable transit option for daily travel.

PATH 2. PROJECT-SPONSORED TRANSIT SERVICE

A sponsored transit service, such as vans, shuttles, or buses, for regular occupants and visitors provides direct and convenient transit options for work and access to amenities for day-to-day needs. Offering a sponsored transit service can reduce the number of private vehicles, which lessens pollution, eases traffic congestion in neighborhoods, and enhances the commuting experience for regular occupants. Additionally, project-sponsored transit services can significantly reduce the need for parking infrastructure. Land that would have been used for parking can be repurposed for other uses, including buildings, green spaces, pollinator habitats, and recreational areas.

Adapting to new routines can take time, which is why a three-year commitment is required for the project-sponsored transit service. This can allow the project adequate time to promote the service and encourage ridership. Also, 0.25 miles (400 meters) walking distance from a functional entry of the project is required because it corresponds roughly to a five-minute walking time, which is convenient for most people.

OPTION 3. WALKABLE LOCATION

Walk Score

Knowing the Walk Score of the project site provides a better understanding of how the building location can encourage physical activity, reduce GHG emissions, and foster social interactions.[23] Walk Score data is categorized as either supported or unsupported. A supported Walk Score is based on verified, publicly available data and can be used to demonstrate compliance with walkability requirements for credit achievement. An unsupported Walk Score, however, relies on incomplete data and cannot be used to achieve this credit.

A high Walk Score means that the location is highly walkable, and that area is likely to have many amenities and services within a short walking distance. A location with a low Walk Score indicates that the location is not as walkable, and building users would likely rely heavily on public or private transit for daily activities.

Projects must be located on a site with a Walk Score of at least 60 to earn a point.

Location efficiency

Locating a project near existing public amenities like parks, restaurants, supermarkets, clinics, and schools can significantly improve residents' quality of life and foster a vibrant, connected community. Increased walkability encourages building users to walk or bike to their destinations, which promotes healthier, active lifestyles. A reduced dependence on cars directly results in reduced VMTs, reduced GHG emissions, and improved outdoor air quality.

A walking distance of 0.5 miles (800 meters) from the project is required because it corresponds roughly to a 10-minute walking time, which most people find reasonable to access the publicly available uses. The goal is to make the number of required publicly available uses equivalent to the corresponding Walk Score ranking. Use Table 8 to determine the types and numbers of uses.

23. Walk Score®, https://www.walkscore.com/.

TABLE 8. Use Types and Categories[24]

Category	Use Type
Food retail	Supermarket
	Grocery with produce section
Community-serving retail	Convenience store
	Farmers market
	Hardware store
	Pharmacy
	Other retail
Services	Bank
	Family entertainment venue (e.g., theater, sports)
	Gym, health club, exercise studio
	Hair care
	Laundry, dry cleaner
	Restaurant, café, diner (excluding those with only drive-thru service)
Civic and community facilities	Adult or senior care (licensed)
	Childcare (licensed)
	Community or recreation center
	Cultural arts facility (e.g., museum, performing arts)
	Education facility (e.g., K–12 school, university, adult education center, vocational school, community college)
	Government office that serves public on-site
	Medical clinic or office that treats patients
	Place of worship
	Police or fire stations
	Post office
	Public library
	Public park
	Social services center
Community anchor uses (BD+C and ID+C only)	Commercial office (e.g., 100 or more full-time equivalent jobs)

SCHOOLS

- Meet Options 1, 2, and/or 3.

24. U.S. Green Building Council, "Community Resources," last accessed April 2, 2025, https://www.usgbc.org/credits/homes-higharise/v4-draft/ltc4.

OPTION 4. SURROUNDING DENSITY AND DEVELOPMENT

PATH 1. SURROUNDING DENSITY

Same as Option 1. Surrounding Density.

PATH 2. CONNECTED SITE

When selecting a project location, it is important to consider whether the parcels directly adjacent or on the other side of the road are previously developed sites. More previously developed land that exists along the border of the project parcel means that existing infrastructure is reused, and the need for new construction materials is reduced, which decreases greenhouse emissions and reduces the environmental impacts from new construction.

The project team must identify parcels adjacent to the project's perimeter on a map and determine whether the project meets the criteria for an adjacent site or infill site. Teams must measure the project's entire perimeter, the length of perimeter segments adjacent to the waterfront (if any) and the longest continuous perimeter segments adjacent to qualifying parcels.

Use Equation 1 to determine the percentage of the project boundary that is continuously adjacent to previously developed parcels. The site qualifies as an adjacent site if the total meets or exceeds 25%, excluding any waterfront adjacency.

Equation 1. Percentage of adjacent boundary for adjacent sites

$$\text{\% adjacent boundary} = \frac{\text{Continuous perimeter adjacent to previously developed parcels}}{(\text{Total perimeter} - \text{Any waterfront perimeter})} \times 100$$

Equation 2 can be used to determine if the site qualifies as an infill site. If at least 75% of the project's boundary is adjacent to previously developed parcels, excluding any waterfront perimeter, the site qualifies as an infill site.

Equation 2. Percentage of adjacent boundary for infill sites

$$\text{\% adjacent boundary} = \frac{\text{Any perimeter length adjacent to previously developed parcels}}{(\text{Total perimeter} - \text{Any waterfront perimeter})} \times 100$$

OPTION 5. ACCESS TO TRANSIT OR PEDESTRIAN ACCESS

PATH 1. ACCESS TO TRANSIT

Same as Option 2. Access to Transit. Transit access to schools ensures that workers, students, and families can access the school facility.

PATH 2. PEDESTRIAN ACCESS

Locating a school within an attendance boundary is important to ensure a safe and accessible environment for students of various grades and ages.

DATA CENTERS, WAREHOUSES, AND DISTRIBUTION CENTERS

- Meet Options 1, 2, and/or 3.

OPTION 6. SURROUNDING DEVELOPMENT AND RESOURCES

PATH 1. DEVELOPMENT AND ADJACENCY

Data centers and warehouses are encouraged to select sites previously developed and adjacent to parcels currently used for industrial or commercial purposes. This helps preserve undeveloped land and conserves natural habitats. Developing on a site next to existing industrial or commercial buildings also means existing infrastructure is in place, which may result in lower greenhouse gas emissions from future infrastructure development.

The project team must identify parcels adjacent to the project's perimeter on a map and determine whether the project is an adjacent site by measuring the project's entire perimeter, the length of perimeter segments adjacent to the waterfront (if any), and the longest continuous perimeter segments adjacent to qualifying parcels. Use Equation 3 to determine the percentage of the project boundary adjacent to previously developed parcels.

Equation 3. Percentage of adjacent boundary for adjacent sites

$$\%\text{ adjacent boundary} = \frac{\text{Continuous perimeter adjacent to previously developed parcels}}{\left(\text{Total perimeter} - \text{Any waterfront perimeter}\right)} \times 100$$

PATH 2. TRANSPORTATION RESOURCES

Locating data centers and warehouses near various transportation resources provides multiple opportunities to transport goods and services. Proximity to the main logistics hub, an on-off ramp to a highway, and access points to an active freight rail line would facilitate efficient movement of people and boost the overall economy by attracting businesses and residents.

Mapping must show the driving routes and distances from the project to any applicable transportation resources. The team must also verify that any transportation resources are completed. Resources that are funded, under construction by the date of certificate of occupancy, and completed within 24 months of that date will also count toward credit compliance.

HEALTHCARE

- Meet Options 1, 2, and/or 3.

OPTION 7. SURROUNDING DENSITY

Healthcare facilities can be significantly sized developments and should contribute to compact and connected development. This is achieved by locating a project in an area that meets surrounding density thresholds in the rating system. However, not all healthcare facilities are in dense areas, especially when they need to serve rural communities. This option also allows flexibility for previously developed existing healthcare facilities by meeting campus development density thresholds instead.

Project teams must identify the building site and buildable land within 0.25 miles (400 meters) offset of the project boundary on a map. Collect information on the surrounding residential and nonresidential building densities, including number of dwelling units and building floor area for all properties within the offset of the boundary, and confirm residential densities and FAR meet or exceed the residential density of 7 DU/acre (17.5 DU/hectare) and nonresidential density (FAR) of 0.5 using separate density. Projects must meet the 22,000 square feet per acre of buildable land for combined density. For previously developed rural healthcare campus sites, achieve a minimum development density of 30,000 sf/acre.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Surrounding Density	Combined density	Vicinity map showing buildable land, any nonbuildable land excluded, project site, and footprints of existing buildings within 0.25 miles (400 meters) offset of project site.
			Surrounding combined density calculation.
			Combined density (in square feet per acre, or square meters per hectare).
		Separate density	Vicinity map showing buildable land, any nonbuildable land excluded, project site, and footprints of existing residential and nonresidential buildings within 0.25 miles (400 meters) offset of project site.
			Surrounding combined density calculation.
	Option 2. Access to Transit	Path 1. Public Transit Service	Vicinity map indicating the project location, location of the transit stop(s), routes serving each stop, and the walking routes (with walking distance noted) between the location of the project functional entry and the stop(s).
			Weekday and weekend route schedules showing the frequency of trips and service.
			For any planned transit service, documentation of planned transit stops and stations sited, funded, and under construction by the date of the project's certificate of occupancy and scheduled for completion within 24 months of that date.
		Path 2. Project-Sponsored Transit Service	Description of project-sponsored service and shelters meeting credit criteria.
			Vicinity map indicating the project location, location of the transit stop(s), route(s) serving each stop, and the walking routes (with walking distance noted) between the location of the project functional entry and the stop(s).
			Weekday and weekend route schedules showing the frequency of trips and service.
			Verification (e.g., letter of assurance or agreement, etc.) that transit service will be provided for at least three years from the project's certificate of occupancy.
	Option 3. Walkable Location	Location efficiency	Walk Score document (e.g., screenshot) showing the score for the project's address.
			Project's Walk Score.
		Public uses	Vicinity map and table of uses identified by type of uses accessible within 0.5 miles (800 meters) walking distances.
Schools	Option 4. Surrounding Density and Development	Path 1. Surrounding Density	Same as Option 1.
		Path 2. Connected Site	Documentation confirming previously developed status of the site.
			Percentage of contiguous project perimeter or boundary bordering previously developed parcels (must be at least 25% for adjacent site status).
			Percentage of project perimeter or boundary bordering previously developed parcels (must be at least 75% for infill status).
			Site plan or vicinity map showing previously developed parcels adjacent to the project boundary with notes confirming adjacent or infill site status.

(continued)

(continued)

Project Types	Options	Paths	Documentation
	Option 5. Access to Transit or Pedestrian Access	Path 1. Access to Transit	Same as Option 2.
		Path 2. Pedestrian Access	Vicinity map showing walkshed boundary for each grade/age level, functional entry of school building and estimated calculations of dwelling units (%) within each walkshed boundary.
			Confirmation of pedestrian access to the site from all residential areas in the attendance boundary.
Data centers, warehouses and distribution centers	Option 6. Surrounding Development and Resources	Path 1. Development and Adjacency	Site plan or vicinity map showing project, its previous development, and use.
			Documentation confirming adjacent site status.
			Calculation showing the percentage of contiguous project perimeter or boundary bordering previously developed parcels (must be at least 25% for adjacent site status).
		Path 2. Transportation Resources	Site plan or vicinity map showing project site, location and type of transportation resources, and driving distance to each.
			If planned transportation resources are counted, verification that they will be sited, funded and under construction by date of project's certificate of occupancy and complete within 24 months of that date.
Healthcare	Option 7. Surrounding Density	All	Documentation in accordance with Option 1. with rural healthcare campus sites demonstrating achievement of minimum development density of 30,000 square feet per acre (6,890 square meters per hectare).

REFERENCED STANDARDS

- Walk Score (https://www.walkscore.com)

FURTHER EXPLANATION

OPTION 1. SURROUNDING DENSITY

- Combined density is the gross floor area of all existing residential and commercial buildings, excluding parking garages, divided by the total area of buildable land within the area offset from the project boundary.
- General tips for determining surrounding density:
  - Plot a 1/4-mile (400-meter) radius around the project site from the project boundary and indicate building site types (residential, nonresidential, mixed-use) and indicate buildable land.
  - Within the radius, determine whether sufficient information is available to calculate residential and nonresidential building densities separately. If the project team cannot determine the number of dwellings units, use the combined density calculation.
  - If the area features a few extremely dense buildings, start with these first to see whether the threshold can be met without further calculations.

  - To calculate separate densities, calculate total dwelling units per acre or hectare for all residential buildings and calculate the total floor-area ratio for all nonresidential buildings.
- Guidance for determining separate densities with mixed uses:
  - If there are mixed-use buildings within the radius, then apportion an amount of residential and nonresidential land area based on the ratios of residential and nonresidential floor area to the total building area. Then, use the allocated land areas in the overall residential and nonresidential density calculations for the property. Use the following guidance:
    - First, apportion the building gross floor according to the following equations:
      - Residential gross floor area of the building divided by the total gross floor area of the building equals the percent of residential building floor area.
      - Nonresidential gross floor area of the building divided by the total gross floor area of the building equals the percent of nonresidential building floor area.
    - Second, apportion the land area of the property as residential or nonresidential according to the following equations:
      - Residential land of the mixed-use building equals the percent of residential building floor area multiplied by the total mixed-use land area (acre/hectare).
      - Nonresidential land of the mixed-use building equals the percent of nonresidential building floor area multiplied by the total mixed-use land area (acre/hectare).
    - Third, include the property land areas in the Residential and Nonresidential Density Calculations. A property will appear twice: once in the residential density calculation and once in the nonresidential calculation.
      - Take the residential land area of the mixed-use building and include it in the Property Land Area in the Residential Density Calculation for the property.
      - Take the nonresidential land area of the same mixed-use building and include it in the Property Land Area in the Nonresidential Density Calculation for the property.
      - Complete the rest of the information for the Residential (number of dwelling units) and Nonresidential (nonresidential building gross area) Density Calculations, for the property (along with any other buildings or tracts of land within the offset).
- Physical education spaces that are part of the school campus but are located outside the project boundary, such as playing fields and ancillary buildings used during sporting events only (e.g., concession stands) and playgrounds with play equipment are excluded from the development density calculations.

OPTION 2. ACCESS TO TRANSIT, PATH 1. PUBLIC TRANSIT SERVICE

- Each point at which a transit vehicle stops to receive or discharge passengers is considered a separate stop. This includes stops facing each other on opposite sides of a street. If a route has two separate stops to serve each direction (e.g., on opposite sides of a street or on separated, one-way streets), choose one stop from which to measure the distance to that route.
- Any transit stop reaching any functional entry of the project within the specified distance can be counted toward the credit; different stops within walking distances of different functional entries may count, provided they meet the credit requirements.
- For a given transit route, count trips in only one direction. If service varies by weekday, count the weekday with the lowest number of trips. If weekend counts are different, use an average.

- Schools are not required to evaluate weekend transit service if students do not commute to schools on weekend days.
- An individual transit stop can be counted only once, regardless of the number of entrances within walking distance of it.
- If weekday or weekend trips meet different point thresholds, then the lowest performing time period (weekday or weekend) determines the number of points earned.

OPTION 2 . ACCESS TO TRANSIT, PATH 2. PROJECT-SPONSORED TRANSIT SERVICE

- Accessible transit stop shelter: A dedicated waiting area with adequate lighting at a transit stop that is covered to protect users from the weather and adjoins accessible routes to a functional entry of the project and the transit boarding area is required. If there is seating or other obstructions, a minimum 30 × 48 inches (76 × 122 cm) of clear space must be maintained for mobility aids. There must be an accessible route around any obstructions and a 60-inch (1.5-meter) turning area. Any entry must be at least 32 inches (81.3 centimeters) wide.

OPTION 3. WALKABLE LOCATION

- The project team must map eligible existing uses or planned uses. Planned uses must be available within one year of the date of a project's certificate of occupancy. Existing and planned uses must be classified with the use types in Table 8, "Use Types and Categories."
- It is recommended that the project team conduct site visits and use satellite imagery or aerial photography to map out the walking routes from the project's main entrance to all eligible uses. To ensure a diverse and balanced mix of amenities and services that cater to different needs (e.g., dining, shopping, recreation, healthcare), the counted uses must represent at least three of the five categories.

SCHOOLS, OPTION 5. PEDESTRIAN ACCESS

- Create a walkshed boundary using mapping software (GIS or CAD) to indicate the areas within a 0.75-mile (1,200-meter) walk (for young students as defined in the rating system) or 1.5-mile (2,400-meter) walk (for older students).
- Compare the walkshed boundary with the attendance boundary map for the school. The attendance boundary map or accompanying analysis generally indicates where concentrations of students live (or are anticipated to live), without indicating precise addresses. Using this information, estimate the required percentage of students who live within the walkshed boundary.

Transportation Demand Management

LTc4
New Construction (1–4 points)
Core and Shell (1–4 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To reduce pollution and land development effects from automobile use through encouraging alternative transportation networks. To promote more livable and healthy communities through reduced vehicle distance traveled and associated emissions.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–4**
Transportation Demand Assessment AND	
Option 1. Parking	1–3
Path 1. Reduce Parking	1–3
AND/OR	
Path 2. Parking Fee	2
AND/OR	
Option 2. Active Travel Facilities	1–3
Path 1. Bicycle Network and Storage	1
AND/OR	
Path 2. Shower and Changing Facilities	1
AND/OR	
Path 3. Bicycle Maintenance	1

TRANSPORTATION DEMAND ASSESSMENT

Assess the number of VMT and carbon emissions associated with the regular building occupants' travel to and from the project building as based on the following:

- Estimate the annual VMT.
- Estimate annual baseline case for carbon emissions.
- Assess low-carbon transportation options.
- Estimate annual proposed case for carbon emissions.
- Estimate the total reduction of carbon emissions between annual baseline case and annual proposed case.

Projects that participate in a local or regional government-mandated TDM program satisfy the transportation demand assessment (TDA) requirement. Residential affordable housing projects in an infill location, or office, mixed-use, residential, or retail projects located within a transit priority area, or within a walking distance of 0.5 miles (800 meters) to an existing or planned major transit stop, are exempt from the preceding requirements.

AND

Implement one or more of the following for up to a total of 4 points.

OPTION 1. PARKING (1–3 POINTS)

PATH 1. REDUCE PARKING (1–3 POINTS)

Provide a reduction in parking capacity, using the base ratios for parking spaces found in the Institute of Transportation Engineers *Parking Generation Manual*, sixth edition, or a comparable resource applied by a qualified transportation engineer or planner or in supplementary LEED guidance. Points are awarded according to Table 1.

TABLE 1. Points for Percentage of Reduced Parking Capacity

Reduced Parking Percentage	Points
30% reduction from base ratios	1
60% reduction from base ratios	2
100% reduction from base ratios (no parking)	3

AND/OR

PATH 2. PARKING FEE (2 POINTS)

Implement a daily, monthly, or annual parking fee at a cost equal to or greater than the local market rate for public or private parking.

AND/OR

OPTION 2. ACTIVE TRAVEL FACILITIES (1–3 POINTS)

PATH 1. BICYCLE NETWORK AND STORAGE (1 POINT)

Bicycle network

Design or locate the project such that a functional entry and/or bicycle storage is within a 600-foot (180-meter) walking distance or bicycling distance of a bicycle network that meets the following criteria:

- It provides a contiguous network that spans a distance of at least three miles (4,800 meters).
- It consists of bicycle paths, lanes, multiuse trails, or streets with a maximum speed limit of 25 mph (40 kph). Sidewalks where local code permits bicycles are acceptable.

Planned bicycle trails or lanes may be counted if they are fully funded by the date of the certificate of occupancy and are scheduled for completion within three years of that date.

Schools

- Provide dedicated bicycle lanes that extend from the student bike-parking location to at least the end of the school property without any barriers (e.g., fences on school property).

AND

Bicycle storage

Provide short-term bicycle storage within 600 feet (180 meters) walking distance to any main entrance, but no fewer than four storage spaces per building.

Provide long-term bicycle storage within 300 feet (90 meters) walking distance from any functional entry, but no fewer than four storage spaces per building, in addition to the short-term bicycle storage spaces.

Points are awarded according to Table 2. Shared micromobility storage, bicycle-sharing stations, and/or publicly available bicycle parking may be counted for up to 50% of the required short-term and long-term storage space if:

- It meets the maximum allowable walking distance.
- It is not double counted (e.g., short-term and long-term storage spaces are counted separately).
- The storage location is communicated to the building occupants and visitors.

TABLE 2. Number of Spaces Required for Short- and Long-Term Bicycle Storage

<table>
<tr><th></th><th>Commercial, Institutional, Schools, Healthcare</th><th>Residential</th><th>Mixed Use</th><th>Retail</th></tr>
<tr><td>Short-term storage</td><td colspan="2">At least 2.5% of all peak visitors but no fewer than four spaces per building.</td><td rowspan="2">Meet the storage requirements for the nonresidential and residential portions of the project separately.</td><td>At least two short-term bicycle storage spaces for every 5,000 square feet (465 square meters) but no fewer than two storage spaces per building.</td></tr>
<tr><td>Long-term storage</td><td>At least 5% of all regular building occupants but no fewer than four storage spaces per building, in addition to short-term storage.</td><td>At least 15% of all regular building occupants but no fewer than one storage space per three dwelling units, in addition to short-term storage spaces.</td><td>At least 5% of regular building occupants but no fewer than two storage spaces per building in addition to the short-term bicycle storage spaces.</td></tr>
</table>

NOTE: For New Construction only: School projects can exclude students in third grade and younger from the regular building occupant count for long-term storage. Healthcare projects can exclude patients from the regular building occupant count for long-term storage.

AND/OR

PATH 2. SHOWER AND CHANGING FACILITIES (1 POINT)

Provide access to on-site showers with changing facilities for 1% of all regular building occupants. Off-site showers and changing facilities are acceptable if they meet the needs of all occupants and are within 0.25 miles (400 meters) walking distance.

Large occupancy projects

Provide at least one on-site shower with a changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter, up to 999 regular building occupants. After that, provide the following:

- One additional shower for every 500 regular building occupants, for an additional 1,000–4,999 regular building occupants.
- One additional shower for every 1,000 regular building occupants, for the additional 5,000+ regular building occupants.

AND/OR

PATH 3. BICYCLE MAINTENANCE (1 POINT)

Provide a permanent and secure bicycle repair station that includes a complete set of tools and an air pump securely fastened to the repair stand in the area dedicated to long-term bicycle storage.

REQUIREMENTS EXPLAINED

This credit provides a holistic approach to reducing transportation impacts and supporting projects' decarbonization efforts by enhancing transportation options. TDM includes both facility-related and behavioral strategies to encourage sustainable transportation choices. TDM strategies may target facilities specifically related to the project, such as bicycle maintenance stations, secure bicycle storage, and access to connected bicycle networks (paths, trails, designated bicycle lanes). These strategies align with behavioral approaches that offer travel incentives or disincentives, such as bus passes and carpooling. TDM helps reduce VMTs, lower parking demand, support ridesharing, and encourages public transit use by addressing the project occupants' current and projected transportation demands. The credit examines commuting patterns and behaviors of the occupants by estimating VMT and assessing alternative mode choices, which makes TDM a comprehensive framework for sustainable transportation planning.

TRANSPORTATION DEMAND ASSESSMENT

Sustainable transportation measures require assessing the number of VMT and carbon emissions associated with the regular building occupants' travel to and from the project building. VMT and carbon emissions are important metrics in evaluating the impacts of efforts toward creating a more sustainable transportation system. It provides critical data for creating effective TDM strategies and insights for designing projects that aim to reduce travel distances, encourage alternative transportation modes, and lead to more sustainable and resilient communities. The USGBC TDM assessment calculator will help complete the calculations in the following equations.

Projects that participate in a local or regional government-mandated TDM program are considered to have met the Transportation Demand Assessment requirement because these programs are typically designed to achieve the same goals of reducing traffic congestion, lowering emissions, and supporting more sustainable transportation systems. Residential affordable housing projects in infill locations, as well as office, mixed-use, residential, or retail projects located within a transit priority area or within 0.5 miles (800 meters) walking distance of an existing or planned major transit stop, are exempt from the TDM assessment requirements because their location inherently promotes sustainable transportation options.

Estimate the annual VMT

Project teams must estimate the annual VMT as part of assessing transportation demand. This process involves determining the number of days the building will be occupied, identifying the number of regular project occupants, and estimating commuting distance to and from the site to calculate the annual VMT. Project teams can refer to federal transportation reports, data from metropolitan planning organizations, and information from local or state transportation departments to estimate commute distances accurately. If local data cannot be found, consider a regional source. These resources provide insights into average commute lengths, which help refine the VMT calculation and support the development of effective TDM strategies.

Equation 1. Calculating the annual VMT

$$\begin{gathered}\text{Annual VMT} = \text{Regular building occupants} \times \\ \text{Estimated number of work days within a year} \times \text{Daily VMT}\end{gathered}$$

Estimate annual baseline case for carbon emissions

Project teams must estimate the annual baseline case for carbon emissions. This baseline emissions estimate assumes the worst-case scenario of all regular building occupants driving SOVs and can use the same emissions factor for simplicity. For example, the U.S. EPA estimates that the average passenger vehicle in the U.S. emits 400 grams/mile. Projects may prefer to take into account differing vehicle types and emissions factors, such as compact cars, typical sedans, light duty passenger trucks and their corresponding emissions factors, but they are not required to do so. The annual baseline case for transportation emissions should be reported in metric tons.

Equation 2. Calculating annual baseline case for transportation emissions

$$\text{Annual baseline case for transportation emissions} = \text{Total annual VMT} \times \text{emission factor}$$

Assess low-carbon transportation options

After the baseline for annual transportation emissions has been established, teams must assess the potential reduction in VMT and related transportation emissions by determining what transportation demand management strategies (TDM) will be used in the project. Walking and biking trips, which assume zero emissions, can significantly offset the overall transportation carbon footprint of the project. Projects should consider the full breadth of TDM strategies as typically used by local government TDM programs and not only those within the *LTc4* credit. Also, LEEDv5 offers a robust set of TDM strategies throughout the rating system that teams are strongly encouraged to prioritize.

Estimate annual proposed case for carbon emissions

After assessing the low-carbon transportation options, project teams must determine the reduction in VMT and commuting emissions based on TDM strategies and a realistic assessment of travel patterns to and from the site. This analysis should also account for various commuting methods, such as walking, cycling, or public transit, and estimate the potential reductions in VMT and commuting emissions. For example, locating within 0.25 miles of a public transit service station might help decrease the VMT and related commuting emissions by a certain percentage.

Equation 3. Calculating the annual proposed case for carbon emissions

$$\text{Annual proposed case for transportation emissions} = \text{Annual baseline case for transportation emissions} \times \text{Estimated \% reduction based on TDM strategies}$$

Estimate the total reduction of carbon emissions

Project teams are required to calculate the difference in emissions to get the total estimated reduction of carbon emissions between the annual baseline and proposed case.

AND

OPTION 1. PARKING

TDM strategies, like reducing parking spaces and implementing parking fees, address broader land use and cost challenges tied to parking. TDM conserves valuable land and reduces infrastructure and maintenance expenses by decreasing the need for large parking facilities and promoting more sustainable transportation options that align with project efficiency and environmental goals.

Limiting parking availability also helps curb induced demand, because fewer parking spaces discourage SOV trips and encourage alternative transportation modes. Another effective approach is unbundling parking, which separates the cost of parking from building rentals or leases. For example, a mixed-use office building leases commercial spaces to businesses. Instead of including parking spaces as part of the standard lease package, the building owner offers parking spaces as a separate, optional service.

PATH 1. REDUCE PARKING

Limiting parking availability also helps curb induced demand, because fewer parking spaces discourage SOV trips and encourage alternative transportation modes. This path uses a parking baseline against which reductions in parking supply can be compared. The baseline should be calculated using industry standards on parking demand, such as the Institute of Transportation Engineers *Parking Generation Manual*[25] or a comparable resource. The significant environmental benefit of less parking has been recognized in LEED v5 by increasing the point allocation from 1 point in LEED v4.1 BD+C: New Construction and LEED v4.1 BD+C: Core and Shell (*LTc4: Reduced Parking Footprint*, Option 1. No Parking or Reduce Parking) to 3 points. Parking capacity must include all existing and new off-street parking spaces that are leased or owned by the project, including parking that is outside the project boundary but is used by the project. On-street parking in public rights-of-way is excluded from these calculations:

Equation 4. Percentage of parking capacity reduction

$$\text{Parking reduction} = \frac{(\text{Total baseline capacity} - \text{Total provided capacity})}{\text{Total baseline capacity}} \times 100$$

PATH 2. PARKING FEE

Implementing a parking fee for public or private parking is a common TDM strategy that aims to reduce the demand for parking spaces and encourage alternative and more sustainable transportation options, such as public transit, biking, walking, or carpooling. More importantly, it helps to address the true cost of parking, which not only includes the construction cost but also considers the land use, maintenance, environmental, social, and economic impacts of building parking facilities.

Project teams must assess the daily, monthly, or annual market rates for parking in the local area, whether for public or private facilities. Project teams are encouraged to compare rates for facilities of the same type (e.g., surface lot, deck, underground deck, or covered parking). Teams must provide justification for the parking fee and set it at a rate that meets or exceeds local market values. The project promotes fair market competition and encourages more sustainable transportation options by establishing parking fees at or above the market rate.

OPTION 2. ACTIVE TRAVEL FACILITIES

Active travel facilities promote sustainable, healthy, and efficient alternatives to car-based transportation by providing comprehensive facilities that support cycling, e-bicycles, scooters, and other eco-friendly travel modes. Essential components include secure storage for bikes, well-designed on-road facilities such as dedicated lanes and paths, access to showers and changing areas, and bike maintenance stations. Together, these elements create an integrated system that enables convenient,

25. ITE, *Parking Generation*, 6th ed., 2023, https://www.ite.org/technical-resources/topics/trip-and-parking-generation-v2/parking-generation-info1/.

active travel options for daily commuting and short trips. In recent years, these modes have surged in popularity; as of 2017, there were around 1,250 bicycle-sharing systems globally with over 10 million bicycles globally.[26] Active transportation not only reduces carbon emissions and improves public health through increased physical activity but also helps manage transportation demand by reducing traffic congestion and lessening the need for vehicle parking infrastructure.

PATH 1. BICYCLE NETWORK AND STORAGE

To promote bicycle-friendly design, this path rewards two items: the provision of long- and short-term bicycle storage and access to a bicycle network, including paths, trails, designated bicycle lanes, and slow-speed roadways. Short-term and long-term bicycle storage capacity is considered separately because visitors and regular project occupants have different bicycle storage needs. Being adjacent to a bicycle network means that project occupants can more easily bicycle to and from the project building.

Bicycle network

Project teams are required to identify a bicycle network within 600 feet (180 meters) walking or bicycling distance of a functional entry and/or bicycle storage in the project and gather information and specifications on distance from the project site and street speed limit for the bicycle network. A bicycle network must be a contiguous network that spans three miles (4,800 meters). The three-mile contiguous path refers to the total length and does not need to span three miles in a single direction. For example, it could consist of one mile to the north and two miles to the south, totaling three miles. The bicycle network must also consist of bicycle paths, lanes, or trails that are at least eight feet (2.5 meters) wide for a two-way path and at least four feet (1.2 meters) wide for a one-way path. Also, any on-street bicycle facilities are to be on streets with a maximum speed limit of 25 mph (40 km/h). Both bicycle lanes and bicycle trails must meet the credit's width requirements. Sidewalks where bicycles are allowed by local code are also acceptable.

Teams must locate the project close to an existing or planned bicycle network that meets credit requirements for use within the specified distance from the project boundary. For planned bicycle trails or lanes, confirm the schedule for funding and completion.

Schools

- This option for schools is in lieu of connecting to a bicycle network. Project teams must ensure safe access to school buildings by providing an on-site bike lane or a multimodal path that is either on-road or off-road and safely connect the edge of school property to school buildings without any barriers.

Bicycle storage

Teams are required to determine the number of expected occupants in the building and determine the number of bicycle storage spaces, shared mobility storage, and stations required. Once the number of bicycle storage spaces is determined, install the short-term and long-term bicycle storage within 600 feet (180 meters) and 300 feet (90 meters) walking distance from any main entrance

26. United Nations, "Sustainable Transport, Sustainable Development," 2021, https://sdgs.un.org/sites/default/files/2021-10/Transportation%20Report%202021_FullReport_Digital.pdf.

and functional entry, respectively. Shared micromobility storage facilities, such as those for e-scooters and e-bicycles, as well as bicycle-sharing stations and publicly accessible bicycle parking, can account for up to 50% of the required short-term and long-term bicycle storage needs. This encourages the use of communal, readily available transportation solutions and reduces the need for dedicated on-site storage infrastructure.

For commercial, institutional, schools, and healthcare projects, use Equation 5 to determine the number of short- and long-term bicycle storage spaces. School projects can exclude students in third grade or younger from regular occupant count for long-term storage. Healthcare projects can exclude patients from regular project occupant count for long-term storage.

Equation 5. Calculating bicycle storage for commercial, institutional, schools, and healthcare projects

$$\text{Short-term storage} = (\text{Peak visitor} \times 0.025) \geq 4$$
$$\text{Long-term storage} = (\text{Regular project occupants} \times 0.05) \geq 4$$

For retail projects, use Equation 6 to determine the number of short- and long-term bicycle storage spaces:

Equation 6. Calculating bicycle storage for retail projects

$$\text{Shortterm storage} = \left\lceil \frac{\left\lfloor \text{Building gross floor area } \left(\text{sf or m}^2\right)\right)}{5000 \text{ or } 465} \right\rfloor \geq 2$$
$$\text{Long-term storage} = (\text{Regular project occupants} \times 0.05) \geq 2$$

Equation 7. Calculating bicycle storage for residential projects

$$\text{Short-term storage} = (\text{Peak visitor} \times 0.025) \geq 4$$
$$\text{Long-term storage} = (\text{Regular project occupants} \times 0.15) \geq$$
$$\text{1 space per three dwelling units}$$

PATH 2. SHOWER AND CHANGING FACILITIES

Providing adequate infrastructure for active commuting, such as lockers and changing/shower facilities, plays a key role in promoting physical activity for all building occupants, not just cyclists. These amenities are especially beneficial for those who may engage in exercise or other physical activities before work. Buildings with these facilities allow occupants to adopt active lifestyles, such as active commuting, and help foster a culture that values health and wellness.

Project teams are required to gather occupant information and determine the number of showers and changing facilities required using Equation 8. Off-site showers and changing facilities are acceptable if they meet the needs of all occupants and are within 0.25 miles (400 meters) walking distance. Teams must provide a vicinity or area map indicating off-site shower and changing facilities.

Equation 8. Calculating the number of showers and changing facilities

$$\text{Number of shower and changing facilities} = \text{Regular project occupants} \times 0.01$$

For large-occupancy projects with regular building occupants up to 999, use Equation 9 to determine the number of showers and changing facilities required.

Equation 9. Calculating the number of shower and changing facilities for large occupancy up to 999

$$\text{If regular project occupants} \leq 100, \text{ Shower facilities} = 1$$

$$\text{If regular project occupants} > 100, \text{ Shower facilities} = 1 + \left[\frac{\text{Regular project occupants} - 100}{150}\right]$$

For large-occupancy projects with an additional 1,000–4,999 regular building occupants, use Equation 10 to determine the number of showers and changing facilities required.

Equation 10. Calculating the number of showers and changing facilities for large occupancy 1,000 or more

$$\text{If regular building occupants} > 1{,}000 \text{ Shower facilities} = 1 + \left[\frac{\text{Regular project occupants} - 1{,}000}{500}\right]$$

$$\text{If regular building occupants} > 5{,}000 \text{ Shower facilities} = 1 + \left[\frac{\text{Regular project occupants} - 5{,}000}{1{,}000}\right]$$

PATH 3. BICYCLE MAINTENANCE

Providing on-site bicycle maintenance facilities promotes and encourages active commuting. Projects can make cycling a more convenient and reliable option for commuters by providing access to basic bicycle repair tools and air pump stations. This proactive support ensures that bicycles remain in good working condition and encourages more individuals to adopt active commuting as part of their daily routine.

The maintenance facility must be permanently secured and should remain in the same place, so users know where to find it when needed. Repairs are often needed at unexpected and inconvenient times, so predictability is key. The tools should accommodate typical repairs and be securely fastened so they do not go missing. Locating the repair facility in the area dedicated to long-term bicycle storage ensures it is conveniently located.

Project teams are required to provide a description of the available bicycle services and facilities, along with a map or site plan indicating the location of the bicycle repair station within the designated long-term bicycle storage area. This ensures that occupants know where to access these resources, makes it easier for them to maintain their bicycles, and promotes a cycling-supportive environment.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	Not claiming TDA exemption	USGBC calculator.
			Description of availability of alternative low-carbon and active travel options.
		Claiming TDA exemption	Documentation showing participation in a local or regional government-mandated TDM program.
			Documentation showing project is an affordable housing project in an infill location.
			Documentation showing project (office, mixed-use, residential, or retail) is located within a Transit Priority Area or within 0.5 miles (800 meters) walking distance of an existing or planned major transit stop.
	Option 1. Parking	Path 1. Reduce Parking	Calculations demonstrating the percent reduction in parking capacity from baseline.
			Parking plan or site plan showing the LEED boundary and the parking used by the project.
		Path 2. Parking Fee	Narrative identifying the parking fees and explaining how the rate charged is equal to or greater than the local market rate for parking.
			Documentation confirming local market parking rate.
	Option 2. Active Travel Facilities	Path 1. Bicycle Network and Storage	Vicinity map showing bicycle network meeting the required criteria and walking/bicycling distance of functional building entrance and/or bicycle storage to existing or planned bicycle network.
			Evidence that any sidewalks contributing to the bicycle network are permitted for bicycle use by local code (e.g., excerpt of the code).
			Site Plan showing main and functional building entrances, short-term bicycle storage and long-term bicycle storage and shared micromobility storage (if applicable), walking distance from short-term storage to the main entrance and from long-term storage to a functional entrance.
			Calculations documenting the percentage of occupants for which short-term bicycle storage is provided.
			Calculations documenting the percentage of occupants for which long-term bicycle storage is provided.
		Path 2. Shower and Changing Facilities	Site plan showing shower and changing facilities location and walking distance within 0.25 miles (400 meters) for off-site facilities.
			Calculation documenting the percentage of regular building occupants with access provided to showers with changing facilities.
			For large occupancy projects, the number of project occupants to determine the number of on-site showers required.
		Path 3. Bicycle Maintenance	Evidence of a permanent, secure bicycle maintenance facility demonstrating complete set of tools and air pump securely fastened to the repair stand (e.g., product information from the manufacturer, photographs, contract specification).
			Evidence of location of the bicycle maintenance facility in the area dedicated to long-term bicycle storage (e.g., contract documents).
Schools		Path 1. Bicycle Network and Storage	Evidence demonstrating dedicated bicycle lanes or sidewalks, if applicable, that extend from the student bicycle parking location to the end of the school property, at minimum, without any barriers.

REFERENCED STANDARDS

- Institute of Transportation Engineers (ITE) Parking Generation Manual, 6th Edition (https://www.ite.org/technical-resources/topics/trip-and-parking-generation-v2/parking-generation-info1)

FURTHER EXPLANATION

Vehicle miles traveled (VMT) is an important transportation metric that measures the total miles driven within a specific area and time period and can be measured at the individual or community level. VMT is a core indicator in transportation planning because it reflects how much people rely on vehicles—especially single-occupancy vehicles—and has strong links to multiple indicators like congestion, air quality, infrastructure costs, public health, and rates for other modes of travel like walking, biking, and transit.

TRANSPORTATION DEMAND ASSESSMENT

Assessment Calculator

- It is recommended that the USGBC calculator be used for these calculations. It was built to take into account the recommended methodology below.

Estimate Annual VMT

- Use trusted sources for VMT data, such as metropolitan planning organizations (MPOs), state departments of transportation (DOTs), and federal reports. Use regional or national averages, if necessary. If VMT data is unavailable, consider other methods of estimating typical travel distances, such as data from a travel demand study for the building, purchasing proprietary data, or surveying occupants.
- Publicly available VMT is often reported at the household level and should be adjusted to reflect only the portion of VMT attributed to work commutes. For example, in the U.S., approximately 30% of daily household VMT is attributed to work commutes.[27]
- For New Construction projects, it is recommended that an occupancy-based approach is used for calculating VMT:
  - Occupancy-based VMT calculations require these essential pieces of data: typical number of regular building occupants within a standard 24-hour period, number of days the building is in operation, and the average daily VMT for a regular building occupant traveling between home and work and back or vice versa (one round trip), assuming no other errands are being added.
  - Occupancy estimates should be based on observation and exclude fully remote workers.
- If occupancy is uncertain, then use USGBC's Default Occupancy Counts to estimate regular building occupants.
- Regular building occupants must be estimated for a typical day of operation (24-hour period beginning at 12 a.m.). If occupants work across dates, count them from the time they arrive.

27. Center for Sustainable Systems, "Personal Transportation Factsheet," University of Michigan, accessed September 9. 2025, https://css.umich.edu/publications/factsheets/mobility/personal-transportation-factsheet.

- Assume single-occupancy vehicle commuting as a worst-case travel pattern for baseline (e.g., one commuter per vehicle) and assume no other modes of travel are used during the commute (e.g., car, then bus, then walking for one trip).
- If it is not possible to find an average VMT reported by a reputable source, then the average VMT for a regular building occupant can be estimated by using an average travel time instead, which is also a commonly reported statistic. Average travel speeds vary by land use and are typically lowest in urban areas and highest in rural areas with suburban travel speeds falling in between.
  - Example Equation for calculating VMT using time and speed:
    - Vehicle Miles Traveled = Average Travel Time × Average Travel Speed
  - Example VMT calculation for rural project with an estimated 20-minute commute:
    - $\frac{1}{3}\text{hour} \times 45\,\frac{\text{miles}}{\text{hour}} = 15\text{ miles VMT}$

Estimate Annual Baseline for Carbon Emissions

- Select a regional- or national-average emissions factor (e.g., grams CO_2 per mile) for a typical passenger vehicle.
- Focus on a representative vehicle type. It is not required to model all potential vehicle and fuel types arriving at the project; however, vehicle mix and fuel types may be taken into account, if desired.

Assess Low-Carbon Transportation Options

- Consider VMT/travel emissions mitigation strategies implemented by the project. Consider transportation demand related credits within LEED, such as:
  - *LTc2: Equitable Development*, Option 2. Housing and Jobs Proximity
  - *LTc3: Compact and Connected Development*, Option 2. Access to Transit, Path 1. Public Transit Service
  - *LTc3: Compact and Connected Development*, Option 2. Access to Transit, Path 2. Project-Sponsored Transit Service
  - *LTc4: Transportation Demand Management*, Option 1. Parking, Path 1. Reduce Parking
  - *LTc4: Transportation Demand Management*, Option 1. Parking, Path 2. Parking Fee
  - *LTc4: Transportation Demand Management*, Option 2. Active Travel Facilities, Path 1. Bicycle Network and Storage
  - *LTc4: Transportation Demand Management*, Option 2. Active Travel Facilities, Path 2. Shower and Changing Facilities
  - *LTc5: Electric Vehicles*, Option 1. Electric Vehicle Supply Equipment
- VMT estimates for a city are already accounting for shorter trips due to location efficiency, so applying a reduction for locating in a dense area would be double counting a VMT reduction.

Estimate Annual Proposed Case for Carbon Emissions

- Apply standard VMT/emission reductions for implemented strategies (e.g., percent reduction in VMT/emissions from bicycle facilities) based on reputable and regionally relevant publications, such as a nearby city's transportation demand management program.
- Consider standard percentage reductions published by reputable organizations or peer-reviewed literature that may be specific to the project region and land use.

Estimate the Total Reduction of Carbon Emissions Between Annual Baseline Case and Annual Proposed Case

- Subtract the annual proposed case from the annual baseline case.

OPTION 1. PARKING, PATH 1. REDUCE PARKING

Baseline Parking Ratios

- For projects with multiple space types, calculate and separately track the base ratio for each use. Use the equations provided to determine the baseline parking capacity for each space type and compute the total project baseline capacity.
- Project teams may use Table 3 to calculate base ratios if applicable uses are listed below; however, if appropriate uses are not listed, then the baseline should be calculated using industry standards on parking demand, such as the latest Institute of Transportation Engineers Parking Generation Manual or a comparable resource.

TABLE 3. Parking Rates

Land Use	Parking Rate ($)
Arena/live theater	0.25 per seat*
Assisted living/nursing home	0.4 per bed*
Colleges	1.66 per 1,000 ft^2, plus 1 per 4 dorm rooms
Convention center (<100,000 ft^2)	4 per 1,000 ft^2 exhibition space
Convention center (100,000–500,000 ft^2)	3 per 1,000 ft^2 exhibition space
Convention center (≥500,000 ft^2)	2 per 1,000 ft^2 exhibition space
Day care center	2.25 per 1,000 ft^2*
Elementary school	0.15 per student*
General office	2 per 1,000 ft^2*
Government office building	2.25 per 1,000 ft^2*
High school	0.25 per student*
Hospital	0.75 per employee*
Hotel	1 per room
Manufacturing	0.80 per employee*
Mini-warehouse/self-storage	1.25 per 100 storage units*
Pharmacy/drugstore with drive-through	2 per 1,000 ft^2*
Pharmacy/drugstore without drive-through	2.5 per 1,000 ft^2*
Residential–multifamily or any affordable housing	1 per dwelling unit*
Residential–condo/apartment, owned	1.25 per dwelling unit*
Restaurant–casual	10 per 1,000 ft^2 for public use*
Restaurant–fine dining	18 per 1,000 ft^2 for public use*
Retail–general (not in shopping center)	2.75 per 1,000 ft^2*

TABLE 3. Parking Rates (*continued*)

Land Use	Parking Rate ($)
Shopping center (>150,000 ft^2)	2.45 per 1,000 ft^{2*}
Shopping plaza or strip retail plaza (<40,000 ft^2)	3 per 1,000 ft^{2*}
Warehousing	0.35 per 1,000 ft^{2*}

NOTES: 1,000 ft^2 = 93 m^2; and all listed floor area is gross floor area.
DISCLAIMER: The rates provided in this table and indicated with an * are based on data from the Institute of Transportation Engineers (ITE) Parking Generation Manual, 6th Edition. These values are intended for general reference only. The Parking Generation Manual contains significantly more detailed information, including variations by day of week, distinctions between weekday and weekend demand, and differentiation based on independent variables such as employees versus seats, or square footage versus beds. Users should consult the ITE Parking Generation Manual, 6th Edition directly to ensure the proper application of this data to specific contexts. Reliance solely on the summarized values presented here, without reference to the manual, may result in inappropriate or inaccurate conclusions.

Calculating Provided Capacity

- The total parking capacity must include all parking made available to project occupants, including parking that is outside the project boundary but is used by the project.
- If parking spaces are shared with two or more buildings/spaces, only the parking allocated to the project must be included in the total parking capacity. It is up to the project team to determine the project's allocation based on the project-specific circumstances and to provide rationale for the parking distribution, if necessary.
- Do not include on-street public parking.

Percent Reduction

- Transportation demand management (TDM) strategies should be integrated to reduce parking demand and support credit achievement.

OPTION 2. ACTIVE TRAVEL FACILITIES, PATH 1. BICYCLE NETWORK AND STORAGE

- Storage spaces must be dedicated to the project pursuing LEED certification and cannot be counted toward another LEED project.
- The following conditions apply to all calculations for short- and long-term bicycle storage:
  - Results must be rounded up to the nearest whole number.
  - Storage spaces must be devoted to the project pursuing LEED certification and cannot be double counted.
  - For mixed-use buildings, identify nonresidential and residential portions of the building and meet the applicable storage requirements for each space type based on prorated occupancy.

OPTION 2. ACTIVE TRAVEL FACILITIES, PATH 1. BICYCLE NETWORK AND STORAGE, SCHOOLS

- School projects must meet both the core credit and the school-specific criteria to comply with Option 2.

OPTION 2. ACTIVE TRAVEL FACILITIES, PATH 2. SHOWER AND CHANGING FACILITIES

- Showers are required for commercial or institutional spaces only. For residential spaces, no additional showers are required beyond those provided inside dwelling units.
- Projects with hotel guests may exclude these occupants from shower calculations.

- Shower facilities should be available to all project occupants without cost during the project's hours of operation.
- For mixed-use buildings, identify the nonresidential portions of the building and meet the applicable shower and changing facility requirements for this space type based on prorated occupancy (see Further Explanation, Example, Mixed-Use Building).

OPTION 2. ACTIVE TRAVEL FACILITIES, PATH 3. BICYCLE MAINTENANCE

Provide a commercial-grade bicycle maintenance stand equipped with an air pump (Presta & Schrader compatible) and a full set of commonly required bicycle tools, such as:

- Repair stand for hanging/securing the bike for repairs
- Various common wrenches, such as hex, torx, open-ended, and adjustable
- Flathead and Phillips screwdrivers
- Tire levers
- Chain tool

Electric Vehicles

LTc5
New Construction (1–2 points)
Core and Shell (1–2 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To encourage the use of EVs and infrastructure. To help reduce the negative health effects on communities by lowering greenhouse gas emissions and other pollutants emitted from conventionally fueled cars and trucks.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–2**
Option 1: Electric Vehicle Supply Equipment AND/OR	1–2
Option 2: Electric Vehicle Readiness	1

OPTION 1. ELECTRIC VEHICLE SUPPLY EQUIPMENT (1–2 POINTS)

Install electric vehicle supply equipment (EVSE) meeting the thresholds listed in Table 1. EVSE must meet the following criteria:

- Provide Level 2 or Level 3 charging capacity per the manufacturer's requirements and the requirements of the *National Electrical Code* (*NFPA 70*).
- Provide 208–240 volts or greater for each required space.
- Meet the connected functionality criteria for ENERGY STAR-certified EVSE and be capable of responding to time-of-use market signals (e.g., price).
- At least one EV charging station has an accessible parking space at least 9 feet (2.5 meters) wide with a 5-foot (1.5-meter) access aisle and have accessibility features for use by people with mobility, ambulatory, and visual limitations.

TABLE 1. Points for Installed EVSE (Percentage of Total Parking Spaces)

Commercial Minimum EVSE Parking	Points
5% or at least two spaces, whichever is greater	1
10% or at least four spaces, whichever is greater	2
Residential Minimum EVSE Parking	**Points**
10% or at least five spaces, whichever is greater	1
15% or at least 10 spaces, whichever is greater	2

AND/OR

OPTION 2. ELECTRIC VEHICLE READINESS (1 POINT)

Make the minimum number of total parking spaces used by the project EV ready as specified in Table 2. EV-ready parking spaces must provide a full-circuit installation, including 208–240 volts, have a 40-amp panel capacity, and a conduit (raceway) with wiring that terminates in a junction box or charging outlet.

Any space with an installed EVSE counted for credit under Option 1 may not be counted for credit as an EV-ready space under Option 2.

TABLE 2. Points for EV-Ready Parking (Percentage of Total Parking Spaces)

Commercial	Residential	Points
At least 10% or at least 10 spaces, whichever is greater.	At least 20% or at least 20 spaces, whichever is greater.	1

REQUIREMENTS EXPLAINED

This credit addresses the advancement of EVSE, or charging stations, which is a significant impediment to the broadscale adoption of EVs. Option 1 rewards projects that implement EVSE for a minimum number of spaces. Option 2 provides an achievement path for projects that install EV-ready spaces. Projects that do not own or lease parking are not eligible for this credit.

Teams may combine Options 1 and 2 for up to 2 points. Projects must meet the requirement for each option and may not use weighted averages or double count spaces to prove compliance with each option.

OPTION 1. ELECTRIC VEHICLE SUPPLY EQUIPMENT

New construction projects offer significant opportunities to expand EV charging solutions for communities. By designing and installing electrical infrastructure during initial construction, projects can ensure adequate electrical capacity for the EVSEs, provide building occupants with EVSE on the first day of occupancy, and eliminate future capital costs from digging up parking lots and replacing electrical panels.

EVSE minimum requirements

All EVSE must have a Level 2 or Level 3 charging capacity, with dedicated services of 208–240 volts for each required space.

ENERGY STAR-certified EVSE are verified to meet performance claims by manufacturers and are fully tested for safety and energy use.[28] EVSE are not required to be ENERGY STAR certified; however, all installed EVSE must meet the ENERGY STAR-connected functionality criteria, including capabilities of responding to time-of-use market signals.

Ensure the equipment has the capability to integrate with industry networks and connect to other devices. Devices may include Wi-Fi routers and electric utility energy management and price signals.

For projects integrating EVSE with demand response programs or load flexibility and management strategies, use guidance in *EAc6: Grid Interactive*.

Minimum number of spaces and accessible parking requirements

Commercial projects must install EVSE for at least 5% of the total vehicle parking capacity and no fewer than two spaces. Parking capacity must include all existing and new off-street parking spaces that are leased or owned by the project, including parking that is outside the project boundary but is used by the project. On-street parking in public rights-of-way is excluded from these calculations.

Use Equation 1 to determine the number of spaces required to meet the 5% threshold. If the equation does not result in a whole number, round up to the next whole number.

Projects must install EVSE for the minimum number of parking spaces per Table 1. Percentage thresholds only apply if the minimum number of installed spaces meets or exceeds those listed in Table 1.

Equation 1. Commercial minimum EVSE parking

$$\text{Min. \# of EVSE parking (commercial)} = \text{Total \# of parking spaces} \times 0.05 \geq 2 \text{ spaces}$$

Residential projects must install EVSE for at least 10% and no fewer than five spaces. Determine the minimum number of spaces using Equation 2.

Equation 2. Residential minimum EVSE parking

$$\text{Min. \# of EVSE parking (residential)} = \text{Total \# of parking spaces} \times 0.10 \geq 5 \text{ spaces}$$

For all projects, provide at least one EV charging station in an accessible parking space. The charging station must have accessibility features that enable equal access to EVSE for a person with mobility, ambulatory, and/or visual limitations.

Example 1

A new office building includes a surface parking lot with 125 spaces. The project targets 1 point, which requires 5% of the total spaces have installed EVSE. Using Equation 1, the project needs

28. ENERGY STAR, "Electric Vehicle Chargers," n.d., retrieved from https://www.energystar.gov/products/ev_chargers.

to install one charging station in an accessible parking space and six additional stations within the parking lot.

The design documents confirm charging stations for one accessible parking space and six additional spaces. The project achieves 1 point. Refer to Calculation 1.

However, if the design includes charging stations for seven parking spaces but does not include charging at an accessible parking space, the project would earn zero points. The project installs the minimum total number of EVSE but does not meet the accessible parking requirement.

Calculation 1: $125 \times 0.05 = 6.25$ spaces = 7 total required spaces, including one space for accessible parking.

OPTION 2. ELECTRIC VEHICLE READINESS

Projects that do not install EVSE during the initial construction phase may still achieve compliance with this credit using Option 2. By designing parking spaces for EV readiness, the building operators ensure the availability of infrastructure for future EVSE. Installing circuits, conduits, and wiring to each space prior to pouring concrete or finishing pavement in a parking lot equates to less future work to install the equipment. In this design, projects can easily install EVSE with minimal construction efforts.

Projects must install all infrastructure to be eligible for this option. The intent is that only the EVSE would require installation when the project purchases the equipment.

EV readiness minimum requirements

Identify the EV-ready spaces and confirm that the design incorporates sufficient infrastructure for the EV-ready spaces, which includes, at minimum, a dedicated electric circuit for each space and conduit and wiring sufficient to provide Level 2 charging (or greater). All infrastructure must terminate at an electrical box or enclosure near each required space.

Minimum number of spaces

Projects must design EV-ready parking for the minimum number of spaces per Table 2. Percentage thresholds only apply if the minimum number of EV-ready spaces meets or exceeds those listed in Table 2.

Commercial projects require EV-ready spaces for at least 10% of the total parking spaces and no fewer than 10 spaces. Parking capacity must include all existing and new off-street parking spaces that are leased or owned by the project, including parking that is outside the project boundary but is used by the project. On-street parking in public rights-of-way is excluded from these calculations. Use Equation 3 to determine the number of spaces required to meet the 10% threshold. If the equation does not result in a whole number, round up to the next whole number.

Equation 3. Commercial minimum EV-ready parking

$$\text{Min. \# of EVready}\left(\text{commercial}\right) = \text{Total \# of parking spaces} \times 0.10 \geq 10 \text{ spaces}$$

Residential projects require EV-ready spaces for at least 20% of the total parking spaces and no fewer than 20 spaces. Determine the minimum number of spaces using Equation 4.

Equation 4. Residential minimum EV-ready parking

$$\text{Min. \# of EVready parking (residential)} = \text{Total \# of parking spaces} \times 0.20 \geq 20 \text{ spaces}$$

Combining Options 1 and 2

LTc5: Electric Vehicles only rewards teams up to 2 points. Therefore, when combining options, projects must meet 1 point for Option 1 and 1 point for Option 2.

Projects cannot double count spaces with installed EVSE as an EV-ready space. Each parking space must meet the characteristics of an EV-ready space or a space with an available EVSE.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Electric Vehicle Supply Equipment	All	Site plan identifying all parking spaces in the project, and all parking spaces with EVSE.
			Confirmation that the accessible parking space is at least 9 feet (2.5 meters) wide, has a 5-foot (1.5-meters) access aisle, and has charging station accessibility features for use by people with mobility, ambulatory, and/or visual limitations.
			Calculation documenting the percentage of parking spaces with EVSE (including breakout for accessible parking spaces).
			Evidence that the EVSE meet all criteria identified in the credit requirements (e.g., product information from manufacturer or contract specification).
	Option 2. Electric Vehicle Readiness	All	Site plan identifying all parking spaces used by the project and the total number of EV-ready spaces, including clear identification of at least 1 EVSE in an accessible parking space.
			Calculation documenting the percentage of parking spaces that are EV-Ready.
			Evidence that each EV-Ready parking space has the necessary electrical infrastructure to support the installation of an EV charger including a 40-amp panel capacity, dedicated 208/240V branch circuit, and conduit (raceway) with electrical wiring terminating at a receptacle or junction box for the parking space (e.g., contract documents).

REFERENCED STANDARDS

- SAE Surface Vehicle Recommended Practice J1772 (https://www.sae.org/standards/content/j1772_201710/)
- SAE Electric Vehicle Conductive Charge Coupler (https://www.sae.org/standards/content/j1772_200111/)

- IEC 62196 of the International Electrotechnical Commission (https://webstore.iec.ch/en/publication/59922)
- ENERGY STAR (https://www.energystar.gov)
- National Electrical Code (NFPA 70) (https://www.nfpa.org/codes-and-standards/nfpa-70-standard-development/70)

FURTHER EXPLANATION

OPTION 1. ELECTRIC VEHICLE SUPPLY EQUIPMENT

- The total parking capacity must include all parking made available to building occupants, including parking that is outside the project boundary and/or property but is used by the project.
- Electric vehicle supply equipment (EVSE) parking spaces must be dedicated to the project pursuing LEED certification and cannot be counted toward another LEED project.
- An access aisle is a marked area dedicated for use by people with mobility impairments that has the following qualities:
  - Adjacent and the same length as an accessible parking space
  - Adjoins an accessible route
  - Does not overlap the vehicular way
  - Measures at least five-feet (1.5-meter) wide
- An accessible parking space is a marked vehicle parking space dedicated for use by people with mobility impairments that has the following qualities:
  - Standard parking space length but at least nine feet (2.5 meters) wide
  - No more than a 1.48 (2.08%) slope in all directions
  - Adjacent to an access aisle on at least one side
- EVSE that meets connected functionality requirements may be identified on the ENERGY STAR® website, and project teams can also use EVSE that otherwise meets connected functionality criteria in the ENERGY STAR Version 1.2 EVSE Final Specification, Section 3.10. If the EVSE is not identified by ENERGY STAR as having connected functionality, then the project team may demonstrate the EVSE's has the following:
  - Network connectivity
  - Security (physical/cyber)
  - Open protocols and interoperability
  - Remote capabilities
  - Payment options
  - Data sharing

OPTION 2. ELECTRIC VEHICLE READINESS

- EV-ready means a designated parking space that has the necessary electrical infrastructure to support the installation of an electric vehicle charger including a 40-amp panel capacity, dedicated 208/240 volt branch circuit, and conduit (raceway) with electrical wiring terminating at a receptacle or junction box for the parking space.

- Project teams are recommended to plan efficient pathways to minimize trenching and disruptions. Make sure the electrical panel can handle future EVSE installation (40 amperes per Level 2 charger), multiple simultaneous users or load sharing systems, and is labeled for circuits for EV-ready spaces. Consider using modular raceways or cable trays for easier future expansion.
- The total parking capacity must include all parking made available to project occupants, including parking that is outside the project boundary but is used by the project.
- If parking spaces are shared with two or more buildings/spaces, only the parking allocated to the project must be included in the total parking capacity. It is up to the project team to determine the project's allocation based on the project-specific circumstances and to provide rationale for the parking distribution, if necessary.
- EV-ready parking spaces must be dedicated to the project pursuing LEED certification and cannot be counted toward another LEED project.
- Do not include on-street public parking.

Sustainable Sites

OVERVIEW

LEED v5 prioritizes biodiversity and resilience through credits that highlight not only conservation but also the restoration of ecosystems and natural water cycles. This updated framework encourages projects to actively benefit the surrounding ecosystems and create green spaces that enable pollinators and wildlife to thrive (*SSc1: Biodiverse Habitat*).

The Sustainable Sites (SS) category acknowledges the powerful impact of healthy ecological infrastructure, such as rich soil, functioning water cycles, and native vegetation, on a project's long-term resilience.[1] In fact, this category contains half of the credits in LEED v5 that address resilience. Moreover, nature-based solutions are consistently a cost-effective approach to mitigating hazards because of their low-maintenance strategies.[2]

With a more holistic and future-oriented approach to the intersections between the building, the site, and their larger ecological context, projects can design spaces that anticipate and reduce heat island effect, withstand and more quickly recover from the impacts of extreme weather, and take a proactive approach to adapting to the challenges posed by our changing climate.

Decarbonization

A more resilient future means a decarbonized future. Urban areas may have up to 50%–90% dark, nonreflective surfaces that absorb and retain heat. These heat-absorbing surfaces create heat islands that significantly warm the surrounding areas and drive up energy consumption as buildings work to maintain comfortable temperatures, which increases their carbon emissions.[3]

To reduce buildings' contributions to these effects, the SS category tackles local temperature increases through shading, increased tree canopy cover and vegetation, and reflective or green roofs (*SSc5: Heat Island Reduction*). These strategies in turn decrease reliance on energy-intensive

1. National Oceanic and Atmospheric Administration (NOAA), "Resilience 101: How America Can Withstand Wild Weather," last updated January 21, 2016, https://www.noaa.gov/resilience-101-science-helps-america-withstand-wild-weather.
2. Marta Vicarelli, Karen Sudmeier-Rieux, Ali Alsadadi, et al., "On the Cost-Effectiveness of Nature-Based Solutions for Reducing Disaster Risk," *Science of the Total Environment* 947 (October 15, 2024): 174524, https://doi.org/10.1016/j.scitotenv.2024.174524.
3. Lennart Olsson, Humberto Barbosa, Suruchi Bhadwal, et al., "Land Degradation," Chap. 4 in *Climate Change and Land: An IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems*, P. R. Shukla, et al. (eds.), 2019, 345–405, https://www.ipcc.ch/site/assets/uploads/sites/4/2022/11/SRCCL_Chapter_4.pdf.

systems like HVAC to maintain indoor temperatures, improve resilience, and mitigate urban heat for the larger community.

Quality of life

These same strategies mitigate the rising temperatures that threaten worker safety and public health. Simply planting trees on city streets would give 77 million people a 1°C reprieve on hot days.[4] By prioritizing the equitable use of shaded, green, and accessible outdoor spaces, projects can become safe havens during rising temperatures and foster community connection and inclusion (*SSc2: Accessible Outdoor Space*). The human health and well-being benefits from access to urban green space are well-documented and underscore this category's far-reaching impacts.[5]

Ecological conservation and restoration

As the credit category most directly tied to ecological health, all seven SS credits offer strategies to minimize impacts on land and wildlife, restore natural habitats that bolster resilience and biodiversity, and steward natural resources for prolonged and responsible use. Low-impact development (LID) practices and GI help prevent flooding, reduce erosion, and improve water quality, while soil restoration, native and adaptive plant use, and bird-friendly glass support healthy ecosystems (*SSc3: Rainwater Management, SSc1: Biodiverse Habitat*).

With global commitments to protect and restore at least 30% of the world's land and seas by 2030, LEED projects are well-positioned to contribute meaningfully to this initiative while planning for long-term resilience.[6]

4. UN Environment Programme (UNEP), "Beating the Heat: A Sustainable Cooling Handbook for Cities," November 3, 2021, https://www.unep.org/resources/report/beating-heat-sustainable-cooling-handbook-cities.
5. Mathew P. White, Lewis R. Elliott, James Grellier, et al., "Associations Between Green/Blue Spaces and Mental Health Across 18 Countries," *Scientific Reports* 11, no. 1 (April 2021): 8903, https://doi.org/10.1038/s41598-021-87675-0.
6. Secretariat of the Convention on Biological Diversity, "Kunming-Montreal Global Biodiversity Framework," n.d., https://www.cbd.int/gbf.

Minimized Site Disturbance

SSp1
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To limit site disturbance from construction activities and preserve existing native vegetation, healthy soils, and wildlife habitats.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Erosion and Sedimentation Control Plan AND	
Site Assessment	

Minimize site disturbance by designing and constructing the project site to meet the following requirements:

EROSION AND SEDIMENTATION CONTROL PLAN

Create and implement an erosion and sedimentation control plan for all construction activities associated with the project. The plan must conform to the erosion and sedimentation requirements of the U.S. EPA's *2022 Construction General Permit (CGP)*; *EU Taxonomy: DNSH, Pollution Prevention, Item 4 Noise and Dust*; or the local equivalent. Projects must apply the CGP regardless of size.

The erosion and sedimentation control plan must also include implementation of the following measures:

- Establishment of construction-exclusion zones demarcated by physical barriers and stormwater controls to protect any identified critical habitat for threatened or endangered species from discharges and discharge-related activities.

- Site inspections for all controls and management practices at least once every seven calendar days, or once every 14 calendar days and within 24 hours of the occurrence of a storm event that produces 0.25 inches (6 millimeters) or more of rain within a 24-hour period. Dewatering inspections must occur once per day on which the discharge of dewatering occurs.
- Immediate corrective actions to repair or replace the controls when failing.

AND

SITE ASSESSMENT

Collect information about the site in a preconstruction survey or assessment that informs design of the site to address the following items, as applicable to the project. The survey or assessment should demonstrate the relationships between the site features and topics listed below and how these features influenced the project design.

- **Special-status vegetation:** Conserve 100% of special-status vegetation located on-site as defined by local, state, or federal entities.
- **Healthy habitat:** Identify healthy plant communities and implement strategies to minimize damage to these areas during construction and ongoing project activities. Establish exclusion zones demarcated by physical barriers to minimize intrusion or disturbance of identified healthy plant communities during construction activities.
- **Invasive vegetation:** Indicate locations of existing invasive vegetation species on-site and address removal and control of invasive species before and during construction. Include only native and adapted vegetation that is not currently listed as invasive.

REQUIREMENTS EXPLAINED

This prerequisite requires projects to focus on protecting and preserving healthy and mature site elements and habitats while promoting environmental protection measures that minimize project construction disturbances. Project teams must preserve special-status vegetation and healthy habitats, manage invasive vegetation, and implement an erosion and sedimentation control (ESC) plan to minimize soil, rainwater systems, and neighboring property disturbance.

EROSION AND SEDIMENTATION CONTROL PLAN

Construction activities can greatly affect the local environment and can speed up natural erosion and sedimentation processes by removing vegetation and leaving soil exposed.

Consequently, stormwater runoff from these sites carries high levels of sediment and associated contaminants. This not only affects water quality but also negatively affects aquatic habitats and wildlife. Soil compaction deteriorates soil structure and fertility and precludes optimal root system development, which leads to long-term degradation of the land.

Each project team must create and implement an ESC plan for all construction activities and then develop the plan based on the unique needs of the project site. The purpose of an ESC plan is to prevent soil erosion and sediment runoff, protect natural resources, ensure regulatory compliance, and promote responsible construction through tailored, site-specific measures and ongoing management. All projects within the U.S., regardless of size, must apply the CGP. A local equivalent may be used if it is equally or more stringent than the CGP, if located in the U.S., *or the EU Taxonomy: DNSH, Pollution Prevention, Item 4 Noise and Dust*, for international projects.

Key CGP requirements for ESC plans

The CGP requires ESC plans to include a detailed site description outlining potential erosion risks, implementation of both temporary and permanent control measures to manage runoff, and a construction sequence that aligns ESC practices with project phases. It mandates regular inspections, maintenance protocols, and corrective actions for failing measures, as well as proper documentation and recordkeeping. Training for construction personnel on ESC responsibilities is essential, alongside plans for stabilizing disturbed areas during and after construction. Projects outside the U.S. do not have to comply with the permitting aspects of the CGP. Each project site is unique, and not all ESC measures identified in the CGP may be applicable or necessary.

SITE ASSESSMENT

A site assessment examines environmental characteristics that can influence the design of a sustainable site and building. It identifies special-status vegetation, healthy habitats, and invasive species. Conducting a site assessment is a key part of an IDP and guides informed design choices when completed before or during the conceptual design phase.

Project teams must conduct a preconstruction survey or assessment to gather essential site information. To complete the assessment, project teams must collect data including a detailed inventory of plant species, habitat characterization, soil and hydrology analysis, and mapping of critical areas. A site plan incorporating all necessary details from the site inventory and the site assessment worksheet are both required.

Special-status vegetation

Special-status vegetation includes plants listed as endangered, threatened, or rare under local, state, or federal acts. These plants provide critical functions, such as maintaining soil stability, which is vital for preventing erosion and supporting environmental health.

Projects must conserve 100% of special-status vegetation as defined by local state or federal entities.

Healthy habitat

A healthy habitat supports a diverse community of plants, animals, and microorganisms while protecting soil health and structure, which are essential for thriving ecosystems. Well-preserved habitats minimize resource use by naturally controlling erosion, stabilizing soils, and reducing the need for replanting or extensive maintenance. Healthy ecosystems also mitigate climate change by acting as carbon sinks and absorbing billions of metric tons of CO_2 annually.[7] They manage water flow, reduce runoff impact, and trap harmful substances before they cause damage. Vegetation

7. United Nations Development Programme, “Forests Can Help Us Limit Climate Change–Here Is How,” *UNDP*, October 25, 2023, https://climatepromise.undp.org/news-and-stories/forests-can-help-us-limit-climate-change-here-how.

slows rainwater and allows it to infiltrate the soil and reduce runoff. Additionally, healthy soil with well-structured porosity filters out nutrients and pollutants before they reach rivers, lakes, or groundwater, and helps retain natural fertility, which reduces the need for chemical fertilizers.

Project teams must establish an exclusion zone around healthy habitats to protect them from construction activities and other disturbances. It is also recommended to include special-status vegetation within the exclusion zone. Teams must establish clear construction boundaries to minimize disturbances to ecosystems.

Invasive vegetation

Preserving vegetation biodiversity enhances a site's ecological value. It is essential to plant adapted and native vegetation to support the health and functionality of ecological systems. Projects must remove invasive species, because these species can prevent ecological systems from recovering and thriving, and they compete with and harm native flora and fauna, crops, fisheries, and forests. Invasive species can be difficult to eliminate and can have severe consequences for natural areas once established. Effective control and management are essential to reduce the spread of established invasive species. The most effective method for managing an invasive species is to prevent its arrival.

Project teams must identify invasive species on a site and only include native and adapted vegetation and ensure that no planted vegetation is classified as an invasive species at the time of installation. It must be native or adapted to the project's EPA Level III ecoregion or local equivalent for projects outside of the U.S.[8] Teams must determine their presence by conducting a thorough site survey and paying attention to areas with special-status vegetation and healthy habitats. If protected areas contain invasive species, teams must remove and control them using strategies that minimize disruptive activities and ensure removal efforts do not harm sensitive ecosystems. It is highly encouraged to consult with the local regulatory agency on best management practices (BMPs) for removal and control.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	Compliance by permit	Confirmation of whether the project site is at least one acre and located in an area where the EPA is the NPDES-permitting authority (or EU Taxonomy).
		Compliance by local equivalent	Confirmation that the local standard/code is equivalent to or more stringent than the *2022 U.S. EPA CGP* or *EU Taxonomy: DNSH, Pollution Prevention, Item 4 Noise and Dust*.
			Name of the local standard/code being applied (if applicable).

8. U.S. EPA, "Level III and IV Ecoregions by State," January 2, 2024, https://www.epa.gov/eco-research/level-iii-and-iv-ecoregions-state.

(*continued*)

Project Types	Options	Paths	Documentation
		All	The project's ESC plan.
			The project's site assessment identifying any special-status vegetation, healthy habitat, exclusion zones demarcated by physical barriers, invasive vegetation, and invasive vegetation removal/control plans.
			Evidence that the project includes only native and adapted vegetation that is not currently listed as invasive (e.g., landscape plan).
			Confirmation that 100% of Special-Status vegetation located on-site, as defined by local, state, or federal entities, is/will be conserved.

REFERENCED STANDARDS

- U.S. EPA 2022 Construction General Permit (CGP) (https://www.epa.gov/npdes/2022-construction-general-permit-cgp)
- National Pollutant Discharge Elimination System (NPDES) (https://www.epa.gov/npdes)
- EU Taxonomy: DNSH, Pollution Prevention, Item 4 Noise and Dust (https://finance.ec.europa.eu/sustainable-finance/tools-and-standards/eu-taxonomy-sustainable-activities_en)
- U.S. National Invasive Species Information Center (NISIC) (https://www.invasivespeciesinfo.gov)
- U.S. National Invasive Species Council (https://www.doi.gov/invasivespecies)
- USDA PLANTS Database (https://plants.usda.gov)
- The Xerces Society for Invertebrate Conservation (https://xerces.org)

FURTHER EXPLANATION

EROSION AND SEDIMENTATION CONTROL PLAN

- This prerequisite applies to all sites, even those smaller than one acre (0.4 hectares).
- In the U.S., the U.S. Environmental Protection Agency or a local authority, depending on the project's location, administers the permitting process associated with the National Pollutant Discharge Elimination System (NPDES) program using the construction general permit (CGP). Projects outside the U.S. may use EU taxonomy, namely, Do No Significant Harm (DNSH), Pollution Prevention, Item 4 Noise and Dust, or a local equivalent to NPDES.
- For all projects, conformance to local standards or code is required in lieu of the CGP when the code is equally or more stringent.
    - To determine equivalence, compare elements of the local code that cover the requirements in the CGP, Section 2, and ensure that all relevant categories listed are covered by local code; if local codes are less stringent, address gaps by following the CGP.
- Zero lot line projects and projects that cause no exterior site disturbance should develop a narrative that describes why no erosion and sediment control (ESC) plan or site assessment is necessary for the site.

SITE ASSESSMENT

Terms Explained

- **Special status vegetation**: Vegetation that is designated as important by local, state or federal entities. Designations may be size, species, age, rare environmental value, unique genetic resources, aesthetics, location, or other unique characteristic (e.g., heritage or legacy trees). Groves and clusters may also be given special status.
- **Healthy plant communities**: This refers to a grouping of existing native or adaptive vegetation that has vibrant, consistent foliage, active growth, and the absence of discoloration, wilting, or unusual patterns or colors on leaves, stems, or roots that would indicate the presence of pests or disease.
- **Invasive vegetation:** An invasive species is a plant or animal that is not native to the ecosystem under consideration and that causes or is likely to cause economic or environmental harm or harm to human, animal, or plant health.

Biodiverse Habitat

SSc1
New Construction (1–2 points)
Core and Shell (1–2 points)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To conserve existing natural areas, enhance biodiversity, restore damaged areas, and provide thriving habitats for local wildlife.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–2**
Option 1. Preserve and Restore Habitat	1–2
Path 1. Greenfield Sites	1
OR	
Path 2. Previously Disturbed Sites	1–2
AND/OR	
Option 2. Bird-Friendly Glass	1

OPTION 1. PRESERVE AND RESTORE HABITAT (1–2 POINTS)

PATH 1. GREENFIELD SITES (1 POINT)

Preserve 40% of the greenfield area on the site by protecting these areas from all development and construction activity.

OR

PATH 2. PREVIOUSLY DISTURBED SITES (1–2 POINTS)

Meet the requirements of Path 1. Greenfield Sites, if such areas exist.

AND

Restore previously disturbed areas of the site (if such areas exist) and follow the soil restoration and vegetation restoration requirements below. Dedicated athletic fields that are solely for athletic use are exempt from counting toward the total site area. These areas may not count toward the protected greenfield or restored habitat areas.

Points are awarded according to Table 1.

TABLE 1. Points for Percentage of Area Restored

Restored Area	Zero Lot Line	Points
20% of previously disturbed area	10%	1
40% of previously disturbed area	20%	2

Soil restoration

Restore all on-site soils disturbed by previous development and soils disturbed by current construction activities that will serve as a final vegetated area. Any imported soils must be reused in a way comparable to their original function and may not include the following:

- Soils defined regionally by the NRCS web soil survey (or local equivalent for projects outside the U.S.) as prime farmland, unique farmland, or farmland of statewide or local importance.
- Soils from other greenfield sites.
- Sphagnum peat moss or organic amendments that contain sphagnum peat.

Engineered growing medium for vegetated roofs are exempt from the soil restoration requirements.

Vegetation restoration

Plant-native and adapted vegetation that is not currently listed as invasive and includes the following:

- At least 10 species that are native or adapted to the project's *EPA Level III* ecoregion (or local equivalent for projects outside of the U.S.).
- Minimum of two of the following categories: trees, shrubs, and ground cover. Zero lot line projects are exempt from this requirement.
- At least 110 square feet (10 square meters) consisting of native flowering plants appropriate for local pollinators. Plants must be in groupings of at least 10 square feet (one square meter). Designate the pollinator habitat area with signage.

AND/OR

OPTION 2. BIRD-FRIENDLY GLASS (1 POINT)

Glass used below specified heights on the exterior of the building and site must have a maximum threat factor of 30, as defined in the American Bird Conservancy's Threat Factor Database.

This applies to all glass, including spandrel glass, when located

- From grade up to 50 feet (15 meters) measured at all points.
- Up to 20 feet (six meters) measured from the finished grade of a green roof.
- At any distance from grade or roof for glass in guardrails and windshields.

REQUIREMENTS EXPLAINED

This credit requires the protection of healthy greenfield areas, restoring disturbed vegetation and soils with non-invasive native and adapted plants, and providing a designated zone for pollinator habitat. This credit also emphasizes integrating bird-friendly design to protect avian species by incorporating bird-friendly materials and design strategies.

OPTION 1. PRESERVE AND RESTORE HABITAT

PATH 1. GREENFIELD SITES

Greenfield areas are important for environmental conservation because they maintain biodiversity and protect the natural habitats for plants, animals, and insects. Greenfield areas have not been previously developed, graded, or disturbed and support open space, habitat, or natural hydrology. Preserving and protecting greenfield areas from development and construction activity within the project boundary is essential for maintaining a diverse ecosystem.

Forty percent of all greenfield areas on the site must be protected from all development and construction activity.

PATH 2. PREVIOUSLY DISTURBED SITES

Restoring previously disturbed sites is crucial to both the ecosystem and building occupants. Projects that reestablish natural conditions and bring back ecosystem life help rebuild degraded areas, protect soils, and enhance biodiversity and local ecosystems. Thoughtfully designed landscapes with greenery and trees also provide building occupants access to nature, which can positively affect mental health and productivity.

Choose areas that are best-suited for restoration. Prioritize restoration strategies in areas with significant environmental damage, including areas with previous grading, compacted soils, equipment storage areas, and parking lots. Restoration strategies can include adding natural elements to the site, such as ponds, other waterbodies, native or adapted vegetation, soil, and rocks, to help support the site's biodiversity. The project must restore at least 20% (for 1 point) or 40% (for 2 points) of previously disturbed areas to minimize environmental impacts and maintain biodiversity preservation for a significant portion of the site. If the project site includes a greenfield area, the project must protect 40% of that greenfield area from all development and construction activity.

Athletic fields solely for athletic use are exempt from the protected greenfield or restored habitat areas. Athletic fields support human health and well-being by providing physical activity and social interaction.

Soil restoration

Recognizing the importance of healthy soil conditions and integrating restoration into a project is vital for maintaining overall ecological balance. Soil restoration focuses on areas that will undergo revegetation and require restoration of the characteristics necessary to support the selected vegetation. Designated areas for rainwater infiltration may be excluded from vegetation or soil restoration requirements.

Additionally, carefully select and use materials that do not come from or contain soils defined regionally as prime farmland, unique farmland, or farmland of statewide or local importance by the NRCS web soil survey (or local/regional equivalent).[9] Prime farmland soil must not be used because of its excellent fertility and suitability for food production. Using prime farmland soil for restoration can also lead to disruption of soil structure and loss of soil health because this soil has well-developed soil layers. Moving and relocating these soils can interrupt the natural structure and reduce its ability to support ecosystem functions.

Imported soils cannot be sourced from greenfield sites, and they must not contain sphagnum peat moss or organic amendments that contain sphagnum peat. Greenfield sites often have rich biodiversity. Removing soil from a greenfield site can disturb the local ecological system and lead to loss of plant and animal species. Using soil containing sphagnum peat moss is problematic for building construction. Peat soils are characterized by a high water table, absence of oxygen, reducing condition, low bulk density and bearing capacity, soft spongy substratum, low fertility, and high acidity. Such soils are not strong enough to support heavy loads and can affect the stability and integrity of the foundations.

Vegetation restoration

Maintaining vegetation biodiversity enhances the ecological value of the site. Planting native and climate-adapted vegetation is crucial for sustaining the health and functionality of ecosystems. When planting species on-site, invasive species must be avoided to prevent threatening native biodiversity.

Planting diverse vegetation species maintains and improves biodiversity, enhances the aesthetic value of natural spaces, and brings greater biophilic impacts. Visual preference studies have shown that people find it more pleasing when several different plant species are visible instead of a single planting (monoculture).[10]

Teams must include at least 10 species that are native or adapted to the project's *EPA Level III* ecoregion or local equivalent for any project outside of the U.S. The team must identify species from at least two of the following categories: tree, shrubs, and ground cover. Native species are crucial in maintaining ecological balance, and including the required amount ensures that construction projects can contribute to ecological health and support of local wildlife.

Pollinators such as birds, butterflies, and bees play a significant role in crop pollination, climate resilience, and the reproduction of many wild plants. Dedicating an area within the project boundary for native flowering plants appropriate for local pollinators creates a pollinator-friendly habitat that promotes plant reproduction and provides support for declining pollinator populations.

Projects must dedicate at least 110 square feet (10 square meters) of habitat, consisting of native flowering plants appropriate for local pollinators. Signage must include information and education about the habitat's purpose and the native species planted.

AND/OR

9. USDA NRCS, "Web Soil Survey," n.d., https://websoilsurvey.nrcs.usda.gov/app/.
10. Marco Costa, "Vegetation Dispersion, Interspersion, and Landscape Preference," *Frontiers in Psychology* 13 (May 2022): 1036864. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.771543/full.

OPTION 2. BIRD-FRIENDLY GLASS

The project team must use elevation plans and section plans to assess all façade materials (glazing, opaque envelope, etc.) above grade or a green roof within the project boundary.

The American Bird Conservancy (ABC) has developed a system, known as the Material Threat Factor (TF), to evaluate and rate materials based on their potential threat to birds. This system assigns scores to materials and provides a relative measure of how well they deter bird collisions. The scores help architects and designers select bird-friendly materials for buildings. The façade material distances analyzed must consist of the first 50 feet (15 meters) above grade, or up to 20 feet (6 meters) from the finished grade of a green roof. All glass, including spandrel glass, must have a maximum TF of 30 under the ABC database if it is located within these distances.[11] These guidelines consider materials with a TF of 30 or below to significantly reduce the risk of bird collisions, estimating at least a 50% reduction in occurrences. This is particularly important for areas like glazed corners and fly-through conditions, where birds are more likely to collide with glass due to reflections or transparency.[12]

Experts consider the material TF a prescriptive criteria for designing bird-friendly buildings. Therefore, no trade-offs are allowed when assessing the façade materials.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Preserve and Restore Habitat	Path 1. Greenfield Sites	Greenfield area calculation.
			Contract document(s) highlighting the location and size of the greenfield area within the LEED project boundary and demonstrating how the site preserves and protects the greenfield area.
		Path 2. Previously Disturbed Sites	Identification of zero lot line project, if applicable.
			Contract document(s) highlighting the location and size of the previously disturbed area within the LEED project boundary, restored soil and vegetation areas, pollinator habitat area, and pollinator habitat signage. Identify the location and size of any dedicated athletic fields exempted from the total site area.
			Restored previously disturbed area calculation.
			Evidence of the original function and content of any imported soils (e.g., the contract document or specification).
	Option 2. Bird-friendly Glass	All	Contract documents identifying all glass used below specified heights on the exterior of the building and site, the relevant grade points, and relevant elevation markers (e.g., exterior elevations, window specifications).
			List of exterior glass types used in the project and their installed height from its relevant grade, identified by type according to the ABC Threat Factor Database.

11. American Bird Conservancy, "Bird Collision Deterrence: Summary of Material Threat Factors," October 2011, https://abcbirds.org/wp-content/uploads/2015/05/Docs10397.pdf.
12. American Bird Conservancy, "About the ABC Rating System," January 1, 2023, https://abcbirds.org/wp-content/uploads/2023/01/What-is-a-Material-Threat-Factor-1_23.pdf.

REFERENCED STANDARDS

- Natural Resources Conservation Service web soil survey (https:websoilsurvey.nrcs.usda.gov/app/)
- EPA Level III and IV Ecoregions (https://www.epa.gov/eco-research/level-iii-and-iv-ecoregions-continental-united-states)
- American Bird Conservancy's Threat Factor Database (https://abcbirds.org/wp-content/uploads/2023/01/What-is-a-Material-Threat-Factor-1_23.pdf)

FURTHER EXPLANATION

OPTION 1. PRESERVE AND RESTORE HABITAT, PATH 1. GREENFIELD SITES

- If a greenfield area exists on the site, work with site professionals to identify at least 40% of the total greenfield area to be preserved and protected from all development and construction activity.
- Completing the assessment prior to beginning construction, while not required, supports the integrative design process and optimizes opportunities to exceed the 40% minimum threshold requirements for preservation and protection.

OPTION 1. PRESERVE AND RESTORE HABITAT, PATH 2. PREVIOUSLY DISTURBED SITES

Soil Restoration

- Include analysis during the site assessment for *SSp1: Minimize Size Disturbance* to allow teams the opportunity to review all areas within the project boundary early in design and optimize decisions related to site development.
- Soil restoration examples include stockpiling and reusing topsoil from the site, amending site soils in place with organic matter, and importing a topsoil or soil blend designed to serve as topsoil.
- Vegetation restoration should involve individuals and organizations that are knowledgeable about pollinator habitats, such as ecologists and botanists, in early planning and design phases. Knowledge of local conditions will create opportunities to optimize the habitat location within the project boundary and maximize the dedicated area.

OPTION 2. BIRD-FRIENDLY GLASS

- Project teams should consider pairing this option with additional bird-friendly strategies and design elements, such as reducing night-time lighting through *SSc6: Light Pollution Reduction*. Also avoid landscape designs that might channel birds toward a glass surface, such as trees that flank either side of a walkway that terminates with a glass wall.
- Qualifying materials include glazing that incorporates physical signals to birds created by fritting and UV coatings; non-glazing, opaque, and non-reflective materials, such as concrete; glazing behind qualifying sunshades and screens; and glazing to which materials such as qualifying window films have been applied.

Accessible Outdoor Space

SSc2
New Construction (1 point)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To create outdoor open space that encourages interaction with the environment, social and physical activities, and passive recreation, and to incorporate elements that celebrate the community served.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1**
Sufficient Outdoor Space Area AND	
Urban Outdoor Space AND	1
Community Outdoor Space	

Comply with the following requirements for 1 point:

SUFFICIENT OUTDOOR SPACE AREA

Provide barrier-free and physically accessible outdoor space for people with limited mobility—space that is greater than or equal to 30% of the total site area (including building footprint). At least 25% of the required outdoor space must be vegetated and planted with two or more types of vegetation or have an overhead vegetative canopy.

AND

URBAN OUTDOOR SPACE

Include one or more of the following elements:

- **Biophilic space:** An area that meets the vegetation restoration requirements of *SSc1: Biodiverse Habitat* and includes elements of human interaction, such as observation platforms or paths.
- **Garden:** Space dedicated to community gardens or urban food production.
- **Recreational area:** Recreation-oriented paving or landscape area that encourages physical activity, such as courts, fields, track, play space, or swimming pools.
- **Social area:** Pedestrian-oriented paving or landscape area that accommodates outdoor social activities and includes seating for 5% of occupants.

AND

COMMUNITY OUTDOOR SPACE

Include one or more of the following elements:

- **Community:** Publicly accessible during daylight hours and open to all members of the community.
- **Cultural:** Include at least two art installations or sculptures by local artists.
- **Acoustics:** Include elements that provide positive soundscapes if located within 0.24 miles (400 meters) of a significant noise source, such as, but not limited to, a roadway, airport, or rail line.

REQUIREMENTS EXPLAINED

This credit awards projects that include vegetated and paved areas and other green spaces that encourage social interactions and recreational activities. Teams must include targeted elements for accessibility and emphasize deliberate attention to enhancing outdoor spaces for community engagement.

SUFFICIENT OUTDOOR SPACE AREA

Outdoor spaces provide building users with opportunities to connect to the outdoors. These spaces play an essential role in enhancing the health and well-being of building users and provide many positive environmental benefits, such as improving air quality, supporting biodiversity, managing stormwater, reducing urban heat island effects, and fostering social interaction.

All projects must provide outdoor spaces that are barrier-free, physically accessible, and cover at least 30% of the total site area, including the building footprint. At least 25% of this outdoor space must feature two or more types of vegetation or include an overhead vegetative canopy, such as a continuous layer of trees or shrubs that create shaded areas. To maintain the ecological function of sensitive areas, such as water bodies or vegetation zones, full accessibility is not required; however, features like boardwalks designed to meet accessibility standards can provide inclusive access.

The outdoor space design must incorporate features that make it accessible to people with disabilities and service animals. This may include providing wheelchair ramps, tactile surfaces, wide pathways, and signage to ensure that everyone, regardless of physical ability, can navigate and use the space comfortably.

URBAN OUTDOOR SPACES

Urban outdoor spaces are pedestrian-oriented paving or landscape areas that facilitate social activities. They also offer users recreational areas where designs incorporate walking paths, playgrounds, and fitness equipment. Additionally, outdoor spaces with vegetation are crucial for the environment. Vegetated outdoor areas help reduce the urban heat island effect, improve air and water quality, and support biodiversity by providing habitats for various species.

Teams pursuing this credit should include multiple, diverse outdoor spaces, which allow for the inclusion of people of all ages and activity levels. At a minimum, the outdoor space must include at least one of the following: biophilic space, garden, recreational area, or social area.

Biophilic space

Biophilic space integrates natural elements into building designs and connects individuals with nature to enhance health, well-being, and productivity. Restoring vegetation in biophilic spaces by planting diverse species helps to sustain and enhance biodiversity while boosting the aesthetic appeal of natural areas, which results in stronger biophilic effects.

Areas that meet the vegetation restoration requirements of *SSc1: Biodiverse Habitat* and include elements of human interaction meet the biophilic space criteria of this credit. These spaces must integrate features that invite engagement, such as seating, walking paths, shaded areas, and educational signage about the local ecological system.

Garden

Community gardens and urban food production spaces offer residents an opportunity to connect with nature and practice environmental stewardship. These areas enable building occupants and neighbors to grow fruits and vegetables while fostering interaction with the environment and building connections within the community.

Project teams that incorporate extensive or intensive vegetated roofs in the design may use these areas to comply with the garden element, if they are physically accessible and include areas dedicated to food production. Maintenance is necessary for the vegetated roof system to keep plants healthy and the supporting structure in good condition.

Recreational area

The design of recreation-focused paved and landscaped areas such as sports courts, fields, tracks, playgrounds, and swimming pools encourages active engagement with the environment. These spaces promote physical activity, social interaction, and active and passive recreation opportunities. Recreational spaces support health and fitness and celebrate and strengthen community connections.

Projects may count turf areas in the total outdoor space calculation if these areas contain physical site elements that support and accommodate outdoor social activities. However, turf is not required to qualify as this space type. Turf areas cannot apply toward meeting the credit's 25% vegetated area requirement under the Sufficient Outdoor Space Area requirement.

Projects may also include ponds or wetlands that occur naturally or are designed to function similarly to natural site hydrology and land cover as outdoor spaces if the average side slope gradients are 1:4 or less and are vegetated.

Social area

Pedestrian-oriented paving or landscape areas qualify as social spaces when intentionally designed to include seating and foster outdoor activities, social interaction, and engagement with nature. They must be accessible to users of all abilities, flexible for diverse activities, and conveniently located near building entrances or amenities.

The space must include sufficient seating to accommodate at least 5% of the daily average occupants, including visitors, to ensure adequate resting space.

COMMUNITY OUTDOOR SPACE

Projects must incorporate one or more elements that enhance community outdoor spaces and focus on aspects such as social gathering, cultural expression, and acoustic quality. These elements might include designated community gathering areas and components that improve the auditory environment, such as natural sound buffers. By integrating these features, projects foster a sense of community, reflect cultural values, and create welcoming, enjoyable outdoor spaces for all users.

Community

Projects that pursue these criteria must include outdoor spaces that are accessible to the public during daylight hours, thereby ensuring all community members can enjoy and use the space. However, facilities that are not open to the public for security reasons (e.g., data centers) are exempt from this requirement. This exemption also applies to international projects with gated apartment complexes, office parks, military bases, manufacturing complexes, research and development campuses, private hotels, resorts, and similar facilities. This approach is designed to promote inclusivity and allow people of all ages and backgrounds to gather and engage in recreational and social activities in a welcoming environment.

Cultural

These outdoor spaces should be designed to represent the community's cultural and social diversity. Project teams can achieve this by incorporating art and sculptures that celebrate and recognize the area's unique cultural identity and demographics. Projects must include at least two art installations, such as paintings on walls, interactive light installations, kinetic art that moves with the wind, or sculptures created by local artists. These additions help to foster a sense of community pride, create meaningful connections to the place, and enhance the overall experience for all users.

Acoustics

Projects located within 0.25 miles (400 meters) of a significant noise source must incorporate design elements that enhance the soundscape and address unwanted or disruptive sounds that negatively affect the comfort, health, or productivity of the building's occupants. A significant noise source is any noise-generating entity or activity that consistently produces sound levels above recognized comfort or safety thresholds according to local regulations. These soundscape enhancements may include sound barriers, strategically positioned vegetation, and water features (e.g., fountains), all designed to mitigate noise pollution. Their effectiveness can be evaluated through metrics such as interior noise levels, Noise Reduction Coefficients (NRC), and Sound Transmission Class (STC) ratings. By dampening external noise, these features contribute to a more pleasant and acoustically balanced environment, which enhances comfort and well-being for building users.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Landscaping plan(s) that include the vegetation key/schedule and the LEED project boundary and highlights the locations and sizes of the barrier-free and physically accessible outdoor space, the vegetated outdoor space, the urban outdoor space, and the community outdoor space. Identify/tag each urban outdoor area as biophilic, garden, recreational, or social area; show and note the project-specific elements that qualify the space (e.g., observational platform, seating, etc.). Identify/tag each outdoor space as community, cultural, or acoustics; show and note the project-specific elements that qualify the space (e.g., sculpture by X), and the source and general direction of any roadway, airport, or rail line.
			Outdoor space calculation.
			Vegetated outdoor space calculation.
			Social area seating calculation.
			Evidence that any outdoor community space is publicly accessible during operating hours and open to all members of the community (e.g., shop drawings showing location and details for publicly posted signage, website content).
			Evidence of any art installations or sculptures by local artists for the project site (e.g., contract documents and/or purchase orders).
			Evidence of any positive soundscapes provided for the project (e.g., contract documents and/or purchase orders, product information from manufacturers).

REFERENCED STANDARDS

- None

FURTHER EXPLANATION

- Extensive or intensive vegetated roofs that are physically accessible can be used toward the minimum vegetation requirement, and qualifying roof-based physically accessible paving areas can be used toward credit compliance.
- Turf areas, including areas of turf grass under overhead tree canopies, can be counted in total open space but do not qualify as vegetated open space.

Rainwater Management

SSc3
New Construction (1–3 points)
Core and Shell (1–3 points)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To reduce runoff volume and improve water quality by replicating the natural hydrology and water balance of the site, based on historical conditions and undeveloped ecosystems in the region, to avoid contributing to flooding downstream in frontline communities.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–3**
Option 1. Percentile of Rainfall Events OR	1–3
Option 2. Natural Land Cover Conditions	3

OPTION 1. PERCENTILE OF RAINFALL EVENTS (1–3 POINTS)

In a manner best replicating natural site hydrology processes, retain the runoff from the associated percentile of regional or local rainfall events on-site. The percentile event volume must be retained (e.g., infiltrated, evapotranspired, or collected and reused) using LID and GI practices. GI and LID strategies can be either structural or nonstructural.

For projects that collect and reuse a portion of the chosen percentile event volume to meet the needs of one or more end uses for the building and grounds, 1 additional point can be earned. Eligible end uses include irrigation, flush fixtures, makeup water systems (e.g., cooling towers or boilers), and other process water demands. Collecting and reusing rainwater within the project can also contribute to points earned in the Water Efficiency credit category. Points are awarded according to Table 1.

TABLE 1. Points for Percentile of Regional or Local Rainfall Events Retained

All Projects	Zero Lot Line Projects	Points	Total Points for Water Reuse
80th percentile	70th percentile	1	2
85th percentile	75th percentile	2	3
90th percentile	80th percentile	3	–

OR

OPTION 2. NATURAL LAND COVER CONDITIONS (3 POINTS)

Calculate the difference between the projected runoff volume under the proposed design conditions and the runoff volume under natural land cover conditions that existed prior to any disturbance. Retain (e.g., infiltrate, evapotranspire, or collect and reuse) on-site the increase in runoff volume using LID and GI practices.

For zero lot line projects only

Project teams may combine on-site and off-site strategies to retain runoff from the associated percentile regional or local rainfall event for points according to Table 1. Engage with local or regional authorities to coordinate off-site rainwater management strategies that meet the credit's intent, such as participating in community-wide rainwater management programs.

REQUIREMENTS EXPLAINED

This credit rewards projects that manage water runoff from the site by using methods that most closely replicate a natural site hydrology process. The system awards points based on increasing thresholds for the percentage of runoff managed on-site, with additional points given if someone collects and reuses rainwater. Projects may also earn points for retaining any excess runoff anticipated due to the new development conditions.

OPTION 1. PERCENTILE OF RAINFALL EVENTS

Data from the percentile rainfall event allows design professionals to study changes in precipitation patterns over time, helping owners and designers understand the impacts of climate change. Using this information, projects can successfully implement strategies to mitigate these impacts.

This option requires that teams gather historical rainfall data for the project area. This data must include the amounts of daily rainfall over a minimum 30-year period. With the data collected, project teams must calculate the rainfall amount (depth) corresponding to the desired percentile of rainfall event and the runoff volume. Using the results of the runoff calculations, the project must implement appropriate stormwater management practices with LID and/or GI measures to maintain the runoff on-site.

For projects encompassing larger watershed boundaries, it is crucial to ensure that stormwater management requirements comply with the chosen percentile event across the entire watershed.

All development within the boundary must meet the same stormwater performance standards and ensure consistent and effective management of runoff. This approach extends beyond the immediate building site boundary and encompasses the entire watershed area associated with the project.

Managing runoff

Projects using this option must retain or manage runoff per the regional/local rainfall percentiles. Percentile calculations help determine how rare or common a particular rainfall event is by comparing it to historical data. By analyzing rainfall percentiles, projects can implement appropriate strategies for water storage, distribution, and retainment.

Table 2 provides a sample list of options that meet the requirements for LID and GI measures.[13]

TABLE 2. LID/GI Measures

LID/GI Measure	Description and Additional Details
Rain gardens and Bioretention gardens	These are decorative gardens that have plants and soil that filter runoff water and encourage infiltration. This practice is ideal for collecting runoff from rooftops, sidewalks, roads, and small parking lots.
Green roofs	Green roofs include plants and soil media that capture and filter water that would have previously been considered runoff.
Rainwater harvesting	Ideal for collecting rooftop runoff, a rainwater cistern captures and stores (e.g., harvests) runoff for later use. Common uses of collected rainwater are for irrigation or for indoor plumbing fixture flushing. Project teams that use any rainwater harvesting strategies with the intent of reuse will earn additional points.
Vegetated swales	The shallow, open channels are ideal for collecting sheet flow runoff from roads, highways, and from subdivisions.
Permeable pavement	Permeable pavement allows stormwater to pass through and into gravel layers, allowing stormwater to soak into the soil. This practice is ideal for developed areas, such as parking lots and driveways.
Exfiltration trenches	Surface runoff collected through drainage inlets and directed into the trench via subsurface perforated pipes. The runoff infiltrates into the ground through the trench's gravel bed, which filters the pollutants.

Rainfall event calculations

The percentile of rainfall events measures the precipitation depth accumulated over 24 hours, typically defined as 12:00:00 a.m. to 11:59:59 p.m., and it relies on the range of all daily event occurrences during the period of record.

Projects in the U.S. may use percentile rainfall events as determined by the National Climatic Data Center. International projects can use the Global Precipitation Climatology Project (GPCP) Daily Precipitation Analysis, provided by the National Centers for Environmental Information (NCEI).[14]

13. Chelsea J. Martin-Mikle, Kirsten M. de Beurs, Jason P. Julian, and Paul M. Mayer, "Identifying Priority Sites for Low Impact Development (LID) in a Mixed-use Watershed," *Landscape and Urban Planning*, 140 (August 2015): 29–41, https://doi.org/10.1016/j.landurbplan.2015.04.002.
14. National Centers for Environmental Information, "Climate Data Records: Global Precipitation Climatology Project (GPCP)—Daily," 2015, https://www.ncei.noaa.gov/products/climate-data-records/precipitation-gpcp-daily.

Using the data from these resources for the project's location, the selected timeline, and other relevant information on the rainfall event from the selected resource, teams must calculate the percentile of the rainfall event by using the USGBC-approved calculator or provide an additional calculator including the necessary information for credit compliance.[15]

Runoff calculations

Teams must calculate the runoff volume using the modified rational method, *the Technical Release 55 (TR-55), Natural Resources Conservation Service* method, the *U.S. EPA Rainwater Management Model* (SWMM), or other runoff methodologies most appropriate for the project.[16,17,18,19] Equation 1 shows an example of a simplified version of the runoff calculation. The runoff calculations must include the watershed boundary, such as buildings, parking lots, landscaping, pervious surfaces, and all other impervious surfaces. This approach ensures a comprehensive method to managing stormwater.

Equation 1. Runoff volume calculation

$$\text{Runoff volume} = \text{Rainfall depth} \times \text{Area} \times \text{Runoff Coefficient}$$

where:
Rainfall depth = Measure the rainfall in inches (or meters) as determined by the percentile rainfall calculator.
Area = Measure the area of both impervious and pervious surfaces within the watershed boundary in square feet (or square meters).
Runoff coefficient = This is an aggregated factor that represents the percentage of rainfall that becomes runoff. For impervious surfaces like concrete or asphalt, this is typically close to 1 (e.g., 0.95).

Rainwater harvesting

Rainwater harvesting contributes to rainwater management. Rainwater harvesting is a sustainable practice that captures and stores rainwater to allow for water reuse within the project site. Projects that collect and reuse rainwater from the chosen percentile event volume for one of the eligible end uses can earn one additional point.

Eligible end uses

- **Irrigation:** Use rainwater to water gardens, lawns, and agricultural fields, which reduces the demand on municipal water supplies.
- **Flush fixtures:** Conserve potable water for drinking and cooking by using harvested rainwater in toilets and urinals.

SS

15. U.S. Green Building Council, "LEED v4.1 Rainfall Events Calculator," updated April 10, 2020, https://www.usgbc.org/resources/leed-v41-rainfall-events-calculator.
16. Bentley Systems, "Modified Rational Method," Bentley SewerCAD SS5, n.d., https://docs.bentley.com/LiveContent/web/Bentley%20SewerCAD%20SS5-v1/en/GUID-85A442CDB33D4B1684EE9E795BA6BABE.html.
17. United States Department of Agriculture, Agricultural Research Service, "Small Watershed Hydrology (WinTR-55)," n.d., https://www.ars.usda.gov/research/software/download/?softwareid=8&modecode=80-42-05-10.
18. Natural Resources Conservation Service, United States Department of Agriculture, "Conservation Planning," n.d., https://www.nrcs.usda.gov/getting-assistance/conservation-technical-assistance/conservation-planning.
19. United States Environmental Protection Agency, "Storm Water Management Model (SWMM)," n.d., https://www.epa.gov/water-research/storm-water-management-model-swmm.

- **Makeup water systems:** Rainwater can serve as makeup water for cooling towers and boilers, which require large volumes of water for operation.
- **Process water demand:** Industries can use harvested rainwater for processes that do not require potable water, such as washing, cooling, and other operational needs.

Additional considerations

Project teams considering rainwater reuse may also earn points under the WE Credit Category. Consider using rainwater for toilet flushing or irrigation to earn additional points for *WEc2: Enhanced Water Efficiency* to pursue these additional points.

OPTION 2. NATURAL LAND COVER CONDITIONS

Option 2 requires that the proposed design retain the increase in runoff compared to the natural land hydrology of the project site. Under this option, projects must maintain runoff on-site and design the site to mimic natural hydrology. This strategy provides absorption and filtration that supports diverse ecosystems by maintaining a balance of water levels in wetlands, rivers, and lakes. Natural land covers, such as forests and grasslands, often allow more water to infiltrate the ground, recharge groundwater supplies, and reduce the volume and speed of runoff.

Natural land hydrology

Natural land hydrology reflects the natural land cover function of water occurrence, distribution, movement, and balance. Teams must determine how quickly water can seep into the existing conditions or soil, which influences groundwater recharge and surface runoff. Site features like topography, soil, and surface water bodies, along with pre-design connections to nearby ecosystems, help maintain natural land hydrology.

Rainfall event

Document historical rainfall data for the project boundary; however, unlike Option 1, specific percentile events are not required for this option. Project teams must use a full range of hydrologic rainfall events over a 10-year period or develop an average representative rainfall year, then use the following process to determine the average runoff volume under natural land cover conditions:

- **Average rainfall calculation:** Calculate the average annual rainfall based on historical data. This can be done using methods like arithmetic means.
- **Monthly distribution:** Distribute the average annual rainfall across months to create a representative year. This helps in understanding seasonal variations and planning for water management.
- **Runoff estimation:** Use the average monthly rainfall data to estimate runoff for each month. This approach provides a simplified view of runoff patterns over a typical year.

By considering both the full range of hydrologic events and an average representative year, teams can develop a comprehensive understanding of stormwater runoff patterns and design an effective management system.

Runoff calculations

Determine runoff volume using the rainfall event as calculated using Equation 1. Natural land hydrology, for this option, is the natural land cover present prior to any development on the site.

Project teams must use the project's natural land hydrology and land use to determine the runoff coefficient. For natural conditions, the runoff coefficient will be lower due to higher infiltration and vegetation cover.

Natural land cover refers to the original vegetation and soil conditions that existed in an area before any development or human activities altered the landscape.

By preserving or restoring natural land cover, projects can help to maintain ecological balance, support biodiversity, and enhance the sustainability of our environment.

Using the proposed design, calculate the design runoff, using the runoff coefficients for the design conditions as indicated in Option 1, Equation 1.

Retention requirements

Projects must retain any increase in runoff within the project site. The retention design strategies in Table 2, "LID/GI Measures," must mimic natural land hydrology.

ZERO LOT LINE PROJECTS

For zero lot line projects, coordinate with regional authorities regarding off-site rainwater management strategies, which may be combined with on-site strategies to retain runoff for the required percentile. Off-site strategies must meet the credit intent of the LID/GI practices.

Given the limited space, it is crucial to maximize the infiltration of stormwater on-site. This can be achieved through permeable pavements, green roofs, and rain gardens. Use underground infiltration systems to store and gradually release stormwater. These features help to reduce runoff and promote natural groundwater recharge prior to consideration of off-site strategies.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Identification of zero lot line project.
			USGBC calculator.
	Option 1. Percentile of Rainfall Event	All	Percentile of rainfall events retained.
			Identify any end uses that the project meets through collection and reuse of rainwater.
			The documents depict and explain the site's design conditions, including the overland flow paths of rainwater, the topography, and the soil conditions. They will also outline how the rainwater will be managed through infiltration, evapotranspiration, capture, reuse, and overflow outlets (e.g., topography plans, landscape plans, plant lists, construction details, cross sections, specifications, product information from manufacturers, and narratives).

(continued)

(continued)

Project Types	Options	Paths	Documentation
All	Option 2. Natural Land Cover Conditions	All	The difference between the projected runoff volume under the proposed design conditions and the runoff volume under natural land cover conditions that existed prior to any disturbance.
			Evidence of the site's natural land cover conditions that existed prior to any disturbance (e.g., historical maps, environmental impact assessments).
			The documents depict and explain the site's design conditions, including the overland flow paths of rainwater, the topography, and the soil conditions. The team will also outline how the increase in runoff will be managed by infiltration, evapotranspiration, capture and reuse (e.g., topography plans, landscape plans, plant lists, construction details, cross sections, specifications, product information from manufacturers, and narratives).

REFERENCED STANDARDS

- Technical Release 55 (T-55) Urban Hydrology for Small Watersheds (https://www.ars.usda.gov/research/software/download/?softwareid=8&modecode=80-42-05-10)
- Natural Resources Conservation Service (https://www.nrcs.usda.gov)
- U.S. EPA Storm Water Management Model (SWMM) (https://www.epa.gov/water-research/storm-water-management-model-swmm)

FURTHER EXPLANATION

RAINWATER HARVESTING AND REUSE

- Projects will need to determine available precipitation, designate a catchment area for rainwater collection, and identify end uses for the rainwater reuse in order to size the cistern or pond appropriately.
- If a stormwater pond is used instead of a cistern, evaporation must be taken into consideration in the drawdown calculations.
- Indoor and outdoor water demand should ideally be calculated daily, but at a minimum monthly.
- Cisterns and other stormwater best management practices (BMPs) that are designed to collect and absorb runoff, such as raingardens and infiltration basins, should be modeled as dry and able to dewater within 72–96 hours to ensure adequate storage before the next precipitation event.

Enhanced Resilient Site Design

SSc4
New Construction (2 points)
Core and Shell (2 points)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

Reduce the risk of catastrophic impacts from natural and climate events on-site and in adjacent landscapes by designing, building, and maintaining sites to be more resilient to observed, projected, and future climate and natural hazards.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**2**
Integrate Requirements for Two High-Priority Hazards	2

Design and construct the site and site structures to meet the following best practices for at least two of the highest priority hazards identified for compliance with *IPp1: Climate Resilience Assessment*.

Drought

Comply with *WEc2: Enhanced Water Efficiency* requirements. Specify native and/or drought-tolerant adapted/appropriate plantings. Water makeup for any created water features must comply with SITES C3.4 or local equivalent.

Core and Shell only

Water makeup for any created water features must not exceed 5,000 gallons (18,927 liters) of potable water per year, or 75% of annual water makeup must come from alternative water sources.

AND/OR

Extreme heat

Integrate two or more additional elements from the following list:

- Provide shaded external spaces adjacent to buildings for use during extreme heat events.
- Provide evaporative cooling solutions (e.g., fountains, misters, water features, etc.).
- Orient buildings and massing to self-shade in summer and extreme heat conditions.
- Provide outdoor cooling stations with emergency backup power.
- Demonstrate proximity to an emergency cooling station within 0.25 miles (400 meters).
- Use paving materials with an initial solar reflectance (SR) value of at least 0.33.
- Use an open-grid pavement system (at least 50% unbound).

AND/OR

Flooding

Integrate two or more of the following strategies, in accordance with *ASCE 24* and *FEMA 543* standards or local equivalent:

- Critical utilities
  - Locate critical utilities in new construction above the design flood elevation (DFE), plus freeboard as recommended.
  - In retrofits, locate critical utilities inside protective, floodproofed enclosures to prevent water intrusion.
  - Design new potable water systems to resist flood damage, infiltration of floodwaters, and discharge of effluent.
  - Elevate on-site wellheads above surrounding landscape to allow contaminated surface water to drain away.
  - Design new sewage systems to avoid infiltration and backup from rising floodwaters.
  - Design and anchor plumbing conduits, water supply lines, gas lines, and electric cables that must extend below DFE to resist the effects of flooding.
  - Design and anchor rainwater storage tanks to resist flood forces.
- Ensure that all structural materials, finish materials, and connectors used below DFE are flood resistant.
- Certify the project under a qualifying flood-resilient design standard(s).

AND/OR

Hail

Design and construct the site and site structures according to *FORTIFIED Commercial High Wind and Hail Specific Design Requirements for Hail*, or local equivalent.

AND/OR

Hurricanes and high winds

For projects in hurricane-prone areas, design and construct the site and site structures according to *FORTIFIED Commercial Wind* standards, or local equivalent. For projects in high-wind areas, design and construct the site and site structures to comply with wind design measures per *ASCE/SEI 7-10* in specified FEMA zones or local equivalent.

- Install backup power systems in hurricane-prone regions.
- Install electrical connections with a transfer switch or docking station (storm switch) to support connection of backup power for critical mechanical and electrical systems.
- Create windbreaks using landscape forms, vegetation, and other locally appropriate natural systems.

AND/OR

Sea level rise

Design and construct the site to accommodate flooding based on sea level rise and storm surge projections for the design service life of the project. In addition, meet two or more of the following:

- Incorporate elevated foundations to minimize projected flood damage to buildings.
- Use materials resistant to projected water damage for construction.
- Apply sealants and coatings to prevent projected water infiltration into structures.
- Install flood barriers to block projected floodwater from entering buildings.
- Design GI solutions to effectively manage projected storm surge and stormwater runoff.
- Ensure backup power systems are in place to maintain critical functions during projected flooding events.
- Develop integrated drainage systems to efficiently manage projected excess flooding.
- Engage in community-level planning, partnerships, and/or design workshops to coordinate flood mitigation efforts to effectively and equitably address the needs of populations vulnerable to projected flooding.
- Retrofit existing structures to enhance their resilience to future flood risks.

AND/OR

Tsunamis

Mitigate the impact of tsunamis through site planning strategies as described in *Designing for Tsunamis* (U.S. National Tsunami Hazard Mitigation Program), or local equivalent. Additionally, incorporate the following elements from the *TsunamiReady® Guidelines,*[20] or local equivalent:

- Install tsunami danger area and evacuation route signage.
- Install Public-Alert-certified NOAA Weather Radio receivers in critical facilities and public venues, or local equivalent.

AND/OR

Wildfires

Follow wildfire management practices pertaining to wildland–urban interface design, vegetation management, debris disposal, and fire safety for equipment referenced in the *National Wildfire*

20. National Weather Service, "TsunamiReady® Guidelines," 2015, https://www.weather.gov/media/tsunamiready/resources/2015TRguidelines.pdf.

Coordinating Group Standards for Mitigation 2023, or local equivalent. Design and construct the site and site structures in compliance with the *SITES v2 rating system credit 4.11: Reduce the risk of catastrophic wildfire*, or local equivalent. Reduce fuel using the zone concept *Safer from the Start*, Appendix E,[21] or local equivalent.

AND/OR

Winter storms

Meet two or more of the following:

- Provide adequate ingress/egress for vehicles and snow removal equipment.
- Provide a snow removal plan, including compatible road materials, areas for accumulated snow, and roof snow removal.
- Ensure safe walking surfaces to exterior parking areas by considering installing heated sidewalks with renewable energy sources.
- Specify native or adapted planting with a capacity for heavy snow loads.

REQUIREMENTS EXPLAINED

This credit requires the design and construction of sustainable and resilient site and site structures based on best practices for at least two of the highest priority hazards identified in the *IPp1: Climate Resilience Assessment*. Addressing additional hazards is highly recommended to create added resilience within the project.

DROUGHT

Climate projections indicate a higher likelihood of more intense droughts in the future. As a slow-onset hazard, droughts can last for months or even years, leading to significant consequences such as increased erosion, water scarcity, and a heightened risk of wildfires.[22] Implementing sustainable practices ensures projects significantly reduce their dependency on freshwater resources, safeguard against water shortages, and contribute to the broader goals of water conservation and climate resilience.

Reduced water use for irrigation

To mitigate the impacts of drought, projects must comply with *WEc2: Enhanced Water Efficiency* requirements to reduce outdoor water use.

Drought-tolerant plant species

Projects must incorporate native and/or drought-tolerant plant species that adapt to the site's conditions and local climate to reduce water demand. Native or drought-tolerant plants help

21. Firewise Landscapes, "Safer from the Start: A Guide to Firewise-Friendly Developments," 2009, https://www.firewise.net/wp-content/uploads/2012/05/Safer-From-the-Start.pdf.
22. IPCC, "Chapter 4: Water," n.d., https://www.ipcc.ch/report/ar6/wg2/chapter/chapter-4/.

conserve water and reduce soil erosion. They require less water, fertilizer, and maintenance, and are a key component of water-efficient design.

Makeup water strategies for water features

Projects must minimize or eliminate potable water, natural surface water, and groundwater withdrawals that are used in water features to reduce short and long-term water use. Designs must also comply with water feature makeup water requirements, as outlined in *SITES Credit 3.4 Reduce outdoor water use*.[23] This means ensuring that the design of water features, including fountains or ponds, minimizes water loss through evaporation or leaks, and efficiently replenishes water, ideally using nonpotable sources such as gray water or harvested rainwater.

LEED BD+C: Core and Shell projects must limit makeup water for any newly created water features to 5,000 gallons (18,927 liters) of potable water annually, or at least 75% of the annual makeup water must come from alternative water sources.

EXTREME HEAT

Improving thermal comfort and reducing extreme heat-related risks in a project is essential for safeguarding public health, enhancing occupant well-being, and ensuring the long-term resilience and sustainability of built environments in the face of rising temperatures. Nature-based solutions are key to achieving these goals, because they leverage natural processes to create cooler, more comfortable spaces while also promoting biodiversity.

Shaded outdoor spaces

Shaded outdoor spaces that use shade from appropriate trees, large shrubs, vegetated trellises, walls, or other exterior structures help cool the surrounding environment and offer protection from direct sunlight. Shaded areas adjacent to buildings can significantly reduce evaporation rates in soil and promote habitats for various species, thereby supporting biodiversity and maintaining the balance of local ecosystems. Providing cooler environments helps decrease the incidences of heat exhaustion and dehydration, particularly among vulnerable populations such as children, the elderly, and those with preexisting health conditions.

Evaporative and outdoor cooling solutions

Projects may also incorporate evaporative and outdoor cooling solutions, such as fountains, misters, and water features. These can significantly enhance outdoor thermal comfort and create a more pleasant microclimate in outdoor spaces by reducing ambient temperatures through the evaporation of water.

Providing outdoor cooling stations equipped with emergency backup power is essential for offering rest and relief during high-temperature events. These stations can include shaded seating and misting systems to create comfortable environments. The backup power ensures that cooling stations remain operational during outages and enhance community resilience and safety by preventing heat-related illnesses.

Locate emergency cooling stations within 0.25 miles (400 meters) of the building for easy access.

23. Sustainable Sites Initiative, https://www.usgbc.org/resources/sites-rating-system-and-scorecard.

Passive cooling strategies

Orienting buildings for passive cooling is a key strategy for enhancing energy efficiency and occupant comfort in sustainable design. Positioning site structures to take advantage of natural airflow and local wind patterns significantly reduces the need for mechanical cooling systems in buildings. The thoughtful placement of windows, doors, and shading devices plays a crucial role in minimizing heat gain from direct sunlight, thereby keeping interior spaces cooler during extreme heat events.

Reducing heat island effects

Using paving materials with high SR can reduce the absorption of heat from the sun and reduce urban heat island effects for site paving and structures (including roads, sidewalks, playgrounds, shelters, and parking lots). Paving materials must have an initial SR value of at least 0.33, as measured in accordance with *ANSI/CRRC S100*.[24] Vegetated surfaces and planted areas, such as an open-grid pavement system, reduce the use of impervious surfaces that can also contribute to heat island effects.

FLOODING

Flooding can lead to significant property damage, infrastructure disruption, and public health risks and result in loss of life and economic hardship for affected communities. Additionally, it causes ecosystem damage, soil erosion, and long-term recovery challenges, which underscores the importance of effective flood management strategies to enhance resilience.

To mitigate the impacts of flooding, projects under flood-resilient design must implement at least two of the indicated flood mitigation strategies in accordance with *FEMA 543*[25] and *ASCE 24*,[26] or their local equivalents. These standards provide minimum requirements and offer critical guidelines that enhance the safety and structural integrity of vulnerable sites and buildings located in flood hazard areas. Projects can choose from any of the strategies listed under "Critical Utilities," as well as flood-resistant materials or certifying under a qualifying design standard.

Critical Utilities

It is essential to elevate critical utilities above the design or base flood Elevation (DFE/BFE) as well as include additional freeboard to prevent water intrusion. This elevation protects essential services such as water supply systems, sewage treatment facilities, communication systems, and electrical infrastructure from submergence, which could lead to costly repairs and service disruptions. Adding freeboard serves as a buffer and an additional safety measure and further enhances the resilience of critical infrastructure. Positioning utilities above the DFE/BFE significantly reduces the risk of damage from floodwater and ensures that communities have continued access to critical services during and after a disaster. This reliability is vital for public health and safety, as well as for emergency response operations.

24. Cool Roof Rating Council (CRRC), "ANSI/CRRC S100 Standard," https://coolroofs.org/resources/ansi-crrc-s100.
25. FEMA, "Design Guide for Improving Critical Facility Safety from Flooding and High Winds," FEMA 543, 2007, accessed April 4, 2025, https://www.fema.gov/sites/default/files/2020-08/fema543_design_guide_complete.pdf.
26. FEMA, "Highlights of ASCE 24-14: *Flood Resistant Design and Construction*," accessed April 4, 2025, https://www.fema.gov/sites/default/files/2020-07/asce24-14_highlights_jan2015.pdf.

Design new potable water and sewage systems to withstand flood conditions, which ensures uncontaminated drinking water and prevents sewage overflow during flood events. Elevating on-site wellheads above the surrounding landscape is essential to allow contaminated surface water to drain away effectively.

Plumbing conduits, water supply lines, gas lines, and electric cables extending below the DFE must be carefully designed and anchored to withstand flooding. In addition, rainwater storage tanks must be designed and secured to resist flood forces so they remain functional and protected during flood events. These measures help minimize damage to critical utilities and enhance flood resilience.

Flood-resistant materials

Projects using this strategy also must ensure that all structural materials, finish materials, and connectors used below DFE are flood resistant. Using flood-resistant structural materials that can withstand water pressure, corrosion, and potential debris impacts during floods, such as reinforced concrete or high-density polyethylene, further enhances a building's resilience to flooding.

HAIL

Hail strikes can cause significant damage to site infrastructure, outdoor storage, and building components such as roofs, siding, equipment, and windows, which can lead to costly repairs. Hail can also damage landscaping, particularly trees, by shredding leaves, breaking branches, and damaging bark, which makes trees vulnerable to disease, pests, and slower growth.

Teams must design and construct the site and site structures according to *FORTIFIED Commercial High Wind and Hail Specific Design Requirements*[27] for hail, or local equivalent. For instance, teams may choose hail guards for air conditioning units and impact-resistant materials for the roofs. Because hail typically occurs during thunderstorms, compromised roofing can allow water infiltration. Ensuring watertightness and hail resistance reduces the risks of hail damage and protects both the structure and occupants.

HURRICANES AND HIGH WINDS

Hurricanes and high winds can cause severe damage to site infrastructure, buildings, landscaping, and safety. High winds can tear off roofs, break windows, and damage siding. Landscaping can also be affected when storms uproot trees or severely damage plants. Flying debris heightens the risk of injury or even death. Implementing wind-resistant design measures, such as reinforced structures and impact-resistant materials, reduces these risks and enhances the site's resilience against such extreme events.

To enhance the resilience of a site against hurricanes and high winds, the project site and site structures must be constructed according to the *FORTIFIED Commercial Wind* standards or a local equivalent.[28] Apply *FORTIFIED Commercial* standards along with federal, state, and local codes, ordinances, and regulations. If there are conflicts between provisions, use the more stringent

27. FORTIFIED, "FORTIFIED Commercial High Wind and Hail Specific Design Requirements," 2023, https://fortifiedhome.org/wp-content/uploads/Fortified_Commercial_Standard-2022-v4.pdf.
28. FORTIFIED, "FORTIFIED Commercial™–Wind Standards," 2020, https://fortifiedcommercial.org/wp-content/uploads/Fortified_Commercial_Wind_Standards_2020.pdf.

regulation. Projects in high-wind areas must comply with the design and construction requirements for site structures as outlined by *ASCE/SEI 7*[29] in specified FEMA zones or local equivalent.

Backup power systems

Projects must install backup power systems in hurricane-prone regions to ensure that critical operations can continue during power outages. Critical operations refer to essential functions, such as HVAC systems, lighting, and communications, which must remain operational continuously to ensure the safety, stability, and proper functioning of the building. Install electrical connections with a transfer switch or docking station (storm switch) to support the connection of backup power. These must be installed for critical mechanical and electrical systems. Backup generators or solar-powered systems with battery storage provide an emergency power supply and ensure the continuation of critical operations during power interruptions.

Windbreaks

A windbreak will reduce wind speed for as much as 30 times the windbreak's height.[30] A windbreak involves strategically planting trees, shrubs, and other vegetation, and using landscape forms and other locally appropriate natural systems to reduce wind speed around buildings and open spaces. Using vegetation also helps prevent topsoil erosion, which is essential for maintaining the structural integrity of buildings and infrastructure.

SEA LEVEL RISE

Sea level rise poses significant threats to buildings, particularly in coastal areas. As global temperatures increase, melting ice caps and expanding ocean waters cause sea levels to rise and lead to more frequent and severe flooding events. These changes put pressure on existing buildings and increase the risk of water infiltration, foundation damage, and structural instability.

To effectively enhance flood resilience, it is essential to design and construct the site to accommodate flooding based on sea level rise and storm surge projections for the design service life of the project. In addition, projects must incorporate at least two measures to limit the impacts of flooding. First, projects may elevate foundations at least 4 feet (1.2 meters) above sea level rise projections. Elevating building foundations significantly reduces flood damage and helps structures stay secure in the face of rising waters.

Projects may ensure that backup power systems are in place to maintain critical functions, such as communications, security, and fire safety systems, during projecting flooding events.

Developing integrated drainage systems to efficiently manage projected excess flooding is another solution. These systems must be designed to work holistically with the landscape and built environment to ensure that floodwaters are managed in ways that minimize damage. An integrated drainage system can combine both traditional and innovative solutions, such as upgrading or expanding existing stormwater systems (e.g., gutters, drains, and culverts) to accommodate larger

29. American Society of Civil Engineers, *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*, ASCE/SEI 7-22 (Reston: VA, ASCE, 2022), https://www.asce.org/publications-and-news/asce-7.
30. U.S. Department of Energy, "Landscaping for Windbreaks," Energy Saver, n.d., https://www.energy.gov/energysaver/landscaping-windbreaks.

volumes of water. Other measures include installing permeable pavements and porous materials, as well as integrating sump pumps in basements or lower levels of buildings to prevent water accumulation.

Retrofitting existing structures is another key strategy to enhance building resilience to future flood risks. Reinforcing foundations, installing flood barriers, and incorporating water-resistant materials will protect critical equipment and prevent water infiltration.

Flood-resistant materials

Using flood-resistant materials according to FEMA standards or local equivalent, such as fasteners, connectors, and products designed to withstand moisture, minimizes damage from water intrusion by preventing rotting and corrosion.[31] The application of sealants and coatings to prevent projected water infiltration into structures is essential.

Flood barriers

Flood barriers around buildings, including permanent walls, portable barriers, or automated systems that activate when floodwaters are detected, can redirect water away from vulnerable areas, thereby protecting properties and ensuring public safety. Restoring or preserving natural landscapes such as wetlands and mangroves enhances the resilience of communities. These ecosystems act as natural buffers, provide habitat for wildlife, and contribute to biodiversity.

Green infrastructure

Designing and incorporating GI solutions is vital to manage storm surge and runoff. GI, which includes features like rain gardens, permeable pavements, green roofs, and wetlands, plays a crucial role in mitigating these risks.

Community engagement

The project team may engage in community-level planning, partnerships, and/or design workshops to coordinate flood mitigation efforts and address the needs of populations vulnerable to projected flooding. Involving community members in the planning process allows the team to gain valuable insights into local conditions, vulnerabilities, and concerns, and leads to more effective and inclusive solutions. Forming partnerships with local organizations, government agencies, and stakeholders further strengthens these efforts and fosters a shared commitment to enhance flood resilience and safeguard the well-being of all community members.

TSUNAMIS

Tsunamis can cause severe damage, especially in coastal areas where the risks are highest. They can lead to soil erosion; undermine foundations; result in loss of life and mass injuries; and damage or destroy homes, businesses, ports, harbors, cultural resources, and critical infrastructure and facilities. Tsunamis can also overwhelm critical services (e.g., water and electricity), contaminate land with saltwater, and cause long-term damage to communities, ecosystems, and agricultural areas.

31. FEMA, "Flood Damage-Resistant Materials Requirements for Buildings Located in Special Flood Hazard Areas in accordance with the National Flood Insurance Program," NFIP Technical Bulletin 2, January 2025, https://www.fema.gov/sites/default/files/documents/fema_tb_2_flood_damage-resistant_materials_requirements_01-22-2025.pdf.

TsunamiReady guidelines

Projects must integrate site planning strategies described in *Designing for Tsunamis* (National Tsunami Hazard Mitigation Program[32]) or local equivalent. Projects must also incorporate elements from the TsunamiReady guidelines, or local equivalent, including installing danger area and evacuation route signage and Public Alert-certified NOAA Weather Radio (NWR) receivers in critical facilities and public venues.[33] Signage must be implemented according to state and local policies and as determined to be appropriate by local authorities.

WILDFIRES

Designing, constructing, and maintaining sites and structures, in compliance with the *SITES v2 Rating System Credit 4.11: Reduce the risk of catastrophic wildfire*,[34] or local equivalent, reduces the risk of catastrophic wildfires on-site and in surrounding landscapes. The project team must also implement wildfire-resistant techniques referenced in the *NWCG Standards for Mitigation 2023*,[35] or local equivalent, and apply the Zone Concept as outlined in the *Firewise Landscaping Checklist*, which is Appendix E of "Safer from the Start: A Guide to Firewise-Friendly Developments."[36]

The *NWGG Standards for Mitigation 2023* specify proper debris disposal and fire safety protocols for equipment. Proper debris disposal helps eliminate flammable material, while fire safety measures for equipment reduce the risk of accidental ignition during construction or maintenance activities. These integrated strategies enhance the site's resilience to wildfires and safeguard both the environment and the built structures.

Projects in wildfire areas must take proactive wildfire management measures, including strategies for managing vegetative biomass, dead plant materials, and fuel loads to safe levels. Clearing flammable vegetation and other fuel sources within a specific distance to create buffer zones around structures reduces wildfire risks. Conducting prescribed burns or other fuel management techniques at frequencies and intensities similar to the natural fire regime for the ecosystem is essential. This proactive approach limits the spread and intensity of wildfires, protecting both the built environment and the surrounding ecosystem.

WINTER STORMS

Winter storms feature heavy snowfall, blowing snow, cold temperatures, strong winds, blizzards, and ice storms. Winter precipitation falling as freezing rain or drizzle can lead to significant ice accumulation, which may cause considerable damage, especially when accompanied by high winds. Heavy snow or ice may damage plants, break branches, and disrupt growth, which can affect landscaping. Snow and ice accumulation can create dangerous conditions for pedestrians and vehicles and increase the risk of accidents and injuries. Projects in areas prone to winter storms must implement at least two of the indicated strategies.

32. National Weather Service, "National Tsunami Hazard Mitigation Program," last accessed April 2, 2025, https://www.weather.gov/nthmp/.
33. National Weather Service, "TSUNAMIREADY® Guidelines," 2015, https://www.weather.gov/media/tsunamiready/resources/2015TRguidelines.pdf.
34. Sustainable Sites Initiative, last accessed April 2, 2025, https://www.sustainablesites.org/.
35. National Wildfire Coordinating Group (NWCG), last accessed April 2, 2025, https://www.nwcg.gov/.
36. Firewise Landscapes, "Safer from the Start."

Adequate ingress/egress

Providing safe access and egress for vehicles and snow removal equipment during winter storms is crucial. Conduct regular maintenance and inspect access points (e.g., entrances, sidewalks, and roads) to ensure they are clear of snow and ice for vehicles and snow removal equipment. Safe walking surfaces in exterior parking areas prevent hazardous walking conditions and related injuries during winter. Heated sidewalks powered by renewable energy sources further enhance safety by preventing snow and ice buildup.

Snow removal plan and safe walking surfaces

Establish protocols before storms hit by developing a snow removal plan. Effective planning enhances safety and minimizes disruption to daily activities and infrastructure. Using appropriate road materials significantly improves snow removal efficiency. Materials that improve traction, such as sand, reduce the risk of accidents during winter storms. Designated areas for accumulated plowed snow and roof snow should be strategically located to prevent the obstruction of roadways, sidewalks, and emergency access points and limit damage to trees and people. Schedule regular inspections during winter months, provide guidelines for safely removing snow, and identify qualified personnel or contractors to conduct the work.

Landscaping considerations

Projects may include adapted plants and native species capable of withstanding significant weight, such as heavy snow loads, which may vary by location.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Identification of the two hazards the site and site structures are designed and constructed to meet.
	Drought	All	Confirmation of whether the project includes any created water features.
		Projects with water features	Evidence that the makeup water for the created water features complies with *SITES C3.4* (e.g., SITES certification and scorecard or evidence that 50% of annual makeup water for site water features comes from nonpotable water sources or that site water features only require a total of 10,000 gallons or fewer, 37,854.12 liters or fewer, of potable water annually).
	Extreme heat	All	Evidence of the two qualifying extreme heat resilience elements included in the project (e.g., contract documents, massing and orientation studies of the building, map to emergency cooling station, product information from paving manufacturers or SR values or open-grid pavement permeability).
	Flooding	All	Evidence that the project is certified under a qualifying flood-resilient design standard (e.g., certificate/stamped drawings) and/or evidence that critical utilities meet the design criteria and/or confirm that all structural materials, finish materials, and connectors used below DFE are flood resistant (e.g., contract documents).
	Hail	All	Evidence that the project site and site structures are designed and constructed according to *FORTIFIED Commercial High Wind and Hail Specific Design Requirements for Hail* (e.g., evidence of the *FORTIFIED* + Hail certification).

(*continued*)

SS

(continued)

Project Types	Options	Paths	Documentation
All	Hurricanes and high winds	All	Evidence that the project site and site structures are designed and constructed according to *FORTIFIED Commercial Wind* standards (e.g., evidence of the *FORTIFIED* certification).
	Sea level rise	All	Evidence of the two qualifying sea level rise resilience elements included in the project (e.g., contract documents, product information from manufacturers; meeting notes documenting engagement in relevant community-level planning, partnerships, and/or design workshops).
	Tsunami	All	Evidence the project has designated a tsunami danger area on-site and installed evacuation route signage and public Alert-notified NOAA Weather Radio (NWR) receivers.
	Wildfire	All	Evidence that the site and site structures are designed and constructed in compliance with *SITES v2 Rating System Credit 4.11: Reduce the risk of catastrophic wildfire* (e.g., SITES certification and scorecard or contract documents demonstrating landscaping design practices in alignment with Firewise, "Safer from the Start: A Guide to Firewise-Friendly Developments," Appendix E).
	Winter storms	All	Evidence of the two qualifying winter storm resilience elements included in the project (e.g., contract documents, snow removal operations and maintenance plan, product documentation from manufacturers, planting information).

REFERENCED STANDARDS

- SITES v2 Rating System (https://www.sustainablesites.org)
- ASCE 24 (https://www.asce.org)
- FEMA 543 (https://www.fema.gov)
- Intergovernmental Panel on Climate Change (IPCC) (https://www.ipcc.ch)
- FORTIFIED (Commercial High Wind and Hail Specific Design Requirements) (https://fortifiedhome.org)
- ASCE/SEI 7 (https://www.asce.org)
- Tsunami-Ready Guidelines (https://www.weather.gov/tsunamiready/guidelines)
- NWCG Standards for Mitigation 2023 (https://www.nwcg.gov)

FURTHER EXPLANATION

EXTREME HEAT

- Extreme heat is a 48-hour period exceeding the 95th percentile of historic average maximum summer temperature for the area.
- Emergency cooling station: A temporary or permanent outdoor shelter to protect people from extreme heat with shade and passive or active cooling features (e.g., evaporative cooling features, breezeways, fans).
- Emergency cooling center: A locally designated, air-conditioned public facility open during heat events (e.g., library, community center).

- Shaded exterior spaces must be large enough to support outdoor use (e.g., seating, gathering).
  - Shaded external spaces shall accommodate seating for a percentage of full-time occupants. That percentage shall match the typical occupancy rate for the building. For example, if the project is a commercial office building with 300 full-time employees (FTEs) and the typical occupancy rate is 60%, that shaded external space shall accommodate 180 people.
- Evaporative cooling features (e.g., splash pads, misting) do not require direct user access. Note that passive cooling is acceptable.
  - High solar reflectance (SR) and open-grid pavement systems need to be adjacent to buildings or shaded exterior spaces. They should be contiguous, and a minimum width of 10 feet (3.05 meters) or to the nearest public sidewalk, whichever is less.

Flooding

- Flood barriers are structures designed to prevent water infiltration from flooding into a protected area.
  - Flood barriers must be permanent installations to meet this requirement.
- Integrated drainage systems are a network of both green and conventional infrastructure (e.g., bioswales and pipes). Alternate terms include blue-green infrastructure, integrated urban drainage (IUD), sustainable urban drainage systems (SUDS), and integrated water management (IWM).
- Each sub-bullet under Critical Utilities counts as a separate eligible strategy.
- Design flood elevation (DFE) shall be no less than the current jurisdictional base flood elevation plus additional flood elevation anticipated over the project design service life and with the selected emissions scenario.
- Installation of a site-wide Integrated Emergency Alert System tied into local and regional jurisdictionally managed emergency management systems will enhance operational resilience for the project site.

HURRICANES AND HIGH WINDS

- High winds are defined as sustained wind speeds of 40 miles per hour (18 meters per second) or greater, and/or wind gusts of 58 miles per hour or greater.
- Windbreaks are defined as single or multiple rows of vegetation or landforms placed perpendicular to wind direction to reduce wind effects. Determine the design that fits the needs of the project best:
  - Dense windbreaks equal greater wind speed reduction.
  - Sparse windbreaks equal broader area of protection.
- Use the latest version of *ASCE/SEI 7-22* with errata and supplement.
  - Earlier versions may require USGBC approval.
  - Use a locally recognized equivalent outside the U.S.
- Use docking stations (a.k.a. generator connection box).
  - Provides pre-installed tie-in point for portable/rental generators.
  - Allows safe, fast backup power deployment without major electrical work.

SUSTAINABLE SITES CREDIT

Heat Island Reduction

SSc5
New Construction (1–2 points)
Core and Shell (1–2 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To mitigate disparate impacts on microclimates and habitats caused by heat islands and extreme heat events.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–2**
Option 1. Nonroof and Roof AND/OR	1
Option 2. Parking Under Cover AND/OR	1
Option 3. Tree Equity	1

Choose one of the following options:

OPTION 1. NONROOF AND ROOF (1 POINT)

Meet the following criteria for the nonroof and roof weighted average approach:

Equation 1. Weighted nonroof and roof calculation

$$\frac{\text{Area of nonroof measures}}{0.5} + \frac{\text{Area of high-reflectance roof}}{0.75} + \frac{\text{Area of vegetated roof}}{0.75} \geq \text{Total site paved area} + \text{Total roof area}$$

Use any combination of nonroof, high-reflectance roof, and vegetative roof strategies so that the weighted sum of site design strategies is greater than or equal to the sum of the total pavement and roof areas. Each surface may only be counted once, even if it is addressed through multiple strategies.

Nonroof measures

- Shade over pavement areas, measured in plain view at noon, with existing or new plants, assuming 10-year canopy width, or vegetated structures. Planting or vegetated structures must be in place at the time of occupancy permit.
- Structures covered by energy generation systems, such as solar thermal collectors, photovoltaics (PVs), and wind turbines.
- Architectural devices or structures. If the device or structure is a roof, it shall have an aged SR value of at least 0.28 as measured in accordance with *ANSI/CRRC S100*. If the device or structure is not a roof, or if aged SR information is not available, at installation it must have an initial SR of at least 0.33 as measured in accordance with *ANSI/CRRC S100.*
- Paving materials with an initial SR value of at least 0.33.
- An open-grid pavement system (at least 50% unbound).

High-reflectance roof

Use roofing materials that have an aged solar reflectance index (SRI) value equal to or greater than the values in Table 1. If aged SRI is not available, the roofing material shall have an initial SRI equal to or greater than the values in Table 1.

TABLE 1. Minimum SRI Value, By Roof Slope

	Slope	Initial SRI	Aged SRI
Low-sloped roof	≤ 2:12	82	64
Steep-sloped roof	> 2:12	39	32

A roof area that consists of functional, usable spaces (e.g., helipads, recreation courts, swimming pools, and similar amenity areas) may meet the requirements of nonroof measures. Applicable roof area excludes roof area covered by mechanical equipment, solar energy panels, skylights, and any other appurtenances.

Vegetated roof

Install a vegetated roof using native or adapted plant species.

AND/OR

OPTION 2. PARKING UNDER COVER (1 POINT)

Place 100% of parking spaces under cover. Any roof used to shade or cover parking must meet at least one of the following criteria:

- Have an aged SRI of at least 32. If aged value information is not available, use materials with an initial SRI of at least 39 at installation.
- Be a vegetated roof.
- Be covered by energy generation systems, such as solar thermal collectors, photovoltaics, and wind turbines.

The credit calculations must include all existing and new off-street parking spaces that are subsidized, leased, or owned by the project, including parking that is outside the project boundary but is used by the project. On-street parking in public rights-of-way is excluded from these calculations.

AND/OR

OPTION 3. TREE EQUITY (1 POINT)

For all U.S. projects only, evaluate the American Forests Tree Equity score for the site location. For projects in areas ranked "high priority" and "highest priority," use the results of the evaluation to inform an increase in on-site canopy cover from the existing condition.

For international projects, refer to *IPp2: Human Impact Assessment* and evaluate the tree cover on-site and in the surrounding community, either by using a local tree census or conducting a site assessment. Analyze the project's local community composition to identify any neighboring underserved and/or disadvantaged populations with lower tree canopy presence. Use the results of the evaluation to inform an increase in on-site canopy cover from the existing condition to provide shade to neighboring underserved and/or disadvantaged areas. Projects with no exterior work are exempt from this requirement.

REQUIREMENTS EXPLAINED

This credit encourages the use of strategies that minimize a project's overall contribution to the heat island effect. Option 1 addresses nonroof and roof measures including reducing hardscapes, incorporating high SRI or high SR materials, increasing tree cover, and implementing vegetation across the site. SRI measures a roofing material's ability to reject solar heat, whereas SR measures the solar heat rejection of hardscape materials.

Unshaded parking lots become mini-heat islands by absorbing the sun's warmth and radiating heat. Option 2 encourages project teams to place parking spaces under cover that have low SRI roofing materials, solar canopies, or are located underground or within a building.

When considering Option 2, projects must account for any existing or new off-street parking that the project leases or owns. If these spaces are not in direct control of the project team, coordinate with additional stakeholders to confirm covered parking is a viable option for these spaces.

OPTION 1. NONROOF AND ROOF

Implementing both nonroof and roof measures is essential to mitigate heat island effects, with each providing significant environmental benefits to the building, site, and occupants. Using a combination of nonroof and roof strategies creates a more sustainable and resilient urban environment.

Nonroof measures

Nonroof measures include shading with new or existing plant material or shading structures, specifying high-reflectance paving and open-grid paving, and including vegetated planters across the site. Using a variety of plant species allows for biodiversity, while tree canopies and shading structures create areas of respite on a hot, sunny day.

Roof measures

Roof measures, including the use of vegetated and high-reflectance roofs, can improve energy efficiency and thermal comfort and can reduce carbon emissions associated with building energy use.

Projects pursuing this option must consider the slope of the roof and both the initial and aged SRI values when selecting compliant materials. For low-sloped roofs, the roofing material must meet the minimum value of an initial SRI of 82 or aged value of 64. For steep-sloped roofs, the minimum required values are an initial SRI of 39 or aged SRI of 32. These specific SRI values are indicative of a material that performs well in reducing heat absorption both when it is new and after it has aged.

Vegetated roof

When incorporating a vegetated roof into the design, projects must prioritize the use of native or adapted plant species. These species are well-adapted to the local environment and typically require less maintenance and support local biodiversity.[37]

Nonroof and roof measures calculation

Projects pursuing this option must confirm compliance with the combined roof and nonroof strategies by calculating the total area associated with each measure (e.g., nonroof, high-reflectance roof, and vegetated roof areas) and dividing it by its weighted value. The total value of implemented strategies must meet or exceed the total site paving area plus the total roof area within the project's boundary.

Equation 1. Weighted nonroof or roof calculation

$$\frac{\text{Area of nonroof measures}}{0.5} + \frac{\text{Area of high-reflectance roof}}{0.75} + \frac{\text{Area of vegetated roof}}{0.75} \geq \text{Total site paved area} + \text{Total roof area}$$

Projects should evaluate and achieve compliance using Equation 1. If the project does not achieve the standard nonroof or roof calculations, teams may use an SRI and SR weighted average approach to calculate compliance. The weighted nonroof or roof equation weighs the SR and SRI for total hardscape and roof area, showing its overall consequence on heat island effect. This equation is useful for projects that have multiple roof angles, and nonroof or roof materials that fall both above and below the required SR and SRI values.

37. Audubon, "Why Native Plants Matter," n.d., https://www.audubon.org/content/why-native-plants-matter#:~:text=Because%20native%20plants%20are%20adapted%20to%20local%20environmental,and%20perhaps%20the%20most%20valuable%20natural%20resource%2C%20water.

Equation 2. Weighted nonroof or roof calculation

$$\left(\frac{\text{Area of high reflectance nonroof A} \times \dfrac{\text{SR of high reflectance nonroof A}}{\text{Required SR}}}{0.5}\right)_1 + \frac{\text{Area of other nonroof measures}}{0.5} + \left(\frac{\text{Area of high reflectance roof} \times \dfrac{\text{SRI of high reflectance roof A}}{\text{Required SR}}}{0.75}\right)_2 + \frac{(\text{Area of vegetated roof})}{0.75} \geq \text{Total site paving area} + \text{Total roof area}$$

where:

1. Summed for all high-reflectance nonroof areas.
2. Summed for all high-reflectance roof areas.

OPTION 2. PARKING UNDER COVER

Unshaded parking areas have become small urban heat islands because most outdoor parking lots use dark colored impervious surfaces, such as asphalt or concrete pavements with low SR value. When parking is under cover, it significantly reduces the exposed hardscape area and minimizes the heat island effect. Covered parking strategies include placing parking underground, underdeck, under roof, or under the building.

Projects pursuing this option must place 100% of the total vehicle parking spaces under cover. Include all new parking in the calculation. Factor existing and new off-street parking spaces that are leased or owned by the project into the calculations.

OPTION 3. TREE EQUITY

The most effective measure in mitigating the climate change effects generated by heat islands is to minimize the hardscape on the project site and increase on-site tree canopy. Given the important role trees play in slowing climate change, American Forests created the Tree Equity Score to focus on addressing the inequity in tree coverage in neighborhoods in the U.S. and the UK.[38] A "high priority" or "highest priority" area identified in the Tree Equity Score map means an area where the community needs trees the most. Incorporating site landscaping can help contribute to a greener and cooler environment.

All projects should aim to increase the on-site tree canopy. Projects identified as "high priority" or "highest priority" must further evaluate the planned tree canopy and increase on-site tree canopy coverage.

Projects in other countries, or in areas without a Tree Equity Score, can use platforms such as Global Forest Watch to help understand tree coverage in the project neighborhood.[39] These projects are

38. American Forests, "Tree Equity Score," accessed June 10, 2025, https://www.treeequityscore.org/.
39. Global Forest Watch, "Forest Monitoring Designed for Action," last accessed April 2, 2025, https://www.globalforestwatch.org/.

required to refer to *IPp2: Human Impact Assessment* during evaluation and should consider development in areas where the local tree census or site assessment identifies neighboring underserved and/or disadvantaged communities with lower tree canopy. Projects must evaluate the tree cover on-site and in nearby areas and use this evaluation to inform an increase in on-site tree coverage.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Nonroof and Roof	All	Heat Island Reduction calculator.
			Site plan or annotated aerial view image identifying nonroof measures, roof areas including high reflectance and vegetated areas, and paving areas.
			Evidence of the SR value for each nonroofing material (or documentation for a similar material), architectural device, or structure.
			The project's roof plan(s) identifying all vegetated roof area, helipads, amenity areas, mechanical equipment, solar energy panels, skylights, other roof appurtenances, roofing materials with area measurements, and roof slope(s).
			Evidence of each roofing materials' SRI value or documentation for a similar material.
	Option 2. Parking under cover	All	Evidence of the location of all existing and new off-street parking spaces that are subsidized, leased, or owned by the project, including parking that is outside the project boundary but is used by the project (e.g., contract documents).
			Evidence that any roof used to shade or cover parking meets the required criteria (e.g., roofing material product information from the manufacturer highlighting SRI, photographs or contract documents depicting the vegetated roof or energy generation systems).
	Option 3. Tree equity	All	Areas of existing condition and design/post-construction tree canopy cover on-site.
			Evidence of increase in tree canopy cover (e.g., pre-design site survey and contract landscape plan).
		Projects in the U.S.	Evidence of the project's Tree Equity Score and ranking (e.g., a screenshot).
		International projects	Analysis of the project's local community composition and identification of any neighboring underserved and/or disadvantaged populations with lower tree canopy presence (e.g., census data from *IPp2: Human Impact Assessment*).
			Evidence of the tree canopy presence in the surrounding community (e.g., a local tree census or a surrounding community site assessment), and identification of any neighboring underserved and/or disadvantaged populations with lower tree canopy presence.

REFERENCED STANDARDS

- ANSI/CRRC S100, "Standard Test Methods for Determining Radiative Properties of Material" (https://coolroofs.org/documents/ANSI-CRRC-S100-2021_Final_Archived.pdf)
- ASTM E1980, "Standard Practice for Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces" (https://astm.org/e1980-11r19.html)

- Cool Roof Ratings Council, Rated Products Directory (https://coolroofs.org/directory)
- U.S. EPA, "What Are Heat Islands" (https://epa.gov/heatislands/learn-about-heat-islands)

FURTHER EXPLANATION

OPTION 1. NONROOF AND ROOF

Nonroof Measures

- Increasing the landscaped portion of the site is the most effective strategy for reducing overall heat island effects and supports achievement of other credits.
- The following typical SR values can be used for standard nonroof materials in lieu of project-specific testing data:

TABLE 2. Typical Solar Reflectance Values for Standard Nonroof Materials

Material	Initial Solar Reflectance	Three-Year Aged Solar Reflectance*
Gray cement concrete	0.26	0.18
White cement concrete	0.70	0.35
Asphalt concrete	0.05	0.10

*NOTE: Three-year aged SR values are based on no cleaning.

ROOF MEASURES

- Incorporate high-reflectance roofing materials and vegetated roof systems in compliance with the credit requirements.
- The top level of multilevel parking structures is considered nonroof surface if it has parking spaces but roof area if it has no parking spaces.
- Project teams can contact manufacturers directly and ask for solar reflectance index (SRI) information. If manufacturers do not provide this information, the project team can identify a similar material from the Cool Roof Rating Council standard for comparison to show that the project's material meets the intent of the credit.

OPTION 2. PARKING UNDER COVER

- The total parking capacity must include all parking made available to project occupants, including parking that is outside the project boundary but is used by the project.
- Parking located below a building or under energy generation systems is compliant with the credit requirements. The roof or deck area located to shade and cover parking must meet the minimum SRI values.
- If parking spaces are shared with two or more buildings/spaces, only the parking allocated to the project must be included in the total parking capacity. It is up to the project team to determine the project's allocation based on the project-specific circumstances and to provide rationale for the parking distribution, if necessary.
- Do not include on-street public parking.

OPTION 3. TREE EQUITY

- For U.S. projects, site locations with Tree Equity Scores other than "high priority" or "highest priority" are not eligible to pursue this option.
- For international projects, the *IPp2: Human Impact Assessment* can be used to inform tree canopy increase on the project site. Consider how on-site increase in tree canopy contributes to the neighboring community:
  - Public right-of-way areas can be used to provide long-term shade benefits.
  - Trees can provide cooling to adjacent areas hundreds of feet away, offering off-site benefits.
  - Consider planting in outdoor areas where building occupants or community members may congregate.
- Projects with no exterior work that increase tree plantings are exempt from using a site assessment to inform on-site canopy cover.

SUSTAINABLE SITES CREDIT

Light Pollution Reduction

SSc6
New Construction (1 point)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To increase night sky access, improve nighttime visibility, and reduce the consequences of development for wildlife and people.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1**
Uplight AND	1
Light Trespass AND	
Internally Illuminated Exterior Signage	

Meet the following uplight, light trespass, and internally illuminated exterior signage requirements for exterior luminaires located inside the project boundary.

UPLIGHT

Do not exceed the following uplight ratings, based on the specific light source installed in the luminaire, as defined in Table 1.

TABLE 1. Maximum Uplight Ratings for Luminaires

Model Lighting Ordinance (MLO) Lighting Zone	Luminaire Uplight Rating
LZ0	U0
LZ1	U0
LZ2	U2

TABLE 1. Maximum Uplight Ratings for Luminaires (*continued*)

Model Lighting Ordinance (MLO) Lighting Zone	Luminaire Uplight Rating
LZ3	U3
LZ4	U4

AND

LIGHT TRESPASS

Do not exceed the following luminaire backlight and glare ratings (based on the specific light source installed in the luminaire), as defined in *IES TM-15-11*, Addendum A, based on the mounting location and distance from the lighting boundary.

TABLE 2. Maximum Backlight and Glare Ratings

	MLO Lighting Zone				
Luminaire mounting	LZ0	LZ1	LZ2	LZ3	LZ4
	Allowed Backlight Ratings				
> 2 mounting heights from lighting boundary	B1	B3	B4	B5	B5
1 to 2 mounting heights from lighting boundary and properly oriented	B1	B2	B3	B4	B4
0.5 to 1 mounting height to lighting boundary and properly oriented	B0	B1	B2	B3	B3
< 0.5 mounting height to lighting boundary and properly oriented	B0	B0	B0	B1	B2
	Allowed Glare Ratings				
Building mounted > 2 mounting heights from any lighting boundary	G0	G1	G2	G3	G4
Building mounted = 1–2 mounting heights from any lighting boundary	G0	G0	G1	G1	G2
Building mounted = 0.5–1 mounting heights from any lighting boundary	G0	G0	G0	G1	G1
Building mounted < 0.5 mounting heights from any lighting boundary	G0	G0	G0	G0	G1
All other luminaires	G0	G1	G2	G3	G4

AND

SS

INTERNALLY ILLUMINATED EXTERIOR SIGNAGE

Do not exceed the maximum luminance level of internally illuminated signage during nighttime hours according to Table 3.

TABLE 3. Maximum Sign Luminance

MLO Lighting Zone	Signage Light Output
LZ0	50 cd/m^2
LZ1	50 cd/m^2
LZ2	100 cd/m^2
LZ3	200 cd/m^2
LZ4	350 cd/m^2

Exemptions to uplight and light trespass requirements

The following exterior lighting is exempt from the requirements, provided it is controlled separately from the nonexempt lighting:

- Specialized signal, directional, and marker lighting for transportation.
- Lighting used solely for façade and landscape lighting in MLO lighting zones 3 and 4 and that is automatically turned off from midnight until 6 a.m.
- Lighting for theatrical purposes for stage, film, and video performances.
- Government-mandated roadway lighting.
- Lighting for hospital emergency departments, including associated helipads.
- Lighting for the national flag in lighting zones 2, 3, or 4.
- Internally illuminated exterior signage.

REQUIREMENTS EXPLAINED

This credit rewards projects that meet the requirements of uplight, light trespass (backlight and glare), and internally illuminated exterior signage for exterior luminaires within the project boundary. Minimizing light pollution preserves the visibility of night skies, protects wildlife, and improves human health. Strategies include the use of light shielding, where fixtures direct light downward and reduce glare and light trespass. Using dimmers, motion sensors, and timers ensure lights are only on when needed.

LUMINAIRES

This credit requires an assessment of all new and existing exterior luminaires within the project boundary, including any building-mounted fixtures. When performing calculations, the photometric characteristics of each luminaire must reflect the design conditions, including mounting orientation and tilt.

DETERMINING THE LIGHTING ZONE

Classify projects under a single lighting zone, as identified in the *Illuminating Engineering Society and DarkSky International (IES/IDA) Model Lighting Ordinance (MLO) User Guide*.[40] Teams shall determine the project's lighting zone based on the property classification, at the time construction begins or in accordance with MLO guidance.

DETERMINING THE LIGHTING BOUNDARY

Assess credit compliance using a lighting boundary. A lighting boundary ensures that projects assess the light trespass from the lighting installed within the project and determine the impacts to the surrounding areas.

40. Illuminating Engineering Society and DarkSky International (IES/IDA), "Model Lighting Ordinance (MLO) User Guide," https://store.ies.org/product/ida-ies-mlo-11-model-lighting-ordinance-mlo-with-users-guide/?v=0b3b97fa6688.

Additional considerations for defining the lighting boundary

Projects must consider the following when developing the lighting boundary for public areas, roadways, and campus properties:

- **Lighting boundary**. The lighting boundary is typically based on the project's property lines and surrounding areas but may not directly align with the project's LEED boundary. Under certain conditions, the lighting boundary may extend beyond the property line.
- **Public areas**. When a public area, including but not limited to a walkway, bikeway, plaza, or parking lot, abuts the property line, the property owner may move the lighting boundary to 5 feet (1.5 meters) beyond the property line.
- **Roadways and transit corridors**. When a property line abuts a public street, alley, or transit corridor, the lighting boundary may move to the center line of that street, alley, or corridor.
- **Campus properties**. For buildings on campuses or shared properties, the lighting boundary can use the campus property line to meet light trespass requirements. Additional properties owned by the same entity must be contiguous on the property. They cannot use off-site properties.

UPLIGHT REQUIREMENTS

Avoiding uplight is an effective strategy to reduce light pollution. Uplighting occurs from site fixtures that direct light upward. Prevent excess light pollution for site luminaires by selecting fully shielded fixtures that direct light downward.

Uplight Ratings

An uplight rating of U0 indicates that the fixture emits zero light upward into the night sky and meets credit intent. This is particularly important for projects aiming to minimize light pollution and comply with Dark Sky standards. A U0 rating ensures that all light directs downward and prevents it from contributing to sky glow.

For projects located in MLO Lighting Zones LZ0 or LZ1, teams must use luminaires with an uplight rating of U0.

For projects in MLO Lighting Zones LZ2, LZ3, or LZ4, luminaire uplight ratings cannot exceed U2, U3, and U4, respectively.

LIGHT TRESPASS REQUIREMENTS

Light trespass is a form of light pollution where unwanted artificial light spills over into areas where it is not intended or needed, often causing disturbances. This can happen when outdoor lighting, such as floodlights or streetlights, illuminate neighboring properties or shine into other property windows.

Fixture Location and Mounting Heights

A common reason for light trespass typically includes poorly shielded lighting located too close to property lines. To minimize lighting trespass, examine luminaire spillage along with the location of the exterior luminaire. This requires a review of the location and mounting height for each luminaire.

Identify the location of all exterior lights on a site lighting plan and measure the horizontal distance from the lighting boundary. Then, determine the mounting height of each luminaire. The mounting height of the exterior luminaire is the vertical distance from the reference plane (i.e., the ground surface).

Determining the Maximum Backlight and Glare Requirements

Table 2 defines the maximum allowed backlight and glare ratings per lighting zone. The team determines backlight and glare ratings per luminaire by using the mounting height and the distance from the lighting boundary. The closer the luminaire is to the boundary, the stricter the requirements.

EXEMPTIONS FOR UPLIGHT AND LIGHT TRESPASS

Specific fixtures are exempt from the light trespass requirements due to their critical functions and unique needs. Identify fixtures that meet these exemptions. Control any exempt fixtures separately from nonexempt fixtures.

Transportation and roadways

Specialized signal, directional, and marker lighting for transportation fixtures ensure safety and proper navigation for vehicles, aircraft, and ships. Government-mandated roadway lighting provides necessary illumination for public safety on roads and highways.

Theatrical lighting

Lighting for theatrical purposes is essential for stage, film, and video performances, and precise lighting is crucial for production quality.

Healthcare

Lighting for hospital emergency departments and helipads are critical for emergency services and ensure visibility for medical personnel and aircraft.

National and state flag lighting

Lighting for the national flag is symbolic and often requires illumination at night, especially in higher lighting zones (2, 3, or 4). When state flags are flown with federal or national flags, they are exempt lighting.

INTERNALLY ILLUMINATED EXTERIOR SIGNAGE

Exterior signage can be a significant source of light pollution and contribute to issues like light trespass, skyglow, and glare. Table 3 provides maximum nighttime luminance levels for any exterior signage. Using photometric calculations, verify that the signage light output does not exceed the luminance levels (cd/m^2) during nighttime hours.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	MLO Lighting Zone.
			Justification of MLO Lighting Zone classification (e.g., aerial image of site and bordering parcels/community).
			Site lighting plan showing all exterior light fixtures within the lighting boundary, measuring the distance from the furthest of each light fixture type to the lighting boundary; include a key/schedule.
			Contract documents showing all the project's internally illuminated signage details.
			Product information from the exterior lighting manufacturers, highlighting the backlight, uplight, and glare (BUG) ratings for each fixture.
			Evidence of the maximum nighttime luminance levels of any internally illuminated signage (e.g., product information from the manufacturer or test results).
			Evidence that all installed exterior lighting in MLO Lighting Zones 3 and 4 is automatically turned off when daylight is available (e.g., contract documents or commissioning reports showing the programmed settings).
			Evidence that any exterior lighting exempt from the requirements is controlled separately from the nonexempt lighting (e.g., contract documents).

REFERENCED STANDARDS

- Illuminating Engineering Society and DarkSky International (IES/IDA) Model Lighting Ordinance (MLO) User Guide (https://store.ies.org/product/ida-ies-mlo-11-model-lighting-ordinance-mlo-with-users-guide)
- Illuminating Engineering Society (IES) TM-15-11, Addendum A (https://www.ies.org/wp-content/uploads/2017/03/TM-15-11BUGRatingsAddendum.pdf)

FURTHER EXPLANATION

- Lighting fixtures must be documented based on their orientation and tilt. If a single fixture has multiple ways, each distinct orientation and tilt must be considered.
- Emergency and safety lighting fixtures are not exempt and must be included in the calculations. If no exterior lighting is installed, the credit is earned.
- If the lighting boundary is extended beyond the LEED Project Boundary, only the fixtures located within the LEED Project Boundary should be included in the calculations.

Water Efficiency

OVERVIEW

LEED v5 integrates water efficiency with new stewardship strategies and redefines water as a valuable and limited resource. The WE category encourages projects to conserve potable water or potable water supplies to safeguard ecosystems, reduce energy use, and boost resilience on-site and in the wider community.

New construction projects can design highly efficient water systems by combining LEED's proven efficiency strategies with innovative water stewardship approaches. Although global water efficiency has improved, water stress and scarcity remain pressing challenges, with approximately 2.4 billion people living in water-stressed regions as of 2020.[1] Climate change and population growth intensify these issues and underscore the need for adaptable, forward-thinking resource management plans.

The connections between efficiency and stewardship show up clearly in the whole project water use strategy (*WEc2: Enhanced Water Efficiency*). Rather than isolating individual components, this approach encourages comprehensive site-wide water consumption assessments. Originally piloted in LEED v4.1, this strategy has become a permanent feature in LEED v5.

This stewardship approach aligns with growing market interest in alternative water use such as is seen in water-limited regions like California.[2] By incorporating alternative water sources, projects can reduce reliance on potable water supplies and alleviate the strain on overburdened systems (*WEc2: Enhanced Water Efficiency*).

Decarbonization

Water efficiency can significantly reduce energy use and carbon emissions. For example, running a faucet for five minutes uses about as much energy as letting a 60-watt light bulb run for 14 hours.[3] LEED v5 advances decarbonization efforts by reducing the energy use linked to inefficiencies within

1. United Nations Sustainable Development, "Goal 6: Ensure Access to Clean Water and Sanitation for All," October 19, 2023, https://www.un.org/sustainabledevelopment/water-and-sanitation/#:~:text=Investments%20in%20infrastructure%20and%20sanitation%20facilities%3B%20protection%20and,efficiency%20is%20one%20key%20to%20reducing%20water%20stress.
2. U.S. EPA, "Water Reuse Case Study: Los Angeles County, California," January 31, 2025, https://www.epa.gov/waterreuse/water-reuse-case-study-los-angeles-county-california.
3. U.S. EPA, "Why Water Efficiency," January 17, 2017, https://19january2017snapshot.epa.gov/www3/watersense/our_water/why_water_efficiency.html.

water treatment, transportation, distribution, and heating (*WEp2: Minimum Water Efficiency, WEc1: Water Metering and Leak Detection*). Additionally, new appliances must meet high-performance requirements. This ensures that future water use hits ambitious performance targets (*WEc2: Enhanced Water Efficiency*).

Quality of life

Conserving potable water enables it to go further for broader community use. Projects can prepare for a resilient future by tracking indoor and outdoor water consumption to identify opportunities for savings (*WEc1: Water Metering and Leak Detection*).

Ecological conservation and restoration

Through water reduction and optimization strategies, projects ease the strain on ecosystems and preserve vital resources. Submetering and leak detection sensors reduce water waste from leaks or system inefficiencies. Through early leak identification, projects can avoid potential water damage and ensure conservation efforts are on track (*WEp1: Water Metering and Reporting*). Building managers and tenants can take immediate action to ensure they meet conservation goals (*WEc1: Water Metering and Leak Detection*).

Embracing the strategies in the WE category protects one of the planet's most essential resources and sets the foundation for a more resilient, sustainable, and equitable future for all.

WATER EFFICIENCY PREREQUISITE

WATER METERING AND REPORTING

WEp1
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To conserve potable water resources, support water management, and identify opportunities for additional water savings by tracking water consumption.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Water Metering	

Install or use existing permanent water meters to monitor, record, and report the total water consumption for potable and alternative water sources for the building and associated grounds. Report whole-project use for each type of water source supplied to the building and associated grounds, with the following additional provisions:

- The facility manager and/or tenant(s) must be able to access the meter data.
- Meter alternative water sources separately from municipally supplied potable water.
- Commit to sharing with USGBC the resulting whole-project water usage data at least annually. This commitment must carry forward for five years or until the building changes ownership or lessee.

The requirements may be applied to the project scope of work and exclude future tenant utility services and submeters that will be installed in the tenant scope of work.

REQUIREMENTS EXPLAINED

This prerequisite requires that projects install or use existing meters and collect water consumption data in gallons or liters from all water sources within the project boundary. This includes potable water sources and alternative water sources. Projects must also provide data access to facility managers, operations managers, tenants, and/or other appropriate people. In addition, projects must commit to reporting the total water consumption to USGBC at least annually and sharing the data for five years or until the building changes ownership or lessee.

Tracking and reviewing data monthly allow the facility manager and tenants ongoing opportunities to identify inefficiencies or anomalies in consumption and immediately address problems, such as leaks and failed valves, before larger issues or excess consumption occur.

Projects in a campus environment must only report usage from within the project boundary.

IDENTIFYING ALL WATER SOURCES AND WATER END USES

Project teams must identify all water end uses in the building and on the project site. Teams should continuously analyze water consumption from each source to identify potential leaks or operational issues. Common end uses include plumbing fixtures, cooling towers, laundry facilities, dishwashers, indoor and outdoor water features, irrigation, and other building and site processes.

For each end use, identify the water source. Potable water sources include public water supply, on-site wells, and on-site potable water treatment systems. Alternative water sources include gray water, rainwater, recycled water, and reclaimed water. When using alternative water sources, meter them separately from municipally supplied potable water.

METERS AND TYPES OF METERS

Install meters only for systems within the scope of work. The prerequisite does not require meters for future tenant utility services and/or submeters not installed within the new construction scope of work.

Specify permanent meter(s) that provide water consumption data in gallons or liters. A utility-owned meter that provides the required data meets the prerequisite requirements. Utility providers often read and bill total water consumption monthly.

Tracking and reporting

Demonstrate that the key person(s) responsible for tracking and reporting water consumption data can access the utility-owned meter. If the location is not accessible, providing access to monthly utility bill(s) achieves the same goal. If projects cannot demonstrate access through direct, visual readings or monthly utility bills, teams must install additional meter(s) to meet the prerequisite.

Multiple sources of potable or alternative water in a project boundary

Teams must identify whether the project boundary uses multiple sources of potable water or alternative water. A single meter per water source can meet the requirement if the design allows for the proper placement of the meter. Projects with campus-level irrigation metering must submit an engineering calculation that accurately reflects the water consumption of the project boundary's landscaped area. Projects can also prorate irrigation data for the project boundary from the campus meter.

Additional considerations

Projects that pursue *WEc1: Water Metering and Leak Detection*, Option 1. Submeters, should consider strategies that meet both the prerequisite and the credit. For example, using a single utility-owned meter for the project's total potable water use meets the prerequisite; however, it does not comply with the credit requirements. For *WEc1: Water Metering and Leak Detection*, Option 1 Submeters, teams must install additional meters to capture each potable water end use as outlined in the credit.

COMMITMENT TO SHARING DATA WITH USGBC

USGBC aims to collect data on water usage from all LEED buildings. Having this data allows USGBC to identify similarities between high-performing projects and recommend solutions with proven results.

Projects must commit to sharing whole-project water usage data with USGBC annually for at least five years, or until the building changes ownership or lessee. Share data using a USGBC-approved data template or an approved third-party data source, such as ENERGY STAR® Portfolio Manager.[4]

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Documentation showing that all water meters are permanently installed.
			Confirmation that the facility manager and/or tenant(s) can access the meter data.
			Confirmation that alternative water sources are metered separately from municipally supplied potable water.
			Commitment from the project owner to share with USGBC the resulting whole-project water usage data at least annually.
			Method of data sharing.

REFERENCED STANDARDS

- None

4. ENERGY STAR, "ENERGY STAR PortfolioManager," accessed March 21, 2025, https://portfoliomanager.energystar.gov/pm/login?testEnv=false.

FURTHER EXPLANATION

METERS AND TYPES OF METERS

- Metering and data collection are the same for Core and Shell projects.
  - Data may be collected from the spaces that the LEED project did not fit out as part of the project scope. Water data sharing must continue following fit out and occupancy of the project unless there is a change of ownership or lessee.
  - Core and Shell projects without fixtures and fittings or water-using systems are exempt from the prerequisite.
- Residential projects must either use a whole-building meter or, provided all potable water uses are accounted for, aggregate data from submeters for each unit and common spaces.
- Additions to existing buildings must be metered separately.

MULTIPLE SOURCES OF POTABLE OR ALTERNATIVE WATER IN A PROJECT BOUNDARY

- If the project is not served by a public water supply or if the project uses multiple sources of potable water, two or more meters may be required. A single meter installed downstream of multiple potable water supply systems may be used if it is upstream of all project water end uses. In some cases, projects may elect to use multiple meters to gain additional information on water use.
- If a building or site uses multiple potable water sources, additional meters may be necessary to record the data accurately. For example, a project requires two meters if it has potable water from a public water supply and an on-site well.
- If a building or site uses multiple sources of alternative water, additional meters may be necessary to capture the required data. For example, if a project uses a municipally supplied reclaimed water and harvested rainwater system for irrigation, a separate meter must be installed for each system.

Minimum Water Efficiency

WEp2
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To reduce potable water consumption and the associated energy consumption and carbon emissions required to treat and distribute water, and to preserve potable water resources through an efficiency-first approach.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Minimum Fixture and Fittings Efficiency Option 1. Prescriptive Path–Maximum Flush and Flow Rates OR Option 2. Performance Path–Calculated Reduction AND	
Minimum Equipment Water Efficiency AND	
Minimum Outdoor Water Use Efficiency Option 1. No Irrigation OR Option 2. Efficient Irrigation	

Meet all minimum water efficiency requirements outlined below.

MINIMUM FIXTURE AND FITTINGS EFFICIENCY

Meet the following minimum water efficiency requirements for fixtures and fittings.

Core and Shell only

These requirements must be met for the base building fixtures and fittings, appliances, equipment, process water, and outdoor water use. For tenant spaces, include manufacturer documentation for the base building's fixtures and fittings, appliances, and equipment in *IPp4: Tenant Guidelines*.

Projects located where standard supply pressure is different than the LEED baseline supply pressure may calculate the water consumption of flow fixtures and fittings at the local standard supply pressure.

OPTION 1. PRESCRIPTIVE PATH—MAXIMUM FLUSH AND FLOW RATES

For all new and existing fixtures and fittings within the project boundary, do not exceed the maximum flush and flow rates listed in Table 1.

TABLE 1. Maximum Installed Flush or Flow Rates for Prescriptive Path

Fixture or Fitting	Maximum Installed Flush or Flow Rate (IP)	Maximum Installed Flush or Flow Rate (SI)
Toilet (water closet)[a]	1.28 gpf[b]	4.8 lpf[b]
Urinal[a]	0.50 gpf	1.9 lpf
Public lavatory (restroom) faucet	0.50 gpm	1.9 lpm
Private lavatory faucets[a]	1.50 gpm	5.7 lpm
Kitchen faucet	1.8 gpm	6.8 lpm
Showerhead[a]	2.00 gpm	7.6 lpm

NOTE: gpf = gallons per flush; gpm = gallons per minute; lpf = liters per flush; lpm = liters per minute.
[a] The WaterSense label is available for this fixture type. WaterSense-labeled fixtures are recommended for projects located in the U.S. and Canada.
[b] For dual-flush toilets, the full-flush volume shall be equal to or fewer than 1.28 gpf/4.8 lpf. A weighted average cannot be used.

OR

OPTION 2. PERFORMANCE PATH—CALCULATED REDUCTION

For all new and existing fixtures and fittings within the project boundary, reduce aggregate water consumption by 20% from the baseline listed in Table 2.

TABLE 2. Baseline Water Consumption of Fixtures and Fittings

Fixture or Fitting	Baseline Installed Flush or Flow Rate (IP)	Baseline Installed Flush or Flow Rate (SI)
Toilet (water closet)[a]	1.6 gpf[b]	6.0 lpf[b]
Urinal[a]	1.0 gpf	3.8 lpf
Public lavatory (restroom) faucet	0.50 gpm at 60 psi	1.9 lpm at 415 kPa
Private lavatory faucets[a]	2.2 gpm at 60 psi	8.3 lpm at 415 kPa
Kitchen faucet	2.2 gpm at 60 psi	8.3 lpm at 415 kPa
Showerhead[a]	2.5 gpm at 80 psi per shower stall	9.5 lpm at 550 kPa per shower stall

NOTE: gpf = gallons per flush; gpm = gallons per minute; psi = pounds per square inch; lpf = liters per flush; lpm = liters per minute; kPa = kilopascals.
[a] The WaterSense label is available for this fixture type. WaterSense-labeled fixtures are recommended for projects located in the U.S. and Canada.
[b] For dual-flush toilets, the full-flush volume shall be equal to or fewer than 1.28 gpf/4.8 lpf. A weighted average cannot be used.

AND

MINIMUM EQUIPMENT WATER EFFICIENCY

Newly installed appliances, equipment, and processes within the project boundary must meet the requirements listed in Tables 3 and 4. Existing appliances and equipment can be excluded.

TABLE 3. Standards for Appliances

Appliance		Requirement	
Residential clothes washers		ENERGY STAR or performance equivalent	
Commercial clothes washers		ENERGY STAR for commercial clothes washers with ≤ 8.0 cubic feet (227 liters) capacity or performance equivalent	
Residential dishwashers (standard and compact)		ENERGY STAR or performance equivalent	
Prerinse spray valves		≤ 1.3 gpm (4.9 lpm)	
Ice machines		ENERGY STAR or performance equivalent and use either air-cooled or closed-loop cooling, such as a chilled or condenser water system	
Commercial Kitchen Equipment		**Requirement (IP)**	**Requirement (SI)**
Dishwasher	Undercounter	≤ 1.6 gal./rack	≤ 6.0 liters/rack
	Stationary, single tank, door	≤ 1.4 gal./rack	≤ 5.3 liters/rack
	Single tank, conveyor	≤ 1.0 gal./rack	≤ 3.8 liters/rack
	Multiple tank, conveyor	≤ 0.9 gal./rack	≤ 3.4 liters/rack
	Flight machine	≤ 180 gal./hr	≤ 680 liters/hr
Food steamer	Boilerless/ connectionless	≤ 2 gal./hr/pan	≤ 7.5 liters/hr/pan
	Steam generator	≤ 5 gal./hr/pan	≤ 19 liters/hr/pan
Combination oven	Countertop or stand	≤ 1.5 gal./pan	≤ 5.7 liters/pan
	Roll-in	≤ 1.5 gal./pan	≤ 5.7 liters/pan

NOTE: gpm = gallons per minute; lpm = liters per minute.

TABLE 4. Standards for Processes

Process	Requirement
Heat rejection and cooling	No once-through cooling with potable water for any equipment or appliances that reject heat.
Cooling towers and evaporative condensers	Equip with all the following: • Makeup water meters • Conductivity controllers and overflow alarms • Efficient drift eliminators that reduce drift to max of 0.001% of recirculated water volume for counterflow towers and 0.002% of recirculated water flow for cross-flow towers.
Discharge water temperature tempering	Where local requirements limit the discharge temperature of fluids into drainage system, use a tempering device that runs water only when the equipment discharges hot water. OR

(*continued*)

TABLE 4. Standards for Processes (*continued*)

Process	Requirement
	Provide a thermal recovery heat exchanger that cools drained discharge water below code-required maximum discharge temperatures while simultaneously preheating inlet makeup water. OR If fluid is steam condensate, return it to boiler.
Venturi-type flow-through vacuum generators or aspirators	Use no device that generates vacuum by means of a water flow-through device into drain.

AND

MINIMUM OUTDOOR WATER EFFICIENCY

OPTION 1. NO IRRIGATION

Show that the landscape does not require a permanent irrigation system beyond a maximum two-year establishment period.

OR

OPTION 2. EFFICIENT IRRIGATION

Reduce the project's irrigation water requirement by at least 30% from the calculated baseline for the site's annual theoretical irrigation requirement (TIR). Reductions must be achieved through plant species selection and irrigation system efficiency, as calculated by the TIR methodology outlined by the U.S. Environmental Protection Agency.

REQUIREMENTS EXPLAINED

The prerequisite sets minimum water efficiency requirements for fixtures, fittings, appliances, process water, and irrigation systems. Two compliance options exist for fixture and fitting efficiency, and two exist for outdoor water use efficiency.

MINIMUM FIXTURE AND FITTING EFFICIENCY

Reducing potable water use for fixtures and fittings begins with conservation efforts. Selecting high-efficiency fixtures reduces both water consumption and demand. This can lead to savings from reductions in pump energy and water heating requirements. For example, selecting high-efficiency fixtures for lavatories, faucets, and showerheads reduces the electric load required for water heating.

For LEED BD+C: Core and Shell projects, the requirements are only for the following items: base building fixtures and fittings, appliances, equipment, process water, and outdoor water use. For tenant spaces, project teams must include manufacturer documentation for the base building's fixtures and fittings, appliances, and equipment, which must be part of *IPp4: Tenant Guidelines*.

Choose either a prescriptive or performance pathway to demonstrate compliance for fixture and fitting efficiencies.

OPTION 1. PRESCRIPTIVE PATH—MAXIMUM FLUSH AND FLOW RATES

The prescriptive path offers a streamlined approach for the prerequisite. Table 1 outlines the maximum allowable flush or flow rate for fixtures and fittings within the project boundary. All fixtures and fittings installed must not exceed these maximum values.

For projects that install dual-flush toilets, the volume of the full-flush must be used when calculating the flush rate. The full-flush rate must not exceed 1.28 gpf (4.8 lpf). A weighted average that demonstrates that the average flow is 1.28 gpf (4.8 lpf) may not be used.

Projects in the U.S. and Canada should use WaterSense-labeled toilets (water closets), urinals, private lavatory faucets, and showerheads. WaterSense-labeled products require testing and verification for efficiency by third-party vendors. These products comply with the U.S. Environmental Protection Agency[5] (U.S. EPA) specifications.

OPTION 2. PERFORMANCE PATH—CALCULATED REDUCTION

Using the performance-based approach, teams can maximize water conservation across all applicable fixtures and fittings within the project boundary. Teams pursuing points under *WEc2: Enhanced Water Efficiency* should consider Option 2. Performance Path—Calculated Reduction, to show prerequisite compliance. Compliance with *WEp2: Minimum Water Efficiency*, Option 2. Performance Path—Calculated Reduction, and *WEc2: Enhanced Water Efficiency*, Option 2. Fixtures and Fittings—Calculated Reduction, requires documented compliance through the USGBC-approved calculator.

Teams must prove a reduction of 20% from the baseline water use to meet the minimum prerequisite requirements.

Determining the baseline and design-case water use

Teams must determine the project's baseline water consumption. Total annual consumption depends on project-specific data, including fitting and fixture types, flush and flow rates, the number of full-time equivalents, and visitors, annual days of operation, and gender ratios. This data must remain consistent across all LEED BD+C: New Construction and LEED BD+C: Core and Shell credits to maintain the integrity of the submission.

The total number of uses for each fixture and fitting remains the same in the baseline and design case calculations. The baseline flush and flow rates must use values from Table 2, which represent the maximum allowed flush and flow rates. The design case must use designed values that represent the fixtures and fittings installed in the project. For projects that have dual-flush toilets, use the full-flush volume in the design case calculations. Do not use the weighted average.

Develop calculations using the USGBC-approved calculator for this option to determine the percentage reduction.

5. U.S. EPA, "WaterSense," accessed on July 14, 2025, https://www.epa.gov/watersense.

MINIMUM EQUIPMENT WATER EFFICIENCY

During design, teams must identify appliances, kitchen equipment, and processes within the project boundary and specify products that meet the requirements of Tables 3 and 4. Projects in the U.S. and Canada must use ENERGY STAR-labeled equipment. For international projects, a performance-based equivalent meets the requirements.

ENERGY STAR-qualified appliances perform better than conventional appliances. For example, ENERGY STAR washing machines and dishwashers use 30% and 18% less water, respectively, than their conventional counterparts. These appliances also consume 10% to 50% less energy than conventional appliances.[6] Commercial kitchens use processes that require high levels of energy and water, such as dishwashing and food preparation.

MINIMUM OUTDOOR WATER EFFICIENCY

OPTION 1. NO IRRIGATION

For projects that do not install permanent irrigation, Option 1 offers a streamlined path to compliance. Teams may use irrigation during the first two years of the initial establishment period but must remove irrigation after that period.

Projects that use alternative water sources for irrigation do not automatically comply with this option. Projects must still prove reductions using *WEp2: Minimum Water Efficiency*, Option 2. Efficient Irrigation.

OPTION 2. EFFICIENT IRRIGATION

For projects with permanent irrigation, teams must design efficient irrigation systems and demonstrate that the installed irrigation system uses at least 30% less water than the baseline. Calculate the baseline water consumption using the site's TIR. Projects may use irrigation system efficiencies, plant species selection, or a combination of strategies to achieve the 30% reduction.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Contract document(s) specifying the project's plumbing fixtures and fittings, including performance specifications.
			Contract document(s) specifying the project's appliances, equipment, and process water equipment, including performance specifications.
			Contract document(s) specifying the project's commercial kitchen equipment water use, including performance specifications.
			Contract document(s) specifying the project's water equipment process, including performance specifications.

6. U.S. Department of Energy, Energy Efficiency and Renewable Energy, "Guide to Home Water Efficiency," October 2010, https://www.energy.gov/sites/prod/files/guide_to_home_water_efficiency.pdf.

(continued)

Project Types	Options	Paths	Documentation
All	All		LEED v5 Fixture and Fittings Efficiency calculator.
			USGBC calculator.
		No Irrigation	Documentation confirming that an irrigation system is not installed or that the landscape does not require a permanent irrigation system beyond a maximum two-year establishment period.
		Irrigation System	Contract document(s) specifying the project's irrigation component details (e.g., pipes, valves, sprinklers, emitters, controllers, sensors), including a key or legend of symbols used to represent the different components.

REFERENCED STANDARDS

- ENERGY STAR (appliance standards) (https://www.energystar.gov/products)
- U.S. EPA Theoretical Irrigation Requirement (TIR) calculation methodology (https://www.epa.gov/sites/default/files/2021-02/documents/watersense_final_technical_evaluation_process_for_home_certification_v1.0.pdf)

FURTHER EXPLANATION

MINIMUM FIXTURE AND FITTINGS EFFICIENCY

- Lavatory faucets must be classified as public or private. The Uniform Plumbing Code, International Plumbing Code, and the National Standard Plumbing Code each define as private those fixtures located in residences, hotel or motel guest rooms, and private rooms in hospitals.
  - Fixtures used by residential occupants and fixtures used by residential-type occupants who use the building for sleeping accommodations fall into the private classification.
  - Resident bathrooms in dormitories, patient bathrooms in hospital and nursing homes, and prisoner bathrooms are considered private use.
  - All other applications are deemed to be public. If it is unclear whether the classification should be public or private, default to public use flow rates in performing the calculations.
- Core and Shell projects must include only the fixtures and fittings in the Core and Shell project's scope for the prerequisite. If no fixtures or fittings are installed, the project is exempt from this requirement.

OPTION 1. PRESCRIPTIVE PATH—MAXIMUM FLUSH AND FLOW RATES

- This option is only applicable to the prerequisite. Projects pursuing points under *WEc2: Enhanced Water Efficiency* must use Option 2. Performance Path—Calculated Reduction.
- For dual-flush toilets, the maximum volume of the full flush must be used for compliance and cannot exceed 1.28 gallons per flush (4.8 liters per flush). A weighted average cannot be used to meet the maximum flush rate.

OPTION 2. PERFORMANCE PATH—CALCULATED REDUCTION

- For all product categories, specify low-flow fixtures. Where possible, select fixtures that meet or exceed a 20% reduction from the baseline listed in Table 2. "Baseline Water Consumption of Fixtures and Fittings."
- The calculations must include all restroom fixtures necessary to meet the occupants' needs (toilets, urinals, lavatories, and showers) when seeking *LTc4: Transportation Demand Management,* Option 2. Active Travel Facilities, Path 2. Shower and Changing Facilities.
- For dual-flush toilets, the maximum volume of the full flush must be used for compliance.
- Using aerators is an acceptable water savings strategy. The installed fixtures in the design case must use the rated flow rate from the manufacturer, and the underlying assumptions must remain consistent between the baseline and design cases.
- If the project occupancy is known, use the actual occupant counts for calculating occupancy. Use occupancy numbers that are a representative daily average over the course of the year. If the occupancy is not known, refer to the "Getting Started" section of the LEED v5 BD+C Reference Guide, under the heading "Project Occupancy" for guidance on calculating project occupancy. Project occupancy must be counted consistently across all LEED credits.
- The default gender mix is half male and half female. Modifications to the 50:50 ratio must be shown to apply for the life of the building. Acceptable special circumstances include projects specifically designed for an alternative gender ratio—for example, a single-gender educational facility. Assumptions that differ from the default must be supported by a narrative and supporting data.

MINIMUM EQUIPMENT WATER EFFICIENCY

- Existing appliances and equipment intended for reuse in the project are not required to meet the requirements.
- Commercial projects with noncommercial, standard-sized dishwashers or clothes washers must comply with the residential requirements. For heat rejection, the requirements apply to systems such as sterilizers, autoclaves, ice machines, X-ray machines, MRI machines, CT scanners, and other medical equipment for which the cooling involves large amounts of water and energy.
- Design equipment-cooling systems to limit or eliminate potable water use and to capture and reuse excess generated heat. Install air-cooled or closed-loop cooling instead of open-loop (e.g., once-through) systems for medical equipment. Redundancy for cooling in critical applications may be required. For emergency backups, consider recirculating systems, draining technology, and holding tanks, as well as nonpotable water sources for air-cooled vacuum pumps and once-through cooling systems.
- For medical equipment, consider designing and installing a dedicated nonpotable water loop to serve multiple pieces.
- If discharge waste temperature is regulated, consider recovering and reusing the system's waste heat for low-temperature heating (e.g., domestic water preheating).

MINIMUM OUTDOOR WATER EFFICIENCY

OPTION 1. NO IRRIGATION

- Projects with no landscape area automatically achieve this option.

OPTION 2. EFFICIENT IRRIGATION

- The entire landscaped area must be included in the calculations. The following landscape types may be included or excluded from landscape calculations: vegetated playgrounds, athletic fields, food gardens, and urban agricultural areas. Obtain monthly precipitation data and evapotranspiration rates (ETo) for the project area to determine the site's potential irrigation needs.
- To reduce irrigation demand, select plants that are appropriate for their intended uses.
  - Identify plant types and coverage that will balance water use efficiency with the area's intended function.
- Select efficient irrigation to reduce the amount of water used on-site.
  - Spray-head nozzles disperse water as a fine mist, often exceeding the soil's absorption rate. Windy conditions further reduce the system's efficiency.
  - Rotary nozzles produce larger water droplets that are less impacted by wind and release water at a slower rate, allowing the soil more time to absorb.
  - Microirrigation provides slower water delivery directly to the plants' root systems, minimizing water waste due to evaporation and runoff.
- When calculating the irrigation water requirement, consider the following:
  - Outdoor water use is calculated annually based on plant and irrigation type using the EPA Theoretical Irrigation Requirement (TIR) methodology.
  - Do not include hardscapes (whether pervious or impervious) or unvegetated softscapes, such as mulched paths and playgrounds.
  - Projects may not enter zero landscape water consumption for any landscaped area. For areas without irrigation, "no irrigation" should be selected in the calculator.
  - Additional savings gained by using alternative water sources and smart sensor technologies is addressed in *WEc2: Enhanced Water Efficiency.*

Water Metering and Leak Detection

WEc1
1 Point
New Construction (1 point)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To conserve potable water resources, support water management, limit potential material waste due to water leak damages, and identify opportunities for additional water savings by tracking water consumption.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1**
Option 1. Submeters OR	1
Option 2. Leak Detection Sensors	1

OPTION 1. SUBMETERS (1 POINT)

Install permanent water meters for each applicable subsystem defined below:

- Indoor plumbing fixtures and fittings. Meter systems serving at least 80% of indoor fixtures and fittings as described in *WEp2: Minimum Water Efficiency.* Fixtures and fittings not addressed in the prerequisite, including janitor sinks, water coolers, and bottle fillers, may be included or excluded from the indoor plumbing fixtures' subsystem at the project team's discretion.
- Irrigation system.
- Each makeup water system (e.g., cold water inlet for domestic hot water, swimming pools, chilled water systems, process water systems).
- Commercial kitchen (if the kitchen serves at least 100 meals per day of operation).
- Laundry (if the project includes commercial laundry equipment that processes at least 120,000 lb [57,606 kg] of laundry per year or if the project includes a public laundry room).

The facility manager and/or tenant(s) must be able to access the submeter data in real time via local network, building management system, cloud service, app, or online database. All submeters must be capable of recording data at least hourly.

Core and Shell only

In addition to the preceding requirement, meters must be installed for future tenant spaces so that tenants will be capable of independently metering water consumption in their spaces. Provide enough meters to capture total potable water use with a minimum of one per floor.

Healthcare only

In addition to the requirements above, install water meters in any five of the following:

- Purified water systems (reverse-osmosis, deionized)
- Filter backwash water
- Water use in the dietary department
- Water use in laundry
- Water use in laboratory
- Water use in central sterile and processing department
- Water use in physiotherapy and hydrotherapy and treatment areas
- Water use in surgical suite
- Closed-loop hydronic system makeup water
- Cold water makeup for domestic hot water systems

If a healthcare project does not include five of the additional subsystems listed above within the project scope, the project may alternatively submeter all water subsystems that are applicable to the project scope.

Residential only

Install a permanent water meter for each residential dwelling unit that measures the total potable water use for the unit. These meters need not be utility owned or utility grade.

OR

OPTION 2. LEAK DETECTION SENSORS (1 POINT)

Install permanent water flow meter or sensors for each applicable subsystem defined below:

- Project irrigation system at the point of entry, if irrigation is included in the project scope.
- At least 50% of the project flush fixtures. Water sensors can be installed on each flush fixture or for a group of flush fixtures (e.g., one per restroom facility). For LEED BD+C: Core and Shell projects, this only applies to flush fixtures within the project's scope of work.
- Each makeup water system (e.g., cold water inlet for domestic hot water, swimming pools, chilled water systems, and process water systems).

The leak detection system should be able to identify a leak triggered by abnormal flow rate above normal range, or physically detect a water leak, and initiate an alarm upon a leak detection.

The facility manager and/or tenant(s) must be able to access the sensor data in real time via local network, BMS, cloud service, app, or online database.

Develop an action plan that addresses how the building manager or tenant will have access to data in real time and how the building manager and/or tenant(s) will address and remedy any detected leak.

REQUIREMENTS EXPLAINED

This credit encourages projects to further develop water submetering beyond the *WEp1: Water Metering and Reporting* requirements. Projects pursuing this credit must permanently install submeters and sensors to report and track water use for applicable subsystems. Option 1 requires submeters on all water-using systems within the project boundary. Healthcare and residential projects have project-specific requirements to meet the Option 1 requirements. Option 2 requires leak detection sensors and data integration with the BMS, or something similar.

Projects can only achieve 1 point for choosing either Option 1 or Option 2 of this credit.

Submetering and leak detection strategies, when developed early in the design, provide significant benefits to the owner and design team. Teams can identify all water-using systems and prioritize submeters for major systems.

OPTION 1. SUBMETERS

Teams must install submeters for all the following water-using subsystems, as applicable to the project: indoor plumbing fixtures and fittings, irrigation systems, makeup water systems, commercial kitchen water use, and laundry water use. *WEp2: Minimum Water Efficiency* addresses cooling tower submeters.

Core and Shell projects must also install meters for future tenant spaces so that tenants will be capable of independently metering water consumption in their spaces. These projects must also install a sufficient number of meters to capture total potable water use with a minimum of one per floor.

Facility managers and/or tenants must have access to the real-time data via the project's local network, BMS, cloud service, web-based application, or an online database.

Indoor plumbing fixtures and fittings

Most projects have indoor plumbing fixtures and fittings. Water use from water closets and lavatories represents a significant amount of a building's total water consumption for many project types. Additionally, leaks from indoor fixtures and fittings often go unnoticed until water damage occurs on walls, ceilings, or floors.

Projects must submeter at least 80% of the total indoor fixtures and fittings, as identified in *WEp2: Minimum Water Efficiency*.

Depending on the distribution piping and the metering strategy, projects can directly meter water consumption from indoor plumbing fixtures and fittings or calculate the consumption by subtracting all other subsystems from the total water consumption of the building and the grounds.

Irrigation systems

Projects that include permanently installed irrigation systems must meter the irrigation water use. This includes any potable or alternative water sources used for the project.

In many cases, irrigation systems require additional submeters to meet the data recording requirements. Although a utility-owned irrigation meter captures the total consumption of the system, hourly recording and reporting to BMS, cloud service, or online database is not typical. Teams must confirm meter capabilities and include additional devices when necessary.

Tracking the water used by irrigation systems allows operators to identify leaking or inefficient sprinkler heads. It can also identify underground pipe leaks, which are often unresolved until visual inspections observe damp areas on the site.

Makeup water systems

Makeup water requirements vary by system and project. Systems that require makeup may include cold water inlets for the domestic hot water system, swimming pools, chilled water systems, or other processes.

Confirm the number of makeup water systems within the project boundary, and meter each system individually. Provide infrastructure capable of reporting and recording hourly consumption.

A single meter that reports total makeup water to a building or site does not meet the credit requirements.

Commercial kitchens

Typical commercial kitchen systems such as dishwashers, food steamers, and combination ovens require large quantities of potable water. Even when using water-efficient equipment, it is critical that projects track consumption from these appliances to ensure efficient operations and identify water supply failures.

The requirement for metering water use in a commercial kitchen depends on the number of daily meals served. Kitchens designed to serve 100 or more daily meals must meter and report all water use from the kitchen operations.

Commercial laundry facilities

Typical commercial laundry systems include large top- or front-load washing machines used to process thousands of pounds or kilograms of laundry annually. Front-load washers use less water and energy than top-load washers; however, even when using ENERGY STAR (or performance equivalent) equipment, it is critical that projects track water consumption from these machines to ensure efficient operations and identify water supply failures.

The requirement for metering water use in a commercial laundry depends on the pounds or kilograms of laundry processed annually. Laundry facilities designed to process 120,000 lb (57,606 kg) of laundry annually must meter and report all water use from the laundry operations.

OPTION 2. LEAK DETECTION SENSORS

Water lost through leaks strains capital and natural resources while creating health and safety risks.[7] Lower system pressures can draw pollutants in from the surroundings, leading to foul drinking water.[8] Capital losses can result from labor costs to identify the location of the leak and the labor hours and material costs necessary to repair major damage. Leaks that are not addressed commonly result in higher bills.

Projects pursuing this option must install permanent water flow meters or sensors on each applicable subsystem (irrigation, flush fixtures, makeup water systems) within the project boundary. The devices must report abnormalities and generate an alarm at a local network, BMS, cloud service, app, or online database accessible by the facility manager and tenants.

Data access and action plan

Providing access to and regularly performing reviews of the data optimizes an operator's ability to address water leaks. Develop an action plan that addresses, at minimum, data access for operators and tenants, the approach for resolving detected leaks, and a communication plan to alert building occupants when repairs impact the building and the site.

Irrigation leak detection requirements

Install devices at the point of entry to the site for permanently installed irrigation systems. This allows for the earliest possible detection of any leaks within the system.

Flush fixture leak detection requirements

For the flush fixtures, install sensors or meters for at least 50% of the flush fixtures identified in *WEp2: Minimum Water Efficiency*. Install the device at the flush fixture for an individual toilet room, such as a unisex or family restroom. A single device that monitors all flush fixtures for group restrooms is acceptable.

Makeup water system leak detection requirements

All makeup water systems must have permanent sensors or meters for leak detection. Leak detection must occur at each system. A sensor or meter that collectively monitors leaks for a single makeup line serving multiple systems does not meet the credit requirement.

7. Megan A. Kilinski, "Overview of Available Leak Detection Technologies: A Summary of Capabilities and Costs," Pacific Northwest National Laboratory, 2019, retrieved from https://www.pnnl.gov/main/publications/external/technical_reports/PNNL-28885.pdf.
8. Richard Collins, Joby Boxall, Marie-Claude Besner, Stephen Beck, and Bryan Karney, "Intrusion Modelling and the Effect of Ground Water Conditions," in *Water Distribution Systems Analysis 2010* (Reston, VA: ASCE, 2011).

Healthcare

OPTION 1. SUBMETERS

Healthcare projects require additional submeters. Many processes typical of healthcare operations, such as sterilization, water use in surgical suites, and purified water systems, require significant amounts of water.

The credit requires that healthcare projects meter an additional five subsystems, as outlined in the rating system. For projects that do not have at least five additional subsystems within the project scope, provide meters for all applicable water subsystems included in the project.

OPTION 2. LEAK DETECTION SENSORS

Same as Option 2, Leak Detection Sensors.

Residential

OPTION 1. SUBMETERS

In addition to any applicable systems in common areas, measure each residential unit's total potable water usage.

Designing a system that tracks and reports water use from each residence meets the intent of the credit.

Meters must report data to the facility manager. Tenants must also have access to the data. Monthly reporting allows facility managers and tenants the opportunity to address excessive water use within living spaces.

OPTION 2. LEAK DETECTION SENSORS

Same as Option 2, Leak Detection Sensors.

DOCUMENTATION

Project types	Options	Paths	Documentation
All	Option 1. Submeters	All	Identification of the permanent water meters installed on the project.
			Metered indoor fixture/fittings calculation.
			Contract documents highlighting the locations and types of the project's permanent water meters for each applicable subsystem (e.g., water supply system drawings).
			Contract documents or narratives demonstrating that the facility manager and/or tenant(s) will be able to access the submeter data in real time via local network, BMS, cloud service, app, or online database.
			Confirmation that all submeters are capable of recording data at least hourly (e.g., product information from the manufacturer and/or contract documents).
Healthcare			Identification of the water systems that are within the project scope and the additional permanent water meters installed on the project.

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Project types	Options	Paths	Documentation
All	Option 2. Leak Detection Sensors	All	Contract documents or narratives demonstrating that the facility manager and/or tenant(s) will be able to access the sensor data in real time via local network, BMS, cloud service, app, or online database.
			Identification of the applicable subsystems (irrigation, flush fixtures, makeup water) included in the project.
			Contract documents highlighting the locations and types of the project's permanent water flow meters and/or sensors for each applicable subsystem (e.g., water supply system drawings).
			Narrative or documents detailing the leak detection system specifications for when and how an alarm is triggered (e.g., the OPR, BOD, or Contract Documents, product information from the manufacturer).
			The project's action plan addresses how the building manager or tenant will have access to data in real time and how the building manager and/or tenant(s) will address and remedy any detected leak.
			Calculation demonstrating the percentage of flush fixtures with leak detection.

REFERENCED STANDARDS

- None

FURTHER EXPLANATION

OPTION 1. SUBMETERS

- In planning for submeters, reviewing the systems during the schematic or design phase allows for design optimizations including, but not limited to, meter placement, water heater or boiler room locations, and distribution piping design.
  - Consider metering the subsystems that consume the most water, are the most expensive to operate, or most closely align with the goals of the building management.
  - Consider not only the number, types, and sizes of meters but also the effort required to take meter readings.
  - Submeters may be manually read or connected to a building information system. The meters may be equipped with data logging capability independent of a building information system.
  - If meters are to be read manually, consider digital indicators, which may reduce errors in recording. Automatic data logging can add to initial cost, but it may be cost-effective if it reduces the effort of obtaining and recording readings.
  - Wet meters will likely be the most cost-effective approach. However, clamp-on meters may be simpler to install.
  - The higher cost of a larger meter may be offset by reduced operations and maintenance costs if facilities staff will be making manual readings.

- For healthcare projects, if less than five of the listed space types or systems are installed, meter the major water subsystems that are applicable to the project scope.
  - For residential projects, designing a system that tracks and reports water use from each residence meets the intent of the credit.

IRRIGATION SYSTEMS

- The meter for reclaimed water may also be the same meter that is counted for irrigation, fixtures, or process water.
- If reclaimed water is used in multiple applications in the project (e.g., for irrigation, flush fixtures, or process water), then all reclaimed water use must be metered. The team may report data from multiple submeters, as applicable.

COMMERCIAL KITCHENS

- For kitchens that have a dedicated water heater, a single meter on the cold-water supply line to the kitchen may capture all kitchen water usage. If the project includes a central water heater for all hot water use, additional meters may be necessary to capture the total hot- and cold-water consumption within the space.

WATER EFFICIENCY CREDIT

Enhanced Water Efficiency

WEc2
New Construction (1–8 points)
Core and Shell (1–7 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To reduce potable water consumption and the associated energy consumption and carbon emissions required to treat and distribute water, and to reward use of alternative water sources that preserve potable water resources.

REQUIREMENTS: NEW CONSTRUCTION

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1–8**
Option 1. Whole-Project Water Use OR	1–8
Option 2. Fixture and Fittings–Calculated Reduction AND/OR	1–3
Option 3. Appliance and Process Water AND/OR	1–2
Option 4. Outdoor Water Use	1–2
Path 1. No Irrigation OR	2
Path 2. Efficient Irrigation AND/OR	1–2
Option 5. Optimize Process Water Use	1–2
Path 1. Limit Cooling Tower Cycles OR	1–2
Path 2. Optimize Water Use for Cooling OR	1–2
Path 3. Process Water Use AND/OR	1–2

(*continued*)

ACHIEVEMENT PATHWAYS	POINTS
Option 6. Water Reuse	1–2
Path 1. Reuse-Ready System	1
OR	
Path 2. Alternative Water Sources	2

Implement a combination of the following strategies for a maximum of 8 points. Projects may either attempt Option 1 or any combination of Options 2–6.

OPTION 1. WHOLE-PROJECT WATER USE (1–8 POINTS)

To pursue this pathway, project teams must develop a water use baseline and create a proposed use model. Points are achieved based on reductions from the baseline in Table 1.

TABLE 1. Points for Reducing Overall Project Water Use

Percent Reduction	Points	Total Points for Alternative Water
30%	1	2
35%	2	3
40%	3	4
45%	4	5
50%	5	6
55%	6	7
60%	7	8
65%	8	Exemplary performance

OR

OPTION 2. FIXTURE AND FITTINGS—CALCULATED REDUCTION (1–3 POINTS)

Further reduce fixture and fitting water use from the calculated baseline in *WEp2: Minimum Water Efficiency,* Minimum Fixture and Fittings Efficiency, Path 2. Performance Path—Calculated Reduction. Additional potable water savings can be earned above the prerequisite level using alternative water sources. Points are awarded according to Table 2.

TABLE 2. Points for Reducing Indoor Water Use

Percentage Reduction	Points
30%	1
35%	2
40%	3

AND/OR

OPTION 3. APPLIANCE AND PROCESS WATER (1–2 POINTS)

Newly installed equipment within the project boundary must meet the minimum requirements in Tables 3, 4, 5, and/or 6. One point is awarded for meeting all applicable requirements in any 1 table for a maximum of 2 points. All applicable, newly installed equipment listed in each table must meet the standard. Existing appliances and equipment can be excluded.

To use Table 3, the project must process at least 120,000 lb (57,606 kg) of laundry per year.

TABLE 3. Compliant Commercial Washing Machines

Washing Machine	Requirement (IP Units)	Requirement (SI Units)
On-premise, minimum capacity 2,400 lb (10,886 kg) per 8-hour shift	Maximum 1.8 gal. per pound*	Maximum 7 liters per 0.45 kg*

* Based on equal quantities of heavily, medium, and lightly soiled laundry.

To use Table 4, the project must serve at least 100 meals per day of operation.

TABLE 4. Standards for Compliant Commercial Kitchen Equipment

Commercial Kitchen Equipment		Requirement (IP)	Requirement (SI)
Dishwasher	Undercounter	ENERGY STAR	ENERGY STAR or performance equivalent
	Stationary, single tank, door	ENERGY STAR	ENERGY STAR or performance equivalent
	Single tank, conveyor	ENERGY STAR	ENERGY STAR or performance equivalent
	Multiple tank, conveyor	ENERGY STAR	ENERGY STAR or performance equivalent
	Flight machine	ENERGY STAR	ENERGY STAR or performance equivalent
Food steamer	Boilerless/connectionless	≤ 1.7 gal./hr/pan including condensate cooling water	≤ 6.4 liters/hr/pan including condensate cooling water
	Steam generator	≤ 2.2 gal./hr/pan including condensate cooling water	≤ 8.3 liters/hr/pan including condensate cooling water
Combination oven	Countertop or stand	ENERGY STAR	ENERGY STAR or performance equivalent
	Roll-in	ENERGY STAR	ENERGY STAR or performance equivalent
Food waste disposer	Disposer	3–8 gpm, full load condition; 10-minute automatic shutoff or 1 gpm, no-load condition	11–30 lpm, full load condition; 10-minute automatic shutoff or 3.8 lpm, no-load condition
	Scrap collector	Maximum 2 gpm makeup water	Maximum 7.6 lpm makeup water
	Pulper	Maximum 2 gpm makeup water	Maximum 7.6 lpm makeup water
	Strainer basket	No additional water usage	No additional water usage

NOTE: gpm = gallons per minute; lpm = liters per minute.

TABLE 5. Compliant Laboratory and Medical Equipment

Lab Equipment	Requirement (IP)	Requirement (SI)
Reverse-osmosis water purifier	75% recovery	
Steam sterilizer	For 60 in sterilizer: 6.3 gal./U.S. tray For 48 in sterilizer: 7.5 gal./U.S. tray	For 1,520 mm sterilizer: 28.5 liters/DIN tray For 1,220 mm sterilizer: 28.35 liters/DIN tray
Sterile process washer	0.35 gal./U.S. tray	1.3 liters/DIN tray
X-ray processor, 150 mm or more in any dimension	Film processor water-recycling unit	
Digital imager, all sizes	No water use	

To use Table 6, the project must be connected to a municipal or district steam system that does not allow the return of steam condensate.

TABLE 6. Compliant Municipal Steam Systems

Steam System	Requirement
Steam condensate disposal	Cool municipally supplied steam condensate (no return) to drainage system with heat recovery system or reclaimed water
OR	
Reclaim and use steam condensate	100% recovery and reuse

AND/OR

OPTION 4. OUTDOOR WATER USE (1–2 POINTS)

PATH 1. NO IRRIGATION (2 POINTS)

Show that the landscape does not require a permanent irrigation system beyond a maximum two-year establishment period.

OR

PATH 2. EFFICIENT IRRIGATION (1–2 POINTS)

Reduce the project's theoretical irrigation requirement (TIR) by at least 50% from the calculated baseline. Points are awarded according to Table 7.

TABLE 7. Points for Reducing Outdoor Water Use

Percentage Reduction	Points
50%	1
100%	2

AND/OR

OPTION 5. OPTIMIZE PROCESS WATER USE (1–2 POINTS)

PATH 1. LIMIT COOLING TOWER CYCLES (1–2 POINTS)

For cooling towers and evaporative condensers, conduct a one-time potable water analysis, measuring at least the five control parameters listed in Table 8.

TABLE 8. Maximum Concentrations for Parameters in Condenser Water

Parameter	Maximum Level
Ca (as $CaCO_3$)	600 ppm
Total alkalinity	500 ppm
SiO_2	150 ppm
Cl^-	300 ppm
Conductivity	3,300 µS/cm

NOTE: ppm = parts per million; µS/cm = micro siemens per centimeter.

Calculate the maximum number of cooling tower cycles by dividing the maximum allowed concentration level of each parameter by the actual concentration level of each parameter found in the potable makeup water analysis. Limit cooling tower cycles to avoid exceeding maximum values for any of these parameters.

The materials of construction for the water system that contact the cooling tower water must be of the type that can operate and be maintained within the cycles established in Table 9. Points are awarded according to Table 9.

TABLE 9. Points for Cooling Tower Cycles

Cooling Tower Cycles	Points
Maximum number of cycles achieved without exceeding any maximum concentration levels or affecting operation of condenser water system.	1
Meet the maximum calculated number of cycles to earn 1 point and increase the number of cycles by a minimum of 25% by increasing the level of treatment and/or maintenance in condenser or makeup water systems. OR Meet the maximum calculated number of cycles to earn 1 point and use a minimum 20% alternative water.	2

Projects whose cooling is provided by district cooling systems are eligible to achieve Path 1 if the district cooling system complies with the preceding requirements.

OR

PATH 2. OPTIMIZE WATER USE FOR COOLING (1–2 POINTS)

To be eligible for Option 2, the baseline system designated for the building using *ASHRAE 90.1-2019 or 90.1-2022*, Appendix G, Table G3.1.1-3, must include a cooling tower (systems 7, 8, 11, 12, and 13).

Achieve increasing levels of cooling tower water efficiency beyond a water-cooled chiller system with axial variable-speed fan cooling towers having a maximum drift of 0.002% of recirculated water volume and three cooling tower cycles. Points are awarded according to Table 10.

TABLE 10. Points for Reducing Annual Water Use Compared to Water-Cooled Chiller System

Percentage Reduction	Points
25%	1
50%	2

Projects whose cooling is provided by district cooling systems are eligible to achieve Path 2 if the district cooling system complies with the preceding requirements.

OR

PATH 3. PROCESS WATER USE (1–2 POINTS)

Demonstrate that the project is using a minimum of 20% alternative water to meet the process water demand for 1 point or that the project is using a minimum of 30% alternative water to meet process water demand for 2 points. Ensure that alternative water is of sufficient quality for its intended end use.

Process water uses eligible for achievement of Path 3 must represent at least 10% of total building regulated water use and may not include water used for cooling.

AND/OR

OPTION 6. WATER REUSE (1–2 POINTS)

PATH 1. REUSE-READY SYSTEM (1 POINT)

Install a water supply system to allow the supply of reclaimed or alternative water to reach one or more of the following end uses. Space shall be provided for treatment equipment as applicable to end uses.

OR

PATH 2. ALTERNATIVE WATER SOURCES (2 POINTS)

Incorporate one of the following water reuse strategies for indoor, outdoor, and/or process water that meets the needs of one or more end uses for the building and grounds:

- On-site water reuse system
- Municipally supplied reclaimed water

Eligible end uses for Paths 1 and 2 include irrigation, flush fixtures, makeup water systems, (such as cooling towers or boilers), or other process water systems.

REQUIREMENTS: CORE AND SHELL

ACHIEVEMENT PATHWAYS	POINTS
Core and Shell	**1–7**
Option 1. Whole-Project Water Use OR	1–7
Option 2. Fixtures and Fittings–Calculated Reduction AND/OR	1–3
Option 3. Appliances and Process Water AND/OR	1–2
Option 4. Outdoor Water Use	1–3
Path 1. No Irrigation OR	3
Path 2. Efficient Irrigation AND/OR	1–3
Option 5. Optimize Process Water Use	1–3
Path 1. Limit Cooling Tower Cycles OR	1–3
Path 2. Optimize Water Use for Cooling OR	1–3
Path 3. Process Water Use AND/OR	1–2
Option 6. Water Reuse	1–2
Path 1. Reuse-Ready System OR	1
Path 2. Alternative Water Sources	2

Implement a combination of the following strategies for a maximum of 7 points. Projects may either attempt Option 1 or any combination of Options 2–6 below.

OPTION 1. WHOLE-PROJECT WATER USE (1–7 POINTS)

To pursue this pathway, project teams must develop a water use baseline and create a proposed use model. Points are achieved based on reductions from the baseline in Table 11.

TABLE 11. Points for Reducing Overall Project Water Use

Percent Reduction	Points	Total Points for Alternative Water
30%	1	2
35%	2	3
40%	3	4

TABLE 11. Points for Reducing Overall Project Water Use (*continued*)

Percent Reduction	Points	Total Points for Alternative Water
45%	4	5
50%	5	6
55%	6	7
60%	7	–

OR

OPTION 2. FIXTURES AND FITTINGS—CALCULATED REDUCTION (1–3 POINTS)

Further reduce fixture and fitting water use from the calculated baseline in *WEp2: Minimum Water Efficiency,* Minimum Fixture and Fittings Efficiency, Path 2. Performance Path—Calculated Reduction. Additional potable water savings can be earned above the prerequisite level using alternative water sources. Points are awarded according to Table 12.

TABLE 12. Points for Reducing Indoor Water Use

Percentage Reduction	Points
30%	1
35%	2
40%	3

AND/OR

OPTION 3. APPLIANCES AND PROCESS WATER (1–2 POINTS)

Newly installed equipment within the project boundary must meet the minimum requirements in Tables 13, 14, 15, and/or 16. One point is awarded for meeting all applicable requirements in any one table for a maximum of 2 points. All applicable, newly installed equipment listed in each table must meet the standard. Existing appliances and equipment can be excluded.

To use Table 13, the project must process at least 120,000 lb (57,606 kg) of laundry per year.

TABLE 13. Compliant Commercial Washing Machines

Washing Machine	Requirement (IP)	Requirement (SI)
On-premise, minimum capacity 2,400 lb (10,886 kg) per 8-hour shift	Maximum 1.8 gal. per pound*	Maximum 7 liters per 0.45 kg*

*Based on equal quantities of heavily, medium, and lightly soiled laundry.

To use Table 14, the project must serve at least 100 meals per day of operation.

TABLE 14. Standards for Compliant Commercial Kitchen Equipment

Commercial Kitchen Equipment		Requirement (IP)	Requirement (SI)
Dishwasher	Undercounter	ENERGY STAR	ENERGY STAR or performance equivalent
	Stationary, single tank, door	ENERGY STAR	ENERGY STAR or performance equivalent
	Single tank, conveyor	ENERGY STAR	ENERGY STAR or performance equivalent
	Multiple tank, conveyor	ENERGY STAR	ENERGY STAR or performance equivalent
	Flight machine	ENERGY STAR	ENERGY STAR or performance equivalent
Food steamer	Boilerless/connectionless	≤ 1.7 gal./hr/pan, including condensate cooling water	≤ 6.4 liters/hr/pan including condensate cooling water
	Steam generator	≤ 2.2 gal./hr/pan, including condensate cooling water	≤ 8.3 liters/hr/pan including condensate cooling water
Combination oven	Countertop or stand	ENERGY STAR	ENERGY STAR or performance equivalent
	Roll-in	ENERGY STAR	ENERGY STAR or performance equivalent
Food waste disposer	Disposer	3–8 gpm, full load condition; 10-minute automatic shutoff or 1 gpm, no-load condition	11–30 lpm, full load condition; 10-minute automatic shutoff or 3.8 lpm, no-load condition
	Scrap collector	Maximum 2 gpm makeup water	Maximum 7.6 lpm makeup water
	Pulper	Maximum 2 gpm makeup water	Maximum 7.6 lpm makeup water
	Strainer basket	No additional water usage	No additional water usage

NOTE: gpm = gallons per minute; lpm = liters per minute.

TABLE 15. Compliant Laboratory and Medical Equipment

Lab Equipment	Requirement (IP)	Requirement (SI)
Reverse-osmosis water purifier	75% recovery	
Steam sterilizer	For 60 in sterilizer: 6.3 gal./U.S. tray For 48 in sterilizer: 7.5 gal./U.S. tray	For 1,520 mm sterilizer: 28.5 liters/DIN tray For 1,220 mm sterilizer: 28.35 liters/DIN tray
Sterile process washer	0.35 gal./U.S. tray	1.3 liters/DIN tray
X-ray processor, 150 mm or more in any dimension	Film processor water-recycling unit	
Digital imager, all sizes	No water use	

To use Table 16, the project must be connected to a municipal or district steam system that does not allow the return of steam condensate.

TABLE 16. Compliant Municipal Steam Systems

Steam System	Requirement
Steam condensate disposal	Cool municipally supplied steam condensate (no return) to drainage system with heat recovery system or reclaimed water
OR	
Reclaim and use steam condensate	100% recovery and reuse

AND/OR

OPTION 4. OUTDOOR WATER USE (1–3 POINTS)

PATH 1. NO IRRIGATION (3 POINTS)

Show that the landscape does not require a permanent irrigation system beyond a maximum two-year establishment period.

OR

PATH 2. EFFICIENT IRRIGATION (1–3 POINTS)

Reduce the project's TIR by at least 50% from the calculated baseline. Points are awarded according to Table 17.

TABLE 17. Points for Reducing Outdoor Water Use

Percentage Reduction	Points
50%	1
75%	2
100%	3

AND/OR

OPTION 5. OPTIMIZE PROCESS WATER USE (1–3 POINTS)

PATH 1. LIMIT COOLING TOWER CYCLES (1–3 POINTS)

For cooling towers and evaporative condensers, conduct a one-time potable water analysis, measuring at least the five control parameters listed in Table 18.

TABLE 18. Maximum Concentrations for Parameters in Condenser Water

Parameter	Maximum Level
Ca (as $CaCO_3$)	600 ppm
Total alkalinity	500 ppm
SiO_2	150 ppm
Cl^-	300 ppm
Conductivity	3,300 µS/cm

NOTE: ppm = parts per million; µS/cm = micro siemens per centimeter.

Calculate the maximum number of cooling tower cycles by dividing the maximum allowed concentration level of each parameter by the actual concentration level of each parameter found in the potable makeup water analysis. Limit cooling tower cycles to avoid exceeding maximum values for any of these parameters.

The materials of construction for the water system that come into contact with the cooling tower water shall be of the type that can operate and be maintained within the cycles established in Table 19. Points are awarded according to Table 19.

TABLE 19. Points for Cooling Tower Cycles

Cooling Tower Cycles	Points
Maximum number of cycles achieved without exceeding any maximum concentration levels or affecting operation of condenser water system.	1
Meet the maximum calculated number of cycles to earn 1 point and increase the number of cycles by a minimum of 25% by increasing the level of treatment and/or maintenance in condenser or makeup water systems. OR Meet the maximum calculated number of cycles to earn 1 point and use a minimum 20% alternative water.	2
Meet the maximum calculated number of cycles to earn 1 point and increase the number of cycles by a minimum of 30% by increasing the level of treatment and/or maintenance in condenser or makeup water systems. OR Meet the maximum calculated number of cycles to earn 1 point and use a minimum 30% alternative water.	3

Projects whose cooling is provided by district cooling systems are eligible to achieve Path 1 if the district cooling system complies with the preceding requirements.

OR

PATH 2. OPTIMIZE WATER USE FOR COOLING (1–3 POINTS)

To be eligible for Path 2, the baseline system designated for the building using *ASHRAE 90.1–2019 or 90.1–2022*, Appendix G, Table G3.1.1–3, must include a cooling tower (systems 7, 8, 11, 12, and 13).

Achieve increasing levels of cooling tower water efficiency beyond a water-cooled chiller system with axial variable-speed fan cooling towers having a maximum drift of 0.002% of recirculated water volume and three cooling tower cycles. Points are awarded according to Table 20.

TABLE 20. Points for Reducing Annual Water Use Compared to Water-Cooled Chiller System

Percentage Reduction	Points
25%	1
50%	2
100%	3

Projects whose cooling is provided by district cooling systems are eligible to achieve Path 2 if the district cooling system complies with the preceding requirements.

OR

PATH 3. PROCESS WATER USE (1–3 POINTS)

Demonstrate that the project is using at minimum 20% alternative water to meet process water demand for 1 point, using at minimum 30% alternative water to meet process water demand for 2 points, or using at minimum 40% alternative water to meet process water demand for 3 points. Ensure that alternative water is of sufficient quality for its intended end use.

Process water uses eligible for achievement of Path 3 must represent at least 10% of total building regulated water use and may not include water used for cooling.

Projects served by district systems are eligible to achieve Path 3 if the district system complies with minimum thresholds for alternative water use.

AND/OR

OPTION 6. WATER REUSE

PATH 1. REUSE-READY SYSTEM (1 POINT)

Install a water supply system to allow the supply of reclaimed or alternative water to reach one or more of the following end uses. Space shall be provided for treatment equipment as applicable to end uses.

OR

PATH 2. ALTERNATIVE WATER SOURCES (2 POINTS)

Incorporate one of the following water reuse strategies for indoor, outdoor, and/or process water that meets the needs of one or more end uses for the building and grounds:

- On-site water reuse system
- Municipally supplied reclaimed water

Eligible end uses for Paths 1 and 2 include irrigation, flush fixtures, makeup water systems (such as cooling towers or boilers), or other process water demand.

REQUIREMENTS EXPLAINED

This credit builds on *WEp2: Minimum Water Efficiency* and rewards teams for additional water conservation strategies. New Construction projects that pursue Option 1 can achieve up to 8 points for the whole project's water use reductions, and those that pursue Options 2–6 can achieve up to 8 points by combining any of the strategies outlined in the rating system. Core and Shell projects that pursue Option 1 can achieve up to 7 points for whole project water use reductions, or up to 7 points for any combination of Options 2–6. Projects cannot combine Option 1 with Options 2–6.

OPTION 1. WHOLE-PROJECT WATER USE

Quantifying whole-project water use and developing strategies to reduce consumption across the entire project can lead to significant water savings. Analyzing all water sources enables teams to identify large consumers and target both conservation and alternative water strategies.

Projects must demonstrate a minimum of 30% reduction from the project's baseline to earn points. Using alternative water sources earns additional points for the calculated reductions.

DEVELOP A WATER BALANCE MODEL

Teams must calculate the building's water demand and develop baseline and design case water balance models. Models must account for all end uses within the project's scope of work, including fixtures and fittings, domestic hot water, appliances, commercial kitchen and laundry equipment, laboratory and medical equipment, process water, HVAC systems, and landscape irrigation.

BASELINE REQUIREMENTS

The baseline model reflects the minimum requirements and typical water use for the project type without any additional water-saving measures.

For fixtures and fittings, determine the baseline using a *USGBC*-approved calculator, as described in *WEp2: Minimum Water Efficiency*, Option 2. Performance Path—Calculated Reduction.

Baseline values for appliances, kitchen equipment, as well as laboratory and medical equipment must align with Tables 13–15 of this credit.

For cooling towers, the baseline water model represents the water use associated with the minimum number of cooling tower cycles, such that parameters do not exceed the values of Table 18.

For outdoor water use, determine the baseline water consumption by calculating the project's TIR as outlined in *WEp2: Minimum Water Efficiency*, Minimum Outdoor Water Use Efficiency, Option 2. Efficient Irrigation. Projects that do not use irrigation in the design compare the as-designed condition to a baseline TIR.

DESIGN CASE REQUIREMENTS

The design case water model must use fixtures and fittings, appliances, kitchen equipment, and laboratory and medical equipment as specified within the contract documents. For outdoor water use, determine the actual water used on-site.

Projects with alternative water sources

Use of alternative water sources can be highly effective in reducing potable water demand and utility costs associated with the building and site. Projects that use alternative water sources achieve one additional point for each threshold met under Table 1. For example, a project that reduces potable water consumption by 30% earns 1 point. A project that reduces potable water consumption by 30% and uses an alternative water source to achieve those reductions earns 2 points.

Teams should always prioritize water efficiency first to reduce consumption and demand before applying alternative water solutions. For any seasonally dependent sources, such as rainwater, calculations must reflect annual seasonal totals to confirm the available quantity of the alternative water source.

Projects pursuing Options 2–6

New Construction projects may earn up to 8 points by combining strategies from Options 2–6, while Core and Shell projects may earn up to 7 points by pursuing Options 2–6. Projects with limited scope, or projects that use targeted reductions by equipment or system type, should review Option 1 and all relevant Options 2–6 to determine the approach that maximizes points for this credit.

OPTION 2. FIXTURES AND FITTINGS—CALCULATED REDUCTION

Option 2 focuses on reducing potable water use from fixtures and fittings. Using the baseline and designed water usage calculations determined from *WEp2: Minimum Water Efficiency*, Minimum Fixture and Fittings Efficiency, Path 2. Performance Path—Calculated Reduction, projects earn points for additional savings beyond the 20% requirements, based on specific fixture flow and flush rates. You must calculate using a USGBC-approved calculator.

Along with the use of high-efficiency fixtures and fittings, projects that use alternative water sources for these systems can report further savings and achieve additional points under this option.

OPTION 3. APPLIANCE AND PROCESS WATER

Option 3 rewards projects that prioritize water-efficient appliances and process equipment. Projects earn 1 point for meeting prescriptive water use requirements in a single equipment category, for up to 2 total points.

Tables 13–16 outline the prescriptive measures for appliances and process equipment. All newly installed equipment must meet the referenced standards, performance equivalents (outside of the U.S.), and/or water use limits. Exclude existing appliances and equipment.

Projects that do not include any applicable systems in their scope of work, or projects that can document compliance with more than two tables, should review the project's whole water balance model from *WEc2: Enhanced Water Efficiency*, Option 1. Whole Project Water Use, as it may optimize points for this credit.

OPTION 4. OUTDOOR WATER USE

There are two paths to earn credit for outdoor water use. Path 1 rewards projects that do not use permanent irrigation beyond the initial two-year establishment period. Projects that include irrigation can earn points by reducing water use from the project's baseline requirements using native or adapted plants and vegetation, alternative water, and/or smart irrigation controls.

PATH 1. NO IRRIGATION

Projects complying with *WEp2: Minimum Water Efficiency*, Minimum Outdoor Water Use Efficiency, Option 1. No Irrigation, automatically earns 2 points.

PATH 2. EFFICIENT IRRIGATION

WEc2: Enhanced Water Efficiency, Path 2. Efficient Irrigation, directly links to *WEp2: Minimum Water Efficiency*, Minimum Outdoor Water Use Efficiency, Option 2. Efficient Irrigation. New Construction projects that achieve a 50% or 100% reduction in irrigation earn up to 2 points, whereas Core and Shell projects that achieve a 50%, 75%, or 100% reduction may earn up to 3 points. Proven solutions include alternative water sources, like reclaimed water or rainwater harvesting, and the use of smart scheduling technology. Calculations completed for *WEp2: Minimum Water Efficiency*, Minimum Outdoor Water Use Efficiency, Option 2. Efficient Irrigation, must prove the reduction from the calculated baseline using the annual TIR.

Include all landscaped areas in the irrigation calculations. However, teams may exclude irrigation for vegetated playgrounds, athletic fields, food gardens, and urban agricultural areas from the calculations.

OPTION 5. OPTIMIZE PROCESS WATER USE

Cooling towers and industrial processes use a significant amount of potable water. Projects that include these systems must consider opportunities to reduce concentration cycles or optimize water use for cooling or select alternative water sources.

Process water uses include, but are not limited to, cooling, humidification, sterilization, dishwashers, clothes washers, and pools. This option offers three pathways, which depend on the type of process water use. Path 3 also requires that process water use meets a minimum percentage of the total building water use.

PATH 1. LIMIT COOLING TOWER CYCLES

This path prioritizes water conservation for cooling towers by limiting the cycles of concentration from the equipment. A cycle of concentration is the number of times water can circulate through the system without creating performance or operational problems. A low cycle of concentration means more single-use water passes through the system, resulting in excess water consumption. However, as we reuse water, dissolved solids remain, which increases the concentration levels of calcium, silicon dioxide (SiO_2), and chloride. Higher cycles of concentration may result in scaling and corrosion issues.

The intent of the credit is not to impact system operations but to inform designers on alternative solutions for reducing water consumption for cooling processes. Finding the correct balance of cooling tower blowdown and chemical treatment maintains system efficiency, reduces maintenance, and conserves potable water.

For each cooling process, conduct a potable water analysis to determine set points for the chemical treatment system and the associated cycles of concentration. Teams must confirm that the systems will operate at the specified cycles of concentration and not exceed parameters outlined in Table 18.

Projects with alternative water sources

Projects using alternative water sources do not require a one-time water analysis. Using a minimum of 20% alternative water can help projects earn 2 points if the maximum calculated number of cycles is also met.

Projects in a campus environment

Projects in a campus environment, or those supplied by a district cooling system, may comply with Path 1 if the district system conducts a potable water analysis and limits the cycles of concentration for its system. Projects can determine compliance at the district level by working with the utility provider.

PATH 2. OPTIMIZE WATER USE FOR COOLING

Path 2 has minimum eligibility requirements. Using *ASHRAE Standard 90.1–2019* (or later), Appendix G. Performance Rating Method, Table G3.1.1–2, confirm that the project's baseline case includes chilled water for cooling and cooling towers in the baseline design. Table 21 outlines the eligible baseline systems from Appendix G.

Projects pursuing this path do not require an Appendix G energy model. Other tools can help perform water-use calculations.

TABLE 21. ASHRAE Standard 90.1–2019, Appendix G Compliant Baseline Systems

System Number	System Description
7	VAV with reheat
8	VAV with parallel fan-powered boxes and reheat
11	Single-zone VAV system with water-cooled chillers
12	Single-zone constant volume system with water-cooled chillers and a hot water fossil fuel boiler
13	Single-zone constant volume system with water-cooled chillers and electric resistance heat

NOTE: VAV = variable air volume.

Projects can demonstrate a 100% reduction from baseline if the Appendix G baseline includes a cooling tower and the final design eliminates the need for a cooling tower.

Projects may benefit from a combination of strategies to reduce water consumption for the cooling system. Strategies include maximized cycles of concentration, increased levels of chemical treatment, smart controls for monitoring and optimization, drift eliminators, flow meters, and water-level controls.

Projects in a campus environment

For campus environments or projects that receive cooling from a district cooling system, projects meet the requirements of Path 2 if the district system meets the reduction thresholds. Projects should consider working with their utility provider to determine compliance at the district system level.

PATH 3. PROCESS WATER USE

Projects pursuing this path must demonstrate that process water use exceeds 10% of the total building regulated water use, excluding cooling water.

Using alternative water sources, such as captured condensate from air handling units, reduces reliance on fresh water for process systems. Diversifying the water sources on a project site increases resilience in buildings. This allows projects to divert freshwater for human consumption instead of processes during a water crisis.

When selecting the alternative water source, ensure that the quality of the water is sufficient for its intended use and that the local AHJ allows that alternative water source per local codes and standards.

OPTION 6. WATER REUSE

Option 6 provides two paths to achieve points for water reuse strategies. Projects must choose a single path and cannot combine these paths for more points.

Path 1 rewards projects that install systems allowing the future supply of reclaimed or alternative water sources to at least one specific end use. Path 2 rewards projects that implement water reuse strategies on-site and/or use reclaimed or alternative water supplied by municipalities.

The project must include water reuse for at least one end use listed below. Implementing alternative or reclaimed water sources for as many systems as possible significantly contributes to potable water conservation efforts.

- Irrigation
- Flush fixtures (urinals, water closets)
- Makeup water systems (including cooling towers and boilers)
- Other process water systems

PATH 1. REUSE-READY SYSTEM

Path 1 rewards a project for preparing for a future transition to water reuse systems.

Planning for water reuse systems early in design allows project teams to optimize solutions for the greatest impact. Through this process, teams determine strategies that reduce the demand and consumption of potable water use in the building and on the project site. The use of reclaimed or alternative water sources allows projects to reduce potable water for end uses not intended for human consumption, such as water closet flushing and makeup water for process systems.

Designing for future infrastructure

Including reclaimed or alternative water supply infrastructure allows for an easier transition when sources become available for public use. The inclusion of these systems in new construction projects also results in economic benefits. Planning for future water treatment equipment enables projects to efficiently design the equipment room to allow for future equipment, which reduces long-term costs to integrate alternative water supply infrastructure. Operational savings result from lower utility costs since alternative water sources are often a lower-cost option.

The credit does not require that projects install physical connections to each flush fixture or makeup line. However, creating a plug-and-play system is highly recommended to limit future major renovations.

PATH 2. ALTERNATIVE WATER SOURCES

Option 2 rewards teams for implementing solutions to reduce reliance on potable water by using alternative water for indoor, outdoor, and/or process water systems.

The project must also meet all metering and commissioning requirements as outlined in *WEp1: Water Metering and Reporting*, *EAp3: Fundamental Commissioning*, and *EAc5: Enhanced Commissioning* (if pursued).

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Whole Project Water Use	All	USGBC calculator.
			Contract documents showing the project's alternative water system details and capacity.
			Contract documents and manufacturer information support the baseline and proposed water use values in the calculator.
	Option 2. Fixtures and Fittings	All	USGBC calculator.
			Contract documents showing the project's alternative water system details, location, and capacity.
	Option 3. Appliance and Process Water All	All	Contract document(s) specifying the project's newly installed commercial washing machines, commercial food waste disposers, commercial laboratory and medical equipment, and/or municipal steam systems and process water equipment, as applicable, including performance specifications.
	Option 4. Outdoor Water Use	Path 2. Efficient Irrigation	Contract documents indicating the project's alternative water system details and capacity, and smart irrigation controls.
	Option 5. Optimize Process Water Use	Path 1. Limit Cooling Tower Cycles	Results from the potable water analysis for cooling towers and evaporative condensers. Include the concentration levels for all five parameters listed in Table 8 of the rating system.
		Path 2. Optimize	Equipment schedule indicating cooling tower type.
			Calculation shows the baseline process water use.

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Project Types	Options	Paths	Documentation
All		Process Water Use	Calculation shows the design process, water use, including any alternative water savings.
		Path 3. Process Water Use	Total percentage of processed water use in building.
			Description of systems included in the project that accounts for more than 10% of the total building regulated water use (excluding cooling water).
			Calculation shows the baseline process of water use.
			Calculation showing the design process water use, including any alternative water savings.
	Option 6. Water Reuse	Path 1. Reuse-ready System	Contract documents that identify the space provided for future treatment equipment.
		Path 2. Alternative Water Source	Identification of the alternative water source (on-site water reuse system or municipally supplied reclaimed water).
			Identification of which eligible end use(s) the water reuse system meets the needs (irrigation, flush fixtures, makeup water systems, such as cooling towers or boilers, or other process water demand).
			Contract documents highlighting the details of the alternative water source system(s) details.

REFERENCED STANDARDS

- ENERGY STAR (https://www.energystar.gov)
- ASHRAE 90.1-2019 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2019-i-p?product_id=2088527)
- ASHRAE 90.1-2022 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2022-i-p?product_id=2522082)

FURTHER EXPLANATION

Projects should consider completing a whole-project water use assessment, even if they pursue Options 2 through 6. Information from the water balance model may provide additional strategies that can optimize water performance for the project.

OPTION 1. WHOLE-PROJECT WATER USE

- Include all necessary restroom fixtures (e.g., toilets, urinals, and lavatories) to meet the project occupants' needs and include showers when seeking *LTc4: Transportation Demand Management,* Path 2. Shower and Changing Facilities.
- Alternatives to potable water include municipally supplied reclaimed water ("purple pipe" water), graywater, rainwater, stormwater, treated seawater, condensate, foundation dewatering water, used process water, and reverse osmosis reject water.

- Untreated water sources ineligible for this credit include raw water from naturally occurring surface bodies of water, streams, rivers, groundwater, well water, and water discharged from an open-loop geothermal system.
- When choosing alternative sources of water, target the uses that require the least treatment first. In most cases, water can be reused outside the building (for irrigation) or inside (for toilet flushing) with minimal treatment, but other uses will require more energy-intensive treatment.

- Core and Shell projects
  - Assume that the as-yet-uninstalled (future) fixtures have the baseline water consumption rates. Kitchen sinks must be included in the credit calculations if installed in the project's scope of work or if addressed in a tenant sales or lease agreement. However, if future kitchen sinks are not installed as part of the project's scope of work or are not addressed in a tenant sales or lease agreement, they may be excluded from the credit calculations. A Core and Shell project can earn credit for the plumbing fixtures installed as part of the project's scope if all fixtures necessary to meet occupants' needs are included in the calculations and if all occupants of the incomplete tenant spaces are included in the calculations.
- Exceptional Calculation—New Technologies

  For a new technology, projects must provide one of the following:
  - Three projects built in the last five years that use the baseline technology
  - Current utility incentive programs for new construction that establish the baseline
  - Published studies justifying the baseline technology as standard practice

  For proprietary manufacturing processes, projects must demonstrate that the production process is more efficient than the company's average production efficiency by providing all of the following:
  - Three projects built in the last five years that manufacture the product.
  - Calculate the process's past average water consumption index (WCI) in units of water per product manufactured to establish the baseline production efficiency.
  - Provide the new process's estimated WCI, anticipated production level, and an explanation of how these numbers were determined.
  - Calculate the annual production process water cost savings using the baseline WCI, proposed WCI, and anticipated production level.

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DEVELOP A WATER BALANCE MODEL

- The baseline water balance model must represent all water sources and uses typically associated with the project type, location, building, and site.
  - Typical sources are potable water from municipal systems or wells. Municipally treated seawater should also be included if this is a typical water source for the project's region.
- Demonstrate that the baseline meets the following:
  - Local building codes for water quality and efficiencies
  - Industrial water consumption meets specific industrial standards established by the American Society for Testing and Materials (ASTM), International Organization for Standardization (ISO),

American Water Works Association (AWWA) standards, United States Pharmacopoeia (USP), the Clinical and Laboratory Standards Institute (CLSI) or other industry-specific water quality and consumption standards

- Illustrate the average amount of water used for each type of equipment to determine an appropriate baseline.
- Water demand and alternative source data should ideally be modeled on an hourly basis, but no greater than daily, to account for fluctuations in water demand and available supply. Extrapolation of data is acceptable if daily data is not available.

BASELINE REQUIREMENTS

- Use *WEp2: Minimum Water Efficiency* for the following water uses:
  - Minimum Equipment Water Efficiency: Table 3. "Standards for Appliances"; and Table 4. "Standards for Processes."
  - Minimum Outdoor Water Efficiency: Option 2. Efficient Irrigation.
- Use *WEc2: Enhanced Water Efficiency* for the following water uses:
  - Appliances and Process Water: Table 3. "Compliant Commercial Washing Machines"; Table 4. "Compliant Commercial Kitchen Equipment"; Table 5. "Compliant Laboratory and Medical Equipment"; and Table 6. "Compliant Municipal Steam Systems."
  - Cooling Towers: Option 5. Optimize Process Water, Path 2. Optimize Water Use for Cooling.

    Note: The requirements in Path 1. Limit Cooling Tower Water Cycles, do not provide a methodology for calculating water use of cooling towers, as the credit addresses filtration parameters and number of cycles. However, water used by cooling towers must be included in the baseline and proposed models.

DESIGN CASE REQUIREMENTS

- The proposed water balance must represent all water sources and proposed water sources for the project building and site.
  - Water sources could include rainwater harvesting, graywater, black water treatment, process water reuse, air handler condensate, retention ponds, municipal water, or well water.
  - Water uses must align with those outlined in the baseline model.
  - Rainwater harvesting sources should include documentation from local weather stations providing hourly or daily precipitation amounts. In the United States of America, Puerto Rico, the U.S. Virgin Islands, and various Pacific Islands, hourly precipitation data for over 5,000 weather stations are available through NOAA.[9]
  - Illustrate the average amount of water used for each type of equipment to demonstrate that the proposed increased efficiency, compared to the baseline, exceeds the minimum threshold.

9. NOAA, "Hourly Precipitation Data (HPD) Publication," accessed September 9, 2025, https://www.ncei.noaa.gov/access/search/data-search/hourly-precipitation-data-publication; and NOAA, "U.S. Hourly Precipitation Data," accessed September 9, 2025, https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.ncdc:C00313.

OPTION 5. OPTIMIZE PROCESS WATER USE, PATH 1. LIMIT COOLING TOWER CYCLES

- If a potable water analysis has already been completed, it must be no more than five years old. At a minimum, the analysis must measure the concentration parameters identified in the credit requirements.
- Calculate the cycles of concentration by determining how many times the cooling tower or evaporative condenser water can circulate through the system without creating performance or operational problems based on the maximum concentrations indicated in the credit requirements.
  - Identify which concentration parameter has the fewest calculated cycles before it exceeds the maximum concentration. That parameter's number of cycles, the limiting factor, will determine the maximum number of cycles for the cooling tower or evaporative condenser.
  - Adjust the cooling tower or evaporative condenser settings for the maximum number of cycles without exceeding concentration levels or affecting condenser operation.
  - Increase the number of cycles by treating the water or, if the project is using 100% potable water, supplementing it with water from a nonpotable source.

OPTION 5. OPTIMIZE PROCESS WATER USE, PATH 3. PROCESS WATER USE

- Projects pursuing this path must demonstrate that process water use exceeds 10% of the total building-regulated water use, excluding cooling water. For example, a building using water for cooling, humidification, sterilization, dishwashers, clothes washers, and/or pools must determine the total process water percentage for all sources, except cooling.

OPTION 6. WATER REUSE

- During early design stages, projects should review all water-using systems within the project boundary and prioritize feasible alternative water sources. Where an alternative water source is not currently available, or solutions are not economically viable, designs should consider back-up water storage solutions with dual piping to allow for future transitions to alternative water sources.
- Alternative water sources include reclaimed wastewater, graywater, swimming pool backwash filter, refrigeration system condensate, captured rainwater, stormwater and foundation drain water, steam system condensate, fluid cooler discharge, food steamer discharge, combination oven discharge, industrial process water, fire pump test water, municipally supplied treated wastewater, and ice machine condensate.

OPTION 6. WATER REUSE, PATH 1. REUSE-READY SYSTEM

- During the initial design efforts for the project, coordinate any future alternative water piping infrastructure with other building systems to avoid collisions and to allow for easier access to the piping in the future.

OPTION 6. WATER REUSE, PATH 2. ALTERNATIVE WATER SOURCES

- Provide the demand flow and the estimated annual volume for each source requiring potable water. When determining alternative options, provide the same information for comparison. For seasonally dependent sources like rainwater, calculations must reflect annual, seasonal totals.

- Prioritizing strategies that meet all or most of the building demand requirements creates the greatest impact; however, any solution that reduces potable water reliance creates significant industry change. Teams should work closely with the plumbing contractor and civil engineer to analyze the complete building and site water balance and the available capacity of the proposed alternative water source(s).

Energy and Atmosphere

OVERVIEW

As of 2025, buildings are responsible for one-third of global energy emissions, accounting for over 34% of energy demand and approximately 37% of energy and process-related carbon emissions.[1] Stabilizing the climate requires the world to decarbonize its new and existing buildings by 2050.

Many regions have identified zero energy and zero carbon goals, targeting 2030 for new construction and 2050 for existing buildings.[2] Thankfully, time-tested and cost-effective strategies now exist to reach these goals. The market now knows how to design and construct the low-carbon buildings of the future.

Well-designed, constructed, and operated buildings use less energy, produce fewer emissions, and increase their resilience to disruptions like power outages or extreme weather events. LEED v5's Energy and Atmosphere (EA) credit category aims to make low-carbon buildings easy to achieve by increasing carbon literacy and providing a clear framework for all buildings to significantly reduce or eliminate emissions, achieve greater energy independence and security, and lower operational energy costs.

As businesses and regulatory agencies prioritize resilience and sustainability in their financial and social continuity planning, decarbonization is becoming an integral priority for leaders worldwide.

Decarbonization

LEED v5 drives decarbonization. Over half of the LEED v5 credits supporting decarbonization are in the EA category. By capitalizing on technological advances and industry expertise, project teams can use EA prerequisites and credits to create more value for owners, occupants, and communities.

First, LEED v5 helps increase the carbon literacy of design teams. In *EAp1: Operational Carbon Projection and Decarbonization Plan*, project teams develop a visual prediction of future carbon emissions showing annual carbon emissions will reduce over time due to the decarbonization of most electric grids. This is different from energy use, which stays constant in the no-change scenario.

1. UN Environment Programme, "2022 Global Status Report for Buildings and Construction," November 9, 2022, https://www.unep.org/resources/publication/2022-global-status-report-buildings-and-construction.
2. Taryn Holowka, "Support the Net Zero Carbon Buildings Commitment," U.S. Green Building Council, August 31, 2022, https://www.usgbc.org/articles/support-net-zero-carbon-buildings-commitment.

In addition, time does not impact all sources equally. The annual emissions from electricity use will nearly vanish with a fully decarbonized grid, while those from on-site combustion will remain unchanged, an essential distinction for approaching neutrality by 2050.

Through the credits of the EA category, LEED v5 provides a simple framework for designing zero carbon–ready buildings. This framework lays out three critical steps and four additional strategies for decarbonization that are inherent to the building itself, such as better envelopes and electrified heating systems, which must be incorporated from the beginning of the project. The four additional strategies build on the primary strategies while providing significant carbon impacts.

The three critical steps are electrification, reduced peak thermal loads, and energy efficiency. *EAc1: Electrification* is a new credit within LEED v5. As electrical grids decarbonize, the carbon emissions from electrical usage will drastically decrease. However, the emissions from fuel-powered systems in buildings—usually for space heating and service hot water—will remain. Replacing those fuel-powered systems with electrically powered equipment, which can provide heat efficiently, will help emissions shrink to near zero by 2050.

The *EAc1: Electrification* credit rewards projects that electrify as many of their systems as possible, while providing compliance options for operations during extreme low temperatures and emergency backup systems.

EAc2: Reduce Peak Thermal Loads is another new credit within LEED v5 and a key step to decarbonization. Grid demand will rise as buildings, vehicles, and industries transition from on-site combustion to electricity. By reducing peak thermal loads, project teams can increase the building's resilience to extreme temperatures and reduce demand on the electrical grid. This credit incentivizes projects to mitigate these peaks.

The third step is energy efficiency—a cornerstone of LEED and high-performing buildings. All LEED v5 projects start with a baseline of energy efficiency by pairing climate zone-appropriate building envelopes with building systems and management practices (*EAp2: Minimum Energy Efficiency*, *EAp4: Energy Metering and Reporting*, and *EAp3: Fundamental Commissioning*).

Energy efficiency provides critical benefits, including lower operational costs, less damage due to the extraction and transport of fuels, and less air pollution and accompanying health issues. Efficiency reduces carbon emissions—even from electricity—because most grids are not carbon neutral yet and will not be soon. For teams that prefer an alternative to energy modeling, LEED v5 offers a new prescriptive option for the full achievement of points—*EAc3: Enhanced Energy Efficiency*.

The additional decarbonization strategies represent industry-leading best practices found in *EAc5: Enhanced Commissioning*, *EAc4: Renewable Energy*, and *EAc6: Grid Interactive*, as well as in *EAp5: Fundamental Refrigerant Management* and *EAc7: Enhanced Refrigerant Management*. Carried over from earlier versions of LEED and refined in LEED v5, these approaches continue to be highly effective in reducing carbon emissions and minimizing energy waste.

LEED v5 Platinum-certified projects will achieve industry best practices for energy efficiency, eliminate on-site combustion (except for emergency and backup needs), use 100% renewable energy, and reduce embodied carbon.

Quality of life

When buildings reduce emissions and energy demand while using technology to communicate with the grid, they will ensure optimal operations. These measures also enhance the value they serve to the community. Teams are encouraged to work toward reducing air leakage from the envelope and mechanical systems while incentivizing energy storage opportunities (*EAc2: Reduce Peak Thermal Loads, EAc6: Grid Interactive*). Combining these strategies with an energy-efficient design and electrified operations can lead to a more resilient and reliable building.

LEED v5 EA prerequisites and credits provide clear paths to greater efficiency and reduced costs and emissions. These tactics help enhance energy and carbon literacy in the building industries and empower communities to achieve energy and carbon neutrality by 2050.

Operational Carbon Projection and Decarbonization Plan

EAp1
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To enable building stakeholders to visualize how their current design decisions will impact their project's long-term operational carbon emissions and to ensure that stakeholders are planning for low-carbon outcomes from the project's inception.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Design Analysis AND	
Site Energy Estimate AND	
Review Carbon Projection AND	
Decarbonization Plan Path 1. Design for Electrification OR Path 2. Plan for Decarbonization	

Comply with the following requirements.

DESIGN ANALYSIS

Analyze efficiency, peak load reduction, and decarbonization measures during the early stages of the design process and account for the results in design decision-making using at least one of the following methodologies:

- Simplified energy modeling
- Analysis from similar projects
- Analysis from published data

AND

SITE ENERGY ESTIMATE

Estimate the amount of each type of energy the project will use annually in terms of site energy and submit the data to USGBC.

AND

REVIEW CARBON PROJECTION

Using the annual energy use data submitted, the project's current grid data, and the project's location, USGBC will generate a business as usual (BAU) projection of the project's carbon emissions from energy use from the present through a 25-year period.

Projects subject to a carbon-based building performance standard (BPS) must create an ordinance-specific BAU with a carbon projection based on the electrical coefficients as defined in the ordinance and with an overlay showing the caps applicable to the project. If applicable, calculate the assessed annual fines or fees that will apply for exceeding the caps and the cumulative fines or fees over a 25-year period.

The building owner or owner's representative shall attest that they have reviewed the BAU carbon projections and fee projection.

AND

DECARBONIZATION PLAN

PATH 1. DESIGN FOR ELECTRIFICATION

Earn 4 or 5 points in *EAc1: Electrification*.

OR

PATH 2. PLAN FOR DECARBONIZATION

Create a plan detailing how decarbonization could be achieved through a 25-year period. The building owner or owner's representative shall attest that they have reviewed the decarbonization plan.

- The plan shall be a narrative no more than two pages in length.
- The narrative shall describe the retrofits to be made, with the approximate timeline and cost of each of the retrofit measures.

EA

- Equipment and/or building materials that will be discarded as a result of the required retrofits should be described along with new equipment to be purchased.
- Electrification readiness strategies incorporated into the initial design should be described along with a rough estimate of the avoided cost, avoided disruption, and avoided materials waste afforded by each readiness measure. Core and Shell projects should incorporate strategies to support tenant build-out and address future retrofits after tenant build-out. Some common readiness strategies include oversizing electrical panels and/or service or installing conduit for future loads, enhanced envelope, or heating distribution systems that can accommodate the lower temperatures of future heat pumps.

REQUIREMENTS EXPLAINED: NEW CONSTRUCTION

To help design teams understand their carbon emissions over time, USGBC provides each project with a visualization of its projected emissions over the next 25 years in the BAU scenario. BAU is the scenario in which the building's operations do not change. The USGBC projection assumes that every grid will decarbonize by 95% over that period, a directionally correct approximation.

This exercise illustrates that carbon emissions from electricity will diminish to near zero over the next 25 years, emissions from on-site combustion will remain constant, and emissions from electricity are not zero now and will not be zero for some time, except for a few unique carbon-neutral grids.

This prerequisite has multiple requirements, working together to help project teams design more energy-efficient projects that should not require expensive retrofits to achieve low-carbon outcomes.

DESIGN ANALYSIS

Teams must analyze energy conservation measures and carbon reduction strategies early in design, creating impactful and cost-effective solutions that integrate into the project. Collaborative discussions with architects, engineers, contractors, and owners can lead to a holistic approach to decarbonization.

Analyze design options and develop alternatives that optimize efficiency measures, reduce peak loads, and prioritize decarbonization. Even a simple box energy model, with minimal zoning and basic project details, can generate valuable feedback on building massing, orientation, HVAC system selection, and lighting power density.

Smaller, less complex projects are not required to have an energy model in hopes of making LEED v5 more accessible. Teams may use prototype models that are specific to the project application. Teams can also use other models for large-scale analysis, such as the publicly available prototype models used to inform ASHRAE energy code development. Another solution would be using data from previous, similar projects. Comparing the project's design to projects of similar size, use, site, and climate zone can provide teams with a target or benchmark for developing a high-performing building.

SITE ENERGY ESTIMATE

Along with the initial design analysis, teams must estimate the project's annual site energy use from each energy source and submit this information to USGBC, which will use the data to create the BAU carbon projection—a fundamental part of this credit.

For teams developing an energy model for *EAp2: Minimum Energy Efficiency* or a simplified energy model, generate the required data from the simulation results. For teams using similar projects or published data, establish estimates based on the predicted design conditions, including any designed optimization strategies.

REVIEW CARBON PROJECTION

USGBC provides all LEED BD+C: New Construction projects with a BAU projection of the project's operational carbon emissions over the next 25 years based on the site energy data provided and the project's current grid emissions factor. This BAU projection assumes that the grid emissions factor starts with the latest national or regional coefficient—such as the subregional eGRID coefficients in the U.S.—and declines in a straight line by 95% over the next 25 years. This step reflects the overall direction of global grid decarbonization and is not an exact prediction of individual project performance.

The projection should educate owners and designers on the carbon impact of their design decisions over the next 25 years. Owners must review the data and attest to the review.

Building performance standards

Depending on the project location, it may be subject to a BPS. These standards are carbon-based or EUI-based performance requirements that many U.S. cities and states have adopted. Typically, BPSs set caps on energy use or carbon emissions that decline over time. The project must meet those caps or incur fees.

Projects subject to a carbon-based or EUI-based BPS must generate a BPS Carbon BAU or BPS Energy BAU, depending on which is applicable. Teams must overlay the BPS caps on the project's BAU projection and calculate the expected fees if they exceed the caps.

Additional requirements

Grid emissions factors, which step down over time, are often mandated in carbon-based BPSs. Use these values in the BPS Carbon BAU rather than USGBC's assumed grid emissions factors used in the BAU carbon projection.

Understanding whether the project will meet the BPS requirements and, if not, the fees it may incur over the next 25 years, can also influence key decisions during the design phase. The owner's review of this information provides awareness of the exposure to future fees.

DECARBONIZATION PLAN

If the current design has not been substantially electrified, projects must create a two-page decarbonization plan. The plan aims to inform design teams and building owners of the future costs and disruptions they may incur when retrofits for all-electric equipment occur post-occupancy.

This requirement has two paths for compliance.

PATH 1. DESIGN FOR ELECTRIFICATION

Projects that have been substantially electrified, as documented by achieving four or five points in *EAc1: Electrification*, are exempt from having to create a plan because they have already been designed for the low-carbon future.

PATH 2. DECARBONIZATION PLAN

Projects without substantial electrification must create a plan outlining the future retrofits required to achieve substantial decarbonization as the grid decarbonizes. Developing this plan is meant to show project teams and building owners how much less expensive it would be to build it right the first time rather than to retrofit it later. The required retrofits would include invasive and expensive envelope improvements. The replacement of new, costly heating equipment and potentially new heating distribution systems might encourage design teams going through this exercise to amend their designs to preclude the highest avoidable future costs on this project or future projects.

The plan must detail a reasonable estimate of the future cost and effort of decarbonization, including engineering, architectural, and structural costs. For example, replacing natural gas heating equipment with electric heating equipment may include costs for power system upgrades and refrigerant and condensate piping in addition to the heat generation equipment.

Electrification readiness

Projects must consider electrification readiness strategies that would reduce costs and could be incorporated in the initial design. Beneficial solutions include adding extra electrical panels or oversizing panels to ensure adequate service for future loads. Installing conduits for future loads will limit the amount of destructive renovation work.

REQUIREMENTS EXPLAINED: CORE AND SHELL

DESIGN ANALYSIS

Additional considerations for Core and Shell

Teams must analyze design options and develop alternatives to optimize efficiency measures, reduce peak loads, and prioritize decarbonization. For Core and Shell projects, perform the analysis based on anticipated occupancy type and associated energy end uses for the planned future tenant(s). For example, suppose the project is a Core and Shell office building. Teams must perform the analysis assuming office occupancy, and account for all lighting, plug and process equipment, and HVAC and service water heating capacity necessary to meet the tenant's needs.

Although the analysis must consider elements of the future tenant, the primary focus should be on the components and systems within the owner's or developer's scope of work. This includes the building envelope, common area HVAC systems, common area domestic hot water, electrical infrastructure, and any base building decisions that can impact the future tenant's design choices for energy-using systems.

DECARBONIZATION PLAN

Additional considerations for Core and Shell

PATH 1. DESIGN FOR ELECTRIFICATION

To qualify for this path, projects must have sufficient systems installed in the base building scope of work. The scope must include enough capacity to meet future HVAC requirements for tenants. It must include systems or power infrastructure that meet service hot water requirements and potential process loads, like kitchen equipment. A project without these systems as part of the Core and Shell scope of work cannot use Path 1.

PATH 2. DECARBONIZATION PLAN

Incorporating assumptions about future tenants into the plan is important, but it is not the plan's main priority. The plan should focus on the base building and equipment elements within the scope of work. It should include some elements of preparing for future equipment electrification and retiring combustion-based equipment. For example, strategies like increased electrical panel capacity, adequate access for future equipment modernization, and identification of retrofit opportunities during tenant build-out/turnover.

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	All	All	Design analysis during the early project design phase analyzed energy efficiency, peak load reduction, and decarbonization strategies for their impact on long-term operational carbon emissions.
			Estimated total annual energy use of each energy source (electricity, natural gas chilled water, steam, etc.) and the annual energy use for each of the following end uses: space heating, service hot water, cooking, cooling, refrigeration, ventilation, plug and process loads, other.
			For projects subject to BPS, provide ordinance-specific BAU carbon projection showing energy and/or carbon caps applicable to the project, and, if applicable, annual fines for exceeding caps over a 25-year period (only for projects subject to a building performance standard).
			Attest that design analysis was performed.
			Confirm building owner or owner's representative has reviewed the carbon projection.
			Provide supplied energy types and estimated annual amounts used.
	Decarbonization plan	Path 1	Achieve four or five points for EAc1: Electrification.
		Path 2	Evidence of a decarbonization plan with the required elements. Plan summary and excerpts are acceptable if the full plan exceeds two pages.

REFERENCED STANDARDS

- None.

FURTHER EXPLANATION

BACKGROUND

For the last 50 years, designers in the real estate industry have focused on improving energy efficiency and achieved impressive results. Along with better design practices and advancements in technology, codes and standards have prioritized increased efficiencies in all building systems. However, as we move closer to 2050, the climate crisis grows increasingly urgent and designers' focus has shifted to reducing carbon emissions, a subject that is new to the industry and much less familiar than energy efficiency. And while energy efficiency is a tool for reducing carbon emissions, carbon and energy are not the same.

Energy consumption can be understood as a static number since the energy used by a building will remain constant if the building's operations or systems do not change. But the carbon emissions from that static building will change. There are two sources of carbon emissions from building operations: electricity and on-site combustion. The emissions from electrical usage will change if the building's electrical grid changes, and electrical grids everywhere are reducing their carbon emission per kilowatt-hour as they incorporate new renewables into their supply mix. This is known as *grid decarbonization* and most grids are expected to be substantially decarbonized by 2050. However, the carbon emissions from the on-site combustion of fuels in that static building will not change over time.

To help design teams understand their carbon emissions over time, U.S. Green Building Council® (USGBC) is providing each project with a visualization of its projected emissions over the next 25 years in the business as usual (BAU) scenario—that is, the scenario in which the building's operations do not change. The USGBC projection assumes that every grid will decarbonize by 95% over that period, an approximation that is directionally correct. This exercise assumes the following: carbon emissions from electricity will diminish to near zero over the next 25 years, emissions from on-site combustion will remain constant, and emissions from electricity are not zero now and will not be for some time (except for a few unique carbon-neutral grids).

To fully decarbonize our built environment, designers must improve their carbon literacy and make better decisions that align with eventually eliminating carbon emissions from our buildings. Energy use remains static; however, as the grid decarbonizes, carbon emissions from buildings will change. Generating awareness of this fact leads to better design decisions from the start.

An operational carbon projection aims to provide design teams with basic carbon emissions information over a 25-year period. In a BAU model, emissions from electric use will decline to near zero during that time. Decisions made for today's design, like removing on-site combustion, allow for immediate action regardless of grid emissions.

Regulations on long-term carbon emissions from buildings are likely to come to many places. Cities like New York and St. Louis, Missouri, already have such regulations, known as building performance standards (BPS). To further help design teams and building owners understand the long-term carbon impacts of their current designs, this credit includes two additional requirements. Buildings in jurisdictions with a BPS in place must create a BPS-specific projection and calculate their anticipated fees for non-compliance. All projects that have not been designed to be near carbon neutral by mid-century must create a narrative explaining the changes required to achieve that end. The aim is to encourage design teams to consider what steps they could take to build it right the first time and avoid the need to expensively retrofit a relatively new building.

For projects subject to BPS, owners can leverage the data to inform on future penalties from inaction or make decisions to delay electrification efforts.

For projects that do not electrify, a future decarbonization plan allows owners to understand the path forward. Addressing future transition costs, like installing conduit for a future all-electric building, saves time, money, and construction efforts.

Minimum Energy Efficiency

EAp2
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To promote resilience and reduce the environmental and economic harms of excessive energy use and greenhouse gas emissions by achieving a minimum level of energy efficiency.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Option 1. *ASHRAE 90.1-2019* OR	
Option 2. *ASHRAE 90.1-2022*	

Projects registering before January 1, 2028, may comply with either Option 1 or Option 2.

Projects registering on or after January 1, 2028, must comply with Option 2.

OPTION 1. *ASHRAE 90.1-2019*

Comply with *ANSI/ASHRAE/IES Standard 90.1-2019* with Addendum cr. Use any applicable compliance path in *ASHRAE 90.1*, Section 4.2.

For projects applying the Normative Appendix G, Performance Rating Method compliance path, the future source energy metric may be used in place of cost:

- Replace all references to cost with future source energy. Use an electric site-to-source energy conversion factor of 2.0 based on future projections for the U.S. A lower national average value may be used as applicable for projects outside of the U.S.
- Replace *ASHRAE 90.1-2019*, Table 4.2.1.1, Building Performance Factors (BPFs), with the BPFs derived for the future source energy metric in Table 1.

TABLE 1. *ASHRAE 90.1-2019*—Equivalent Building Performance Factors for a Future Source Energy Metric

Building Type	Climate Zone																		
	0A	**0B**	**1A**	**1B**	**2A**	**2B**	**3A**	**3B**	**3C**	**4A**	**4B**	**4C**	**5A**	**5B**	**5C**	**6A**	**6B**	**7**	**8**
Multifamily	0.74	0.69	0.73	0.70	0.73	0.70	0.71	0.70	0.63	0.70	0.71	0.69	0.68	0.70	0.70	0.68	0.68	0.68	0.74
Healthcare/ hospital	0.72	0.72	0.73	0.73	0.74	0.71	0.72	0.74	0.71	0.72	0.73	0.71	0.74	0.73	0.80	0.73	0.77	0.78	0.79
Hotel/motel	0.72	0.71	0.72	0.71	0.71	0.70	0.71	0.73	0.72	0.71	0.73	0.73	0.71	0.73	0.74	0.70	0.72	0.70	0.70
Office	0.62	0.63	0.61	0.62	0.58	0.60	0.57	0.62	0.55	0.55	0.61	0.57	0.58	0.61	0.59	0.58	0.60	0.54	0.58
Restaurant	0.65	0.62	0.63	0.61	0.62	0.58	0.63	0.63	0.63	0.67	0.66	0.66	0.70	0.70	0.68	0.73	0.72	0.74	0.77
Retail	0.57	0.54	0.53	0.53	0.48	0.47	0.47	0.47	0.47	0.52	0.50	0.56	0.57	0.53	0.59	0.58	0.56	0.53	0.60
School	0.57	0.57	0.58	0.57	0.55	0.54	0.57	0.51	0.49	0.48	0.51	0.52	0.51	0.53	0.51	0.53	0.50	0.51	0.58
Warehouse	0.28	0.30	0.24	0.27	0.23	0.24	0.27	0.23	0.20	0.33	0.26	0.28	0.40	0.32	0.29	0.44	0.38	0.40	0.44
All others	0.65	0.62	0.64	0.62	0.57	0.54	0.57	0.56	0.58	0.59	0.57	0.60	0.60	0.59	0.65	0.62	0.62	0.61	0.64

OR

OPTION 2. *ASHRAE 90.1-2022*

Comply with *ANSI/ASHRAE/IES Standard 90.1-2022*. Use any applicable compliance path in *ASHRAE 90.1*, Section 4.2.

For projects applying the Normative Appendix G, Performance Rating Method compliance path, one of the following metrics may be used in place of cost:

- **Future source energy**
  - Replace all references to cost with future source energy. Use an electric site-to-source energy conversion factor of 2.0 based on future projections for the U.S. A lower national average value may be used as applicable for projects outside of the U.S.
  - Replace *ASHRAE 90.1-2022*, Table 4.2.1.1, Building Performance Factors (BPFs), with the BPFs derived for the future source energy metric in Table 2.
- Site energy or source energy documented using *ASHRAE 90.1-2022*, Informative Appendix I.

TABLE 2. *ASHRAE 90.1-2022*—Equivalent Building Performance Factors for a Future Source Energy Metric

Building Type	Climate Zone																		
	0A	**0B**	**1A**	**1B**	**2A**	**2B**	**3A**	**3B**	**3C**	**4A**	**4B**	**4C**	**5A**	**5B**	**5C**	**6A**	**6B**	**7**	**8**
Multifamily	0.64	0.59	0.62	0.60	0.61	0.59	0.61	0.60	0.49	0.57	0.59	0.56	0.55	0.57	0.57	0.55	0.55	0.55	0.60
Healthcare/ hospital	0.64	0.64	0.66	0.65	0.66	0.63	0.64	0.65	0.63	0.64	0.65	0.62	0.64	0.62	0.69	0.63	0.68	0.69	0.70
Hotel/motel	0.65	0.63	0.64	0.63	0.62	0.61	0.62	0.63	0.62	0.59	0.60	0.60	0.57	0.58	0.59	0.56	0.58	0.56	0.56
Office	0.54	0.54	0.53	0.54	0.49	0.52	0.49	0.52	0.45	0.46	0.52	0.47	0.48	0.51	0.48	0.48	0.50	0.45	0.49

(continued)

TABLE 2. *ASHRAE 90.1-2019*—Equivalent Building Performance Factors for a Future Source Energy Metric (*continued*)

Building Type	Climate Zone																		
	0A	0B	1A	1B	2A	2B	3A	3B	3C	4A	4B	4C	5A	5B	5C	6A	6B	7	8
Restaurant	0.61	0.58	0.58	0.57	0.57	0.54	0.58	0.59	0.57	0.62	0.61	0.61	0.65	0.64	0.63	0.67	0.66	0.69	0.72
Retail	0.47	0.45	0.44	0.44	0.40	0.39	0.37	0.39	0.36	0.40	0.41	0.42	0.45	0.43	0.46	0.44	0.43	0.42	0.46
School	0.52	0.53	0.53	0.53	0.51	0.51	0.53	0.48	0.46	0.43	0.48	0.47	0.45	0.49	0.46	0.46	0.44	0.44	0.48
Warehouse	0.25	0.25	0.21	0.24	0.20	0.21	0.24	0.20	0.17	0.30	0.22	0.25	0.36	0.28	0.25	0.40	0.34	0.36	0.40
All others	0.58	0.56	0.56	0.56	0.50	0.47	0.49	0.48	0.48	0.49	0.49	0.50	0.51	0.50	0.55	0.52	0.52	0.52	0.55

REQUIREMENTS EXPLAINED

Widely referenced in building codes and regulations, *ASHRAE Standard 90.1* determines the minimum energy efficiency required for prerequisite compliance.

REQUIRED *ASHRAE STANDARD 90.1* VERSION

The required efficiency increases for project registrations beginning in 2028, stepping up from the *ASHRAE Standard 90.1-2019* to the more stringent *ASHRAE Standard 90.1-2022*. Compliance with *ASHRAE 90.1-2022* achieves a net average site energy savings of 14% compared to *ASHRAE 90.1-2019* when including savings for prescriptively required renewable energy.

OPTION 1. *ASHRAE 90.1-2019*

Only projects registered before January 1, 2028, can use *ASHRAE 90.1-2019*. Projects must apply Addendum cr, which requires a building envelope backstop for projects documented using the Energy Cost Budget (ECB) Method or Appendix G, Performance Rating Method. All other *ASHRAE 90.1-2019* addenda are optional.

OPTION 2. *ASHRAE 90.1-2022*

Projects registered after January 1, 2028, must use *ASHRAE 90.1-2022*.

Projects registered before January 1, 2028, can use the 2022 version of the standard to earn points under *EAc3: Enhanced Energy Efficiency*.

SUMMARY OF *ASHRAE 90.1*, SECTION 4.2.1 COMPLIANCE PATHS

Projects must choose from one of the following compliance paths from *ASHRAE 90.1*, Section 4.2.1:

- Prescriptive Method
  - *ASHRAE 90.1-2019*, Section 5–10 (for projects applying Option 1)
  - *ASHRAE 90.1-2022*, Section 5–11 (for projects applying Option 2)
- ECB Method
- Appendix G, Performance Rating Method (PRM)

ASHRAE 90.1 further distinguishes these three compliance paths for each type of construction:

- Section 4.2.1.1: New Buildings
- Section 4.2.1.2: Addition to Existing Buildings
- Section 4.2.1.3: Alterations of Existing Buildings (applicable to major renovations)

The path commonly referred to as the prescriptive method requires individual compliance with each referenced *90.1* section (building envelope, HVAC, service water heating, electrical power, lighting, other equipment, and in *ASHRAE 90.1-202*, additional efficiency requirements).

The ECB Method and Appendix G, Performance Rating Method offer greater flexibility in trading off performance between different systems. These methods rely on whole-building energy modeling to demonstrate that the proposed building performs at least as well as a project meeting the prescriptive requirements. For both methods, the building envelope backstop constrains building envelope performance.

To pursue additional credit for regulated energy efficiency in *EAc3: Enhanced Energy Efficiency*, project teams must apply either the prescriptive method or the Appendix G, Performance Rating Method.

KEY CHANGES IN *ASHRAE 90.1-2022*

The following table briefly summarizes key changes to the *ASHRAE Standard* between the *90.1-2019* and *2022* publications. For a more comprehensive summary, refer to *ASHRAE 90.1-2022*, Foreword and *ASHRAE 90.1-2022*, Informative Appendix M Addenda Description.

TABLE 3. Key Changes in *ASHRAE 90.1-2022*

Section	Key Change
Additional efficiency requirements Section 11	For the prescriptive method of compliance, designers select from a list of 33 energy efficiency measures to satisfy the total required Energy Credits for a given building type and climate zone, resulting in an average of 4%–5% savings. These additional savings are also accounted for when determining the ECB Method requirements and in the BPFs referenced for the Appendix G Performance Rating Method.
On-Site Renewable Energy Section 10.5.1.1	On-site renewable energy is required for the prescriptive method, averaging over 4% savings. The ECB Method and Appendix G Performance Rating Method factors in this prescriptive renewable contribution in determining compliance, although other efficiency measures can make up the difference for project designs without renewable energy. **Exceptions to On-Site Renewable Requirements** *ASHRAE 90.1*, Section 10.5.1.1 exempts buildings with inadequate access to solar energy and small buildings with less than 10,000 sq. ft. (929 sq. m.) of conditioned floor area for the three largest floors. The following additional guidance applies to LEED projects: • Where regulatory limitations restrict maximum rooftop solar capacity, use the lesser of the Section 10.5.1.1 criteria or the maximum capacity or area stipulated by regulatory limitations. • Canadian provinces and U.S. eGRID regions are not required to have on-site renewable. It is also not required in other countries where current annual electric grid emissions average less than 20 grams/kWh CO_2eq. • *ASHRAE 90.1-2022*, Addendum k provides additional alternatives for projects to find off-site renewable energy meeting the prescriptive requirements in lieu of the on-site renewable power.
Thermal Bridges Section 5.5.5	A new prescriptive section addresses thermal bridging in the building envelope. For projects using Section 5.6, Building Envelope Trade-off Compliance Path, the ECB Method, or the Appendix G, Performance Rating Method, the proposed design model must account for thermal bridges that do not meet prescriptive requirements.

EA

ASHRAE 90.1 MANDATORY PROVISIONS

All projects must meet applicable mandatory provisions from the referenced version of *ASHRAE 90.1*, found in Sections 5.4 (Building Envelope), 6.4 (HVAC), 7.4 (Service Water Heating), 8.4 (Power), 9.4 (Lighting), and 10.4 (Other Equipment).

Early in the design process, the project architect, engineer, and lighting designer should review these provisions and ensure they are integrated into the project design.

Exceptions to Mandatory Provisions

Project-specific exceptions or Project Priority Library alternatives to the mandatory provisions may apply to:

- Projects outside the United States where variations in equipment rating methodologies or limited availability of the required equipment or controls preclude compliance.
- Provisions exempted by the local authority having jurisdiction in areas regulated by codes of similar stringency to the referenced version of *ASHRAE 90.1*.
- Provisions available in *ASHRAE 90.1*, Appendix G PRM energy simulation.

FURTHER DESCRIPTION OF *ASHRAE 90.1*, SECTION 4.2.1 COMPLIANCE PATHS

PRESCRIPTIVE METHOD

For the prescriptive method, projects must meet all applicable requirements from each referenced *ASHRAE 90.1* section. The provisions of each section do not permit trade-offs.

- Section 5. Building Envelope
- Section 6. Heating Ventilation and Air Conditioning (HVAC)
- Section 7. Service Water Heating
- Section 8. Power
- Section 9. Lighting
- Section 10. Other Equipment

For *ASHRAE 90.1-2022*, Section 11. Additional Efficiency Requirements

This method provides a straightforward, easy-to-follow path to compliance, especially for smaller projects or those with less complex designs where flexibility is less of a concern.

The prescriptive method specifies minimum requirements for various building components, such as insulation levels, window performance, lighting power densities, HVAC system efficiencies, and system controls.[3]

Additional Considerations

EAc3: Enhanced Energy Efficiency, Option 1. Prescriptive Method, Path 1

CASE 1. *ASHRAE 90.1-2019*

Projects that comply with Option 1 of the prerequisite using the *ASHRAE 90.1-2019* prescriptive method may earn points for additional improvements to regulated systems by documenting

3. U.S. Department of Energy, Building Energy Codes Program, "ASHRAE Standard 90.1 Performance Based Compliance (Section 11 and Appendix G)," https://www.energycodes.gov/performance_based_compliance.

EA

achievement of energy credits using the methodology referenced in *ASHRAE 90.1-2022*, Section 11, Additional Efficiency Requirements.

CASE 2. *ASHRAE 90.1-2022*

Projects that comply with Option 2 of the prerequisite using the *ASHRAE 90.1-2022* prescriptive method are awarded 4 points under the New Construction rating system, or 2 points under the Core and Shell rating system.

Additional points may be awarded for incremental energy efficiency credits beyond the minimum required by *ASHRAE 90.1-2022*, Section 11, Additional Efficiency Requirements.

PERMISSIBLE TRADE-OFFS FOR PRESCRIPTIVE METHOD

Some sections provide an option for limited trade-offs within the section:

- Section 5.6, Building Envelope Trade-off Compliance Path, assesses the overall envelope performance compared to a prescriptively compliant envelope and permits trade-offs between building envelope components. For example, improved wall assembly U-factors may compensate for window-to-wall ratios that exceed the 40% prescriptive maximum. The project complies when the proposed envelope performance factor does not exceed the base envelope performance factor determined following the simplified modeling protocol in *ASHRAE 90.1*, Normative Appendix C. Software, such as the freely available COMcheck tool, automates this Appendix C energy simulation by building envelope data entry by the architect or design professional and completing these calculations in a matter of minutes.
- Section 6.6.2, Mechanical System Performance Path (*90.1-2022* only), uses simplified energy modeling to calculate the Total System Performance Ratio (TSPR) for the project's HVAC systems, which is typically used for office, retail, hotel/motel, multifamily/dormitory, and school or education buildings. Professional mechanical engineers without building energy modeling experience can calculate the TSPR using a software tool that automates the analysis.

ASHRAE 90.4 for spaces matching the *ASHRAE 90.1* definition of computer room such as data centers or server rooms:

- Section 6.6.1, Computer Room System Path
- Section 8.6.1, Computer Room Systems

OPTION 2. PRESCRIPTIVE METHOD, *ASHRAE 90.1-2022*, SECTION 11, ADDITIONAL EFFICIENCY REQUIREMENTS

For the *ASHRAE 90.1-2022* prescriptive method, designers must also select from a list of additional efficiency measures in Section 11, Additional Efficiency Requirements, to earn the minimum number of Energy Credits required for the project's building type and climate zone per Table 11.5.1-1. This gives the project team more flexibility to select the additional measures that are most feasible and appropriate for their project application.

Each of the efficiency measures referenced in Section 11 is awarded a specific number of base Energy Credits per building type and climate, which equates to approximately 0.1% savings per credit (See *ASHRAE 90.1-22*, Tables 11.5.3-1 through 11.5.3-9). Section 11.5.2 outlines opportunities for further adjustments to augment these base credits for certain efficiency measures.

For example, an office project in climate zone 4A earns 8 base credits for achieving a 5% reduction in lighting power (L06), but this increases to 16 credits for a 10% reduction based on the adjustment described in the detailed summary of this measure.

Combined credits for renewable and load management measures are limited to 60% of the total required energy credits per *ASHRAE 90.1-2022*, Section 11.5.2.

ASHRAE 90.1-2022, Addendum j allows projects to use the TSPR to demonstrate overall improvement in HVAC performance rather than applying individual system efficiency measures.

Additional considerations

For projects with multiple building types, the minimum required credits and credits achieved are weighted by the gross floor area of each building type.

Minimum required Energy Credits are adjusted lower than the default thresholds in Table 11.5.1.1-1 for certain project applications.

Core and Shell projects with central HVAC or service water heating must achieve 50% of the credits from Table 11.5.1.1-1. Other Core and Shell projects must achieve 33% of these credits.

Major renovations, referred to as substantial alterations in *ASHRAE 90.1*, must achieve 50% of the credits referenced in Table 11.5.1.1-1.

Projects without roof availability for PV or meeting Section 10.5.1 exceptions may use the referenced equation to adjust required credits below the default threshold per Section 11.5.1(e).

Unconditioned spaces, semiheated spaces, and parking garages must achieve 50% of the credits referenced for other building types in Table 11.5.1.1-1.

ASHRAE 90.1 4.2.1 COMPLIANCE PATH: ECB METHOD

This approach compares the annual energy cost of the proposed design to that of a budget building. The budget building is essentially a clone of the proposed design but adjusted to meet the prescriptive requirements. The proposed design achieves *ASHRAE 90.1-2019* compliance if the energy cost does not exceed the budget. For *ASHRAE 90.1-2022*, on-site renewable must be included in the budget building model when prescriptively required and meet an additional improvement below the ECB based on an adjustment referencing the prescriptively required energy credits from Section 11. Refer to the Building Envelope Backstop section referenced below for further guidance addressing limitations on envelope trade-offs when applying the ECB Method.

Avoid using the ECB Method to demonstrate improved regulated energy savings for *EAc3: Enhanced Energy Efficiency*.

ASHRAE 90.1 4.2.1 COMPLIANCE PATH: APPENDIX G, PERFORMANCE RATING METHOD

ASHRAE 90.1, Appendix G, Performance Rating Method, uses a stable baseline methodology that supports comparison of building performance across versions of *ASHRAE 90.1* using a variety of performance metrics. Using the PRM, an energy modeler can develop a single set of *ASHRAE 90.1*,

Appendix G, Baseline Building Design and Proposed Building Design models to document compliance with this prerequisite, with *EAc3: Enhanced Energy Efficiency*, and with any code requirements linked to *ASHRAE 90.1-2016* or later or *IECC-2018* or later.

The stable baseline methodology in the PRM requires a performance index (PI) less than or equal to the performance index target (PI_t), with further adjustments and limitations addressing on-site renewable energy. The scale for the PI ranges from one to zero, where one represents a baseline building that minimally complies with *ASHRAE 90.1-2004* requirements, and zero represents a net-zero building.

Calculate the performance index target using the results of the baseline building model completed under the PRM protocol, and the BPF for the project type and climate zone. The BPFs are provided in *ASHRAE 90.1,* Table 4.2.1.1, for the energy cost metric.

Refer to the Building Envelope Backstop section for further guidance addressing limitations on envelope trade-offs when applying the PRM method. Also, refer to *ASHRAE 90.1-2022*, G1.2.1(b) for similar requirements limiting trade-offs from interior lighting power.

Major renovations

Major renovations have slightly less stringent performance index targets than New Construction and are determined by multiplying published BPFs by a factor of 1.05 (see *ASHRAE 90.1-2019,* Addendum cr or *ASHRAE 90.1-2022*, 4.2.1.3).

APPENDIX G PRM. ALTERNATIVE METRICS IN LIEU OF ENERGY COST

Although the *ASHRAE 90.1*, Appendix G PRM references a cost metric, the energy modeler may instead demonstrate *EAp2: Minimum Energy Performance* compliance for the PRM using the future source energy metric referenced in *EAc3: Enhanced Energy Efficiency*, or a site energy metric, or a current source energy metric. Substitute all PRM references to cost with the new metric. Building performance factors to assess compliance must be specific to the metric and the referenced version of *ASHRAE Standard 90.1*.

EAc3: Enhanced Energy Efficiency only uses the future source energy metric and does not use energy cost, site energy, or source energy metrics (See *EAc3: Enhanced Energy Efficiency*, Option 2).

Energy modelers can limit the documentation level of effort by calculating both prerequisite and credit compliance using the future source energy metric.

OPTION 1. *ASHRAE 90.1-2019*, APPENDIX G PRM, ALTERNATIVE METRICS

- **Future source energy**. Calculate the PI_t for both the prerequisite and *EAc3: Enhanced Energy Efficiency* with BPFs from the prerequisite Table 1. 90.1-2019, Equivalent Building Performance Factors for a Future Source Energy Metric.
- **Site energy or source energy metric**. (Prerequisite only). Apply *ASHRAE 90.1-2019*, Addendum ch for a source energy metric matching those listed in Addendum ch, Table X4-1, follow Addendum ch, Section X5, Methodology for BPF Adjustment To Account for Localized Conversion Factors.

OPTION 2. *ASHRAE 90.1-2022*, APPENDIX G PRM, ALTERNATIVE METRICS

- **Future source energy metric**. Calculate the PI_t for the prerequisite with BPFs from the prerequisite *ASHRAE 90.1-2022*, Table 2. Equivalent Building Performance Factors, for a Future Source Energy Metric.
  - **Core and Shell projects**. Use the same BPFs to document the prerequisite and *EAc3: Enhanced Energy Efficiency*.
  - **New Construction projects**. Calculate the less stringent PI_t for *EAc3: Enhanced Energy Efficiency* using *ASHRAE 90.1-2019*, Table 6. Equivalent Building Performance Factors, for a future source energy metric.
  - **Site energy or source energy metric**. (Prerequisite only). Apply *ASHRAE 90.1-2022*, Informative Appendix I. For the source energy metric, use *ASHRAE 90.1-2022*, Appendix I5. Methodology for BPF Adjustment to Account for Localized Conversion Factors, if the project source energy conversions do not match those listed in Addendum Table I4-1.

APPENDIX G PRM. TREATMENT OF ON-SITE RENEWABLE ENERGY

OPTION 1. *ASHRAE 90.1-2019* PRM. TREATMENT OF ON-SITE RENEWABLE ENERGY

Per *ASHRAE 90.1-2019*, Section 4.2.1.1, the renewable energy contribution toward meeting PRM requirements is limited to 5% of baseline building performance.

This varies from *EAc3: Enhanced Energy Efficiency*, Option 2, which either includes or excludes the entire renewable contribution from the determination of credit compliance.

OPTION 2. *ASHRAE 90.1-2022* PRM. TREATMENT OF ON-SITE RENEWABLE ENERGY

Per *ASHRAE 90.1-2022*, Section 4.2.1.1, the modeler must perform three calculations of proposed building performance to account for the renewable energy contribution prescriptively required in Section 10.5.1, Renewable Energy Resources:

- Proposed building performance without any credit for the on-site renewable energy contribution.
- Proposed building performance including the on-site renewable energy contribution prescriptively required from Section 10.5.1.
- Proposed building performance including all on-site renewable energy contributions.

The renewable energy contribution toward meeting PRM requirements over and above the amount prescriptively required from Section 10.5.1 is limited to 5% of baseline building performance.

This varies from *EAc3: Enhanced Energy Efficiency*, Option 2, which either includes or excludes the entire renewable contribution from determination of credit compliance.

APPENDIX G PRM. EXCEPTIONS TO MANDATORY MEASURES

For mandatory measures, the modular may document the associated savings in the energy simulation and account for the measure's absence instead of complying with the mandatory provisions. Model the Baseline Building Performance (BBP) per Appendix G requirements and the Proposed Building Performance (PBP) as designed for measures that are not required for inclusion in the BBP, such as daylighting controls. For measures where Appendix G requires the PBP to match the BBP, model the measure as present in the baseline BBP and absent in the proposed PBP.

ECB AND APPENDIX G PRM. BUILDING ENVELOPE BACKSTOP

New buildings using the ECB Method or the Appendix G, Performance Rating Method, must limit building envelope trade-offs per *ASHRAE 90.1-2019,* Addendum cr or the associated requirements embedded in *ASHRAE 90.1-2022*. Doing so prioritizes building envelope performance that has a lifetime impact over more short-lived efficiency measures.

Projects must either meet the prescriptive building envelope requirements of *ASHRAE 90.1,* Section 5.5 or apply *ASHRAE 90.1,* Section 5.6, Building Envelope Trade-off Compliance Path, to demonstrate that the building envelope does not significantly underperform compared to a prescriptively compliant building envelope. (See more details for each path under the prescriptive method section of this guidance.)

If using the Building Envelope Trade-off Compliance Path, the proposed envelope performance factor cannot exceed the baseline envelope performance factor by more than 15% for residential occupancies or more than 7% for nonresidential occupancies.

For projects that cannot prescriptively comply with *ASHRAE 90.1* building envelope criteria, the design team should evaluate Building Envelope Trade-Off compliance early in the design process and make any necessary design changes to meet these minimum requirements. Compliance may prove particularly challenging for projects with high window-to-wall ratios.

Additional considerations

Projects pursuing points for *EAc2: Reduce Peak Thermal Loads* and using the Building Envelope Trade-Off Compliance Path must have a proposed envelope performance factor that does not exceed the baseline envelope performance factor.

EQUIVALENCE TO *ASHRAE 90.1*

Refer to the Project Priorities Library for regional paths addressing equivalence to *ASHRAE 90.1*.

For U.S.-based projects, state or local codes are equivalent to *ASHRAE 90.1-2019* or *ASHRAE 90.1-2022* when the U.S. Department of Energy Building Energy Codes Program status of state energy code adoption indicates a commercial code efficiency category matching the referenced version of *ASHRAE 90.1* or later for the project location in effect at the time of project permit application.

For projects documenting equivalence with *ASHRAE 90.1*, provide additional documentation to demonstrate compliance with the provisions of the envelope backstop referenced in *ASHRAE 90.1-2019* Addendum cr and *ASHRAE 90.1-2022*. Examples include the following:

- The project meets the prescriptive envelope requirements for the referenced code.
- The project complies with *ASHRAE 90.1-2019,* Addendum cr G1.2.1(c) requirements or *ASHRAE 90.1-2022* Section G1.2.1(d) requirements.
- The project complies with Thermal Energy Demand Intensity (TEDI) criteria specified in the referenced code requirements.

- The envelope weighted average UA (U-factor x Area) in all climate zones and the weighted average Solar Heat Gain Coefficient (SHGC) in climate zones 1 through 4 does not exceed the prescriptive maximum by an established percentage. Replace fenestration area exceeding the prescriptive maximum with opaque assemblies to determine the prescriptive maximum UA.
- The energy consumption model for the proposed design with a proposed envelope does not exceed the simulated energy consumption for the proposed design with a prescriptively compliant envelope by a given percentage.

The *2021 International Energy Conservation Code* (IECC 2021) and *ASHRAE 90.1-2019* with Addendum cr are equivalent.

The *2024 International Energy Conservation Code* (IECC 2024) with additional envelope backstop provisions and *ASHRAE 90.1-2022* are equivalent.

For both versions of the standard:

- *IECC,* Section C407, Total Building Performance may only be used to document prerequisite compliance with additional documentation demonstrating compliance with the envelope backstop.
- Use *IECC Prescriptive Compliance* (C402 through C406) instead of the *ASHRAE 90.1-2019* prescriptive method.

DISTRICT ENERGY SYSTEMS

ASHRAE 90.1 refers to district energy systems (DES) as purchased chilled water or purchased heat, even when this energy is part of a campus distribution plant and not directly purchased from a utility or municipality. Because DES systems are outside the project scope of work, DES efficiency cannot contribute toward achieving prerequisite compliance.

For projects that use the ECB method, model purchased heat and/or purchased chilled water as independent energy sources using the same utility rates per unit of energy for the energy cost budget and design energy cost models.

For projects that use the PRM, use one of the following modeling methods to document the DES:

METHOD A. *ASHRAE 90.1-2022* ADDENDUM A (REMOVES INHERENT DES PENALTY)

The project may apply *ASHRAE 90.1-2022*, Addendum a to either the *ASHRAE 90.1-2019* or *ASHRAE 90.1-2022*, Appendix G criteria as applicable to avoid the inherent penalty in the Appendix G, PRM Performance Index Targets when modeling purchased heat and purchased chilled water. Model HVAC systems for the baseline building design from the criteria in *ASHRAE 90.1-2022*, Addendum a, as if all heating and cooling generation equipment is on-site.

Model the proposed design with natural gas-forced draft boilers in place of district heating and water-cooled chillers in place of district cooling, matching the type and number specified in Addendum a. For projects using *ASHRAE 90.1-2019*, replace all *ASHRAE 90.1-2022*, Addendum a references to Section 6 prescriptive criteria for the proposed building design with *ASHRAE 90.1-2019*, Section 6.

Additional considerations: Linked credits

EAc3: Enhanced Energy Efficiency. Use Method A for prerequisite compliance if crediting DES efficiency toward the project performance in *EAc3: Enhanced Energy Efficiency*.

EAc4: Renewable Energy. In the energy simulation, use submetering to distinguish on-site fuel from the modeled fuel for the district hot water plant. This information informs the *EAc4: Renewable Energy*, Renewable Energy Attributes, Project Energy Source criteria.

METHOD B. DIRECTLY MODEL PURCHASED HEAT AND/OR PURCHASED CHILLED WATER

Apply the PRM requirements to model purchased heat and/or purchased chilled water as independent energy sources in the baseline and proposed design, with the same utility rates or source energy conversion factors for the baseline and proposed design. Use published utility rates or source energy conversion factors when available.

Otherwise, if purchased energy rates or source energy conversion factors are not published for the district energy sources serving the project, derive these purchased energy rates and/or conversion factors as follows:

- **Purchased chilled water:** Multiply the utility rate or source energy conversion factor for electricity by a factor of 0.325 to estimate the chilled water utility rate or source energy conversion factor.
- **Purchased heat from district hot water:** Multiply the utility rate or source energy conversion factor by a factor of 1.5 for the predominant fossil fuel source used to generate the district hot water or natural gas if unknown.
- **Purchased heat from district steam:** Multiply the utility rate or source energy conversion factor by a factor of 1.7 for the predominant fossil fuel source used to generate the district hot water or natural gas if unknown.

TABLE 4. Summary of EA Credit Linkages to EAp2: Minimum Energy Efficiency

	Option 1 ***ASHRAE 90.1-2019***	**Option 2** ***ASHRAE 90.1-2022***
Applicability	Limited to projects registered before January 1, 2028.	Available to all projects. Required for projects registered on or after January 1, 2028.
Linked Credits		
EAc3: Enhanced Energy Efficiency		
Prescriptive method (Option 1. Path 1 of the credit)	Required as a precondition to document points for additional efficiency. (See Case 1. *ASHRAE 90.1-2019*).	Points automatically awarded for prerequisite compliance using Prescriptive Method. (New Construction: 4 points. Core and Shell: 2 points). Further points for incremental efficiency beyond the minimum per *ASHRAE 90.1-2022* Section 11. Additional Efficiency Requirements (see Case 2. *ASHRAE 90.1-2022*).
Appendix G PRM (Option 2 of the credit) NOTE: Treatment of on-site renewable energy varies from prerequisite.	Incremental performance improvement documented using the future source energy metric earns points.	New Construction. Points rewarded for improvement beyond a *90.1-2019* equivalent-baseline using the future source energy metric. Performance target for prerequisite and credit are determined separately using simple calculations adjusting referenced BPF.
		Core and Shell: 2 points rewarded for meeting *ASHRAE 90.1-2022* using the future source energy metric. Further points for additional incremental improvements.

(continued)

TABLE 4. Summary of EA Credit Linkages to EAp2: Minimum Energy Efficiency (*continued*)

	Option 1 ***ASHRAE 90.1-2019***	**Option 2** ***ASHRAE 90.1-2022***
EAc2: Reduce Peak Thermal Loads		
Precondition for all credit options	Building envelope cannot perform worse than a prescriptively compliant envelope per *ASHRAE 90.1-2019* 5.5 or 5.6	Building envelope cannot perform worse than a prescriptively compliant envelope per *ASHRAE 90.1-2022 5.5* or *5.6*.
Option 3. Thermal Bridging		Comply with the prescriptive thermal bridging requirements of *ASHRAE 90.1-2022*, Section 5.5.5(a) without exceptions.
Prescriptive method, ECB, or Appendix G PRM (Option 4. Path 2, Envelope)	Rewards improvements to peak thermal loads documented using the *ASHRAE 90.1* Section 5.6 Building Envelope Trade-Off Method (used for the prerequisite for the prescriptive method or for the envelope backstop in the ECB Method, and PRM Method).	Rewards improvements to peak thermal loads documented using the *ASHRAE 90.1* Section 5.6 Building Envelope Trade-Off Method. (Used for the prerequisite for the prescriptive method or for the envelope backstop in the ECB Method, and PRM Method).
Prescriptive method (Option 4. Path 2, Ventilation)	TSPR Same documentation may be used to demonstrate peak thermal load reductions for ventilation, prerequisite HVAC compliance using the prescriptive method, and further improvements for *EAc3: Enhanced Energy Efficiency*.	TSPR. Same documentation may be used to demonstrate peak thermal load reductions for ventilation, prerequisite HVAC compliance using the prescriptive method, and further improvements for *EAc3: Enhanced Energy Efficiency*.
Appendix G PRM (Option 4. Path 3)	Same documentation may be used to demonstrate peak thermal load reduction for this credit and energy efficiency meeting the efficiency requirements for *EAp2: Minimum Energy Efficiency* and *EAc3: Enhanced Energy Efficiency*.	Same documentation may be used to demonstrate peak thermal load reduction for this credit and energy efficiency meeting the efficiency requirements for *EAp2: Minimum Energy Efficiency* and *EAc3: Enhanced Energy Efficiency*.

DOCUMENTATION

Project Types	**Options**	**Paths**	**Documentation**
New Construction and Core and Shell	All	All	Document compliance with mandatory measures from *ASHRAE Standard 90.1*.
		Prescriptive method	*ASHRAE 90.1* compliance forms or COMcheck compliance report confirming a system-by-system approach.

(continued)

Project Types	Options	Paths	Documentation
		Energy simulation (*ASHRAE 90.1* Appendix G PRM or ECB Method)	Input-output reports from modeling software.
			Energy consumption and demand for each building end use and energy source.
			Exceptional calculation and supporting documentation (if applicable).
			USGBC LEED v5 Energy Simulation Report.
			Supporting documentation for metrics, as applicable, including published reference for source energy conversion factors, published reference for utility rates or tariffs per energy source, and for projects attempting Enhanced Energy Efficiency, published reference for greenhouse gas emissions factors. Appendix I site or source energy calculations as applicable for projects using *90.1-2022.*
District Energy System	All	All	Documentation on DES including energy source serving the project (as applicable).

REFERENCED STANDARDS

- ASHRAE 90.1-2019 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2019-i-p?product_id=2088527)
- ASHRAE 90.1-2022 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2022-i-p?product_id=2522082)
- ASHRAE 90.4 (https://store.accuristech.com/ashrae/standards/ashrae-90-4-2022?product_id=2524333)
- IECC 2021 (https://codes.iccsafe.org/content/IECC2021P3)
- IECC 2024 (https://codes.iccsafe.org/content/IECC2024P1)

ENERGY AND ATMOSPHERE PREREQUISITE

Fundamental Commissioning

EAp3
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To improve energy performance and limit greenhouse gas emissions by verifying that systems are operating per the owner's project requirements.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Comply with Commissioning Requirements	

Comply with *ANSI/ASHRAE/IES Standard 90.1* commissioning requirements for building systems, controls, and the building envelope, with the following additional provisions:

- All projects shall provide fundamental commissioning. Section 4.2.5.2 exceptions shall not apply.
- The referenced version of *Standard 90.1* with errata shall be:
  - 2019 or later for projects registered before January 1, 2028.
  - 2022 or later for projects registered on or after January 1, 2028.
- By the end of the design development phase, the owner shall designate a commissioning provider (CxP) with experience completing commissioning on at least two projects of equal or larger scope and complexity.
- In addition to the requirements of the applicable version of *ASHRAE 90.1*, the CxP shall:
  - In predesign or as early as possible, assist in the development of the OPR, reviewing and updating the OPR through design and construction. OPR must include HVAC, service water heating, power, lighting, other equipment (include on-site renewable energy), and envelope.
  - During design, review the BOD for compliance with the OPR, and attend at least one meeting focused on mechanical, electrical, and plumbing, and one focused on envelope, which may be separate or combined, to discuss review comments and commissioning.

- During construction, review submittals and substitutions for design deviations that impact the OPR, attend milestone meetings at 50% and 100% completion, and perform a sample review (minimum 10%) of completed contractor documentation for quality assurance/quality control. For envelope, include testing in the commissioning documents and witness a sample of tests (not required for Core and Shell projects).
- Occupancy/operations phase: Develop an ongoing commissioning plan.

REQUIREMENTS EXPLAINED

The prerequisite requires that projects perform commissioning for building systems, controls, and the building envelope in compliance with the minimum requirements of *ASHRAE Standard 90.1* and additional LEED BD+C: New Construction and LEED BC+C: Core and Shell rating system provisions.

AUTOMATIC ACHIEVEMENT OF PREREQUISITE THROUGH *EAC5: ENHANCED COMMISSIONING*, OPTION 1

Projects that achieve all *EAc5: Enhanced Commissioning*, Option 1 requirements for the building enclosure and MEP systems automatically comply with the prerequisite requirements since the credit requirements encompass all commissioning requirements from the *ASHRAE 90.1* standard referenced in *EAp3: Fundamental Commissioning*.

Therefore, when planning the commissioning scope, review the *EAc5: Enhanced Commissioning*, Option 1 requirements, paying special attention to the required timing for CxP engagement during predesign or very early in the commissioning process to accomplish the broader commissioning scope of work required for credit compliance.

Tables 1 and 2 compare the required tasks for *EAp3: Fundamental Commissioning* and *EAc5: Enhanced Commissioning*, Option 1.

COMPLIANCE USING *EAP3 FUNDAMENTAL COMMISSIONING* PATH

REQUIRED *ASHRAE* STANDARD 90.1 VERSION

Projects registered before January 1, 2028, can reference *ASHRAE Standard 90.1-2019* or any later version. Projects registered on or after January 1, 2028, must use *ASHRAE Standard 90.1-2022* or later.

The commissioning requirements for the 2019 and 2022 versions of *ASHRAE 90.1* are similar. One notable addition to *ASHRAE 90.1-2022* that influences the commissioning scope is the requirement for whole-building air leakage testing for buildings less than 25,000 square feet (2,320 square meters).

Additional considerations

Using a single version of *ASHRAE Standard 90.1* for *EAp2: Minimum Energy Efficiency, EAp3: Fundamental Commissioning*, and *EAp4: Energy Metering and Reporting* streamlines documentation efforts.

Approved Equivalent Standards

Projects may use *IECC 2021* instead of *Standard 90.1-2019* and *IECC 2024* instead of *ASHRAE Standard 90.1-2022*. Projects that use IECC instead of ASHRAE must still comply with all additional LEED BD+C: New Construction and LEED BD+C: Core and Shell rating system requirements.

SELECTING THE COMMISSIONING PROVIDER

As buildings and systems become more complex and the systems required for commissioning (Cx) expand beyond traditional mechanical equipment, many CxPs will consist of a team instead of one person. Therefore, either an entity or an individual can act as the CxP. For projects that use a CxP entity, designating a single person as the Cx Project Lead or Manager ensures consistency in documentation and quality control of the process.

Minimum qualifications

The CxP must have direct commissioning experience from the design phase through the construction phase for at least two projects with equal or larger scope and complexity. The previous experience should address buildings of similar types and size range, similar types and capacities of HVAC and service water heating equipment, and controls with similar complexity, and the building envelope unless this scope of work is completed independently by a Building Envelope Commissioning Provider (BECxP).

Experience documented for a CxP entity must reflect the team performing the commissioning work for the project.

Eligible entities

Per *ASHRAE Standard 90.1-2019*, Section 4.2.5.2, the CxP must be completely independent of the design or construction team. Consider the following when selecting a CxP:

- The CxP can be a third-party entity not currently contracted for any design or construction aspects of the building or site.
- The owner can employ the CxP, provided the entity meets the CxP minimum qualifications.
- The CxP can be employed by the company performing the design or construction of the building; however, they must be a completely independent member of the design team and not directly associated with any aspect of the design and/or installation of the building systems.
- If the CxP employs any company that has direct influence on the design and construction, the Commissioning Plan must clearly address any potential conflicts of interest and demonstrate that the CxP acts and operates solely on behalf of the owner, reports directly to the owner, and works entirely independently from the design and construction team.

Verification and Testing Providers

ASHRAE Standard 90.1 also requires that the CxP includes Verification and Testing (V&T) providers. A V&T provider is an entity that completes the activities needed to implement the building functional performance testing (FPT) activities or verify that elements of the building project meet stated requirements. In many cases, the entity acting as the CxP fulfills this requirement. Confirm that an individual qualified in verification and FPT execution is part of the commissioning team.

CxP requirements for Building Envelope Commissioning

The CxP must include qualified individuals who can perform the building envelope design review and air barrier inspection or a V&T provider who can perform air leakage testing.

V&T PROVIDER FOR BUILDING ENVELOPE COMMISSIONING

ASHRAE Standard 90.1-2019, Section 5.4.3.1.1 requires an independent third party to test whole-building pressurization for air leakage. Alternately, a continuous air barrier design and installation verification program conducted by an independent third party meets the requirements when using Section 5.4.3.1.1, Exception 3. *ASHRAE Standard 90.1-2019*, Section 5.9.1.2, requires V&T, including a design review, periodic field inspection, and reporting.

Teams must confirm the compliance path for testing and ensure qualified individuals conduct each element of the commissioning efforts.

Timing for CxP engagement

Identify a CxP no later than the end of the design development phase. The CxP is primarily responsible for leading the design and construction team through all aspects of the MEP systems and building envelope systems commissioning, respectively.

Additional considerations

A single entity can perform all MEP and Building Envelope system commissioning efforts, provided the entity meets the minimum requirements.

For projects that use different qualified individuals to perform various Cx tasks, ensure sufficient collaboration within the team to provide continuity from design through operations. For example, an entity may have different CxP team members review the design documents than the team witnessing testing.

SCOPE OF FUNDAMENTAL COMMISSIONING

The prerequisite requires that projects conform with *ASHRAE Standard 90.1* requirements for all referenced systems and provisions outlined in the LEED BD+C: New Construction and LEED BD+C: Core and Shell rating system.

Projects using *ASHRAE Standard 90.1-2022* for compliance guidance must include relevant verification, testing, and commissioning for the additional efficiency measures and thermal break requirements from *ASHRAE Standard 90.1-2022*.

Systems requiring commissioning

At a minimum, commissioned systems must include all energy-using systems within the project boundary that are referenced in *ASHRAE 90.1*, Sections 5–10, including the building envelope, HVAC, service water heating, power, lighting, on-site renewable energy, energy monitoring systems, refrigeration equipment, energy storage systems, load management systems, and other energy-using systems within the project's boundary.

Owner's Project Requirements

The OPR documents the functional requirements of the building. It also details the expectations of the building's use and operation. The OPR includes objectives for the project, which verify that all stated goals integrate with the building design, construction, and operation.

The owner or a design professional can develop the OPR. However, the owner must provide input during the development, ensuring that the OPR captures the project's critical elements, like sustainability goals and targets. If a CxP's engagement begins before the OPR development, the CxP may provide input on the initial development efforts. The OPR is a living document requiring ongoing updates throughout the design phases.

Additional considerations

The owner plays a critical role in developing and updating the OPR. The OPR establishes a clear vision for the project, identifying expected outcomes and goals for sustainable building development. As the project progresses, decisions should align with the OPR. The owner must remain a key stakeholder and ultimate approver of the document's final version.

Basis of Design

The project's design professionals typically create the basis of design (BOD). The BOD explains how the design and construction team will execute the OPR. Include processes and assumptions made early in the design phases to achieve the OPR's intent in the BOD, along with relevant project information. A BOD, at minimum, addresses performance criteria, general building characteristics (e.g., envelope, HVAC, water), and governing codes and standards.

The BOD is a living document requiring updates throughout the design and construction phases.

Cx Plan

The Cx Plan, developed by the CxP, outlines the goals and objectives, general project information, and all systems included in the commissioning scope of work. The plan details the complete Cx process, including roles and responsibilities, key tasks and milestones performed by each responsible party, and FPT or verification procedures for all systems verified, commissioned, or tested.

Design Reviews

Design reviews by the CxP are critical elements of the commissioning process. Reviews support the energy efficiency goals of *ASHRAE Standard 90.1* by verifying that the design meets the standard's requirements. Early reviews allow teams to correct areas of the design that do not meet the requirements before construction begins, thereby avoiding costly change orders during construction.

The CxP must review the OPR, BOD, and design reviews and confirm that the construction documents include the required commissioning information.

The design reviews should confirm that the design meets relevant energy efficiency, energy metering and reporting, peak thermal load reduction, renewable energy, and grid-interactive requirements documented for the LEED Energy and Atmosphere (EA) credit category.

During the design phase, the CxP must participate in at least one coordination meeting to discuss design review comments.

Submittal Reviews and Functional Performance Test Development

During construction, the CxP must review submittals for equipment included in the Cx scope of work. This review allows the CxP to identify deviations from the design and OPR and provide comments to the owner, engineer, and contractor on any significant issues. Addressing these issues before procurement saves time and money by avoiding incorrect equipment purchases.

The CxP is the primary one responsible for developing the FPTs. FPTs written specifically for the equipment and systems designed for the project provide the most value. Therefore, CxPs must use the design team's approved submittals to develop any testing procedures for the project. The FPTs should cover all modes of operations, including seasonal testing.

Pre-commissioning Site Visits and Contractor Documentation Review

Determining commissioning readiness before execution is essential for minimizing potential delays from failed testing efforts. Through visual inspections and a sample review of the contractor's completed documents, the CxP can confirm the timing for FPT execution efforts.

The CxP must review at least 10% of the contractor's completed Cx documents. This quality assurance review allows the CxP to understand the quality of documentation efforts and identify any gaps in the process. Performing this review before the Cx readiness site visit helps the CxP to determine timing for the required site visit.

Before Cx execution, the CxP must complete at least one site visit to verify Cx readiness.

> **Additional considerations**
>
> Projects with phased construction and phased Cx testing should consider multiple Cx readiness site visits that align with each phase of the construction efforts.

Functional Performance Tests—Execution

The CxP must witness FPT executed by the contractors and subcontractors. Perform testing when all system components are installed, energized, programmed, balanced, and checked for functionality.

> **FPT SAMPLING**
>
> A sampling strategy is acceptable for functional testing of projects with a large number of similar system types, like an office with multiple variable air volume (VAV) boxes or a multifamily residential building with individual heat pumps for each tenant.

An acceptable sampling rate is typically 10%. The CxP should consider the testing procedure's failure rate when using a sampling rate. If multiple failures occur for the same equipment or system type, determine if there is a systemic issue.

> **Additional example—sampling rate and failure rate**
>
> A hotel has 200 fan coil units. The owner and CxP agree to a 10% sampling strategy. During testing, the CxP tests 20 fan coil units. If 10 of the 20 tests fail, conduct additional testing. Failures of that magnitude would cause more concern.

SEASONAL OR DEFERRED TESTING

When necessary, the CxP can use seasonal or deferred testing. For example, if the initial FPT effort occurs in the summer, tests for heating mode can occur during colder months. Once testing is complete, the CxP is responsible for amending the final report and other documents.

Meetings

During the construction phase, the CxP participates in 50% and 100% milestone meetings to discuss the commissioning findings and work toward resolving identified issues.

Final Cx Report

The CxP is the primary one responsible for authoring the final Cx report. The final report should include, at minimum, an executive summary of the Cx process and the results of the project's testing efforts, an updated issues and resolution log that identifies items that are closed and proposed resolutions for outstanding items, copies of the final versions of the OPR and BOD, design review logs, copies of the approved submittals used for the FPTs, and copies of the completed FPTs.

Provide a preliminary Cx report for projects that finalize the LEED application before completing Cx. The Report must address all major envelope, MEP, renewable, and grid-interactive systems, confirm system installation, and indicate that Cx has commenced for all systems.

The CxP must provide the Final Cx Report to the owner once the Cx is complete.

Ongoing Cx Plan

An ongoing Cx Plan ensures systems remain operationally efficient throughout the building's life. The plan should provide facility managers with procedures, blank FPTs, and a recommended schedule for ongoing Cx activities.

The ongoing Cx plan should address requirements for continuous documentation and updates. Building operations change over time, including retrofits or equipment replacement projects. Ensure the ongoing Cx plan reflects the most current information for the building.

KEY TASKS AND MILESTONES FOR COMMISSIONING

The Commissioning Provider is responsible for completing the following tasks to comply with *EAp3: Fundamental Commissioning* requirements:

- **Pre-design (or immediately upon engagement of the CxP no later than the end of design development)**
  - Assist in the development of the OPR.
  - Develop Cx Plan.
- **Design phase**
  - Review BOD.
  - Develop or approve Cx specifications.
  - Design document reviews (design drawings and specifications).
  - Attend at least one coordination meeting.
  - Assist in updating the OPR.

- **Construction phase**
  - Perform focused submittal reviews for design deviations that impact the OPR.
  - Perform field reviews.
  - Review/witness performance testing.
  - Attend milestone meetings at 50% and 100% completion.
  - Review sampling of QA/QC documentation (checklist and tests).
  - Track identified issues to resolution (I/R Log).
  - Develop a preliminary commissioning report.
- **Occupancy/operations phase**
  - Review training program.
  - Develop final commissioning report.
  - Develop or review the ongoing Cx plan.

COMPARISON OF FUNDAMENTAL *EAP3: FUNDAMENTAL COMMISSIONING* AND *EAC5: ENHANCED COMMISSIONING*

Table 1 for MEP systems and Table 2 for the Building Envelope provide a detailed comparison of key tasks and milestones for *EAp3: Fundamental Commissioning* and *EAc5: Enhanced Commissioning*.

TABLE 1. MEP System Tasks for *EAp3: Fundamental Commissioning* and *EAc5: Enhanced Commissioning*

Phase	Task Descriptions	Fundamental *ASHRAE 90.1-2019*	Enhanced *ASHRAE 202-2018*	Minimum Requirements as Listed
	Identification of CxP	Section 4.2.5.2, with timing required by LEED	Section 5.1.1, with timing required by LEED	**Fundamental:** By the end of design development phase. **Enhanced:** During predesign or very early in the design phase.
Pre-design For CxP engaged later than predesign, tasks must be completed immediately upon CxP engagement.	Assist in development/ review and update opr to include hvac, service water heating, power, lighting and other equipment	Required by LEED	Section 6.2, 6.3	**Enhanced:** OPR shall list and define the systems and assemblies to be commissioned, including sampling strategies accepted by the Owner. It should clearly define objectives, Cx scope and requirements and identify the number, format and scheduling of design and submittal reviews.
	Develop Cx plan	Section 4.2.5.2.1 Section 4.2.5.2.2	Section 7.2, 7.3	**Fundamental and Enhanced:** Development of Cx Plan. **Enhanced:** Update Cx plan at least once per phase (DD, CD, Construction).
Design phase	Review BOD	Required by LEED	Section 8.2, 8.3	**Enhanced:** Review BOD for compliance with OPR.
	Update Cx plan	Not required	Section 7.2, 7.3	**Enhanced:** Update Cx plan at least once per phase (DD, CD, Construction).
	Develop Cx specification	Section 4.2.5.1.1, 4.2.5.2.1, 6.9.2	Section 9.2, 9.3	

(*continued*)

EA

TABLE 1. System Tasks for *EAp3: Fundamental Commissioning* and *EAc5: Enhanced Commissioning* (*continued*)

Phase	Task Descriptions	Fundamental *ASHRAE 90.1-2019*	Enhanced *ASHRAE 202-2018*	Minimum Requirements as Listed
	Design document reviews (design drawings and specifications)	Section 4.2.5.2, 4.2.5.2.2	Section 10.2, 10.3	Full Drawing and Specification Review for systems to be commissioned. **Fundamental and Enhanced:** Detail compliance with the OPR and provisions in respective standards. Each design review to include issues log for tracking/resolution of issues. **Enhanced**: Back-check review to confirm if recommendations and comments have been addressed.
	Attend coordination/ design meetings to discuss review comments and commissioning	Required by LEED	Required by LEED	**Fundamental**: Minimum of one coordination or design review meeting discussing design review comments. **Enhanced**: Minimum of one additional coordination or design review meeting discussing review comments.
	Update to OPR	Required for LEED	Section 6.2, 6.3	**Fundamental and Enhanced**: Update OPR as needed prior to end of Design Phase.
Construction phase	Preconstruction kick-off meeting	Not required	Section 12.2.4	**Enhanced**: CxP conducts a Cx kick-off and scoping meetings with the Project Team to explain Cx procedures and coordinate Cx Activities throughout the Construction Phase.
	Update Cx plan	Not required	Section 7.2, 7.3	**Enhanced**: Update Cx plan at least once per phase (DD, CD, Construction).
	Submittal review	Required by LEED	Section 11.2, 11.3	**Fundamental**: Review submittals or substitutions for design deviations impacting the OPR. **Enhanced**: Thorough review of relevant building system submissions for compliance with the Design Documents and OPR.
	Schedule	Required relative to other tasks	Section 7.2.3.d	**Fundamental**: Ensure Cx requirements/ milestones are included in the project construction schedule. **Enhanced**: Detailed description of Cx activities and a schedule of activities. Schedule is included in the Cx plan.
	Field reviews	Section 4.2.5.1	Section 12.2.6	**Fundamental**: Minimum of one site visit to verify Cx readiness **Enhanced**: Minimum of one site visit to review contractor completed construction checklists. A checklist for each major system type should be reviewed during the site visit.
	Testing (Review/ witness performance testing)	Section 4.2.5.2	Section 12.2.6	**Fundamental and Enhanced**: Minimum of one site visit to witness execution of FPT, per the scope of the project.

TABLE 1. System Tasks for *EAp3: Fundamental Commissioning* and *EAc5: Enhanced Commissioning* (*continued*)

Phase	Task Descriptions	Fundamental *ASHRAE 90.1-2019*	Enhanced *ASHRAE 202-2018*	Minimum Requirements as Listed
	Meetings	Required by LEED	Required by LEED	**Fundamental and Enhanced**: Milestone meetings at 50% and 100%
	QA/QC documentation (checklist and tests)	Required by LEED	Section 12.2.2a, 12.2.6.c	**Fundamental**: Sample review of completed contractor documentation (i.e., 10%) **Enhanced**: Additional reviews of completed contractor documentation (i.e., 25%)
	Track identified issues to resolution (I/R Log)	4.2.5.1, 4.2.5.2	Section 13.2, 13.3	**Fundamental**: Include I/R Log in the preliminary Cx report. **Enhanced**: Maintain a formal I/R log throughout the project until the owner resolves or accepts all issues. The final I/R log, with all items closed, is included in the final Cx report.
	Systems manual	Not required	Section 14.2, 14.3	**Enhanced**: Compile the Systems Manual, which includes all information needed to understand, operate and maintain the building's systems and assemblies.
	Operations and maintenance manual	Not required	Required for LEED	**Enhanced:** Compile an Operations and Maintenance Manual from contractor submissions.
	Preliminary commissioning report	4.2.5.1.2, 4.2.5.2.2	Section 17.2.1	**Fundamental and Enhanced:** Completion of LEED Online documentation as well as report summarizing Cx activities to end of Construction Phase, Including OPR, Cx Plan and reports.
Occupancy/operations phase	Review training program	4.5.2.2.2.c.5	Section 15.2, 15.3	**Fundamental:** Review the training plan. **Enhanced:** Review the training plan and confirm that it has been implemented. Include training plan in the Systems Manual.
	Post-occupancy review	Not required	Section 16.2, 16.3	**Enhanced:** Conduct minimum of one in person, post-occupancy site visit with Facility Maintenance staff (or similar) prior to end of the warranty period.
	Final commissioning report	4.2.5.2.2	Section 17.2.3	Full report summarizing Cx activities, including Occupancy Phase activities.
	Ongoing Cx plan	Required for LEED	Required for LEED	**Fundamental and Enhanced:** Provide an ongoing Cx plan that allows building operators to maintain a building's high performance. At minimum, include a set of blank forms for future use by the O&M team.

TABLE 2. Building Enclosure (Envelope) Tasks for *EAp3: Fundamental Commissioning* and *EAc5: Enhanced Commissioning*

Phase	Task Descriptions	Fundamental *ASHRAE 90.1-2019*	Enhanced ASTM E2947-2021	Minimum Requirements as Listed
	Identification of CxP	Section 4.2.5.2, with timing required by LEED	Section 5.1.1, with timing required by LEED	**Fundamental:** By the end of design development phase. **Enhanced:** During predesign or very early in the design phase.
Predesign (or immediately upon CxP engagement)	Assist in development/review and update OPR to include envelope	Required by LEED	Section 6.3	**Fundamental and Enhanced:** Include building envelope requirements in OPR. OPR to be updated as needed during design and construction phases.
Design phase	Review BOD	Required by LEED	Section 7.2.3	**Fundamental and Enhanced:** Review BOD for compliance with OPR related to building envelope.
	Develop BECx plan	Section 4.2.5.2.1.a Section 4.2.5.2.2.a	Section 6.5 7.3.6 (DD phase) 7.4.5 (CD phase) 9.6 (Construction Phase)	**Fundamental:** Development of BECx plan. **Enhanced:** Update BECx plan at least once per phase (DD, CD, Construction).
	Design document reviews (design drawings and specifications)	Section 4.2.5.2.2.b	Section 7.1.1 7.2.3 (SD phase) 7.3.4 (DD phase) 7.4.3 (CD phase)	Full Drawing and Specification Review for critical barrier design and continuity. **Fundamental:** Focus on air and thermal barrier continuity and performance. **Enhanced:** Review air, thermal, moisture and vapor barrier continuity and performance. Each design review to include issues log for tracking/resolution of issues. **Enhanced:** Back-check review confirming if recommendations/comments have been addressed.
	Develop BECx specification	Section 4.2.5.1.1 Section 4.2.5.2.1.d	Section 7.4.6	
	Attend key building envelope focused design meetings	Required by LEED	Section 7.2.2 Section 7.3.2 Section 7.4.2	Fundamental: Minimum of one coordination or design review meeting to discuss design review comments. Enhanced: Number of meetings increased to each design phase (SD, DD, CDs)
	Update to OPR	Required by LEED	Sections 7.2.5, 7.3.5, 7.4.4	Fundamental and Enhanced: Update OPR as needed prior to end of Design Phase.
Construction phase	Pre-construction kick-off meeting	Not required	Section 9.2.1	Enhanced: BECxA to lead meeting, prepare agenda to minimally include discussion on roles/responsibilities, BECx plan and spec overview, schedule, summary of mock-up, site reviews and testing.
	Review submittals/ substitutions	Required by LEED	Section 9.2.2	Fundamental: Review submittals/substitutions for design deviations that impact the OPR. Enhanced: Thorough review of relevant building envelope submissions for compliance with the Design Documents and OPR.

TABLE 2. Building Enclosure (Envelope) Tasks for *EAp3: Fundamental Commissioning* and *EAc5: Enhanced Commissioning* (*continued*)

Phase	Task Descriptions	Fundamental *ASHRAE 90.1-2019*	Enhanced ASTM E2947-2021	Minimum Requirements as Listed
	Schedule	Required relative to other tasks	Section 9.2.2.3	Fundamental: Ensure BECx requirements and milestones are in the project Construction Schedule. Enhanced: Review and comment on Construction Schedules as required.
	Mock-ups	Not required	Section 9.3.3	Enhanced: Attend mock-up reviews (laboratory, factory, Performance mock-ups (PMU), in place mock-ups) as applicable.
	Field reviews	Section 4.2.5.2.2.c.3	Section 9.3.1	Fundamental: Minimum of one (1) per envelope assembly type. Enhanced: should include more regularly scheduled /periodic site reviews with a focus on early reviews (approx. 10% of installation).
	Testing (Review/ witness performance testing)	Section 4.2.5.2.2.c.3	Section 9.3.1	Fundamental: N/A Enhanced: Witnessing a sampling of envelope tests. May also include laboratory and mockup testing.
	Meetings	Required by LEED	Section 9.3.4	**Fundamental**: Milestone meetings at 50% and 100%. **Enhanced**: Additional milestone meetings (i.e., 25%, 50%, 75%, and 100% of envelope schedule minimum).
	QA/QC Documentation (Contractor's checklists)	Required by LEED	Section 9.2.2	**Fundamental**: Sampling review of building envelope contractor checklists (i.e., 10%). **Enhanced**: Review a minimum of 25% sampling of installer checklists.
	Operations and maintenance manual	Not required	Section 9.4	Confirm compliance with OPR, BOD, and BECx Plan.
	Construction phase BECx report	Section 4.2.5.2.2.c	Section 9.7	**Fundamental and Enhanced**: Completion of LEED Online documentation as well as report summarizing BECx activities to end of Construction Phase, including OPR, BECx Plan, and reports.
Occupancy/ operations phase	Review building envelope training program	4.5.2.2.2.c.5	Section 10.2	Review that training plan has been implemented.
	Post-occupancy review	Not required	Section 10.3	**Enhanced**: Conduct minimum of one in-person post-occupancy site visit with Facility Maintenance staff (or similar) before the end of envelope warranty period.
	Final BECx report	Section 4.2.5.2.2.d	Section 10.1	Full report summarizing BECx activities, including Occupancy Phase activities.
	Ongoing BECx plan	Not required	Section 10.4	**Enhanced**: Contribute to development of Preventive Maintenance Plan for Building Envelope (i.e., Ongoing BECx Plan) for owner.

CORE AND SHELL

For LEED BD+C: Core and Shell projects, complete commissioning for systems within the core and shell scope of work. Follow all BD+C: New Construction requirements with the following guidance.

SCOPE OF WORK

The commissioning scope of work for a Core and Shell project varies depending on the energy and water-using systems included in the design. For example, a project may consist of base building systems, like air-source heat pumps and central air handling units. Alternatively, the developer may limit the scope of work and provide a cold shell with no central HVAC equipment and minimal levels of lighting. Per the project scope, the CxP must verify and test systems.

DESIGN REVIEW FOR CORE AND SHELL PROJECTS

During the design reviews, include review comments on LEED BD+C: Core and Shell systems and how any incomplete system can meet *ASHRAE Standard 90.1* requirements for tenant fit-outs. This includes both energy efficiency measures and tenant metering requirements.

FINAL COMMISSIONING REPORT

Along with the required elements listed in the Final Commissioning Report section above, identify and defer any tests until base building systems connect with future tenant equipment (e.g., a central VAV air handling unit with controls testing that must be deferred until after installing tenant VAV terminal equipment).

TENANT GUIDELINES FOR COMMISSIONING

IPp4: Tenant Guidelines allow owners to inform all tenants of the building's sustainable design and construction features. Project teams are encouraged to include a section on commissioning that addresses any interconnection between base building and tenant-installed systems.

DOCUMENTATION

Project Types	Options	Documentation
All	All	If the report is a draft, include a plan for the completion of commissioning and training, including climatic and other conditions required for performance of any deferred tests.
		Confirmation of compliance with *ANSI/ASHRAE/IES Standard 90.1* commissioning requirements for building systems, controls, and the building envelope (Section 4.2.5.2 exceptions shall not apply).
		Confirmation of design phase meeting.
		Commissioning Plan and sample FPT Test scripts (one sample per discipline).
		OPR and BOD.
		Identification of Commissioning Provider, including key personnel (CxP) and V&T providers (as applicable).
		Qualifications of CxP and V&T providers.
		Ongoing Cx Plan (post-occupancy).
		Confirmation of construction phase milestone meetings at 50% and 100% completion.
		Confirmation that submittals were reviewed and at least 10% of the contractor's documents were QA/QC'd.

REFERENCED STANDARDS

- *ANSI/ASHRAE/IES Standard 90.1* commissioning requirements for building systems, controls, and the building envelope, with the following additional provisions:
- The referenced version of *Standard 90.1* with errata shall be:
  - 2019 for projects registered before January 1, 2028.
  - 2022 for projects registered on or after January 1, 2028.
- ASHRAE 90.1-2019 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2019-i-p?product_id=2088527)
- ASHRAE 90.1-2022 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2022-i-p?product_id=2522082)
- IECC 2021 (https://codes.iccsafe.org/content/IECC2021P3)
- IECC 2024 (https://codes.iccsafe.org/content/IECC2024P1)

Energy Metering and Reporting

EAp4
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To support energy management practices and facilitate identification of ongoing opportunities for energy and greenhouse gas emissions savings by tracking and reporting building energy use and demand.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Energy Monitoring and Recording AND	
Report Energy Data	

Comply with the following requirements:

Energy Monitoring and Recording

- Install (or use existing) devices to monitor and record energy use per *ANSI/ASHRAE/IES Standard 90.1.* The version of *Standard 90.1* shall be:
  - 2019 or later for projects registered before January 1, 2028.
  - 2022 or later for projects registered on or after January 1, 2028.
- Install (or use existing) devices to monitor and record energy use for the following, meeting the same monitoring and reporting requirements as required in *ASHRAE* for electrical end uses:
  - On-site renewable electricity generation.
- Major renovations and buildings eligible for exceptions to *ASHRAE 90.1-2019*, Section 10.4.6, or *ASHRAE 90.1-2022*, Section 10.4.7, must install measurement devices capable of monitoring whole-building energy use for each building energy source and building peak electricity demand at least monthly.

AND

Report Energy Data

Commit to reporting the following data to USGBC at least annually: monthly energy data for 12 consecutive months of total energy consumption for each energy source, on-site renewable energy generation, and peak electrical demand. This commitment must carry forward for five years or until the building changes ownership or lessee.

Exception for Core and Shell projects

Future tenant utility services and meters that will be installed in the tenant scope of work.

REQUIREMENTS EXPLAINED

This prerequisite requires the installation of devices to monitor and record monthly energy use per energy source and peak electric demand and to report monthly energy use data to USGBC post-occupancy for all buildings.

New construction projects subject to the *ASHRAE Standard 90.1* provisions referenced by this prerequisite must also provide metering and submetering of electricity at 15-minute intervals and additional reporting capabilities. Electrical system designers should evaluate these requirements early in the project design to ensure the electrical distribution, circuitry, and wiring necessary to accommodate the required submetering. Addressing these requirements too late in the design can substantially escalate costs.

Record and report data so owners and facility managers can access and use it to make informed decisions on energy efficiency and carbon emission reduction strategies.

Refer to Table 1 for a summary of the prerequisite energy metering and reporting requirements.

TABLE 1. Summary of LEED v5 EAp4: Energy Metering and Reporting Requirements

	Projects Subject to *ASHRAE 90.1* Monitoring and Reporting Requirements			
Monitoring	**Referenced *90.1* Sections OR LEED-Specific Requirements**	**New Buildings (Except Small Buildings)**	**Tenant Spaces ≥ 10,000 sq. ft. (Excludes Shared Building Systems)**	**Small Buildings and Major Renovations LEED-Specific Requirements**
Total energy use by energy source (except residential dwelling units) • Natural gas • Fuel oil • Propane • District chilled water • District steam • District hot water	*90.1-2019* 10.4.6.1 *90.1-2022* 10.4.7.1	Monthly		Monthly

(continued)

TABLE 1. Summary of LEED v5 EAp4: Energy Metering and Reporting Requirements (*continued*)

Monitoring	Projects Subject to *ASHRAE 90.1* Monitoring and Reporting Requirements			
	Referenced *90.1* Sections OR LEED-Specific Requirements	**New Buildings (Except Small Buildings)**	**Tenant Spaces ≥ 10,000 sq. ft. (Excludes Shared Building Systems)**	**Small Buildings and Major Renovations LEED-Specific Requirements**
Total electricity (except residential dwelling units)	*90.1-2019* 8.4.3.1 *90.1-2022* 8.4.3.1	15-minute	15-minute	Monthly
Submetered electricity (except residential dwelling units) • HVAC • Interior lighting • Exterior lighting • Receptacle circuits • Refrigeration systems (*90.1-2022*)	*90.1-2019* 8.4.3.1 *90.1-2022* 8.4.3.1	15-minute	15-minute	N/A
Large, chilled water plant electricity and efficiency in kW/ton	*90.1-2019* 6.4.3.11 *90.1-2022* 6.4.3.11	15-minute	15-minute	
On-site renewable electricity	LEED-Specific	15-minute		Monthly
Reporting				
Commit to annually sharing monthly data with USGBC for: • Energy consumption by energy source • On-site renewable energy generation • Peak electrical demand	LEED-Specific	x		x
Capable of creating user reports for consumption and demand at least hourly, daily, monthly, and annually, with a system capable of maintaining all data collected for 36 months.	*90.1-2019* 10.4.6.2 *90.1-2022* 10.7.4.2	x	x	
Visual Display				
Graphic display of electricity data for buildings with any of the following: • AHUs with fans > 10 hp • Chilled water plants • Hot water plants	*90.1-2019* 8.4.3 *90.1-2022* 8.4.3	x		
Graphic display of chilled water plant data for large, chilled water plants	*90.1-2019* 6.3.4.11.2 *90.1-2022* 6.3.4.11.2	x		

NOTE: Small buildings refer to commercial buildings less than 25,000 sq. ft. (2,323 sq. m.) or residential projects with less than 10,000 sq. ft. (929 sq. m.) of common space. All other new buildings are subject to *ASHRAE 90.1* monitoring and reporting requirements.

ENERGY MONITORING AND RECORDING

SMALL BUILDINGS AND MAJOR RENOVATIONS

ASHRAE 90.1 energy monitoring and reporting requirements do not apply to major renovations and small buildings eligible for exceptions to *ASHRAE 90.1-2019*, Section 10.4.6 or *ASHRAE 90.1-2022*, Section 10.4.7, including commercial buildings under 25,000 sq. ft. (2,323 sq. m.), and new residential buildings with less than 10,000 sq. ft. (929 sq. m.) of common space.

For these project applications, provide measurement devices capable of monitoring whole-building energy use for each building energy source and building peak electricity demand at least monthly. The prerequisite compliance does not require further submetering or interval metering.

Provide monitoring for all energy sources supplied to the project from outside the building boundary, including utility usage and energy supplied from a campus utility plant or adjacent building.

ASHRAE STANDARD 90.1 MONITORING AND RECORDING REQUIREMENTS

All other projects must comply with *ASHRAE Standard 90.1* energy monitoring and recording requirements:

Whole-Building Energy Monitoring Requirements

Electricity meters must be capable of metering and recording total project electricity use at 15-minute intervals. Refer to *ASHRAE 90.1*, Sections 8.4.3.1.

For all other fuels, provide measurement devices capable of monitoring whole-building energy use for each building energy source at least monthly. Include all energy sources supplied to the building from outside the project boundary. Refer to *ASHRAE 90.1-2019* 10.4.6.1 or *ASHRAE 90.1-2022* 10.4.7.1.

Electricity Submetering Requirements (*ASHRAE 90.1*, Section 8.4.3.1)

Projects must submeter end-use electricity data at 15-minute intervals for HVAC, interior lighting, exterior lighting, and receptacle; and if using *ASHRAE 90.1-2022*, refrigeration systems.

Combine electricity end uses less than 10% of the whole-building electrical load with other categories. For example, if exterior lighting loads are less than 10% of the whole-building load, teams can report exterior lighting with interior lighting. Use the energy end-use estimation from *EAp1: Operational Carbon Projection and Decarbonization Plan* to determine applicable loads.

ASHRAE Standard 90.1 Tenant Electricity Submetering (*ASHRAE 90.1*, Section 8.4.3.1)

Tenant spaces larger than 10,000 sq. ft. (929 sq. m.) require electricity submetering at 15-minute intervals, both for total tenant electricity use and for direct tenant loads for HVAC, interior lighting, exterior lighting, receptacle, and if using *ASHRAE 90.1-2022*, refrigeration systems. For acceptable grouping of end uses, reference the building-level 10% exception.

Exclude electricity from shared HVAC equipment (e.g., a central air handling unit providing supply air to the tenant space) when determining the tenant submetering requirements. The estimated load must include electricity for system components in the tenant space such as fan coil units and VAV terminals.

ASHRAE Standard 90.1 Reporting

Refer to *ASHRAE 90.1*, Sections 8.4.3.2 and *90.1-2019* 10.4.6.1 or *90.1-2022* 10.4.7.1.

The monitoring system must include the capability to report total and submetered electricity data at least hourly, daily, monthly, and annually, to report whole-building energy consumption monthly and annually, and maintain all data collected for 36 months.

The monitoring system must include functionality for tenants to access their electricity data.

Use third-party energy monitoring services or applications to comply with the data reporting and storage requirements.

Graphically display electricity data in buildings that are required to have digital control systems (buildings with air handling units with fans > 10 hp (7.5 kW), chilled water plants, or hot water plants).

ASHRAE Standard 90.1 Version

Projects registered before January 1, 2028, must reference the energy monitoring and recording requirements in *ASHRAE Standard 90.1-2019* or later. Projects registered on or after January 1, 2028, must use the slightly augmented energy monitoring and recording requirements from *ASHRAE Standard 90.1-2022*. For example, *ASHRAE 90.1-2022* adds a submetering requirement for refrigeration systems.

> **Additional considerations**
> Teams that use a single version of *ASHRAE Standard 90.1* for *EAp2: Minimum Energy Efficiency*, *EAp3: Fundamental Commissioning*, *and EAp4: Energy Metering and Reporting* can streamline documentation efforts.

International Energy Conservation Codes (IECC) Equivalent Standard

IECC is an approved equivalent standard for this prerequisite. *IECC 2021* requirements can replace *ASHRAE Standard 90.1-2019*. *IECC 2024* requirements can replace *ASHRAE Standard 90.1-2022*.

On-site renewable energy

All projects shall separately monitor on-site renewable energy generation. Projects subject to *ASHRAE Standard 90.1* monitoring and recording requirements must record renewable electricity data in 15-minute intervals. Major renovations or small buildings eligible for exceptions to *ASHRAE 90.1-2019*, Section 10.4.6 or *ASHRAE 90.1-2022*, Section 10.4.7 require only monthly data collection.

Energy use on a shared site

Projects in a campus application or shared site can address site lighting or other site energy use within the project's metered data or in a separate meter dedicated to the campus. For example, if there is a shared parking lot with exterior lighting, project teams can include metered data from all exterior lighting within the project's reported data. Teams can exclude exterior lighting if they confirm that the campus energy meter reports the required data.

District heating and district cooling sources supplied from outside the project boundary must be separately metered for the project and not through a shared campus energy meter.

Additional considerations: EVSE metering recommended
Consider submetering EVSE for vehicle charging to exclude this energy from the total energy consumption modeled for *EAc3: Enhanced Energy Efficiency* or used to assess compliance for LEED O+M: Existing Buildings.

REPORT ENERGY DATA

Report data annually to USGBC for at least five years post-occupancy. Data must include monthly peak electrical demand and monthly energy consumption from each energy source (including on-site renewable energy generation). Provide data using the USGBC-provided platform, which includes third-party interfaces with tools such as ENERGY STAR Portfolio Manager.

This valuable data enhances the understanding of building performance for project owners and managers. It also educates occupants and building users on behaviors that impact energy consumption and how positive behavioral changes can create better buildings.

USGBC aims to collect data from all LEED BD+C projects. Comparing data across similar project types allows for ongoing benchmarking of high-performing buildings within the LEED portfolio. The data influences refinements and enhancements to future LEED rating system requirements. Data shared with USGBC gives critical insight into the industry on the design, construction, and operation of high-performing buildings.

CORE AND SHELL PROJECTS

ENERGY MONITORING AND RECORDING FOR CORE AND SHELL
Comply with the Energy Monitoring and Recording criteria above for the project scope of work.

Provide meters for each utility connection installed in the project scope, each district energy source supplied to the project, and all on-site renewable energy.

If the project is a new commercial building that is at least 25,000 sq. ft. (2,323 sq. m.) or a residential project with at least 10,000 sq. ft. (929 sq. m.) of common space, comply with the *ASHRAE 90.1* monitoring and reporting provisions:

- At 15-minute intervals, provide electricity metering capable of monitoring total electrical use associated with the project scope of work. Include all common area and shared space electrical use, plus electric usage for tenant systems and equipment installed in the project scope.
- Provide electricity end-use submetering at 15-minute intervals for HVAC, interior lighting, exterior lighting, and receptacles installed within the project scope. Include all common area and site energy use, all shared HVAC equipment, and any HVAC, lighting, or receptacles installed in tenant spaces.
- For each tenant space ≥ 10,000 sq. ft. (929 sq. m.) with electrical systems in scope, provide electricity submetering at 15-minute intervals for total tenant electricity in scope, and

for HVAC, lighting, and receptacle end uses in scope. If the future tenant configuration is unknown, install the necessary circuitry, wiring, and hardware to accommodate the required submetering upon tenant build-out.

- Provide required recording and reporting functionality. If networking functionality is not operational upon substantial project completion, design the monitoring system to comply with the requirements upon activation of networking capabilities.
- If the required monitoring, data storage, and/or reporting functionalities are in the tenant scope, these may be excluded from the core and shell project scope.

REPORTING FOR CORE AND SHELL

Annually report energy data to USGBC for at least five years for the meters controlled by the owner.

Additional considerations

Teams can report all data, including tenant energy use. However, this is not required to meet the prerequisite.

Projects pursuing *IPc2: Green Leases*

Consider including information on how energy consumption for the whole-building energy use and in common areas is shared with tenants.

Consider requesting annual tenant energy disclosures, even if not pursuing *IPc2: Green Leases*.

RESIDENTIAL

The following are acceptable in place of reporting whole-building energy usage to USGBC:

- Provide a separate electric meter for each nontransient dwelling unit.
- Provide central metering per energy source for all common area uses and shared services. Report this metered data to USGBC.

Additional considerations

It is encouraged for residential dwelling units to aggregate and report whole-building energy consumption, including residential dwelling units, when not precluded by utility service restrictions or regulatory provisions. This data must be available for residential projects pursuing LEED EB:O+M certification.

EA

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	All	All	Confirmation of compliance with *90.1-2019* monitoring or *90.1-2022* requirements.
			Documentation showing monitoring and recording devices of all utilities, including renewable energy, district energy, electrical plans, schedules, or other documents that detail the required monitoring and recording devices.
			Confirmation from the project owner or responsible party that the required energy data will be shared with USGBC.
			List of energy sources delivered to the building.

REFERENCED STANDARDS

- ASHRAE 90.1-2019 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2019-i-p?product_id=2088527)
- ASHRAE 90.1-2022 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2022-i-p?product_id=2522082)
- IECC 2021 (https://codes.iccsafe.org/content/IECC2021P3)
- IECC 2024 (https://codes.iccsafe.org/content/IECC2024P1)

Fundamental Refrigerant Management

EAp5
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To reduce greenhouse gas emissions from refrigerants by accelerating the phaseout of refrigerants with high global warming potential and by reducing refrigerant leakage.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Option 1. No Refrigerants OR	
Option 2. Refrigerants	

OPTION 1. NO REFRIGERANTS

Do not use refrigerants in the project.

OR

OPTION 2. REFRIGERANTS

Meet the following requirements:

- Complete refrigerant inventory. Complete an inventory of the refrigerant-containing equipment installed within the project scope of work and any existing equipment owned by the building owner. The inventory shall include the refrigerant type, GWP, amounts of refrigerants contained in each, and the total GWP of all refrigerants.
- Do not use hydrochlorofluorocarbon refrigerants in new equipment.

- Evaluate available alternatives during the design process for any refrigerants with GWP > 700.
- Leak check and repair. Prior to substantial completion, check both new and existing refrigerant-containing equipment for refrigerant leaks and repair all identified leaks. For systems with field-assembled joints, perform a leak check, vacuum check, and pressure check prior to charging with refrigerant.

REQUIREMENTS EXPLAINED

Option 1 applies only to projects without refrigerants. Projects with refrigerants must follow Option 2.

OPTION 1. NO REFRIGERANTS

Buildings with no refrigerant-containing equipment automatically meet the prerequisite. Option 1 criteria does not preclude the use of equipment containing less than 0.5 lb (225 g), such as standard residential refrigerators, small wine coolers, or portable space dehumidifiers.

OPTION 2. REFRIGERANTS

Teams pursuing this path must avoid hydrochlorofluorocarbon (HCFC) refrigerants, analyze alternatives for refrigerants with a GWP greater than 700, inventory all refrigerant-using equipment, and ensure there are no leaks from refrigerant-containing equipment.

NO HCFC REFRIGERANTS

Projects must not use HCFC refrigerants, which cause damage to the ozone layer and often have very high GWP.

Developed and developing countries

In developed countries, government regulations have already phased out HCFC refrigerants for new equipment per the Kigali Amendment to the Montreal Protocol.

In developing countries, teams must take precautions to limit selections to equipment that does not use HCFCs. Equipment specifications must disallow HCFCs such as R-22 used in air conditioners, or R-123, commonly used in chillers.

EVALUATION OF ALTERNATIVE REFRIGERANTS

Global warming potential measures the relative contribution of a substance toward heating the atmosphere compared to the same mass of carbon dioxide (CO_2). For example, R-410A with a GWP of 2,088 traps 2,088 times more heat in the atmosphere than CO_2. This prerequisite references 100-year GWPs assessed in accordance with the IPCC Fourth Assessment Report or later.

Reviewing refrigerant properties during the design process allows teams to address high GWP refrigerants and find suitable alternatives before construction begins. As the industry continues developing new alternative refrigerants, projects can find cost-effective solutions that meet efficiency and environmental goals.

EQUIPMENT WITH GWP > 700

During the design process, if the specifications reference equipment with refrigerant GWP exceeding 700 or do not specify the refrigerant(s) to use, develop a list of alternative equipment options with refrigerant GWP less than or equal to 700. Review all proposed options with the owner.

Refrigerant properties vary in efficiency, toxicity, flammability, volumetric capacity, and pressure ratings. Not all refrigerants are interchangeable within a piece of equipment or system. Therefore, completing this evaluation early in design provides the most benefit.

For applications where a GWP less than 700 is impractical, consider using reclaimed refrigerant instead of newly manufactured virgin refrigerant to limit the overall impact.

Table 1 provides a common list of refrigerants and their GWP, adapted from the Net Zero Carbon Guide and The Heating, Refrigeration and Air Conditioning Institute of Canada (HRAI).

TABLE 1. Common Refrigerants and Their Applications

Refrigerant	Classification	GWP	Common System Applications
R404A	HFC blend	3,920	Low-medium temperature commercial refrigeration Low-medium temperature industrial refrigeration Ice Machines
R410A	HFC	2,088	Conventional VRF systems Heat pumps Chillers
R22	HCFC	1,810	Commercial refrigeration Industrial refrigeration Commercial air conditioning Residential air conditioning
R134a	HFC	1,430	Heat pumps Chillers
R32	HFC	633	Hybrid VRF systems Heat pumps Chillers
R513A	HFO	573	Medium temperature commercial refrigeration Medium temperature industrial refrigeration Chillers Air conditioning units Heat pumps
R600 (Butane)	HC	4	Heat pumps Chillers

TABLE 1. Common Refrigerants and Their Applications (*continued*)

Refrigerant	Classification	GWP	Common System Applications
R290 (Propane)	Natural	3	Commercial Refrigeration Heat Pumps Chillers
R1234ze	HFO	<1	Heat Pumps Chillers
R744 (CO_2)	Natural	1	Heat Pumps
R717 (NH_4, ammonia)	Natural	0	Heat Pumps

NOTE: VRF = variable refrigerant flow.

EQUIPMENT INVENTORY

Project teams must identify all refrigerant-containing equipment included in the scope of work, including any existing equipment within the project boundary owned or controlled by the project owner and/or facilities manager. Equipment that contains less than 0.5 pounds (225 grams) of refrigerant, such as standard residential refrigerators in dwelling units, can be excluded from the calculations.

Manage the inventory during the Construction Phase. If equipment substitutions occurred during the submittal and procurement phases, update the inventory to reflect the actual installed equipment.

Table 2 is a sample and noncomprehensive list of the types of refrigerant-using equipment that a project's scope of work may include.

Additional considerations: Equipment inventory

Completing the inventory in the Construction Documents Phase provides owners and design professionals with a complete understanding of the future climate impacts.

TABLE 2. Refrigerant-Containing Equipment

Application	Equipment/System Type
HVAC, space cooling equipment	Stationary air conditioners and heat pump Chillers Computer room air conditioning units
Service water heaters	Heat pump service water heaters
Retail	Food refrigeration Cold storage
Commercial	Vending machines Ice machines
Industrial process refrigeration	Process chillers Ice rink chillers Other process refrigeration

Data Collection

Document the refrigerant properties for each type of equipment, including type of refrigerant, refrigerant GWP ($GWP_{REFRIGERANT}$), and refrigerant charge (R_c).

When the project design includes field-assembled refrigerant piping with long pipe lengths or large pressure drops (e.g., variable refrigerant flow [VRF] systems, industrial process equipment), teams must account for additional required R_c in the calculations per the manufacturer's specifications or confirm that the manufacturer's default charge or referenced submittals already account for this additional charge.

Calculate the total equipment GWP for each equipment using Equation 1.

Equation 1. $GWP_{EQUIPMENT}$ calculation

$$GWP_{EQUIPMENT} = R_C \times GWP_{REFRIGERANE}$$

The project's total GWP is the sum of the GWPs for all refrigerant-using equipment in the project:

Equation 2. GWP_{TOTAL} calculation

$$GWP_{TOTAL} = \sum GWP_{EQUIPMENT}$$

Determine the weighted average GWP for the project by dividing the project's total GWP by the sum of Refrigerant Charge for all equipment:

Equation 3. Weighted average GWP calculation

$$\text{Weighted average GWP} = \frac{GWP_{TOTAL}}{\sum R_C}$$

LEAK CHECK AND REPAIR

Perform leak checks for all refrigerant-containing equipment in the project, including new and existing equipment. Field-installed piping requires vacuum and pressure testing during the installation process per the International Mechanical Code Chapter 11, *EPA Clean Air Act Section 608, European Union F-Gas Regulations*, or similar referenced standards.

For existing systems or self-contained systems, leak check inspections may leverage electronic leak detectors, data from the Building Automation System, visual inspections for oil residue on joints or for bubbling from leaks after applying soapy water, audible detection of hissing or bubbling sounds, and/or pressure testing.

RENOVATION PROJECTS

For major renovations, properly decommission refrigerant-containing equipment that has been removed or disposed of during construction. The U.S. EPA regulations (40 CFR Part 82, Subpart F) require refrigerant recovery and proper recycling, reclamation, or destruction of refrigerants classified as ozone-depleting substances. International projects must comply with other regional regulations, like the European Union F-gas regulation.

- **Recovery:** Extract all refrigerant from the equipment, including refrigerant in refrigerant piping. Store in a leak-free container.
- **Disposal options:** Recycle, reclaim, or destroy the recovered refrigerant as follows:
  - **Recycling**. Clean the refrigerant and reuse it on-site in other equipment owned by the same owner.
  - **Reclamation:** Clean the refrigerant for resale. The refrigerant must meet specific purity requirements. Reclamation efforts commonly occur in a dedicated processing facility.
  - **Destruction:** Incineration or other technologies break down the refrigerants into less harmful components that will not contribute to ozone depletion or high GWP.

Core and Shell only

For Option 2, address all refrigerant-containing equipment in the Core and Shell scope of work and any existing equipment within the boundary owned or controlled by the project owner and/or facilities manager. The inventory may exclude tenant equipment not installed as part of the Core and Shell scope of work.

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	Option 1. No Refrigerants	All	Description of the cooling and heating systems used for the project. Confirmation that no refrigerants are used within the project boundary and how the project meets cooling, heating, and other project loads without refrigerants.
	Option 2. Refrigerants		Narrative summarizing the evaluation of available alternatives for any refrigerant with a GWP >700. Include a list of the original selected refrigerants as compared to the alternatives considered.
			USGBC calculator.
			Narrative describing the refrigerant leak check evaluation. Include confirmation that the leak check has occurred for all equipment.
			Attestation that all leaks are repaired prior to charging with refrigerant.

REFERENCED STANDARDS

- EPA 2023 AIM ACT Technology Transitions Rule (https://www.epa.gov/climate-hfcs-reduction/regulatory-actions-technology-transitions)
- EPA regulations: 40 CFR Part 82, Subpart F (https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-82/subpart-F)
- 2021 International Mechanical Code Chapter 11 (https://codes.iccsafe.org/content/IMC2021P3)
- EPA Clean Air Act Section 608 (https://www.epa.gov/section608/section-608-clean-air-act)
- European Union F-Gas Regulations (https://eur-lex.europa.eu/eli/reg/2024/573/oj)

ENERGY AND ATMOSPHERE CREDIT

Electrification

EAc1
New Construction (1–5 points): 5 points are required for LEED BD+C: New Construction Platinum projects Core and Shell (1–4 points): 4 points are required for LEED BD+C: Core and Shell Platinum projects

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To encourage building design that does not depend on burning fuel on-site, which leads to better indoor and outdoor air quality and to low-carbon operations as the grid decarbonizes.

REQUIREMENTS: NEW CONSTRUCTION

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1–5**
Option 1. No On-Site Combustion OR	5
Option 2. No On-Site Combustion Except at Low Temperatures	1–4
Path 1. Space Heating AND/OR	2
Path 2. Service Water Heating AND/OR	1
Path 3. Cooking and Other Process Loads	1

OPTION 1. NO ON-SITE COMBUSTION (5 POINTS)

Design and operate the project from start-up with no on-site combustion except for emergency support systems.

Combined weighted average equipment efficiency for space heating and service water heating (SWH) must be at least 1.8 coefficient of performance (COP).

The following equipment may be excluded from the COP determination:

- Space-heating equipment in climate zones 0–2.
- Supplemental heating equipment designed only for operation at low temperatures.

- SWH equipment in nonresidential spaces complying with the point-of-use water heater criteria in *ASHRAE 90.1-2022*, Section 11.5.2.3.3, W05, without exceptions.

OR

OPTION 2. NO ON-SITE COMBUSTION EXCEPT AT LOW TEMPERATURES (1–4 POINTS)

Pursue any combination of the following paths for a maximum of 4 points:

PATH 1. SPACE HEATING (2 POINTS)

Design space heating to be capable of operating without on-site combustion except in low temperatures. Projects in climate zones 3 and above must have a weighted average space heating equipment efficiency of at least 1.8 COP.

The following equipment may be excluded from the COP determination:

- Supplemental or auxiliary heating equipment designed only for operation at low temperatures.

AND/OR

PATH 2. SERVICE WATER HEATING (1 POINT)

Design SWH systems to be capable of operating without on-site combustion except at low temperatures. Projects with total SWH capacity exceeding 34,000 Btu/hr (10 kW) must have a weighted average service hot water equipment efficiency of at least 1.8 COP OR domestic hot water solar fraction of at least 0.4.

The following equipment may be excluded from the COP determination:

- SWH equipment in nonresidential spaces complying with the point-of-use water heater criteria in *ASHRAE 90.1-2022*, Section 11.5.2.3.3, W05, without exceptions.
- Supplemental or auxiliary heating equipment designed only for operation at low temperatures.

AND/OR

PATH 3. COOKING AND OTHER PROCESS LOADS (1 POINT)

Design cooking, laundry, process equipment, and on-site power generation except emergency support systems to be capable of operating without on-site combustion. Projects that do not have these systems automatically earn this point.

The following equipment may be excluded:

- Process heating equipment designed for operation at low temperatures.

Equipment efficiency: Determine weighted average COP using either of the following:

- **Equipment efficiencies at rated conditions**: For equipment with multiple rated conditions, use the rating closest to 17°F (–9°C) outside air dry-bulb temperatures (OA db), 32°F (0°C) entering liquid temperature, or 44°F (6°C) heating source leaving liquid temperature.
- Annual average COP calculated with an energy simulation.

District energy: Projects with district energy must comply with the requirements of this credit at the district facility or see additional guidance for interpretation of credit requirements.

Fuel cells: Fuel cells using fossil fuel are ineligible for credit.

Low temperatures: The term *Low temperatures* refers to OA db below 20°F (–6.5°C).

REQUIREMENTS: CORE AND SHELL

ACHIEVEMENT PATHWAYS	POINTS
Core and Shell	**1–4**
Option 1. Electrification	2–4
Path 1. No On-Site Combustion	3–4
OR	
Path 2. No On-Site Combustion Except at Low Temperatures	1–3
Case 1. Space Heating	1–2
AND/OR	
Case 2. Service Water Heating	1
OR	
Path 3. No On-Site Combustion—Limited Scope	1–2
AND/OR	
Option 2. Electrification Readiness	1

OPTION 1. ELECTRIFICATION (2–4 POINTS)

For Paths 1 and 2:

- Include all heating and service hot water systems necessary to meet total building heating and SWH load in the calculations of weighted average COP.
- Future heating or SWH systems must be included in the calculations with a COP of 1.0.
- Future equipment may be excluded from the calculations and deemed as compliant when the applicable building code, or construction drawings for projects with tenancy, confirms a weighted average COP of at least 1.8 for future installed equipment.

PATH 1. NO ON-SITE COMBUSTION (3–4 POINTS)

Design and operate the project from start-up with no on-site combustion except for emergency support systems.

Combined weighted average equipment efficiency for space heating and SWH must be at least 1.8 COP for 4 points and at least 1.3 COP for 3 points.

The following equipment may be excluded from the COP determination:

- Space heating equipment in climate zones 0–2.
- Supplemental heating equipment designed only for operation at low temperatures.
- SWH equipment with a total project SWH capacity less than 34,000 Btu/hr (10 kW).
- SWH equipment in nonresidential spaces complying with the point-of-use water heater criteria in *ASHRAE 90.1-2022*, Section 11.5.2.3.3, W05, without exceptions.

OR

PATH 2. NO ON-SITE COMBUSTION EXCEPT AT LOW TEMPERATURES (1–3 POINTS)
Pursue any combination of the following cases for a maximum of 3 points:

Case 1. Space Heating (1–2 points)
Design space heating to be capable of operating without on-site combustion except at low temperatures. Projects must have a weighted average space heating equipment efficiency of at least 1.8 COP for 2 points and 1.3 COP for 1 point.

The following equipment may be excluded from the COP determination:

- Space heating equipment in climate zones 0–2
- Supplemental heating equipment designed only for operation at low temperatures

AND/OR

Case 2. Service Water Heating (1 point)
Design SWH systems to be capable of operating without on-site combustion except at low temperatures. Projects with total SWH capacity exceeding 34,000 Btu/hr (10 kW) must have a weighted average service hot water equipment efficiency of at least 1.8 COP or domestic hot water solar fraction of at least 0.4.

The following equipment may be excluded from the COP determination:

- SWH equipment in nonresidential spaces complying with the point-of-use water heater criteria in *ASHRAE 90.1-2022*, Section 11.5.2.3.3, W05, without exceptions.
- Supplemental heating equipment designed only for operation at low temperatures.

OR

PATH 3. NO ON-SITE COMBUSTION—LIMITED SCOPE (1–2 POINTS)
Do not install on-site combustion equipment in the project except for emergency support systems.

Combined weighted average equipment efficiency for space heating and SWH must be at least 1.8 COP. Points are awarded per Table 1 based on the qualifying minimum project scope of work.

The following equipment may be excluded from the COP determination:

- Space heating equipment in climate zones 0–2.
- Supplemental heating equipment designed only for operation at low temperatures.

- SWH equipment with a total project SWH capacity less than 34,000 Btu/hr (10 kW).
- SWH equipment in nonresidential spaces complying with the point-of-use water heater criteria in *ASHRAE 90.1-2022*, Section 11.5.2.3.3, W05, without exceptions.

TABLE 1. Points for No On-Site Combustion, Limited Scope

Minimum Project Scope of Work	Points
One or more heating, SWH, or process heating systems	1
At least 30% of the project's peak combined heating and SWH load	2

AND/OR

OPTION 2. ELECTRIFICATION READINESS (1 POINT)

Provide building infrastructure that ensures the capability of operating the building without on-site combustion except at low temperatures and of installing heating and SWH systems that will have a weighted average COP of at least 1.8.

Include the details for electrification readiness in the project plans and the tenant guidelines and include tenant guidance for designing and installing efficient electrified systems. Provide the following infrastructure as applicable to the project application and sized to ensure the capability to meet the requirements in Table 1:

- Dedicated physical space for future electric space heating, SWH, or process heating equipment. Provide designated spaces of sufficient size for outdoor heat pump equipment.
- Chase ways with space for refrigerant lines, condensate drainage, or other required piping.
- When electrical distribution systems are installed within the project scope, provide a junction box in the same physical space as the space allocated for the future electric equipment. Also provide dedicated electrical panel space for an appropriately phased branch circuit sized to accommodate the future electric equipment or appliances to meet the specified load.
- For portions of the building where ventilation air is not installed within the project scope of work, provide space and accommodations capable of supporting energy recovery ventilation for at least 50% of ventilation air.

For all options

EQUIPMENT EFFICIENCY

Determine weighted average COP using either of the following:

- **Equipment efficiencies at rated conditions**:
  - For equipment with multiple rated conditions, use the rating closest to 17°F (–9°C) OA db, 32°F (0°C) entering liquid temperature, or 44°F (6°C) heating source leaving liquid temperature.
  - Annual average COP calculated with an energy simulation.
- **District energy**: Projects with district energy must comply with the requirements of this credit at the district facility or see additional guidance for interpretation of credit requirements.

- **Fuel cells**: Fuel cells using fossil fuel are ineligible for credit.
- **Low temperatures**: Low temperatures refer to outside air dry-bulb temperatures below 20°F (–6.5°C).

REQUIREMENTS EXPLAINED: NEW CONSTRUCTION

This credit rewards decarbonization achieved through electrifying building systems traditionally fueled with on-site combustion, including space heating, SWH, cooking, and other process equipment.

Under Option 1, maximum points are available to projects that fully electrify all building systems. Option 2 selectively rewards electrification per system category for heating, SWH, cooking, and other process loads and affords flexibility for a hybrid design capable of limiting on-site combustion to low-temperature operation.

For both options, electrified space heating and SWH equipment must meet efficiency criteria to limit undue burden on the electric power grid. For further guidance, refer to the Weighted Average COP section.

OPTION 1. NO ON-SITE COMBUSTION

Electrification

Projects designed to operate entirely without on-site combustion for building energy use or for district energy supplied to the building offer the most significant emission reduction through electrification. Projects pursuing this option must eliminate on-site combustion from the building system design and operations. Refer to the Exemptions section below for limited exceptions.

Efficiency

The combined weighted average equipment efficiency for applicable space heating and SWH equipment must be at least 1.8 COP per the guidance in the Weighted Average COP section below.

OPTION 2. NO ON-SITE COMBUSTION EXCEPT AT LOW TEMPERATURES

For Option 2, teams may apply any combination of Path 1, Path 2, and Path 3, which separately address electrification of space heating, service hot water heating, and cooking and other process systems, respectively.

PATH 1. SPACE HEATING

Electrification

All space heating systems must operate without on-site combustion, except in low-temperature operating mode at or below 20°F (–6.5°C).

Design electrified space heating equipment with sufficient capacity to meet the entire project space heating load at the system-, zone- and space-level for outdoor temperatures above 20°F (–6.5°C), or the project's design heating temperature. Hybrid designs with fuel/electric equipment must have a sequence of operations with at least one all-electric operating mode above 20°F (–6.5°C).

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Credit achievement is automatic for projects that neither install space heating equipment in the project scope nor require space heating for occupant thermal comfort.

Efficiency

Projects in climate zones 3 and above must design space heating to achieve a weighted average equipment efficiency of at least 1.8 COP. Refer to the Weighted Average COP section for calculations and exclusions from COP determination.

PATH 2. SERVICE WATER HEATING

Electrification

All SWH must operate without on-site combustion, except in low-temperature operating mode at or below 20°F (–6.5°C).

Service water heating supplies hot water for purposes other than space heating and process applications. It is primarily for handwashing, showering, and cleaning.

Design electrified SWH equipment with sufficient capacity and distribution capability to provide all necessary SWH at outdoor temperatures above 20°F (–6.5°C) or the project's design heating temperature.

Path 2 is unavailable to projects without SWH.

Efficiency

If the total project SWH capacity exceeds 34,000 Btu/h (10 kW), design the SWH system with efficient heat pump technology to achieve a weighted average SWH equipment efficiency of at least 1.8 COP or generate at least 40% of the building's total SWH load with solar thermal energy.

Point-of-use SWH equipment in nonresidential spaces may be excluded from the weighted average COP determination if it meets *ASHRAE 90.1-2022*, Section 11.5.2.3.3, W05, without exceptions.

Refer to the Weighted Average COP section for calculations and exclusions from COP determination.

PATH 3. COOKING AND OTHER PROCESS LOADS

Electrification

This path encourages design teams to eliminate on-site combustion from cooking, laundry, pool or spa heating, power generation, and all other process applications.

Other process applications commonly addressed through electrification include process heating, process drying, pre-conditioned air or 400 Hz systems in airports, and powering of vehicles or equipment operated exclusively on the project site (e.g., forklifts or golf carts).

All process systems must operate without on-site combustion except in low-temperature operating mode for outdoor dry-bulb temperatures at or below 20°F (–6.5°C).

Projects automatically comply with Path 3 when electricity powers all building systems and equipment except space conditioning systems, SWH systems, and systems referenced in the Exemptions section below.

REQUIREMENTS EXPLAINED: CORE AND SHELL

To achieve full electrification of a core and shell project upon final build-out with efficient heat pump technologies, the core and shell scope of work must either include the design and installation of these systems or incorporate electrification readiness strategies to support future installation of these systems.

The credit rewards projects by the degree of electrification and readiness addressed in the core and shell scope of work.

OPTION 1. ELECTRIFICATION

Path 1 and Path 2 require a substantial proportion of heating and SWH equipment installed in the project scope of work, included in current tenant construction drawings, or dictated by local code.

Refer to the Weighted Average COP section for further background and guidance.

PATH 1. NO ON-SITE COMBUSTION

Refer to New Construction Option 1 above.

For a Core and Shell project scope that addresses only a portion of combined space heating and SWH loads, the 1.3 COP weighted average COP threshold worth 3 points is more attainable than the 1.8 COP threshold worth 4 points.

> **Example**
> A dedicated outside air system with 3.0 COP heat pump supplies 15% of total combined heating and SWH required for the project, achieving a weighted average COP of 1.3. No other systems are in scope.

PATH 2. NO ON-SITE COMBUSTION EXCEPT AT LOW TEMPERATURES

Case 1. Space Heating

Refer to New Construction Option 2, Path 1 above.

For a Core and Shell project scope that addresses only a portion of space heating loads, the 1.3 COP weighted average COP threshold worth 1 point is more attainable than the 1.8 COP threshold worth 2 points.

Case 2. Service Water Heating

Refer to New Construction Option 2, Path 2 above.

PATH 3. NO ON-SITE COMBUSTION—LIMITED SCOPE

This path applies to projects that have inadequate scope to achieve points under Path 1 or Path 2.

Projects must fully electrify any space heating or SWH systems in scope and achieve a combined weighted average equipment efficiency of at least 1.8 COP for these systems per the guidance in the Weighted Average COP section below.

Minimum scope must include at least one space heating, SWH, or process heating system for 1 point; and at least 30% of the project's combined heating and SWH load for 2 points.

OPTION 2. ELECTRIFICATION READINESS

Electrification of building heating systems directly reduces carbon emissions in buildings. For Core and Shell projects, the exact project scope may vary, and project teams may only provide the infrastructure without selecting and installing all heating generation equipment. Under this circumstance, demonstrate that the provided building infrastructure supports future equipment that does not rely on combustion for space heating, SWH, and process heating. Refer to the rating system language for specific electrification readiness strategies that must be included in the project design if applicable to the project application.

Provide tenant guidelines that explain the electrification readiness strategies and address the design and installation of efficient electrified systems.

REQUIREMENTS EXPLAINED: BOTH NEW CONSTRUCTION AND CORE AND SHELL (ALL OPTIONS)

WEIGHTED AVERAGE COP

The minimum weighted average COP criteria is predicated on a design that uses heat pump technology instead of inefficient electric resistance heating to meet most of the project's space heating and SWH loads. This COP criteria limits increased grid peak demand associated with electrification.

EXCLUSIONS FROM WEIGHTED AVERAGE COP DETERMINATION

Optionally exclude the following equipment from the weighted average COP determination:

- Space heating equipment in climate zones 0–2. This equipment is not required because electric resistance heating contributes much less to peak grid load in hot climates than in cooler climates. Refer to *ASHRAE Standard 169*, "Climatic Data for Building Design Standards"[4] to determine the project's climate zone.
- Supplemental heating equipment designed only for operation at or below 20°F (–6.5°C).
- This equipment can skew the average efficiencies calculated using capacity weightings of rated efficiencies. This exclusion applies to supplemental or auxiliary electric heating used for space heating or SWH. It is also fuel for low-temperature operation for space heating or SWH in Option 2 (Core and Shell Option 1, Path 2).
- Point-of-use SWH equipment in nonresidential spaces meeting *ASHRAE 90.1-2022*, Section 11.5.2.3.3, W05, without exceptions. Electric point-of-use water heaters are often more appropriate than centralized heat pump equipment and negligibly increase peak electric demand for nonresidential projects with low SWH demand distributed throughout the building. Therefore, teams may exclude point-of-use water heaters in nonresidential spaces

EA

4. ASHRAE, "Climatic Data For Building Design Standards," ASHRAE Standard 169, 2021, https://www.ashrae.org/file%20library/technical%20resources/standards%20and%20guidelines/standards%20addenda/169_2020_a_20211029.pdf.

from the COP determination if they comply with *ASHRAE 90.1-2022*, 11.5.2.3.3, W05 without exception.

- Nonresidential SWH equipment that does not comply with the *ASHRAE 90.1-2022*, 11.5.2.3.3 criteria must be included in determining the weighted average COP. Examples include storage water heaters supplying showers or commercial kitchen operations.
- This exception does not apply to equipment in residential spaces due to higher SWH demand for these space types.
- Service water heating equipment for projects with a total SWH capacity of less than 34,000 Btu/h (10 kW). For New Construction, this refers to the installed capacity. For Core and Shell, this refers to the total capacity necessary to meet project loads, regardless of whether the equipment is in the project scope.

For Core and Shell, future equipment where applicable building code or construction drawings for projects with tenancy confirm a weighted average COP of at least 1.8.

Projects do not need to calculate the weighted average COP if all equipment meets one of the criteria above. For example, there is no required minimum COP for a nonresidential project in climate zone 1 with point-of-use water heating for 100% of its SWH load per *ASHRAE 90.1-2022*, 11.5.2.3.3.

When installing heat pump space heating for a project in climate zones 0–2 or heat pump SWH is installed for projects with a total SWH capacity of less than 34,000 Btu/h (10 kW), the analyst may include all heating and SWH equipment in the calculations to demonstrate the required COP.

INCLUDED EQUIPMENT FOR WEIGHTED AVERAGE COP DETERMINATION

- Include all equipment not specifically excluded above in the weighted average COP determination.
- **Option 1. No On-Site Combustion**. (Core and Shell Option 1, Path 1) Include all space heating and SWH not specifically excluded above.
- **Option 2. No On-Site Combustion Except at Low Temperatures**. (Core and Shell Option 1, Path 2)
  - **Path 1. Space Heating**. (Core and Shell, Case 1) Include all space heating equipment not specifically excluded above.
  - **Path 2. Service Water Heating**. (Core and Shell, Case 2) Include all SWH equipment not specifically excluded above.
- Projects pursuing both Path 1 and Path 2 may optionally show a combined weighted average equipment efficiency for space heating and SWH of at least 1.8 COP rather than a weighted average COP per system.
- **Core and Shell, Option 1. Path 3. No On-Site Combustion—Limited Scope**. Include all space heating and SWH equipment included in the project scope of work not specifically excluded above.

For New Construction, assess the weighted average COP based on total installed equipment capacity.

Core and Shell only

Path 1. No On-Site Combustion and Path 2. No On-Site Combustion Except at Low Temperatures. Assess weighted average COP based on the total capacity necessary to meet project loads, regardless of whether the project scope installs the equipment. Use a COP of 1.0 for future heating or SWH capacity that is not in the project scope.

Path 3. No On-Site Combustion—Limited Scope. Assess weighted average COP based on the project scope of work.

METHODOLOGIES FOR WEIGHTED AVERAGE COP DETERMINATION

Projects may determine weighted average COP using a streamlined method, rated capacities, or energy simulation.

Method 1. Streamlined Weighted Average COP Determination

The streamlined method conservatively estimates compliance based on *ASHRAE 90.1*, Section 6.8 mandatory rated heating efficiencies of at least 2.0 COP for heat pumps and heat recovery chillers.

- To confirm a weighted average COP of at least 1.8, document that at least 80% of equipment capacity consists of heat pumps, heat recovery chillers, or solar heating.
- To confirm a weighted average COP of at least 1.3, document that at least 30% of equipment capacity consists of heat pumps, heat recovery chillers, or solar heating.

Method 2. Energy Simulation

Calculate the COP by dividing total annual heating generation by total annual heating energy consumption (using consistent units in numerator and denominator) per Equation 1.

Equation 1. COP calculation using energy simulation data

$$\text{COP} = \frac{\text{Total annual heating generation}}{\text{Total annual heating consumption}}$$

Include all applicable space heating and SWH energy used at the plant, system, and zone levels.

Projects may use any of the following to document the weighted average COP:

- Modeling used for *EAp2: Minimum Energy Efficiency*
  - *ASHRAE 90.1*, Appendix G Performance Rating Method, proposed model.
  - *ASHRAE 90.1*, Energy Cost Budget Method, design energy cost model.
  - TSPR. Simplified model from *ASHRAE 90.1-2022*, Section 6.6.2.2, Mechanical System Performance Rating Method. This only shows space heating efficiency. Demonstrate SWH compliance using one of the other methods.
- Energy simulation used to document local code compliance.
- Simplified energy simulation used to estimate energy consumption for *EAp1: Operational Carbon Projection and Decarbonization Plan*, provided the model inputs include sufficient detail relevant to equipment efficiencies, capacities, and loads estimations.

For Core and Shell Paths 1 and 2, modifications to these models or post-processing of modeled results may be necessary to show a COP of 1.0 for all future capacities not in the project scope of work.

Method 3. Rated Capacities

Calculate the weighted average COP based on the capacity-weighted average rated equipment efficiency per Equation 2.

Equation 2. Weighted average COP

$$\text{Weighted Average COP} = \frac{\sum \text{rated capacity of each equipmentrated COP of each equipment}}{\text{total rated equipment capacity of all equipment}}$$

If equipment has more than one rated condition, calculate the weighted average COP using the rated conditions closest to the following:

- **Air-source heat pumps**. 17°F (–9°C) OA db
- **Ground source heat pumps**. 32°F (0°C) entering liquid temperature
- **Liquid source heat pump and heat recovery water-chilling packages**. 44°F (6°C) heating source leaving liquid temperature.

Reference *ASHRAE 90.1*, Section 6.8 Tables, to identify applicable rated conditions.

For equipment with efficiency ratings using heating season performance factors (HSPF), annual fuel utilization efficiency (AFUE), or any rating other than COP or coefficient of performance heating (COPH), convert these ratings to COP using Table 2 before calculating the weighted average COP.

Additional considerations

For heat pump water-chilling packages or heat recovery water-chilling packages rated per *ASHRAE 90.1*, Table 6.8.1-16, adjust COP using the equations from Table 2 to align equipment ratings for entering/leaving heating liquid temperature at medium, high, or boost conditions with the default low ratings.

TABLE 2. Determination of Equipment COP for Calculation of Weighted Average COP

Heating Equipment Type	Heating Equipment Efficiency Rating	Equation to Convert to COP	Test Procedure
Electrically operated air-cooled unitary heat pumps	HSPF2	$= -0.0296 \times \text{HSPF22} + 0.7134 \times \text{HSPF2}$	AHRI 210/240-2023
	SCOP2H	$= -0.3446 \times \text{SCOPH22} + 2.434 \times \text{SCOPH2}$	AHRI 210/240-2023
	COPH at 17°F db/15°F wb (-8.3°C db/–9.4°C wb)	= COPH	AHRI 340/360
PTHP	COPH	= COPH	AHRI 310/380
SPVHP	COPH	= COPH	AHRI 390
VRF air cooled	HSPF	$= -0.0296 \times \text{HSPF2} + 0.7134 \times \text{HSPF}$	AHRI 1230
	COPH at 17°F db/15°F wb (-8.3°C db/–9.4°C wb)	= COPH	AHRI 1230
	SCOPH	$= -0.3446 \times \text{SCOPH2} + 2.434 \times \text{SCOPH}$	AHRI 210/240-2023

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TABLE 2. Determination of Equipment COP for Calculation of Weighted Average COP (*continued*)

Heating Equipment Type	Heating Equipment Efficiency Rating	Equation to Convert to COP	Test Procedure
VRF water source	COPH 68°F (20°C) entering water	= COPH	AHRI 1230
VRF groundwater source	COPH 50°F (10°C) entering water	= COPH	AHRI 1230
VRF ground source	COPH 32°F (0°C) entering water	= COPH	AHRI 1230
Electrically operated DX-DOAS air-source heat pump or water-source heat pump	ISCOP	= ISCOP	AHRI 920
Electrically operated water-source heat pump, water-to-water, water loop	COPH 68°F (20°C) entering water	= COPH	ISO 13256-1
Electrically operated water-source heat pump, water-to-air, groundwater	COPH 50°F (10°C) entering water	= COPH	ISO 13256-1
Electrically operated water-source heat pump, brine-to-air, ground loop	COPH 32°F (0°C) entering water	= COPH	ISO 13256-1
Air-source heat pump and heat recovery chiller packages	COPH at 17°F db/15°F wb (–8.3°C db/–9.4°C wb), Low leaving heating water temperature = 105°F (40°C)	= COPH	AHRI 550/590
	COPH at 17°F db/15°F wb (–8.3°C db/–9.4°C wb), Medium leaving heating water temperature = 120°F (50°C)	= 1.14 × COPH	
	COPH at 17°F db/15°F wb (–8.3°C db/–9.4°C wb), High leaving heating water temperature = 140°F (60°C)	= 1.37 × COPH	
Water source electrically operated positive displacement. (COPH evaluated at 54°F (19°C) source water entering temperature/44°F (7°C) source water leaving temperature)	COPH Low leaving hot water temperature = 105°F (40°C)	= COPH	AHRI 550/590
	COPH Medium leaving hot water temperature = 120°F (50°C)	= 1.26 × COPH	
	COPH High leaving hot water temperature = 140°F	= 1.73 × COPH	

TABLE 2. Determination of Equipment COP for Calculation of Weighted Average COP (*continued*)

Heating Equipment Type	Heating Equipment Efficiency Rating	Equation to Convert to COP	Test Procedure
Water source electrically operated positive displacement or centrifugal. (at 75°F/65°F source entering/leaving water temperature)	COPH Boost leaving hot water temperature = 140°F (60°C)	= 1.31 × COPH	AHRI 550/590
Electric furnace	AFUE	= 1.0 × AFUE	10 CFR 430 Appendix N
Electric boiler	None listed	COP = 0.96 if not rated	10 CFR 430
Electric service hot water heaters	COP 50°F (10°C) entering air and 60°F (21°C) entering water	= COP	AHRI Standard 1301
	UEF	= UEF × 1.3	10 CFR 430 Appendix E

NOTE: HSPF = heating season performance factors; SCOP = seasonal coefficient of performance; COPH = coefficient of performance heating; PTHP = package terminal heat pump; SPVHP = single package vertical heat pump; VRF = variable refrigerant flow; db = dr bulb; wb = wet bulb; ISCOP = integrated season coefficient of performance; AFUE = annual fuel utilization efficiency; UEF = uniform energy factor
NOTE: Equations to convert HSPF, HSPF2, and SCOP2 to COP are from *ASHRAE 90.1-2019*, 11.5.2(c) and *90.1-2022*, 12.5.2(c). The remaining conversions are rough approximations not accounting for variability in standby losses or other factors referenced in the test procedures.

DISTRICT ENERGY SYSTEMS (APPLICABLE FOR NEW CONSTRUCTION AND CORE AND SHELL)

If the project has thermal energy from a DES, either refer to DES compliance paths in the Project Priorities Library or demonstrate that the DES complies with the following credit requirements at the district facility:

- **New Construction Option 1. No On-Site Combustion**. (Core and Shell Option 1, Path 1)
  - No on-site combustion may be used in the district energy facility to generate heating, cooling, or electricity supplied to the project. Average district heating efficiency must be at least 1.8 COP.
- **New Construction Option 2. No On-Site Combustion Except at Low Temperatures**. (Core and Shell Option 1, Path 1)
  - If district heating supplies any of the project's space heat (New Construction Path 1 or Core and Shell Case 1), service water heating (New Construction Path 2 or Core and Shell Case 2), or process energy (New Construction Path 3), the district heating facility must be capable of generating the entire required district heating capacity without using on-site combustion above 20°F (–6.5°C). Average district heating efficiency must be at least 1.8 COP for New Construction Paths 1 and 2 (Core and Shell Cases 1 and 2).

FUEL CELLS AND COMBINED HEAT AND POWER INELIGIBLE

Projects with fuel cells that use fossil fuel to generate electricity or combined heat and power systems (CHP) that use fuel to generate heat and electricity are ineligible for credit.

Exemptions (Applicable for New Construction and Core and Shell)

On-site combustion may be used for the following limited circumstances:

- **Emergency support systems:** Emergency support systems generate electricity, heating, or cooling upon failure of the primary system in a power outage or extreme temperature event.

    - To apply the exemption, the sequence of operations must specifically limit emergency support system operations to emergency events associated with power disruptions or extreme temperatures falling outside of ASHRAE 99% design conditions for a prolonged period.
    - Emergency power systems are not exempt for locations where power outages commonly occur for more than 200 hours per year.
- **Portable equipment:** Exclude portable equipment for outdoor cooking amenities or outdoor patio heating not in the project scope when limited to less than 200 hours per year. Do not permanently pipe fuel lines to the equipment.
- **Special circumstances:** Use limited on-site combustion for specialty cases where on-site combustion is integral to system function. Examples include
    - Portable laboratory equipment
    - Wood-fired ovens in commercial kitchens fitted with emissions control devices
    - Vocational schools using fossil fuels solely for the purpose of training
    - Electric incinerators for medical waste
    - Fireplaces, hearths, and fire pits exclusively designated to support ceremonial practices (such as those unique to indigenous cultures or religious practices that require the use of fire)

These special circumstances do not extend more broadly to process equipment except where approved on a project-specific basis for systems with very low greenhouse gas emissions.

PLATINUM REQUIREMENTS

Projects aiming to achieve LEED Platinum certification must meet the following criteria applicable to the project type.

- **New Construction:** Achieve 5 points (no on-site combustion).
- **Core and Shell:** Achieve 4 points (no on-site combustion).
- **Projects served by DES:** Refer to the Project Priorities Library for DES compliance paths.

The Platinum requirements intentionally encompass industrial and manufacturing processes. To obtain Platinum certification, on-site boilers used for manufacturing production must comply with the credit criteria.

Exceptions to Platinum requirements, such as for renewable fuels, may be included in the Project Priorities Library.

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	All		Documentation of all heating, SWH, solar water heating, and process heating equipment within the project. Include information on the energy source, equipment quantity, and the capacity and efficiency for each piece of equipment (equipment cutsheets or schedules).
			Description of emergency support and on-site generation equipment, including how it is used on-site. Include an estimated annual run time for any combustion equipment.
			Document emergency support systems and on-site generation equipment including system type, fuel source, and capacity.
			Documentation that the project is subject to local code that requires full electrification down to 20°F or lower. Provide relevant code language and applicability (if applicable).
			Weighted average COP calculation, as applicable for SWH and space heating. Projects may determine weighted average COP using a streamlined method, rated capacities, or energy simulation.
	Option 1		Narrative or mechanical drawings showing that systems used for heating, SHW, and cooking and other process loads are not fueled by on-site combustion.
New Construction	Option 2	All	Sequence of operations for hybrid electric/nonelectric systems and evidence that electric equipment (electric mode) can meet space heating, SHW, and process heating loads >20°F (as applicable). Evidence may include equipment capacities and accompanying design load calculations, or energy simulation reports, or other.
			For projects attempting no on-site combustion except at low temperatures, document how the applicable systems can operate without on-site combustion at outside air temperatures above 20°F (–6.5°C).
		Path 1	Total building space heating load at OA Temp 20°F or below.
			Provide a description of the basis of analysis for space heating loads.
			Sequence of operations for all space heating equipment.
		Path 2	Sequence of operations for all SWH equipment.
			Sequence of operations for all solar water heating equipment.
		Path 3	Confirmation that the project's emergency support systems (if installed) use on-site combustion, and that the system will only run during power outages and for less than 200 hr/year.
		Path 3	Systems in scope of work (cooking, laundry, other process).
		Path 3	Schedules or cutsheets showing electrified equipment.

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Project Types	Options	Paths	Documentation
Core and Shell Only	Option 1	Path 2	For projects attempting no on-site combustion except at low temperatures, document how the applicable systems can operate without on-site combustion at outside air temperatures above 20°F (–6.5°C).
	Option 1	Path 3	Calculation for the installed space heating and SWH equipment capacity as a percentage of the project's peak combined heating and SWH load.
			Description of emergency support and on-site generation equipment and how it is used on-site. Include estimated annual run time for any combustion equipment.
	Option 2		Documentation of base building design showing applicable infrastructure, including floor plans, mechanical plans, electrical plans, and one-line diagrams.
			Narrative describing the sizing and capabilities to meet future needs.
			Tenant guidelines describing electrification readiness and design and installation of efficient electrified systems.
New Construction and Core and Shell			District Energy System Relevant documentation and calculations demonstrating that the relevant DES criteria have been met (as applicable).

REFERENCED STANDARDS

- ASHRAE 90.1-2019 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2019-i-p?product_id=2088527)
- ASHRAE 90.1-2022 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2022-i-p?product_id=2522082)

Reduce Peak Thermal Loads

EAc2
New Construction (1–5 points)
Core and Shell (1–5 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To minimize demand on grid resources and improve the resilience of buildings.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–5**
Option 1. Infiltration and Balanced Ventilation AND/OR	1
Option 2. Ventilation Energy Recovery AND/OR	1
Option 3. Thermal Bridging AND/OR	1
Option 4. Peak Thermal Load Reductions	1–3
Path 1. Peak Load Intensity OR	1–3
Path 2. *ASHRAE 90.1* Trade-Off Methods OR	1–3
Path 3. Energy Simulation	1–3
Residential	**1–5**
Option 1. Infiltration and Balanced Ventilation AND/OR	2
Options 2, 3, and/or 4	1–5

Comply with any combination of Options 1–4 for a maximum of 5 points.

For all options, the building envelope must meet the requirements of *ASHRAE 90.1*, Section 5.5, Prescriptive Building Envelope Compliance Path or *ASHRAE 90.1*, Section 5.6, Building Envelope Trade-Off Compliance Path per the version of *ASHRAE 90.1* referenced in *EAp2: Minimum Energy Efficiency*. Building envelope efficiency shall not be traded off with other building systems.

OPTION 1. INFILTRATION AND BALANCED VENTILATION (1 POINT)

Comply with both of the following:

BALANCED VENTILATION

Design the ventilation and exhaust airflows within 10% of each other and include a testing, adjusting, and balancing (TAB) report demonstrating balanced ventilation in the commissioning scope. This requirement does not apply to Core and Shell projects.

AND

INFILTRATION

Use an air leakage test to demonstrate a measured air leakage of the building envelope less than or equal to Table 1. Buildings smaller than 25,000 square feet (2,322 square meters) must use a whole-building air leakage test.

- Complete air leakage testing using ASTM *E779*, *ANSI/RESNET/ICC 380*, *ASTM E3158*, *ASTM E1827*, or equivalent.
- For buildings greater than 5,000 sq. ft. (465 sq. m.), maximum air leakage is determined per square feet or square meter of building envelope area (including exterior walls, roofs, and base floor/slab).
- For projects that include both new construction and major renovation, use the weighted average maximum air leakage.

TABLE 1. Caps on Air Leakage Rates

Building Conditioned Floor Area (CFA)	Pressure Test Conditions Across the Building Envelope	Maximum Air Leakage	
		New Construction	Major Renovation
≥ 5,000 sq. ft. (465 sq. m.)	At pressure difference of 50 pascals (0.2 in H_2O)	0.13 cfm/sq. ft. (0.65 L/s/m²)	0.20 cfm/sq. ft. (1.0 L/s/m²)
	At pressure difference of 75 pascals (0.3 in H_2O)	0.18 cfm/sq. ft. (0.90 L/s/m²)	0.27 cfm/sq. ft. (1.35 L/s/m²)
< 5,000 sq. ft. (465 sq. m.)	At 50 pascals (0.2 in in H_2O)	1 ACH	1.5 ACH
	At 75 pascals (0.3 in H_2O)	1.35 ACH	2 ACH

NOTE: cfm = cubic feet per meter; L/s/m² = liters per second per square meter; ACH = air changes per hour.

OR

Residential (2 points)

Balanced Ventilation. Meet the New Construction requirements.

AND

INFILTRATION

- Compartmentalize each residential dwelling unit to minimize leakage between units. Perform a blower door test of residential dwelling units, following the procedures in *ANSI/RESNET/ICC 380* or equivalent. For each unit tested, demonstrate a maximum leakage of enclosure area that is no more than 1.5 times the thresholds identified in Table 1 (enclosure area refers to all surfaces enclosing the dwelling unit, including exterior and party walls, floors, and ceilings). Demonstrate a weighted average leakage of the enclosure area for the building, including dwelling units, that complies with the caps in the limits identified in Table 1.

AND/OR

OPTION 2. VENTILATION ENERGY RECOVERY (1 POINT)

Each fan system supplying outdoor air must have an energy or heat recovery system with a minimum 70% enthalpy recovery ratio or a minimum of 75% sensible heat recovery ratio. Provisions must be made to bypass or control the energy recovery system during moderate outside air conditions.

In aggregate, fan systems supplying less than 15% of the project's total outdoor air can be excluded.

Core and Shell only

Fan systems must supply a minimum of 50% of the ventilation air required for the project per *ASHRAE Standard 62.1-2022*, Ventilation Rate Procedure, to qualify for this option.

AND/OR

OPTION 3. THERMAL BRIDGING (1 POINT)

Comply with the prescriptive thermal bridging requirements of *ASHRAE 90.1-2022*, Section 5.5.5(a), including all applicable requirements of Sections 5.5.5.1–5.5.5.5, without applying exceptions for projects in climate zones 0–3.

AND/OR

OPTION 4. PEAK THERMAL LOAD REDUCTIONS (1–3 POINTS)

Comply with the following:

- Ventilation loads must be included in the determination of peak coincident loads.
- Measure building envelope air leakage using air leakage testing and use the measured air leakage to calculate the peak loads for Path 1, Path 2 (envelope), and Path 3. (Meeting the leakage rates in Option 1 is not required to pursue this option.)
- Demonstrate balanced ventilation meeting the criteria in Option 1.

AND

PATH 1. PEAK LOAD INTENSITY (1–3 POINTS)

Limit the sum of peak heating load and peak cooling load per unit of treated floor area to be less than or equal to the thresholds specified in Table 2. Calculate peak loads using one of the following:

- WUFI Passive Design Tool, following the Passive House Institute U.S. protocol.
- Passive House Planning Package, following the Passive House Institute protocol to determine maximum heating load and maximum cooling load.

TABLE 2. Points for Meeting Caps on the Sum of Peak Heating and Cooling Loads

Points	New Construction	Major Renovation
1	16 Btu-h/sq. ft. (50 W/sq. m.)	20 Btu-h/sq. ft. (63 W/sq. m.)
2	12 Btu-h/sq. ft. (38 W/sq. m.)	15 Btu-h/sq. ft. (47 W/sq. m.)
3	8 Btu-h/sq. ft. (25 W/sq. m.)	10 Btu-h/sq. ft. (32 W/sq. m.)

OR

PATH 2. ASHRAE 90.1 TRADE-OFF METHODS (1–3 POINTS)

Comply with envelope and/or HVAC improvements for a maximum of 3 points.

ENVELOPE LOADS (1–2 POINTS)

Demonstrate a percent improvement in the sum of system peak heating loads and system peak cooling loads associated with the *proposed envelope performance factor* compared to the *base envelope performance factor* determined in accordance with the *ASHRAE 90.1-2022*, Building Envelope Trade-off Option (Normative Appendix C). Points are awarded according to Table 3.

TABLE 3. Points for Percentage Improvement in Peak Thermal Loads from Envelope

Points	Percent Improvement
1	10%
2	20%

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AND/OR

Ventilation Loads (1 point)

Demonstrate a minimum 10% improvement in the sum of building peak coincident heating loads and building peak coincident cooling loads for the TSPR proposed building design versus the product of the TSPR reference building design and *ASHRAE 90.1-2022*, Table L5-4, Mechanical Performance Factors (MPF), for the project's location and climate zone determined in accordance with the *ASHRAE 90.1-2022*, Mechanical System Performance Rating Method (Normative Appendix L).

OR

PATH 3. ENERGY SIMULATION (1–3 POINTS)

Demonstrate a PI calculated per *ASHRAE 90.1-2019* or later from Normative Appendix G's Performance Rating Method, replacing all references to cost with the sum of building peak coincident heating loads and building peak coincident cooling loads.

Points are awarded according to Table 4.

TABLE 4. Points for PI for Peak Heating and Cooling Loads

Performance Index	Points
0.5	1
0.4	2
0.3	3

REQUIREMENTS EXPLAINED: NEW CONSTRUCTION

The credit rewards reduced peak heating and peak cooling loads, primarily achieved through enhanced building envelope performance and lower ventilation loads.

This limits strain on the grid during peak summer and winter operations when grid capacity and associated grid emissions are highest.

Options 1, 2, and 3 are distinct peak thermal load reduction strategies. Option 4 rewards overall peak thermal load reductions achieved from the strategies referenced in Options 1–3 and any further peak load reduction measures used for the project.

For credit eligibility, the peak thermal load reduction measures designed for the building must be fully implemented during construction. The construction phase verification is required to confirm measured air leakage (Options 1 and 4), balanced ventilation (Options 1 and 4), and reduced thermal bridging (Options 3).

New construction projects may apply any combination of Options 1–4 for up to 5 points.

FOR ALL OPTIONS

The building envelope must comply with *ASHRAE 90.1* prescriptive method requirements as a pre-condition for all options.

Do not claim trade-offs between the envelope and other energy-using systems. This sets a higher bar for the building envelope than *EAp2: Minimum Energy Efficiency* for projects documented using *ASHRAE 90.1* Energy Cost Budget (ECB) or *ASHRAE 90.1* Appendix G, Performance Rating Method.

Envelope performance must comply with one of the following:

- *ASHRAE Standard 90.1*, Section 5.5, Prescriptive Building Envelope Compliance Path.
- *ASHRAE Standard 90.1*, Section 5.6, Building Envelope Trade-off Compliance Path. (Refer to *EAp2: Minimum Energy Efficiency* guidance on this topic).

Use the same version of *ASHRAE 90.1* referenced for *EAp2: Minimum Energy Efficiency* (i.e., *ASHRAE 90.1-2019* or *ASHRAE 90.1-2022*).

OPTION 1. INFILTRATION AND BALANCED VENTILATION

This strategy reduces heat gains and losses associated with infiltration and exfiltration. Projects must provide testing during the construction phase to verify the achievement of both the balanced ventilation and measured air leakage criteria.

Balanced ventilation

Design the building's mechanical system and controls to support balanced ventilation airflow that maintains the building's total supply ventilation airflow within 10% of the total exhaust airflow.

During construction, conduct a TAB of building air handling systems that verifies a ratio of ventilation to exhaust airflow between 90% and 110% on a whole-building basis.

Infiltration

Design and construct the building's air barrier to minimize air leakage through the building enclosure. During construction, test for air leakage to confirm the project achieves targeted performance levels (referenced in Table 1 of the credit requirements). The testing must conform to one of the referenced air leakage testing standards.

ASTM testing criteria

ASTM defines air leakage testing criteria. *ASTM E779* uses the fan pressurization method for testing.[5] *ASTM E3158* provides a standard method for testing large or multizone buildings. *ASTM E1827* determines air tightness using an orifice blower door. Residential spaces in mixed-use buildings may also apply *ANSI/RESNET/ICC 380*.

Residential buildings

In residential buildings, compartmentalize each unit, incorporating measures to limit air leakage between units. Determine credit compliance by conducting blower door testing of units per *ANSI/RESNET/ICC 380*. Sample rates may apply, depending on the number of units within the building.

Maximum leakage rates for each unit cannot exceed 150% of the Table 1 thresholds for the enclosure area.

Furthermore, weighted average building leakage rates must be less than the Table 1 thresholds. Calculate the weighted average using the measured results for the individual dwelling units.

Failed testing

If a unit fails testing, corrective action is recommended to reduce air leakage in the space. Determine compliance with the credit by the weighted average of leakage rates reported for the project. If

5. ASTM, "Standard Test Method for Determining Air Leakage Rate by Fan Pressurization," ASTM E779-19, January 23, 2019, https://www.astm.org/e0779-19.html.

a single unit fails in a project with a large number of dwelling units, the project may still show compliance with this option.

OPTION 2. VENTILATION HEAT RECOVERY

Energy recovery ventilators (ERV) or heat recovery ventilators (HRV) reduce peak heating and cooling loads associated with ventilation by pretreating the incoming outside air with heated or cooled air recovered from the exhaust stream.

For each system that supplies outdoor air to the building, include an ERV or HRV with at least a 70% enthalpy recovery ratio or 75% sensible heat recovery ratio.

Controls for ERVs and HRVs

ERVs must have controls to disable energy exchange or bypass air during economizer operation. The requirement does not apply to systems with design outdoor airflow rates of less than 80% and design outdoor air volume of less than 10,000 cubic feet/minute (cfm). For details, refer to *ASHRAE 90.1-2022* Section 6.5.6.1.2.2, Provision for Air Economizer or Bypass Operation.

Exceptions

In aggregate, fan systems supplying less than 15% of the project's total outdoor air can be excluded.

Exclude the following systems from the requirements in addition to this 15% exclusion:

- Kitchen exhaust demand ventilation systems meeting the provisions of *ASHRAE 90.1*, 6.5.7.2.3(b).
- Laboratory exhaust systems meeting the provisions of *ASHRAE 90.1*, Section 6.5.7.3 (a) or (b).

Kitchen or laboratory systems must meet these criteria to be eligible for this exclusion, even if the project's total kitchen or laboratory exhaust volumes are less than those referenced in Section 6.5.7.

OPTION 3. THERMAL BRIDGING

Projects must document prescriptive compliance with each thermal bridging requirement in *ASHRAE 90.1-2022*, Section 5.5.5a, and design and construct a continuous thermal barrier that minimizes heat conductance associated with thermal bridges. A thermal bridge is an element that penetrates the building insulation, such as a wall and roof intersection, a wall and window intersection, an exterior cladding support, or a beam penetrating the exterior wall assembly.

Do not use the envelope trade-off method to show compliance with this option.

Climate zones 0–3

Projects in warmer climate zones (0–3) cannot apply any exceptions and must comply entirely with Section 5.5.5. Providing thermal breaks, continuous insulation, and using reflective exterior coatings on exterior surfaces can reduce solar heat gain and prevent moisture intrusion.

OPTION 4. PEAK THERMAL LOAD REDUCTIONS

Option 4 rewards overall peak thermal load reduction quantitatively assessed for the project using one of three methods:

- Path 1 leverages passive building modeling tools to document peak thermal load intensities below the required thresholds.
- Path 2 leverages *ASHRAE 90.1* trade-offs for the prescriptive method to show a reduction in peak thermal loads below a referenced baseline.
- Path 3 leverages the *ASHRAE 90.1*, Appendix G Performance Rating Method, to show a reduction in peak thermal loads below a referenced baseline.

For all three paths, the referenced tools can be used to inform an IDP and make design decisions that holistically reduce peak thermal loads. The analysis should reflect the savings from the strategies referenced in Options 1–3 and further savings for any other load reduction strategies incorporated in the building design, such as improved insulation, improved window performance, or lower internal loads.

Preconditions applicable to all paths

Projects pursuing Option 4 must comply with all the following:

- **Ventilation loads:** When calculating peak heating and peak cooling loads, account for ventilation loads. Use the design of outdoor air from the project's design documents.
- **Measured Air Leakage:** During construction, provide air leakage testing to measure the building's air leakage and use the measured air leakage to calculate peak loads. If evaluating credit compliance during the design phase, perform the peak load calculations using the targeted air leakage that aligns with project design documents. Recalculate during construction if measured air leakage exceeds targeted air leakage.

Measured air leakage must be less than or equal to the *ASHRAE 90.1* required air leakage rates.

For Path 2, this measured air leakage requirement only applies to envelope loads calculations, not ventilation loads calculations.

- **Balanced Ventilation**. Maintain the project's ventilation and exhaust airflows within 10% of each other (see Option 1. Balanced Ventilation requirements).

PATH 1. PEAK LOAD INTENSITY

Path 1 requires projects to limit peak thermal loads below the specified thresholds.

Calculate the sum of the peak heating load and peak cooling load per unit of treated floor area using either the WUFI Passive Design Tool or the Passive House Planning Package.

Additional considerations

Under this option, projects designed to achieve Passive House compliance (using either Passive House Institute U.S. [PHIUS] or Passive House Institute [PHI]) will likely achieve 3 points.

PATH 2. *ASHRAE 90.1* TRADE-OFF METHODS

For projects documenting *EAp2: Minimum Energy Efficiency* using the prescriptive method or ECB Method, Path 2, provides the most streamlined method for showing compliance. However, due to the limitations of the referenced *ASHRAE 90.1* Trade-off Methods, it may not be possible to show loads reductions for some buildings using this path. For example, projects with minimal envelope contribution to peak thermal loads cannot achieve the required envelope loads thresholds. Similarly, the TSPR calculations referenced for the Ventilation Loads requirements are only available for a subset of building types.

Envelope loads

Use the *ASHRAE Standard 90.1*, Section 5.6, Building Envelope Trade-off Compliance Path analysis to show credit compliance. (Refer to *EAp2: Minimum Energy Efficiency* guidance on this topic.) Amend the proposed inputs to reference measured air leakage. If this is not possible in the software interface, perform the air leakage testing as required, but use the *ASHRAE 90.1* prescriptively required air leakage as a conservative savings estimate.

Extract the system peak heating loads and the system peak cooling loads from the modeled outputs for the proposed envelope factor and the base envelope factor.

Show the project achieves the required percentage improvement in the sum of system peak heating loads and system peak heating loads comparing the results for the two models.

The peak load metric used here is distinct from the metric of annual energy cost used to assess compliance with *ASHRAE 90.1*.

Ventilation loads

This case rewards projects for a 10% improvement in the sum of building peak coincident heating loads and building peak coincident cooling loads achieved through HVAC control strategies addressing ventilation loads, such as demand control ventilation or energy recovery ventilation.

Use the *ASHRAE Standard 90.1*, Section 6.6.2 Mechanical System Performance Path analysis to show credit compliance. (Refer to *EAp2: Minimum Energy Efficiency* guidance on this topic.)

Extract the building peak coincident heating loads and the building peak coincident cooling loads from the TSPR modeled outputs for the referenced building design and for the proposed building design. Calculate performance using the following equations:

Equation 1. Reference peak (based on modeled outputs from the referenced building design)

$$\text{REFERENCE}_{\text{peak}} = \text{Bldg peak coincident heating loads} + \text{Bldg peak coinci dent cooling loads}$$

Equation 2. Proposed peak (based on modeled outputs from the proposed building design)

$$\text{PROPOSED}_{\text{peak}} = \text{Bldg peak coincident heating loads} + \text{Bldg peak coincident cooling loads}$$

Equation 3. Performance target

$$\text{Performance target} = \text{REFERENCE}_{peak} \times \text{MPF}$$

where:
MPF = Mechanical performance factor for project's location and climate zone per *ASHRAE Standard 90.1-2022, Table L5-4*

Equation 4. Percentage improvement

$$\text{Percentage improvement} = 100\% - \frac{\text{PROPOSED}_{peak}}{\text{Performance target}}$$

PATH 3. ENERGY SIMULATION
Projects that comply with *EAp2: Minimum Energy Efficiency* using the *ASHRAE 90.1* Appendix G, Performance Rating Method, can use the PRM modeled results to show compliance with this path.

This path is the most appropriate selection for projects with high process loads or projects with significant quantities of laboratory or kitchen exhaust. Unlike Path 1 and Path 2, this path allows credit for lower plug and process loads documented using *ASHRAE 90.1*, Section G2.5 Exceptional Calculation Methods (See *EAc3: Enhanced Energy Efficiency* guidance for further information).

Performance index

For the *ASHRAE 90.1* Appendix G baseline and proposed models, extract the building peak coincident heating load and the building peak coincident cooling load.

Calculate the BBP, the PBP, and the PI by replacing the cost metric with the sum of building peak coincident heating load and building peak coincident cooling load.

Equation 5. Performance index

$$\text{PI} = \frac{\text{PBP}}{\text{BBP}}$$

For this path, neither BPFs nor performance index targets (PI_t) are used. Per Table 4 of the rating system, points achievement is directly linked to the PI equal to the ratio of peak thermal loads for the proposed building compared to the baseline building that meets *ASHRAE 90.1-2004* prescriptive criteria.

REQUIREMENTS EXPLAINED: CORE AND SHELL

OPTION 1. INFILTRATION

Same as LEED BD+C: New Construction, Option 1, except:

- Core and Shell projects are not required to meet the Balanced Ventilation requirements of this option.

OPTION 2. VENTILATION ENERGY RECOVERY

Same as LEED BD+C: New Construction, Option 2, if:

- Option 2 is only available to LEED BD+C: Core and Shell projects that include fan systems within the project scope of work. At least 50% of the total ventilation air required by the *ASHRAE Standard 62.1-2022*, Ventilation Rate Procedure must be installed as part of the Base Building scope of work.

OPTION 3. THERMAL BRIDGING

Same as LEED BD+C: New Construction, Option 3, except:

- LEED BD+C: Core and Shell projects are not required to meet the Balanced Ventilation requirements of this option.

OPTION 4. PEAK THERMAL LOAD REDUCTIONS

Same as LEED BD+C: New Construction, Option 4.

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	All	All	Documentation of compliance with *ASHRAE 90.1*, Section 5.5, Prescriptive Building Envelope Compliance Path or *ASHRAE 90.1*, Section 5.6, Building Envelope Trade-Off Compliance Path.
New Construction and Core and Shell	Option 1	All	Confirmation that supply and exhaust flows are designed within 10% of each other.
New Construction	Option 1	All	Balanced ventilation design calculations.
New Construction and Core and Shell	Option 1	All	Air leakage test report describing method, conditions, and results. Note: Include which Path (New Construction or Major Renovation; larger or smaller than 5,000 sq. ft.; Pressure Test Condition) is followed.
	Option 2	All	Documentation showing outdoor air delivery systems and flow rates, including energy recovery devices and efficiencies, and OA bypass controls (e.g., mechanical schedules, specifications, submittals, controls diagram).
	Option 3	All	Comcheck or *ASHRAE 90.1-2022* prescriptive thermal bridging compliance forms. Include Details of envelope and calculations showing that intersections and edges of each type meet *ASHRAE 90.1 2022*, Section 5.5.5.
	Option 4	All	Ventilation loads must be included in the determination of peak coincident loads.
			Air leakage test report describing method, conditions, and results.
			Confirmation that supply and exhaust flows are designed within 10% of each other.
		Path 1	Input/output report from PHPP or WUFI.
			Sum of peak sensible heating and cooling load.

(*continued*)

(continued)

Project Types	Options	Paths	Documentation
		Path 2. Envelope AND Path 2. Ventilation	Input/output reports from *ASHRAE 90.1-2022*, Building envelope trade-off option and/or TSPR as applicable.
			Envelope peak load reduction and/or TSPR peak load reduction.
			Sum of peak heating loads and peak cooling loads (Base/Reference Case).
			Sum of peak heating loads and peak cooling loads (Proposed Case).
		Path 3	Calculation of Performance index using sum of peak heating and peak cooling.

REFERENCED STANDARDS

- ASHRAE 90.1-2019 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2019-i-p?product_id=2088527)
- ASHRAE 90.1-2022 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2022-i-p?product_id=2522082)
- ASHRAE 62.1-2022 (scope requirements for Core and Shell), (https://store.accuristech.com/ashrae/standards/ashrae-62-1-2022?product_id=2501063)
- Passive House Planning Package (PHPP) following the Passive House Institute (PHI) protocol (https://passivehouse.com/04_phpp/04_phpp.htm#PH10)
- WUFI Passive Design Tool, following the Passive House Institute U.S. (PHIUS) protocol (https://www.phius.org/resources/tools-and-support/wufi-passive-design-tool)
- ASTM E779 Standard Test Method for Determining Air Leakage Rate by Fan Pressurization (https://astm.org/e0779-19.html)
- ASTM E3158 Standard Test Method for Measuring the Air Leakage Rate of a Large or Multizone Building (https://www.astm.org/e3158-18.html)
- ASTM E1827 Standard Test Methods for Determining Airtightness of Buildings Using an Orifice Blower Door (https://astm.org/e1827-11r17.html)
- ANSI/RESNET/ICC 380 Standard for Testing Airtightness of Building, Dwelling Unit, and Sleeping Unit Enclosures; Airtightness of Heating and Cooling Air Distribution Systems; and Airflow of Mechanical Ventilation Systems (https://www.resnet.us/wp-content/uploads/Std380-2022_Strk-UndrIn_blk_wCover_cln5.pdf)

ENERGY AND ATMOSPHERE CREDIT

Enhanced Energy Efficiency

EAc3
New Construction (1–10 points): 8 points are required for LEED BD+C: New Construction Platinum projects Core and Shell (1–7 points): 5 points are required for LEED BD+C: Core and Shell Platinum projects

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To design buildings that minimize energy use to reduce the environmental damage caused by resource extraction, air pollution, and greenhouse gas emissions and to facilitate the transition to a clean energy future.

REQUIREMENTS: NEW CONSTRUCTION

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1-10**
Option 1. Prescriptive Path	1–10
Path 1. Regulated Loads	1–7
Case 1. *ASHRAE 90.1-2019*	1–5
OR	
Case 2. *ASHRAE 90.1-2022*	4–7
AND/OR	
Path 2. Plug and Process Loads (PPLs)	1–4
Case 1. Plug Load Management	1
AND/OR	
Case 2. Efficient Plug and Process Load Equipment	1–4
OR	
Case 3. Plug and Process Load Exceptional Calculation	1–4
OR	
Option 2. Energy Simulation	1–10
Path 1. Percentage Reduction Excluding On-Site Renewable Contribution	1–10
OR	
Path 2. Percentage Reduction Including On-Site Renewable Contribution	1–10

OPTION 1. PRESCRIPTIVE PATH (1–10 POINTS)

Pursue Path 1 and/or Path 2 for a maximum of 10 points.

PATH 1. REGULATED LOADS (1–7 POINTS)

Points are awarded according to Table 1 below, using either Case 1 or Case 2.

Case 1. *ASHRAE 90.1-2019* (1–5 points)
Available only to projects registered before January 1, 2028.

Comply with the provisions of *ASHRAE 90.1-2019*, Sections 5–10.

Implement Additional Efficiency Requirements credits calculated per *ASHRAE 90.1-2022*, Section 11, from the list of eligible measures referenced below. Where *ASHRAE 90.1-2022*, Section 11, references the prescriptive requirements of *ASHRAE 90.1-2022*, Sections 5–10, such as lighting power density or equipment efficiency, replace those references with the matching prescriptive values in *ASHRAE 90.1-2019*.

OR

Case 2. *ASHRAE 90.1-2022* (4–7 points)
Comply with the provisions of *ASHRAE 90.1-2022*, Sections 5–11.

Implement incremental *ASHRAE 90.1-2022*, Section 11, credits above the minimum required from the list of eligible measures below.

Eligible measures from *ASHRAE 90.1-2022*, Section 11.5.2, for LEED points:

- HVAC measures (H01 to H07)
- Service water heating measures (W01 to W09)
- Lighting measures (L01 to L06)
- G07 building mass/night flush

TABLE 1. Points for *ASHRAE 90.1-2022*, Section 11, Credits

Points	Case 1. ***ASHRAE 90.1-2019***	Case 2. ***ASHRAE 90.1-2022***
1	25 credits	N/A
2	50 credits	
3	75 credits	
4	100 credits	Min. required by 90.1-2022
5	125 credits	Min. required by 90.1-2022 plus 25 credits
6	N/A	Min. required by 90.1-2022 plus 50 credits
7	N/A	Min. required by 90.1-2022 plus 75 credits

AND/OR

PATH 2. PLUG AND PROCESS LOADS (1–4 POINTS)

Case 1. Plug Load Management (1 point)

Implement the following:

- Provide a plug load dashboard that is accessible through an application to all regular occupants of the building if tenants can opt out of displaying their plug loads to other tenants.
- For building types and/or tenant types with IT departments, implement policies for PCs, monitors, and visual displays to be controlled off when not in use, except during scheduled maintenance periods.

AND/OR

Case 2. Efficient Plug and Process Load Equipment (1–4 points)

Install or reuse eligible plug and process equipment meeting the criteria in Table 2 for 90% of applicable equipment by quantity or rated load. Either include or exclude all eligible equipment reused in the project from the calculations.

- For one, Table 2 equipment category (1 point)
- For two, Table 2 equipment categories (2 points)
- For three or more, Table 2 equipment categories (3 points)

OR

For process-intensive buildings, install or reuse eligible plug and process equipment meeting the criteria in Table 2 for at least 90% of total applicable equipment-rated load. Rated load of compliant equipment must total at least:

- 0.5 Watt (W)/sq. ft. (5.4 W/sq. m.) (3 points)
- 1.0 W/sq. ft. (10.8 W/sq. m.) (4 points)

TABLE 2. Plug, Process, Refrigeration, and Conveyance Equipment Criteria

Equipment Category	Applicable Equipment	Criteria
ENERGY STAR products: plug loads and small appliances	Office equipment Appliances Electronics Other (e.g., vending machines, pool pumps, water coolers)	ENERGY STAR rated or approved equivalent with at least 0.1 W/sq. ft. (1.1 W/sq. m.) of total rated load.
ENERGY STAR products: process loads	Commercial food service equipment Data center/server equipment Commercial laundry equipment Electric vehicle chargers (EVSE) Other (e.g., laboratory-grade refrigerators and freezers)	ENERGY STAR rated or approved equivalent with at least 0.1 W/sq. ft. (1.1 W/sq. m.) of total rated load.
People conveyance	Elevators Escalators Moving walkways	*ISO 25745.* At least Class A rated.

(*continued*)

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TABLE 2. Plug, Process, Refrigeration, and Conveyance Equipment Criteria (*continued*)

Equipment Category	Applicable Equipment	Criteria
Data center electrical system	Electrical system design	*ASHRAE 90.4-2022*. Design electrical loss component is at least 20% lower than the maximum design electrical loss.
Refrigeration systems	Referenced in *ASHRAE 90.1*, Section 6.8 tables, AND not ENERGY STAR eligible	10% improvement beyond *ASHRAE 90.1*, Section 6.8 tables.
	Refrigerated warehouse	*California Title 24-2022*, Section 120.6, refrigerated warehouse requirements.
Airport equipment	Baggage handling equipment	Individual carrier systems with variable frequency drive.
	Aircraft and jetway air conditioning	Preconditioned air systems with efficiencies meeting *ASHRAE 90.1* prescriptive efficiencies for HVAC equipment.

OR

Case 3. Plug and Process Load Exceptional Calculation (1–4 points)

Using the *ASHRAE 90.1*, Section G2.5, Exceptional Calculation Method, demonstrate a minimum percentage improvement in total project plug and process, refrigeration, and conveyance loads. Points are awarded according to Table 3.

TABLE 3. Points for Percent Improvement in Plug and Process Loads

Percent Improvement	Points
10%	1
20%	2
30%	3
40%	4

OR

OPTION 2. ENERGY SIMULATION (1–10 POINTS)

Demonstrate an improvement in future source energy calculated per *ASHRAE Standard 90.1*, Normative Appendix G, Performance Rating Method, with the following additional provisions:

- Use the *ASHRAE 90.1* version applied for *EAp2: Minimum Energy Efficiency.*
- Replace *ASHRAE 90.1-2019* or *90.1-2022*, Table 4.2.1.1, Building Performance Factors (BPF), with Table 5 below. For major building renovation areas, multiply the BPF by 1.05.
- Replace all references to cost with future source energy. Use an electric site-to-source energy conversion factor of 2.0 based on future projections for the U.S. A lower national average value may be used as applicable for projects outside of the U.S.

- Model energy efficiency measures for plug and process loads using the Section G2.5, Exceptional Calculation Method, or approved calculations.
- Calculate the performance index (PI) and percentage improvement with and without the plug and process savings.
- Calculate the PI and PI target as follows:
  - $PI_{nre} = PBP_{nre} / BBP$
  - $PI = PBP / BBP$
  - $PI_t = [BBUE + (BPF \times BBRE)] / BBP$

  where:
    - PI_{nre} = performance index for future source energy excluding on-site renewable contribution
    - PI = performance index for future source energy including on-site renewable contribution
    - PI_t = performance index target for future source energy use
    - BBP = baseline building performance for baseline building future source energy use
    - BBUE = baseline building unregulated future source energy use
    - BBRE = baseline building regulated future source energy use
    - PBP_{nre} = proposed building performance without any credit for reduced annual future source energy from on-site renewable energy generation systems
    - PBP = proposed building performance, including the reduced annual future source energy associated with all on-site renewable energy generation systems

Points are awarded according to Table 4, using either Path 1 or Path 2.

TABLE 4. Points for Percentage Improvement in PI Below PI_t

Path 1. Percentage Reduction Excluding On-Site Renewable Contribution $(100\% - PI_{nre} / PI_t)$	OR	Path 2. Percentage Reduction Including On-Site Renewable Contribution $(100\% - PI / PI_t)$	Points
3%		10%	1
6%		20%	2
9%		30%	3
12%		40%	4
15%		50%	5
18%		60%	6
21%		70%	7
24%		80%	8
27%		90%	9
30%		100%	10

TABLE 5. *ASHRAE 90.1-2019*: Equivalent Building Performance Factors for a Future Source Energy Metric

Building Type	Climate Zone																		
	0A	0B	1A	1B	2A	2B	3A	3B	3C	4A	4B	4C	5A	5B	5C	6A	6B	7	8
Multifamily	0.74	0.69	0.73	0.70	0.73	0.70	0.71	0.70	0.63	0.70	0.71	0.69	0.68	0.70	0.70	0.68	0.68	0.68	0.74
Healthcare/ hospital	0.72	0.72	0.73	0.73	0.74	0.71	0.72	0.74	0.71	0.72	0.73	0.71	0.74	0.73	0.80	0.73	0.77	0.78	0.79
Hotel/motel	0.72	0.71	0.72	0.71	0.71	0.70	0.71	0.73	0.72	0.71	0.73	0.73	0.71	0.73	0.74	0.70	0.72	0.70	0.70
Office	0.62	0.63	0.61	0.62	0.58	0.60	0.57	0.62	0.55	0.55	0.61	0.57	0.58	0.61	0.59	0.58	0.60	0.54	0.58
Restaurant	0.65	0.62	0.63	0.61	0.62	0.58	0.63	0.63	0.63	0.67	0.66	0.66	0.70	0.70	0.68	0.73	0.72	0.74	0.77
Retail	0.57	0.54	0.53	0.53	0.48	0.47	0.47	0.47	0.47	0.52	0.50	0.56	0.57	0.53	0.59	0.58	0.56	0.53	0.60
School	0.57	0.57	0.58	0.57	0.55	0.54	0.57	0.51	0.49	0.48	0.51	0.52	0.51	0.53	0.51	0.53	0.50	0.51	0.58
Warehouse	0.28	0.30	0.24	0.27	0.23	0.24	0.27	0.23	0.20	0.33	0.26	0.28	0.40	0.32	0.29	0.44	0.38	0.40	0.44
All others	0.65	0.62	0.64	0.62	0.57	0.54	0.57	0.56	0.58	0.59	0.57	0.60	0.60	0.59	0.65	0.62	0.62	0.61	0.64

REQUIREMENTS: CORE AND SHELL

ACHIEVEMENT PATHWAYS	POINTS
Core and Shell	**1-7**
Option 1. Prescriptive Path	1-7
Path 1. Regulated Loads	1-6
Case 1. *ASHRAE 90.1-2019*	1-5
OR	
Case 2. *ASHRAE 90.1-2022*	2-6
AND/OR	
Path 2. Plug and Process Loads (PPLs)	1-3
Case 1. Plug Load Management	1
AND/OR	
Case 2. Efficient Plug and Process Load Equipment	1-3
OR	
Option 2. Energy Simulation	1-7
Path 1. Percentage Reduction Excluding On-Site Renewable Contribution	1-7
OR	
Path 2. Percentage Reduction Including On-Site Renewable Contribution	1-7

OPTION 1. PRESCRIPTIVE PATH (1–7 POINTS)

Pursue Path 1 and/or Path 2 for a maximum of 7 points.

PATH 1. REGULATED LOADS (1–6 POINTS)

Points are awarded according to Table 6, using either Case 1 or Case 2.

Case 1. *ASHRAE 90.1-2019* (1–5 points)

Available only to projects registered before January 1, 2028.

- Comply with the provisions of *ASHRAE 90.1-2019*, Sections 5–10.
- Implement Additional Efficiency Requirements credits calculated per *ASHRAE 90.1-2022*, Section 11, from the list of eligible measures referenced below. Where *ASHRAE 90.1-2022*, Section 11, references the prescriptive requirements of *ASHRAE 90.1-2022*, Sections 5–10, such as lighting power density or equipment efficiency, replace with the matching prescriptive values in *ASHRAE 90.1-2019*.
  - **Central systems**. Earn 1 point for every 13 energy credits where the LEED BD+C: Core and Shell project includes a central HVAC system or service water heating system that includes chillers, boilers, service water heating equipment, or loop pumping systems with heat rejection.
  - **All others**. Earn 1 point for every nine energy credits for all other LEED BD+C: Core and Shell projects.

OR

Case 2. *ASHRAE 90.1-2022* (2–6 points)

- Comply with the provisions of ASHRAE 90.1-2022, Sections 5–11.
- Implement incremental ASHRAE 90.1 2022, Section 11 credits above the minimum required from the list of eligible measures below.
  - Central systems: 1 point for every 13 incremental energy credits where the LEED BD+C: Core and Shell project includes a central HVAC system or service water heating system that includes chillers, boilers, service water heating equipment, or loop pumping systems with heat rejection.
  - All others: 1 point for every nine incremental energy credits for all other LEED BD+C: Core and Shell projects.

Eligible measures from *ASHRAE 90.1-2022*, Section 11.5.2, for LEED points

- HVAC measures (H01 to H07)
- Service water heating measures (W01 to W09)
- Lighting measures (L01 to L06)
- G07 building mass/night flush

TABLE 6. Points for *ASHRAE 90.1-2022*, Section 11 Credits

Points	Compliance Path			
	Case 1. *ASHRAE 90.1-2019*		Case 2. *ASHRAE 90.1-2022*	
	Central Systems	All Others	Central Systems	All Others
1	13 credits	9 credits	N/A	N/A
2	26 credits	18 credits	Min. required by *ASHRAE 90.1-2022*	Min. required by *ASHRAE 90.1-2022*
3	39 credits	27 credits	Min. required + 13 credits	Min. required + 9 credits
4	52 credits	36 credits	Min. required + 26 credits	Min. required + 18 credits
5	65 credits	45 credits	Min. required + 39 credits	Min. required + 27 credits
6	–	–	Min. required + 52 credits	Min. required + 36 credits

AND/OR

PATH 2. PLUG AND PROCESS LOADS (1–3 POINTS)

Case 1. Plug Load Management (1 point)

Implement the following:

- Provide to all regular occupants of the building a plug load dashboard accessed through an application that displays base building plug loads and allows tenants to choose whether to display plug loads to their occupants.
- Configure the plug load monitoring system with the capability to expand to monitor future plug loads for each floor and for individual tenant spaces.

AND/OR

Case 2. Efficient Plug and Process Load Equipment (1–3 points)

Install or reuse equipment meeting the criteria in Table 7. Points are awarded according to Table 7 up to a maximum of 3 points for each equipment category where the criteria are met.

TABLE 7. Plug, Process, Refrigeration, and Conveyance Equipment Criteria

Equipment Category	Applicable Equipment	Criteria	Points
People conveyance	Elevators Escalators Moving walkways	*ISO 25745* At least Class A-rated.	2
Data center electrical system	Electrical system design	*ASHRAE 90.4-2022* Design electrical loss component is at least 20% lower than the maximum design electrical loss.	1
Refrigeration systems	Referenced in *ASHRAE 90.1*, Section 6.8 tables, AND not ENERGY STAR eligible	10% improvement beyond *ASHRAE 90.1*, Section 6.8 tables.	1
	Refrigerated warehouse	*California Title 24-2022*, Section 120.6, refrigerated warehouse requirements.	

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TABLE 7. Plug, Process, Refrigeration, and Conveyance Equipment Criteria (*continued*)

Equipment Category	Applicable Equipment	Criteria	Points
Airport equipment	Baggage handling equipment	Individual carrier systems with variable frequency drive.	2
	Aircraft and jetway air-conditioning	Preconditioned air systems with efficiencies meeting *ASHRAE 90.1* prescriptive efficiencies for HVAC equipment.	

Data centers

Data centers that comprise at least 40% of the project's total area with the electrical system in the project scope earn 2 points for complying with the data center electrical system requirements.

Warehouses

Refrigerated warehouses that comprise at least 20% of the project's gross floor area with the refrigeration systems in the project scope earn 2 points for complying with the refrigeration system requirements.

OR

OPTION 2. ENERGY SIMULATION (1–7 POINTS)

Demonstrate an improvement in future source energy calculated per *ASHRAE Standard 90.1*, Normative Appendix G, Performance Rating Method, with the following additional provisions:

- Use the *ASHRAE 90.1* version applied for *EAp2: Minimum Energy Efficiency*.
  - **Case 1**. *ASHRAE 90.1-2019*. Replace *ASHRAE 90.1-2019*, Table 4.2.1.1, Building Performance Factors, with Table 4. For major building renovation areas, multiply the BPF by 1.05.
  - **Case 2**. *ASHRAE 90.1-2022*. Replace *ASHRAE 90.1-2022*, Table 4.2.1.1, Building Performance Factors, with Table 5. For major building renovation areas, multiply the BPF by 1.05.
- Replace all references to cost with future source energy. Use an electric site-to-source energy conversion factor of 2.0 based on future projections for the U.S. A lower national average value may be used as applicable for projects outside of the U.S.
- Model energy efficiency measures for plug and process loads using the Section G2.5 exceptional calculation method or approved calculations. Calculate the PI and percentage improvement with and without the plug and process savings.
- Calculate the PI and PI Target as follows:
  - $PI_{nre} = PBP_{nre} / BBP$
  - $PI = PBP / BBP$
  - $PI_t = [BBUE + (BPF \times BBRE)] / BBP$

  where:

  - PI_{nre} = performance index for future source energy excluding on-site renewable contribution

- PI = performance index for future source energy including on-site renewable contribution
- PI_t = performance index target for future source energy use
- BBP = baseline building performance for baseline building future source energy use
- BBUE = baseline building unregulated future source energy use
- BBRE = baseline building regulated future source energy use
- PBP_{nre} = proposed building performance without any credit for reduced annual future source energy from on-site renewable energy generation systems
- PBP = proposed building performance, including the reduced annual future source energy associated with all on-site renewable energy generation systems

Points are awarded according to Table 8, using either Path 1 or Path 2.

TABLE 8. Points for Percentage Improvement in PI Below PI_t

Path 1. Percentage Reduction Excluding On-Site Renewable Contribution (100% – PI_{nre} / PI_t)		OR	Path 2. Percentage Reduction Including On-Site Renewable Contribution (100% – PI / PI_t)		
Case 1. *ASHRAE 90.1-2019*	**Case 2. *ASHRAE 90.1-2022***		**Case 1. *ASHRAE 90.1-2019***	**Case 2. *ASHRAE 90.1-2022***	**Points**
2%	N/A		10%	N/A	1
5%	N/A		20%	0%	2
8%	0%		30%	13%	3
11%	3%		40%	26%	4
14%	6%		50%	39%	5
17%	9%		60%	52%	6
20%	12%		70%	65%	7

TABLE 9. *ASHRAE 90.1-2019:* Equivalent Building Performance Factors for a Future Source Energy Metric

	Climate Zone																		
Building Type	**0A**	**0B**	**1A**	**1B**	**2A**	**2B**	**3A**	**3B**	**3C**	**4A**	**4B**	**4C**	**5A**	**5B**	**5C**	**6A**	**6B**	**7**	**8**
Multifamily	0.74	0.69	0.73	0.70	0.73	0.70	0.71	0.70	0.63	0.70	0.71	0.69	0.68	0.70	0.70	0.68	0.68	0.68	0.74
Healthcare/ hospital	0.72	0.72	0.73	0.73	0.74	0.71	0.72	0.74	0.71	0.72	0.73	0.71	0.74	0.73	0.80	0.73	0.77	0.78	0.79
Hotel/motel	0.72	0.71	0.72	0.71	0.71	0.70	0.71	0.73	0.72	0.71	0.73	0.73	0.71	0.73	0.74	0.70	0.72	0.70	0.70
Office	0.62	0.63	0.61	0.62	0.58	0.60	0.57	0.62	0.55	0.55	0.61	0.57	0.58	0.61	0.59	0.58	0.60	0.54	0.58
Restaurant	0.65	0.62	0.63	0.61	0.62	0.58	0.63	0.63	0.63	0.67	0.66	0.66	0.70	0.70	0.68	0.73	0.72	0.74	0.77
Retail	0.57	0.54	0.53	0.53	0.48	0.47	0.47	0.47	0.47	0.52	0.50	0.56	0.57	0.53	0.59	0.58	0.56	0.53	0.60
School	0.57	0.57	0.58	0.57	0.55	0.54	0.57	0.51	0.49	0.48	0.51	0.52	0.51	0.53	0.51	0.53	0.50	0.51	0.58
Warehouse	0.28	0.30	0.24	0.27	0.23	0.24	0.27	0.23	0.20	0.33	0.26	0.28	0.40	0.32	0.29	0.44	0.38	0.40	0.44
All others	0.65	0.62	0.64	0.62	0.57	0.54	0.57	0.56	0.58	0.59	0.57	0.60	0.60	0.59	0.65	0.62	0.62	0.61	0.64

TABLE 10. *ASHRAE 90.1-2022*: Equivalent Building Performance Factors for a Future Source Energy Metric

Building Type	Climate Zone																		
	0A	0B	1A	1B	2A	2B	3A	3B	3C	4A	4B	4C	5A	5B	5C	6A	6B	7	8
Multifamily	0.64	0.59	0.62	0.60	0.61	0.59	0.61	0.60	0.49	0.57	0.59	0.56	0.55	0.57	0.57	0.55	0.55	0.55	0.60
Healthcare/ hospital	0.64	0.64	0.66	0.65	0.66	0.63	0.64	0.65	0.63	0.64	0.65	0.62	0.64	0.62	0.69	0.63	0.68	0.69	0.70
Hotel/motel	0.65	0.63	0.64	0.63	0.62	0.61	0.62	0.63	0.62	0.59	0.60	0.60	0.57	0.58	0.59	0.56	0.58	0.56	0.56
Office	0.54	0.54	0.53	0.54	0.49	0.52	0.49	0.52	0.45	0.46	0.52	0.47	0.48	0.51	0.48	0.48	0.50	0.45	0.49
Restaurant	0.61	0.58	0.58	0.57	0.57	0.54	0.58	0.59	0.57	0.62	0.61	0.61	0.65	0.64	0.63	0.67	0.66	0.69	0.72
Retail	0.47	0.45	0.44	0.44	0.40	0.39	0.37	0.39	0.36	0.40	0.41	0.42	0.45	0.43	0.46	0.44	0.43	0.42	0.46
School	0.52	0.53	0.53	0.53	0.51	0.51	0.53	0.48	0.46	0.43	0.48	0.47	0.45	0.49	0.46	0.46	0.44	0.44	0.48
Warehouse	0.25	0.25	0.21	0.24	0.20	0.21	0.24	0.20	0.17	0.30	0.22	0.25	0.36	0.28	0.25	0.40	0.34	0.36	0.40
All others	0.58	0.56	0.56	0.56	0.50	0.47	0.49	0.48	0.48	0.49	0.49	0.50	0.51	0.50	0.55	0.52	0.52	0.52	0.55

REQUIREMENTS EXPLAINED

This credit rewards decreased regulated and unregulated energy usage below the *ASHRAE 90.1-2019* requirements referenced in *EAp2: Minimum Energy Efficiency*, Option 1. To limit documentation level of effort, use the same *ASHRAE 90.1* compliance method for prerequisite and credit compliance, either the prescriptive method or Appendix G, Performance Rating Method (PRM).

For the prescriptive method, use the *ASHRAE 90.1-2022* standard for maximum achievement of points and to earn 4 New Construction points or 2 Core and Shell points automatically per *EAc3: Enhanced Energy Efficiency*, Option 1, Path 1, Case 2.

TABLE 11. Linked Prerequisite and Credit Compliance Options for Energy Efficiency

EAp2: Minimum Energy Efficiency	Linked EAc3: Enhanced Energy Efficiency (Choose Option 1 OR Option 2)	Available Points	
		New Construction	Core and Shell
Option 1. *ASHRAE 90.1-2019*, Prescriptive method	Option 1. Prescriptive Path	1–9	1–7
	Path 1. Regulated Loads, Case 1. *ASHRAE 90.1-2019*. Implement additional efficiency measures for HVAC, service water heating, lighting, and/or building mass with night flush using *ASHRAE 90.1-2022*, Section 11, method	1–5	1–5
	AND/OR	+	+
	Path 2. Plug and Process Loads	1–4	1–3
Option 2. *ASHRAE 90.1-2022*, Prescriptive Method	Option 1. Prescriptive Path	1–10	1–7
	Path 1. Regulated Loads, Case 2. *ASHRAE 90.1-2022* Four New Construction or two Core and Shell points automatically rewarded for prerequisite compliance. Up to three additional points to implement incremental efficiency measures for HVAC, service water heating, lighting, and/or building mass with night flush per *ASHRAE 90.1-2022*, Section 11.	4–7	2–6

(continued)

TABLE 11. Linked Prerequisite and Credit Compliance Options for Energy Efficiency (*continued*)

EAp2: Minimum Energy Efficiency	**Linked EAc3: Enhanced Energy Efficiency (Choose Option 1 OR Option 2)**	**Available Points**	
		New Construction	**Core and Shell**
	AND/OR	+	+
	Path 2. Plug and Process Loads	1–4	1–3
Option 1 or Option 2. *90.1-2019 or 90.1-2022* Energy cost budget method (ECB)	Option 1. Prescriptive Path Path 2. Plug and Process Loads	1–4	1–3
New Construction and Core and Shell Option 1. *ASHRAE 90.1-2019*: and New Construction Option 2. *ASHRAE 90.1-2022*, Appendix G, Performance Rating Method (PRM)	Option 2. Energy Simulation Implement efficiency measures to achieve a percentage reduction in future source energy below a performance index target (PI_t) referenced to *ASHRAE 90.1-2019* equivalent performance.	1–10	1–7
Core and Shell Option 2. *ASHRAE 90.1-2022*, Appendix G, Performance Rating Method (PRM)	Option 2. Energy Simulation Implement efficiency measures to achieve a percentage reduction in future source energy below a performance index target (PI_t) referenced to *ASHRAE 90.1-2022* equivalent performance. Two to three points automatically rewarded for future source energy that minimally complies with *ASHRAE 90.1-2022* (0% reduction).		2–7

OPTION 1. PRESCRIPTIVE PATH

Option 1. Prescriptive Path rewards improved efficiency for regulated loads under Path 1 and for plug and process loads under Path 2, with points weighted for each path based on the typical distribution of regulated versus plug and process loads in buildings. Combined points for Paths 1 and 2 cannot exceed 10 for LEED BD+C: New Construction or seven for LEED BD+C: Core and Shell.

For less complex buildings, the prescriptive path offers a streamlined approach to achieving energy efficiency without requiring complex energy modeling by implementing predefined efficiency measures. It is particularly beneficial for projects with limited resources or for those that do not require the flexibility of the energy simulation path.

PATH 1. REGULATED LOADS

Path 1 is available to projects that comply with *EAp2: Minimum Energy Efficiency* using the prescriptive method.

- Case 1 applies to projects complying with Prerequisite Option 1, using *ASHRAE 90.1-2019*, Sections 5–10.
- Case 2 applies to projects complying with Prerequisite Option 2, using *ASHRAE 90.1-2022*, Sections 5–11.

Projects registered beginning January 1, 2028, must use *ASHRAE 90.1-2022*, and earlier registered projects may use either *ASHRAE 90.1-2019* or *ASHRAE 90.1-2022*.

For both Case 1 and Case 2, projects must use the protocol in *ASHRAE 90.1-2022*, Section 11, Additional Efficiency Requirements, to achieve energy credits. See guidance from *EAp2: Minimum Energy Efficiency* and *ASHRAE 90.1-2022*, Section 11, Additional Efficiency Requirements.

Case 1. ASHRAE 90.1-2019

ASHRAE 90.1-2022, Section 11 energy credits must be achieved from the list of eligible HVAC, lighting, service water heating, or building mass/night flush measures rather than the full list of measures from *ASHRAE 90.1-2022*, Section 11.5.2.

Use the less stringent *ASHRAE 90.1-2019* prescriptive references in lieu of *ASHRAE 90.1-2022* prescriptive references for the following measures:

- H02: HVAC Heating Performance Improvement (*ASHRAE 90.1-2019, 11.5.2.2.2*).
  - Refer to the heating equipment efficiencies in *ASHRAE 90.1-2019*, Section 6.8.1 tables.
- H03: HVAC Cooling Performance Improvement (*ASHRAE 90.1-2019, 11.5.2.2.3*).
  - Refer to the cooling equipment efficiencies in *ASHRAE 90.1-2019*, Section 6.8.1 tables.
- L06: Reduce Interior Lighting Power (*ASHRAE 90.1-2019, 11.5.2.5.6*).
  - Determine the interior lighting power allowance using the *ASHRAE 90.1-2019*, Section 9.6, Alternative Compliance Path: Space-by-Space Method.

A maximum of five points can be awarded, depending on project scope:

- **New Construction:** One point for each 25 *ASHRAE 90.1-2022*, Section 11 energy credits documented for eligible measures.
- **Core and Shell—projects with central systems:** One point for each 13 *ASHRAE 90.1-2022*, Section 11 energy credits documented for eligible measures. This encompasses any project with Core and Shell scope of work that includes central boilers, chillers, service water heating equipment, or loop pumping systems with heat rejection, such as water- or ground-source heat pump loops.
- **Core and Shell—projects without central systems:** One point for each nine *ASHRAE 90.1-2022*, Section 11 energy credits documented for eligible measures.

Example: Path 1. Regulated Loads, Case 1. ASHRAE 90.1-2019

A New Construction retail project in climate zone 4A achieves 52 energy credits from three eligible measures, earning two LEED points.

- H02: HVAC Heating Performance Improvement. 20% or greater weighted average improvement in *ASHRAE 90.1-2019* heating efficiency achieves 28 energy credits per *90.1-2022*, Table 11.5.3-6, and adjustment equations from Section 11.5.2.2.2.
- L04: Increased Daylighting Control Area. 65% of total daylighting area with continuous daylight dimming achieves four energy credits per *ASHRAE 90.1-2019*, Table 11.5.3-6.
- L06: Reduce Interior Lighting Power. 10% or greater improvement in *ASHRAE 90.1-2019* regulated lighting power achieves 20 energy credits per *ASHRAE 90.1-2019*, Table 11.5.3-6, and allowed adjustments from Section 11.5.2.5.6.

$$EC_{eligible} = EC\_H02 + EC\_L04 + EC\_L06$$
$$EC_{eligible} = 28 + 4 + 20 = 52$$

Ineligible measures such as E01: Improved Envelope Performance or R01: On-site Renewable Energy do not contribute toward achievement of points under Case 1.

Case 2. ASHRAE 90.1-2022

Minimum Required by ASHRAE 90.1-2022

Projects earn four New Construction points or two Core and Shell points for complying with *EAp2: Minimum Energy Efficiency*, Option 2. *ASHRAE 90.1-2022* using the prescriptive method. Any *ASHRAE 90.1-2022*, Section 11.5.2 efficiency measures may be used to achieve the minimum energy credits required by *2022,* Section 11 for the project's building type and climate zone. However, per *ASHRAE 90.1-2022,* Section 11.5.2*,* the combined contribution of renewable and load management measures is limited to 60% of the total required energy credits.

Incremental energy credits

For additional LEED points, projects must achieve incremental energy credits above the minimum required for prescriptive method compliance. These incremental energy credits must be from the list of eligible HVAC, lighting, service water heating, or building mass/night flush measures rather than the full list of measures from *ASHRAE 90.1-2022*, Section 11.5.2.

To assess achievement of incremental energy credits (EC_{inc}):

- Identify the minimum energy credits required by *ASHRAE 90.1-2022*, Section 11 (EC_{req}).
- Determine the total combined energy credits achieved using *AHRAE 90.1-2022*, Section 11 from eligible measures and from noneligible measures (EC_{total}).
- Determine the total combined energy credits achieved from the list of eligible measures ($EC_{eligible}$).
- The quantity of incremental energy credits achieved is equal to the lesser of the energy credits from eligible measures or the total combined energy credits minus the energy credits required by *ASHRAE 90.1-2022*, Section 11.

$$EC_{inc} = \text{Minimum }(EC_{eligible}, EC_{total} - EC_{req})$$

Points are awarded based on project scope:

- **New Construction:** Earn one point for each 25 incremental *ASHRAE 90.1-2022*, Section 11 energy credits documented for eligible measures up to a maximum of three additional points.
- **Core and Shell projects with central systems:** Earn one point for each 13 incremental *ASHRAE 90.1-2022*, Section 11 energy credits documented for eligible measures up to a maximum of four additional points. Central systems refer to LEED BD+C: Core and Shell scope of work that includes central boilers, chillers, service water heating equipment, or loop pumping systems with heat rejection, such as water- or ground-source heat pump loops.
- **Core and Shell projects without central systems:** Earn one point for each nine incremental *ASHRAE 90.1-2022*, Section 11 energy credits documented for eligible measures up to a maximum of four additional points.

Example 1

A new construction two-story retail project in climate zone 4A earns five LEED points: four points for complying with *ASHRAE 90.1-2022* using the prescriptive method and one additional point for achieving at least 25 incremental energy credits above the minimum required from the list of eligible measures.

Per *ASHRAE 90.1-2022*, Table 11.5.3-6, the project must achieve at least 50 energy credits to comply with *ASHRAE 90.1-2022* using the prescriptive method.

$$EC_{req} = 50$$

- The project documents achievement of 97 total energy credits per *ASHRAE 90.1-2022*, Section 11.
- H02: HVAC Heating Performance Improvement. 20% or greater weighted average improvement in *ASHRAE 90.1-2022* heating efficiency achieves 28 energy credits.
- L04: Increased Daylighting Control Area. 65% of total daylighting area with continuous daylight dimming achieves four energy credits.
- L06: Reduce Interior Lighting Power. 10% or greater improvement in *ASHRAE 90.1-2022* regulated lighting power achieves 20 energy credits.
- E01: Improved Envelope Performance. 4.5% improvement in envelope performance factor achieves 45 energy credits.

$$EC_{total} = EC\_H02 + EC\_L04 + EC\_L06 + EC\_E01$$
$$EC_{total} = 28 + 4 + 20 + 45 = 97$$

- 52 of these energy credits are from measures eligible for LEED points (H02, L04, and L06).

$$EC_{eligible} = EC\_H02 + EC\_L04 + EC\_L06$$
$$EC_{eligible} = 28 + 4 + 20 = 52$$

- The project achieves 47 incremental energy credits above the minimum required for prescriptive method compliance.

$$EC_{inc} = \text{Minimum}\ (EC_{eligible},\ EC_{total} - EC_{req})$$
$$EC_{inc} = \text{Minimum}\ (52,\ 97 - 50) = 47$$

Example 2

The project referenced in Example 1 adds a roof-mounted photovoltaic array with a rated capacity of at least 0.75 W/sq. ft. (8.1 W/sq. m.) of the project's gross floor area, achieving 30 energy credits for measure R01: On-site Renewable Energy, and maximizing the combined allowable contribution from *ASHRAE 90.1-2022*, Section 11, for renewable and load management credits. This results in achievement of six total LEED points for Path 1: four points for complying with *ASHRAE 90.1-2022* using the prescriptive method, and two additional points for achieving at least 50 incremental energy credits above the minimum required from the list of eligible measures.

- $EC_{req} = 50$
- The project documents achievement of 127 total energy credits per *ASHRAE 90.1-2022,* Section 11.

$$EC_{total} = EC\_H02 + EC\_L04 + EC\_L06 + EC\_E01 + EC\_R01$$
$$EC_{total} = 28 + 4 + 20 + 45 + 30 = 127$$

- 52 of these energy credits are from measures eligible for LEED points (H02, L04, and L06).

$$EC_{eligible} = EC\_H02 + EC\_L04 + EC\_L06$$
$$EC_{eligible} = 28 + 4 + 20 = 52$$

- The project achieves 52 incremental energy credits above the minimum required for prescriptive method compliance.

$$EC_{inc} = \text{Minimum } (EC_{eligible}, EC_{total} - EC_{req})$$
$$EC_{eligible} = \text{Minimum } (52, 127 - 50) = 52$$

Eligible Measures for Path 1

TABLE 12. Eligible Measures From *ASHRAE 90.1-2022*, Section 11.5.2 for LEED Points

HVAC Measures	Lighting Measures
H01: HVAC System Performance Improvement (*ASHRAE 90.1-2022*, Addendum j). (cannot be used in conjunction with H02, H03, or H06)	L01. Lighting System Performance (not included in *ASHRAE 90.1-2022*, but may be in future ASHRAE addenda)
H02: HVAC Heating Performance Improvement	L02: Lighting Dimming and Tuning
H03: HVAC Cooling Performance Improvement	L03: Increase Occupancy Sensor
H04: Residential HVAC Controls	L04: Increase Daylight Area
H05: Ground-Source Heat Pump	L05: Residential Light Controls
H06: DOAS/Fan Controls	L06: Light Power Reduction
H07: Guideline 36 Sequences	
Service Water Heating (SWH) Measures	**Load Management Measures**
W01: SWH Preheat Recovery	G07: Building Mass/Night Flush
W02: Heat-Pump Water Heater	
W03: Efficient Gas Water Heater	
W04: SWH Pipe Insulation	
W05: Point-of-Use Water Heaters	
W06: Thermostatic Balancing Valves	
W07: SWH Submeters	
W08: SWH Distribution Sizing	
W09: Shower Drain Heat Recovery	

The remaining measures referenced in *ASHRAE 90.1-2022*, Section 11.5.2 are ineligible for incremental LEED points, because these measures are separately rewarded in other LEED credits:

- E01: Improved Envelope Performance is rewarded in *EAc2: Reduce Peak Thermal Loads*

- P01: Energy Monitoring is rewarded under *EAc5: Enhanced Commissioning*, Option 2. Monitoring-Based Commissioning (MBCx)
- R01: Renewable Energy is rewarded in *EAc4: Renewable Energy*
- Q01: Efficient Elevator Equipment and Q02: Efficient Kitchen Equipment are rewarded in *EAc3: Enhanced Energy Efficiency*, Option 1. Path 2. Plug and Process Loads
- G01 to G06: Load Management measures are rewarded in *EAc6: Grid Interactive*

Maximum achievable points

Lower quantities of available *ASHRAE 90.1-2022* energy credits from eligible measures for some project types and climate zones place further limitations on the maximum available LEED points for these applications, as shown in Table 13.

TABLE 13. Maximum LEED Points Availability for Option 1. Prescriptive Path, Path 1. Regulated Loads (New Construction)

	Case 1. *ASHRAE 90.1-2019*		**Case 2. *ASHRAE 90.1-2022***	
Building Type	**Maximum Points**	**Climate Zones**	**Maximum Points**	**Climate Zones**
Multifamily	5	All	7	All
Healthcare	5	0B, 1B, 2A, 0A, 2B, 3A, 3B, 3C, 4A, 5C, 6A, 6B, 7, 8	7	All
	4	1A, 4B, 4C, 5A, 5B		
Office	4	0A, 0B, 1A, 1B, 2A, 2B, 6A, 6B, 7, 8	7	All
	3	3A, 3B, 3C, 4A, 4B, 4C, 5A, 5B, 5C		
Hotel/motel	5	0A, 0B, 1A, 1B, 2A, 3A, 6A, 6B, 7, 8	7	All
	4	2B, 3B, 3C, 4A, 4B, 4C, 5A, 5B, 5C		
Restaurant	5	0A, 0B, 1B, 4A, 5A, 5B, 6A, 6B, 7, 8	7	All except 3C
	4	1A, 2A, 2B, 3A, 3B, 4B, 4C, 5C	6	3C
	2	3C		
Retail	5	0A, 0B, 1A, 1B2A, 2B, 3A, 3B, 4A, 6A, 6B, 7, 8	7	All
	4	3C, 4B, 4C, 5A, 5B, 5C		
Warehouse	5	6A, 8	7	0A, 0B, 1A, 1B, 4A, 5A, 5B, 6A, 6B, 7, 8
	4	5A, 6B, 7	6	2A, 2B, 3A, 3B, 3C, 4B, 4C, 5C
	3	0A, 0B, 1A, 1B, 4A, 5B		
	2	2A, 2B, 3A, 3B, 3C, 4B, 4C, 5C		
Education	5	0A, 0B, 1B, 1A, 2A, 2B, 3A, 8	7	All
	4	3B, 3C, 4A, 4B, 4C, 5A, 5B, 5C, 6A, 6B, 7		
Other	2	All except 0A	6	All except 0A
	1	0A	5	0A

Equivalence to ASHRAE 90.1

Refer to the Project Priorities Library for regional paths addressing equivalence to *ASHRAE 90.1*.

To be eligible to apply *EAc3: Enhanced Energy Efficiency*, Option 1. Prescriptive Path, Path 1. Regulated Loads, projects using an *ASHRAE 90.1*-equivalent standard must use a prescriptive method from the equivalent standard. Compliance must be demonstrated per category of building systems (building envelope, HVAC, lighting, service water heating, etc.) without trade-offs between system categories.

Projects that use a standard other than *ASHRAE 90.1* or *International Energy Conservation Code (IECC)* for prerequisite compliance must apply the methodology in *ASHRAE 90.1-2022*, Section 11 to achieve points under Path 1:

- **Case 1**. Prescriptive values from the *ASHRAE 90.1-2019*-equivalent standard for efficiencies, such as lighting power density allowance or equipment efficiency, may replace *ASHRAE 90.1-2022*, Section 11 referenced values.
- **Case 2**. Four New Construction points or two Core and Shell points are automatically rewarded for prescriptive compliance with an *ASHRAE 90.1-2022*-equivalent standard. For additional points, projects must document achievement of *ASHRAE 90.1-2022*, Section 11 credits, including
  - The minimum required credits for the project type and climate zone, AND
  - Incremental credits above the minimum required from the list of eligible measures.

For projects that use the *IECC* prescriptive method for prerequisite compliance, energy credits from *IECC-2024*, Section C406 Additional Efficiency Renewable and Load Management Requirements may directly replace energy credits from *ASHRAE 90.1-2022*, Section 11, Additional Efficiency Requirements. Projects must exclusively use either *ASHRAE 90.1-2022* or *IECC-2024* to document achievement of energy credits. Eligible measures from *IECC-2024*, C406 are limited to HVAC (H01 to H05), Lighting (L01 to L06), and Service Water Heating (W01 to W10).

- **Case 1:** Implement additional efficiency credits calculated per *IECC-2024,* C406 from the list of eligible measures. Prescriptive values from *IECC-2021*, Sections C402–C405, such as lighting power density allowance and equipment efficiency, may replace *IECC-2024*, Section 11 referenced values.
- **Case 2:**
  - Four New Construction points or two Core and Shell points are automatically rewarded for compliance with *IECC-2024*, Sections C402–C406.
  - Implement incremental C406 credits, above the minimum required from the list of eligible measures.

EA

District energy

Projects may optionally account for District Energy System (DES) efficiency when assessing energy credits for Case 1 and incremental energy credits beyond the minimum required for Case 2:

- H02: HVAC Heating Performance Improvement, or H03: HVAC Cooling Performance Improvement. Calculate the heating and/or cooling capacity-weighted average improvement separately for the DES system and for the building systems. Use the peak DES capacity supplied to the building and the sum of equipment capacity on-site to calculate the heating and/or cooling capacity-weighted average improvement for the combined systems.
- H05: Ground-Source Heat-Pump System. Demonstrate that the credit requirements are met by a Ground-Source Heat-Pump District Energy System serving the project.

- W02: Heat-Pump Water Heater, or W03: Efficient Gas Water Heater. Demonstrate that the credit requirements are met for a District Heating System serving the project.

PATH 2. PLUG AND PROCESS LOADS

Path 2 places a spotlight on plug, process, cooking, refrigeration, and elevator/escalator system energy use that represents 30%–50% of building energy usage yet are only partially or peripherally addressed through *ASHRAE 90.1* standard requirements.

Projects may apply Case 1 in conjunction with Case 2 or document compliance with Case 1, Case 2, or Case 3 independently. (Case 3 is only available for New Construction.)

Case 1. Plug Load Management (New Construction and Core and Shell)

A plug load dashboard empowers building occupants to actively engage in reducing building energy consumption from plug loads. For projects with required monitoring and recording of receptacle use at 15-minute intervals per *EAp4: Energy Metering and Reporting*, the dashboard should visualize this data and compare receptacle energy consumption to the prior interval annually, monthly, daily, and hourly. The dashboard is only required to show usage for receptacle circuits but may optionally address other process usage (such as elevators) or other building end uses (such as lighting or HVAC energy).

New Construction

The dashboard must be accessible to all regular building occupants. For buildings with multiple tenants or multiple floors, the dashboard may be configured so that an individual user sees only the receptacle usage for common areas and for their space.

Buildings with information technology (IT) departments operating in the building must also develop and implement policies for monitors, visual displays, personal computers, and laptops to be controlled off or in a very low power mode when not in use, except during scheduled maintenance periods.

Core and Shell

The dashboard must be configured with the capability for all regular building occupants to view common area receptacle usage and include functionality enabling each tenant manager to provide their occupants with access to plug load data specific to the tenant space. The project design must include a plug load monitoring system feeding into the plug load dashboard, including the capability to expand the monitoring of future plug loads for each floor and for each individual tenant spaces.

Case 2. Efficient Plug and Process Load Equipment (New Construction and Core and Shell)

Case 2 provides a streamlined path for rewarding plug and process equipment efficiency. The path is best suited for projects where a significant proportion of the project's plug and process load consists of equipment referenced in Table 2.

New Construction

One point is rewarded for each equipment category where at least 90% of applicable project equipment in the project scope meets the efficiency criteria. Projects can earn up to a maximum of three points.

Up to 90% of applicable equipment may be assessed using either equipment quantity or rated load:

- **Equipment quantity:** Divide the total quantity of equipment that meets the efficiency criteria for the equipment category by the total quantity of applicable equipment within the project scope for the equipment category.
- **Rated load:** For applicable equipment in the equipment category, divide the sum of rated load for equipment that meets the efficiency criteria by the sum of rated load for all equipment within the project scope.

Either include or exclude all applicable equipment reused in the project from the calculations. Reused ENERGY STAR products are deemed compliant even when not meeting current ENERGY STAR specifications.

To ensure a measurable impact on project performance, Table 2 criteria for both ENERGY STAR products and categories specify a minimum 0.1 W/sq. ft. (1.1 W/sq. m.) of eligible equipment per unit of gross floor area. Credit gets rewarded as one consolidated Table 2 equipment category when the project has less than 0.1 W/sq. ft. (1.1 W/sq. m.) of rated load for each individual ENERGY STAR Products Equipment Category but exceeds this value for the sum of the two categories.

New Construction: Process-intensive buildings

Process-intensive buildings, such as data centers, restaurants, or refrigerated warehouses, have greater potential to achieve substantial building energy savings from a single equipment category. Therefore, projects with combined equipment load from the equipment categories in Table 2 totaling at least 0.5 W/sq. ft. (5.4 W/sq. m.) of gross floor area may assess compliance based on 90% of total applicable equipment-rated load rather than per equipment category. All applicable equipment from all Table 2 equipment categories must be included in the assessment of credit compliance.

Core and Shell

The credit rewards efficiency measures implemented for plug and process equipment commonly within the Core and Shell scope of work. Points are rewarded per Core and Shell Table 2 for each equipment category with project scope meeting the credit criteria, up to a maximum of three points. To be eligible for credit, all equipment installed within the project scope of work for the referenced equipment category must meet the credit criteria. For example, for a project with multiple elevators and escalators, each elevator and escalator that is eligible for an *ISO 25745* rating must achieve at least a Class A rating to qualify for points under the People Conveyance Equipment category.

Equivalence to ENERGY STAR products

Refer to the Project Priorities Library for regional paths addressing equivalence to ENERGY STAR products.

Case 3. Plug and Process Load Exceptional Calculation (New Construction only)

Case 3 primarily applies to project applications with unique plug and process loads largely unaddressed by Table 2 equipment categories (such as manufacturing or laboratory) or projects for which the streamlined methodology from Cases 1 and 2 insufficiently reveals the magnitude of impact for plug and process efficiency measures implemented for the project. Case 3 cannot be combined with Case 1 or Case 2.

Project analysts must use the *ASHRAE 90.1*, Section G2.5, Exceptional Calculation Method, to demonstrate a minimum percentage improvement in total plug and process energy use compared to a baseline representative of standard practice for a similar newly constructed building.

Perform a detailed assessment to determine the total estimated annual building energy consumption from all plug and process equipment, even when regulated under *ASHRAE 90.1*:

- Receptacle equipment
- Cooking equipment
- Refrigeration equipment
- Conveyance equipment, including elevators, escalators, or moving walkways
- Process heating or process cooling (e.g., for manufacturing processes)
- Data center IT equipment and Electrical Loss Component
- All other process energy used to support a manufacturing, industrial, or commercial activity other than conditioning spaces and maintaining comfort and amenities for the occupants of a building.

Projects for which start-up plug and process energy use projections are lower than estimated full build-out usage, apply either the full build-out usage or a lesser value representing the maximum plug and process usage possible from the building power and thermal energy generation capacity installed in the project scope of work.

For each process efficiency measure implemented in the project, document that the efficiency measure is not conventional practice. Examples include

- A recent study with researched tabulations or monitored data establishing standard practice for the given application in similar newly constructed facilities.
- A new construction utility or government program that provides incentives for the measure.
- A document showing the systems used to perform an analogous function in similar facilities built or reconstructed within the last 10 years.
- Applicable prescriptive requirements from the version of *ASHRAE 90.1* or equivalent standard used for *EAp2: Minimum Energy Efficiency*.

Use the conventional practice references to define the baseline systems. Provide detailed calculations and supporting narrative justification for any variations in baseline and proposed energy use.

EA

OPTION 2. ENERGY SIMULATION

Option 2 rewards future source energy improvement below a performance index target (PI_t) documented per *ASHRAE 90.1*, Appendix G, PRM. The credit structure prioritizes efficiency over on-site renewable energy, setting the performance improvement thresholds more than three times higher for Path 2 (including on-site renewable contribution) than for Path 1 (excluding on-site renewable contribution).

For credit compliance assessment, apply simple additional calculations to the outputs from the *ASHRAE 90.1,* Appendix G, PRM, energy models used to document *EAp2: Minimum Energy Efficiency*, adjusting the metric, treatment of renewable energy, and/or BPF.

- **Future source energy.** The future source energy metric must be used to assess credit compliance even when a different metric is used to document prerequisite compliance.
- **BPF.** PI_t must be calculated using LEED-published BPFs derived for the future source energy metric.
- **Treatment of on-site renewable energy.**
  - The on-site renewable energy contribution must either be fully excluded for Path 1 or fully included for Path 2.
  - PI_t includes no adjustments for renewable energy, unlike *ASHRAE 90.1-2022*, which adjusts for prescriptively required on-site renewable energy.

Further energy savings may also be documented for plug and process and/or district energy efficiency measures that do not contribute toward prerequisite compliance.

Future source energy metric

All projects must use the future source energy metric for credit compliance. This metric reflects the average environmental impact of building energy consumption through 2050, considering the primary energy sources and their associated emissions.

Future source energy conversion factor (primary energy factor)

The terms *source energy* and *source energy conversion factor* may be used interchangeably with the corresponding terms *primary energy* and *primary energy factor* that are commonly used in the European Union (EU).

Source energy is defined as the site energy plus the estimated energy consumed or lost in the extraction, processing, and transportation of primary energy forms such as coal, oil, natural gas, biomass, and nuclear fuel; energy consumed in conversion to electricity or thermal energy; and energy consumed or lost in transmission and distribution to the building site.

Source energy conversion factors must be at least one for all electricity and combustible fuel sources.

Electricity

For electricity, future source energy is determined using a national average electric site-to-source conversion factor rather than a more granular determination by grid region or province to enable broader comparison of building energy efficiency across the entire spectrum of projects, and to account for the interconnectedness of electric grids.

For projects located in the U.S., use a national average electricity source energy conversion factor of 2.0 based on projections through 2050.

Projects located in other countries must use this same source energy conversion factor of 2.0 or provide data supporting a lower average source energy conversion factor for the project's country.

European Union average values may be used instead of the national average for projects in the EU. The source energy conversion factor must be one of the following:

- Current published national or EU average source energy conversion factor.
- National or EU average source energy conversion factor from the present through 2050 or earlier, determined based on published policy-based grid renewable projections. Calculate by averaging the current published national or EU average source energy conversion factor and the predicted source energy conversion factor for the year 2050 or earlier or by an average that accounts for year-to-year source energy projections.

For example, projects in Europe may use the current default factor from the EU directive (1.9 as of this publication[6]), a lower published factor from the project's EU member state, or a factor determined from policy-based published grid renewable projections through 2050 for the EU or the project's member state.

More granular future source energy conversion factors per state, province, or eGRID region are not allowed, because these necessitate greater complexity of the *EAc3: Enhanced Energy Efficiency* credit requirements and increase ambiguity in comparative results.

Fuel

Use one of the following references for conversion factors:

- ENERGY STAR Portfolio Manager Technical Reference: Source Energy
- Natural Gas source energy conversion factor = 1.09 per *ASHRAE 90.1-2022*, Table I5-1
- *ASHRAE 100-2024*, Table 5-2, or *ASHRAE 228*, Table 4
- Published national or EU average source energy conversion factor conforming to the definition of source energy provided earlier

District Energy Systems

For projects that directly model purchased heat and/or purchased chilled water as independent sources, refer to *EAp2: Minimum Energy Efficiency,* District Energy Systems, Method B.

Source energy factors published by the DES provider or calculated for a campus DES must conform to the definition of source energy provided earlier.

Future source energy metric

The future source energy metric prioritizes energy efficiency and limits trade-offs with decarbonization measures recognized in other LEED credits. Other metrics used for code compliance or *EAp2: Minimum Energy Efficiency* are not applicable for *EAc3: Enhanced Energy Efficiency*:

- Energy costs can fluctuate due to market conditions, subsidies, and other economic factors and skew the representation of the environmental footprint of energy use for the cost metric.

6. "Directive (EU) 2023/1791 of the European Parliament and of the Council of 13 September 2023 on energy efficiency and amending Regulation (EU) 2023/955 (recast) (Text with EEA relevance)," https://eur-lex.europa.eu/eli/dir/2023/1791/.

EA

- A site energy metric overemphasizes the decarbonization already credited under *EAc1: Electrification.*
- A greenhouse gas emissions metric overemphasizes the decarbonization already credited under *EAc1: Electrification* and *EAc4: Renewable Energy*.

Building performance factor

PI_t must be calculated using LEED-published BPFs derived for the future source energy metric.

For major renovations or projects with existing building area in the project boundary, multiply the LEED-published BPFs by 1.05 for the proportion of existing building area associated with each building area type.

NEW CONSTRUCTION

Reference *ASHRAE 90.1-2019*, New Construction Table 5, "Equivalent Building Performance Factors for a Future Source Energy Metric." For projects that document prerequisite compliance using *ASHRAE 90.1-2022*, this directly rewards the differential future source energy savings from *ASHRAE 90.1-2019* to *ASHRAE 90.1-2022*.

CORE AND SHELL

Case 1. *ASHRAE 90.1-2019*: If the prerequisite was documented using *ASHRAE 90.1-2019,* Appendix G, PRM, use Building Performance Factors from Core and Shell Table 4, "Equivalent Building Performance Factors for a Future Source Energy Metric."

Case 2. *ASHRAE 90.1-2022*: If the prerequisite was documented using *ASHRAE 90.1-2022,* Appendix G, PRM, use Core and Shell Table 5, "Equivalent Building Performance Factors for a Future Source Energy Metric." Future source energy less than or equal to the PI_t automatically achieves three points without crediting on-site renewable energy (Path 1) or two points when counting on-site renewable energy (Path 2).

Treatment of on-site renewable energy

The PI_t for *EAc3: Enhanced Energy Efficiency* omits the *ASHRAE 90.1-2022,* Section 4.2.1.1 Prescriptive Renewable Energy (PRE) adjustment for prescriptively required on-site renewable energy, matching the *ASHRAE 90.1-2019* equation for PI_t :

$$PI_t = [BBUE + (BPF \times BBRE)] / BBP$$

To determine the PI, either fully exclude the renewable contribution for Path 1 or fully include the renewable contribution for Path 2.

PATH 1. PERCENTAGE REDUCTION EXCLUDING ON-SITE RENEWABLE CONTRIBUTION

Path 1, the default energy simulation path, focuses solely on energy efficiency and requires energy savings at or near achievable technical potential for maximum achievement of points.

This path does not recognize any on-site renewable contribution for the project (striking out *ASHRAE 90.1,* G2.4, on-site renewable energy guidance). Calculate the proposed building performance with total proposed design energy for all electricity, fuel, and district energy use, regardless whether this energy is purchased or generated from on-site renewable systems.

PATH 2 PERCENTAGE REDUCTION INCLUDING ON-SITE RENEWABLE CONTRIBUTION

Path 2 applies primarily to project types that have limited opportunities for incremental energy savings beyond the referenced standard, such as unconditioned warehouses, and to projects that generate a large proportion of total building energy from on-site renewables.

Consistent with *ASHRAE 90.1*, G2.4, this path credits the on-site renewable contribution for the project. Prior to calculating the PBP, subtract eligible on-site renewable energy generation from the proposed design energy consumption. Total savings documented for on-site renewable electricity generation can be up to 100% of building electricity use on an annual basis. To qualify for credit compliance, the on-site renewable energy must be installed and commissioned within the project scope of work or an earlier scope of work on the site of a contiguous campus with all renewable attributes allocated to the project.

Plug and process efficiency

For *EAc3: Enhanced Energy Efficiency*, projects may document credit for plug and process efficiency using *ASHRAE 90.1*, G2.5, Exceptional Calculation Method.

Document that each process efficiency measure is not conventional practice. Provide detailed calculations and narrative justification that supports the future source energy savings claimed. (See preceding guidance from Option 1, Path 2, Case 3 Plug and Process Load Exceptional Calculation.) To convey the magnitude of impact associated with process efficiency measures, the energy analyst must separately report the performance index, performance index target, and all associated terms with and without the process efficiency savings.

District energy

For *EAc3: Enhanced Energy Efficiency*, energy analysts may opt to replace prescriptive purchased heat and purchased chilled water efficiencies modeled per *ASHRAE 90.1-2022*, Addendum a, with improved virtual DES efficiencies representative of the DES purchased heat and purchased chilled water systems serving the project.

Provide an engineering analysis based on monitored data and/or energy simulation to justify the improved virtual DES efficiencies modeled for the project. For each DES source, virtual DES efficiency must account for the total annual energy required to generate and distribute the district energy. Include all pump energy use from the DES and within the project, thermal distribution losses, heat rejection, and all operational effects influencing efficiency, such as standby losses, equipment cycling, equipment staging, and partial-load operation. When thermal distribution losses are not measured or modeled, estimate default losses of 5% for chilled water, 10% for hot water, 15% for closed-loop steam, and 25% for open-loop steam.

No further adjustments are required to a baseline building performance model with purchased heat and purchased chilled water documented per *ASHRAE 90.1-2022*, Addendum a, in *EAp2: Minimum Energy Efficiency.*

Combined heat and power

To limit undue credit for site-recovered energy from combined heat and power (CHP) with on-site combustion emissions, projects must model CHP systems using one of the following methods instead of *ASHRAE 90.1*, G2.4.2.1:

- **CHP in baseline and proposed.** Model CHP systems, including all fuel inputs and associated site-recovered energy, identically in the baseline design and the proposed design.
- **Purchased electricity in baseline and proposed.** Model purchased electricity instead of the on-site electricity generation. Either credit site-recovered energy from the CHP toward the thermal loads for the baseline and proposed design identically or ignore the site-recovered energy contribution in the baseline and proposed design.
- **CHP in proposed; purchased electricity in baseline.** Model CHP systems, including all fuel inputs and associated site-recovered energy, in the proposed design. Model the baseline design per *ASHRAE 90.1* with purchased electricity and with no credit for site-recovered energy.

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	Option 1	Path 1: Case 1 and 2	Demonstrate compliance with *EAp2: Minimum Energy Efficiency*.
		Path 1: Case 1 and 2	List options showing prescriptive compliance with *ASHRAE 90.1 2019*.
		Path 2: Case 1	IT department policies for plug load controls.
			Evidence of a plug load dashboard accessible to all regular occupants of the building (specifications, photos, screenshots, etc.)
		Path 2: Case 2	Provide documentation demonstrating the *ISO 25745* rating for people conveyance equipment (as applicable).
			Provide the design electrical loss component (ELC) calculation following methods in Informative Appendix C, and calculation for the percentage of reduction from *ASHRAE 90.4-2022* maximum design electrical loss (as applicable).
			Provide refrigeration equipment cut sheets (as applicable).
			Provide refrigerated warehouse design documents to verify compliance with *California Title 24-2022 Section 120.6* (as applicable).
			Provide documentation for the Baggage Handling Equipment Individual Carrier System, such as specifications or cut sheets, to verify the installation of variable frequency drives (as applicable).
			Provide documentation for the Preconditioned Air (PCA) Systems, such as specifications or cut sheets, to verify efficiencies meeting *ASHRAE 90.1* prescriptive efficiencies for HVAC equipment (as applicable).
	Option 2	Path 1 and Path 2	Link to *EAp2: Minimum Energy Efficiency*.
			USGBC LEED v5 Energy Simulation Report.

(continued)

Project Types	Options	Paths	Documentation
New Construction	Option 1	Path 2: Case 2	Evidence of ENERGY STAR equivalency documentation for nonrated equipment (as applicable).
			Applicable plug and process load equipment power density and calculated percentage meeting the required criteria. Including equipment, rated power, and quantity (either including or excluding all reused equipment).
		Path 2: Case 3	Plug and process load exceptional calculation and documentation.

REFERENCED STANDARDS

- *ASHRAE 90.1-2019* (store.accuristech.com/ashrae/standards/ashrae-90-1-2019-i-p?product_id=2088527).
- *ASHRAE 90.1-2022* (store.accuristech.com/ashrae/standards/ashrae-90-1-2022-i-p?product_id=2522082).
- *ASHRAE 100-2024* (ashrae.org/technical-resources/bookstore/standard-100)

FURTHER EXPLANATION

ENERGY STAR® EQUIVALENCE

The ENERGY STAR program distinguishes energy efficient products verified through third-party testing. Plug loads, small appliances, and process equipment are recognized as approved equivalents to the ENERGY STAR label under *EAc3: Enhanced Energy Efficiency* when certified under an eco-label administered by an independent third-party organization or government entity with transparent processes and standards in place, demonstrating at least one of the following:

- Most energy efficient tier for the product category—for example, the most efficient ranking of “A” in a scale from A to F, or the highest ranking of three stars in a scale of 1 to 3.
- Rating demonstrating the lowest 30th percentile of energy consumption for the product category in the project’s country or geographical region—for example, a ranking of “B” corresponding to product energy usage lower than at least 70% of similar products.
- Labeling specifications corresponding to a minimum 20% weighted average improvement in annual energy consumption compared to standard practice in the project’s country or geographical region:
  - For all products certified by the eco-label, OR
  - For the tier of products certified by the eco-label, such as a ranking of “B” corresponding to at least 20% weighted average improvement in energy consumption compared to standard practice across all rated products.

Renewable Energy

EAc4

New Construction (1–5 points): 100% of site energy use from any combination of Tier 1, Tier 2, and Tier 3 renewable energy is required for LEED Platinum projects.

Core and Shell (1–4 points): 100% of base building energy use from any combination of Tier 1, Tier 2, and Tier 3 renewable energy is required for LEED Platinum projects.

IMPACT AREA ALIGNMENT:

- [x] Decarbonization
- [ ] Quality of Life
- [ ] Ecological Conservation and Restoration

INTENT

To encourage and recognize the use of renewable energy to reduce environmental and economic impacts associated with fossil fuel energy use and increase the supply of new renewable energy within the electrical grid, thereby fostering a just transition to a green economy.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1-5**
Option 1. Renewable Energy Supply or Procurement	1-5
Core and Shell	**1-4**
Option 1. Renewable Energy Supply or Procurement AND/OR	1-4
Option 2. Renewable Energy Readiness	1

OPTION 1. RENEWABLE ENERGY SUPPLY OR PROCUREMENT (1–5 POINTS NEW CONSTRUCTION, 1–4 POINTS CORE AND SHELL)

Supply or procure renewable energy meeting the renewable energy criteria referenced below. Points are rewarded according to Table 1.

Points documented for Tier 1, Tier 2, and/or Tier 3 renewable energy may be added together for a maximum of 5 points:

TABLE 1. Points for Renewable Energy Procurement for New Construction Projects

Points	Tier 1			Tier 2	Tier 3
	Minimum Rated Capacity*	**OR**	**Percent of Annual Site Energy**	**Percent of Annual Site Energy**	**Percent of Annual Site Energy**
1	A × 1 W/sq. ft. (A × 10.8 W/sq. m.)	OR	5%	20%	50%
2	A × 2 W/sq. ft. (A × 21.6 W/sq. m.)	OR	10%	40%	100%
3			20%	60%	
4			35%	80%	
5			100% Tier 1 and/or Tier 2 renewable energy		

*A = sum of gross floor area of all floors up to the three largest floors.

Core and Shell only

Points documented for Tier 1, Tier 2, and/or Tier 3 renewable energy may be added together for a maximum of 4 points, based on the percentage of total annual base building site energy use, where base building energy use is defined as the greater of the estimated site energy consumption from base building energy meters or 25% of the total estimated building energy use:

TABLE 2. Points for Renewable Energy Procurement for Core and Shell Projects

Points	Tier 1			Tier 2	Tier 3
	Minimum Rated Capacity*	**OR**	**Percent of Annual Base Building Site Energy**	**Percent of Annual Base Building Site Energy**	**Percent of Annual Base Building Site Energy**
1	A × 1 W/sq. ft. (A × 10.8 W/sq. m.)	OR	15%	35%	100%
2	A × 2 W/sq. ft. (A × 21.6 W/sq. m.)	OR	30%	70%	200%
3			65%	100%	
4			100%	200%	

*A = sum of gross floor area of all floors up to the three largest floors.

RENEWABLE ENERGY CRITERIA

Renewable energy classifications

TIER 1: ON-SITE RENEWABLE ENERGY GENERATION OR SOCIAL IMPACT PROJECT

The renewable generation equipment may be located:

- On the project site.
- On the campus on which a project is located.
- On the site of a social impact project, provided that the renewable power system is provided, installed, and commissioned at no cost to the social impact entity, that the ownership of the renewable power system is transferred to the social impact entity, and that the rights to the power provided be given to the social impact entity.

TIER 2: NEW OFF-SITE RENEWABLE ELECTRICITY

Off-site renewable electricity produced by new generation asset(s):

- Contracted to be operational within two years of building occupancy, OR
- Contracted no more than five years after commercial operations date

TIER 3: OFF-SITE RENEWABLE ENERGY

- Off-site renewable electricity that is Green-e® Energy certified or equivalent.
- Renewable fuels that are Green-e Energy certified or equivalent.

Renewable energy contract length

- Contract length shall be 10 years or prorated across 10 years for shorter contract lengths.

Renewable energy environmental attributes

- **Ownership:** All environmental attributes—energy attribute certificates (EACs) or renewable energy certificates (RECs)—associated with renewable energy generation must be retired on behalf of the LEED project for the renewable energy procurement to contribute to credit achievement.
- **Project energy source:** Renewable electricity generation and EAC/REC procurement can only be applied to project electricity use or district energy use up to 100% of annual electricity plus district energy use. Renewable fuels can only be applied to project fuel use or district heat up to 100% of annual fuel plus district heat use.

Core and Shell only

Renewable electricity generation and EAC procurement can only be applied to project electricity use or district energy use up to 100% of total estimated building annual electricity plus district energy use. Renewable fuels can only be applied to project fuel use or district heat up to 100% of total estimated building annual fuel plus district heat use.

- **Vintage:** EACs credited to the project must be generated no earlier than 18 months before the LEED project's initial application submission date.
- **Location:** Tier 2 and Tier 3 renewable assets must be in the same country or region where the LEED project is located.
- **Tier 2 bulk purchase:** Green-e Energy certification or equivalent is required for one-time purchase or annual purchase of EACs or renewable power totaling more than 100% of the annual electricity use for New Construction, or 100% of base building annual electricity use for Core and Shell.

AND/OR

OPTION 2. RENEWABLE ENERGY READINESS (1 POINT)—CORE AND SHELL ONLY

Design the project for renewable energy readiness. Document renewable energy readiness features in the project design and provide tenant guidelines indicating how tenants can leverage these features to install renewable energy capacity for their tenant space. Tenant guidelines shall include a copy of the construction documents or a comparable document indicating the information below. Projects shall either document solar readiness in accordance with the criteria below or an equivalent solar readiness standard, or document equivalent on-site renewable energy readiness for another qualifying renewable energy source.

SOLAR READINESS

Solar zone

A designated solar zone shall be included in the project design.

- Designate a dedicated solar zone area equal to at least 40% of the gross roof area.
- Conduct an analysis to determine the most appropriate location for optimal location of the solar zone, avoiding shading from trees, buildings, and so forth, and accounting for future construction that may result in shading.
- Perform a wind and load analysis and confirm that the roof or other structure encompassing the solar zone is designed to accommodate all mounting configurations identified in the tenant guidelines.
- Total area shall be comprised of areas with no dimension less than five feet and no less than 160 square feet.
- No obstruction, such as vents, chimneys, or roof-mounted equipment, shall be in the solar zone or planned for future installation in the solar zone.
- Any obstruction located on the roof or other part of the building that projects above a solar zone shall be located at least twice the distance, measured in the horizontal plane, of the height difference between the highest point of the obstruction and the horizontal projection of the nearest point of the solar zone, measured in the vertical plane. (Exceptions: projects located within 10 degrees of the earth's equatorial plane, or any obstruction located north of all points of the solar zone in the northern hemisphere, or any obstruction located south of all points of the solar zone in the southern hemisphere.)
- All sections of the solar zone located on steep-sloped roofs shall have an azimuth range between 90 degrees and 300 degrees of true north.
- If the project is intended to include future mechanical or electrical equipment in or near the solar zone, tenant guidelines shall include a sample plan showing how the equipment can be installed to maintain the integrity of the solar zone.

Mounting considerations

- Identify the panel-mounting options most likely to be implemented in a future solar panel installation. Provide documentation confirming that the roof warranty is not affected by the future installation of solar panels. Install roof-penetrating mounts at the time of roof installation if analysis indicates roof-penetrating mounts are best-suited for the project application. Roof-penetrating mounts must be designed to limit thermal bridging and included in the envelope commissioning.

Interconnection pathways

- Construction documents shall indicate locations reserved for inverters and metering equipment, and the location of a pathway reserved for routing conduit from the solar zone to the point of interconnection with the electrical service. For projects with high SWH loads where there is the potential for the solar zone to be used for water heating, construction documents shall indicate a pathway for routing plumbing from the solar zone to the water heating system.

Electrical service

- The main electrical service panel shall have a minimum busbar rating of 200 amps.
- The main electrical service panel shall have a reserved space to allow for the installation of a double pole circuit breaker for a future solar electric installation. The reserved space shall be permanently marked as "For future solar electric."

REQUIREMENTS EXPLAINED: NEW CONSTRUCTION

This credit promotes renewable energy generation and procurement, reduces the effects of fossil fuel use, and supports grid decarbonization.

OPTION 1. RENEWABLE SUPPLY OR PROCUREMENT

This credit establishes a three-tier hierarchy for renewable energy and preferentially rewards renewable energy supply and procurement that has the most direct and long-term effects on building decarbonization:

- **Tier 1**. On-site renewable energy generation or social impact project
- **Tier 2**. New off-site renewable electricity
- **Tier 3**. Off-site renewable energy

Projects may choose to supply or procure renewable energy from Tier 1, Tier 2, and Tier 3 renewable energy for a maximum of 5 points.

For all three tiers of renewable energy, the project team should first confirm that the project will comply with the credit requirements for renewable energy environmental attributes and renewable contract length before proceeding with procurement and/or installation of the renewable energy.

RENEWABLE ENERGY CLASSIFICATIONS

Tier 1: On-site Renewable Energy Generation or Social Impact Project

On-site renewable energy generation, when strategically integrated with grid-interactive strategies, can increase the building resilience and support effective grid management.

To qualify as a Tier 1 renewable system, the renewable energy must be produced and generated either on the project site, the site of the contiguous campus where the project is located, or the site of a social impact project.

ELIGIBLE TIER 1 RENEWABLE RESOURCE TYPES

Eligible Tier 1 renewable energy resources include:

- Solar electric (photovoltaic)
- Solar thermal (i.e., for SWH or hot water heating)
- Wind
- Recovered heat from municipal wastewater

Only usable energy generated from the renewable system shall be considered toward the Tier 1 renewable energy contribution. Usable energy is defined as the output energy from the system less any transmission and conversion losses, such as standby heat loss, loss when converting electricity from DC to AC, or waste heat that is exhausted to the atmosphere. Sell the excess energy beyond the building's energy demand at a given point to the utility company (net metering) when all associated renewable attributes are retained by the project owner. Net metered electricity may count toward the renewable contribution up to 100% of annual electricity and district energy use.

Additional considerations: nonqualifying systems

Renewable fuels harvested, produced, or refined off-site and used to generate thermal energy or electricity on-site are classified as Tier 3 renewable energy and do not count as Tier 1 renewable energy.

TIER 1 SOCIAL IMPACT PROJECT

Project owners may opt to install renewable energy on the site of a social impact project with a capital investment like that incurred for installing a new renewable system on their own project site. A social impact project is defined as a building or project site providing housing and/or community services to historically marginalized communities. Examples include but are not limited to affordable housing projects, community centers, schools, or recreational facilities serving historically marginalized communities.

For social impact projects, the social impact project owner who owns, operates, and/or occupies the building will have no financial burden for the renewable equipment, the installation, or the commissioning of the renewable system. The social impact project owner must gain ownership of the system. They will have the right to power generated from the new system. This provides affordable clean power that will result in permanent cost savings to members of historically marginalized communities.

For residential social impact projects, residents responsible for paying their own electricity bills must receive proportionate cost savings for the renewable power generation. Renewable generation may be allocated first to central water heating and HVAC equipment serving the residential units before proportioning the remainder to the residents.

Additional considerations: Nonqualifying Tier 1 social impact projects

Most community renewable energy installations do not comply with the social impact project requirements for location of the renewable system and permanent transfer of ownership and rights to the power to the social impact project owner.

TIER 1 COMMISSIONING

Tier 1 renewable systems must be installed and commissioned per *EAp3: Fundamental Commissioning.* For projects pursuing *EAc5: Enhanced Commissioning*, Option 1, Path 1. Enhanced Commissioning for MEP Systems, Tier 1 renewable systems must also comply with the commissioning criteria for that credit. Qualifying for the credit requires completing the functional testing of renewable before the final LEED Certification application.

The commissioning provider for an equity or campus renewable project may be different than the project's commissioning provider. Documentation of previous commissioning of equity or campus renewable projects is acceptable, provided it complies with the *EAp3: Fundamental Commissioning* and *EAc5: Enhanced Commissioning* requirements. For existing campus or equity systems that were not commissioned during the original system design and construction, recommissioning of the system is required.

TIER 1 METHODS FOR DEMONSTRATING COMPLIANCE

To achieve points in Tier 1 on-site renewable energy systems a project must either install the minimum rated capacity of on-site renewable energy as a function of project area for the three largest floors or install qualifying renewable energy that will generate the specified percent of the project's annual site energy. Projects may quickly calculate compliance using either method and apply the method that leads to greatest achievement of points.

TIER 1 MINIMUM RATED CAPACITY METHOD FOR DEMONSTRATING COMPLIANCE

The minimum rated capacity method is most appropriate for multistory building projects or projects with high process loads not capable of supplying a significant proportion of building site energy use through on-site renewable energy.

The area (A) used to calculate the minimum rated capacity is the sum of the gross project floor area of all the floors up to the three largest floors. This value refers to the project dimensions, regardless of whether the renewable system is installed on the project site, on the campus, or on the site of a social impact project.

- For projects with three or fewer floors, A is equal to the total gross floor area of the project.
- For multistory buildings with equal floor plates across all floors, A is equal to three times the floor plate area.
- For all other projects, A is determined by identifying the three largest floors and summing the area for these three floors.

Use A to calculate the required minimum rated capacity of renewable energy for up to 2 points. For solar photovoltaic panels, use the Direct Current (DC) rated capacity, without degrading for system losses.

TABLE 3. Points Awarded for Minimum Rated Capacity

Points	Minimum Rated Capacity	
	IP Units	SI Units
1	A × 1.0 W/sq. ft.	A × 10.8 W/sq. m.
2	A × 2.0 W/sq. ft.	A × 21.6 W/sq. m.

For a building taller than three stories, the minimum required rated capacity corresponds to:

- Approximately 20% of gross roof area covered by solar photovoltaics for 1 point, OR
- Approximately 40% of gross roof area covered by solar photovoltaics for 2 points.

The 1-point threshold for minimum rated capacity is double the value of on-site renewable energy prescriptively required by *ASHRAE 90.1-2022* without exceptions.

TIER 1 PERCENT OF ANNUAL SITE ENERGY METHOD FOR DEMONSTRATING COMPLIANCE

This method is most appropriate for projects with three or fewer floors that have relatively low process loads. Projects must use the percentage of annual site energy method when documenting more than 2 points for Tier 1 renewable energy.

Tier 2: New Off-site Renewable Electricity

Age of the renewable generator marks the key difference between Tier 2 and Tier 3 qualified electricity generation resources. Tier 2 requires new off-site renewable power either from generators contracted to be built and operational within two years of building occupancy or from generators with a commercial operations date (COD), no more than five years before the execution of the purchase contract.

OLDER CONTRACTS

It is acceptable to use older long-term purchase contracts to comply with the COD requirement, provided that the contract shows the COD for the generators occurred less than five years before the contract was executed, and the allocated energy generation from the contract meets all Renewable criteria below. For example, a 20-year purchase contract for newly installed wind power executed 10 years ago allocated to the project in accordance with Renewable criteria below qualifies as Tier 2 renewable energy.

Tier 3: Off-site Renewable Energy

Tier 3 encompasses both renewable electricity that is Green-e Energy certified or equivalent, and renewable fuels certified to the *Green-e Renewable Fuels* standard or equivalent.

For renewable electricity, the COD of the renewable power generator may be up to fifteen years old to meet the Green-e Energy Generator Age and New Date criteria.

Eligible renewable electricity resource types: Tiers 2 and 3

Eligible renewable power generation resources for Tier 2 and Tier 3 electricity include:

- Solar electric (photovoltaics)
- Wind
- Geothermal energy (electricity or heat generated from subterranean steam or hot water)
- Ocean-based energy (such as wave or tidal energy conversion)*
- Low-impact hydropower*
- Biomass production*

Nearly all solar electric, wind, and geothermal power generation systems that meet the Green-e New Date criteria qualify as Green-e renewable resource types.

In contrast, many hydropower, biomass power generation, and ocean-based energy systems do not meet the Green-e Framework criteria governing those system types. If considering a renewable

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* These renewable electricity generation sources should meet the criteria in *Green-e Framework for Renewable Energy Certification*, Section IIIA, Renewable Resource Types, including any applicable location-specific criteria (e.g., Section II. Eligible Sources of Supply from the *Green-e Renewable Energy Standard for Canada and the United States*).

resource that is not wind or solar and is not Green-e certified, review applicable Green-e criteria to confirm resource eligibility.

For instance, in the United States, hydropower must meet one of the following criteria per the *Green-e Renewable Energy Standard for Canada and the United States*, Section II. Eligible Sources of Supply:

- New generation capacity on a non-impoundment, OR
- New generation capacity on an existing impoundment from a hydropower facility certified by the Low Impact Hydropower Institute (LIHI), or from a hydropower facility consisting of a turbine in a pipeline or in an irrigation canal.

Additional considerations: Geoexchange systems ineligible

Geoexchange systems such as geothermal heat pumps that use vapor compression cycles are not considered a renewable energy resource. These systems are credited in *EAc3: Enhanced Energy Efficiency*.

Eligible Tier 3 Renewable Fuel Resource Types

For any fuel used on the project site or for district heating, the project may procure renewable fuel that is Green-e certified or equivalent. The Green-e renewable fuels standard certifies biomethane—also called renewable natural gas (RNG)—that meets specific production facility and feedstock criteria and is purified to meet gas pipeline specifications.

Annual Site Energy Determination (For Tier 1, Tier 2, and Tier 3 Percent of Annual Site Energy)

For New Construction, annual site energy refers to the total building annual site energy use including electricity, on-site fuel use, and district thermal energy. This includes all regulated and unregulated energy use.

Exclusion of energy for electric vehicle charging

Exclude energy for electric vehicle charging of vehicles used for off-site transportation purposes if this EVSE is separately metered from the main building energy use.

Renewable energy procurement may be contractually linked to a percentage of monthly metered data for each project energy source, which together sum to the total renewable percentage of annual site energy claimed for the project. Alternatively, the project's annual site energy use may be estimated.

ANNUAL SITE ENERGY ESTIMATION: ENERGY SIMULATION

Projects using energy modeling to demonstrate compliance with *EAp2: Minimum Energy Efficiency* must use the PBP before accounting for renewable energy credit as the basis for annual site energy.

- For projects referencing the *ASHRAE 90.1*, Appendix G, Performance Rating Method, use the site energy consumption for the PBP without any credit for the on-site renewable energy contribution.
- For projects referencing the *ASHRAE 90.1*, Energy Cost Budget Method (ECB), use the site energy consumption for the proposed design without any credit for the on-site renewable energy contribution.

ANNUAL SITE ENERGY ESTIMATION: PRESCRIPTIVE PATH

Projects using the prescriptive method to show compliance with *EAp2: Minimum Energy Efficiency* must rely on estimations from *EAp1: Operational Carbon Projection and Decarbonization Plan* to determine total annual site energy for renewable energy credit calculations. Break down the site energy into electric, fuel, and district energy consumption.

Additional considerations

Refer to the District Energy Guidance for optional site energy adjustments that may be applied for district energy use.

RENEWABLE ENERGY CRITERIA

Contract Length

Projects must retain EACs for the annual renewable energy generation for a minimum of 10 years. Contractual documentation must show ownership of the EACs for the required duration. Examples include a ten-year contract for renewable power from:

- **Tier 1**. Third party-owned on-site renewable energy system
- **Tier 2**. Virtual Power Purchase Agreement (VPPA)
- **Tier 3**. Green Tariff

For contract durations shorter than 10 years, prorate the renewable energy across 10 years. For a one-time bulk purchase of renewable energy, the annual renewable energy quantity allocated to the project is the total purchase quantity divided by 10.

For older contracts, only count the remaining time left in the contract no earlier than 18 months before the initial submission date for LEED certification (consistent with the Vintage criteria below).

For Tier 3 renewable energy, where a ten-year contract is not available, project teams may show compliance with the 10-year minimum contract term by demonstrating the following:

- The project has an executed contract for a minimum of one year, or where contracts are not available per regulatory requirements, document that the project has been enrolled in the Green-e or equivalent utility tariff for a minimum of one month, AND
- The building owner must provide a signed letter of commitment indicating that the project will remain continuously enrolled in the 100% renewable Green-e or equivalent utility tariff, or alternate 100% Green-e or equivalent procurement source for a minimum of 10 years (or the number of years documented for credit if less than 10 years).

Environmental Attributes

When procuring off-site electricity, environmental attributes must meet specific requirements for ownership, source, vintage, and location.

An EAC is a transferrable certificate, record, or guarantee used to track the environmental attributes for a unit of energy and the rights to those attributes. Examples of EACs include RECs and Guarantees of Origin (GOs), where one REC or one GO corresponds to one Megawatt-Hour (MWH) of renewable electricity.

OWNERSHIP

Ownership of the renewable energy environmental attributes must reside with the LEED BD+C project and be demonstrated through retirement of the EACs on behalf of the LEED project.

If the renewable attributes are not retained by the project owner, the renewable project is disqualified from credit compliance. For example, if the project cedes ownership of the RECs from on-site photovoltaics in exchange for a utility incentive, the system is ineligible for credit.

The renewable energy contract shall not permit replacement of EACs from one project with that of a different renewable energy project (referred to as REC Arbitrage) unless the contract specifies that the replacement EACs meet all relevant LEED criteria. For example, the contract shall not allow replacement of Tier 2 EACs with those of an asset older than five years at the time of contract execution.

PROJECT ENERGY SOURCE

Renewable electric generation from Tier 1, Tier 2, and/or Tier 3 can only be applied to electricity or district thermal energy up to 100% of total combined electricity and district thermal energy.

Tier 3 renewable fuels can only be applied to project fuel use or district heat up to 100% of the total combined fuel and district heat.

Therefore, if the project uses any fuel on-site:

- The qualifying combined Tier 1 and Tier 2 energy use as a percentage of annual site energy will be less than 100%.
- The project must procure both Tier 3 electricity and Tier 3 renewable fuel to achieve 100% Tier 3 renewable energy required for 2 points.

VINTAGE

Renewable energy cannot be generated more than 18 months before the initial submission date for LEED certification.

A one-time purchase of EACs or RECs cannot occur more than 18 months before the initial submission date for LEED Certification unless the terms of the purchase agreement ensure renewable energy generation occurs no earlier than the referenced date.

Allocation of renewable power to the project from a multiyear contract must be limited to power generation beginning 18 months prior to LEED initial submission.

LOCATION

For projects in large countries such as the U.S., India, and China, the renewable energy must be generated in the same country as the project. For projects in smaller countries such as those in the European Union, the renewable energy must be generated in the same multicountry geographical region as the project, provided that these countries share an interconnected electric utility grid or that EACs are unavailable in the project's country.

TIER 2 BULK PURCHASES

Tier 2 bulk purchases totaling more than 100% of the project's total combined annual electricity and district energy usage require Green-e Energy certification (or equivalent). This ensures the proper level of transparency and verification necessary to confirm additionality and environmental impact associated with the EACs.

GREEN-E EQUIVALENCE

Projects not using Green-e certified products for Tier 2 bulk purchases or for Tier 3 electricity or fuel must demonstrate equivalency to the Green-e requirements.

For electricity, the EACs retired on behalf of the LEED Project must:

- Be certified under an eco-label or similar program developed by an independent organization or government entity with transparent accounting process and standards in place.
- Be from an eligible renewable energy resource (see *Green-e Framework for Renewable Energy Certification*, Section IIIA, Renewable Resource Types, and additional regional requirements as applicable, that is, *Appendix D: Green-e Renewable Energy Standard for Canada and the United States*, Section II (Eligible Sources of Supply).
- Be from renewable assets that have come online within the last 15 years, or for projects outside the U.S., the eco-label program may instead include provisions ensuring incremental environmental benefits for assets older than 15 years.
- Have a verifiable chain of custody.
- Have a mechanism to prevent double counting.

For Tier 3 fuel, the EACs retired on behalf of the LEED Project must have a mechanism to prevent double counting and meet one of the following criteria:

- Certified under an eco-label or similar program developed by an independent organization with transparent accounting process and standards in place, OR
- Officially recognized as a renewable fuel source in the country, province, state, or locality in which the project is located.

DISTRICT ENERGY SYSTEMS

District Energy Systems Fueled by Renewable Energy

For District Energy sources fueled by renewable energy, projects may either allocate this renewable energy toward credit compliance or exclude the renewable energy proportion from the project's annual site energy determination:

EA

ALLOCATION OF DES RENEWABLE ENERGY TO THE PROJECT

For each district energy source serving the project, assign the project the proportion of DES input energy that meets the LEED requirements for each Tier of renewable energy:

- **Tier 1:** On-site renewable electricity generation or solar thermal energy generation at the District Energy Plant.
- **Tier 2:** New off-site renewable electricity.
- **Tier 3:** Green-e or equivalent off-site renewable electricity for DES electricity inputs, or Green-e or equivalent fuel for DES fuel inputs.

For example, allocate the project Tier 3 renewable fuel totaling 60% of the project's annual district heating energy consumption for a district heating system with 60% of annual energy inputs from Green-e or equivalent fuel. EACs must be retained by either the DES supplier or the project owner to be eligible for this approach.

In place of documentation showing a 10-year contract for renewable fuels, the project may submit evidence of annual DES renewable percentage achieved for the most recent three years of operation and provide narrative confirmation justifying that ongoing achievement is anticipated at or above the specified level.

OR

EXCLUSION OF DES RENEWABLE ENERGY PROPORTION FROM ANNUAL SITE ENERGY DETERMINATION

For each district energy source fueled by renewable energy, exclude this proportion of renewable energy from the annual site energy determination. To be eligible for exclusion, the renewable energy shall either be an Eligible Renewable Electricity Resource Type or an Eligible Tier 3 Renewable Fuel Resource Type per the descriptions above, or shall be classified as renewable energy by national, state, or local policy governing the project location.

For example, for a district heating system fueled by 70% biofuel classified as renewable in the project's country, include only the 30% of project district heating associated with nonrenewable fuel inputs in the annual site energy determination used for credit compliance.

Annual Site Energy Adjustments for District Energy Systems (optional)

For projects where the high efficiency associated with district chilled water generation artificially inflates total estimated site energy consumption, projects may optionally apply either the DES Multiplier or the Virtual DES Efficiency to all DES sources serving the project:

DES Multiplier

- Multiply total reported site energy consumption for purchased chilled water by 0.325.
- Multiply total reported site energy consumption for purchased heat by 1.2.

Virtual DES efficiency in *EAc3 Enhanced Energy Efficiency*

Projects crediting DES efficiency toward the project performance in *EAc3: Enhanced Energy Efficiency* by modeling the proposed district energy use as virtual on-site chilled water and hot water plants may optionally use the total modeled site energy consumption from this proposed design rather than separating out the site energy consumption for purchased chilled water and purchased hot water.

In the energy simulation, use submetering to distinguish fuel used on-site from the modeled fuel use for the district hot water plant. Per the Renewable Attributes, Project Energy Source criteria, either renewable electricity generation or renewable fuel may be applied to the submetered fuel and/or electricity use associated with the district heating system, whereas only renewable fuel may be applied to fuel used on the project site.

DES Site energy adjustments are not applicable to projects modeled using *ASHRAE Standard 90.1-2022*, Addendum a. Projects applying Addendum a should use submetering to distinguish fuel used on-site from the modeled fuel used for the district hot water plant.

REQUIREMENTS EXPLAINED: CORE AND SHELL

Core and Shell options afford enhanced flexibility for developers to implement renewable energy supply, procurement, and/or readiness strategies. Projects may achieve all 4 available points through Option 1. Renewable Energy Supply or Procurement, or 1 point for Option 2. Renewable Energy Readiness and up to 3 points under Option 1.

OPTION 1. RENEWABLE ENERGY SUPPLY OR PROCUREMENT (CORE AND SHELL)

Core and Shell projects may choose to supply or procure renewable energy from Tier 1 or Tier 2 for a maximum of 4 points, from Tier 3 for a maximum of 2 points, or from any combination of Tier 1, Tier 2, and Tier 3 renewable energy for a maximum of 2 points.

The Tier 1 Minimum Rated Capacity Method for Demonstrating Compliance is identical for New Construction and Core and Shell, with no differences in the required thresholds.

In contrast, for Core and Shell, the percentage of annual site energy is calculated using the base building site energy rather than the total building site energy referenced in New Construction, which more closely aligns the required percent procurement with the core and shell scope of work.

Annual base building site energy determination (for Tier 1, Tier 2, and Tier 3)

To calculate the annual base building site energy use:

- Perform the total annual site energy determination described above. Include all building energy consumption inclusive of base building energy use and tenant use.
- Estimate the total annual site energy consumption from base building energy meters, including:
  - Meters for equipment fully contained within the project scope of work, such as elevators, parking garage lighting and ventilation, site lighting, common area lighting, common area HVAC, and common area receptacles and process equipment.
  - Meters for shared building systems serving tenants (e.g., chiller plant, boiler plant, central water heating, shared VAV air handling units, and dedicated outside air handling units).

Annual base building site energy use is equal to the total annual site energy consumption from base building energy meters when this comprises more than 25% of estimated total annual site energy use.

Otherwise, the base building site energy use is 25% of total annual site energy use.

OPTION 2. RENEWABLE ENERGY READINESS

For projects where financial and leasing considerations preclude large on-site renewable energy installations within the Core and Shell scope of work, Option 2 incentivizes developers to incorporate renewable energy readiness features into the project design to enable future on-site renewable energy installation by tenants.

Projects must design for solar readiness in accordance with the solar readiness credit criteria or a similar standard, or document equivalent on-site renewable readiness for another qualifying renewable energy source, such as wind energy or solar thermal energy.

Tenant guidelines must integrate guidance on how tenants can use this available infrastructure to apply on-site renewables to their project.

SOLAR READINESS

A design for solar readiness must include one or more solar zones and address mounting considerations, interconnection pathways, and electrical capacity per the detailed criteria in the credit requirements.

Solar zones

Solar zones set aside designated spaces to house future solar energy generation equipment and provide the necessary structural support to accommodate the equipment. The design team must perform a wind and load analysis to inform the solar zone design and design the roof or other supporting structure with the structural integrity to accommodate the solar installation. The solar zone should avoid shaded areas or obstructions from other equipment to maximize the potential for solar collection.

Mounting considerations

The solar zone must be designed to accommodate all panel-mounting options identified in the tenant guidelines. It may be necessary to install roof-penetrating mounts within the Core and Shell scope of work to ensure the future integrity of the roofing system. Envelope commissioning must address any potential impact of the roof mounts on the building enclosure integrity, including thermal bridging and/or moisture intrusion.

Interconnection pathways

Dedicated areas must be designed for future inverters, metering equipment, and pathways reserved for conduit routing. These interconnection pathways must specifically address anticipated tenant design configurations. For example, for distributed net-metering of each tenant's electricity, the design should accommodate conduit pathways to the projected location for each tenant electrical panel and should reserve nearby space for inverters and other equipment. For projects where solar thermal hydronic systems may be installed, the design should address pathways for routing of plumbing.

Electrical service

Sufficient electrical space must be permanently reserved and marked for future solar electric use within the main electrical service panel.

LEED PLATINUM REQUIREMENTS

To achieve LEED Platinum, projects must supply 100% of energy usage from any combination of Tier 1, Tier 2, and/or Tier 3 renewable energy:

- **New Construction:** 100% of total annual site energy usage
- **Core and Shell:** 100% of total annual base building site energy usage

Refer to the Project Priorities Library for regional compliance alternatives.

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	All except Core and Shell Option 2	All tiers	USGBC calculator.
			Confirmation of renewable attribute ownership and renewable energy criteria relevant to each tier.
	Third-party-owned systems	All tiers	Excerpts of purchase letter or contract showing renewable energy for the targeted point threshold, including confirmation of renewable attribute ownership, quantity of renewable energy, type of renewable energy, location, contract execution date, duration of contract, vintage, and if applicable, Tier 2 COD.
	Shared renewable attributes	All tiers	Confirmation of allocation of attributes to the project.
	All Except Core and Shell Option 2	Tier 1	Plans or documentation confirming Tier 1 renewable systems and their rated capacity (DC and AC).
		Tier 1 on the site of social impact project	Documentation showing the system meets the requirements for a social impact project.
		Tier 1 percentage of annual site energy method: New system	Inputs and description of calculation methodology or tool for determining annual renewable energy generation.
		Tier 1 percentage of annual site energy method: Existing system	Summary of measured data used to determine annual renewable energy generation.
		Tier 2 exceeding 100% of annual electricity, and all Tier 3	Evidence that EACs are Green-e® or equivalent.
Core and Shell	Option 2		Confirmation of analysis conducted for the solar zone location, including considerations for wind and load, and explaining how the solar zone meets the credit requirements. Additionally, identification of selected mounting option.
			Confirmation that roof warranty is not affected by the future installation of solar panels.

(continued)

(continued)

Project Types	Options	Paths	Documentation
			Confirmation that tenant guidelines have been provided and developed, including dimensioned construction documents (e.g., building roof plans, elevations, and site plan) showing the designated solar zone, obstructions, and location reserved for inverters and metering equipment, pathways, future mechanical or electrical equipment, etc.
			Confirmation of main electrical service panel's busbar rating and the reserved space allocated for the installation of a double pole circuit breaker in the main electrical service panel to accommodate a future solar electric installation.

REFERENCED STANDARDS

- Green-e Framework for Renewable Energy Certification (https://www.green-e.org/programs/energy/documents)

FURTHER EXPLANATION

BD+C CORPORATE RENEWABLE ENERGY PROCUREMENT (TIER 2 AND TIER 3)

When corporations or campuses participate in renewable energy initiatives that procure off-site renewable energy certificates (RECs) for all projects within the portfolio, the eligible Tier 2 and Tier 3 renewable resources contribute toward project credit compliance. Refer to the Project Priorities library for further alternatives for demonstrating compliance at scale using a corporate or campus renewable energy procurement policy.

Enhanced Commissioning

EAc5
New Construction (1–4 points)
Core and Shell (1–3 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To further ensure that the building systems function as designed, and that they continue to maintain energy performance over time.

REQUIREMENTS: NEW CONSTRUCTION

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1–4**
Option 1. Enhanced Commissioning	1–3
Path 1. Enhanced Commissioning for MEP Systems	2
AND/OR	
Path 2. Enhanced Commissioning for Building Enclosure	1
AND/OR	
Option 2. Monitoring-Based Commissioning (MBCx)	1–2
Path 1. Basic Software	1
OR	
Path 2. Enhanced Software	2

OPTION 1. ENHANCED COMMISSIONING (1–3 POINTS)

Owner must designate an independent commissioning provider (CxP) during predesign or very early in the design phase.

PATH 1. ENHANCED COMMISSIONING FOR MEP SYSTEMS (2 POINTS)

- Comply with *ANSI/ASHRAE/IES Standard 202-2024*, Commissioning Process, for mechanical, electrical, plumbing, control, data center, process, building monitoring, and renewable energy systems.

- Comply with the following additional requirements:
  - During the design phase, attend at least two coordination/design meetings to discuss review comments and commissioning.
  - Prior to or during occupancy, review the training materials to confirm that they meet the training plan, and confirm that the training occurred.

AND/OR

PATH 2. ENHANCED COMMISSIONING FOR BUILDING ENCLOSURE (1 POINT)

Comply with all tasks and deliverables referenced with *ASTM E2947-21a*, "Standard Guide for Building Enclosure Commissioning," except Sections 7.2.4 and 7.4.3.

Comply with the following field-testing requirements:

- Building air leakage testing per *ASTM E783*, *ASTM E779*, *ASTM E1186*, or *ASTM E3158*.
- Water penetration testing per *ASTM E1105* or *AAMA 501.2*.
- Infrared imaging per *ASTM C1153* or *ASTM C1060*.

During occupancy, review the training materials to confirm that they meet the training requirements provided in the building enclosure commissioning (BECx) plan or specification, and confirm that the training occurred.

AND/OR

OPTION 2. MONITORING-BASED COMMISSIONING (MBCX) (1–2 POINTS)

PATH 1. BASIC SOFTWARE (1 POINT)

Process and communications

Commit to implementing MBCx for a minimum of three years, through a contract with an MBCx provider or qualified monitoring-based commissioning provider (MBCxP) staff person. MBCx shall commence no later than building occupancy and shall be fully coordinated between the commissioning provider, facilities management, and MBCxP.

Develop a monitoring-based commissioning plan summarizing the process including all of the following:

- Roles and responsibilities
- Software technology description, including frequency and duration for trend monitoring
- Review and reporting criteria, including:
  - Training of facilities staff
  - Expeditious communication of major anomalies or faults identified to facilities staff
  - At least quarterly, MBCxP summary of anomalies and faults detected and communication with facilities staff to discuss and prioritize issues
  - At least annually, MBCxP summary reporting of trends, benchmarks, faults, energy savings opportunities, corrective actions taken, and planned actions
  - At least two MBCxP reviews of building systems, equipment, and operational controls

Energy information system

Provide a remotely accessible platform with software functionality to perform smart analytics and visually present all metered data referenced in *EAp4: Energy Metering and Reporting*.

Include the following functionality:

- Annual energy benchmarking of energy use intensities
- Comparison of energy consumption to the prior interval annually and monthly; for electric interval meters, daily and hourly
- For electric interval data, hourly load shape with comparisons
- Visualization and reporting of hourly electric submetered data

In addition, provide hourly monitoring and visualization of electric energy use for:

- Elevators, escalators, and/or moving walkways.
- Commercial kitchen equipment in spaces with more than 10 kW of rated capacity.
- Process equipment in spaces with more than 10 kW of rated capacity.

OR

PATH 2. ENHANCED SOFTWARE (2 POINTS)

Comply with Path 1 AND provide the following enhanced monitoring and software technology functionality:

- Fault detection and diagnostics (FDD) for projects with large HVAC or refrigeration capacity. For total project installed capacity of cooling, heating, or refrigeration systems exceeding 7,200 kBtu/hr (600 tons or 2,110 kW), provide a remotely accessible FDD system that addresses at least 60% weighted by capacity of:
  - Air handling equipment AND
  - Large hydronic or commercial refrigeration equipment (chillers, boilers, etc.)
- The FDD system must include the following functionality:
  - Perform smart analytics and visually present FDD data
  - Direct link from reported fault to view relevant trend data
  - Fault sorting and filtering
  - Exporting of fault reports (summary reports and detailed individual faults)
  - Data historian capable of storing critical trend data for at least three years
- Energy information system (EIS)
  - For major renovations and buildings less than 25,000 sq. ft., comply with *ASHRAE 90.1*, Section 8.4.3 requirements, for measurement devices in new buildings, without exceptions. Include visualization of this data per Path 1 requirements above and provide automated reporting of energy use anomalies.
  - For all other buildings, include the following additional functionality for the EIS:
- Automated reporting of energy use anomalies
- Normalization of energy consumption
- Greenhouse gas emissions reporting

REQUIREMENTS: CORE AND SHELL

ACHIEVEMENT PATHWAYS	POINTS
Core and Shell	**1-3**
Option 1. Enhanced Commissioning	1–2
Path 1. Enhanced Commissioning for Building Enclosure	1
OR	
Path 2. Enhanced Commissioning for Building Enclosure and MEP Systems	2
AND/OR	
Option 2. Monitoring-Based Commissioning (MBCx)	1

OPTION 1. ENHANCED COMMISSIONING (1–2 POINTS)

Owner must designate an independent commissioning provider (CxP) during predesign or very early in the design phase.

PATH 1. ENHANCED COMMISSIONING FOR BUILDING ENCLOSURE (1 POINT)

Comply with all tasks and deliverables referenced with *ASTM E2947-21a*, “Standard Guide for Building Enclosure Commissioning,” except Sections 7.2.4 and 7.4.3.

Comply with the following field-testing requirements:

- Building air leakage testing per *ASTM E783, ASTM E779, ASTM E1186*, or *ASTM E3158*
- Water penetration testing per *ASTM E71105* or *AAMA 501.2*
- Infrared imaging per *ASTM C1153* or *ASTM C1060*

During occupancy, review the training materials to confirm that they meet the training requirements provided in the BECx plan or specification, and confirm that the training occurred.

OR

PATH 2. ENHANCED COMMISSIONING FOR BUILDING ENCLOSURE AND MEP SYSTEMS (2 POINTS)

Comply with Path 1, AND:

- Comply with *ANSI/ASHRAE/IES Standard 202-2024*, Commissioning Process, for mechanical, electrical, plumbing, control, data center, process, building monitoring, and renewable energy systems.
- Provide commissioning for components installed within the project scope of work. For any systems or controls that require future interconnection with tenant systems, provide templates for future commissioning of these interconnections that address the most likely tenant interconnection scenarios in the systems manual and tenant guidelines. At a minimum, provide these templates for design review checks, functional tests, and systems manual addenda.
- The CxP must comply with the following additional requirements:

- Attend at least two coordination meetings during the design phase and at least four milestone meetings during the construction phase to discuss review comments and commissioning.
- Provide an ongoing commissioning plan.
- During occupancy, review the training materials to confirm that they meet the training plan, and confirm that the training occurred.

AND/OR

OPTION 2. MONITORING-BASED COMMISSIONING (MBCX) (1 POINT)

PROCESS AND COMMUNICATIONS

Commit to implementing MBCx for a minimum of three years through a contract with an MBCx provider or qualified monitoring-based commissioning provider (MBCxP) staff person. MBCx shall commence no later than building occupancy and shall be fully coordinated between the commissioning provider, facilities management, tenants, and MBCxP.

Develop a monitoring-based commissioning plan summarizing the process including all the following:

- Roles and responsibilities
- Software technology description, including frequency and duration for trend monitoring review, and reporting criteria including:
  - Training of facilities staff
  - Expeditious communication of major anomalies or faults identified to facilities staff
  - At least quarterly, MBCxP summary of anomalies and faults detected and communication with facilities staff to discuss and prioritize issues
  - At least annually, MBCxP summary reporting of trends, benchmarks, faults, energy savings opportunities, corrective actions taken, and planned actions
  - At least two MBCxP reviews of building systems, equipment, and operational controls

AND

ENERGY INFORMATION SYSTEM

Provide a remotely accessible platform with software functionality to perform smart analytics and visually present all metered data referenced in *EAp4Energy Metering and Reporting*, and to be expandable to include all tenant data referenced in *EAp4: Energy Metering and Reporting*. Include the following functionality:

- Annual energy benchmarking of energy use intensities
- Comparison of energy consumption to the prior interval annually and monthly, and for electric interval meters, daily and hourly
- For electric interval data, hourly load shape with comparisons
- Visualization and reporting of hourly electric submetered data
- For tenant spaces > 10,000 sq. ft. (930 sq. m.), tenant portal with capability to provide visualization and reporting of:
  - Base building data for shared systems serving the tenants and
  - Tenant electricity energy use (excluding shared systems)

In addition, provide hourly monitoring and visualization of electric energy use for:

- Elevators, escalators, and/or moving walkways.
- Commercial kitchen equipment in spaces with more than 10 kW of rated capacity.
- Process equipment in spaces with more than 10 kW of rated capacity.

REQUIREMENTS EXPLAINED: NEW CONTRUCTION

The credit rewards projects that provide commissioning beyond the *EAp3: Fundamental Commissioning* requirements. Option 1 requires early engagement of a Commissioning Provider to lead a comprehensive commissioning process spanning from pre-design through the warranty period.

Option 2 requires the implementation of a monitoring-based commissioning process that verifies ongoing performance post-occupancy that leverages automated data analytics and reporting.

Projects can combine Options 1 and Option 2 to achieve up to 4 points.

OPTION 1. ENHANCED COMMISSIONING

Option 1 offers two compliance paths. Projects can combine Paths 1 and 2 to achieve a total of 3 points. Enhanced commissioning activities for MEP systems or the building enclosure will lead to optimized system performance and further integration of the CxP into the design and post-occupancy efforts in a new construction project.

Enhanced commissioning provides substantial value for the limited additional efforts beyond *EAp3: Fundamental Commissioning*. *EAp3: Fundamental Commissioning* Tables 1 and 2 provide a comparison of the *ASHRAE Standard 90.1* Commissioning Requirements for the prerequisite versus the credit requirements from *ASHRAE Standard 202* for MEP systems and *ASTM E2947-2021* for the building enclosure, along with typical milestones for key tasks to occur.

Projects that achieve both paths under Option 1 will automatically achieve *EAp3: Fundamental Commissioning*.

PATH 1. ENHANCED COMMISSIONING FOR MEP SYSTEMS

ASHRAE Standard 202-2024

ASHRAE Standard 202-2024 outlines the commissioning process for mechanical, electrical, plumbing, control, data center, process, building monitoring, and renewable energy systems. It is a systematic process that begins in the early design phase and continues through the warranty or post-occupancy phase. Along with the *ASHRAE Standard 202-2024* requirements, teams must also comply with incremental LEED BD+C requirements.

EAp3: Fundamental Commissioning, Table 1 provides a detailed comparison of the *ASHRAE Standard 90.1*, Commissioning Requirements for the prerequisite and the *ASHRAE Standard 202* requirements for Enhanced Commissioning of MEP systems. The table also provides timing for each task.

Timing of CxP Engagement

ASHRAE 202 requires CxP review of the OPR and initial development of the Commissioning Plan during pre-design, which necessitates very early engagement of the CxP in the design process. If the CxP is engaged after pre-design, take alternative measures ensuring alignment with the intent of the *ASHRAE 202* timeline.

Examples of acceptable CxP engagement after pre-design

- Portfolio applications where the OPR and Cx Plan at the predesign phase are like other projects.
- A qualified employee of the owner provides the initial review of the OPR and initial draft of the Cx Plan, and the CxP is designated early in design development to continue the analysis.

Commissioning Scope

The enhanced MEP commissioning scope must comprehensively address alignment with the OPR by expanding the focus beyond energy and GHG emissions to address water efficiency, air quality, and thermal comfort. Commissioned systems must include the following if in the project scope:

- Mechanical (HVAC and refrigeration, including any process heating or cooling systems in the project).
- Electrical (lighting, receptacle power).
- Plumbing (indoor fixtures, SWH, pool equipment, etc.).
- Data center (electrical, cooling, humidity; identify a mechanism for evaluating whether server equipment efficiency targets are met).
- Building monitoring. Include all monitoring systems required by *ASHRAE 90.1*, as well as any EIS and FDD systems referenced in Option 2.
- On-site renewable systems, including all Tier1 systems credited in *EAc4: Renewable Energy*.
- Controls. Include systems credited in *EAc6: Grid Interactive.*

Additional LEED Tasks

LEED appends additional required tasks to those referenced in *ASHRAE 202-2024*:

MINIMUM OF TWO DESIGN PHASE MEETINGS

Early CxP involvement in design phase meetings aids complete integration of commissioning requirements and recommendations from the CxP into the final construction documents. These early interactions also lead to collaborative discussions and collectively address the CxP's design review comments.

REVIEW OF TRAINING MATERIALS

Facility staff training represents a critical step between the construction phase and post-occupancy. Before or during occupancy, the CxP must review training material to confirm that the training documents meet the training plan and sufficiently address the OPR and BOD. The CxP must also confirm that training occurred.

PATH 2. ENHANCED COMMISSIONING FOR BUILDING ENCLOSURE

Addressing the performance of the building enclosure (also referred to as the building envelope) is as important as the MEP systems. For certain project types, the building enclosure may be the most important factor for continued building energy performance. The credit requires projects to follow a comprehensive process that includes all tasks referenced in *ASTM E2947-21a*, “Standard Guide for Building Enclosure Commissioning,” except Section 7.2.4 and 7.4.3.

Building Enclosure Commissioning Provider

ASTM E2974-21a requires a BECxP to lead and implement the Building Enclosure Commissioning Process. The entity serving as the BECxP performs each required task or delegates it to the appropriate member. The BECxP must be proficient in core competencies for enclosure-related design, construction, and performance.

The CxP and the BECxP can be the same entity or distinct entities.

TIMING OF BECXP ENGAGEMENT

E2947-21a, Section 6 describes the pre-design phase tasks, which require BECxP engagement by pre-design. If the BECxP is engaged after pre-design, take alternative measures ensuring alignment with the intent of *E2947-21a* timeline.

Examples of acceptable BECxP engagement after pre-design

- Portfolio applications where the OPR and Cx Plan at the predesign phase are like other projects.
- A qualified employee of the owner provides the initial review of the OPR and initial draft of the Cx Plan, and the CxP is designated early in design development to continue the analysis.

Minimum field testing

At minimum, field testing must include air leakage testing, water penetration testing, and infrared imaging, if applicable to the project scope:

- **Air leakage testing:** Provide field testing of air leakage using one of the referenced ASTM or ANSI standards. *ASTM E783* evaluates air leakage through windows and doors. *ASTM E779* uses the fan-pressurization method for testing. *ASTM E1186* provides an array of methods for evaluating air leakage. *ASTM E3158* provides a standard method for testing large or multizone buildings. *ASTM E1827* determines air tightness using an orifice blower door. Residential spaces in mixed-use buildings may also apply *ANSI/RESNET/ICC 380*.
- **Water penetration testing:** For projects with scope of work including fenestration or exterior doors, apply *ASTM 1105* or *AAMA 501.2* to provide water penetration of installed exterior windows, skylights, doors, and curtain walls. Otherwise, water penetration testing is not required.
- **Infrared imaging:** *ASTM C160* is appropriate for envelope scope of work that includes framed members. For scope that includes a roof with insulation above deck, *ASTM C1153* can be used to locate wet insulation. Otherwise, infrared imaging is not required.

EAp3: Fundamental Commissioning, Table 2 compares the *ASHRAE 90.1* Commissioning Requirements for the prerequisite and the *ASTM E2947-21a* requirements for Enhanced Commissioning of the Building Enclosure. The table also provides the timing for each task.

OPTION 2. MONITORING-BASED COMMISSIONING

Monitoring-based commissioning enables building operators to identify operational issues as they occur, facilitating achievement of the project's performance goals on an ongoing basis.

Both Paths 1 and 2 require an EIS that enables visualization, analytics, and automated reporting of the metered data referenced in *EAp4: Energy Metering and Reporting*; and a minimum three-year commitment to implement monitoring-based commissioning (MBCx) informed by the EIS. Path 2 requires enhanced monitoring and software functionality.

SELECTING AN MBCX PROVIDER

Contract with a third-party MBCx Provider (MBCxP) for a minimum three-year time frame or designate a qualified MBCxP on the building operations team. The MBCxP must have direct experience on similar projects. Many MBCxP's have a programming or controls integration background and extensive experience with EIS and FDD technologies.

The MBCxP can be the same as the CxP or can be a different entity. If the CxP and the MBCxP are different entities, a communication plan must be established so both entities can coordinate during the construction phase and the warranty period.

MBCX PLAN

Teams must develop a comprehensive MBCx Plan that summarizes the complete process building operators will follow during the term of the MBCx contract. Refer to the rating system language for the specific details that must be included in the plan.

Path 1. Basic Software

An EIS empowers operators to review trends, identify anomalies, perform preventive and predictive maintenance, and reduce energy consumption and greenhouse gas (GHG) emissions over the building's lifespan.

The EIS system must provide visualization and analytics of the metered data required in *EAp4: Energy Metering and Reporting*.

- For new commercial buildings that are at least 25,000 sq. ft. (2,323 sq. m.) and residential projects with at least 10,000 sq. ft. (929 sq. m.) of common space, the metered data includes monthly energy use for each nonelectric energy source, and hourly energy use for electricity recorded for the whole project, for each tenant ≥ 10,000 sq. ft. (929 sq. m.), for specific end uses (HVAC, interior lighting, exterior lighting, receptacles, and refrigeration), and for on-site renewable electricity generation. Marginal additional EIS visualization and analytics capabilities are necessary beyond the *ASHRAE 90.1* energy monitoring and recording requirements.
  - For major renovations and smaller new buildings, the metered data only includes monthly energy consumption for each energy source, and monthly peak electric demand. Unlike *EAp4: Energy Metering and Reporting*, this data must be automatically transmitted for use in the EIS platform analytics.

The credit further requires hourly monitoring of large electric power uses (e.g., elevators and commercial kitchen equipment) and incorporation of this data into the EIS reporting.

The EIS system must include all visualization and analytic capabilities referenced in the credit language.

Path 2. Enhanced Software

Projects that comply with Path 1. Basic Software requirements can achieve an additional point for providing enhanced monitoring and software functionality to inform the monitoring-based commissioning.

The path requires the most incremental functionality for projects with large HVAC or refrigeration capacity, or major renovations, or small new buildings. For all other project applications, this path only requires three additional functions for the EIS system.

FAULT DETECTION AND DIAGNOSTICS FOR PROJECTS WITH LARGE HVAC OR REFRIGERATION CAPACITY

FDD Software is only required for projects with large HVAC or refrigeration capacity, where any of the following apply:

- Total project installed capacity of cooling exceeds 7,200 kBtu/h (600 tons or 2,110 kW)
- Total project installed capacity of heating exceeds 7,200 kBtu/h (2,110 kW)
- Total project installed capacity of refrigeration exceeds 7,200 kBtu/h (600 tons or 2,110 kW)

FDD is a program procedure for identifying and isolating system operational flaws. FDD uses data-driven or knowledge-driven techniques. Data-driven techniques include artificial intelligence (AI) and machine learning. Knowledge-driven techniques include having an FDD specialist use qualitative methods to analyze fault scenarios.[7] Refer to the rating system language for minimum required FDD software functionality.

Include fault detection algorithms that address at least 60% of total air handling unit capacity. Additionally, include fault detection algorithms that address at least 60% of total combined capacity for large commercial refrigeration systems, large hydronic heating systems, and large hydronic cooling systems where large systems are defined as a system with total installed capacity exceeding 7,200 kBtu/h (600 tons or 2,110 kW).

Faults assessed may include improper economizer or energy recovery operation, faulty sensor readings, improper valve and damper operation, improper equipment schedules, improper operation of control system reset algorithms (e.g., set point always at maximum value), non-optimal zone temperature set points (e.g., lower than recommended deadband; same values for occupied and unoccupied set points), equipment short cycling, improper chiller and boiler plant lockouts, and unstable/hunting control loop.

7. Maryam Mirnaghi and Fariborz Haghighat, "Fault Detection and Diagnosis of Large-Scale HVAC Systems in Buildings Using Data-Driven Methods: A Comprehensive Review," *Energy and Buildings*, 229 (December 2020): 110492, https://doi.org/10.1016/j.enbuild.2020.110492.

ADDITIONAL MONITORING FOR MAJOR RENOVATIONS AND SMALL BUILDINGS

For major renovations and small buildings less than 25,000 sq. ft. (2,323 sq. m.), provide additional hourly electricity metering for the whole project, for each tenant ≥ 10,000 sq. ft. (929 sq. m.), for specific end uses (HVAC, interior lighting, exterior lighting, receptacles, and refrigeration), and for on-site renewable electricity generation. (See *ASHRAE 90.1*, Section 8.4.3 criteria, ignoring exceptions addressing building area.)

The EIS shall provide visualization and analytics of this metered data.

ADDITIONAL EIS SOFTWARE FUNCTIONALITY

For new construction of buildings larger than 25,000 square feet (2,323 square meters), provide additional functionality for the EIS:

- Automated reporting of energy use anomalies
- Normalization of energy consumption
- Greenhouse gas emissions reporting

REQUIREMENTS EXPLAINED: CORE AND SHELL

OPTION 1. ENHANCED COMMISSIONING

PATH 1. ENHANCED COMMISSIONING FOR BUILDING ENCLOSURE

Same as BD+C: New Construction Option 1, Path 2.

PATH 2. ENHANCED COMMISSIONING FOR BUILDING ENCLOSURE AND MEP SYSTEMS

To achieve 2 points under Option 1, Core and Shell projects must address both Building Enclosure and MEP systems. This encourages a focus on the envelope, which often has the greatest impact on building lifetime performance for the Core and Shell scope of work. Refer to the additional guidance above for BD+C: New Construction Option 1, Paths 1 and 2.

For Core and Shell projects, commission systems and equipment installed in the base building scope of work.

Base buildings often install central air handling units or central plant equipment that require future interconnections to tenant-provided systems or components, such as VAV terminal units, fan coils, or water loop heat pumps. For these systems, the CxP must provide templates that support the tenant in commissioning of these interconnections. These should provide a listing of key considerations to be evaluated during design review, sample functional test procedures, and sample systems manual content that the tenant would include to address the tenant design.

OPTION 2. MONITORING-BASED COMMISSIONING

Meet the same requirements as BD+C: New Construction Option 2, plus:

- For Core and Shell projects, the base building EIS must have expansion capability, to ensure all data identified in *EAp4: Energy Metering and Reporting* is accessible through the EIS. This includes metering requirements for future tenants.

Tenant requirements

If the project anticipates future tenants will lease space that is greater than 10,000 sq. ft. (929 sq. m.), install submeters for compliance with *EAp2: Minimum Energy Efficiency*. The base building EIS must include a tenant portal where tenants can access reports, including visual representations of the energy consumption. At minimum, tenants must have access to the base building data for shared systems that serve the tenant space and the electricity energy use associated with their space.

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	All	All	Final (or Draft) Commissioning Report. If the report is a draft, include a plan for the completion of commissioning and training, including climatic and other conditions required for performance of any deferred tests.
	Option 1	Path 1 and/or Path 2	Confirmation of compliance with *ANSI/ASHRAE/IES Standard 202-2024* and/or *ASTM E2947-21a* "Standard Guide for Building Enclosure Commissioning," except Section 7.2.4 and 7.4.3 (as applicable).
			Confirmation of design phase/milestone meetings.
			Provide Commissioning Plan and sample FPT Test scripts (1 sample per discipline).
			OPR and BOD.
			Identification of Commissioning Provider, including key personnel (CxP) and V&T providers (as applicable).
			Qualifications of CxP and V&T providers.
			Confirmation that submittals were reviewed and at least 25% of the contractors' documents were QA/QC'd.
	Option 1	Path 1. Enhanced Cx for MEP Systems	Provide evidence, such as contract or other documentation, confirming the involvement of CxP during pre-design or very early in the design phase.
		Path 2. Enhanced Cx for Building Enclosure	Field report or completed test that proves building air leakage testing, water penetration testing, and infrared imaging was completed.
	Option 2	Path 1. And Path 2. MBCx–Basic Software	Documentation of owner commitment to at least three years of MBCx for the building and identification of key individual(s) responsible for MBCx (contract, letter signed by owner, job descriptions or other evidence).
			MBCx Plan.
			Narrative describing the EIS, including functionality, accessibility, and sample graphics. Identify systems included in the MBCx plan.
	Option 2	Path 2. MBCx–Enhanced EIS	Narrative describing the EIS, including functionality, accessibility, and sample graphics. Identify systems included in the MBCx plan. As applicable, confirm compliance with *ASHRAE 90.1*, Section 8.4.3 or additional functionality of EIS.
		Path 2. MBCx–Enhanced Software	Schedules, drawings, or other documentation confirming FDD devices installed in the system and verification that 60% threshold is met for systems exceeding 600 tons.
			Narrative describing the FDD system functionality, accessibility, and sample graphics, reports or trends from the system.
Core and Shell	Option 1	Path 1. Enhanced Cx for MEP Systems	Confirmation of all construction phase milestone meetings

REFERENCED STANDARDS

- ASHRAE 90.1-2019 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2019-i-p?product_id=2088527)
- ASHRAE 90.1-2022 (https://store.accuristech.com/ashrae/standards/ashrae-90-1-2022-i-p?product_id=2522082)
- ANSI/ASHRAE/IES Standard 202-2024 (https://store.accuristech.com/ashrae/standards/ashrae-202-2024?product_id=2908468)
- ASTM E2947-21a "Standard Guide for Building Enclosure Commissioning" (https://astm.org/e2947-21a.html)
- ANSI/ASHRAE/IES Standard 202-2024 (https://ashrae.org/technical-resources/bookstore/commissioning)

Grid Interactive

EAc6
New Construction (1–2 points)
Core and Shell (1–2 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To enhance power resilience and position buildings as active partners contributing to grid decarbonization, reliability, and power affordability through integrated management of building loads in response to variable grid conditions.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–2**
Option 1. Energy Storage AND/OR	1–2
Option 2. Demand Response Program AND/OR	1
Option 3. Automated Demand-Side Management	1
Path 1. System-Level Controls	1
OR	
Path 2. Building Automation System	1
AND/OR	
Option 4. Power Resilience	1

All projects must evaluate grid-interactive measures concerning the current and forecasted grid context, location, building type, and ownership structure and account for the results in decision-making.

Interval recording meters and equipment capable of accepting an external signal must also be provided.

OPTION 1. ENERGY STORAGE (1–2 POINTS)

Provide on-site electric storage and/or thermal storage meeting the criteria in Table 1.

Include automatic load management controls capable of storing the electric or thermal energy during off-peak periods or periods with low grid carbon intensity and using stored energy during on-peak periods or periods of high grid carbon intensity.

TABLE 1. Peak Storage Capacity Relative to Peak Demand

Storage	1 Point	2 Points
Electric storage capacity Relative to peak electric demand	0.2 kWh/kW	0.4 kWh/kW
Thermal storage capacity Relative to peak coincident thermal demand (heating + cooling + SWH + process heat)	1.0 kWh/kW or Btu/Btu/hr or ton-hrs/ton	2.0 kWh/kW or Btu/Btu/hr or ton-hrs/ton

AND/OR

OPTION 2. DEMAND RESPONSE PROGRAM (1 POINT)

Enroll in a minimum one year demand response (DR) contract with a qualified DR program provider, with the intention of multiyear renewal.

On-site electricity generation and fuel combustion cannot be used to meet the demand-side management criteria.

AND/OR

OPTION 3. AUTOMATED DEMAND-SIDE MANAGEMENT (1 POINT)

On-site electricity generation and fuel combustion cannot be used to meet the demand-side management criteria.

PATH 1. SYSTEM-LEVEL CONTROLS (1 POINT)

Provide automated demand response controls for at least two of the following systems installed within the project scope of work:

- HVAC systems (50% of rated capacity)
- Lighting systems (50% of power)
- Automatic receptacle controls
- Service water heating (90% of capacity)
- Electric vehicle supply equipment

OR

PATH 2. BUILDING AUTOMATION SYSTEM (1 POINT)

Develop a plan for shedding at least 10% of the project's peak electricity demand for a minimum of one hour. The plan must address both winter and summer peaks considering electrified grid projections.

Have in place a control system that automatically sheds electricity demand in response to triggers denoting strain on the grid or high grid emissions. For example:

- A signal from a DR program provider
- Data obtained through an API indicating high grid emissions
- Peak demand tariff period when the grid is operating in the highest demand window
- Time-of-use rate when pricing is highest

AND/OR

OPTION 4. POWER RESILIENCE (1 POINT)

Identify critical equipment that requires continuous operation. Design the project to be able to island and operate independently from the grid to power the critical loads with the project's on-site renewable and energy storage systems for at least three days.

REQUIREMENTS EXPLAINED

This credit rewards projects that implement solutions to reduce stress on the grid and increase building resilience. Projects are encouraged to combine the strategies from Options 1–3 to optimize resilient solutions for the project.

All options

EVALUATE GRID-INTERACTIVE MEASURES

Projects that pursue this credit must complete an evaluation of potential grid-interactive measures that are viable for the project. Using this research, teams can make informed decisions on solutions that can be integrated into the design.

Helpful guidance supporting this evaluation is provided in New Buildings Institute (NBI) GridOptimal Buildings Initiative,[8] ASHRAE's Grid-Interactive Building Guide[9] or the U.S. Department of Energy's Grid-Interactive Efficient Building guide.[10] Projects outside the U.S. are encouraged to reference local technical reports that include further context for the regional grid context.

INTERVAL RECORDING AND EXTERNAL SIGNAL

For all options, the design must include electric meters with interval recording capabilities. Grid-interactive equipment installed as part of the project's scope must be capable of receiving an external signal directly or indirectly to indicate when load management should be enabled.

8. NBI, "The GridOptimal Buildings Initiative," *New Buildings Institute*, October 26, 2021, https://newbuildings.org/resource/gridoptimal/.
9. ASHRAE, "ASHRAE Releases Guide on the Role of Grid Interactivity in Decarbonization," November 2, 2023, https://www.ashrae.org/about/news/2023/ashrae-releases-guide-on-the-role-of-grid-interactivity-in-decarbonization.
10. U.S. Department of Energy, "Grid-Interactive Efficient Buildings," https://www.energy.gov/eere/buildings/grid-interactive-efficient-buildings.

PEAK LOAD DETERMINATION—OPTION 1 AND OPTION 3, PATH 2

For options that reference a reduction in peak thermal or peak electricity demand, use the modeled coincident peak demand for the proposed design for projects documented using the ECB Method or Appendix G, Performance Rating Method, in *EAp2: Minimum Energy Efficiency*. For projects documented using the prescriptive method, generate estimates of peak thermal or peak electricity demand using the approach outlined in *EAp1: Operational Carbon Projection and Decarbonization Plan*.

The peak demand contribution from EVSE for recharging vehicles used for off-site transportation may be excluded from the peak load determination if separate metering is provided for the EVSE.

OPTION 1. ENERGY STORAGE

Projects are awarded points for installing on-site electric and/or thermal storage that meets capacity requirements relative to peak demand per Table 1.

Electricity storage refers to large batteries that store electricity until it is needed.

Thermal energy storage (TES) stores heating or cooling energy for later reuse. Examples include ice storage, chilled water storage, and hot water storage.

Electrical peak demand differs from thermal peak demand, especially in an all-electric building. In an all-electric building, the peak demand includes the contribution of thermal demand and other loads such as lighting, plug and process loads, pumps, and fans. For an all-electric building, the Table 1 thermal storage capacity thresholds that compare to peak coincident thermal demand are expected to achieve similar electricity demand reductions to the Table 1 electric storage capacity thresholds that compare to peak electric demand.

Determine the total capacity required to meet thresholds for the storage type by using the peak electric or peak thermal demand determined consistent with the previously described method.

Energy simulation results typically report the peak thermal load separately for each category of peak thermal load (i.e., heating, cooling, SWH, and process heating or cooling).

To assess peak coincident heating load for heating + cooling + SWH + process heating or cooling:

- Determine the category with the highest peak load (the primary load).
- Determine the time (month, day, and hour) when this highest peak load occurs.
- Calculate peak coincident load by adding the primary load to the simultaneous load for all other categories with substantive use. Ignore categories with peak loads or annual energy use less than 10% of the primary load.

Provide automatic load management controls for the thermal or electric storage systems.

For projects that include both an electric and thermal storage system on-site, points can be prorated to achieve the minimum required thresholds of Table 1.

Example: Thermal storage

A project with peak space heating loads of 1,000,000 Btu/h (290 kW) and coincident cooling loads occurring at the same time are negligible. The energy modeling results indicate peak space heating occurs January 10 at 6 a.m., based on the modeled weather data. Peak space cooling loads occurring in the summer are lower than the space heating peak. The analyst uses the simulation outputs to identify simultaneous SWH loads of 200,000 Btu/h (60 kW), which results in peak coincident loads of 1,2000,000 Btu/h (350 kW). This requires the project to install 1,2000,000 Btu (350 kWh) of thermal storage capacity to earn 1 point.

OPTION 2. DEMAND RESPONSE PROGRAM

Established DR programs and contracting with a qualified provider offers a streamlined path to credit compliance. Projects can contract directly with the utility or with a DR program provider.

Project teams must clearly identify what systems will be included in the program during a DR event. Teams should work with the DR provider to determine the best strategy for the specific project and contract. For example, teams can commit to a reduction of a specified percentage, when a signal is received. Teams may also commit to automated reductions in select equipment or systems, in response to a direct, automated signal from the DR program provider.

Contract length

Execute contracts for at least one year and commit to ongoing renewal of the contract.

OPTION 3. AUTOMATED DEMAND-SIDE MANAGEMENT

Projects must select between Paths 1 and Path 2.

On-site electricity generation and fuel combustion cannot be used to meet the demand-side management criteria. This includes renewable electricity generation, which is separately credited in *EAc4: Renewable Energy*.

PATH 1. SYSTEM-LEVEL CONTROLS

Path 1 is most suitable for small buildings or buildings without a Building Automation System.

Projects must provide Automated Demand Response (ADR) controls for at least two systems and select from HVAC, lighting, automatic receptacle controls, SWH, or EVSE.

HVAC systems

Provide ADR for at least 50% of the total rated capacity. Examples include smart thermostats that adjust the cooling and heating set points or controllers for variable-speed equipment that limit maximum speed during a DR event.

Lighting systems

Provide ADR for at least 50% of installed lighting power. For example, provide automated dimming for 50% of installed lighting power.

Automatic receptacle controls

Provide ADR to turn off automatic receptacle controls as defined in *ASHRAE 90.1*, Section 8.4.2.

Service water heating

Provide ADR for at least 90% of installed SWH capacity. For example, provide storage water heaters with ADR technology that controls the heating cycle off during a DR event.

Electric vehicle supply equipment

Provide ADR capable of curtailing and scheduling vehicle charging for all EVSE. The project must have at least some EVSE equipment in scope.

PATH 2. BUILDING AUTOMATION SYSTEM

Path 2 requires building-level controls that automatically shed at least 10% of building electricity in response to triggers denoting strain on the grid. A signal from a DR provider is the most common trigger. Refer to the rating system for other examples.

Develop a comprehensive plan that provides clear direction for implementing the automated load shedding, both in summer and in winter. Address the following in the plan:

- Individual assignments.
- Communication protocols.
- Project total peak electricity demand.
- Systems and end uses targeted for peak load shedding.
- Justification for why systems and end uses were selected.
- Triggers for initiating automated load shedding, and rationale for selecting these triggers. Address both current grid context and future projections that account for renewables and electrification trends.
- Total percentage of load included in the load-shedding program and a description of the method used to estimate this percentage. Address both winter and summer peaks.

OPTION 4. POWER RESILIENCE

This option requires on-site renewable generation paired with energy storage capable of powering the building's critical equipment operation for at least 72 hours. Critical equipment consists of components most essential to maintain functionality during a power outage, often related to life safety or business continuity. Examples include servers, communication equipment, life support equipment, security systems, emergency lighting, or minimum HVAC capacity needed to maintain life support conditions. To be eligible for this option, the project must have at least some critical equipment.

The design must include an automatic transfer switch and controls that enable the project to operate the building's on-site renewable systems, energy storage, and critical equipment in the event of a power outage (referred to as islanding).

CORE AND SHELL PROJECTS

For Core and Shell projects, peak electric demand or coincident peak thermal demand must be determined for the whole building energy use, inclusive of expected tenant energy.

Owners may provide an option for future tenants to opt out of participation in the DR program. It is highly recommended that owners include educational information on the importance of the DR program, including environmental and resilience benefits.

Option 1. Energy Storage

Same as LEED BD+C: New Construction, Option 1.

Option 2. Demand Response Program

Same as LEED BD+C: New Construction, Option 2.

Option 3. Automated Demand-side Management

Same as LEED BD+C: New Construction, Option 3, plus:

For Core and Shell projects, the tenant lease agreement must describe details of what systems are included in the automatic DR control. Communicating this information is important for both the base building systems and for any systems that may impact the tenant spaces.

Additionally, include language describing any tenant efforts to integrate their systems with the DR controls, as applicable.

Option 4. Power Resilience

Same as BD+C: New Construction, Option 4.

District Energy Systems

For projects documenting energy efficiency savings for a campus district energy system in *EAc3: Enhanced Energy Efficiency*, the district energy must be included in the determination of peak electric demand.

Grid-interactive strategies applied to the DES system may be used to document achievement of Option 1, Option 3, and Option 4 at the building level.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Description of the grid interactive measures evaluated and the relevant context (e.g., forecasted grid conditions, project location, building type, ownership structure, and other relevant factors), and confirmation that the results have been accounted for in decision-making.
			Confirmation the project has interval recording meters and equipment capable of accepting an external signal.
			Documentation that the technology and controls in place for energy storage, DR, automated demand-side management, and power resiliency, as applicable, are within the CxP scope of work.

(continued)

Project Types	Options	Paths	Documentation
			Affirmation that the technology and controls in place for energy storage, DR, automated demand-side management, and power resiliency, as applicable, are documented in the project systems manual, or Current Facilities Requirements (CFR) and Operations and Maintenance (O&M) plan.
	Option 1		Calculation showcasing achievement of point threshold. (Estimated Energy use, Peak Demand, Storage Capacity, Peak Storage Capacity Relative to Peak Demand).
			Narrative documenting the automatic load management controls.
	Option 2		Proof of enrollment in DR program.
	Option 3	Path 1	Identification of systems with automatic DR controls and calculation showing required thresholds have been met.
		Path 2	Project total peak electricity demand and total percentage of load included in the load-shedding program.
			Description of how the project will shed 10% of the peak demand for one hour and what triggers the event. (Examples include short narrative, Sequence of Operations, etc.).
			BAS sequence of operations illustrating demand management for building equipment.
	Option 4		Define and give examples of critical equipment that requires continuous operation. Provide evidence of controls capable of meeting power resiliency.
			Narrative describing the on-site renewable and energy storage system design and operation, and calculation to demonstrate islanding capabilities for the critical infrastructure for at least three days.
Core and Shell	Option 3	Path 1 and Path 2	Tenant Lease Agreement showing the inclusion of automatic DR control within their scope of work.
		Path 1. System-Level Controls	Building systems manual, or CFR and O&M plan that address the technology and controls in place for ADR system-level controls.

REFERENCED STANDARDS

- NBI GridOptimal (The GridOptimal Buildings Initiative) (https://newbuildings.org/resource/gridoptimal)
- ASHRAE Grid-Interactive Building Guide (https://store.accuristech.com/ashrae/standards/grid-interactive-buildings-for-decarbonization-design-and-operation-resource-guide)
- DOE Grid-Interactive Efficient Buildings (https://energy.gov/eere/buildings/grid-interactive-efficient-buildings)

FURTHER EXPLANATION

In line with *EAp3: Fundamental Commissioning*, any automated demand-side management via the building automation system (Option 3, Path 2) must be commissioned as part of the commissioning scope.

ENERGY AND ATMOSPHERE CREDIT

Enhanced Refrigerant Management

EAc7
New Construction (1–2 points)
Core and Shell (1–2 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To reduce greenhouse gas emissions by accelerating the use of refrigerants with low global warming potential and promoting better refrigerant management practices.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–2**
Option 1. No Refrigerants or Low GWP	1–2
Path 1. No Refrigerants	1
OR	
Path 2. Low GWP Refrigerants	1–2
AND/OR	
Option 2. Limit Refrigerant Leakage	1
AND/OR	
Option 3. GreenChill Certification for Food Retailers	1–2

OPTION 1. NO REFRIGERANTS OR LOW GWP (1–2 POINTS)

PATH 1. NO REFRIGERANTS (1 POINT)

Do not use refrigerant-containing equipment in the project.

Core and Shell only

Projects without future equipment necessary to meet the project’s heating or cooling load are not eligible for this option.

OR

PATH 2. LOW GWP REFRIGERANTS (1–2 POINTS)

The maximum total weighted average refrigerant GWP in all new refrigerant-containing equipment is less than or equal to 80% (1 point) or 50% (2 points) of the total weighted average GWP of refrigerants meeting the benchmarks in Table 1.

Projects that limit effective refrigerant GWP by reducing refrigerant charge per unit of capacity relative to comparable equipment may use adjusted benchmarks.

TABLE 1. Refrigerant GWP Benchmarks

GWP Benchmark*	Equipment and Systems
1,400	Heat pump service hot water heaters
700	HVAC
	Data centers, computer room air conditioning, and information technology equipment cooling
	Process chiller equipment or ice rink refrigeration equipment
300	All other process refrigeration for retail, industrial, or cold storage

*GWP benchmarks are based on a 100-year time horizon GWP relative to CO_2.

AND/OR

OPTION 2. LIMIT REFRIGERANT LEAKAGE (1 POINT)

Core and Shell only

This option is only applicable to projects with at least 20% of total estimated heating and cooling generation capacity in the project scope of work.

Design, construct, and operate the project's refrigerant-using equipment to minimize refrigerant leakage.

DESIGN

- Refrigerant-using equipment shall be self-contained, with no field-installed piping:
  - For equipment with refrigerants > 700 GWP, AND
  - For at least 80% of the total GWP of refrigerants used in the project
- Specify an automatic leak detection system in fully enclosed spaces with equipment that has an overall refrigerant charge exceeding 100 tCO_2e.

INSTALLATION

- Field-installed refrigerant piping shall use brazed or press-type fittings.

OPERATION

- Have in place a refrigerant maintenance plan and designate a responsible oversight party. The plan shall include standards for recordkeeping and the following protocols:
  - Updating the refrigerant inventory.
  - Tracking and recording refrigerant charge and leakage rates for all refrigerant-using equipment.
  - Ensuring that installation, maintenance, and removal of refrigeration-containing equipment is performed by appropriately certified refrigeration personnel, including in tenant spaces.
  - Performing an annual audit and calibration of automatic leak detection systems.
  - For equipment without automatic leak detection systems, checking pressure loss and leaks at least as frequently at the following minimum intervals for equipment containing refrigerant with total GWP as follows: every 24 months for 50 tCO_2e or less; every 12 months for 50 to 500 tCO_2e; every 3 months for more than 500 tCO_2e.
  - Identifying the maximum time frame for repairing leaks.
  - Making leakage testing and repair twice as frequent if the total annual refrigerant recharge/leakage exceeds 1%.

AND/OR

OPTION 3. GREENCHILL CERTIFICATION FOR FOOD RETAILERS (1–2 POINTS)

Available to projects where food retailing constitutes more than 20% of the project's gross area.

Demonstrate achievement of the EPA's GreenChill Certification program for projects in the U.S. For international projects, comply with the relevant GreenChill requirements for the certification level.

- GreenChill Silver certification (1 point)
- GreenChill Gold or Platinum certification (2 points)

For all options

DISTRICT ENERGY

Projects with district energy must comply with the requirements of this credit at the district facility or see additional guidance for interpretation of credit requirements.

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REQUIREMENTS EXPLAINED

The credit builds on the *EAp5: Fundamental Refrigerant Management* requirements and rewards teams who further minimize or eliminate refrigerant impacts of their projects. Refrigerants used in equipment that provides thermal comfort, SWH, process heating or cooling, or refrigeration for food storage or other process application in buildings are powerful greenhouse gases and typically cause over one thousand times the detrimental impact than carbon dioxide. As projects electrify heating and SWH systems with heat pumps, mitigation of refrigerant impact becomes increasingly important.

This credit rewards strategies for reducing refrigerant impact by limiting refrigerant GWP under Option 1 and by reducing refrigerant leakage under Option 2. Projects may achieve a maximum of 2 points using either Option 1, Path 2, or by combining Option 1, Path 2 and Option 2.

Option 3 is available for food retailers and comprehensively addresses the high refrigerant emissions associated with refrigeration equipment for cold storage.

OPTION 1. NO REFRIGERANTS OR LOW GWP REFRIGERANTS

Path 1 rewards the elimination of refrigerants for the few project applications that can accomplish this without jeopardizing thermal comfort or total greenhouse gas emissions.

Path 2 requires the selection of low-impact refrigerants and is more appropriate for most projects. Path 2 supports a design that comprehensively addresses decarbonization through electrification using efficient heat pump technology.

PATH 1. NO REFRIGERANTS

To pursue this path, the project cannot use refrigerants in the building or in DES serving the building.

Alternative design solutions, including passive strategies, present opportunities to remove refrigerants from buildings and promote further decarbonization. Consider using passive cooling strategies such as natural ventilation, night flushing, and thermal massing solutions or passive heating strategies such as solar storage and added insulation. Review opportunities to minimize infiltration and ventilation losses through the building envelope.

> **Additional considerations**
>
> Projects are encouraged to pursue a design that includes electrification of space heating, SWH, and process heating systems with efficient heat pump technology per *EAc3: Enhanced Energy Efficiency* rather than using electric resistance or fuel heating to meet the requirements for this path. Therefore, Path 1 is limited to 1 point.

PATH 2. LOW GWP REFRIGERANTS

Path 2 requires the use of refrigerants with weighted average global warming potential that average at least 20% lower than the Refrigerant GWP benchmarks in Table 1 of the credit.

These GWP benchmarks are primarily derived from GWP limits in the *EPA 2023 AIM Act Technology Transitions Rule*, which are like those in the *EU F-Gas Regulations* and other regulations following the Montreal Protocol and Kigali Agreement.

Calculations

Use equipment data and the project's total weighted average GWP reported in the refrigerant inventory completed for *EAp5: Fundamental Refrigerant Management*.

For each refrigerant-using equipment, determine the GWP benchmark using the equipment's refrigerant charge reported in the refrigerant inventory, and the Table 1 GWP benchmark for the equipment:

Equation 1. GWP benchmark for each piece of equipment

$$GWP_{Equipment\_Benchmark} = Rc_{Equipment} \times GWP_{Benchmark}$$

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Calculate the total benchmark GWP by summing the GWP benchmark for each piece of equipment.

Equation 2. Total GWP benchmark

$$GWP_{Total\_Benchmark} = \sum GWP_{Equipment\_Benchmark}$$

Calculate the weighted average GWP benchmark by dividing the total GWP benchmark by the sum of refrigerant charge for all equipment.

Equation 3. Weighted average $GWP_{Benchmark}$ calculation

$$\text{Weighted average } GWP_{Benchmark} = \frac{GWP_{Total\_Benchmark}}{\sum R_c \text{ for all equipment}}$$

Total weighted average GWP for the project cannot exceed 80% of the weighted average GWP benchmark for 1 point; it cannot exceed 50% of the weighted average GWP to achieve both points.

Equation 4. Percentage (%) threshold calculations

$$\text{\% of benchmark} = \frac{\text{Weighted average GWP}}{\text{Weighted average } GWP_{Benchmark}}$$

Adjusted Benchmarks for Limiting Effective GWP

For equipment that has been specifically designed to limit effective refrigerant GWP by minimizing refrigerant charge, projects may reference comparable equipment to establish an adjusted benchmark instead of using the refrigerant charge for the project equipment.

To be eligible to apply adjusted benchmarks, neither the GWP for the comparable equipment, nor the GWP for the referenced project equipment may exceed the GWP benchmark in Table 1. Comparable equipment shall have the same equipment type description and size category as the referenced project equipment (see Section 6.8 tables in *ASHRAE 90.1* for equipment types and size categories).

Calculate the adjusted benchmark by multiplying the original Table 1 benchmark by the ratio of refrigerant charge per unit of capacity for the comparable equipment versus the project equipment.

Example: Adjusted benchmark calculation

The project team evaluates two water-cooled centrifugal chiller alternatives with GWP less than 700—the first with a refrigerant charge per unit of capacity equal to 3.0 lb/ton (0.38 kg/kW), the second with a refrigerant charge per unit of capacity equal to 1.5 lb/ton (0.19 kg/kW). The project team selects the second chiller.

From Table 1, the GWP Benchmark for HVAC equipment is 700. The adjusted GWP Benchmark is 1,400, calculated as $700 \times 3.0/1.5$.

OPTION 2. LIMIT REFRIGERANT LEAKAGE

Combine design and construction strategies with operational best practices to effectively manage refrigerant leakage for the life of a building.

Design

Field-installed piping experiences much higher leakage rates than self-contained equipment. During design, prioritize self-contained equipment. At minimum, projects must specify self-contained equipment for systems that use refrigerants with a GWP ≥ 700. Additionally, teams must use self-contained equipment for at least 80% of the total refrigerant GWP. Self-contained equipment is less prone to leakage and better accommodates leakage detection measures than equipment with field-installed piping.

Installation

For projects that include field-installed piping, install piping in a manner that limits leakage. Use brazed or press type fittings to minimize the potential for refrigerant leaks.

Install automatic leak detection systems in any fully enclosed space that houses equipment with an overall refrigerant charge greater than 100 tCO_2e. (tCO_2e is a metric ton of carbon dioxide equivalent, where a metric ton equals 1,000 kg or 2,205 lb.)

Operations

Maintaining systems during operations provides continued assurance that refrigerant leaks are identified as soon as possible and reduces GWP for leakage. Teams must develop a refrigerant maintenance plan that requires updates to the refrigerant inventory, tracking and recording of refrigerant charge and leakage rates, routine pressure testing on required systems, annual audits, and calibration of automatic leak detection system devices.

Major leaks identified during operations require immediate corrective action. Additionally, where leakage exceeds 1% of the total annual refrigerant recharge, teams must conduct additional testing and repairs to reduce the total leakage of the system. This ensures systems operate as intended and minimizes global warming associated with leakage.

Teams must designate a key individual or the appropriate management team to manage and enforce the plan.

CORE AND SHELL PROJECTS

OPTION 1. NO REFRIGERANTS OR LOW GWP REFRIGERANTS

Path 1. No Refrigerants

To be eligible for Path 1, demonstrate that additional equipment is not required to meet the project's heating and cooling load.

This path is not available to projects where mechanical heating or cooling are likely to be installed during tenant build-out, even for a single space such as an office space in a large unconditioned warehouse.

Projects that do not install equipment as part of the base scope are ineligible for this option.

Path 2. Low GWP Refrigerants

Teams must include any new refrigerant-containing equipment installed during the project's scope of work. Future tenant equipment may be excluded. Core and Shell projects may also include refrigerant-using equipment installed as part of the tenant scope of work if tenants specify the equipment on their construction documents. When including tenant equipment, include all refrigerant-using equipment specified on the tenant drawings.

OPTION 2. LIMIT REFRIGERANT LEAKAGE

Include at least 20% of the total predicted heating and cooling generation capacity in the project's scope of work for the base building.

Example 1: Eligible project

An owner pursues LEED BD+C: Core and Shell certification for a speculative office and laboratory base building. The base building systems include chillers, central air handling units, and laboratory exhaust fans. Through energy modeling or load calculations, the project team confirms that the total heating and cooling generation capacity exceeds 20% of the total predicted capacity. The building is eligible to pursue this credit and document compliance for the systems within the scope of work.

Example 2: Non-eligible project

An owner pursues LEED BD+C: Core and Shell certification for a speculative office building. The scope of work is limited to a cold, dark shell. The project excludes base building systems and cannot pursue this credit.

DISTRICT ENERGY SYSTEMS

OPTION 1. NO REFRIGERANTS OR LOW GWP REFRIGERANTS

Path 1. No Refrigerants

Refrigerants cannot be used to generate any district energy source serving the project.

Teams connected to a DES for any energy source should work directly with their provider to determine if compliance is met for this option.

Path 2. Low GWP Refrigerants

Teams can elect to include the refrigerant impacts from the DES when compliance cannot be demonstrated based solely on new equipment within the project scope.

Where teams include DES equipment within the calculations, account for all new and existing equipment containing refrigerant from all district energy sources serving the project. Teams must work with the district energy provider to determine the equipment type, refrigerant type, and refrigerant charge.

For projects that also have refrigerant-containing equipment within the building, the weighted average GWP can be determined using the percentage capacity from each source. For example, if 90% of the energy comes from a DES and 10% from systems on-site, teams can apply those percentages to respective DES and on-site equipment weighted average GWP values.

OPTION 2. LIMIT REFRIGERANT LEAKAGE

Demonstrate that at least 50% of the combined peak heating and cooling capacity generates on-site. Alternatively, projects demonstrate compliance with the specified criteria for all refrigerant-using systems in the building and all refrigerant-using systems in the district energy system serving the project.

Retail Only

- Meet Option 1 and/or Option 2 under the LEED BD+C: New Construction criteria.

AND/OR

OPTION 3. GREENCHILL CERTIFICATION FOR FOOD RETAILERS

Option 3 offers another path that is eligible for up to 2 points for projects with 20% or more gross floor area of food retail space. Food retail spaces include grocery stores, convenience stores, big-box wholesalers, and general merchandise retailers with large refrigeration sections. Additional spaces may include bakeries, seafood and meat markets, and juice and smoothie bars.

In 2007, the U.S. EPA launched a voluntary partnership program called GreenChill that works cooperatively with the food retail industry to reduce refrigerant emissions and decrease their impact on the ozone layer and climate change.[11] A food retailer achieves Silver-, Gold-, or Platinum-Level certification.

U.S. projects that document GreenChill certification earn points based on the level of certification achieved. International projects must document compliance with each requirement outlined on the EPA GreenChill website for the targeted certification level.

DOCUMENTATION

Project Types	Options	Paths	Documentation
New Construction and Core and Shell	All	All	Mechanical schedules (or similar) that show equipment type, capacity, quantity, refrigerant type, and maximum refrigerant charge.
	Option 1	Path 1	Attestation that no refrigerants are used on the project (*EAp5: Fundamental Refrigerant Management* requires a narrative describing how the building loads will be met if this option is pursued).
		Path 2	USGBC calculator.
			Determine the GWP benchmark and weighted average GWP utilizing the Refrigerant calculator.

(continued)

11. U.S. EPA, "GreenChill Program," November 4, 2024, https://epa.gov/greenchill.

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(continued)

Project Types	Options	Paths	Documentation
	Option 2	All	Documentation of compliance with field-installed piping limits.
			Contractor documentation demonstrating that installation complies with credit criteria.
			Refrigerant Maintenance Plan that describes how the refrigerant-using systems will be maintained. The plan must identify the designated responsible party for implementing the plan.
			Plans, specifications, or field photos confirming automatic leak detection, if applicable.
	Option 3	U.S. projects	Demonstrate that the project has achieved EPA's GreenChill Store Certification Program.
		International projects	Compliance documentation demonstrating that the project meets relevant GreenChill requirements for the certification level specified.
Core and Shell	All	All	Confirmation that the Core and Shell project is eligible for this option.

REFERENCED STANDARDS

- ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems (https://www.ashrae.org/technical-resources/standards-and-guidelines/read-only-versions-of-ashrae-standards)
- EPA Green Chill (https://www.epa.gov/greenchill)
- EPA 2023 AIM Act Technology Transitions Rule (https://www.epa.gov/climate-hfcs-reduction/regulatory-actions-technology-transitions)
- European Union F-gas regulations (https://eur-lex.europa.eu/eli/reg/2024/573/oj)

Materials and Resources

OVERVIEW

The Materials and Resources (MR) category in LEED v5 focuses on critical areas that reduce embodied carbon, protect human and environmental health, and foster a circular economy. Along with these critical areas of focus, the credits target data availability, transparency, and supply chain improvements, and ultimately increase the accessibility of compliant materials for all.

The MR credits support LEED v5's materials strategy by furthering the shift toward multi-attribute product selection and procurement. This approach evaluates materials based on a variety of key metrics—from sourcing to manufacturing processes, and overall environmental and social impacts—to guide projects toward well-rounded material choices that go beyond single-issue solutions.

Another key focus of the MR credit category is embodied carbon, or the emissions generated during the extraction, manufacturing, transportation, installation, and disposal of products. The impact of opportunities to reduce embodied carbon go beyond design phase, Environmental Product Declaration (EPD) analysis, and whole-building life-cycle assessment (WBLCA), with material reuse and key waste management practices playing an important role in the global effort to minimize the impacts of building materials on the environment. Because embodied carbon from building materials accounts for at least 11% of annual global emissions, LEED v5 targets strategies for high-impact actions like supply chain decarbonization, low-embodied carbon material selection, and building reuse to help project teams achieve meaningful carbon reductions immediately.

LEED v5 simplifies strategies to maximize impact and promote industry alignment. The MR category harmonizes terminology and standards across systems and aligns with initiatives like the Embodied Carbon Harmonization and Optimization (ECHO) project,[1] the Mindful Materials Common Materials Framework,[2] and the AIA Architecture and Design Materials Pledge.[3] These efforts reduce complexity and make it easier for manufacturers and project teams to meet sustainability goals and establish workflows that will keep industry advancement moving forward.

1. ECHO Project, "Embodied Carbon Harmonization and Optimization (ECHO) Project," n.d., https://www.echo-project.info/.
2. "Mindful MATERIALS," 2024, https://www.mindfulmaterials.com/.
3. AIA, "AIA Materials Pledge," n.d., https://www.aia.org/design-excellence/climate-action/zero-carbon/materials-pledge.

Decarbonization

The MR category equips projects to reduce embodied carbon across the supply chain using strategies like WBLCA, analysis of EPDs, and jobsite emissions tracking (*MRc2: Reduce Embodied Carbon, MRp2: Quantify and Assess Embodied Carbon*). Reducing construction waste and promoting circularity lessen demand for virgin resources and extend material life, which also directly reduce embodied carbon emissions from the supply chain (*MRc5: Construction and Demolition Waste Diversion, MRp1: Planning for Zero Waste Operations*).

Embodied carbon could account for half of new construction's carbon footprint by 2050. LEED v5 plans for a different outcome by rewarding manufacturing innovations that decarbonize new materials coupled with circular strategies that preserve resources and cut emissions.

Quality of life

The MR category enhances indoor environmental quality by promoting low-emitting materials that reduce occupant exposure to harmful chemicals (*MRc3: Low-Emitting Materials*, *MRc4: Building Product Selection and Procurement*). Improved air quality supports health, cognitive function, and overall well-being, which all benefit building occupants. Upstream and downstream impacts from product manufacturing can also affect fence line communities, supply chain actors, and installers, which make the selection of materials focused on green chemistry and ecological protections a priority (*MRc4: Building Product Selection and Procurement*).

Ecological conservation and restoration

Prioritizing reuse and diverting waste from landfills decrease reliance on virgin material, reduce methane emissions, preserve natural resources, and reduce environmental harm (*MRc5: Construction and Demolition Waste Diversion*, *MRp1: Planning for Zero Waste Operations*). When product manufacturing shifts to support a circular economy, improvements to ecological conservation and restoration can be significant (*MRc4: Building Product Selection and Procurement*).

Ultimately, LEED v5 empowers project teams to make practical, high-impact choices that cut embodied carbon emissions, improve health outcomes, and advance a sustainable market, thereby building a future where both people and the planet can thrive.

MATERIALS AND RESOURCES PREREQUISITE

Planning for Zero Waste Operations

MRp1
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To reduce the amount of waste that is generated by building occupants and hauled to and disposed of in landfills and incinerators through reduction, reuse, and recycling services and education, and to conserve natural resources for future generations. To set the building up for success in pursuing zero waste operations.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Storage and Collection of Recyclables AND	
Zero Waste Operations Planning	

Comply with the following requirements:

STORAGE AND COLLECTION OF RECYCLABLES

Provide dedicated areas accessible to waste haulers, janitorial staff, and building occupants for the collection and storage of recyclable materials for the entire building.

- Collection and storage areas may be separate locations.
- Recyclable materials must include organics/food waste, mixed paper, corrugated cardboard, glass, plastics, and metals.
  - Mixed recyclables are acceptable for paper, corrugated cardboard, glass, plastics, and metals if required by local conditions.
  - Space for the storage of organics/food waste recycling is required even if service is not available at the time of building occupancy.

- Take appropriate measures for the safe collection, storage, and disposal of batteries, mercury-containing lamps, and electronic waste.

AND

ZERO WASTE OPERATIONS PLANNING

Include design details, maintenance manuals, and/or other resources from the design and construction team that help facilitate building occupants and operators to meet high-performance waste prevention and recycling goals once in operation.

Core and Shell only

Communicate the building's infrastructure and service options information in the tenant guidelines.

REQUIREMENTS EXPLAINED

The goal for the prerequisite is to minimize waste generated by building occupants and to implement strategies in reducing, reusing, and recycling waste throughout the building's life-cycle. This includes incorporating design measures that prioritize waste prevention, material reuse, and effective waste management. It also encourages project teams to thoughtfully consider and plan for the access requirements of all individuals who will service and use the building. Additionally, teams should anticipate and accommodate the operational needs necessary for achieving zero waste by enabling effective waste diversion practices. This includes ensuring proper sorting, storage, and access solutions to facilitate recycling, composting, and other waste diversion methods while prioritizing reuse as a key strategy.

STORAGE AND COLLECTION OF RECYCLABLES

Municipal solid waste has become a growing concern as the volume of waste generated in the U.S. continues to increase.[4] An obstacle to effective recycling in buildings is the lack of convenient, dedicated physical spaces for collection. Incorporating recycling infrastructure early in the design process encourages successful recycling once operations begin. Well-designed and accessible waste management infrastructure is intuitive (easy to locate, reach, and use) for all occupants, regardless of physical ability or mobility, anticipating how and where waste will be discarded by occupants. Address concerns for noise, odor, and vectors.

Recycling includes traditional materials like glass, plastic, and metals, as well as organic materials like food scraps, paper products, and landscape materials. Organic materials collected in buildings can be composted on-site or off-site. Composting can occur at multiple scales and locations. Small-scale systems might include basic compost piles or bins, and large-scale operations involve centralized, commercial facilities that process organic waste from an entire region.

4. University of Michigan Center for Sustainable Systems, "U.S. Municipal Solid Waste," 2024, https://css.umich.edu/sites/default/files/2025-02/CSS04-15.pdf.

To meet the prerequisite, teams must provide dedicated area(s) for recyclable items including organics (compostables), mixed paper, corrugated cardboard, glass, plastics, and metals. Teams should work with the owner and architects to provide sufficient collection and storage space for all required recyclables based on the building size, occupants, and local recycling markets. In many places, recycling various materials is required by laws or regulations. Project teams should check for local requirements and service vendors to ensure maximal waste diversion.

Given the substantial greenhouse gas emissions from discarded organic waste in landfills (from methane production), planned space for composting organic materials generated by occupants is required, even if composting services are not immediately available at time of building occupancy. For organics collection, some material processors will encourage paper products to be recycled separately (such as office paper placed in mixed-paper recycling bins and excluded from food scraps collection). Check with local authorities for guidance on best practices for mixed paper or cardboard and other forms of organics composting.

To meet the prerequisite, project teams must provide an adequate amount of dedicated space with the appropriate infrastructure to handle such recycling. This includes planning for the installation of collection systems or bins for recyclable materials that are collected by the building and sent for recycling at time of occupancy. Base these bins or collection systems on current service offerings in the project region.

Commingled recycling bins (excluding organics) are acceptable if the local municipality or recycling vendor allows commingled recycling, although commingled recycling tends to reduce the quality of diverted materials and leads to lower overall recovery rates. Therefore, source separation of recycled material types is encouraged to maximize diversion rates and help meet zero waste goals, but is not required unless separate streams are required by local regulations or guidelines.

There is a growing environmental concern for the increasing volume of electronic waste (e-waste), such as computers, cameras, printers, and keyboards. The e-waste disposal procedure is more hazardous than cardboard, glass, plastic, metals, and paper. Therefore, identifying safe storage areas, recycling facilities, and haulers that can process e-waste is important. Teams must indicate space dedicated to the storage and collection of recyclables, composts, and e-waste areas on a floor plan and describe how these spaces will be serviced and accessed safely by building occupants and staff.

Project teams should follow the TRUE waste hierarchy framework that prioritizes waste management actions to minimize environmental impact and conserve resources. It ranks waste management strategies from most to least preferred and focuses on reducing waste generation and maximizing resource recovery before resorting to disposal, in this order:

- **Reduce**. Minimize waste at the source (e.g., use fewer materials and resources).
- **Reuse**. Extend the life of products by using them more than once.
- **Recycle**. Transform waste material into new products.
- **Compost**. Decompose organic waste into nutrient-rich soil.
- **Anaerobic Digestion**. Break down organic material into a form of nutrient-rich liquid for soil application (does not necessarily equal composting).

ZERO WASTE OPERATIONS PLANNING

Project teams must include design details, maintenance manuals, and other resources in a plan to help building operators minimize waste and implement recycling practices post-occupancy. Consider implementing reusable infrastructure, such as refillable dispensers, reusable service ware, and reusable event items, while also establishing comprehensive systems for waste diversion, including take-back programs for specialty items and user-friendly sorting stations with adequate storage. The plan should consist of materials for training staff and contractors, such as literature, presentations, and onboarding training resources. These resources may cover topics like designated recycling and composting areas, waste separation procedures, and proper disposal guidelines. Include strategies to help with waste prevention, materials recovery, and operational procedures that support zero waste objectives.

Core and Shell only

Communicate the building's infrastructure and service options information in the Tenant Guidelines.

Core and Shell project teams may have limited influence on how the final tenant space is fitted out. The tenant guidelines help tenant design teams understand and design features to achieve significant reductions in waste generation. To effectively communicate details of waste management, the tenant guidelines should provide clear, actionable information that outlines the building's waste management system and encourages responsible waste disposal practices. This allows tenants to remain informed about the building's systems and their responsibilities. Teams should include information on the waste collection and disposal collection area, waste minimization initiatives, waste pick-up schedule, training materials, and contact information.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Evidence of the size, locations, and accessibility of the dedicated collection and storage areas for all required recyclable materials and hazardous waste (e.g., contract documents, photographs).
			Comparison of the expected volume of waste for the project by streams to the provided amount of storage and collection/pick-up frequencies.
			Zero waste operations plan, including design details, maintenance manuals, and/or other resources from the design and construction team that will help facilitate zero waste post occupancy.
Core and Shell			Tenant guidelines communicating the building's infrastructure and service options information: waste and disposal collection area, waste minimization initiatives, waste pick-up schedule, training materials, and contact information (This may be provided under *IPp4: Tenant Guidelines*).

REFERENCED STANDARDS

- Compostable BPI standard (https://bpiworld.org)
- TRUE diversion rate guidelines (https://true.gbci.org/true-diversion-data-additional-guidance)

FURTHER EXPLANATION

BATTERIES, MERCURY-CONTAINING LAMPS, AND ELECTRONIC WASTE

The minimum measures to be taken to ensure safe collection, storage, and disposal of batteries, mercury-containing lamps, and electronic waste include the following:

- Identification and record of local hazardous waste drop-off locations or collection programs
- Inclusion of dedicated storage areas in the project for each hazardous waste stream that are securely located away from children and animals, have labeled waste stream type, and have hazard warnings
- Mercury-containing products with a tight-fitting lid
- Posted signage with handling instructions
- Handling materials (e.g., tape to cover lithium battery terminals; original packaging or bubble wrap for lamps)

STORAGE AND COLLECTION ACCESSIBILITY

Storage areas must be accessible to waste haulers, janitorial staff, and building occupants and must be designed and sized to hold the project's operations recycling waste, separated by stream (aligned with the local recycling infrastructure), until the contents are taken to be recycled. Storage areas may sometimes also be a point of collection.

Collection areas (or point of collection) must be accessible to waste haulers, janitorial staff, and building occupants. Collection areas must be designed to collect the project's operations recycling waste, commonly in the form of appropriately sized bins located near the sources of waste, in which recycling waste is grouped by stream (aligned with the local recycling infrastructure) until the contents are moved to a different, larger container for longer term storage or until the contents are taken to be recycled or diverted. The size of the bin, how often it is emptied, and whether a larger storage container is also necessary should be based on the anticipated volume of waste to be generated in the amount of time between emptying (to storage or recycling/diversion).

WASTE PREVENTION

During design, it is recommended that project teams identify waste minimization goals and strategies for the project and consider any reusable infrastructure necessary to prevent the need for ongoing disposables. Any such infrastructure shall be communicated to the building occupants and operators. Examples of infrastructure that may have been designed into the space to minimize ongoing waste include the following:

- Pre-plumbed or pre-installed dishwashing machines for washing reusable silverware and dishes
- Dedicated space in a closet allocated for storage of infrequent e-waste
- Dedicated storage areas allocated for holding reusable pallets or shipping containers

MATERIALS AND RESOURCES PREREQUISITE

Quantify and Assess Embodied Carbon

MRp2
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To quantify the embodied carbon impacts of the structure, enclosure, and hardscape of a project and assess the top sources of embodied carbon.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Embodied Carbon AND	
High-Priority Embodied Carbon Sources	

Comply with the following requirements:

EMBODIED CARBON

- Quantify the embodied carbon impacts—global warming potential (GWP)—of the structure, enclosure, and hardscape materials for the project. All ancillary structures, such as parking structures or outbuildings within the LEED project boundary, must be included in the calculations. At a minimum, include asphalt, concrete, masonry, structural steel, insulation, aluminum extrusions, structural wood and composites, cladding, and glass.
- Quantify the cradle-to-gate (A1–A3) embodied carbon emissions for each material, defined as the product's GWP/unit times the amount of material used.
  - Alternatively, projects using life-cycle assessment or embodied carbon software tools may report A1–A3 results from their tool.

AND

HIGH-PRIORITY EMBODIED CARBON SOURCES

Identify the top three sources of embodied carbon on the project and describe how project-specific strategies were considered to reduce the impacts of these hot spots.

Core and Shell only

Communicate the embodied carbon measurements and material suppliers in the tenant guidelines.

REQUIREMENTS EXPLAINED

This prerequisite ensures that all projects become acquainted with embodied carbon and gain a basic understanding of how to quantify and measure it. It also highlights the critical role of material selection and emphasize that informed choices across major material categories are essential to achieving meaningful reductions in embodied carbon. The prerequisite aims to raise awareness of the upfront embodied carbon associated with key materials used in the structure, enclosure, and hardscape of a project. This prerequisite does not mandate reductions in embodied carbon.

Project teams can achieve this prerequisite in two ways:

- As a stand-alone assessment conducted for projects not attempting the *MRc2: Reduce Embodied Carbon*, OR
- As an output from projects that are attempting the *MRc2: Reduce Embodied Carbon*.

The intention is that many projects will attempt to earn points from the *MRc2: Reduce Embodied Carbon* and use the analysis as the documentation for this prerequisite with no further analysis needed.

In addition, all projects will need to summarize the top three sources of embodied carbon in their project and describe what strategies in the project were considered to reduce the impact of these hot spots.

EMBODIED CARBON

The extraction and manufacturing phases of building materials account for a substantial portion of embodied carbon emissions, primarily due to energy-intensive raw material extraction, transportation from manufacturers to construction sites, and the waste produced during manufacturing. Projects that seek carbon reductions in the early design phases (schematic design and design development) can make the most significant decisions to reduce embodied carbon early in projects, not after design is complete when material substitutions may not be allowed or become cost prohibitive. The owner, designers, and contractors can collectively make decisions to reduce the impact a building's materials will have on the environment by using a fully integrated and collaborative design process.

QUANTIFY EMBODIED CARBON IMPACTS

Quantifying the embodied carbon impacts provides a holistic assessment across major material groups within the project's structure, enclosure, and hardscape. Calculate the GWP for all materials within each subcategory. For example, within the structural system, the GWP of each type of concrete, steel, or other material is individually assessed and then combined to determine the total GWP for the structure. Similarly, for the enclosure, materials like insulation, cladding, and glazing are evaluated, and components, such as paving and landscaping materials, are analyzed for the hardscape. By summing up the impacts of all subcategory materials, the project team can achieve a detailed understanding of the embodied carbon for each major group and enable informed decisions to reduce the project's overall environmental impact.

Teams must use building project documents, including construction drawings and specifications or software tools, to group materials into a bill of materials (BOM) of structure, enclosure, and hardscape materials used on the project. The BOM must encompass all elements within the LEED project boundary, including any ancillary structures like attached or detached parking garages and other components with significant embodied carbon impacts. Quantities may come from as-built data or estimated quantities from the design phase.

QUANTIFY THE CRADLE-TO-GATE EMBODIED CARBON EMISSION

Quantifying the cradle-to-gate (A1–A3) embodied carbon emissions for each material focuses on emissions associated with raw material extraction (A1), transportation to the manufacturing site (A2), and the manufacturing processes (A3). Calculate the GWP based on the cradle-to-gate (A1–A3 stages) impacts for each applicable material used in the project. This involves multiplying the GWP/unit (from EPDs or default values) by the quantity of each material.

Teams will locate EPDs to determine the embodied carbon values for each material. If EPDs are unavailable, teams may use industry-standard defaults provided by regional data sources or integrated within qualifying software tools. Projects should follow a hierarchy of data:

- Use specific EPDs for the product as published by the manufacturer. If a product-specific EPD is not available, use the U.S. EPA default values when available.
- Refer to the most recent Carbon Leadership Forum (CLF) Material Baselines report.[5]
- Consult other widely used, well-established, authoritative publications or databases in the industry that are supported by extensive peer-reviewed research.
- Use industry-wide EPDs relevant to the project region (Table 1).

TABLE 1. Common Industry-Wide EPD Data for North America

Material	Organization
Concrete	National Ready Mixed Concrete Association (NRMCA)
Steel	American Institute of Steel Construction (AISC)
	Steel Recycling Institute (SRI)
Masonry	National Concrete Masonry Association (NCMA)
Wood	American Wood Council (AWC)
Insulation	North American Insulation Manufacturers Association (NAIMA)

5. Carbon Leadership Forum (CLF), "Carbon Leadership Forum Material Baselines for North America / August 2023," https://carbonleadershipforum.org/clf-material-baselines-2023/.

Tools and databases

A variety of free tools and databases are available for finding EPDs and calculating embodied carbon as part of common building design software/tools. These tools enhance usability in design workflows.

The data sources within these databases vary from EPDs, manufacturer-provided data, academic research, industry statistics, government publications, and other databases. These differences in data sources can influence the accuracy and comparability of the results. Some databases account for regional variations in life-cycle inventory (LCI) data, baseline LCA values, and EPD information, whereas others may only support analysis within specific countries or regions. Table 2 lists credible and widely recognized LCA tools, supported by robust, up-to-date databases.

RESULTS OF THE EMBODIED CARBON QUANTIFICATION

When tallied up, the results of the embodied carbon quantification represent the total embodied carbon calculated from the cradle-to-gate (A1–A3) stages of materials used in the project's structure, enclosure, and hardscape. With the results of this quantification, teams can identify the high GWP materials early in design and prioritize alternatives or modifications.

TABLE 2. Common LCA Software and Reporting Metrics

LCA Software	Reporting Metrics
Athena	A1–A3 stages (cradle-to-gate)
Tally	A1–A5, B1–B7 stages, and C1–C4
OneClick LCA	A1–A5, and optional B and C

AND

High-priority embodied carbon sources

High-priority embodied carbon sources, such as concrete, steel, and insulation, are responsible for significant embodied carbon emissions. An incredible opportunity to reduce embodied carbon in these high-impact materials exists through policy, design, material selection, and specification. For example, optimizing a structure to minimize the use of concrete can lower demand for concrete and reduce emissions by 22%.[6] In addition, projects can explore mandates to procure structural steel produced in the U.S. to reduce transportation emissions.

Projects must identify the three primary contributors to embodied carbon using the analysis from this prerequisite. Use strategies within design and procurement to reduce carbon emissions from the structure, enclosure, and/or hardscape materials. A hot spot analysis is mandatory to identify the most carbon-intensive materials and allow teams to focus on areas with the greatest potential for impactful reductions.

For each of these key sources, explain how the team evaluated and implemented project-specific strategies to reduce their environmental impact. For instance, structural materials such as concrete and steel often emerge as hot spots due to their high embodied carbon. An acceptable analysis would

6. Carbon Smart Materials Palette®, "Increase Material Efficiency and Reduce Use," n.d., https://www.materialspalette.org/increase-material-efficiency-and-reduce-use/.

include examining options like low-carbon concrete mixes, recycled steel, or changes in design that reduce material use and are ideally part of an early design analysis to ensure the greatest impact.

Core and Shell only

Projects must include the embodied carbon measurements and material supplier information in the tenant guidelines. This involves sourcing EPDs, conducting a WBLCA to calculate embodied carbon impacts, and documenting suppliers' sustainability attributes like recycled content or certifications. The guidelines should offer recommendations for low-impact material selection and outline opportunities for material reuse. This ensures that tenants have information about the environmental impact of the project's materials and helps to promote sustainable decision-making during fit-out and renovation processes.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	N/A	All	The data sources used to determine the embodied carbon values for each material (when available: specific EPDs for each product, as published by the manufacturer; when EPDs are unavailable: the industry-standard defaults provided by regional data sources or integrated within qualifying software tools).
			When EPDs are not available, justification of the baselines used, demonstrating that the required hierarchy of data has been followed.
			Project documents, including drawings and specifications documenting the structure, enclosure, and hardscape materials used on the project.
			Identification of the project's three primary contributors (from structure, hardscape, and enclosure) to embodied carbon.
			Explanation of how the team evaluated project-specific strategies to reduce the environmental impact.
		Projects attempting *MRc2: Reduce Embodied Carbon*	Results from the whole-building life-cycle assessment tool (or other embodied carbon emissions analysis software tool) showing A1–A3 impacts by material.
		Projects not attempting *MRc2: Reduce Embodied Carbon*	Bill of Materials (worksheet) that lists the quantities of major structure, enclosure, and hardscape materials used in the project, along with the cradle-to-gate (A1–A3) embodied carbon emissions for each material.
Core and Shell	All	All	Tenant guidelines communicating the embodied carbon measurements and material suppliers (this may be provided under *IPp4: Tenant Guidelines*).

REFERENCED STANDARDS

- Buy Clean/GSA/EPA (https://www.gsa.gov/real-estate/gsa-properties/inflation-reduction-act/lec-program-details/material-requirements)
- Carbon Leadership Forum (benchmarks) (https://carbonleadershipforum.org/clf-material-baselines-2023)
- NRMCA average EPDs (https://www.nrmca.org/association-resources/sustainability/environmental-product-declarations/)

FURTHER EXPLANATION

The analysis for this prerequisite is best performed as early in the design process as possible and in concert with the integrated design timeline. It should include the full design and construction team with the intention that the carbon goals of the project can be established together and met at project completion. Project teams should utilize this analysis to inform their design and material selection to meet the intended global warming potential (GWP) reduction target. Project teams will use this assessment to identify the top three sources of embodied carbon.

MATERIALS TO INCLUDE

Include in this analysis what is available at the time of the assessment. Only include materials within the LEED project boundary:

- Include building envelope and structural elements, including the material components of footings and foundations, structural wall assembly (from cladding to interior finishes), structural floors and ceilings (not including finishes), and roof assemblies (e.g., concrete slabs, structural timber, and wood framing).
- Include hardscape and site development (e.g., concrete and asphalt paving and pathways, concrete retaining walls).
- Include parking structures and parking lots.

REPORTING FOR LEED

Project teams will gather their material quantities from the most accurate and updated project documents.

If conducting a whole-building life-cycle assessment (WBLCA), include the data from the report for product stages A1–A3 (see below).

TABLE 3. WBLCA Results

Environmental Impact Totals	Product Stage A1–A3
Global Warming (kg CO_2eq)	1.200E+007
Acidification (kg SO_2eq)	43,609
Eutrophication (kg Neq)	2,380
Smog Formation (kg O_3eq)	723,306
Ozone Depletion (kg CFC-11eq)	0.003318
Primary Energy (MJ)	1.327E+008
Non-renewable Energy (MJ)	1.327E+008
Renewable Energy (MJ)	1.682E+007
Global Warming (kg CO_2eq/m^2)	560.8

(continued)

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TABLE 3. WBLCA Results (*continued*)

Environmental Impacts/Area	Product Stage A1–A3
Acidification (kg SO_2eq/m^2)	2.038
Eutrophication (kg Neq/m^2)	0.1112
Smog Formation (kg O_3eq/m^2)	33.80
Ozone Depletion (kg CFC-11eq/m^2)	1.550E-007
Primary Energy (MJ/m^2)	6,984
Non-renewable Energy (MJ/m^2)	6,200
Renewable Energy (MJ/m^2)	785.7

If not conducting a WBLCA, project teams will generate a bill of materials (BOM), see Table 4 below.

TABLE 4. Sample Bill of Materials with Embodied Carbon Impacts

Material	Quantities	GWP (Stages A1–A3)	Total Embodied Carbon Emissions	Source
Portland cement (U.S.)	5 metric tons	919 (per 1 metric ton)	4,595	2025 CLF Material Baselines for North America Report
Steel (structure)	–	–	–	–
FOAMULAR® NGX® XPS Insulation (enclosure)	500 m^2	6.58E+00 (per m^2)	3,290	Product-specific EPD
Plywood (enclosure)	16 m^3	231 (per m^3)	–	2025 CLF Material Baselines for North America Report
Wood exterior cladding (enclosure)	–	–	–	Product-specific EPD
Asphalt paving (hardscape)	–	–	–	–

USING DEFAULTS

When there are gaps in the data or no environmental product declarations (EPDs) available, project teams shall note this in their submittal and utilize an industry average value in place of a product-specific value. Project teams shall provide documentation of where this data came from and justification of why it was necessary to include in place of EPD data.

Reused and Salvaged Materials

For materials that are reused or salvaged when documenting this prerequisite, project teams shall denote materials as reused or salvaged within their BOM. These materials are to be marked as a GWP value of zero for stages A1–A3 because the reuse of building materials removes the initial "one time" raw resource extraction and manufacturing impacts.

Building and Materials Reuse

MRc1
New Construction (1–3 points)
Core and Shell (1–5 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To incorporate reused materials into new building design, thereby reducing embodied carbon, keeping materials in circularity, reducing demand for virgin material sourcing, preserving resources and histories, and increasing demand for reused materials.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1-3**
Option 1. Building Reuse AND/OR	1–3
Option 2. Materials Reuse	1–2
Core and Shell	**1-5**
Option 1. Building Reuse AND/OR	1–5
Option 2. Materials Reuse	1–2

OPTION 1. BUILDING REUSE (1–3 POINTS NEW CONSTRUCTION, 1–5 POINTS CORE AND SHELL)

Maintain the existing building structure, including floor decking, roof decking, and enclosure. Calculate reuse of the existing project area according to Table 1.

Portions of buildings deemed structurally unsound or hazardous are excluded from the credit calculations.

TABLE 1. Points for Reuse of Existing Building Structure and Enclosure Elements for New Construction Projects

Percentage of Existing Structure and Enclosure Reuse by Project Area	Points
20%	1
35%	2
50%	3

TABLE 2. Points for Reuse of Existing Building Structure and Enclosure Elements for Core and Shell Projects

Percent of Existing Structure and Enclosure Reuse by Project Area	Points
10%	1
20%	2
30%	3
40%	4
50%	5

AND/OR

OPTION 2. MATERIALS REUSE (1–2 POINTS)

- Survey and identify opportunities for materials reuse and/or procurement of reused materials from off-site.
- Reuse materials by keeping them in place or acquiring them from applicable salvage sources or reuse markets and incorporating the materials into the new project design. Specific targeted materials are valued higher because they have high impacts (embodied carbon or pollution), are hard to recycle, and significant amounts of these materials end up in landfill.
- For projects with deconstruction or demolition in scope, conduct a salvage assessment prior to deconstruction or demolition activities and identify materials that can be retained on-site or diverted off-site to reuse markets.
  - Salvaged materials sent for off-site reuse contribute to *MRc5: Construction and Demolition Waste Diversion*. Materials retained on-site contribute to this credit option.

Calculate the percentage of reused per material type according to Equation 1.

Points are achieved according to Table 3.

Equation 1. Reuse % per material type

Reuse % per material type =

Amount of material type reused /
Total amount of material type in New Construction scope

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TABLE 3. Points for Incorporating Reused Materials

Reuse Materials Threshold	Points
Reuse at least 30% of 1 targeted material type OR Reuse at least 15% of 2 targeted material types OR Reuse at least 15% of 4 other material types OR Reuse an equivalent weighted average of targeted and other material types	2 points

TABLE 4. Reuse Material Types and Correlating Units

Material Type	Unit
Targeted materials	
Carpeting	Surface area
Ceilings	Surface area
Furniture (ancillary and systems)	Pieces, weight, volume, or floor area
Interior walls	Linear or surface area
Other materials	
Dimensional lumber	Board foot or linear
Doors	Count
Casework	Linear
Floor-covering materials (not including carpet)	Surface area
Lighting fixtures	Count
Plumbing fixtures	Count
Mechanical equipment	Count
Door hardware	Count
Project defined other	Project defined

REQUIREMENTS EXPLAINED

This credit encourages projects to reuse existing buildings and building materials. Option 1 rewards projects that reuse structural and enclosure elements of existing buildings. Option 2 is focused on the reuse of nonstructural products by keeping materials in place or acquiring materials from salvaged sources. This option encourages projects to focus on reusing targeted materials that have high environmental impacts. Points are earned based on the percentage of targeted and other materials that are reused in the project.

Reusing materials in new building designs benefit the environment by lowering the demand for virgin material sourcing, decreasing embodied carbon, and extending the life-cycle of products. Materials diverted through construction activities and sent off-site for reuse contribute to the *MRc5: Construction and Demolition Waste Diversion*.

OPTION 1. BUILDING REUSE

Building reuse extends the lifespan of structures by revitalizing existing buildings or adapting properties when their original function changes. Demolishing an old building and rebuilding with new materials increases a building's carbon footprint due to the energy required for both processes. Reusing a building conserves resources by reducing the need for new construction materials and minimizes waste. This option can be more cost-effective and reduces the environmental impact associated with construction.

Option 1 addresses the on-site reuse of existing structure and enclosure materials, including materials left in-situ as well as those procured off-site and incorporated into the building's structure/enclosure. To earn points, projects maintain portions of the existing building structure and enclosure and/or incorporate off-site reuse from the early design phase and work with architects and structural engineers to identify structural elements of the existing space that can be reused.

Table 5 outlines how different types of reuse are recognized within the LEED v5 Materials and Resources category, including their contributions to *MRc5: Construction and Demolition Waste Diversion* and *MRc4: Building Product Selection and Procurement*.

TABLE 5. Types of Building/Material Reuse and Credit Contributions

	On-site Reuse (Keep Materials in Place or Within the Same Project)	Procure Off-site Salvaged Materials and Incorporate Them into the Project	Divert On-site Materials to Salvage Markets
Credits/Options/Paths	(On-Site/On-Site)	(Off-Site/On-Site)	(On-Site/Off-Site)
MRc1: Building and Material Reuse, Option 1. Building Reuse (structural materials)	x	x	
MRc1: Building and Material Reuse, Option 2. Materials Reuse (nonstructural materials)	x	x	
MRc5: Construction and Demolition Waste Diversion			x
MRc4: Building Product Selection and Procurement (nonstructural materials)	x	x	

Teams are awarded points by salvaging 20% or more of the existing building structural elements, including enclosure elements. Key existing structural elements include floor and roof decking, load-bearing walls, columns, and beams that provide the primary support framework for the building. Enclosure elements include the building's exterior skin and structural framing that form the thermal and weather barrier of the building. Enclosure materials in this credit include facades, cladding, and exterior walls but exclude nonstructural components like roofing membranes, shingles, window assemblies, and other elements that are not essential to the building's integrity or enclosure.

Teams must exclude hazardous and unsound materials (e.g., remediated as part of the project for historic, abandoned, or unsafe buildings) from the calculation because these materials cannot be safely reused or incorporated into the project. Nonhazardous materials are encouraged to be diverted from landfill or incineration and can contribute to *MRc5: Construction and Demolition Waste Diversion*.

For projects with new building components or additions, base the calculations on the existing floor area, not including additional floor area. Per Equation 2, the reuse calculation is based on the surface areas of major existing structural and enclosure elements. Teams should prepare a calculation table or spreadsheet listing all enclosure and structural elements within the existing building prior to construction or renovation. Teams should quantify each item, listing the square footage of both the existing area and the retained area. Determine the percentage of existing elements that are retained by dividing the square footage of the total retained materials area by the square footage of the total existing materials area. The reused area in the calculation should include any salvaged or reused materials sourced off-site and integrated into the project.

Equation 2. Percentage of existing building reuse

$$\text{Existing building reuse} = \frac{\text{Area reused onsite} + \text{Area reused from off-site}}{\text{Existing building area}} \times 100$$

OPTION 2. MATERIALS REUSE

Option 2 addresses the on-site reuse of nonstructural materials. This option also encourages searching for and procuring of off-site reused materials that are incorporated into the building.

Deconstruction, which involves carefully dismantling structures to salvage materials, supports environmental outcomes by reducing waste, air pollution, carbon emissions, and resource use while conserving energy from the creation of otherwise new materials.[7]

Teams must conduct a salvage assessment to identify opportunities for reusing materials within the project. The assessment must also scan local or regional sources for reclaimed materials from off-site locations that could be incorporated into the project. A salvage assessment is typically conducted prior to any construction taking place and is done by deconstruction professionals, sustainability consultants, or salvage experts. It includes an inventory of materials categorized by type, quantity, and reuse potential. An assessment is considered adequate if it thoroughly covers all relevant materials, addresses safety concerns, and provides actionable guidance for stakeholders to implement reuse or recycling strategies effectively.

The assessment also should evaluate the existing building or site for salvageable components and explore local markets, suppliers, or deconstruction initiatives for reusable materials. Architectural salvage stores, reuse websites/databases, and Habitat for Humanity ReStores are sources for reused building materials.

7. Delta Institute, "Deconstruction and Building Material Reuse: A Tool for Local Governments and Economic Development Practitioners," May 2018, https://delta-institute.org/wp-content/uploads/2018/05/Deconstruction-Go-Guide-6-13-18-.pdf.

Table 2 of the requirement outlines the points awarded for incorporating reused materials into a project. Points are based on the reuse percentage per material type, by quantity, relative to the total amount of the material type. Higher percentages of reused materials and reusing more material types earn more points and encourage teams to prioritize salvaging and reusing structural elements, enclosure components, and other building materials.

Reusing materials

Reusing existing materials, whether for a different purpose than originally intended, for the same purpose, or with modifications for reinstallment can reduce waste and extend their useful life, which provides economic and environmental benefits to owners, contractors, building occupants, and communities. The reuse of existing materials also leads to direct savings on materials and transportation costs. Reusing materials on-site helps reduce the overall embodied carbon on the project because it eliminates the need to produce new materials that generate greenhouse gas emissions. A reduction in construction and demolition waste will decrease the environmental burden on waste management systems.

This credit option rewards projects for incorporating reused elements into the building. The sources for reused materials can be from on-site or gathered off-site from vendors, other projects, salvage yards, donations, and more. Early coordination with contractors is recommended to help identify salvageable materials and align salvage efforts with the construction schedule. Teams should photograph materials before, during, and after salvage, as well as images of off-site-sourced items for verification if requested by reviewers.

Targeted materials found in Table 3 for this credit include carpeting, ceilings, furniture (ancillary and systems), and interior walls. Prioritizing these materials is based on several criteria, including high embodied carbon, toxic impacts in landfills, and significant potential for recovery in existing or emerging salvage and reuse markets, despite current low participation levels. Teams should refer to the EPA WARM tool[8] and the Build Reuse association for more on the impacts of targeted materials in LEED.[9] These targeted materials receive a 2× multiplier compared to other reused materials. However, all forms of reuse are recognized in this credit and are eligible for additional rewards in other LEED credits.

Salvaging materials

Successful salvaging begins with careful planning and requires a thorough audit of the existing materials and structures to identify which materials can be reclaimed. Conduct an early salvage assessment during building design to determine which tools and methods will be most valuable and effective for removal and preservation.

Teams must calculate the percentage of salvaged materials per material type categories listed in Table 3. Calculations must be based on the unit specified per the materials type. Project teams are welcome to include unlisted material type(s) in the table as “project defined other.”

8. U.S. EPA, “Waste Reduction Model (WARM),” n.d., https://www.epa.gov/warm.
9. Build Reuse, n.d., https://www.buildreuse.org/.

Conduct salvage assessment for projects with deconstruction or demolition in scope

For projects with deconstruction or demolition in scope, conduct a salvage assessment to document the quantity of materials that could be salvaged and reused on-site or off-site. A salvage assessment is most useful when conducted before construction activities begin. Conducting a salvage assessment identifies valuable materials and components that can be reclaimed and reused, thus reducing waste and disposal costs. Examples of salvageable materials include clean wood, siding, roofing materials, plumbing, finishes, and lighting fixtures.

Teams should connect with local/regional reuse organizations and/or visit reuse facilities to understand available materials that could be incorporated into the project. Teams can gain a better understanding of local supply and demand for salvaged materials to help target materials for deconstruction (e.g., structural elements like wood, steel, brick, and concrete, and nonstructural components such as doors, windows, cabinetry, and flooring).

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	USGBC calculator.
	Option 1. Building Reuse	All	Evidence of the existing building structure, including floor decking, roof decking, and enclosure that is to be demolished or renovated (e.g., contract documents); identifying any portions that have been deemed structurally unsound or hazardous and identifying portions/areas that are to be kept or maintained for reuse.
	Option 2. Materials Reuse		For projects with deconstruction or demolition in scope, the salvage assessment conducted prior to deconstruction or demolition activities that identifies materials that can be retained on-site or diverted off-site to reuse markets (e.g., contract documents or photographs and a narrative).

REFERENCED STANDARDS

- U.S. EPA WARM (https://www.epa.gov/warm)
- Build Reuse Association (https://www.buildreuse.org)

FURTHER EXPLANATION

SALVAGE ASSESSMENT

Salvage assessment refers to an assessment performed prior to deconstruction or demolition activities to identify materials that can be retained on site or salvaged and diverted off site to reuse markets. Note that salvaged materials diverted to off-site uses/markets count toward *MRc5: Construction and Demolition Waste Diversion*.

Successful salvage assessments involve salvage experts, such as deconstruction contractors, that know how to safely remove building materials and are knowledgeable of which materials have salvage value. Identifying possible destinations for salvaged material during the salvage assessment can also increase successful salvage and reuse. Further, contacting salvage retailers and verifying that they will accept or purchase a particular salvaged material will ultimately save time and resources in the long term. Note that some salvage organizations offer pickup services depending on quantity, distance, and value of salvaged material and products.

Online marketplaces for salvaged material can be a viable alternative to brick-and-mortar salvage retailers. They also present the opportunity to sell salvaged materials in advance of actual removal, thereby avoiding some transportation impacts and solidifying a diversion strategy early in the project.

- Examples of salvage assessments can be found in the section "Referenced Standards."
- Local salvage locations and additional reuse resources can be found on the Reuse Ecosystem Map.[10]

SALVAGED MATERIALS

Salvaged materials refer to materials, components, or assemblies recovered from a structure for the purpose of being reused.

- Materials sent off site for reuse, such as selling salvaged materials to a salvage retailer or donating materials to a non-profit that accepts salvaged building materials, are counted toward meeting *MRc5: Construction and Demolition Waste Diversion*.
- Materials salvaged and reused on site count toward meeting *MRc1: Building and Materials Reuse*.

EXCLUDED FROM CREDIT—STRUCTURALLY UNSOUND OR HAZARDOUS

Some portions of existing buildings can be unsafe and impractical to preserve, including structural integrity concerns, fire damage, and/or past manufacturing facilities that included hazardous substances. For purposes of credit calculations, exclude portions of the building deemed structurally unsound or hazardous.

In determining if areas of the building are unsound or hazardous and therefore excluded from calculations, consult with a structural engineer, local police and fire departments, and/or hazardous material professionals.

While hazardous materials (e.g., asbestos or lead-based paint) must be properly handled/disposed, their presence in the building does not automatically mean the materials cannot be salvaged or reused. Check with local laws and professionals to verify if materials can be recovered.

10. All For Reuse, "Reuse Ecosystem Map," accessed September 9. 2025, https://www.allforreuse.org/ecosystem-map.

RESOURCES

Example salvage assessments are available through the City of Seattle,[11] the City of Shoreline,[12] and the City of Edmonds.[13]

Salvage/deconstruction service providers and location of salvage retailers can be found within the All for Reuse Ecosystem Map.[14]

11. City of Seattle, https://usgbc.sharepoint.com/:b:/s/LEEDTechDevTeam/EeNqB47Nn1hLntwb-pnqEYgBFFAKLTRbCQojcBQ3srXLKA?e=oqFBfG.
12. City of Shoreline, "Waste Diversion Plan," accessed September 9, 2025, https://www.shorelinewa.gov/home/showpublisheddocument/18696/636818681163130000.
13. City of Edmonds, "Salvage Assessment," accessed September 9, 2025, https://cdnsm5-hosted.civiclive.com/UserFiles/Servers/Server_16494932/File/Services/Permits%20Development/General%20Permit%20Assistance/Informational%20Handouts/B90-Salvage_Assessment.pdf.
14. All For Reuse, "Reuse Ecosystem Map," accessed September 19, 2025, https://www.allforreuse.org/ecosystem-map.

MATERIALS AND RESOURCES CREDIT

Reduce Embodied Carbon

MRc2
New Construction (1–6 points): A 20% reduction in embodied carbon is required for LEED Platinum projects Core and Shell (1–8 points): A 20% reduction in embodied carbon is required for LEED Platinum projects

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To track and reduce embodied carbon of major structural, enclosure, and hardscape materials from construction processes on new construction and renovation projects.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1–6**
Option 1. Whole-Building Life-Cycle Assessment	1–6
AND/OR	
Option 2. Environmental Product Declaration (EPD) Analysis	1–3
Path 1. Project-Average Approach	1–3
OR	
Path 2. Materials-Type Approach	1–2
AND/OR	
Option 3. Track Carbon Emissions from Construction Activities	1–2
Core and Shell	**1–8**
Option 1. Whole-Building Life-Cycle Assessment	1–7
AND/OR	
Option 2. Environmental Product Declaration (EPD) Analysis	1–4
Path 1. Project-Average Approach	1–4
OR	
Path 2. Materials-Type Approach	1–3
AND/OR	

(continued)

ACHIEVEMENT PATHWAYS	POINTS
Option 3. Track Carbon Emissions from Construction Activities	1–2
Path 1. Track All Fuel and Utility Usage for Contractor Jobsite Operations	1
OR	
Path 2. Track All Fuel and Utility Usage for Contractor and Subcontractor Jobsite Operations	2

Quantify the reduction of embodied carbon of major structure, enclosure, and hardscape materials. All ancillary structures, such as parking structures or outbuildings within the LEED project boundary, must be included in the calculations.

Both baseline projects and final results may use as-designed or as-constructed final quantities, provided that quantities did not change more than 10% from design through construction. Results must be based on embodied carbon intensities of materials as constructed.

Points are awarded according to Table 1 for reductions in embodied carbon. Projects may earn up to 6 points total.

TABLE 1. Points for Embodied Carbon Reductions in Options 1 and 2 for New Construction Projects

	Option 1. Whole-building Life-cycle Assessment	AND/OR	Option 2. EPD Analysis		
			Path 1. Project-Average Approach	OR	Path 2. Materials-type Approach
Meet baseline or industry average	2		1		Three material categories for 1 point OR Five or more material categories for 2 points
10% reduction in GWP	3		–		–
20% reduction in GWP	4		2		–
30% reduction in GWP	5		–		–
40%+ reduction in GWP	6		3		–

NOTE: Meeting the baseline or industry average in Table 1 can achieve no more than 2 points.

TABLE 2. Points for Embodied Carbon Reductions in Options 1 and 2 for Core and Shell Projects

	Option 1. Whole-building Life-cycle Assessment	AND/OR	Option 2. EPD Analysis		
			Path 1. Project-Average Approach	OR	Path 2. Materials-type Approach
Meet baseline or industry average	2		1		Two material categories for 1 point OR Four material categories for 2 points OR Six or more material categories for 3 points

(continued)

TABLE 2. Points for Embodied Carbon Reductions in Options 1 and 2 for Core and Shell Projects (*continued*)

	Option 1. Whole-building Life-cycle Assessment	AND/OR	Option 2. EPD Analysis		
			Path 1. Project-Average Approach	OR	Path 2. Materials-type Approach
10% reduction in GWP	3		–		–
20% reduction in GWP	4		2		–
30% reduction in GWP	5		–		–
40%+ reduction in GWP	6		3		–
50%+ reduction in GWP	7		4		–

NOTE: Meeting the baseline or industry average in Table 2 can achieve no more than 2 points.

OPTION 1. WHOLE-BUILDING LIFE-CYCLE ASSESSMENT (1–6 POINTS NEW CONSTRUCTION, 1–7 POINTS CORE AND SHELL)

Conduct a cradle-to-grave (modules A–C, excluding operating energy and operating water-related energy) whole-building life-cycle assessment (WBLCA) of the project's structure, enclosure, and hardscape materials. Compare results to a baseline developed for the project and earn points according to Table 1.

Include results for the following impact categories in the WBLCA report:

- GWP (greenhouse gases), in kg CO_2e
- Depletion of the stratospheric ozone layer, in kg CFC-11e
- Acidification of land and water sources, in moles H^+ or kg SO_2e
- Eutrophication, in kg nitrogen equivalent or kg phosphate equivalent
- Formation of tropospheric ozone, in kg NOx, kg O_3e, or kg ethene
- Depletion of nonrenewable energy resources, in megajoules (MJ) using CML/depletion of fossil fuels in TRACI

AND/OR

OPTION 2. ENVIRONMENTAL PRODUCT DECLARATION (EPD) ANALYSIS (1–3 POINTS NEW CONSTRUCTION, 1–4 POINTS CORE AND SHELL)

PATH 1. PROJECT-AVERAGE APPROACH (1–3 POINTS)

Earn points for reducing embodied carbon of the project based on EPD data for the procured materials compared to industry average values. Points are awarded according to Table 1 for the whole-project weighted average of applicable material categories. Industry averages for material categories are defined by the U.S. EPA, the most recent Carbon Leadership Forum (CLF) Material Baselines report, or similarly robust and widely recognized publications, and industry-wide EPDs applicable to the project region.

Projects must track the GWP/unit of the materials installed and reconcile the design-phase embodied carbon intensities if materials or GWP values have changed. The reconciliation of material quantities is not necessary unless quantities have changed more than 10% from design through construction. Projects must use project-specific material quantities and identify product-specific or facility-specific Type III EPDs for covered materials to demonstrate reductions. Biogenic carbon may only be included for calculations that include C-stage emissions.

OR

PATH 2. MATERIALS-TYPE APPROACH (1–2 POINTS NEW CONSTRUCTION, 1–3 POINTS CORE AND SHELL)

Earn points according to Table 1 by demonstrating that structural, enclosure, and hardscape materials for targeted material types have lower embodied carbon impacts than industry benchmarks as demonstrated by product-specific Type III EPDs. Track the GWP per unit of the materials installed, reconciling the design-phase embodied carbon intensities if materials or GWP values have changed. The reconciliation of material quantities is not necessary unless quantities have changed more than 10% from design through construction.

A weighted average approach can be used to calculate average embodied carbon intensity values within a product category.

Industry averages for embodied carbon intensity values are defined by the U.S. EPA, the most recent CLF Material Baselines report, or similarly robust and widely recognized publications and industry-wide EPDs applicable to the project region.

AND/OR

OPTION 3. TRACK CARBON EMISSIONS FROM CONSTRUCTION ACTIVITIES (1–2 POINTS)

Earn points for tracking carbon emissions during construction activities according to Table 3.

TABLE 3. Points for Tracking Emissions During Construction Activities

Pathway	Type of Construction-Phase Emissions to Track	LCA Modules	Points
Path 1	Track all fuel and utility usage for contractor jobsite operations	A5	1
Path 2	Track all fuel and utility usage for contractor and subcontractor jobsite operations	A5	2

MR

REQUIREMENTS EXPLAINED

The goal of *MRc2: Reduce Embodied Carbon* is to implement strategies that reduce embodied carbon through the various stages of a project, from early design development through construction activities and procurement. There are several pathways to reduce embodied carbon. Each way can be tailored to different project stages and objectives.

The intention of this credit is to provide flexibility but increase carbon literacy through incentivizing completing a WBLCA alongside product research through EPD analysis. This allows project teams to earn a maximum of 6 points; however, meeting the baseline or industry average can achieve no more than 2 points in this credit.

For example, if a project team aims to achieve the maximum 6 points for this credit but are only able to achieve a 20% reduction in GWP as identified in their WBLCA, they can take either Path 1 or Path 2 in Option 2 to earn the additional 2 points. Path 1 and Path 2 are not allowed to be combined. Project teams could also pursue Option 3 and track construction phase emissions to earn points in action or exclusion of Option 1 or Option 2.

OPTION 1. WHOLE-BUILDING LIFE-CYCLE ASSESSMENT

This pathway rewards projects seeking carbon reductions in the early design phases (Schematic Design and Design Development). The most significant decisions can be made early on in projects to reduce embodied carbon rather than after the design is complete when material substitutions may not be allowed or become cost prohibitive. When using a fully integrated and collaborative design process, decisions can collectively be made by the owner, designers, and contractors to substantially cut embodied carbon emissions.

A WBLCA allows projects to demonstrate reductions in life-cycle stages from raw material extraction and manufacturing, through construction, demolition, and disposal, and provides owners and design teams with a better understanding of the full life-cycle impacts of design decisions. WBLCAs assess all stages of a building's life-cycle in their calculation, including the product stage (Modules A1–A3), construction stage (Modules A4–A5), use stage (Modules B1–B7), end-of-life stage (Modules C1–C4), and the benefits and loads beyond the system boundary stage (Modules D).[15]

The baseline and proposed buildings must be of comparable size, function, orientation, and operating energy performance as defined in *EAp2: Minimum Energy Efficiency*. The service life of the baseline and proposed buildings must be the same and at least 60 years to fully account for maintenance and replacement. Baseline assumptions must be based on standard design and material selection for the project location and building type. Use the same life-cycle assessment software tools and data sets to evaluate both the baseline building and the proposed building and report all listed impact categories. Data sets must be compliant with *ISO 14044*.[16]

When developing a baseline model (also known as reference building), use recommended modeling software to generate a model that is comparable in size, function, orientation, building geometry, and structural and thermal performance. If a team iterates early in design and makes design changes to create a lower embodied carbon design, they may use their early design iteration as a baseline given that it aligns with the comparative requirements listed above. A team may also make a copy of their proposed design that includes low embodied carbon implementation and replace materials with the regional commonly used materials. For further guidance on developing a baseline model, see the following resources:

15. Carbon Leadership Forum (CLF), "Measuring Embodied Carbon" (Figure 1), 2023, https://carbonleadershipforum.org/toolkit-2-measuring/.
16. ISO, "Environmental management—Life cycle assessment—Principles and framework," ISO 14040, 2006, https://www.iso.org/standard/37456.html.

- Whole Building Life-cycle Assessment: Reference Building Structure and Strategies[17]
- National Guidelines for whole-building life-cycle assessment[18]
- City of Vancouver Embodied Carbon Guidelines[19]

OPTION 2. EPD ANALYSIS

EPDs are a standardized way of communicating the environmental impacts associated with a product's raw material extraction; energy use; chemical makeup; waste generation; and air, soil and water emissions, among other end points. Project teams who analyze EPDs can more accurately compare and evaluate similar products and improve their decisions when selecting materials during design.

PATH 1. PROJECT-AVERAGE APPROACH

Projects can earn points by reducing embodied carbon, which is measured by comparing the project's total embodied carbon of procured materials to industry average values through an EC3 comparison or similar tool. For industry average values, project teams should use EPA values, the most recent CLF Material Baselines Report, or similar. If no values are found in these given sources, regionally appropriate industry-wide EPD's are acceptable. Project teams must provide a narrative comparing the impact between the procured materials and the industry averages as well as a justification of the sources used for comparison if they vary from those recommended.

PATH 2. MATERIAL-TYPE APPROACH

This path is also meant to align with state and federal procurement policies, specifically the *Federal Buy Clean Act*,[20] a U.S. federal government initiative that prioritizes the use of locally made, lower-carbon construction materials in federal procurement and federally funded projects to allow flexibility in choosing products to reduce embodied carbon. Project teams will compare their structure, enclosure, and hardscape targeted material types to product-specific Type III EPDs to demonstrate that the products they have procured have a lower embodied carbon intensity than typical across the industry.

Project teams can research products with EPDs in online databases (Table 4). In addition to these databases, many manufacturers publish EPDs directly on their website. Industry-specific associations may also have resources available for searching EPDs related to their trade.

17. WBLCA Guide Special Project Working Group of the Sustainability Committee, *Whole Building Life Cycle Assessment: Reference Building Structure and Strategies*, (Reston, VA: ASCE, 2018), https://sp360.asce.org/personifyebusiness/Merchandise/Product-Details/productId/239605051.
18. Matthew Bowick, Jennifer O'Connor, James Salazar, Jamie Meil, and Rob Cooney, "National Guidelines for Whole-Building Life Cycle Assessment," National Research Council Canada, 2022, https://doi.org/10.4224/40002740.
19. City of Vancouver, "Embodied Carbon Guidelines," October 2023, https://vancouver.ca/files/cov/embodied-carbon-guidelines.pdf.
20. Office of the Federal Chief Sustainability Officer, Council on Environmental Quality, "Federal Buy Clean Initiative," n.d., accessed April 2, 2025, https://www.sustainability.gov/archive/biden46/buyclean/index.html..

TABLE 4. Resources for Finding EPDs

Material and Product Databases
Embodied Carbon in Construction Calculator (EC3)[21]
LCA Digital Commons[22]
Quartz[23]
Athena[24]
EPD Library[25]
The ICE Database[26]
Open LCA Nexus[27]
The International EPD System[28]
UL Spot[29]
Environmental Product Declaration Australasia[30]
Institut Bauen and Umwelt e.V.[31]
Sustainable Minds[32]

OPTION 3. TRACK CARBON EMISSIONS FROM CONSTRUCTION ACTIVITIES

Tracking construction emissions is an emerging practice among leading contractors that aims to reduce environmental impact and achieve sustainability goals. This option provides incentives for projects to begin tracking and reporting carbon emissions. It is not required for projects to show reduction of carbon emissions to earn points.

Option 3 aligns with aspects of the Sustainable Construction Leaders Contractor Commitment,[33] where the guidelines are intended to set a sustainability benchmark specific to the construction industry. The guidelines also encourage companies to procure sustainable materials through the submittal process. Contractors are encouraged to follow the *AGC Playbook on Decarbonization and*

21. Building Transparency, "Embodied Carbon in Construction Calculator," n.d., https://buildingtransparency.org/auth/login.
22. Federal LCA Commons, "LCA Digital Commons," n.d., https://www.lcacommons.gov/.
23. Pharos, "Quartz Countertops," 2025, https://pharos.habitablefuture.org/common-products.
24. Athena Sustainable Materials Institute, "Athena Research Teams Follow Common Building Materials from Cradle-To-Grave to Calculate the Environmental Effects at Each Stage in the Product's Life Cycle," 2025, https://www.athenasmi.org/our-software-data/lca-databases/.
25. The International EPD System, "Search the EPD Library," n.d., https://environdec.com/library.
26. Circular Ecology, "Embodied Carbon–The ICE Database," 2025, https://circularecology.com/embodied-carbon-footprint-database.html.
27. Open LCA Nexus, "openLCA Nexus," n.d., https://nexus.openlca.org/.
28. The International EPD System, EPD International AB, n.d., https://portal.environdec.com/.
29. UL Spot, UL LLC, https://spot.ul.com/.
30. The EPD Portal, The International EPD System, "Register and Manage Your EDPS Online," n.d., https://epd-australasia.com/.
31. Institut Bauen and Umwelt e.V., http://ibu-epd.com/.
32. Sustainable Minds, https://www.sustainableminds.com/.
33. Building Green, "Contractor's Commitment to Sustainable Building Practices," 2021, https://www.buildinggreen.com/sites/default/files/Contractors-Committment-Sustainability.pdf.

Carbon Reporting for the Construction Industry,[34] a document created by contractors for contractors to help address carbon emissions for the projects they build.

PATH 1. TRACK ALL FUEL AND UTILITY USAGE FOR CONTRACTOR JOBSITE OPERATIONS

Fuel usage tracking may include the type and amount of fuel consumed by construction equipment, vehicles, and machinery. Utility usage tracking may include electricity and water consumption on-site for construction activities.

PATH 2. TRACK ALL FUEL AND UTILITY USAGE FOR CONTRACTOR AND SUBCONTRACTOR JOBSITE OPERATIONS

See Path 1 for fuel and utility usage for contractor jobsite operations. Path 2 additionally requires that subcontractors also track their fuel and utility usage, including equipment operation and specific tasks. This can include fuel used to transport materials to or from the jobsite.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	USGBC calculator.
	Option 1. Whole-Building Life-Cycle Assessment AND Option 2. EPD Analysis	All	Confirmation of approach to determine actual material quantities for the project: • as-constructed quantities used, or
			When EPDs are not available as a baseline data source, justification of the baselines used, demonstrating that the required hierarchy of data has been followed.
			Description of strategies that resulted in reductions, if applicable.
	Option 1. Whole-Building Life-Cycle Assessment	N/A	Baseline LCA report and raw WBLCA data output file (e.g., CSV file).
			Proposed Design LCA report and raw WBLCA data output file.
			Description of LCA assumptions, scope, and process.
	Option 2. EPD Analysis	Path 1. Project-Average Approach	List and sum of the industry average embodied carbon emissions values, by material, used in the analysis.
			List and sum of the project-specific embodied carbon emissions values, by material, used in the analysis.
		Path 2. Materials-Type Approach	List of the project's embodied carbon emissions values, by material, from the product-specific Type III EPDs.
	Option 3. Track Carbon Emissions from Construction Activities	All	Description of tracking process implemented and how the carbon emissions were monitored and logged.
			Evidence of emissions tracking (such as monthly tracking logs, photos or log of shipments inbound/outbound from jobsite, and associated emissions estimates).

34. Association of General Contractors of America, *AGC Playbook on Decarbonization & Carbon Reporting*, 2024, https://www.agc.org/climate-change-playbook.

REFERENCED STANDARDS

- None

FURTHER EXPLANATION

MATERIALS TO INCLUDE IN ANALYSIS

- Include the complete building envelope and structural elements, including the material components of footings and foundations, structural wall assembly (from cladding to interior finishes), structural floors and ceilings (not including finishes), and roof assemblies (e.g., concrete slabs, structural timber and wood framing, insulation, coverboard, membranes).
- Exclude electrical and mechanical equipment and controls, plumbing fixtures, fire detection and alarm system fixtures, elevators, and conveying systems.
- Include hardscape and site development (e.g., concrete and asphalt paving and pathways, retaining walls).
- Include parking structures and parking lots.
- Additional building elements, such as interior nonstructural walls or finishes, may be included at the discretion of the project team.

AS-CONSTRUCTED VERSUS AS-DESIGNED

Project teams shall update their analysis for both the proposed and baseline for Options 1 or 2 if the material quantities change more than 10% from design to construction in order to accurately represent what was installed on the project. If materials are substituted and their embodied carbon intensity changes, project teams must update this information in the analysis.

BIOGENIC CARBON

Project teams must account for biogenic carbon storage and sequestration separately when performing embodied carbon analysis for this credit and only if the analysis includes life-cycle stage C.

OPTION 1. WHOLE BUILDING LIFE-CYCLE ASSESSMENT

Creating a Baseline

Collect information needed to perform a life-cycle assessment (LCA) of the structure, enclosure, and hardscape of the building. Follow the standard process associated with performing a typical WBLCA:

- Define goal and scope of assessment
- Collect information about materials and scenarios
- Perform calculations for impacts using reliable LCA assessment tools
- Understand and interpret results
- Document process and produce detailed assessment reports

Ensure that the scope of the analysis is a cradle-to-grave assessment that includes environmental impacts associated with the life-cycle stages A–C for the building structure, enclosure, and hardscape.

If the design team wants to use a material, product, or assembly that is not in the selected LCA tool's dataset, the results of a critically reviewed LCA or a verified EPD can be used, provided the results cover the required full set of impact indicators for that component. The material, product, or assembly in the model must then be removed and the impact measures for the replacement added as a side calculation, taking account of all related ancillary product use. Any such additions must be documented and the documentation included in the submittal. Include the rationale for the change and the source of the replacement impact measures.

The system boundary of the analysis must include a cradle-to-grave scope (modules A–C). However, some gaps in submodules may exist due to the materials or dataset chosen and design optimizations attempted for the project. Gaps in submodules are allowed so long as the system boundary in total encompasses a cradle-to-grave assessment. The required modules for a compliant whole building life-cycle analysis include life-cycle stages A–C.

If the LEED project boundary includes detached ancillary structures, such as parking structures, parking lots, or outbuildings, those structures must be included in the WBLCA calculations.

For projects demonstrating impact reductions compared to a baseline, the LCA software or tool used for the baseline and proposed design must be the same and contain the same modules and impact categories evaluated. Note that LCA software or tools must have *ISO-14044*-compliant datasets and conform to *ISO 21931-2017* and/or *EN 15978:2011,* and their data must meet the requirements of *ISO 21930-2017* and *EN 15804*. Typically, the software tool providers will document that they meet these criteria in the LCA output report.

The baseline and proposed buildings must be of comparable size, function, orientation, and operating energy performance. The service life of the baseline and proposed buildings must be the same; their service life should be at least 60 years to fully account for maintenance and replacement. Use the same LCA software tools and datasets to evaluate both the baseline building and the proposed building, and report all listed impact categories. Datasets must be compliant with *ISO 14044*.

While projects are only required to document reduction for GWP, they must also include the other LCA impact areas from the credit requirements in the WBLCA report.

FUNCTIONAL EQUIVALENCE

Project teams must demonstrate structural functional equivalence between the baseline and proposed building. Functional equivalence of a building design includes loading, life safety, energy performance, and/or serviceability requirements (deflection, vibration, durability, noise transmission).

REUSED AND SALVAGED MATERIALS IN WBLCA

For materials that are reused or salvaged when performing a WBLCA, project teams shall mark materials within an assembly as reused or salvaged in their software of choice and zero out the impacts of the manufacturing life-cycle stages (A1–A3) for the reused materials. Other impacts from reused materials (A3, A4, and B–C stages) shall be included in the analysis as appropriate.

Note: For both Path 1 and 2, baseline and proposed quantities must remain the same for reused materials.

RESOLVING DATA GAPS WITH EPDS

If life-cycle inventory data is not available for certain products, project teams can incorporate product EPDs into their LCA tools if the following criteria have been met:

- The EPD has not expired.
- EPD scenarios are representative of contemporary technologies and/or practice and are relevant to the project location.
- The EPD reports all indicators and system boundary information modules required by the WBLCA tool.
- The EPD characterizes the impact categories reported according to the same LCA methodology as the WBLCA tool.
- The EPD can be applied to the study period of the assessment.
- The EPD clearly indicates which product (including manufacturer and product name) or geographical region it reflects in comparison to the industry-wide weighted average results of a material or fuel already available in the tool.

WBLCA RESOURCES AND GUIDES

Guidance and examples for WBLCAs can be found through the City of Vancouver,[35] the Structural Engineering Institute,[36] the Government of Canada,[37] and RMI reports.[38]

OPTION 2. EPD ANALYSIS

The following guidelines are applicable to both pathways in this credit.

Bill of Materials

Project teams shall compile the material quantities of their structure, enclosure, and hardscape from the most up-to-date project documents at the time of analysis, updating them if they have changed by more than 10% from design to construction. While project teams may use other methods, it is highly recommended that a project team create a BOM for tracking purposes. Project teams may use the BOM format from the Materials and Resources prerequisite, *MRp2: Quantify and Assess Embodied Carbon*.

Comparison to Industry Average Data

Project teams will compare their analysis with industry average values. The quality and applicability of industry average values can vary, so project teams shall follow this hierarchy of quality to determine the industry average values for a product type of category:

- U.S. Environmental Protection Agency (EPA)
- The most recent Carbon Leadership Forum (CLF) Material Baselines report (or similarly robust and widely recognized publications)
- Industry-wide EPDs applicable to the project region

35. City of Vancouver, "City of Vancouver Embodied Carbon Guidelines," 2023, https://vancouver.ca/files/cov/embodied-carbon-guidelines.pdf.
36. Structural Engineering Institute (SEI), "Design Guidance for Reducing Embodied Carbon in Structural Systems," accessed September 9, 2025, https://se2050.org/resources-overview/structural-materials/lean-design-guidance/.
37. Government of Canada, "National Whole-Building Life Cycle Assessment Practitioner's Guide," accessed September 9, 2025, https://nrc-publications.canada.ca/eng/view/object/?id=533906ca-65eb-4118-865d-855030d91ef2.
38. Tracy Huynh et al., "Driving Action on Embodied Carbon in Buildings," RMI and USGBC, 2023.

Eligible Environmental Product Declarations

Eligible product-specific Type III EPDs are also called manufacturer-specific EPDs, which includes manufacturer-average and/or facility-specific data that represent one manufacturer's product(s). When using product-specific Type III EPDs, compliant documents must include all of the following in the report:

- The covered product name(s), product type(s), and description
- Each document shall have a unique ID number
- The declaration holder or producer with contact information (typically the manufacturer)
- The program operator (entity that creates and registers the EPD)
- LCA independent verifier's name and statement of external verification
  - External verification requires the person conducting the third-party verification of the underlying LCA data to be independent and outside of the organization (as per *ISO 14025* and *EN 15804* or *ISO 21930*) in which the EPD is developed.
  - Third-party LCA/EPD verification must be done in conformance with or certified to *ISO 17065*.
- Conformance to *ISO 14025* and either *ISO 21930* or *EN 15804* (Type III EPD)
- Functional unit
- Manufacturing location(s) indicated in document and global regions covered
- URL link to official publication of the document or link to the website or online source where the EPD can be found
- Report results from all relevant impact categories as outlined in the *MRc4: Building Product Selection and Procurement,* Additional Guidance for EPDs
- Validity period with expiration date
- Indicate type of LCA software used
- Reference for post-consumer recycled (PCR) content is cited
- Confirmation of *ISO 14044* conformance

OPTION 2, PATH 1. PROJECT AVERAGE APPROACH

Reporting and Documentation

This path may be achieved by either compiling a BOM of all major materials or by utilizing software tools, such as the Building Transparency's Embodied Carbon in Construction Calculator (EC3) tool.

In the analysis, project teams shall include the GWP of each material from its product-specific or facility-specific Type III EPDs, along with its quantity, and develop a whole-project weighted average to compare against industry average to earn points based on the reduction from baseline or industry average achieved.

If there is no EPD available for a specific material, project teams shall utilize an industry average value for both baseline and proposed scenarios. The average data shall come from a source listed above and will gain no reductions from industry average for that given material.

If utilizing a tool like EC3, project teams can actively compare EPD data of materials selected to average product data. Project teams may compile a report of their findings using this tool and show reductions to meet the credit requirements.

OPTION 2, PATH 2. MATERIALS TYPE

Project teams shall generate an embodied carbon intensity per product category to show reductions.

Project teams shall include the GWP per unit of each material from product-specific Type III EPDs to compare against industry average to earn points based on the reduction from baseline or industry average achieved. Project teams shall utilize industry averages as found in EPA sources, the most recent CLF Material Baselines report (or similarly robust and widely recognized publications), and industry-wide EPDs applicable to the project region. Teams can either ensure that each material in the product category meets the defined limits or use a weighted average approach by square footage to demonstrate compliance for the whole category.

OPTION 3. CONSTRUCTION PHASE TRACKING

Reporting and Documentation

Project teams will compile the types and amounts of fuel consumed by jobsite operations performed by the contractor for equipment, vehicles, and machinery. If pursuing Path 2, include, quantify, and calculate all the subcontractors' operations as well. Use a tracking sheet to calculate the anticipated emissions by creating a line item for each piece of equipment used, the associated emissions factor, type and amount of fuel, as well as the duration of the noted operation. Project teams shall utilize emissions factors for given equipment that is regionally appropriate. If defaults are used for emissions factors, describe where the defaults came from and why they are relevant to the project. Project teams shall include all relevant equipment, vehicles, and machinery associated with the construction of the building when quantifying the jobsite emissions.

Low-Emitting Materials

MRc3
New Construction (1–2 points)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To reduce concentrations of chemical contaminants that can damage air quality and the environment. To protect human health and the comfort of installers and building occupants.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1–2**
Low-Emitting Material Criteria	1–2
Core and Shell	**1**
Low-Emitting Material Criteria	1

Specify and install permanently installed products, paints, coatings, adhesives, sealants, flooring, walls, ceilings, insulation, furniture, and/or composite wood products that meet the low-emitting criteria. Points are awarded according to Table 1.

TABLE 1. Thresholds for Low-Emitting Materials for New Construction Projects

Pathway	Product Categories	Threshold	Points
Path 1	Achieve all three categories: • Paints and coatings • Flooring • Ceilings	>90% of all products in each product category	1
Path 2	Achieve Path 1, plus any two of these additional categories: • Adhesives and sealants • Walls • Insulation • Composite wood	>80% of each additional product category	2
Path 3	Achieve Path 1 plus the furniture category	>80% of the furniture product category	2

TABLE 2. Thresholds for Low-Emitting Materials for Core and Shell Projects

Product Categories	Threshold	Points
Achieve any three categories: • Paints and coatings • Flooring • Ceilings • Adhesives and sealants • Walls • Insulation • Composite wood	>90% of all products in each product category	1

Core and Shell only

Products to be specified and installed by tenants may be excluded. Communicate the base building's product list in the tenant guidelines.

PRODUCT CATEGORIES

The following products and materials are not applicable to the low-emitting materials product categories: equipment related to fire suppression, HVAC (including ductwork), plumbing, electrical, conveying and communications systems, poured concrete, structural framing, structural insulated panels (SIPs), and water-resistive barriers (material installed on a substrate to prevent bulk water intrusion).

Paints and coatings

- Paints and coatings, by volume, cost, or surface area, must meet the volatile organic compounds (VOC) emissions evaluation criteria.
- The paints and coatings product category includes all interior paints and coatings wet-applied on-site.
- Exclude foamed-in-place and sprayed insulation (include in insulation category).

Adhesives and sealants

- Adhesives and sealants, by volume or cost, must meet the VOC emissions evaluation criteria.
- The adhesives and sealants product category includes all interior adhesives and sealants wet-applied on-site, including those used to install air or vapor barrier membranes and floor-setting materials.

Flooring

- Nonstructural flooring materials, by surface area or cost, must meet the VOC emissions evaluation criteria.
- The flooring product category includes all types of hard and soft surface flooring finishes (e.g., carpet, ceramic tile, vinyl, rubber, engineered wood, solid wood, stone, or laminate), raised flooring systems, entryway ("walk-off") systems, area rugs, wood subflooring, underlayments, sandwich panels, and air barrier membranes and vapor barrier/vapor retarder membranes (if used inside an air barrier membrane).
- Exclude poured concrete, composite wood subflooring (include in the composite wood category, if applicable), and wet-applied products applied on the floor.

Walls

- Nonstructural wall materials, by surface area or cost, must meet the VOC emissions evaluation criteria.
- The walls product category includes all finish wall treatments (e.g., wall coverings or wall tile), finish carpentry (e.g., millwork, paneling, railings, or trim/moldings), gypsum wallboard, wall base/skirting, interior and exterior doors, nonstructural wall framing, and nonstructural sandwich panels.
- Exclude wet-applied products applied on the wall, case goods, cabinetry (included in the furniture category), countertops (included in the furniture category), bathroom accessories, door hardware, and curtain wall and storefront systems.

Ceilings

- Nonstructural ceiling materials, by surface area or cost, must meet the VOC emissions evaluation criteria.
- The ceilings product category includes all types of ceiling finishes (e.g., ceiling panels and ceiling tile), suspension grids, surface ceiling structures (such as gypsum wallboard or plaster), suspended systems (including canopies and clouds), and nonstructural sandwich panels.
- Exclude wet-applied products applied on the ceiling and corrugated metal decking.

Insulation

- Insulation products, by surface area or cost, must meet the VOC emissions evaluation criteria.
- The insulation product category includes all thermal and acoustic boards, batts (faced and unfaced), rolls, blankets, sound attenuation fire blankets, and foamed-in-place, loose-fill, blown, and sprayed insulation.
- Exclude insulation installed outside an air barrier membrane.

Furniture

- Furniture in the project scope of work, by cost, area, or number of units, must meet the furniture emissions evaluation criteria or VOC emissions evaluation criteria.
- The furniture product category includes all permanently installed office furniture, cubicles/systems furniture, seating, desks, tables, filing/storage, specialty items, beds, case goods, casework, countertops, movable/demountable partitions, bathroom/toilet partitions, shelving, lockers, retail fixtures (including slatwall), window treatments, and furnishing items (such as nonfixed area rugs, cubicle curtains, and mattresses) purchased for the project.
- A custom item in the furniture category is considered to meet the low-emitting criteria if all components of the finished piece, applied on-site or off-site, are declared under the furniture category and meet the VOC emissions evaluation criteria. Alternatively, a custom piece meets the criteria if the finished piece meets the furniture emissions evaluation or VOC emissions evaluation criteria.
- Exclude office and bathroom accessories, art, recreational items (such as game tables), cabinet and drawer hardware, and planters from the credit.

Composite wood

- Composite wood products, by surface area or cost, must meet the formaldehyde emissions evaluation criteria.
- The composite wood product category includes all particleboard, medium density fiberboard (both medium density and thin), hardwood plywood with veneer, composite or combination core, and wood structural panels or structural wood products.

LOW-EMITTING CRITERIA

VOC emissions evaluation criteria

- **Third-party certification:** Product has a qualifying third-party certification, valid at the time of product purchase, that demonstrates testing and compliance according to the *California Department of Public Health (CDPH) Standard Method v1.2-2017* using the private office scenario. Products used in classrooms may be modeled using the schools or private office scenario.

OR

- **Qualified independent laboratory report:** Product has a qualifying laboratory report (or summary) demonstrating the product has been tested no more than three years prior to the product's purchase, according to the *California Department of Public Health (CDPH) Standard Method v1.2-2017*. Products must meet the VOC limits in Table 4-1 of the private office scenario. Products used in classrooms may be modeled using the schools or private office scenario.

OR

- Product is inherently non-emitting, salvaged, or reused.

Furniture emissions evaluation criteria

- Product has a qualifying third-party certification, valid at the time of product purchase, that demonstrates testing according to *ANSI/BIFMA Standard Method M7.1-2011* (R2021) and complies with specific sections of the *ANSI/BIFMA e3-2014* or *e3-2024*, "Furniture Sustainability Standard." Statements of product compliance must include the exposure scenario(s).
- Seating products must be evaluated using the seating scenario. Classroom furniture must be evaluated using the standard school classroom scenario. Other products should be evaluated using the open plan or private office scenario, as appropriate. The open plan scenario is more stringent.

OR

- Product is inherently non-emitting, salvaged, or reused.

FORMALDEHYDE EMISSIONS EVALUATION CRITERIA

Product has a qualifying third-party certification from a California Air Resources Board (CARB)-approved/EPA-recognized third-party certifier (TPC), valid at the time of product purchase, that demonstrates the product is one of the following:

- Certified as ultra-low-emitting formaldehyde (ULEF) product under the *EPA Toxic Substances Control Act*, *Formaldehyde Emission Standards for Composite Wood Products* (*TSCA, Title VI*) (*EPA TSCA Title VI*), or *CARB Airborne Toxic Control Measure* (ATCM).

- Certified as no added formaldehyde resins (NAF) product under *EPA TSCA Title VI* or *CARB ATCM*.
- Wood structural panels manufactured according to *PS 1-09* or *PS 2-10* (or one of the standards considered by *CARB* to be equivalent to *PS 1* or *PS 2*) and labeled bond classification Exposure 1 or Exterior.
- Structural wood product manufactured according to *ASTM D5456* (for structural composite lumber), *ANSI A190.1* (for glued laminated timber), *ASTM D5055* (for I-joists), *ANSI PRG 320* (for cross-laminated timber), or *PS 20-15* (for finger-jointed lumber).

OR

- Product is inherently non-emitting, salvaged, or reused.

REQUIREMENTS EXPLAINED

Installing low-emitting products can significantly reduce the quantity of indoor air contaminants in buildings. Coupled with adequate ventilation and filtration, specifying and installing low-emitting materials is an important strategy toward improving indoor air quality.

This credit is awarded to projects with permanently installed products that meet established low-emitting criteria. There are multiple pathways for earning credit compliance. In New Construction, product categories are grouped to reflect the significant impact emissions can have from walls, ceiling, and flooring due to the large surface area these categories cover, as well as progress in the market. In the Core and Shell and ID+C rating systems, product categories may be attempted individually, based on project scope.

IDENTIFYING AND SPECIFYING LOW-EMITTING PRODUCTS

The easiest way to find products with a VOC emissions evaluation may be to search third-party certification program databases from the qualified TPCs or programs listed on the CDPH website.

Other sources for finding compliant products include online aggregated product databases including Ecomedes,[39] the Sustainable Minds® Transparency Catalog™,[40] Building Ease,[41] and UL SPOT®.[42]

Save certificates for the specified products to ensure that the specified products and the certificates match. Note any certificates expected to expire before the time of purchase. Certification periods that begin after the product's date of purchase do not demonstrate compliance with the installed product. Track progress toward credit achievement using the LEED materials calculator.

There are international third-party programs and low-emitting third-party standards that can be used for this credit. See the Low-emitting Materials resource document on USGBC's website.[43]

39. ecomedes, https://www.ecomedes.com/.
40. Sustainable Minds, "Sustainable Minds® Transparency Catalog™," 2025, https://transparencycatalog.com/.
41. BuildingEase, "Welcome to the BuildingEase Platform," n.d., https://app.buildingease.com/.
42. UL Solutions, "SPOT®–Sustainable Product Database," 2025, https://spot.ul.com/.
43. U.S. Green Building Council, "Low-Emitting Materials," n.d., https://www.usgbc.org/credits/new-construction-core-and-shell-a-new-construction-retail-new-construction-data-38.

CALCULATING PRODUCT CATEGORY ACHIEVEMENT

Project teams can decide how many product categories to attempt to earn points. To achieve 1 point, the project demonstrates meets or exceeds the threshold for each product category. This can be based on cost, surface area, volume, or number of units, depending on the measurement methods available for each product category. Project teams can choose different measurement types to measure progress toward achievement if the measurement method is consistent in each product category. For example, a project could use "surface area" to demonstrate achievement of the Flooring category, "number of units" for the furniture category, and "volume" for the Adhesives and Sealants category.

Project teams are advised to set project-achievable category goals and research, specify, and track low- or non-emitting products in those categories according to the low-emitting criteria appropriate for the products. A targeted approach focusing on specific products, or product categories, is likely to be more manageable and successful than amassing documentation for all products in every category and determining attempted categories post-construction. Additionally, aiming for 100% compliance within a category, when possible, may simplify the process by eliminating the need to track individual units.

This credit will be documented by product category using the LEED materials calculator. Note that this calculator is combined with the *MRc4: Building Product Selection and Procurement* calculator. Teams are encouraged to combine submittal reviews and product vetting with the criteria found in both credits to maximize credit achievement and harmonize product selection, specification, and documentation processes.

Except for the overall product exclusions stated in the Requirements, all permanently installed, nonstructural products—within and inclusive of the project's air barrier membrane—must be in the calculation for the attempted categories. These products are expected to impact indoor air quality and can be tested in alignment with the low-emitting criteria. Products installed in parking garages and basements are to be included, because these spaces are occupied by people, even if intermittently.

Product categories that have no applicable products installed (i.e., they are not in the project scope of work) are not eligible to attempt the category.

PAINTS AND COATINGS

This category applies to products covered in *CSI MasterFormat 09 90 00 Painting and Coating*. Exterior painting, staining, and finish can be excluded. Exterior paint cannot be excluded from the calculation if used indoors. Aerosol products are included. Coatings also includes sealers, which are products applied to either block materials from penetrating into or leaching out of a substrate, to prevent subsequent coatings from being absorbed by the substrate, or to prevent harm to subsequent coatings by materials in the substrate (*SCAQMD Rule 1113*).

VOCs in paints and coatings may be ingredients that are included to enhance product performance and shelf life, added by the contractor, or by-products of the paint drying process. Water-based acrylic latex paints generally have lower VOCs than solvent-based paints. Lime and mineral silicate paints are most likely to be compliant with VOC limits. Paints that are advertised as antimicrobial, recycled, specialty paints (chalk, dry-erase, magnetic), and paints containing alkylphenol ethoxylates or PFAS may have compliant emissions evaluations but introduce additional human and/or environmental hazards not addressed by this credit that the project team may wish to consider.

ADHESIVES AND SEALANTS

Common adhesives and sealants used in construction are defined in *SCAQMD Rule 1168*. An adhesive is any substance used to bond one surface to another surface by attachment. A sealant is any material with adhesive properties that is designed to fill, seal, waterproof, or weatherproof gaps or joints between two surfaces. Sealers are different types of products and are to be categorized as coatings. Aerosol products are also included in this product category for LEED calculation purposes.[44]

FLOORING

In most buildings, the flooring category represents a significant source of indoor emissions due to the large amount of surface area covered in relation to the project. Consider reusing existing floors, where possible. When reuse is not available, solid wood floors, ceramic tiles, cork floors (especially pre-finished without a PVC/vinyl layer), linoleum sheet and tile, are likely to have compliant VOC emissions evaluations, as are many carpet and vinyl flooring products. Evaluate products holistically, like the presence of contaminants or additives like lead in recycled content products, additives included in sealants, and those used for cleaning flooring materials.

Other concerns can relate to the project team's environmental priorities, such as the lack of recovery and circularity options for vinyl products at end of life or potential toxic emissions released during a product's production. These additional multi-attribute considerations may not be addressed by product emissions criteria but are considered in the aligned *MRc4: Building Product Selection and Procurement*.

WALLS

In many buildings, the walls category represents a significant source of indoor emissions from products due to the large surface area. For instance, gypsum wallboard and doors often compose most of the surface area in this category. Look for compliant gypsum wallboards especially those made with natural gypsum or post-consumer recycled content. As it relates to the *MRc5: Construction and Demolition Waste Diversion*, consider how a project can separate unpainted gypsum wallboard cut-offs for manufacturer take-back and recycling.

CEILINGS

Like the walls category, the ceilings category is likely to be strongly influenced by surface area. (See the Walls category for notes on gypsum wallboard.) Acoustic ceiling systems are also a popular material option and are likely to have compliant VOC emissions evaluation. Be sure to include ceiling suspension grids/components, noting that powder-coated metal components are most likely to be compliant.

44. U.S. EPA, "Controlling Pollutants and Sources: Indoor Air Quality Design Tools for Schools," n.d., https://www.epa.gov/iaq-schools/controlling-pollutants-and-sources-indoor-air-quality-design-tools-schools#WallsandCeilingMaterials.

INSULATION

Insulation products with compliant testing typically include both natural and synthetic products. Products include:

- Expanded cork
- Blown-in wood fiber
- Cellulose
- Fiberglass or mineral wool
- Hemp or wood fiber batts and boards
- Unfaced fiberglass batts
- Formaldehyde free mineral wool batts and boards

Plastics and foam insulation products can also meet the emission criteria. Even if products meet the emissions evaluation criteria, they may still include problematic ingredients like formaldehyde and fire retardants. Consider these when selecting products and seek synergies for product optimization with the *MRc4: Building Product Selection and Procurement*.

FURNITURE

The furniture category includes both systems furniture as well as ancillary furniture. Typically, the ability to find compliant furniture will be more available from systems furniture manufacturers as opposed to free standing or custom furniture. A convenient way to find products with a furniture emissions evaluation is to search product databases that list qualified third-party verified programs and reports. (See the Low-Emitting Materials resource document on USGBC's website.)[45]

The Business and Institutional Furniture Manufacturers Association (BIFMA) writes standards for furniture safety, ergonomics, and sustainability. Qualifying furniture products in LEED will meet the *ANSI/BIFMA M7.1-2011 (R2021) Standard Test Method for Determining VOC Emissions from Office Furniture Systems, Components, and Seating*. In addition, products must comply with *ANSI/BIFMA e3-2024e Furniture Sustainability Standard*, Section 7.6.2. Laboratories that conduct the tests must be accredited under *ISO/IEC 17025* for the test methods they use.

COMPOSITE WOOD

The composite wood product category includes all particleboard; medium density fiberboard (both medium density and thin); hardwood plywood with veneer, composite, or combination core; and wood structural panels or structural wood products. Products in this category must meet the formaldehyde emissions evaluation requirements in the rating system.

MR

Formaldehyde emissions evaluation

The composite wood category typically includes products that adhere to classifications and standards defined by leading industry organizations and frameworks, such as the *CARB* standards and ensure low formaldehyde emissions and compliance with stringent air quality requirements. *CARB* maintains a list of composite wood mills that have been approved by third-party certifiers,[46] and this list can be

45. See the Low-Emitting Materials resource document at https://www.usgbc.org/resources/cdph-list-certifications-use-cdph-standard-method-v12.
46. CARB, "CARB Composite Wood Mills," n.d., https://ww2.arb.ca.gov/resources/documents/certified-mills-list-august-25-2025.

used to help find and specify compliant composite wood products. Certificates demonstrating a product is certified as NAF or ULEF for products must be from a CARB-approved Third-Party Certifier.[47] The certification period must cover the date of purchase.

This credit does not refer to the minimum requirements of the *CARB 93120 ATCM* or *EPA TSCA Title VI*. It uses the more stringent requirements for ULEF resins or NAF resins as defined in the *CARB ATCM*. These criteria are some of the strongest available for formaldehyde emissions from composite wood. The certificate must confirm this threshold is met.

The CARB composite wood definition includes wood structural panels, structural composite lumber, glued laminated timber, I-joists, cross-laminated timber, and finger-jointed lumber. These products are subject to other standards. APA–The Engineered Wood Association website[48] can be used to source compliant products.

Goods containing composite wood components like doors with a composite wood core do not belong in the composite wood category. They are subject to the more comprehensive emissions evaluations of other categories.

VOC EMISSIONS EVALUATION CRITERIA

CDPH Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers, v.1.2-2017[49] is also known as the emission testing method for California Specification 01350, which is widely recognized as a leadership standard for its stringent scientific criteria and detailed specificity. It uses the chronic reference exposure levels established by the California Office of Environmental Health Hazard Assessment, which include some of the most stringent criteria in use, although they do not account for non-cancer risks.

Compliant products can come from qualified third-party product certifications or from a qualified independent laboratory. (See USGBC resources for a list of qualifying third-party certifications.)[50]

Qualifying independent laboratory reports that are provided by the manufacturer may be used to demonstrate VOC emissions evaluation, although because they are not third-party verified, the project team must confirm all criteria are reported on the report, including:

- Declaration that the product has been tested according to *CDPH Standard Method v1.2-2017* and complies with the VOC limits in Table 4-1 of the method.
- Total volatile organic compounds (TVOC) results at 14 days measured as specified in *CDPH Standard Method v1.2-2017.*
- Test date (less than three years from date of purchase).

MR

47. CARB, “CARB-Approved Third-Party Certifiers Executive Orders,” 2025, https://ww2.arb.ca.gov/resources/documents/carb-approved-third-party-certifiers-executive-orders.
48. APA, “About Us,” n.d., https://www.apawood.org/about-us.
49. Indoor Air Quality Section Environmental Health Laboratory Branch Division of Environmental and Occupational Disease Control, California Department of Public Health, “Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers, version 1.2,” January 2017, https://www.cdph.ca.gov/programs/ccdphp/deodc/ehlb/iaq/cdph%20document%20library/cdph-iaq_standardmethod_v1_2_2017_ada.pdf.
50. See USGBC resources at https://www.usgbc.org/resources/cdph-list-certifications-use-cdph-standard-method-v12.

- The name of the laboratory that performed the evaluation and documentation (such as accreditation number or certificate with scope of accreditation), thereby demonstrating the accreditation under *ISO/IEC 17025* for the test method.
- The modeling scenario used (must be private office unless the product is installed in a classroom).
- For wet-applied products, the amount of product applied in mass per surface area (during testing).

INHERENTLY NON-EMITTING CRITERIA

Inherently non-emitting products are building materials that, owing to their composition or use in construction, do not emit VOCs and therefore do not require an emissions evaluation. Products that are inherently non-emitting include stone, ceramic, powder-coated metals, plated or anodized metal, and unfinished or untreated solid wood. For the purposes of this credit, untreated and unfinished solid wood (not engineered wood) can also be considered non-emitting even though such materials will likely naturally emit some formaldehyde. Ceramic and powder-coated metals meet the criteria for being inherently non-emitting when their manufacturing processes result in chemically stable and inert surfaces that do not release VOCs into the environment after production. These materials are compliant without any VOC emissions testing if they do not include additives, surface coatings, binders, or sealants, as such products would emit VOCs.

If a product applied to the inherently non-emitting material has a separate manufacturer and cost to the end user from the original material, the applied product may be documented as a separate product subject to the applicable low-emitting criteria, even if applied off-site.

If a product applied to the inherently non-emitting material does not have a separate manufacturer and cost to the end user, the result is considered a new finished product that no longer qualifies as an inherently non-emitting material and is subject to the applicable low-emitting criteria.

SALVAGED OR REUSED MATERIALS CRITERIA

Products that are salvaged and reused and more than one year old will automatically comply with the VOC emission evaluation and do not require emissions evaluations. For salvaged or reused composite wood products, project teams must account for any off-site applied finishes or treatments in the composite wood category. These must comply with the VOC emissions evaluation criteria to ensure comprehensive assessment of the product's total environmental impact.

Some salvaged or reused materials will have products applied to them (such as sealants or finishes). For instance:

- If a product is applied to the salvaged or reused material and has a separate manufacturer and cost to the end user from the original material, the applied product may be documented as a separate product subject to the applicable low-emitting criteria, even if applied off-site.
- If a product is applied to the salvaged or reused material but does not have a separate manufacturer and cost to the end user, the result is considered a new finished product that no longer qualifies as a salvaged or reused material and is subject to the applicable low-emitting criteria.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	N/A	All	USGBC calculator.
			Contract documents, including key/schedules, identifying all installed products (inclusive of any salvage and reuse) in the attempted categories (e.g., floor finish plans, furniture plans, reflected ceiling plans, interior elevations, specifications, wall details).
		Paints and coatings, flooring, ceilings, adhesives and sealants, walls, insulation	Qualifying VOC emissions evaluation for every product in the calculator, in the attempted categories, that is indicated as meeting the low-emitting criteria, AND/OR Report(s) from USGBC-approved tools for documenting a suite of eligible products.
			Qualifying VOC emissions evaluation for any eligible product applied off-site to an inherently non-emitting, salvaged, or reused product.
		Composite wood (Path 2)	Qualifying formaldehyde emissions evaluation for every composite wood product in the calculator that is indicated as meeting the low-emitting criteria, AND/OR Report(s) from USGBC-approved tools for documenting a suite of eligible products.
			Qualifying VOC emissions evaluation for any eligible product applied off-site to a composite wood product.
		Furniture (Path 3)	Qualifying furniture emissions evaluation for every furniture product in the calculator that is indicated as meeting the low-emitting criteria, AND/OR Report(s) from USGBC-approved tools for documenting a suite of eligible products.
			Alternately, for custom items, qualifying VOC emissions evaluations for all components of the finished piece applied on-site or off-site.
Core and Shell	N/A	N/A	Tenant guidelines communicating the base building's product list (this may be provided under *IPp4: Tenant Guidelines*).

REFERENCED STANDARDS

- CDPH Standard Method v1.2 (https://www.cdph.ca.gov/programs/ccdphp/deodc/ehlb/iaq/cdph%20document%20library/cdph-iaq_standardmethod_v1_2_2017_ada.pdf)
- ANSI/BIFMA Standard M7.1, etc. (https://www.bifma.org)
- SCAQMD Rule 1113 (https://www.aqmd.gov/home)
- SCAQMD Rule 1168 (https://www.aqmd.gov/home)
- CARB ATCM 93120 (https://ww2.arb.ca.gov)
- EPA TSCA Title VI (https://www.epa.gov/formaldehyde/formaldehyde-emission-standards-composite-wood-products)
- ASTM D5456, D5055 (https://www.astm.org)

FURTHER EXPLANATION

COMPLIANT THIRD-PARTY-VERIFIED VOC EMISSIONS EVALUATION STANDARDS

Qualifying third-party-verified VOC emissions evaluations that follow the California Department of Public Health (CDPH) Standard Method v1.2 are available to view at their website.[51]

PRODUCT CATEGORIES (AIR BARRIER)

Each category identifies some general criteria for establishing the products/materials that are to be included and excluded from a category. A supplemental resource, to help determine whether a product is to be included or excluded and in which category it belongs, can be found in the credit resources. One general rule has guided the inclusions and exclusions for this credit: products inclusive of and inside an air barrier are most likely to impact indoor air quality. Therefore, envelope products outside of the air barrier may, but do not have to be, included in this credit. Projects without a qualifying continuous air barrier should include all envelope products in the attempted categories.

An air barrier is a continuous plane of materials that prevents air and pollutants carried in the air from passing through the building's envelope. Air barriers can help keep pollutants out of the building, which improves air quality and minimizes air leakage. In most climate zones, an air barrier is required by code. Where one is not required for insulative properties (such as in Climate Zone 2B), project teams may still choose to implement air barriers for their other performance benefits. More details about air barriers can be found in the 2015 International Energy Conservation Code, Section C402.5.1.

An air barrier may take the form of a sheet or fluid-applied air barrier membrane product but can also be made up of common building products that are air impermeable. In either case, joints and seams must be sealed to ensure a continuous and airtight barrier surrounding the entire building. Adhesives and sealants used in the air barrier installation must be reported under the Adhesives and Sealants category, if attempted. Other components of the air barrier must also be reported in the appropriate categories, when attempted (unless they are listed as an exclusion). Note that a vapor barrier serves a different function and is not an air barrier.

The following products are air impermeable and therefore may be part of the building's continuous air barrier, provided that all joints/seams are sealed.

- Poured concrete slabs
- 3/8 inch (or thicker) oriented strand board (OSB) sheathing
- 1/2 inch (or thicker) plywood sheathing
- Extruded polystyrene (XPS) board
- Rigid foam insulation
- Asphalt-based or polymer-based roof membranes

51. California Department of Public Health (CDPH), "List of Certifications and Programs that Use CDPH Standard Method v1.2 (2017)," accessed September 9, 2025, https://www.cdph.ca.gov/Programs/CCDPHP/DEODC/EHLB/IAQ/CDPH%20Document%20Library/List%20of%203rd%20party%20certifications%20for%20CDPH%20v1.2-Oct-10-2019%20ADA.pdf.

- 1/2 inch (or thicker) gypsum board
- 1/2 inch (or thicker) foil-faced urethane insulation board
- 8 inch (or thicker) concrete masonry units (CMU)
- Metal decking
- Reinforced plastic sheeting
- Aluminum
- Stainless steel
- Spray polyurethane foam
- Glass
- Other materials that have an air permeability not greater than 0.004 cfm/ft2 (0.02 L/s-m^2) under a pressure differential of 0.3-inch water gauge (75 pascals) when tested in accordance with *ASTM E2178*.

Building Product Selection and Procurement

MRc4
New Construction (1–5 points)
Core and Shell (1–5 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To encourage the use of products and materials that have sustainability information available and that have environmentally, economically, and socially preferable impacts in alignment with industry momentum. To reward project teams for selecting products from manufacturers who have disclosed sustainability information about their products and have optimized their products across multiple criteria areas.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–5**
Product Categories	1–5

Select nonstructural building products that demonstrate achievement in one or more of five criteria areas:

- Climate health
- Human health
- Ecosystem health
- Social health and equity
- Circular economy

Products that achieve two or more criteria areas are considered multi-attribute. Products that achieve higher levels of use and/or that are across additional criteria areas will be given a higher value in credit calculations.

Achievement is demonstrated through eligible compliant manufacturer product documentation, which includes third-party product certifications, ecolabels, declarations, and standards. A single product document can demonstrate multiple benefits and/or achievement levels, or the product can earn multi-attribute criteria through a combination of separate eligible product documents.

There are three achievement levels for products:

- **Level 1**: A product in this level achieves a first step toward sustainability for a criteria area. Widespread achievement of these practices drives market transformation toward sustainability outcomes within the criteria area. Products scored at this level earn a 1× multiplier.
- **Level 2**: This level represents a leadership position in the marketplace for a given sustainability attribute. Products at this level are optimized and demonstrate a level of sustainability that peers aspire to achieve. Products scored at this level earn a 2× multiplier.
- **Level 3**: Products that earn this level are elite and represent the forefront of sustainability. Products scored at this level earn a 3× multiplier.

This credit rewards the selection of eligible interior and enclosure materials from the following product categories:

- Paints and coatings
- Adhesives and sealants
- Flooring
- Walls
- Ceilings
- Insulation
- Furniture
- Composite wood
- Plumbing fixtures

Eligible products meet the achievement levels and are scored as 1, 2, or 3. These scores are added across criteria areas to add up to a maximum score of 5 per product. This cumulative score is called the product multi-attribute score (MAS).

Each individual product's value (cost, area, volume, or unit) is adjusted based on its MAS:

$$\text{Product value} \times \text{MAS} = \text{Adjusted product value for LEED}$$

To determine total compliant product value per category, follow Equation 1.

Equation 1. Calculate the multi-attribute adjusted value of a product category

$$\text{Product category adjusted value for LEED} = 100 \times \left[\frac{\begin{pmatrix}\text{Product A MAS} \times \\ \text{Product A value}\end{pmatrix} + \begin{pmatrix}\text{Product B MAS} \times \\ \text{Product B value}\end{pmatrix} + \begin{pmatrix}\text{Product C MAS} \times \\ \text{Product C value}\end{pmatrix} + \begin{pmatrix}\text{Product D MAS} \times \\ \text{Product D value}\end{pmatrix}}{(\text{Total value of all products in the product category})} \right]$$

Any product category adjusted value for LEED that exceeds 100% earns 1 point. Points are awarded for achievement of whole product categories up to a maximum of 5 points according to Table 1.

TABLE 1. Points for Multi-Attribute Achievement of Product Categories

Number of Product Categories	Points
1 product category	1
2 product categories	2
3 product categories	3
4 product categories	4
5 product categories	5

Please see the resources section of the credit library for additional details on this credit.

REQUIREMENTS EXPLAINED

This credit incentivizes projects to prioritize more environmentally responsible materials and choose products with multiple eco-friendly attributes. It focuses on finish materials, such as paints, coatings, flooring, and walls, and considers their impact on the overall environmental performance of the project.

Some structure, enclosure, and hardscape materials are not included in this credit but are addressed in the embodied carbon credits, including *MRp2: Quantify and Assess Embodied Carbon* and *MRc2: Reduce Embodied Carbon*. Other materials can earn rewards for multi-attribute considerations within the Project Priorities library.

In this credit, products are evaluated based on how they perform according to five criteria areas: climate health, human health, ecosystem health, social health and equity, and circular economy. Within each criteria area, there are three achievement levels that products can meet. Evaluation will be based on how products demonstrate achievement in each of the criteria areas with respect to the three levels. A product that reaches achievement levels in multiple criteria areas is considered a multi-attribute product and will earn a higher value within this credit.

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SELECTING BUILDING PRODUCTS

The goal is to recognize products that are optimized across multiple attributes, including promoting human and environmental health, supporting regenerative sourcing practices, and fostering a circular economy. These criteria areas are meant to align with the impact areas in the *Mindful Materials Common Materials Framework* (CMF) and the *AIA Architecture and Design (A&D) Materials Pledge*.

The *AIA A&D* Materials Pledge[52] provides a framework to encourage the use of building materials that prioritize sustainability throughout their life cycles, including aspects like green chemistry,

52. AIA, "AIA Materials Pledge," n.d., https://www.aia.org/design-excellence/climate-action/zero-carbon/materials-pledge.

responsible sourcing, and end-of-life management. The *Mindful Materials Common Materials Framework*[53] standardizes product evaluations and emphasizes environmental and health impacts. AIA has also introduced reporting requirements, and the CMF is expanding its focus to include data integration and related advancements. Both initiatives aim to enhance transparency and optimization in building materials and function independently of specific certifications by providing a structured framework that allows various standards to align with key impact areas. The five criteria areas help connect different ecolabels by offering a consistent and holistic approach to material evaluation where certifications are scored based on disclosure, verification, and optimization, among other criteria.

PRODUCT CATEGORIES

To achieve a point, the project must demonstrate they meet or exceed the threshold for each product category. This can be based on cost, surface area, volume, or number of units, depending on the measurement methods available for each product category. Project teams can choose different measurement types to measure progress toward achievement as long as the measurement method is consistent in each product category. For example, a project could use surface area to demonstrate achievement of the flooring category, number of units for the furniture category, and volume for the adhesives and sealants category.

This credit will be documented by product category using the LEED materials calculator. This calculator is combined with the *MRc3: Low-Emitting Materials* calculator. Teams are encouraged to combine submittal reviews and product vetting with the criteria found in both credits to maximize credit achievement and harmonize product selection, specification, and documentation processes.

PRODUCT MULTIPLIERS

Multipliers are awarded for products that earn any level of achievement in one or more criteria areas. A product does not have to be multi-attribute (meet achievement levels in multiple criteria areas) to have a multiplier. Products that achieve a first step toward sustainability for a criteria area are categorized as Level 1 and will receive a 1× multiplier. Products in Level 2 represent a leadership position in the marketplace for a given sustainability attribute and will receive a 2× multiplier. Products in Level 3 are elite and represent the forefront of sustainability and will receive a 3× multiplier. These scores are added across criteria areas to add up to a maximum score of five per product. This value is called the multi-attribute score for the product.

MULTI-ATTRIBUTE SCORING OF PRODUCTS

Product documentation provided by manufacturers will be eligible for reward in this credit. The documentation must meet the USGBC-approved list of eligible product documentation. Individual products selected for compliance with *MRc4: Building Product Selection and Procurement* may have more than one eligible product documentation. Multiple document scores can be added together for a combined MAS for that product, but only the highest value in each criteria area will be awarded (double counting is not allowed).

53. Mindful MATERIALS, "CMF Reference Guide," n.d., https://www.mindfulmaterials.com/cmf-reference-guide.

MULTI-ATTRIBUTE ADJUSTED VALUE

Each eligible product earns an MAS based on its level of achievement in each criteria area. The MAS is multiplied by the product's value to find each product's adjusted product value for LEED. In a single product category, each eligible product's adjusted value is added together and divided by the total unadjusted value of all products in the product category. This value is the product category adjusted value for LEED. This value is how to determine credit achievement. Any product category adjusted value for LEED that exceeds 100% earns one point.

Equation 2. Calculating product category adjusted value

$$\text{Product category adjusted value for LEED} = 100 \times \left[\frac{\begin{pmatrix}\text{Product A MAS} \times \\ \text{Product A value}\end{pmatrix} + \begin{pmatrix}\text{Product B MAS} \times \\ \text{Product B value}\end{pmatrix}}{(\text{Total value of all products in the product category})} \right].$$

NUMBER OF PRODUCT CATEGORIES

A total of nine product categories to consider include paints and coatings, adhesives and sealants, flooring, walls, ceilings, insulation, furniture, composite wood, and plumbing fixtures. Projects may earn one point for each category up to a maximum of five product categories.

Projects can earn points based on the number of categories in which they select eligible materials that meet the required criteria. Nonstructural products that do not clearly align with one of the product categories listed in Table 2 may still be eligible to be included for assessment. This credit is designed to offer flexibility in evaluating different product types that fall outside the nine listed categories. The tiered point system incentivizes broader and deeper integration of sustainable materials across multiple categories and encourages projects to enhance their overall environmental performance by incorporating products that support health, sustainability, and reduced environmental impact. Table 2 lists the product categories and some product examples found within the categories.

TABLE 2. Product Categories and Product Examples

Product Categories	Description and Examples
Paints and coatings	Paints and coatings are materials applied to surfaces for protection and decoration. Coatings are generally chosen for their enhanced protective properties and functional capabilities, whereas paints are chosen for their aesthetic appeal. Product examples: • Primers • Sealers • Topcoats • Specialized dyes • Specialized sealers • Specialized hardeners • Specialized toppings for concrete floors • Plasters

TABLE 2. Product Categories and Product Examples (*continued*)

Product Categories	Description and Examples
Adhesives and sealants	Adhesives and sealants are substances used to bond two materials together and are widely used in construction, manufacturing, and various other industries. The main difference is that adhesives are focused on creating strong bonds between surfaces, whereas sealants are designed to fill gaps and prevent the infiltration or leakage of fluids, gases, or other substances. Product examples: • Wood bonding adhesive • Tile bonding adhesive • Carpet bonding adhesive • Sealants for joints and gaps in walls, floors, and ceilings • Specialty adhesives for flooring • Specialty adhesives for panel • Sealants for HVAC system
Flooring	Flooring materials are used to cover the ground surface of a building or structure to provide a functional walking surface for building users. The flooring product category encompasses a wide range of materials, including both hard and soft surface finishes. Product examples: • Carpet • Ceramic tile • Vinyl flooring • Rubber flooring • Engineered wood flooring • Solid wood flooring • Stone flooring • Terrazzo flooring • Laminates flooring • Raised flooring systems • Wall base • Transition strips • Stair nosing • Entryway systems • Area rugs • Wood and composite wood subflooring • Underlayment • Other types of floor coverings
Walls	Wall products are designed to provide crucial functions within a building and refer to materials and finishes used to provide structural support, insulation, and protection within a building. These products also help regulate indoor temperatures and maintain comfort levels by reducing heat transfer between the indoor and outdoor environments. Wall products serve as barriers for protection against sound, fire, and moisture. Product examples: • Wall coverings • Wall paneling • Wall tile • Surface wall structures (e.g., gypsum wallboard, or plaster) • Cubicle wall • Curtain wall • Partition wall • Trim • Interior and exterior doors • Wall frames • Interior and exterior windows • Window treatments

(*continued*)

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TABLE 2. Product Categories and Product Examples (*continued*)

Product Categories	Description and Examples
Ceilings	Ceiling products are materials and systems used to construct, finish, or enhance the ceilings of a building. Ceilings play a key role in acoustics, lighting, insulation, and the overall functionality of a space. Product examples: • Ceiling panels • Ceiling tile • Surface ceiling structures (e.g., gypsum or plaster) • Suspended or drop ceiling systems (e.g., grid systems, canopies, and clouds) • Glazed skylights
Insulation	Insulation is any type of material that provides a barrier within the walls, ceilings, and floors of a home and helps regulate temperature and noise.[54] It plays an important role in heat transfer and maintaining indoor temperatures in buildings by providing thermal resistance. Product examples: • Thermal and acoustic boards • Batt insulation • Roll insulation • Blanket insulation • Sound attention fire blankets • Foamed-in place insulation • Loose-fill insulation • Blown insulation • Sprayed insulation
Furniture	Furniture refers to movable objects that support various human activities, such as seating, eating, sleeping, and storing items. Furniture is both functional and decorative and plays a significant role in the design and use of interior spaces. It can be made from a variety of materials, including wood, metal, plastic, glass, and fabric, and comes in many styles, shapes, and sizes to suit different needs and tastes. Product examples: • Seating • Desks • Tables • Filing/storage • Free-standing cabinetry • Systems furniture • Partitions • Bathroom partitions • Shelving • Lockers • Specialty and custom fixtures
Composite wood	Composite wood is an engineered wood product made by combining wood fibers, particles, or veneers with adhesives or resins to create a material that is often used in place of solid wood. Product examples: • Particleboard • Medium density fiberboard • Hardwood plywood with veneer • Composite or combination core • Wood structural panels or structural wood products
Plumbing fixtures	A plumbing fixture is connected to the plumbing system and is designed to deliver and drain water. Product examples: • Water closets • Urinals • Lavatory and kitchen faucets • Showerheads

54. U.S. Department of Energy, "Powering Today, Transforming Tomorrow," n.d., https://www.energy.gov/energysaver/insulation#:~:text=Insulation%20in%20your%20home%20provides,costs%2C%20but%20also%20improves%20comfort.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	N/A	All (Paints and coatings, adhesives and sealants, flooring, walls, ceilings, insulation, furniture, composite wood, plumbing fixtures)	USGBC calculator.
			Individual USGBC-approved product documentation (found under Resources and Tips) for every product indicated in the calculator as meeting BPSP criteria; up to 5 documents per product. AND/OR Report(s) from USGBC-approved tools for documenting a suite of eligible products.
			Contract documents (if not already provided in Project Information or other credits), including key/schedules, identifying all installed products (inclusive of any salvage and reuse) in the attempted categories (for example, floor finish plans, furniture plans, reflected ceiling plans, interior elevations, specifications, wall details, or fixture schedules).

REFERENCED STANDARDS

- AIA A&D Materials Pledge (https://www.aia.org/design-excellence/climate-action/zero-carbon/materials-pledge)
- Mindful Materials CMF (https://www.mindfulmaterials.com)

FURTHER EXPLANATION

The five criteria areas—namely, climate health, human health, ecosystem health, social health and equity, and circular economy—are intended to align with the Mindful Materials Common Materials Framework and the American Institute of Architects Materials Pledge. The three levels of achievement for each criteria area (Levels 1, 2, and 3) are defined by USGBC and can be found in the "Resources" section of the credit library. These definitions and the accompanying product documentation scoring table will be updated over time through the LEED addenda process. Individual products are scored based on compliance with required documentation for each level. Project teams earn points within an entire product category using the weighted values of individual products.

CRITERIA AREAS

See the "Resources" section of the credit library to find the specific requirements to achieve Levels 1, 2, and 3 for each criteria area. The document defines each criteria area and lists eligible product documentation and reporting requirements.

Brief summaries of each criteria area and the levels of achievement within them are provided. Note that this does not provide a comprehensive list of requirements and eligible product documentation. See the full additional resource in the credit library for complete requirements.

HUMAN HEALTH

The human health criteria area refers to strategies for selecting products that are verified to minimize the use and generation of harmful substances. Building products eligible for Level 1 achievement have disclosed product ingredients following accepted methodologies to the public or a trusted third party. To be rewarded under Levels 2 or 3, products have been screened to minimize negative human health impacts and avoid a wide range of harmful substances.

Some examples of eligible product documentation for human health are health product declarations prechecked for LEED, Declare labels, Cradle-to-Cradle Material Health certificate, and Green Seal GS-11 certified.

CLIMATE HEALTH

The climate health criteria area refers to impacts or strategies that quantify, reduce, or reverse the embodied carbon emissions associated with a building product's life cycle. To be eligible for Level 1 achievement, building products have quantified and disclosed embodied carbon impacts. Levels 2 and 3 require products to show improvements that reduce carbon impacts and aspire to be carbon neutral or negative.

Level 1 achievement in climate health prioritizes the quantification and disclosure of life-cycle GWP through product-specific Type III EPDs. Levels 2 and 3 reward reduction in GWP through an EPD Comparative Assessment report. Material reuse is also rewarded within the Climate Health criteria area.

ECOSYSTEM HEALTH

The ecosystem health criteria area rewards the selection of materials that disclose, reduce, and reverse impacts on the environment throughout the materials supply chain. Eligible building products may be rewarded under Level 1 by disclosing raw material sources and/or protecting against the worst environmental harms during manufacturing. Under Levels 2 or 3, products are recognized for minimizing environmental impacts and rigorous verification of claims and aspire to regenerate ecosystems affected throughout the supply chain.

Eligible product documentation for ecosystem health criteria includes Cradle-to-Cradle certified, wood products that are legally sourced, and optimized EPDs.

SOCIAL HEALTH AND EQUITY

The social health and equity criteria area rewards the selection of products from manufacturers that uphold human rights throughout the supply chain, therefore striving to protect workers and communities impacted by material extraction and production. This criteria area aims to harmonize the efforts for social health and equity product excellence with sources such as the Mindful Materials Common Materials Framework, the Grace Farms Design for Freedom guidance, and Cradle-to-Cradle product standards.

An example of eligible documentation for this criteria area is Cradle-to-Cradle certification. As additional standards are adopted by the industry to verify that products come from manufacturers who protect human rights in the supply chain and support other goals in this area, eligible documentation will be added.

CIRCULAR ECONOMY

The circular economy criteria area refers to products that are designed to be cycled after use, contribute to waste minimization, reduce reliance on virgin materials, extend the life cycle of building materials, or shift the burden of disposal to product manufacturers. Products rewarded under Level 1 include those that have incorporated recycled or qualified bio-based content. To earn Level 2 or Level 3, products preserve more materials through reuse and maximize value through closed-loop circular systems.

Eligible products include Cradle-to-Cradle certified, reused, bio-based content, recycled content, or are eligible for take-back by an approved USGBC EPR program.

CALCULATIONS

To determine points under this credit, project teams will focus on optimizing individual product categories. Note that if a product does not directly fit into a specific product category, please submit it in the category that the project team believes it most aligns with.

EXAMPLE EQUATION:

$$\text{Product category adjusted value for LEED} = 100 \times \frac{(200.2) + (100.5) + (300.2)}{(200 + 100 + 300 + 400)}$$
$$= 100 \times \frac{1{,}500}{1{,}000} = 150\%$$

Where:

- The flooring category includes four products total with a combined value of 1,000 ft^2
- Product A has a value of 200 ft^2
  - Multi-attribute score (MAS) = 2
- Product B has a value of 100 ft^2
  - MAS = 5
- Product C has a value of 300 ft^2
  - MAS = 2
- Product D has a value of 400 ft^2
  - MAS = 0

One point is earned, as the product category adjusted value for LEED exceeds 100%.

Construction and Demolition Waste Diversion

MRc5
New Construction (1–2 points)
Core and Shell (1–2 points)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☐ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To reduce construction and demolition waste disposed of in landfills and incineration facilities and pollution to the environment. To reduce the environmental impacts and embodied carbon of manufacturing new materials and products. To delay the need for new landfill facilities that are often located in frontline communities. To create green jobs and materials markets for building construction services.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–2**
Construction and Demolition Materials Management Plan AND	
Diversion	1–2

Comply with the following requirements:

CONSTRUCTION AND DEMOLITION MATERIALS MANAGEMENT PLAN

Develop and implement a construction and demolition (C&D) materials management plan and achieve points through diversion and recycling.

AND

DIVERSION (1–2 POINTS)

Follow the materials management plan and provide a final waste management report detailing all waste generated, including disposal and diversion rates for the project. Calculations can be by weight or volume but must be consistent throughout. Points are awarded according to Table 1.

Divert C&D waste materials by using strategies including off-site salvage, source-separation for single-material recycling, mixed C&D recycling, and industry/manufacturer take-back programs.

- Source-separated materials are considered 100% diverted for credit calculation purposes. These include:
  - Recovered materials sent to a single-material recycler
  - Recovered materials sent for off-site salvage/reuse
  - Materials sent to a qualifying manufacturer or industry take-back program
  - Salvaged materials, which are valued at twice the diversion rate (200%) of other diverted materials for credit calculation purposes and include recovered materials sent off-site for reuse
    NOTE: Materials reused on-site contribute to *MRc1: Building and Materials Reuse*.
- Mixed C&D materials sent to a processing facility for recovery must take the facility average recycling rate. Recycling rates not verified by a third party must assume a maximum of 35% diversion rate.
- Materials destined for alternative daily cover or incineration/energy recovery are considered waste (0% diverted).
- Exclude hazardous waste from calculations. Exclude on-site reuse from credit calculations (include in *MRc1: Building and Materials Reuse*).
- Exclude excavated soil and land-clearing debris from calculations.

TABLE 1. Points for C&D Diversion

Thresholds	Points
Divert at least 50% of the total construction and demolition material. At least 10% of diverted materials must be salvaged or source-separated and sent to single-material recycler(s).	1
Divert at least 75% of the total construction and demolition material. At least 25% of the total diverted materials must be salvaged or source-separated and sent to single-material recycler(s).	2

Core and Shell only

Include the building's approved construction and demolition waste management plan in the tenant guidelines.

REQUIREMENTS EXPLAINED

This credit encourages projects to plan and make design changes that reduce waste during construction. It rewards behavior change that leads to increased quality of recycling and a higher potential for materials to be recovered during construction.

CONSTRUCTION AND DEMOLITION MATERIALS MANAGEMENT PLAN

Creating the C&D material management plan early in the design phase allows sufficient time for planning and coordination, identifying effective strategies, and establishing contractual agreements to maximize waste prevention and diversion. It also educates project teams, construction site workers, and waste haulers on the importance of following the plan for successfully diverting materials from landfills and incinerators. The salvage assessment featured in *MRc1: Building and Materials Reuse* is also useful to identify materials that can be diverted off-site to reuse markets. A well-structured plan can minimize costs and maximize returns by lowering disposal costs, recovering value from scrap materials, and identifying materials for reuse.

General contractors are required to develop a customized C&D material management plan for the deconstruction/demolition and construction phases. This plan should begin in the project design phase prior to construction. The plan must include a summary of materials targeted for diversion from landfills or incineration and identify recycling haulers, single-material recycling facilities, mixed C&D processing facilities, data collection, and reporting procedures. Teams must indicate in the plan whether the selected recycling facilities that process mixed C&D materials have third-party verification of their recycling rates. Recycling rates not verified by a third party must assume a maximum of 35% diversion rate. The 35% cap serves as a baseline assumption for mixed-material facilities without verification, reflecting an approximate average recycling rate for facilities in the U.S. If the project team uses a recycling facility for which recycling rates have been independently certified by an approved third-party process, such as the Recycling Certification Institute, then the project team can use the verified recycling rate. This third-party verification of recycling rates provides assurance that diversion rates are accurate and that materials are being diverted from the landfill.[55]

The plan must also include strategies targeted to reduce the total amount of waste generated during construction, renovation, or demolition activities.

DIVERSION

WASTE TRACKING

Teams are required to develop a method for tracking the amount of all waste and recyclable materials generated during demolition activities. Web-based tools can provide contractors with an easy, step-by-step process for electronically tracking and submitting waste management and recycling plans. Electronic tracking can also save time and money by identifying materials that can be recycled, locating the nearest recycling facilities, following recycling progress in real time, gathering comprehensive statistics, and creating reports regarding waste generation and recycling for projects. Waste tracking systems can also identify opportunities for recycling, off-site reuse, and salvage. Examples of waste tracking software include Green Halo,[56] Waste Management's Diversion and Recycling Tracking Tool (DART),[57] SmartWaste, and Enviance.

DIVERSION RATE

Project teams are required to calculate total waste generated and diverted to determine the C&D waste diversion rate. Contractors are recommended to keep all tickets/paperwork in a safe location (if not online) and track the diversion rate periodically (e.g., monthly or bimonthly) so that adjustments can be made to meet diversion goals. Teams must ensure that calculations for

55. RCI, "Recycling Facility Certification Program," accessed March 31, 2025, https://www.recyclingcertification.org/.
56. Green Halo Systems, Inc., "Green Halo Systems," accessed March 31, 2025, https://www.greenhalosystems.com.
57. Waste Management, "Our Online Diversion and Recycling Tracking Tool (DART) Puts Your Environmental Data at Your Fingertips," accessed March 31, 2025, https://www.wm.com/documents/pdfs-for-services-section/001983_DARTSales%20Sheet%20(9-14-11).pdf.

all materials are done by weight. Many waste management facilities use scales to weigh loads of materials as they enter and exit the site. However, not all facilities have scales available. In such cases, a volume-based calculation is used instead. When a facility does not have scales, use a volume-to-weight conversion factor if volume is provided. If local conversion rates are not available, projects may use national averages, such as those found in Table 2.

TABLE 2. Default Volume-to-Weight Conversion Factors for Common C&D Waste

C&D Materials	
Asphalt paving (with or without rebar)	1 cubic yard = 773 lbs
Concrete (with or without rebar)	1 cubic yard = 860 lbs
Gypsum board	1 cubic yard = 467 lbs
Wood	1 cubic yard = 169–268 lbs
Metal	1 cubic yard = 143–225 lbs
Roofing	1 cubic yard = 731–860 lbs
Mixed C&D (bulk)	1 cubic yard = 484 lbs
Aggregate (rock)	1 cubic yard = 999 lbs
Cardboard (flat)	1 cubic yard = 106 lbs
Cardboard (baled)	1 cubic yard = 700–1,100 lbs

SOURCE: U.S. Environmental Protection Agency Volume-to-Weight Conversion Factors, April 2016.[58]

Excluding Alternative Daily Cover

All materials that are recycled, salvaged, reused, or donated are included in the project's diversion rate. However, projects must exclude certain materials from the diversion total while still accounting for them in the total C&D waste calculations. Specifically, alternative daily cover (ADC) cannot be counted as diverted waste because it is a disposal method rather than a true form of recycling, as the material is used for landfill operations rather than being repurposed into new products. To obtain ADC values from a mixed recycling or certified facility, request detailed documentation of material processing and their average ADC rates per month. If they do not have the ADC rates monthly, then quarterly, semiannual, or annual rates are acceptable.

Excluding Hazardous Waste, Land-Clearing Debris, Soil, and Landscaping Materials

Hazardous waste, land-clearing debris, soil, and landscaping materials must be excluded from diversion totals. Soil is excluded because clean soil is rarely landfilled due to its high cost and other suitable uses, which could skew results. Contaminated soil, meanwhile, is typically classified as hazardous and managed under strict regulations, often requiring specialized disposal. Similarly, land-clearing debris, such as rocks and trees, is generally not landfilled due to its weight and is commonly diverted. Hazardous waste must follow regulatory guidelines for safe handling, disposal in lined landfills, or destruction to prevent environmental harm. This includes proper identification, labeling,

58. U.S. Environmental Protection Agency, "Volume-to-Weight Conversion Factors," 2016, accessed March 31, 2025, https://www.epa.gov/sites/default/files/2016-04/documents/volume_to_weight_conversion_factors_memorandum_04192016_508fnl.pdf.

and containment of hazardous materials, as well as transportation by certified handlers to facilities equipped to manage such waste.

Equation 1. Diversion rate

$$\text{Diversion rate} = \left[\frac{\text{Total waste diverted from landfill}}{\text{Total waste produced by project}}\right] \times 100$$

SOURCE-SEPARATED MATERIAL

Project teams must identify materials that will be diverted from landfill and incineration facilities. Common C&D waste materials include concrete, metals, brick, wood, and cardboard. Depending on the project's scope of work, additional sources may include carpet, ceilings, gypsum board, and furniture.

The project must account for source separation or salvage as a percentage of the total diversion in the achievement thresholds. This represents a percentage of the overall diversion amount for the project and is not in addition to the overall diversion rate.

Teams should target source separation where each homogeneous material is collected and sent to a specific recycling facility (or is sent for reuse). Source-separated materials are not mixed with other materials, which significantly reduces the contamination in recycling streams and results in higher overall diversion rates for those recovered materials. Source separation involves segregating recyclable materials from mixed waste at the point of generation. This practice involves sorting materials, such as metals, wood, and concrete, directly at the construction or demolition site before they are commingled in a central recycling area or bin. Contractors should consider setting up dedicated areas on construction sites and clearly label and monitor bins for each source-separated material to ensure proper collection.

Teams are encouraged to prioritize the source separation of materials like carpet, ceiling tiles, gypsum board, and furniture. Although these materials may not be specifically targeted, their significant environmental impacts make their diversion particularly important for reducing overall environmental harm.

SALVAGING MATERIALS

Successful salvaging begins with careful planning and requires a thorough audit of the existing materials and structures to identify which materials can be reclaimed. It is recommended to conduct an early salvage assessment during building design to determine which tools and methods will be most valuable and effective for removal and preservation.

Projects that salvage materials off-site must send materials from the job to legitimate off-site salvage and reuse vendors or markets. Destinations must be locations that either directly reuse the materials or place them into a marketplace for distribution, sale, or reuse. Materials must not be stockpiled without the intention of being cycled back into use. Stockpile locations are acceptable only if they actively work to move materials through reuse cycles and provide documentation detailing what actions will be taken if the materials remain unused for an extended period. Even with best intentions, some salvaged materials do not find a home in a new project for various reasons and ultimately get recycled or disposed of. This entropy of salvaged materials is acceptable so long as most materials sent for salvage are intended to remain in circulation.

MIXED C&D MATERIALS

Mixed C&D materials, or commingled waste, is recyclable material mixed in a single container that is sorted and processed at an off-site recycling facility.

Projects must obtain diversion rates from each commingled or mixed waste processing facility used. Facilities must operate legally and be regulated by state and local authorities. However, these authorities may not oversee diversion rates or the reporting of such rates. Therefore certifications, like Recycling Certification (RCI) or equivalent (as determined by USGBC), to ensure accurate tracking and reporting of diversion rates are necessary. Project teams are encouraged to use facilities verified by an approved third party to achieve higher diversion rates. Facilities whose recycling rates are not third-party verified can only claim a maximum diversion rate of 35%.

Certified recycling facilities

Projects must use a recycling facility that processes and recycles commingled (mixed) construction and demolition waste materials that have received independent third-party certification of their recycling rates. Qualified third-party organizations who certify facility average recycling rates include these minimum program requirements:

- The certification organization follows guidelines for environmental claims and third-party oversight, including *ISO/IEC 17065*[59] and relevant portions of the *ISO 14000*[60] family of standards.
- The certification organization is an independent third party who continuously monitors certified facilities to ensure that they are operating legally and meeting the minimum program requirements for facility certification and recycling rates.
- Certification organizations shall certify to a protocol that was developed on a consensus basis for recycling facility diversion rates that is not in a draft or pilot program.

The methodology for calculating facility recycling rates must

- Be developed with construction and demolition recycling industry stakeholders and be specific to the construction and demolition recycling industry;
- Include a methodology that is applicable across broad regions (e.g., nationally); and
- Refer to a published and publicly available standard.

Data submitted by the facilities to the certification organization in support of the recycling rate is audited. At a minimum, the audit includes the evaluation of recyclable sales records, verification of facility sales into commodity markets, an assessment of downstream materials and how these materials are managed after they leave the site, monitoring off-site movement of materials, and a review of the facility's customers' weight tags information.

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59. International Organization for Standardization, "ISO/IEC 17065, Edition 1," 2012, accessed March 31, 2025, https://www.iso.org/standard/46568.html.
60. International Organization for Standardization, "ISO 14000 family," accessed March 31, 2025, https://www.iso.org/standards/popular/iso-14000-family.

- Facilities submit data to the certification organization that supports the recycling rate, such as a mass balance recycling rate (tons in/tons out) for a 12-month period, or quarterly sorts completed and verified by an independent third-party entity.
- Breakdown of materials (by type and by weight), including analysis of supporting data relating to amounts (in tons) and types of materials received and processed at the facility.
- At a minimum, the third-party certifying organization conducts an on-site visit of the facility for the first-year certification, with subsequent site visits occurring at least once every two years, unless additional visits are deemed necessary by the certification organization. The site visit will
  - Examine how materials enter, are measured, deposited, processed/sorted, and exit the facility.
  - Conduct interviews with key personnel and discuss how materials are managed after they leave the site.
  - Confirm equipment types and capacity.
  - Observe and verify load/materials sorting and accuracy.
  - Verify use and accuracy of scales including calibration frequency.
- Diversion rates shall adhere to these requirements:
  - Measurements must be based on weight (not volume), using scales.
  - Diversion rates must be available on a website and viewable by the general public.
  - Methodology for calculating diversion and recycling rates must be publicly available and applicable to national or country-level accounting standards for construction and demolition waste recycling facilities.

Facility recycling data submitted to a certification program will be analyzed for recycling rates using a mass balance formula or quarterly sorts completed and verified by an independent third-party entity.

Final recycling rate will include overall facility diversion rates, with and without ADC/Beneficial Reuse, and will include separate recycling rates by material type as well as combined average, including wood-derived fuel/biofuel separate from other waste to energy or incineration end markets.

Project Type Variations

For projects with incomplete or speculative spaces, the building owner must commit to preserving the approved C&D waste management plan and apply it during future phases of construction to ensure consistency in waste reduction efforts. Additionally, they may pursue certification under LEED AP Interior Design + Construction (LEED AP ID+C) for the unfinished spaces, which promotes sustainable practices, such as material reuse, recycling, and overall resource efficiency.

For projects with incomplete or speculative spaces completed by the tenant, project teams must include the building's approved C&D waste management plan in the tenant guidelines, which serves as a best-practice example to guide tenants in managing their own construction or fit-out activities. This inclusion not only promotes uniformity in waste management practices but also encourages tenants to adopt sustainable strategies that align with the broader environmental goals of the building.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	N/A	N/A	The project-specific C&D materials management plan.
			USGBC calculator.
			Documentation of all waste materials (such as waste tickets, hauler reports or invoices, receipts, etc.) that corresponds to the USGBC calculator (facility name, material type, weight, date/tracking number).
			Documentation of all diverted materials (such as recycling facility tickets, hauler reports or invoices, receipts, etc.) that corresponds to USGBC calculator (facility name, material type, weight, date/tracking number, diversion percentage, percentage of ADC, if applicable).
			For projects in the U.S., confirmation that materials destined for ADC are excluded from diverted waste calculations but included in total construction and demolition waste generated total.
			For mixed C&D materials sent to a processing facility for recovery, documentation from the processing facility of the facility average diversion rate (excluding ADC).
			Evidence of third-party verified certification for the processing facility's diversion rate(s), if applicable.
			Evidence that each source-separated material is sent to a specific recycling facility or sent for reuse.
Core and Shell	N/A	N/A	Tenant guidelines including the base building's approved construction and demolition materials management plan (this may be provided under *IPp4: Tenant Guidelines*).

REFERENCED STANDARDS

- EPA volume-to-weight conversion factors (https://www.epa.gov/smm/volume-weight-conversion-factors-solid-waste)
- Recycling Certification Institute (https://www.recyclingcertification.org)
- TRUE diversion criteria (https://true.gbci.org/true-diversion-data-additional-guidance)

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FURTHER EXPLANATION

FINAL WASTE MANAGEMENT REPORT

Project teams may use the USGBC calculator for the final materials management report or use their own calculator or material tracker. If teams choose to use and submit their own calculator, at a minimum the report must include all the same information and data points found in the USGBC calculator.

Project teams are recommended to track all the construction and demolition (C&D) waste leaving the site and retain waste hauler reports for documentation. Keep records of estimated weight or volume of materials that are reused on site or salvaged for reuse on other projects by subcontractors or vendors. Also keep receipts and estimates of weight or volume for materials donated to charities, reuse retailers, or other recipients that can verify and track incoming and outgoing materials.

TERMINOLOGY

Salvaged materials: This refers to materials, components, or assemblies recovered from a structure for the purpose of being reused.

- For purposes of calculating material diversion, materials salvaged on site and sent for off-site reuse are valued at 200% of the weight in this credit.
  - Off-site reuse examples include selling salvaged materials to a salvage retailer or donating materials to a non-profit that accepts salvaged building materials.
- Materials salvaged on site can be reused on site; however they do not count toward C&D diversion calculations and instead count toward *MRc1: Building and Materials Reuse.*

Reused materials: This refers to salvaged material from onsite or off site that is used again for the same or different purpose, inclusive of materials that are modified and reinstalled.

- Reused material does not include materials or products containing recycled content.
- Reused material does not include surplus or overstock.

Recovered materials: This refers to C&D waste that is captured and targeted for diversion from the landfill or incineration (e.g., reuse and/or recycling).

- Materials recovered through salvage and reused off site are weighted at 200% given the importance of diverting materials to reuse versus recycling.
- Materials recovered through salvage and reused on site are not included in *MRc5: Construction and Demolition Waste Diversion* and are instead included in *MRc1: Building and Materials Reuse*.

Mixed C&D materials/recycling: This is also known as commingled recycling and refers to C&D material placed in a single container on site and/or that is later sorted and processed for recycling at an off-site facility commonly referred to as a material recovery facility (MRF).

- To determine a commingled materials recycling rate, projects must use an average diversion rate for the facility or recycling line that corresponds to the time materials were generated on the project and sent to the facility/line for diversion.
- The average recycling rate for commingled materials must exclude alternative daily cover (ADC).
- In LEED v5 there is a 35% cap on commingled recycling diversion rates if the recycling rate is not verified by a USGBC-approved third party.

Source separation/separated: This refers to segregating recyclable materials by material type before they are sent to a single-material recycler.

- Source separation of materials happens at the construction or demolition site and materials remain separated until they enter a recycling process that turns the materials into a new product. Also referred to as sole-source recycling and single-source recycling.
  - Typically, source-separated materials achieve high rates of recycling (typically 90% or higher diversion rates) by avoiding contamination associated with mixing with other material types.
  - Examples include clean wood processed for mulch, crushed concrete processed for aggregate, clean gypsum board scraps recycled into new gypsum board, and recovered carpet that is incorporated into new carpet.

- Source separation is intended to recover materials as clean as possible, ensuring higher quality and greater potential for recycling or reuse than contaminated or mixed materials. Therefore, the following examples illustrate some common scenarios that would not be considered source separation:
  - Materials that are source separated at the site but are then subsequently sent to a mixed recovery facility and mixed with other materials for recovery is not considered source separation. In this case, the materials would be considered commingled and must take the facility average diversion rate for commingled materials.
  - If materials are collected as mixed recyclable materials on site, then sent to a facility that separates the materials and recycles them, that is not considered source separation. In this case, the materials would be considered commingled and must take the facility average diversion rate for commingled materials.

Third-party-verified recycling rates: If a recycling rate for mixed C&D materials (commingled) has been third-party verified by an approved USGBC program, project teams can use the verified rate for LEED calculation purposes.

USGBC-approved recycling rates:

- Recycling Certification Institute's Certification of Real Rates (CORR) protocol meets the requirements for third-party-verified recycling rates. Their website provides facilities with certified recycling rates.[61]
- San Francisco Department of the Environment maintains a list of verified recycling rates that has been approved by USGBC. The list can be found online.[62]

Alternative daily cover: This refers to a residual product resulting from the sorting and processing of recyclable C&D debris. In places where ADC is commonplace, it is typically applied on top of exposed areas of a landfill (as an alternative to soil) at the end of each operating day to control vectors, fires, odors, blowing litter, and scavenging.

Biomass, Waste-to-Energy, and Incineration: The combustion of recovered wood materials from recycling processing (also known as wood-derived fuel or biomass) is an acceptable means of diversion and is not considered waste-to-energy for LEED project diversion reporting purposes. All other forms of waste-to-energy or incineration (other than recovered/recycled wood from the construction waste recycling process) are not considered diversion under any circumstances.

Connections//Overlaps to Other Credits

- *MRc1: Building and Materials Reuse*—Excludes on-site reuse from credit calculations (included in *MRc1: Building and Materials Reuse*)
- *MRc2: Reduce Embodied Carbon*—Reused and salvaged materials are rewarded in WBLCA.

61. Recycling Certification Institute, "Certified Facilities," accessed September 9, 2025, https://www.recyclingcertification.org/certified-facilities.
62. San Francisco Environment Department, "Construction and Demolition Registered Facilities," accessed September 9, 2025, https://www.sfenvironment.org/construction-demolition-registered-facilities.

Indoor Environmental Quality

OVERVIEW

Buildings are more than shelters—they offer stable environments with the power to enable human activities, foster health, and cultivate safety and comfort. Through the Indoor Environmental Quality (EQ) credit category, LEED v5 offers a framework to create places where more people can thrive. The rating system includes established practices for air quality, thermal comfort, daylight and views, and acoustics and incorporates holistic design considerations such as biophilia, accessibility, adaptability, and responsiveness. Using these strategies, project teams can develop buildings that welcome and care for all occupants more effectively, adapt to changing conditions, and drive long-term value.

Decarbonization

Decarbonization is integral for creating a more stable and predictable climate as well as lasting social and economic value. The reduction in fossil fuel use from energy efficiency and renewable energy measures has the co-benefit of improved air quality, especially in neighborhoods close to sources like power plants and highways. Through an IDP and collaborative planning, project teams can create spaces that are energy and resource efficient, and human-centric.

Quality of life

Human-centric design is interwoven throughout EQ, fostering diverse environments that enhance occupant well-being, improve health outcomes, and create more memorable, delightful spaces. LEED v5 builds on established approaches and advances new innovative pathways to address a broader range of human experiences and bolster occupants' quality of life.

Good indoor air quality is a cornerstone of the EQ credit category. LEED v5 offers best practices for responding and adapting to regular or episodic indoor and outdoor air pollution to reduce exposures and protect the health of occupants. Key methods to achieve that goal include improved filtration (*EQp2: Fundamental Air Quality*), designing management modes for wildfire smoke or respiratory diseases (*EQc4: Resilient Spaces*), and testing and monitoring air quality (*EQc5: Air Quality Testing and Monitoring*).

EQ credits provide additional options to support the well-being of workers and building users, including older adults and children, caregivers, and people with disabilities. For example, *EQc3: Accessibility and Inclusion* encourages careful design with best practices for physiological and neurological inclusivity, whereas *EQp1: Construction Management* outlines comprehensive construction management practices to reduce construction workers' exposure to harmful pollutants and extreme heat. Through these and other strategies, occupants benefit from better health and cognitive outcomes and increased levels of comfort and satisfaction.

Together, EQ credits and prerequisites help indoor spaces remain conducive to health and well-being even during adverse conditions.

Ecological conservation and restoration

Finally, the EQ category emphasizes the importance of dynamic spaces that foster emotional connections between people and their environments. With credit strategies that enhance access to high-quality daylight, views, and biophilic design, EQ credits incentivize ecological placemaking. Aligning building systems with natural environmental patterns, for example, through lighting or thermal patterns, can contribute to a positive occupant experience while also improving building efficiency (*EQc2: Occupant Experience*).

EQ prerequisites and credits empower project owners, occupants, and the building community to create buildings where occupants can experience a sense of belonging and stewardship toward their built environment, community, and natural world.

CROSS-CUTTING ISSUES

Floor area calculations and floor plans

For many of the credits in the EQ category, compliance is based on the percentage of floor area that meets the credit requirements. In general, floor areas and space categorization should be consistent across EQ credits. Any excluded spaces or discrepancies in floor area values should be explained and highlighted in the documentation. See Space Categorization below for additional information on which floor area should be included in which credits.

Space categorization

The EQ category focuses on the interaction between the occupants of the building and the indoor spaces in which they spend their time. For this reason, it is important to identify which spaces are used by the occupants, including any visitors (transients), and what activities they perform in each space. Depending on the space categorization, the credit requirements may or may not apply (Table 1).

Occupied versus unoccupied space

All spaces in a building must be categorized as either occupied or unoccupied. Occupied spaces are enclosed areas intended for human activities. Unoccupied spaces are places intended primarily for other purposes; they are occupied only occasionally and for short periods of time. In other words, they are inactive areas. Examples of spaces that are typically unoccupied include the following:

- Mechanical and electrical rooms
- Egress stairway or dedicated emergency exit corridor

- Closets in a residence (but a walk-in closet is occupied)
- Data center floor area, including a raised floor area
- Inactive storage area in a warehouse or distribution center

For areas with equipment retrieval, the space is unoccupied only if the retrieval is occasional.

Regularly versus irregularly occupied spaces

Occupied spaces are further classified as regularly occupied or irregularly occupied based on the duration of the occupancy. Regularly occupied spaces are enclosed areas where people normally spend time and are defined as more than one hour of continuous occupancy per person per day, on average. The occupants may be seated or standing as they work, study, or perform other activities.

For spaces that are not used daily, the classification should be based on the time a typical occupant spends in the space when it is in use. For example, a computer workstation may be largely vacant throughout the month, but when it is occupied, a worker spends 1–5 hours there. It would then be considered regularly occupied because that length of time is sufficient to affect the person's well-being, and they would have an expectation of thermal comfort and control over the environment.

Occupied spaces that do not meet the definition of regularly occupied are irregularly occupied—that is, areas that people pass through or areas used an average of less than one hour per person per day.

Examples of regularly occupied spaces include the following:

- Airplane hangar
- Auditorium
- Auto service bay
- Bank teller station
- Conference room
- Correctional facility cell or day room
- Data center network operations center
- Data center security operations center
- Dorm room
- Exhibition hall
- Facilities staff office
- Facilities staff workstation
- Food service facility dining area
- Food service facility kitchen area
- Gymnasium
- Hospital autopsy and morgue
- Hospital critical-care area
- Hospital dialysis and infusion area
- Hospital exam room
- Hospital waiting room
- Hospital diagnostic and treatment area
- Hospital laboratory
- Hospital nursing station
- Hospital operating room
- Hospital patient room
- Hospital recovery area
- Hospital staff room
- Hospital surgical suite
- Hospital solarium
- Hospital waiting room
- Hotel front desk
- Hotel guest room
- Hotel housekeeping area
- Hotel lobby
- Information desk
- Meeting room
- Natatorium
- Open-office workstation
- Private office
- Reception desk
- Residential bedroom
- Residential dining room
- Residential kitchen
- Residential living room

- Residential office, den, workroom
- Retail merchandise area and associated circulation
- Retail sales transaction area
- School classroom
- School media center
- School student activity room
- School study hall
- Shipping and receiving office
- Study carrel
- Warehouse materials-handling area

Examples of irregularly occupied spaces include the following:

- Break room
- Circulation space
- Copy room
- Corridor
- Fire station apparatus bay
- Hospital linen area
- Hospital medical record area
- Hospital patient room bathroom
- Hospital short-term charting space
- Hospital prep and cleanup area in surgical suite
- Interrogation room
- Lobby (except hotel lobby)*
- Locker room
- Residential bathroom
- Residential laundry area
- Residential walk-in closet
- Restroom
- Retail fitting area
- Retail stock room
- Shooting range
- Stairway

TABLE 1. Space Types in EQ Credits

Space Category	Prerequisite or Credit
Occupied space	EQp2: Fundamental Air Quality EQc1: Enhanced Air Quality EQc2: Occupant Experience EQc5: Air Quality Testing and Monitoring
Regularly occupied space	EQc2: Occupant Experience EQc1: Enhanced Air Quality EQc4: Resilient Spaces EQc5: Air Quality Testing and Monitoring
Quiet space	EQc2: Occupant Experience
Classroom and core learning spaces	EQc2: Occupant Experience

* Hotel lobbies are considered regularly occupied because people often congregate, work on laptops, and spend more time there than they do in an office building lobby.

INDOOR ENVIRONMENTAL QUALITY PREREQUISITE

Construction Management

EQp1
Required
New Construction

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To promote the well-being of construction workers and building occupants by minimizing environmental quality problems associated with construction and renovation.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Construction Management	

Develop and implement construction management practices for the construction and preoccupancy phases of the building. The practices must address all of the following:

- **No smoking:** Prohibit smoking during construction except in designated smoking areas located at least 25 feet (7.5 meters) from the building. Install signage that prohibits smoking during construction.
- **Extreme heat protection:** Implement measures that protect construction workers from extreme heat.
- **HVAC protection:** Keep contaminants out of the HVAC system. Do not run permanently installed equipment, if possible, but if it must be used, maintain proper filtration. Replace all air filtration media after completion of construction and before occupancy. Confirm that testing and balance work is completed with new filtration.
- **Source control:** Keep sources of contaminants out of the building and have a plan to eliminate any that are introduced.
    - Store carpets, acoustic ceiling panels, fabric wall coverings, insulation, upholstery and furnishings, and other absorptive materials in a designated area protected from moisture damage.

- **Pathway interruption:** Prevent circulation of contaminated air when cutting concrete or wood, sanding drywall, installing volatile-organic-compound-emitting materials, or performing other activities that affect indoor air quality in other workspaces.
  - Isolate areas of work to prevent contamination of other spaces, whether they are finished or not. Seal doorways and windows, or tent off areas as needed using temporary barriers.
  - Use walk-off mats at entryways to reduce introduced dirt and pollutants.
  - Use dust guards and collectors on saws and other tools.
- **Housekeeping:** Maintain a clean jobsite. Use vacuum cleaners with high-efficiency particulate filters and use sweeping compounds or wetting agents for dust control when sweeping.
- **Scheduling:** Sequence construction activities to reduce air quality problems. For renovation projects, coordinate construction activities to minimize or eliminate disruption of operations in occupied areas.

REQUIREMENTS EXPLAINED

Using established best practices during construction can protect construction workers from poor air quality and extreme heat.

The prerequisite requires projects to develop and implement construction management practices for the building's construction and preoccupancy phases. The required practices are primarily adapted from the *SMACNA IAQ Guidelines for Occupied Buildings under Construction, 2nd edition, 2007, ANSI/SMACNA 008–2008, Chapter 3*.[1] The extreme heat requirement is adapted from OSHA prevention guidance for preventing heat-related illness.[2]

The practices must address all the following criteria:

NO SMOKING

Prohibiting smoking during construction supports a healthier and safer work environment. Smoking is a fire hazard. It creates odors and elevated levels of airborne contaminants that are associated with respiratory, cardiovascular, and other health problems.[3] Although cigarette smoking has declined among U.S. workers overall, its prevalence remains high among construction workers.[4] Prohibiting smoking preserves the integrity and longevity of building materials that can absorb smoke, such as insulation and drywall.

Projects must prohibit smoking on the entire jobsite during the construction phase except in designated outdoor smoking areas. This applies to conventional cigarettes (cigarettes, cigars, pipes), cannabis (medical or recreational), and electronic smoking devices (e-cigarettes).

1. SMACNA, "IAQ Guidelines for Occupied Buildings Under Construction," n.d., https://store.smacna.org/iaq-guidelines-for-occupied-buildings-under-construction.
2. OSHA, "Heat: Prevention," accessed February 3, 2025, https://www.osha.gov/heat-exposure/prevention.
3. Mattias Öberg, et al., "Worldwide Burden of Disease from Exposure to Second-Hand Smoke: A Retrospective Analysis of Data from 192 Countries," *The Lancet* 377, no. 9760 (2021): 139–146.
4. Girija Syamlal, Brian A. King, and Jacek M. Mazurek, "Tobacco Product Use Among Workers in the Construction Industry, United States, 2014–2016," *American Journal of Industrial Medicine* 61, no. 11 (2018): 939–951.

Projects may elect to provide an outdoor designated smoking area on-site. An outdoor smoking area can be a covered pavilion with safe disposal bins for cigarettes. The area must be at least 25 feet (7.5 meters) away from the building.

Temporary signage

Projects must communicate the no-smoking policy with temporary signage that is displayed until construction completion. The exact design and content of the signs is up to the project team and can be tailored to the project location and circumstances, including accommodating safety sign guidelines.

PROTECTION FROM EXTREME HEAT

Construction workers exposed to hot environments are at risk for heat-related illnesses and injuries.[5] Construction management practices must address actions that employers and workers can take to prevent heat-related illnesses, which include heat stroke, exhaustion, cramps, and fainting.

Preventive measures include providing cool, shaded, or air-conditioned areas for rest; implementing required rest breaks; and scheduling labor-intensive activities in cooler parts of the day. Scheduling must accommodate reduced workdays for workers who are new to working in a warm environment (or returning to work) and during seasonal changes or abrupt weather changes. Provide workers with proper attire, like light-colored, breathable clothing.

Train workers on extreme heat measures to increase awareness and likelihood of successful implementation. Refer to the *IPp2: Human Impact Assessment* and *IPp1: Climate Resilience Assessment* findings to ensure the training and preventive measures are guided by a thorough understanding of the social context of the local community and workforce.

HVAC PROTECTION

Construction activities release contaminants that may unintentionally enter the building's HVAC system. Safeguard against this by avoiding use of HVAC systems during construction or, when use of the system is necessary, by ensuring the equipment has proper filtration during use.

Replace all HVAC filters prior to occupancy and after all construction activities are complete. Additionally, complete all tests and balance efforts after installing the new filters.

SOURCE CONTROL

Building materials that are exposed to the environmental conditions during construction can be soiled or degraded prior to installation. Proper storage and material handling can ensure they are protected from contaminants, dirt, debris, and moisture during the construction process.

If there is enough storage capacity on-site, keep materials in a separate, ventilated or conditioned building or storage area. Keep materials away from heavy traffic areas to limit exposure to dirt,

5. National Institute for Occupational Safety and Health, "Heat Stress and Workers," *Heat Stress*, July 11, 2024, https://www.cdc.gov/niosh/heat-stress/about/index.html#:~:text=Workers%20who%20are%20exposed%20to,heat%20storage%20within%20the%20body.

debris, and dust. Absorptive materials like carpet, ceiling panels, wall coverings, and insulation can trap moisture and cause mold and mildew growth. Cover or raise these materials off the floor.

PATHWAY INTERRUPTION

Certain construction activities, such as cutting, sawing, sanding, and painting, can result in emissions of airborne contaminants into the interior space. Their migration to adjacent spaces can result in inadvertent exposure to contaminated air, dust, debris, and odors. Proper hazard identification and appropriate control measures are necessary to safeguard health.

Control measures include the use of personal protective equipment and the use of temporary barriers to isolate emissions and prevent their spread into adjacent spaces. Examples of isolation techniques include sealing doors and windows, tenting areas with high levels of activity, or using dust guards or collectors on power tools. Additionally, when installing manufactured countertops, implement dust control measures and use personal protective equipment when sawing or sanding.[6]

For entryways and indoor pathways between construction areas and other interior spaces, use walk-off mats to minimize migration of dirt and pollutants into clean areas.

HOUSEKEEPING

Regular and thorough housekeeping can help control or eliminate workplace hazards.[7]

Sweeping

Sweep finished and hard surfaces using sweeping compounds or wetting agents, which can be oil-based, gritted, or gritless, to help control dust.

Vacuums

Use vacuums with high-efficiency filters to trap fine particles that would otherwise escape through the vacuum's exhaust, which helps to maintain a cleaner job site with better air quality.

SCHEDULING

Construction activities can be sequenced to minimize exposure and prevent adverse impacts for workers not directly involved in the construction activity.

Schedule construction activities that generate significant dust or emissions at different times or places. For example, schedule drywall finishing and carpet installation for different days or different sections of the building.

Whenever possible, install absorptive-finish materials after wet-applied materials have fully cured. For example, install carpet and ceiling tile after paints and stains are completely dry.

6. OHSA NIOSH, "Worker Exposure to Silica during Countertop Manufacturing, Finishing and Installation," OSHA NIOSH Hazard Alert 2015-106, n.d., https://osha.gov/sites/default/files/publications/OSHA3768.pdf.
7. Canadian Centre for Occupational Health and Safety, "Workplace Housekeeping," February 10, 2024, https://www.ccohs.ca/oshanswers/hsprograms/housekeeping.

In currently occupied buildings, consider relocating them before disruptive activities start in those areas to reduce their exposure to air and noise pollution. If the building is operational, communicate the construction activity schedule with workers and occupants. This may minimize foot traffic and encourage avoidance of the area, as necessary and feasible.

Schedule high-intensity activities during cool hours of the day and plan for work/rest periods and other scheduling modifications in line with the extreme heat protections.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	USGBC checklist.

REFERENCED STANDARDS

- Sheet Metal and Air-Conditioning National Contractors Association (SMACNA) IAQ Guidelines for Occupied Buildings under Construction, 2nd edition, 2007, ANSI/SMACNA 008–2008 (Chapter 3) (https://store.smacna.org/iaq-guidelines-for-occupied-buildings-under-construction).

FURTHER EXPLANATION

PROTECTION FROM EXTREME HEAT

For guidance on how to properly acclimatize workers to heat conditions, refer to the Occupational Safety and Health Administration website on "Protecting New Workers."[8]

8. Occupational Safety and Health Administration, "Heat: Protecting New Workers," accessed September 8, 2025, https://www.osha.gov/heat-exposure/protecting-new-workers.

Fundamental Air Quality

EQp2
Required
New Construction

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To design for above-average indoor air quality to support occupant health and well-being.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Investigate Regional and Local Air Quality AND	
Ventilation and Filtration Design AND	
Entryway System Design	

INVESTIGATE REGIONAL AND LOCAL AIR QUALITY

Investigate outdoor air quality in accordance with *ASHRAE Standard 62.1-2022*, Sections 4.1–4.3.

AND

VENTILATION AND FILTRATION DESIGN

Meet the requirements of *ASHRAE Standard 62.1-2022*, Sections 5 and 6. Use the ventilation rate procedure, the IAQ procedure, the natural ventilation procedure, or a combination thereof. Where the ventilation rate procedure or IAQ procedure are used, comply with the following additional provisions:

- **Filtration:** Each central HVAC system that supplies outdoor air and/or recirculated air to regularly occupied spaces must meet one of the following:
  - Minimum efficiency reporting value (MERV) of 13, in accordance with *ASHRAE Standard 52.2–2017*; OR

  - Equivalent filtration media class of ePM1 50%, as defined by *ISO 16890-2016*, *Particulate Air Filters for General Ventilation–Determination of the Filtration Performance*; OR
  - In-room air cleaning systems;
  - Systems tested for effectiveness and safety per *ASHRAE Standard 241-2023*, Section 7.4 (and Normative Appendix A). If treating for particles and gases, use systems tested for effectiveness per *ASHRAE 62.1-2022*, Addendum n. If treating for infectious aerosols, use systems tested for effectiveness per *ASHRAE Standard 241-2023*, Section 7.
- **Outdoor air measurement:** Provide outdoor airflow measurement devices for all mechanical ventilation systems with outdoor air intake flow greater than 1,000 cfm (472 L/s).

Healthcare

- For healthcare spaces, meet the requirements of Sections 6–10 of *ASHRAE Standard 170-2021*.

Residential

- For residential spaces, follow the additional dwelling unit provisions below.

Dwelling Unit Provisions

If the project building contains residential units, each dwelling unit must meet all the following requirements:

- Design and install a dwelling-unit mechanical ventilation system that complies with *ASHRAE 62.2-2022*, Sections 4, 6.6, and 6.7. Supply and balanced mechanical ventilation systems must be designed and constructed to provide ventilation air directly from the outdoors. Mechanical ventilation systems are not required when the project meets the exception detailed in *ASHRAE 62.2-2022*, Section 4.1.1.
- Design and install local mechanical exhaust systems in each kitchen and bathroom, including half baths, that comply with *ASHRAE 62.2-2022*, Sections 5 and 7. Exhaust air to the outdoors. Do not route exhaust ducts to terminate in attics or interstitial spaces. Recirculating range hoods or recirculating over-the-range microwaves do not satisfy the kitchen exhaust requirements. For exhaust hood systems capable of exhausting in excess of 400 cubic feet per minute (188 liters per second), provide makeup air at a rate approximately equal to the exhaust air rate. Makeup air systems must have a means of closure and be automatically controlled to start and operate simultaneously with the exhaust system. Use ENERGY STAR labeled bathroom exhaust fans in all bathrooms (including half baths) or performance equivalent for projects outside the U.S. An HRV or ERV may be used to exhaust single or multiple bathrooms if it has an efficacy level meeting the *ENERGY STAR Technical Specifications for Residential Heat-Recovery Ventilators and Energy-Recovery Ventilators* (H/ERVs), version 2.3 as certified by the Home Ventilating Institute.
- Unvented combustion appliances (ovens and ranges excluded) are not allowed.
- A carbon monoxide (CO) monitor must be installed on each floor of each dwelling unit, hard-wired with a battery backup. CO monitors are required in all types of units, regardless of the type of equipment installed in the unit.
- Any indoor fireplaces and woodstoves must have solid glass enclosures or doors that seal when closed. Any indoor fireplaces and woodstoves that are not closed combustion or power-vented must pass a backdraft potential test to ensure that depressurization of the combustion appliance zone is less than 5 Pa.

- Space- and water-heating equipment that involves combustion must be designed and installed with closed combustion (i.e., sealed supply air and exhaust ducting) or with power-vented exhaust or located in a detached utility building or open-air facility.

AND

ENTRYWAY SYSTEM DESIGN

Install permanent entryway systems to capture dirt and particulates entering the building at primary exterior entrances. There is no length requirement for entryway systems.

REQUIREMENTS EXPLAINED

This prerequisite requires the project team to research regional and local air quality and provide ventilation systems and design elements that effectively support air quality within the building. Healthcare and residential projects have additional considerations.

INVESTIGATE REGIONAL AND LOCAL AIR QUALITY

Indoor air quality is significantly influenced by outdoor air quality, which can be highly localized, and varies over time, season, and throughout the project site.

To understand outdoor air quality, the project must investigate regional air quality and local air quality, and consider seasonal variations. *ASHRAE 62.1* provides a template for documenting this investigation. In many regions, spring months have higher pollen levels from flowering plants and trees. Summer months in some dry, hot regions have higher levels of PM2.5 due to wildfires and ozone from photochemical smog. Air quality for the project's location will likely change over time due to climate change. For example, the periods with higher pollen levels and wildfires may increase or intensify. For this reason, information collected during the *IPp1: Climate Resilience Assessment* should be included in this investigation.

REGIONAL AIR QUALITY

Regional air quality is partially determined by reviewing compliance with national ambient air quality standards. In the U.S., use available air quality monitoring data to determine the region's status relative to acceptable levels for six regional outdoor air quality pollutants (particulate matter, carbon monoxide, ozone, nitrogen dioxide, lead, and sulfur dioxide).

LOCAL AIR QUALITY

Local air quality is typically determined through observations by walking around the project site and reviewing the neighborhood context. Examples of items to survey include facilities on-site and on adjacent properties, description of sources of vehicle exhaust on-site and adjacent properties, and identification of potential contaminant sources on the site and from adjoining properties. Most of this information will be gathered in the *IPp2: Human Impact Assessment*.

The results of the outdoor air quality investigation inform the design of critical elements of the mechanical system, including the air intake locations on the building, the filtration levels used, or the use of air cleaning devices. The investigation also helps determine exhaust and equipment locations to minimize impacts to neighboring buildings or building occupants.

VENTILATION AND FILTRATION

ASHRAE Standard 62.1-2022 is the referenced ventilation standard for most commercial buildings. The Standard establishes minimum ventilation requirements and other measures for indoor air quality that are acceptable to human occupants and that minimize adverse health effects.[9]

The Standard involves designing for indoor air quality using one of three available procedures: the ventilation rate procedure (VRP), the indoor air quality procedure (IAQP), or the natural ventilation procedure (NVP). Any combination of options may be used for compliance with this prerequisite.

IAQP VERIFICATION

If the IAQP is used to comply with this prerequisite, an extra verification step after building completion is required that involves air quality testing and conducting a subjective occupant evaluation. These verification steps are outlined in *ASHRAE 62.1-2022*, Section 7.3, Indoor Air Quality Procedure Verification.

FILTRATION

This prerequisite has additional filtration requirements beyond those in *ASHRAE 62.1. MERV 13* filtration (and ePM1 50%) is becoming standard practice. Most HVAC systems can be designed to easily accommodate this level of filtration. The filtration requirements apply to all central HVAC systems that supply outdoor air, recirculated air, or outdoor air and recirculated air to regularly occupied spaces.

MERV 13 filtration is not required for systems that supply air to warehouses or other areas addressed in the *ASHRAE* exemption for outdoor air treatment.

> **Exemption to 6.1.4**
> Systems supplying air for enclosed parking garages, warehouses, storage rooms, janitor's closets, trash rooms, recycling areas, and shipping/receiving/distribution areas are exempt.

An alternative approach that uses in-room air cleaning systems offers flexibility in meeting this prerequisite requirement for situations where design constraints make the central system-level filtration requirement infeasible or impractical. For this prerequisite, "In room air cleaning systems" are HVAC-independent systems that are integrated into the building's controls and systems. They can be located in the roomor placed in the mechanical room or nearby closet.

AIR CLEANING SYSTEMS

Air cleaning systems may be used in the design to meet the prerequisite requirements. Before selecting the air cleaning system, ensure the manufacturer has a safety testing report and manufacturer certification that the product is safe. Verified effectiveness values also provided by the manufacturer must be used in the design calculations as applicable.

9. ASHRAE, "Standards 62.1 & 62.2," 2022, https://www.ashrae.org/technical-resources/bookstore/standards-62-1-62-2.

- **Safety:** All air cleaning systems require safety testing according to *ASHRAE Standard 241-2023*, Section 7.4 and Normative Appendix A. This standard has the most up to date language to assess safety which includes addressing chemical emissions and some potential by-products, ultraviolet radiation, combustion by-products, and noise generated during operation. Testing is conducted in a specialized test chamber with specific environmental controls.[10]
- **Effectiveness for particle filtration efficiency or gaseous removal efficiency:** For systems that treat particles and gases, use only systems that have a verified effectiveness determined according to *ASHRAE Standard 62.1-2022*, Addendum n.
- **Effectiveness for infectious aerosols:** For systems being selected to treat infectious aerosols and meet the minimum equivalent clean airflow rates outlined in *ASHRAE 241-2023*, Section 5.1, for compliance with the *EQc4: Resilient Spaces* credit Option 2. Management Mode for Respiratory Diseases, to treat infectious aerosols, use only systems that have a verified effectiveness according to *ASHRAE Standard 241-2023*, Section 7.

OUTDOOR AIR MEASUREMENT

With proper outdoor airflow monitoring, facility managers can identify ventilation issues and correct problems before they impact IAQ. This credit requires airflow monitoring for larger systems—that is, those with more than 1,000 cfm (472 L/s) of outside air. Airflow monitoring is encouraged for smaller systems but not required.

Specific alarm and system control capabilities are not addressed in this prerequisite and may be designed to suit the project's specific needs.

Healthcare

- Healthcare projects must comply with *ASHRAE Standard 170-2021* for ventilation design and filtration requirements. *ASHRAE 170*, Table 7-1 specifies the minimum outdoor air changes per hour (ACH) and minimum total ACH for each healthcare space type. Both requirements must be met. For space types not covered in *Standard 170*, use *ASHRAE Standard 62.1-2022*.

ENTRYWAY SYSTEMS

Permanent entryway systems prevent dirt and particulates from entering the building. The entryway system requirement for this prerequisite is intentionally broad to accommodate more project situations and to ensure feasibility as a prerequisite requirement.

Acceptable entryway systems include permanently installed grates, grilles, slotted systems that allow for cleaning underneath, or rollout mats.

Entryway systems are required at all primary exterior entrances**.**

PRIMARY VERSUS NON-PRIMARY ENTRANCES

Primary exterior entrances are the main, designated entry points to the building. These entrances typically have the most visibility and accessibility. Design elements, like canopies or biophilic elements, often attract entry.

10. ASHRAE, "Control of Infectious Aerosols," 2023, https://www.ashrae.org/file%20library/technical%20resources/standards%20and%20guidelines/standards%20addenda/241_2023_a_20241021.pdf.

Non-primary entrances are less visible and often have limited access or are used less frequently. A non-primary entrance includes service access points or lift lobbies, side or back doors, garage and parking level entries, emergency exits, connections between concourses, and atrium entries.

Although the prerequisite only requires entryway systems at primary exterior entrances, projects may benefit from installing entryway systems at all exterior entrances of the building.

ENTRYWAY SYSTEM LENGTH AND DESIGN

There are no length requirements for the entryway systems, but it is highly recommended to use the best-practice length of at least 10 feet (3 meters) long in the primary direction of travel which allows for approximately two full steps per shoe from an average person.

Design the entryway system to accommodate and withstand specific climate conditions. Areas with high precipitation, for example, may need more absorbent mats made with mold- and mildew-resistant materials. If using rollout mats, consider selecting ones that have a solid backing. A nonporous backing captures dirt and moisture and helps prevent contaminants from collecting underneath the mat.

Regular cleaning and maintenance will extend the integrity of the entryway system. Projects are encouraged to provide routine care (typically weekly) for these systems.

Residential

- Residential projects have additional provisions for dwelling units.

DWELLING-UNIT MECHANICAL VENTILATION

ASHRAE Standard 62.2-2022 establishes minimum requirements for acceptable indoor air quality in residential buildings. Requirements for dwelling-unit mechanical ventilation systems are specified for different system types that consist of supply systems, exhaust-only systems, or balanced systems. Exhaust-only ventilation systems are only permitted in certain situations. Supply and balanced mechanical ventilation systems must provide ventilation air directly from the outdoors via a dedicated outdoor air supply duct to the unit or the HVAC equipment serving the unit. All systems must be tested and balanced before occupancy.

MECHANICAL EXHAUST SYSTEMS

ASHRAE Standard 62.2-2022 requires mechanical exhaust systems for kitchens, bathrooms, and half baths. Recirculating range hoods do not meet the requirements of the Standard.

Fans with ENERGY STAR labels (or performance equivalent for projects outside the U.S.) are required in all bathrooms (including half baths). These fans use about 50% less energy than standard models.[11] Performance equivalent for international projects means that the product meets the ENERGY STAR key product criteria for the product type outlined on the ENERGY STAR website.

UNVENTED COMBUSTION APPLIANCES

Unvented combustion appliances such as charcoal and propane grills are a significant source of air pollution and are not permitted within the project. Lease agreements should clearly identify this requirement for all residents.

11. ENERGY STAR, “Ventilation Fans,” n.d., https://energystar.gov/products/ventilation_fans.

CARBON MONOXIDE MONITORING

Hard-wired CO monitors are required for each dwelling unit. A monitor must be installed on each level for dwelling units with multiple floors. Specify CO monitors with battery backup to ensure continuous functionality in the event of a power outage.

INDOOR FIREPLACES AND WOOD STOVES

Indoor fireplaces and wood stoves must include solid glass enclosures or doors that seal when closed to reduce the risk of smoke exposure for residents.

For projects that do not include smoke control elements (closed combustion or power-venting), perform a backdraft potential test for each appliance. Testing must confirm the depressurization of the combustion appliance zone is less than 5 Pa. This is the pressure difference considered safe to ensure proper combustion air intake for natural draft appliances and to prevent dangerous back drafting while still allowing for efficient operation.[12]

DWELLING UNIT SPACE-HEATING AND WATER-HEATING EQUIPMENT

Many residences have their own water-heating equipment in the dwelling unit, including equipment that uses combustion (i.e., gas-fired water heaters). When including these systems, design and confirm that the installed systems use a closed combustion design.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Results of regional and local air quality investigation, at minimum, date and time of observations, a description of the site, observed odors or irritants, and conclusions regarding acceptability of the outdoor air quality.
			Indication of whether the building is in an area where the national guideline is exceeded (outdoor air treatment is required per *ASHRAE 62.1-2022*, Section 6.1.4).
			Calculation documents for mechanical ventilation (VRP), natural ventilation (NVP) and IAQP.
			Air cleaning systems (if used): supporting documentation for safety and effectiveness.
			IAQP: Air Quality Test Plan and Occupant Survey Methodology.
			IAQP: Completed air quality testing report including time, date, testing methods complying with credit requirements, results and limits of the tested contaminants in all locations, and lab accreditation scope for VOCs.
			IAQP: Subjective Occupant Evaluation Results.
			Floor plans or photos of entryway systems.
			Design documents confirming filter grade and implementation.

12. J. Fitzgerald and D. Bohac, "Measure Guideline: Combustion Safety for Natural Draft Appliances Through Appliance Zone Isolation," The National Renewable Energy Laboratory and U.S. Department of Energy's Building America Program, NorthernSTAR, and Center for Energy and Environment, 2014, https://www1.eere.energy.gov/buildings/publications/pdfs/building_america/measure_guide_combustion_safety_appliancezone.pdf.

(*continued*)

Project Types	Options	Paths	Documentation
			Control diagrams showing outdoor air measurement devices (where required).
			Confirmation the outdoor air measurement devices measure airflow rates.
Residential	All	All	Confirmation there are CO monitors in all units.
			Description of how each system complies with *ASHRAE 62.2-2022*.

REFERENCED STANDARDS

- ASHRAE standard 62.1-2022 (https://www.ashrae.org/technical-resources/bookstore/standards-62-1-62-2)
- ASHRAE standard 62.2-2022 (https://www.ashrae.org/technical-resources/bookstore/standards-62-1-62-2)
- ASHRAE standard 170-2021 (https://www.ashrae.org/technical-resources/standards-and-guidelines/standards-addenda/ansi-ashrae-ashe-standard-170-2017-ventilation-of-health-care-facilities)

FURTHER EXPLANATION

LIMITATIONS OF EXHAUST-ONLY VENTILATION SYSTEMS

- Exhaust-only ventilation systems are not permitted for newly constructed attached dwelling units that open directly to an enclosed corridor per *ASHRAE 62.2-2022*, Section 4.2.
- The required outdoor airflow may not be sufficiently provided. The pressure drops from outdoor air intake openings, and the space content between the openings and exhaust outlet lead to decreased outdoor air intake flow compared to exhaust fan airflow rates.
- Exhaust-fan-driven ventilation would draw air from the least resistance path and outdoor air sources of different air quality will be drawn in, which may lead to unsatisfactory indoor air quality.
- It is difficult to treat outdoor air when outdoor air quality is not appropriate for ventilation. Installing air treatment equipment is prohibitive when the outdoor air sources are not easily tracked.
- Using an isolated bathroom exhaust fan, such as a fan located in a primary bathroom, to also provide dwelling-unit ventilation might not achieve ideal distribution of ventilation air.
- Exhaust ventilation contributes to depressurization of the building with respect to outdoors. Attention should be given to depressurization and potential back drafting of combustion systems (*ASHRAE 62.2-2016* User's Manual).

No Smoking or Vehicle Idling

EQp3
Required
New Construction
Core and Shell

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To minimize exposure to tobacco smoke, smoke from tobacco substitutes or cannabis, and vehicle emissions.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**N/A**
Prohibit Smoking AND	
Prohibit Vehicle Idling	

Comply with the following requirements:

PROHIBIT SMOKING

- **Indoor smoking:** Prohibit smoking inside the building with limited exceptions (see below).
- **Outdoor smoking:** Prohibit smoking outside the building except in designated smoking areas located at least 25 feet (7.5 meters) (or the maximum extent allowable by local codes) from all entries, outdoor air intakes, and operable windows.
- **School projects:** Prohibit all smoking on-site.

AND

PROHIBIT VEHICLE IDLING

- Prohibit vehicle idling on-site.

Communicate the no-smoking and vehicle idling prohibition policy to occupants. Have in place provisions for enforcement or prohibitive signage.

Residential

- Meet the preceding requirements for all areas inside and outside the building except dwelling units and private balconies.
- For residential projects that do not prohibit smoking, each dwelling unit where smoking will be permitted must meet the following compartmentalization requirements:
  - Perform a blower door test of residential dwelling units, following the procedures in *ANSI/RESNET/ICC 380* or equivalent.
  - For each unit tested, demonstrate a maximum leakage of enclosure area that is no more than 1.5 times the thresholds identified in Table 1 ("enclosure area" refers to all surfaces enclosing the dwelling unit, including exterior and party walls, floors, and ceilings).
  - Demonstrate a weighted average leakage of enclosure area for the building, including dwelling units, that complies with the caps in the limits identified in Table 1.

TABLE 1. Caps on Air Leakage Rates

Building Conditioned Floor Area (CFA)	Pressure Test Conditions Across the Building Envelope	Maximum Air Leakage	
		New Construction	**Major Renovation**
≥ 5,000 sq. ft. (465 sq. m.)	At pressure difference of 50 pascals (0.2 in H_2O)	0.13 cfm/sq. ft. (0.65 L/s*sq. m.)	0.20 cfm/sq. ft. (1.0 L/s*sq. m.)
	At pressure difference of 75 pascals (0.3 in H_2O)	0.18 cfm/sq. ft. (0.90 L/s*sq. m.)	0.27 cfm/sq. ft. (1.35 L/s*sq. m.)
< 5,000 sq. ft. (465 sq. m.)	At 50 pascals (0.2 in H_2O)	1 ACH	1.5 ACH
	At 75 pascals (0.3 in H_2O)	1.35 ACH	2 ACH

NOTE: cfm = cubic feet per minute; L/s = liters per second; ACH = air changes per hour; L/s*sq = liters per second per square meter.

REQUIREMENTS EXPLAINED

Establishing smoke-free and idle free policies may minimize exposure to tobacco smoke, smoke from tobacco substitutes and cannabis, and vehicle emissions.

This prerequisite requires the project to prohibit smoking inside the building and its immediate vicinity, restrict smoking on-site, and prevent vehicle idling.

PROHIBIT SMOKING

Smoke-free policies effectively reduce tobacco use, protect people from secondhand smoke exposure, prevent tobacco-related illnesses and deaths, and help more people successfully quit smoking.

Smoking is prohibited for conventional cigarettes (cigarettes, cigars, pipes), cannabis (medical or recreational), and electronic smoking devices (e-cigarettes). The intent is to keep the air inside the building free from pollutants and contaminants associated with smoking.

INDOOR SMOKING

Smoking must always be strictly prohibited inside the building. Evidence of this prohibition can be obtained through a policy from the owner, facility manager, or smoke-free indoor air law.

OUTDOOR SMOKING

Smoking must be prohibited on the project site except in areas specifically designated for smoking. No smoking is permitted within 25 feet (7.5 meters) of all building openings, such as doors, windows, and ventilation intakes, to minimize the likelihood of smoke entering the building. Emergency exits do not qualify as building openings if the doors are alarmed, because alarmed doors will rarely be opened. Emergency exits without alarms qualify as building openings.

Smoking is not allowed in programmable spaces (e.g., outdoor cigar lounges or casino areas, courtyards, outdoor cafes or sidewalk seating, spaces used for business purposes).

A designated smoking area is a specific outdoor location where smoking is permitted. This can be an unenclosed pavilion with safe disposal bins for cigarettes. Business cannot be conducted in this area, but design strategies to make people feel comfortable, such as covered seating, are encouraged.

Schools

- Schools must prohibit all smoking on-site to ensure no secondhand smoke exposure to students, staff, and visitors. Banning smoking on school premises also sets a strong example for students and encourages them to adopt and maintain healthy, smoke-free lifestyles. Signage must be posted on the property line to indicate the no-smoking policy. Signage at the school helps to ensure public awareness and compliance with smoke-free environments, especially in areas where children's health is a priority. The signs serve as a clear visual reminder for staff, visitors, and passersby.

Residential

- Smoking is prohibited for all areas inside and outside residential buildings except in dwelling units and on private balconies. If smoking is permitted in a residential project, each dwelling unit where smoking will be permitted must meet the compartmentalization requirements that follow the procedures in *ANSI/RESNET/ICC 380* or equivalent. For each unit tested, demonstrate a maximum leakage of enclosure area that is no more than 1.5 times the thresholds identified in Table 1.
- Prohibiting smoking on private residential balconies is a best practice for protecting nearby nonsmoking units and balconies from environmental tobacco smoke (ETS) infiltration. Consider prohibiting smoking on balconies in lease agreements.

Exclusions

The enforcement of no-smoking policies does not extend to areas within residential healthcare projects, such as long-term care facilities, where residents may have a clinical need to smoke.

This prerequisite is not intended to prohibit or deter indigenous or other cultural ceremonial practices (e.g., smudging), which may include the combustion of tobacco and other ceremonial materials. LEED projects that accommodate cultural ceremonial practices may still pursue this prerequisite.

NO SMOKING SIGNAGE

The project team determines the placement and design of signage, which allows for flexibility to address site-specific considerations.

When communicating a no-smoking policy, use signage that includes illustrations, photographs, or clear and concise wording. Consider using explicit language such as "No smoking allowed within × feet" or "Smoking is allowed in designated smoking areas only." Signs should clearly indicate the designated smoking areas.

PROHIBIT VEHICLE IDLING

Vehicle idling is prohibited to minimize the likelihood of vehicle exhaust entering the building and to provide better air quality on-site and for the neighboring community. An idling vehicle emits harmful chemicals and gases into the air. Exposure to these chemicals can aggravate asthma, allergies, and cardiovascular and respiratory diseases. Note that some cities, campuses, and other institutions may already have vehicle idling policies in place as part of their efforts to reduce emissions.

A vehicle idling policy is a set of guidelines designed to minimize the unnecessary running of vehicle engines while stationary. Such a policy typically prohibits idling beyond a specified duration, except in certain situations where exceptions may apply. For example, exceptions might be allowed for conditions where engine idling is necessary to maintain vehicle performance or occupant comfort, during emergency or safety situations, or for vehicles with specific operational needs, such as refrigeration trucks or vehicles running specialized equipment. The policy should require all vehicles and equipment to be turned off when not in active use.

NO IDLING SIGNAGE

It is the responsibility of the project team to determine signage content and the best locations for placement. For best results, place signage where drivers are most likely to idle, such as near vehicle waiting or parking areas. The signage should inform drivers of the policy upon arrival, departure, and waiting for parking spaces or picking up passengers.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	All	Site plan or map showing the location of designated outdoor smoking areas, vehicle idling signage, location of property line, and site boundary, and indicating 25-foot (7.5-meter) distance from building openings.
			Description of project's no-smoking policy, including information on how the policy is communicated to building occupants and enforced.
			Description of project's vehicle idling policy, including information on how the policy is communicated to building occupants and enforced.
Core and Shell	All Core and Shell projects	All Core and Shell projects	Tenant guidelines communicating the building's indoor and outdoor smoking and vehicle idling policies and the locations and details of signage installed to communicate the policies.

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Project Types	Options	Paths	Documentation
Residential		Projects that permit smoking	Air leakage calculations and blower door test report.
			Confirmation that the blower door test of residential dwelling units follows the procedures in *ANSI/RESNET/ICC 380* or equivalency of testing method demonstrated.

REFERENCED STANDARDS

- None

Enhanced Air Quality

EQc1
New Construction (1 point)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To design for increased indoor air quality (IAQ) to better protect the health of building occupants.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1**
Option 1. Increased Ventilation OR	1
Option 2. Enhanced Indoor Air Quality Design	1

Design the building to exceed the requirements of *ASHRAE 62.1-2022*, Section 6. If using the VRP to comply with *EQp2: Fundamental Air Quality*, use Option 1 or Option 2; if using the IAQP, use Option 2.

OPTION 1. INCREASED VENTILATION (1 POINT)

Increase breathing zone outdoor air ventilation rates by at least 15% above the minimum rates (for 1 point, or 30% for exemplary performance) as determined in *EQp2: Fundamental Air Quality*.

Increased outdoor air rates should be provided to 95% of all regularly occupied spaces.

OR

OPTION 2. ENHANCED INDOOR AIR QUALITY DESIGN (1 POINT)

In addition to the design compounds and design limits outlined in *ASHRAE 62.1-2022*, Tables 6-5 and 6-6, design and verify enhanced IAQ using the lower design limits listed below in Table 1.

TABLE 1. Additional Design Limits for Enhanced IAQ Design

Design Compound or PM2.5	Enhanced IAQP Design Limit
PM2.5	10 μg/m^3
Formaldehyde	20 μg/m^3
Ozone	10 ppb

NOTE: ppb = parts per billion.

REQUIREMENTS EXPLAINED

The credit incentivizes designing systems that continuously provide enhanced air quality during building occupancy. For measures that provide enhanced air quality only in specific circumstances, refer to *EQc4: Resilient Spaces* for more guidance.

Both options in this credit use measures described in *ASHRAE Guideline 42-2023*, Enhanced Indoor Air Quality in Commercial and Institutional Buildings, with the goal of providing enhanced IAQ. Projects are encouraged to address additional measures from *ASHRAE Guideline 42* beyond those included in this credit.

NOTE: If the VRP is being used for *EQp2: Fundamental Air Quality*, it is expected that Option 1 for this credit will be used; however, it is not required. Review both options and choose the one that most aligns with the project goals for enhanced air quality.

OPTION 1. INCREASED VENTILATION

Providing additional outdoor airflow, above the minimum requirements for ventilation and building pressurization, can further dilute and reduce indoor air pollutants. Research has shown increased ventilation may improve cognitive performance and associated productivity and income, reduce absenteeism in commercial offices,[13] and sleep quality in residential applications.[14]

INCREASED VENTILATION THRESHOLDS

Projects must demonstrate a 15% increase in outdoor airflow rates over the minimum requirements of *ASHRAE Standard 62.1-2022*. Most large HVAC systems can accommodate a 15% increase with minimal impacts to the design and to energy consumption.

REGULARLY OCCUPIED SPACES REQUIREMENT

At least 95% of all regularly occupied spaces must have increased ventilation. This provides flexibility for projects that cannot meet the higher thresholds in every space.

EQ

13. Dusan Licina, Pawel Wargocki, Christopher Pyke, and Sergio Altomonte, "The Future of IEQ in Green Building Certifications," *Buildings and Cities*, 2, no. 1 (2021): 907–927, https://doi.org/10.5334/bc.148.
14. Pawel Wargocki, Mizuho Akimoto, Xiajoun Fan, Shin-ichi Tanabe, Chandra Sekhar, and Li Lan, "Ventilation and Sleep Quality," AIVC, 2023, https://www.aivc.org/resource/ventilation-and-sleep-quality.

DETERMINING THE INCREASED VENTILATION REQUIREMENT

The following VRP calculations determine the increased ventilation requirement:

- **Single-zone or 100% outdoor air system:** Multiply the calculated minimum outdoor airflow for the system (V_{ot}) by 1.15.
- **Multiple-zone recirculating system:** Multiply the uncorrected outdoor airflow for the system (V_{ou}) by 1.15. Multiply the breathing zone outdoor airflow for the critical zone (V_{bz} for critical zone) by 1.15. Calculate the new system ventilation efficiency (E_v) using the updated values for V_{ou} and the critical zone V_{bz} and recalculate the required outdoor air intake flow for the system (V_{ot}) using these values.

OPTION 2. ENHANCED INDOOR AIR QUALITY DESIGN

The option requires using more stringent design limits for ozone, formaldehyde, and PM2.5 to achieve enhanced levels of IAQ beyond *EQp2: Fundamental Air Quality*. This credit uses LEED-specific design targets selected by the LEED indoor environmental quality technical advisory group, with the following basis:

OZONE

A design target of 10 ppb was selected as referenced in the *Environmental Health Committee Emerging Issue Report*.[15] This number reflects the thinking that ozone indoors is harmful, and the lower the concentration the better. Studies indicate that any safe threshold would exist at very low concentrations.

FORMALDEHYDE

A design target of 20 μg/m^3 (16 ppb) was selected based on the National Institute for Occupational Safety and Health (NIOSH) recommended airborne exposure limit (REL) and FEMA goal for emergency housing.[16] Formaldehyde is a carcinogen that can irritate the skin and eyes. Long-term exposure has been associated with increased allergic sensitivity and asthma. Many building products contain formaldehyde as addressed in the *MRc3: Low-Emitting Materials*.

PM2.5

A design target of 10 μg/m^3 was selected based on the World Health Organization (WHO) global air quality guidelines, PM2.5 interim target 4.[17] This number reflects the thinking that adverse effects from PM2.5 can occur at low concentrations approaching zero, so PM2.5 should be maintained as low as reasonably achievable in interior spaces.

If air cleaning systems are used to achieve the enhanced IAQ design targets, refer to the guidance on air cleaning devices in *EQp2: Fundamental Air Quality*.

15. ASHRAE, "Emerging Issue: Ozone and Indoor Air Chemistry," 2011, https://www.ashrae.org/file%20library/communities/committees/standing%20committees/environmental%20health%20committee%20(ehc)/ehc_emerging_issue-ozoneandindoorairchemistry.pdf.
16. Chemical Insights, "Formaldehyde—A Common Air Pollutant," 2021, https://chemicalinsights.org/wp-content/uploads/FactSheet_Formaldehyde.pdf.
17. World Health Organization, "WHO Global Air Quality Guidelines: Particulate Matter (PM2.5 and PM10), Ozone, Nitrogen Dioxide, Sulfer Dioxide," 2021, https://iris.who.int/bitstream/handle/10665/345329/9789240034228-eng.pdf.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Increased Ventilation	All	Calculations documented under the fundamental air quality prerequisite.
	Option 2. Enhanced Indoor Air Quality Design	All	Documentation provided under the fundamental air quality prerequisite.

REFERENCED STANDARDS

- ASHRAE 62.1-2022 (https://store.accuristech.com/ashrae/standards/ashrae-62-1-2022?product_id=2501063)

Occupant Experience

EQc2
New Construction (1–7 points)
Core and Shell (1–7 points)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To move beyond neutral or sufficient spaces toward human-centered design that supports customization, enjoyment, and emotional connections between people and the building, thus increasing the likelihood of consistent satisfaction and ongoing stewardship.

REQUIREMENTS: NEW CONSTRUCTION

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1-7**
Option 1. Biophilic Environment	1–4
Path 1. Integrated Biophilic Design	1
AND/OR	
Path 2. Quality Views	2–3
AND/OR	
Option 2. Adaptable Environment	1
AND/OR	
Option 3. Thermal Environment	1
AND/OR	
Option 4. Sound Environment	1–2
Path 1. Mapping Acoustic Expectations for Indoor and Outdoor Spaces	1
OR	
Path 2. Acoustic Criteria for Indoor and Outdoor Spaces	2
AND/OR	
Option 5. Lighting Environment	1–6
Path 1. Solar Glare	1
AND/OR	

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ACHIEVEMENT PATHWAYS	POINTS
Path 2. Quality Electric Lighting	1
AND/OR	
Path 3. Proximity to Windows for Daylight Access	1
AND/OR	
Path 4. Daylight Simulation	1–4

OPTION 1. BIOPHILIC ENVIRONMENT (1–4 POINTS)

PATH 1. INTEGRATED BIOPHILIC DESIGN (1 POINT)

Integrate biophilic design that demonstrates each of the following five principles adapted from *The Practice of Biophilic Design* by Kellert and Calabrese[18]:

- Biophilic design requires repeated and sustained engagement with nature.
- Biophilic design focuses on human adaptations to the natural world that, over evolutionary time, have advanced people's health, fitness, and well-being.
- Biophilic design encourages an emotional attachment to the building and building location.
- Biophilic design promotes positive interactions between people and nature that encourage an expanded sense of relationship and responsibility for the human and natural communities.
- Biophilic design encourages mutual reinforcing, interconnected, and integrated architectural solutions.

AND/OR

PATH 2. QUALITY VIEWS (2–3 POINTS)

Provide occupants in the building with a view to the outdoor natural or urban environment for 75% (for 2 points, 90% for 3 points) of all regularly occupied floor area. Auditoriums, conference rooms dedicated to video conferencing, and gymnasiums may be excluded. Views into interior atria may be used to meet up to 30% of the required area.

- Views must be through glass with a visible light transmittance above 40%. If the glazing has frits, patterns, or tints, the view must be preserved. Neutral gray, bronze, and blue-green tints are acceptable.
- Views must include at least one of the following:
  - Nature, urban landmarks, or art; OR
  - Objects at least 25 feet (7.5 meters) from the exterior of the glazing.
- Occupants must have direct access to the view and be within three times the head height of the glazing.

AND/OR

18. Stephen R. Kellert and Elizabeth F. Calabrese, *The Practice of Biophilic Design*, 2015, https://www.biophilic-design.com.

OPTION 2. ADAPTABLE ENVIRONMENT (1 POINT)

Allow occupants choice and flexibility and/or the capability to adapt the space to meet their individual needs. Provide variability and/or optionality for thermal, sound, and lighting environments that invite occupants to either alter their experience and/or move between sensory zones. Include at least one accessible quiet space that allows occupants to retreat from high levels of sensory stimulation. Projects must also demonstrate at least one of the listed additional strategies.

Additional strategies

- Provide socializing, meeting, dining, and/or working areas where occupants can sit outside the main action and have permanent architectural features at their backs that creates a comfortable, semi-protected space that overlooks the larger area (prospect). Provide alternative paths that enable travel around the perimeter of the area so that people are not required to travel across a large open space.
- Provide choice in furniture configuration and a variety of seating to accommodate a wide range of body types, including seating with back rests and without arm rests.
- Provide height variety for permanently installed fixtures, like counters and sinks, and/or height-adjustable tables and desks, where appropriate.
- Provide outdoor or transitional space that encourages interaction with nature and is flexible or multiuse. Ensure the space is easily accessible for all occupants from within the building or located within 200 feet (60 meters) of a building entrance or access point.

AND/OR

OPTION 3. THERMAL ENVIRONMENT (1 POINT)

Design indoor occupied spaces to meet the requirements of *ASHRAE Standard 55-2023*, *Thermal Environmental Conditions for Human Occupancy* with errata. Investigate thermal conditions in and around the project and explain how the design considers the following:

- Thermal conditions that align and adjust with changing seasons.
- Overcooling during nontemperate seasons.
  - Design solutions for newly arrived occupants or occupants transitioning between different thermal environments to adjust to the space while maintaining an appropriately warm environment for those already in the building.
  - Design solutions for long-term occupants in transition spaces to customize their working area.
- Support for occupants carrying out different tasks that require varying levels of movement.
  - Cooling solutions for those completing high-movement tasks.

AND/OR

OPTION 4. SOUND ENVIRONMENT (1–2 POINTS)

PATH 1. MAPPING ACOUSTIC EXPECTATIONS FOR INDOOR AND OUTDOOR SPACES (1 POINT)

Determine the desired sound environment early in the design process by mapping the acoustic expectations for each primary indoor and outdoor space that is specific to the use of the space and occupant needs. Categories to consider include noise exposure, acoustic comfort and noise sensitivity, acoustic privacy, communication, and soundscape, with the following possible classifications:

- **Noise exposure zones:** High risk, medium risk, low risk, or no risk.
- **Acoustic comfort:** Loud zone, quiet zone, mixed zone, circulation, sensitive, and no specific expectations.
- **Acoustic privacy:** High speech security, confidential speech privacy, normal speech privacy, marginal speech privacy, or no privacy.
- **Communication zones:** Excellent, good, marginal, and none or no specific expectations.
- **Soundscape management:** Preserve, improve, restore, mitigate, specialize (e.g., for wellness, therapeutic, or agency in equity), or no specific expectations.

Define acoustic criteria and potential design strategies and solutions to meet the acoustic expectations for each space. Categories to consider include internally generated background noise, externally intrusive background noise, electronically generated masking sound, outdoor acoustic environment, airborne sound reverberation, sound insulation, vibration insulation, and impact noise.

OR

PATH 2. ACOUSTIC CRITERIA FOR INDOOR AND OUTDOOR SPACES (2 POINTS)

Through calculations, modeling, and/or measurements, demonstrate that the mapping exercise completed in Path 1 informed design strategies and solutions to meet acoustic criteria for at least 75% of the occupied spaces and all classrooms and other core learning spaces.

AND/OR

OPTION 5. LIGHTING ENVIRONMENT (1–6 POINTS)

PATH 1. SOLAR GLARE (1 POINT)

Provide manual or automatic (with manual override) glare-control devices in all regularly occupied spaces that will receive direct or reflected sun penetration. Spaces designed intentionally for direct sunlight may be excluded.

AND/OR

PATH 2. QUALITY ELECTRIC LIGHTING (1 POINT)

Comply with the following requirements for regularly occupied spaces:

Electric light glare control

Each luminaire shall meet one of the following requirements:

- Have calculated luminance of less than 6,000 candela per square meter (cd/sq. m.) between 45 and 90 degrees from nadir.
- Achieve a unified glare rating (UGR) of 19 or lower using the UGR tabular method for each space.

- Achieve a UGR rating of 19 or lower using software modeling calculations of the designed lighting. (Modeling must be performed as outlined in the NEMA White Paper on Unified Glare Rating[19]).

Color rendering

Use luminaires that have a color rendering index of at least 90 or that meet the color rendering requirements in Table 1, in accordance with *Illuminating Engineering Society (IES) TM-30-20.*

TABLE 1. Color Rendering Requirements Using *IES TM-30-20*

Measure		Requirement
Fidelity index	R_f	78 or higher
Gamut index	R_g	95 or higher
Red Local Chroma Shift	$R_{cs,h1}$	–1% to 15%

AND/OR

PATH 3. PROXIMITY TO WINDOWS FOR DAYLIGHT ACCESS (1 POINT)

Design the interior layout to provide at least 30% of the regularly occupied area to be within a 20-foot (6-meter) horizontal distance of envelope glazing. The glazing must have a visible light transmittance above 40%. Regularly occupied areas with visual obstructions (incapable of providing a view to envelope glazing) should be excluded from the compliant area.

OR

PATH 4. DAYLIGHT SIMULATION (1–4 POINTS)

Perform a daylight simulation analysis for the project to understand and optimize access to daylight and visual comfort. Use the calculation protocols in *IES LM-83-23* with the following clarifications:

- Calculate spatial daylight autonomy$_{300/50\%}$ ($sDA_{300/50\%}$) and annual sunlight exposure$_{1000,250}$ ($ASE_{1000,250}$) as defined in *IES LM-83-23* for each regularly occupied space in the project. $sDA_{150/50\%}$ may be used for areas without visual tasks with design targets of 225 lux.
- For any regularly occupied spaces with $ASE_{net(1000,250h)}$ greater than 20%, identify how the space is designed to address glare.
- Calculate the average $sDA_{300/50\%}$ or $sDA_{150/50\%}$ for the total regularly occupied floor area. Do not exclude spaces based on annual sunlight exposure (ASE). Points are awarded based on this calculation, according to Table 2.

TABLE 2. Points for Daylight Simulation

Average $sDA_{300/50\%}$ or $sDA_{150/50\%}$ Value	Points
≥ 40%	1
≥ 55%	2
≥ 65%	3
≥ 75%	4

19. NEMA, "White Paper on Unified Glare Rating (UGR)," 2021, accessed March 21, 2025, https://www.nema.org/standards/view/white-paper-on-unified-glare-rating-(ugr).

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REQUIREMENTS: CORE AND SHELL

ACHIEVEMENT PATHWAYS	POINTS
Core and Shell	**1-7**
Option 1. Biophilic Environment	1–4
Path 1. Integrated Biophilic Design	1
AND/OR	
Path 2. Quality Views	2–3
AND/OR	
Path 3. Outdoor Connections	1
AND/OR	
Option 2. Lighting Environment	1–6
Path 1. Solar Glare	1
AND/OR	
Path 2. Quality Electric Lighting	1
AND/OR	
Path 3. Proximity to Windows for Daylight Access	1
AND/OR	
Path 4. Daylight Simulation	1–4

OPTION 1. BIOPHILIC ENVIRONMENT (1–4 POINTS)

PATH 1. INTEGRATED BIOPHILIC DESIGN (1 POINT)

Integrate biophilic design that demonstrates each of the following five principles adapted from *The Practice of Biophilic Design* by Kellert and Calabrese[20]:

- Biophilic design requires repeated and sustained engagement with nature.
- Biophilic design focuses on human adaptations to the natural world that, over evolutionary time, have advanced people's health, fitness, and well-being.
- Biophilic design encourages an emotional attachment to the building and building location.
- Biophilic design promotes positive interactions between people and nature that encourage an expanded sense of relationship and responsibility for the human and natural communities.
- Biophilic design encourages mutual reinforcing, interconnected, and integrated architectural solutions.

AND/OR

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20. Kellert and Calabrese, *Practice of Biophilic Design*.

PATH 2. QUALITY VIEWS (2–3 POINTS)

Provide occupants in the building with a view to the outdoor natural or urban environment for 75% for 2 points or 90% for 3 points of all regularly occupied floor area. Auditoriums, conference rooms dedicated to video conferencing, and gymnasiums may be excluded. Views into interior atria may be used to meet up to 30% of the required area.

- Views must be through glass with a visible light transmittance above 40%. If the glazing has frits, patterns, or tints, the view must be preserved. Neutral gray, bronze, and blue-green tints are acceptable.
- Views must include at least one of the following:
  - Nature, urban landmarks, or art; OR
  - Objects at least 25 feet (7.5 meters) from the exterior of the glazing.
- Occupants must have direct access to the view and be within three times the head height of the glazing.

AND/OR

PATH 3. OUTDOOR CONNECTIONS (1 POINT)

Allow occupants the choice and flexibility to easily transition between indoor and outdoor environments. Provide outdoor or transitional space that encourages interaction with nature. Ensure the space is accessible for all occupants from within the building or located within 2,000 feet (60 meters) of a building entrance or access point.

AND/OR

OPTION 2. LIGHTING ENVIRONMENT (1–6 POINTS)

PATH 1. SOLAR GLARE (1 POINT)

Provide manual or automatic (with manual override) glare-control devices in all regularly occupied spaces that will receive direct or reflected sun penetration. Spaces designed intentionally for direct sunlight may be excluded.

AND/OR

PATH 2. QUALITY ELECTRIC LIGHTING (1 POINT)

Comply with the following requirements for regularly occupied spaces:

Electric light glare control

Each luminaire shall meet one of the following requirements:

- Have calculated luminance of less than 6,000 candela per square meter (cd/sq. m.) between 45 and 90 degrees from nadir.
- Achieve a UGR of 19 or lower using the UGR tabular method for each space.
- Achieve a UGR of 19 or lower using the software modeling calculations of the designed lighting. Modeling must be performed as outlined in the NEMA White Paper on Unified Glare Rating.[21]

21. NEMA, "White Paper on Unified Glare Rating."

Color rendering

Use luminaires that have a color rendering index of at least 90 or that meet the color rendering requirements in Table 3, in accordance with *Illuminating Engineering Society (IES) TM-30-20*.

TABLE 3. Color Rendering Requirements Using *IES TM-30-20*

Measure		Requirement
Fidelity index	R_f	78 or higher
Gamut index	R_g	95 or higher
Red Local Chroma Shift	$R_{cs,h1}$	–1% to 15%

AND/OR

PATH 3. PROXIMITY TO WINDOWS FOR DAYLIGHT ACCESS (1 POINT)

Design the building floorplates and interior layout to provide at least 30% of the regularly occupied area to be within a 20-foot (6-meter) horizontal distance of envelope glazing. The glazing must have a visible light transmittance above 40%. Regularly occupied areas with visual obstructions (incapable of providing a view to envelope glazing) should be excluded from the compliant area.

OR

PATH 4. DAYLIGHT SIMULATION (1–4 POINTS)

Perform a daylight simulation analysis for the project to understand and optimize access to daylight and visual comfort. Use the calculation protocols in *IES LM-83-23* with the following clarifications:

- Calculate spatial daylight autonomy$_{300/50\%}$ (sDA$_{300/50\%}$) and annual sunlight exposure$_{1000,250}$ (ASE$_{1000,250}$) as defined in *IES LM-83-23* for each regularly occupied space in the project. sDA$_{150/50\%}$ may be used for areas without visual tasks with design targets of 225 lux.
- For any regularly occupied spaces with ASE$_{net(1000,250h)}$ greater than 20%, identify how the space is designed to address glare.
- Calculate the average sDA$_{300/50\%}$ or sDA$_{150/50\%}$ for the total regularly occupied floor area. Do not exclude spaces due to ASE. Points are awarded based on this calculation, according to Table 4.

TABLE 4. Points for Daylight Simulation

Average sDA$_{300/50\%}$ or sDA$_{150/50\%}$ Value	Points
≥ 40%	1
≥ 55%	2
≥ 65%	3
≥ 75%	4

REQUIREMENTS EXPLAINED

This credit promotes spaces that are designed to enhance occupant experience through multisensory experiences, connections with nature and natural systems, spatial variability and opportunities for personalization, as well as a broader view of thermal, sound, and lighting design. Although multiple options exist to achieve the credit—and strategies will (and should) look significantly different between projects—all approaches should aid in catalyzing enjoyment and memorability of the space, which in turn, increases the likelihood of sustained satisfaction and ongoing stewardship of the building.[22]

Core and Shell projects do not have options for thermal or sound environment, because there are fewer opportunities for projects to address these topics within the Core and Shell scope. These topics may be included in the *IPp4: Tenant Guidelines*.

OPTION 1. BIOPHILIC ENVIRONMENT

PATH 1. INDOOR BIOPHILIC DESIGN

Biophilic design is based on the ethical imperative to promote human and environmental health and well-being by reconnecting people to nature and to each other within the built environment. Incorporating nature, both directly or indirectly, can offer significant benefits to physical, mental, and social health.[23] Biophilic design can also improve how we connect with others to enhance a sense of community.[24] Effective biophilic design considers cultural, geographic, and ecological contexts. Refer to the *IPp2: Human Impact Assessment* findings for relevant contextualization.

Strategies must contribute to an integrated experience and should not exist in an individual or fragmented manner. As such, there is no minimum threshold for the number of required biophilic design strategies. Instead, projects must demonstrate compliance with each of the five principles adapted from *The Practice of Biophilic Design* by Kellert and Calabrese[25]:

- **Biophilic design requires repeated and sustained engagement with nature**. Projects must incorporate nature, or natural patterns or systems, throughout multiple facets of the building.
- **Biophilic design focuses on human adaptations to the natural world that over evolutionary time have advanced people's health, fitness, and well-being**. Complying with this principle requires creating spaces that support fundamental needs of both human and natural communities, including access to daylight and fresh air, connections to nature through views and materials, opportunities for movement and sensory engagement, spaces for both social interaction and quiet refuge, and integration with local ecosystems and natural patterns.
- **Biophilic design encourages an emotional attachment to particular settings and places**. Projects must understand, embrace, and celebrate the specific ecology, culture, climate, and/or region of a project so that it is culturally and ecologically responsive and unique.
- **Biophilic design promotes positive interactions between people and nature that encourage an expanded sense of relationship and responsibility for the human and natural communities**. Complying with this principle requires design that elicits a positive emotional response and increases the likelihood of ongoing enjoyment, belonging, and ultimately, stewardship of fellow occupants and of the building.

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22. Mark DeKay and Gail Brager, *Experiential Design Schemas* (Novato, CA: ORO Editions, 2023).
23. Catherine O. Ryan, William D. Browning, and Dakota B. Walker, *The Economics of Biophilia: Why Designing with Nature in Mind Makes Financial Sense,* 2nd ed. (New York: Terrapin Bright Green, 2023), http://www.terrapinbrightgreen.com/report/eob-2.
24. Oliver Heath, Victoria Jackson, and Eden Goode, *Creating Positive Spaces by Designing for Community* (Interface, 2019), https://www.interface.com/content/dam/interfaceinc/interface/publications/brochures-collateral/emea/design-guides/community-design-guide/DesignGuide_community_emea_EN.pdf.
25. Kellert and Calabrese, *Practice of Biophilic Design*.

- **Biophilic design encourages mutual reinforcing, interconnected, and integrated architectural solutions**. Projects must apply an ecosystem approach to the design process in which the solution is greater than the sum of its parts, and nature and natural systems or processes are holistically integrated directly and indirectly throughout the project.

Any biophilic design framework may be used to demonstrate compliance. Other widely respected frameworks include the *14 Patterns of Biophilic Design* by Terrapin Bright Green[26] and the *Biophilic Design Framework*[27] developed by Stephen Kellert, Judith Heerwagen and Martin Mador.

TABLE 5. Frameworks and Strategies for Biophilic Design

Frameworks	Strategy Category	Specific Strategy
14 Patterns of Biophilic Design (Terrapin Bright Green)	Nature in the space	Visual connection with nature
		Nonvisual connection with nature
		Nonrhythmic sensory stimuli
		Thermal and airflow variability
		Presence of water
		Dynamic and diffuse light
		Connection with natural systems
	Natural analogues	Biomorphic forms and patterns
		Material connection with nature
		Complexity and order
	Nature of the space	Prospect
		Refuge
		Mystery
		Risk/Peril
The Practice of Biophilic Design (Kellert and Calabrese)	Direct experience of nature	Light
		Air
		Water
		Plants
		Animals
		Weather
		Natural landscapes and ecosystems
		Fire
	Indirect experience of nature	Images of nature
		Natural materials

26. Terrapin Bright Green, "14 Patterns of Biophilic Designs," n.d., https://www.terrapinbrightgreen.com/reports/14-patterns/.
27. Stephen R. Kellert, Judith H. Heerwagen, and Martin L. Mador, *Biophilic Design: The Theory, Science and Practice of Bringing Buildings to Life,* (Hoboken, NJ: Wiley, 2011), https://www.usgbc.org/resources/biophilic-design-theory-science-and-practice-bringing-buildings-life.

TABLE 5. Frameworks and Strategies for Biophilic Design (*continued*)

Frameworks	**Strategy Category**	**Specific Strategy**
		Natural colors
		Simulating natural light and air
		Naturalistic shapes and forms
		Evoking nature
		Information richness
		Age, change, and the patina of time
		Natural geometries
		Biomimicry
	Experience of space and place	Prospect and refuge
		Organized complexity
		Integration of parts to wholes
		Transitional spaces
		Mobility and wayfinding
		Cultural and ecological attachment to place
Biophilic Design Framework (Kellert, Heerwagen, and Mador)	Prospect	Unobstructed views
		Visual access to the horizon
		Elevated positions
	Refuge	Concealed or protected spaces
		Shelter from environmental conditions
		Secluded seating areas
	Mystery	Curving paths
		Partial views into other spaces
		Elements that entice exploration
	Complexity	Diverse textures and patterns
		Richness in visual detail
		Layered views
	Coherence	Logical organization of space
		Clear pathways
		Consistent design elements
	Change and variability	Seasonal changes
		Natural lighting variations
		Presence of water

(*continued*)

TABLE 5. Frameworks and Strategies for Biophilic Design (*continued*)

Frameworks	Strategy Category	Specific Strategy
	Risk/peril	Elements that evoke thrill or excitement
		Overlooks or balconies
		Stepping stones or bridges
	Security and safety	Easily accessible escape routes
		Clear lines of sight
		Sturdy construction

PATH 2. QUALITY VIEWS

Light, shadow, color, and patterns can create engaging spaces that enhance occupant well-being.[28] Visual connections to the outdoors, particularly through windows offering natural views, provide documented psychological and emotional benefits.[29] These benefits extend across building types and sectors, from improved patient recovery in healthcare settings to increased productivity in offices to better learning outcomes in schools.[30] Views and daylight are essential components of healthy human habitats, but this does not require excessive glazing. Strategic window placement and thoughtful interior layouts can maximize outdoor connections while maintaining building performance.

Consider findings from the *IPp2: Human Impact Assessment* that are related to the project's physical context. Identify exterior site elements that meet the view quality requirements of this credit: nature, urban landmarks, arts, or objects at least 25 feet (7.5 meters) from the exterior of the glazing.

Occupants must have direct access to the view and be within three times the head height of the glazing. For example, if the top of a window is 8 feet high, occupants must be positioned no more than 24 feet (8×3) away from that window. Account for any permanent interior obstructions in the calculations. For example, identify interior features that may block the view to the window, such as structural columns. Vertical columns smaller than 1 foot (0.3 meters) wide and horizontal features smaller than 1 foot (0.3 meters) high typically do not block views. Analysis must consider occupant positions throughout all regularly occupied areas to confirm that quality views are present for at least 75% of the total area.

Exterior views through glazing, or vision glazing, must be clear and undistorted. Projects should use bird-friendly glass—or glazing with elements visible only to birds—to maintain clear views (refer to *SSc1: Biodiverse Habitat*, Option 2. Bird-friendly Glass). Although some patterns are permitted if they maintain visibility, avoid frits, fibers, patterned glazing, or added tints that distort color balance or obstruct the views. Neutral gray, bronze, and blue-green tints typically do not distort the color balance.

Gymnasiums and Auditoriums

- Gymnasiums and auditoriums may be excluded from the quality views requirements.

28. Lisa Heschong, *Visual Delight in Architecture: Daylight, Vision, and View* (Routledge, 2021).
29. Kellert and Calabrese, *Practice of Biophilic Design*.
30. Ryan, et al., *Economics of Biophilia*.

OPTION 2. ADAPTABLE ENVIRONMENT

OPTIONALITY OR FLEXIBILITY

Adaptable environments empower occupants to control and customize their surroundings and foster a greater sense of comfort and belonging. This can take two forms: by providing occupants with the ability to move between spaces with different sensory characteristics and to adjust conditions of the space itself. Although adaptable spaces have the potential to benefit all occupants, sensory customization opportunities are particularly important for neurodivergent individuals (approximately 15%–20% of the population) who may experience heightened sensitivities to environmental factors or stimuli like noise, light, and textures.[31,32]

Rating system requirements are flexible to encourage highly specific design strategies. Projects must demonstrate optionality between zones or flexibility through personal comfort options for categories outlined in Table 6.

TABLE 6. Optionality or Flexibility Strategies

Category	Examples
Thermal environment	Personal control systems, warm or cool enclaves, radiant heating or cooling, transitional spaces that act as buffer zones between indoor and outdoor areas, or spaces with varying sun and shade exposure.
Sound environment	Combining active, high-sensory collaboration zones with quieter areas featuring sound masking, white noise, nature sounds, or acoustic alcoves for focused work, which aligns with the requirements under Option 4. Sound Environment.
Lighting environment	Combination of task lighting, dimmable lighting systems, circadian lighting that mimics natural daylight cycles in intensity and color temperature, zoned lighting controls, natural light zones, or gradual transition lighting designed to change in intensity to help eyes adjust.

Project-specific strategies that do not fit within the three categories but meet the intent of Option 2, to create diverse sensory spaces, may be submitted for compliance. This may include spatial character, degree of stimulation, or other strategies to enable people to manage their own sensory needs.[33]

QUIET ZONES

Quiet zones are required for all projects pursuing this option. Quiet zones are crucial for neurologically inclusive spaces because many neurodivergent individuals, including people with autism, experience hypersensitivity or sensory processing differences that make them more sensitive to environmental stimuli, particularly sound. For these individuals, everyday sounds that some individuals might easily filter out (like HVAC systems, conversations, or equipment noise) can trigger sensory overload and lead to increased stress, decreased focus, and difficulty communicating effectively.[34] Quiet zones must be accessible to all occupants and comfortable for extended use. Restrooms are not applicable for compliance.

31. HOK, "Designing a Neurodiverse Workplace," 2019, accessed October 26, 2021, https://www.hok.com/ideas/publications/hok-designing-a-neurodiverse-workplace/.
32. Nancy Doyle, "Neurodiversity at Work: A Biopsychosocial Model and the Impact on Working Adults," *British Medical Bulletin* 135 (October 2020): 108–125, https://doi: 10.1093/bmb/ldaa021.
33. HOK, "Designing a Neurodiverse Workplace."
34. Ana Margarida Gonçalves and Patricia Monteiro, "Autism Spectrum Disorder and Auditory Sensory Alterations: A Systematic Review on the Integrity of Cognitive and Neuronal Functions Related to Auditory Processing." *Journal of Neural Transmission* 130 (March 2023): 325–408, https://doi.org/10.1007/s00702-023-02595-9.

Adaptability Strategies

There is a menu of adaptability strategies intended to increase the number of people who can not only successfully use the space but also enjoy it. Findings from the *IPp2: Human Impact Assessment* must be used to better understand unique occupant needs to inform strategy selection. An alternative strategy may be acceptable in place of the provided strategies if it meets this intent.

OPTION 3. THERMAL ENVIRONMENT

More closely aligning indoor conditions with outdoor environments by adjusting temperature set points to correspond with exterior conditions can positively impact thermal comfort, because research indicates that occupants' thermal comfort preferences and expectations vary seasonally in relation to climate conditions.[35,36] Designing in connection with natural patterns can increase occupant comfort and result in more efficient buildings by reducing reliance on air conditioning and heating.

To prevent thermal discomfort and wasted energy, projects must carefully manage cooling systems to avoid overcooling spaces during warmer seasons. Project designs must consider seasonal temperature changes, potential overcooling during intemperate months, and the needs of occupants performing tasks with high metabolic rates. *ASHRAE Standard 55-2023* outlines methods to determine acceptable thermal conditions in mechanically conditioned spaces and in occupant-controlled naturally conditioned spaces, considering occupants' anticipated metabolic rate (activity level) and clothing as well as environmental variables such as temperature and air speed. Projects must comply with *ASHRAE Standard 55–2023* using the applicable method. To address the risk of overcooling in intemperate/warm seasons, teams must refer to *ASHRAE 55-2023,* Informative Appendix E, Sections 8.1 and 8.2.

To avoid discomfort or physiological stress for people transitioning between thermal environments, specifically occupants entering from the outdoors, create zones with intermediate temperatures to allow gradual acclimatization and consider using air movement as a cooling strategy. Transition spaces include spaces at the indoor/outdoor boundary (e.g., lobby entrance). Projects must also account for long-term occupants of these areas including, but not limited to, receptionists or security guards, and so forth. Provide adaptability options such as local air speed reduction to allow long-term occupants to customize their microclimate. (These design strategies may also contribute to Option 2. Adaptable Environment.)

Provide thermal comfort support for occupants carrying out tasks requiring varying levels of movement. This must include considerations for occupants completing metabolically demanding tasks.

35. Lyu Yue and Chen Zhongqing, "Seasonal Thermal Comfort and Adaptive Behaviours for the Occupants of Residential Buildings: Shaoxing as a Case Study," *Energy and Buildings* 292 (August 2023): 113165, "https://doi.org/10.1016/j.enbuild.2023.113165.
36. Charles Munonye, "The Influence of Seasonal Variation of Thermal Variables on Comfort Temperature in Schools in a Warm and Humid Climate," *Open Access Library Journal* 7, no. 9 (September 2020): e6753, https://doi.org/10.4236/oalib.1106753.

OPTION 4. SOUND ENVIRONMENT

PATH 1. MAPPING ACOUSTIC EXPECTATIONS FOR INDOOR AND OUTDOOR SPACES

Mapping involves establishing acoustic expectations for the project spaces and identifying design targets that increase the likelihood of achieving those expectations. Performing this mapping early in the design process minimizes the potential of locating incompatible spaces adjacent to each other.

The mapping exercise must include documenting acoustic expectations based on intended space function and related occupant needs. Standardized classifications for typical acoustic expectations may be used for this mapping exercise (Table 7). Teams may refer to the USGBC worksheet to help guide this mapping process.

TABLE 7. Mapping Acoustic Expectations

Noise Exposure Zones	Acoustic Comfort	Acoustic Privacy	Communication Zones	Soundscape Management
High risk, Medium risk, Low risk, No risk.	Loud zone, Quiet zone, Mixed zone, Circulation, Sensitive, No specific expectations	High speech security, Confidential speech privacy, Normal speech privacy, Marginal speech privacy, No privacy	Excellent, Good, Marginal, None, No specific expectations	Preserve, Improve, Restore, Mitigate, Specialized (e.g., Wellness, Therapeutic, Agency in equity), No specific expectations

For each primary indoor and outdoor space**,** the mapping exercise is continued to identify acoustic criteria and subsequent design strategies or solutions that if implemented in the design increase likelihood of the project to meet the desired expectations. Example acoustic criteria and design targets are listed in Table 8.

TABLE 8. Common Acoustic Criteria

Acoustic Criteria Category	Example Design or Performance Target	
	Acoustic Criteria Threshold	Threshold Reference
Internally generated background noise	e.g., 35 dBA	e.g., *ANSI S12.60-2010*
Externally intrusive background noise		
Electronically generated masking sound	None	N/A
Outdoor acoustic environment	e.g., Daytime: 55 dBA, Nighttime: 50 dBA	e.g., local code
Airborne sound reverberation	e.g., 0.7s	e.g., *AS/NZS 2107:2016*
Sound insulation	e.g., STC 45	e.g., FGI Guidelines
Vibration insulation	e.g., IIC 50	e.g., WELL Beta Feature Impact Noise Management
Impact noise	e.g., NISR	

PATH 2. ACOUSTIC CRITERIA FOR INDOOR AND OUTDOOR SPACES

This pathway builds off the mapping exercise in Path 1. It requires using generally accepted engineering practices to demonstrate the project design is likely to meet the acoustic expectations outlined for the space. The acoustic consultant will select calculations, modeling methods, or measurements as appropriate for the acoustic criterion. For example, if the acoustic criterion is internally generated background noise below 35 dBA, the project might calculate sound pressure levels or measure sound pressure levels in the completed space.

The acoustic environment is particularly important to consider when designing classrooms and other core learning spaces because it can affect students' learning and teacher health and well-being. For this reason, to earn this path all classroom or core learning spaces must comply with the acoustic criteria defined in Path 1.

For other project types, more flexibility is provided for projects prioritizing better acoustics in targeted environments. In occupied spaces, 75% must comply with the acoustic criteria defined in Path 1.

OPTION 5. LIGHTING ENVIRONMENT

PATH 1. SOLAR GLARE

When designing spaces with daylight, consider solar glare, which can significantly disrupt visual comfort and impact thermal comfort. Evaluate how natural light might create uncomfortable reflections on digital screens and potentially interfere with tasks requiring visual concentration.

Projects must provide glare-control devices for all transparent glazing in regularly occupied spaces. The requirement applies to transparent glazing, so diffused and translucent glazing systems do not require glare-control devices. All glare-control devices must be operable by the building's occupants to address unwanted glare and to allow for active participation between the building and the building occupant. Automatic devices with user override and exterior shading designs such as awnings, louvers, and shading screens are acceptable.

Glare-control devices are not required for spaces designed specifically for direct sunlight such as atriums or solar collection areas where direct sunlight is part of the design intent. In these cases, teams must establish a clear rationale, articulate the benefits, and ensure alignment with project goals. For example, the space may be intentionally designed to support Option 1. Integrated Biophilic Design, or Option 2. Adaptable Environments.

TABLE 9. Glare Control Devices

Acceptable Glare-Control Devices	Unacceptable Glare-Control Devices
Interior window blinds	Fixed fins
Interior shades	Fixed louvers
Curtains	Dark color glazing
Movable exterior louvers	Frit glazing treatment
Movable screens	Additional glazing treatments
Movable awnings	

PATH 2. QUALITY ELECTRIC LIGHTING

Properly focused lighting at the right level and quality enhances concentration and minimizes distractions. Poorly lit spaces can cause headaches and eye discomfort while well-illuminated work environments can help reduce stress levels and improve the overall well-being of employees. Electric lighting glare can significantly impact specific populations (e.g., aging adults and neurodivergent individuals) by causing discomfort, visual strain, and even emotional or cognitive disruptions.[37]

Projects must meet both Electric Light Glare Control and Color Rendering requirements within all regularly occupied spaces. Exceptions to the electric glare requirements include wallwash fixtures properly aimed at walls, as specified by manufacturer's data, indirect uplighting fixtures, provided there is no view down into these uplights from a regularly occupied space above, and any other specific applications (i.e., adjustable fixtures). Exceptions to the color rendering requirements include non-white light sources used for decorative color effects that are in addition to the general illumination.

Electric light glare control can be documented based on individual luminaire specifications (luminance) or for the space as a whole (UGR).

Luminance

Minimizing light fixture luminance helps reduce disability and discomfort glare. The threshold, 6,000 candela per square meter (cd/sq. m.) between 45 and 90 degrees from nadir, was selected to align with *WELL v2 Feature L04–Electric Light Glare Control*.[38] Luminance information for the luminaire can be found in manufacturer specifications.

Unified Glare Rating

The UGR is a measure of the discomfort produced by a lighting system along a psychometric scale of discomfort. The value of 19 corresponds to just acceptable glare. The UGR approach requires software modeling calculations of the designed lighting. Modeling must be performed as outlined in the NEMA White Paper on Unified Glare Rating.[39]

Color Rendering: Color Rendering Index

Color Rendering Index (CRI) is the most widely adopted method for evaluating color fidelity. The CRI requirement of 90 or above indicates that the light source closely mimics natural light, which allows colors to appear vibrant and true to their actual hue. This is especially important in environments where color differentiation is critical, such as in medical facilities, retail spaces, or galleries. CRI information for the luminaire can be found in manufacturer specifications. The luminaire must have a CRI of at least 90.

Color Rendering: *TM-30-20*

TM-30 is a newer method developed by the IES for measuring color fidelity. The luminaire requirements for this credit are based on the Priority Level 1 criteria for the Preference design

37. Silfia Aryani, Afif Kusumawanto, Jatmika Suryabrata, and Dinta Wijaya, "The Correlation of Lighting and Mood in the Workplace: Digital Image-Based Research," *Journal of Graphic Engineering and Design* 15, no. 1 (2024): 23-31, https://doi.org/10.24867/JGED-2024-1-023.
38. WELL Standard v2, "Electric Light Glare Control," n.d., https://v2.wellcertified.com/en/wellv2/light/feature/4.
39. NEMA, "White Paper on Unified Glare Rating."

intent recommendation in *TM-30-20 Annex E* (Table 10). This threshold was selected to align with *WELL v2 Feature L08–Electric Light Quality.*[40]

TABLE 10. Color Rendering Requirements for TM-30 Method

Measure		Requirement
Fidelity index	R_f	78 or higher
Gamut index	R_g	95 or higher
Red Local Chroma Shift	$R_{cs,h1}$	–1% to 15%

PATH 3. PROXIMITY TO WINDOWS FOR DAYLIGHT ACCESS

This path involves designing the building floorplates and interior layout to have regularly occupied areas located in close proximity—within 20 feet (6 meters)—of envelope glazing (e.g., windows, curtain walls, or other transparent elements in the building façade). The 30% threshold serves as a baseline or entry-level standard that aligns with *WELL v2 Precondition L01–Light Exposure* and encourages projects to incorporate daylight access into the design.[41] Projects aiming for more extensive daylighting strategies should explore Path 4. Daylight Simulation.

PATH 4. DAYLIGHT SIMULATION

Daylight is dynamic and highly dependent on local climate and site conditions, with daily and seasonal variations. These dynamic qualities can be explored during the design process using daylight simulation tools and daylight performance metrics standardized by the daylight design community. Incorporating simulation early in the design phase supports the optimization of building form, window placement, façade elements, and interior configurations to achieve the best possible balance of natural light.

Occurrences of direct sunlight can be minimized with thoughtful design, but daylight glare and reflections will likely still be needed and desired by occupants for certain parts of the day or year. For this reason, it is highly recommended to also pursue Option 5. Path 1. Solar Glare.

This path uses a tiered point system with four thresholds of increasing stringency based on spatial Daylight Autonomy (sDA) calculations, which indicates how much of a space receives sufficient daylight throughout the year. According to research conducted under the *Illuminating Engineering Society's LM-83* standard,[42] spaces with sDA values of 75% or higher provide preferred levels of daylight, whereas spaces with sDA values between 55% and 75% achieve nominally acceptable daylight levels. The credit's highest point threshold aligns with this research by requiring the preferred sDA level of 75% or greater.

Building Model

The model must be sufficiently detailed and complete to ensure accurate predictions of daylight performance. A simulation checklist must be used to support high-quality modeling practices. Checklist details are outlined in *IES Standard LM-83*, Section 4. Details include but are not limited to:

40. WELL Standard v2, "Electric Light Quality," https://v2.wellcertified.com/en/wellv2/light/feature/8.
41. WELL Standard v2, "Light Exposure," https://v2.wellcertified.com/en/wellv2/light/feature/1.
42. Illuminating Engineering Society LM-83 Standard, https://webstore.ansi.org/preview-pages/IESNA/preview_IES+LM-83-12.pdf?srsltid=AfmBOooGs6Y6LIYypeU1Kcroi6lXg3hKS6PcEH-TjlwvLLgnEQnTg9rm.

- Exterior details (e.g., neighboring buildings/obstructions, trees, ground plane)
- Orientation of the building, in relation to true north, is as designed
- Geometry of the space is accurately modeled (e.g., wall thicknesses, angled ceilings, walls, fenestration surfaces, interior partitions and furniture)
- Blinds and window groupings and glazing properties
- Interior surface reflectance/material properties
- Local climate data (e.g., typical meteorological year [TMY] weather data files)

Daylight Performance Metrics

Two daylight performance metrics must be calculated: spatial daylight autonomy (sDA) and ASE. Both metrics are outlined in the IES standard *LM 83-23: Approved Method: IES Spatial Daylight Autonomy (sDA)* and *Annual Sunlight Exposure (ASE).*

The sDA method assesses the prevalence of daylight over the course of a year. Calculate sDA for each regularly occupied space and calculate an average sDA across the total regularly occupied floor area. Include all regularly occupied spaces regardless of the ASE results. This approach intentionally differs from the calculation procedure outlined in the *LM-83* standard to accommodate the wide range of project types and locations that pursue LEED. LEED associates points to daylit areas, despite the glare risk. Some designers find it confusing to exclude overlit areas from daylight calculations. Use $sDA_{300/50\%}$ for all spaces except areas without visual tasks.

REGULARLY OCCUPIED AREAS WITHOUT VISUAL TASKS

Some spaces in the indoor environment are used for less critical visual tasks like walking through spaces or performing nonvisually demanding activities. Adequate daylight in these spaces is 150 lux in *LM-83*. These spaces may be identified by reviewing IES horizontal ambient illuminance design targets for spaces with illuminance targets of 225 lux or less. For these spaces, calculate sDA using $sDA_{150/50\%}$ as an alternative to the standard $sDA_{300/50\%}$.

ASE assesses the risk of visual discomfort from too much sunlight in the space. Two variations of ASE are introduced in the 2023 version of the *LM-83* standard. ASE_{net} is used for this LEED credit to encourage LEED projects to use automated glare control if desired. Calculate $ASE_{net1000,250}$ for each regularly occupied space. ASE is only calculated on a room-by-room basis.

Regularly occupied spaces with $ASE_{net(1000,250h)}$ greater than 20%

Based on IES's experience to date and analysis of the study data, an ASE_{net} of 20% or higher is a level of concern. For spaces where ASE_{net} exceeds 20%, projects must work with the designer to consider architectural methods to reduce the risk of sunlight penetration or consider the use of an automated daylight management system in as many window areas necessary to guarantee that it can avoid excessive occurrence of direct sunlight within the space.

Auditoriums

- Auditoriums may be excluded from the daylight requirements.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Biophilic Environment	Path 1. Integrated Biophilic Design	Report/narrative identifying, classifying, and explaining each of five biophilic design principles USGBC Template.
			Evidence of the project's indoor biophilic design features (e.g., any one of the following: contract documents, photographs, renderings, architectural mood board).
		Path 2. Quality Views	Percentage of regularly occupied area with access to views (%).
			USGBC calculator or a quality view simulation report.
			Architectural drawings that demonstrate direct access to the view and qualifying distance from glazing or a quality view simulation report and report checklist.
		Path 3. Outdoor Connection Core and Shell only	Evidence of occupant access to outdoor or transitional space and opportunities for interaction with nature (e.g., floor and site plans, contract documents, photographs, narrative).
	Option 2. Adaptable Environment	All	Report/narrative identifying, classifying, and explaining each variability and/or optionality strategy for thermal, sound, and lighting environments.
			Optional evidence of variability and/or optionality strategy for thermal, sound, and lighting environments (e.g., contract documents, photographs, product information from the manufacturer).
			Identification of one or more accessible quiet space.
			Identification of the additional adaptability strategy in the project (prospect areas, furniture, permanently installed fixtures, outdoor transitional space).
		Prospect areas	Evidence of the project's prospect spaces and alternative paths of travel (e.g., contract documents, photographs, narrative).
		Furniture	Evidence of the project's furniture configuration choices and variety of seating (e.g., contract documents, photographs, product information from the manufacturer).
		Permanently installed fixtures	Evidence of the height variety for permanently installed fixtures (e.g., contract documents, photographs, product information from the manufacturer).
		Outdoor transitional space	Evidence of occupant access to outdoor or transitional space and opportunities for interaction with nature (e.g., floor and site plans, contract documents, photographs, narrative).
		Alternative strategies	Evidence of an alternative adaptability strategy meeting the intent of the credit.
	Option 3. Thermal Environment	All	Design documentation per *ASHRAE* 55-2023.
			Identify transition spaces and spaces with occupants carrying out different activities and provide design documents that show solutions for these spaces to address various thermal comfort requirements.

(continued)

Project Types	Options	Paths	Documentation
All	Option 4. Sound Environment	Path 1. Mapping Acoustic Expectations for Indoor and Outdoor Spaces	USGBC calculator.
			Potential design strategies and solutions the project could use to meet the acoustic expectations for each space.
		Path 2. Acoustic Criteria for Indoor and Outdoor Spaces	Calculations, modeling, and/or measurements to demonstrate how the strategies and solutions contribute to accomplishment of the acoustic criteria identified in the USGBC calculator.
	Option 5. Lighting Environment	Path 1. Solar Glare	Contract documents highlighting provision of glare-control devices in all qualifying regularly occupied spaces.
			Floor plans highlighting regularly occupied spaces.
		Path 2. Quality Electric Lighting	Lighting specifications with luminance values.
			Documentation demonstrating required UGR using UGR Tabular Method or software modeling calculations of the designed lighting.
			Lighting specifications with color rendering information.
		Path 3. Proximity to Windows for Daylight Access	USGBC calculator or a daylight simulation report.
			Evidence of proximity to envelope glazing (e.g., floor plans and sections, showing furniture [at least within scope], identifying all space types and whether they are regularly or irregularly occupied, indicating the horizontal distance from regularly occupied areas to the envelope glazing).
			Evidence of the visible light transmittance for each regularly occupied space (e.g., contract documents).
		Path 4. Daylight Simulation	Average sDA value for all regularly occupied floor area (%).
			Daylight simulation report and model simulation checklist.
			Evidence of the simulation inputs including model simulation checklist.
			Explain which regularly occupied spaces have a ASEnet(1000,250h) greater than 20%. How was the space designed to address glare?
			Explain which areas used $sDA_{150/50\%}$ in the simulation, and the tasks that take place there, justifying that design targets of only 225 lux are needed.

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REFERENCED STANDARDS

- *The Practice of Biophilic Design* by Kellert and Calabrese (https://www.biophilic-design.com/_files/ugd/21459d_81ccb84caf6d4bee8195f9b5af92d8f4.pdf)
- *14 Patterns of Biophilic Design* by Terrapin Bright Green, (https://www.terrapinbrightgreen.com/reports/14-patterns)
- Biophilic Design Framework developed by Stephen Kellert, Judith Heerwagen and Martin Mador, (usgbc.org/resources/biophilic-design-theory-science-and-practice-bringing-buildings-life)
- ASHRAE 55 (https://www.ashrae.org/technical-resources/bookstore/standard-55-thermal-environmental-conditions-for-human-occupancy)

- Illuminating Engineering Society (IES) TM-30-20 (https://www.lightingservicesinc.com/sites/default/files/2023-04/ies_technical_memorandum_tm-30-20-e1_0.pdf)
- WELL v2 (https://wellcertified.com/en/wellv2/overview.
- LM-83 (https://webstore.ansi.org/preview-pages/IESNA/preview_IES+LM-83-12.pdf)

FURTHER EXPLANATION

OPTION 1. BIOPHILIC ENVIRONMENT

CORE AND SHELL ONLY

Path 3. Outdoor Connections (1 point)

- Outdoor connections can take on two forms: access to direct outdoor space (such as those outlined in *SSc2: Accessible Outdoor Space*) or certain transitional spaces.
- Outdoor spaces do not need to meet the spatial requirements in *SSc2: Accessible Outdoor Space*; however, they must demonstrate design that allow occupants access to nature or natural elements with a low level of effort. Examples include terraces, porches, and rooftop gardens. If the space cannot be directly accessed from the building interior, locate it within the specified 200-foot radius of any building entrance, connected by accessible pathways.
- Transitional spaces may connect the inside to the outside or any one area to another. Regardless, transitional space must actively engage occupants with natural systems through features like natural ventilation, daylighting strategies, or corridors demonstrating biophilic design. Examples include breezeways, sliding walls, or garage doors. This creates a sensory bridge that allows occupants to gradually acclimate between environments—from the controlled interior to the dynamic exterior—while providing opportunities for restoration, social interaction, and connection to seasonal and diurnal cycles.

OPTION 2. ADAPTABLE ENVIRONMENT

- Projects may use a combination of strategy categories (choice and flexibility) but must apply them to, at least, each of the three required environments: thermal, sound, and lighting.
- Providing occupants the ability to move between sensory zones (choice).
  - Sensory zones are spaces perceived by the senses to have different characteristics that, when implemented effectively, allow people to manage their own sensory needs, particularly individuals who are hyper- or hypo-sensitive to environmental factors or stimuli. Projects must demonstrate a combination of more than one sensory zone, including low, medium, and/or high stimuli/sensory zones. For guidance on sensory zones, refer to the table from "Creating Positive Spaces by Designing for Cognitive and Sensory Wellbeing," by Interface in collaboration with Oliver Heath Design, shown below.[43]

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43. Interface, "Creating Positive Spaces by Designing for Cognitive and Sensory Wellbeing," pp. 42–44, accessed September 8, 2025, https://www.interface.com/content/dam/interfaceinc/interface/publications/brochures-collateral/emea/design-guides/cognitive-and-sensory-guide/DesignGuide_Cognitive%20and%20Sensory%20Wellbeing_EN.pdf.

TABLE 11. Design Features for Cognitive and Sensory Wellbeing in the Workplace

Low Sensory Threshold *Need little or no noise, no scenting, and muted colors; visual and acoustic privacy; and overall tactile comfort (which includes vestibular and proprioception).*	
Visual comfort	• Provide shielded and segregated areas to work • Positioning of desks to minimize number of sightlines and views over movement in a space (consider angle & layout of furniture) • High backed furniture and refuge pods • Visually lowered ceiling heights • Plain partitions, desk dividers and curtains for screening • Use low level circadian lighting systems through the space with adequate task lighting provided by individual desk lamps • Use pastel colours • Reduce clutter using enclosed storage • Focused spaces away from risk and peril/ drops and heights i.e. when looking out of skyscraper (vertigo)
Acoustic comfort	• Establish spaces as quiet zones • Reduce reverberation time using acoustic curtains, ceiling and wall panels, soft furnishings and insulation • Use double or triple glazing, or silent glass, to block sound from internal meeting rooms or external sounds such as traffic and roadworks • Reduce impact of intermittent mechanical noises through your specification of HVAC systems, and avoid loud clunky keyboards • Zones and booths for concentrated work e.g.: soundproof modular meeting pods or call booths
Tactile comfort	• Ensure furniture isn't wobbly (spinning chairs made optional) and flooring isn't uneven or squeaky • Cushioned mouse mats on desks • Hand rails to support on stairs • Use soft, non-abrasive, and natural materials • Give occupants a level of control over thermal comfort e.g.: heated desks, accessible thermostats, operable windows • Reduce proximity issues–larger desks that aren't too close together, ensure walkways are kept to a minimum and away from desks to prevent people brushing past each other • Zone & create paths using flooring to distinguish between circulation routes and work areas • Position focused work spaces away from airflow changes
Medium Sensory Threshold *Can handle some sensory stimulation but generally prefer low background noise, harmonious colour, not necessarily concerned with total privacy, but cannot handle to much activity in the space.*	
Visual comfort	• Permeable screening • Subtly planted partitions or desk dividers • Positioning of desks to reduce number of sightlines and views over movement in a space (consider angle & layout of furniture) • Include some high-backed seats • Use diffused lighting, window films, adjustable blinds, brise soleil, and deciduous planting outside windows • Use circadian lighting systems throughout the space • Using tones of the same colour to create a harmonious colour palette • Include macro images of nature or artworks that show simple natural forms • Use simple nature inspired patterns in the harmonious colour palette • Use simple biomorphic shapes and forms - organic shaped furniture and curved edges • Introduce plants on desks in desk dividers and on shelves
Acoustic comfort	• Reduce reverberation time using plants and planting schemes, as well as acoustic curtains, ceiling and wall panels, soft furnishings and insulation • Incorporate ways to signal 'do not disturb' (i.e. red/green lights above desks) • Nature sounds: Incorporate speakers within a space to enable low level positive nature sounds to be played • Reflective surfaces: include wood as well as soft furnishings
Tactile comfort	• Give occupants a level of control over thermal comfort e.g.: heated desks, accessible thermostats, operable windows • Include a choice of large and small desks, some in clusters and others more isolated on the periphery • Use flooring to zone paths away from desks

(continued)

TABLE 11. Design Features for Cognitive and Sensory Wellbeing in the Workplace (*continued*)

High Sensory Threshold *Need sensory stimulation, background noise is good, can handle brighter colours, not necessarily concerned with privacy, prefer general hubbub and activity*	
	• Position focused work spaces away from airflow changes • Organically shaped handrails and door handles (or anything else occupants might touch and interact with) • Incorporate natural materials within seating, table tops etc. • Encourage movement with adjustable standing desks
Visual comfort	• NRSS, such as kinetic sculptures, fish tanks & dappled light • Plenty of windows, glass partitions to allow natural light & circadian lighting systems throughout the space • Use more intense pops of colour in places, and some contrasting tones to visually stimulate (don't overdo it - solid red walls will be too much, even for high thresholds) • A mixture of nature videos, images and artworks of luscious landscapes • Use more complex nature inspired patterns in places • Include biomorphic shapes and forms, such as fractals, organic shaped furniture and tree-like columns • Different types of plants at a range of heights, with contrasting scale, structure and leaf size—potted plants, planting systems or green walls • Create prospect views—raised platforms at the edge of a room, long sightlines through a space, desks perpendicular to windows to allow views of the outdoors & onto natural elements
Acoustic comfort	• Nature sounds: create a subtle soundscape to mimic the acoustics of a natural setting by varying heights and positions of speakers according to what they are playing • Reflective surfaces: encourage live acoustic environments (hubbub) by incorporating surfaces such as wood, stone, and glass • In kitchenettes, cafes and social areas include low rhythmic music without vocals/lyrics
Tactile comfort	• Use plants to line walking routes; these can be brushed past as occupants move around the space • Incorporate tactile natural materials within seating, table tops etc. that excite the senses when touched, according to area (within seating, table tops, handles, handrails, shelving) • Organically shaped handrails and door handles (or anything else occupants might touch and interact with) • Use waney/ live edged wooden countertops • Cool and tactile marble in bathrooms and kitchens • Standing/walking desks • Create a 'no shoe' zone with pebbles in areas of the floor and textured mats under desks • Mats under desks to act as wobble boards, space for Pilates and yoga, balance balls instead of chairs • Encourage movement e.g. clear circulation routes, winding paths (mystery), routes for walking meetings, destination points such as outdoor spaces, stairs rather than lifts • Encourage positive interactions: collision spaces such as townhall steps, eating places, triangulation • Create dense zones for people to work in closer proximity to one another and encourage interactions i.e. smaller desks closer together • Create subtle sensory contrasts throughout (think woodland) e.g. variation in flooring surfaces or materials; use hard wooden or leather surfaces in combination with softer furnishings

- Providing occupants the ability to adjust the conditions of the space itself (flexibility).
  - To increase comfort and feelings of belonging in the space, adaptive behaviors are highly beneficial. Adaptive behaviors include, but are not limited to, actions like adjusting the lighting or thermostats, opening windows, or turning on fans. Therefore, the indoor environment must provide provisions for adaptive opportunities.
  - Personal comfort systems contribute to this option. This does not include portable space heaters.
- Providing quiet zones.
  - Quiet zones, or quiet spaces, include designated areas with low sensory stimuli, accessible to any occupant within a project, that serve as a comfortable respite from high stimuli areas, including, but not limited to, areas with HVAC or equipment noise, conversational noise, and consistent sensory stimuli.

- For guidance on quiet and restorative spaces, refer to Jean Hewitt's "Neurodiversity and the Built Environment–PAS 6463:2022."[44]

OPTION 5. LIGHTING ENVIRONMENT

SPATIAL DAYLIGHT AUTONOMY: FURTHER DEFINITION IN THIS CONTEXT

- Spatial daylight autonomy (sDA) is a metric that describes the sufficiency of ambient daylight levels in interior environments across the year, while also accounting for the operation of daylight management devices (e.g., blinds) that limit direct sunlight penetration into the space (*ANSI/IES LM-83-23*). For LEED, daylight sufficiency is selected at a minimum illuminance level of 300 or 150 lux.

OPTION 4. SOUND ENVIRONMENT

ACOUSTIC PROFESSIONALS

- Professionals expected to be capable of serving as acoustical consultants for this mapping exercise typically have professional designations in the field of acoustics and are affiliated with one of the acoustical organizations outlined in Table 11. The International Commission for Acoustics also serves as a reference point for the acoustic community and maintains an extensive, but not exhaustive, list of national and regional societies focused on acoustics.[45]

TABLE 12. Acoustic Organizations that Provide Professional Designations

Country	Organization
North America	Institute of Noise Control Engineering
	Acoustical Society of America
	National Council of Acoustical Consultants
South America	IberoAmerican Federation of Acoustics (FIA)
Europe	European Acoustics Associations
	Institute of Acoustics
Asia	Western Pacific Acoustics Commission
Oceania	Australian Acoustical Society
	Association of Australasian Acoustical Consultants
	Acoustical Society of New Zealand

44. Jean Hewitt, "Neurodiversity and the Built Environment–PAS 6463:2022," BSI, 2022.
45. International Commission for Acoustics, "Our Members," accessed September 8, 2025, https://www.icacommission.org/our-members/.

Accessibility and Inclusion

EQc3
New Construction (1 point)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To support the diverse needs of occupants and increase widespread usability of the building to foster an individual and collective sense of belonging.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1**
Accessibility and Inclusion Strategies	1

COMPLY WITH LOCAL ACCESSIBILITY CODES

All projects must support access for those with physical disabilities through designs meeting all locally applicable accessibility codes identified in *IPp2: Human Impact Assessment.* If there is no code in place, include the following strategies:

- Accessible routes or regularly used exterior building entrances have ramps to accommodate elevation change.
- All doors meant for human passage have a minimum clear width of 32 inches (0.86 meters).
- Reception desks, security counters, and service counters all have a front approach, wheelchair accessible section.

AND

Include at least 10 of the following accessibility and inclusion strategies most relevant to the project that go beyond the locally applicable accessibility code.

ACCESSIBILITY FOR PHYSICAL DIVERSITY

- Provide wave-to-open or vertical hand/foot press door operators at all regularly used building entrances.

- Design meeting spaces to accommodate mobility devices for at least 10% of occupants.
- Incorporate accessible and inclusive equipment and activities in fitness facilities. Ensure an open and accessible route to and around the equipment.
- Where inaccessible routes are provided (e.g., stairs), provide an alternate accessible route that starts and terminates at the same location.

ACCESSIBILITY FOR SAFETY AND AGING

- Provide nonslip flooring.
- Fix underside of area rugs to floor and provide transition strips at all edges.
- Provide visual indication or railing at all full-height glazing, except in private residences.
- Provide audible and visual alerts for emergency alerts.
- Provide closed risers (visually and physically) in all stairwells.
- Use visual contrast between walls and floors, walls and doors, and walls and casework.
- Provide visual, tactile, contrasting, or photoluminescent warnings at floor-level changes.

ACCESSIBILITY FOR SOCIAL HEALTH

- Provide lactation room pods.
- Provide at least one fully accessible, all-gender, single-use restroom OR one multiuse, all-gender restroom on each floor of the building.
- Include at least one adult changing station or table in a designated, accessible restroom or family restroom, or in one men's and one women's restroom.
- Provide signage in all languages spoken by more than 5% of the local population.
- Support neurodivergent users by achieving *EQc2: Occupant Experience*, Option 1. Biophilic Environments, Path 1. Integrated Biophilic Design.

ACCESSIBILITY FOR NAVIGATION

- Provide wayfinding signage that clearly indicates exits, entrances, and major functions in the project.
- Provide nontext diagrams and symbols at signage.
- Provide braille, visual and auditory cues, and/or continuous linear indicators on paths of travel.
- Use patterns and color blocking to identify key access spaces.
- Provide haptic/tactile maps for wayfinding.

REQUIREMENTS EXPLAINED

This credit encourages design that embraces the principles of accessibility and inclusive design and that considers the physical, sensory, and cognitive needs of occupants.[46] The goal is to go beyond basic accessibility measures to create environments that not only accommodate individuals with disabilities but also consider how all people interact, socialize, and move through spaces.

46. Matteo Zallio, and P. John Clarkson, "Inclusion, Diversity, Equity and Accessibility in the Built Environment: A Study of architectural Design Practice," *Building and Environment* 206 (December 2021): 108352, https://doi.org/10.1016/j.buildenv.2021.108352.

COMPLY WITH LOCAL ACCESSIBILITY CODES

All projects must ensure accessibility at a foundational level by demonstrating compliance with all relevant local accessibility codes identified in *IPp2: Human Impact Assessment*. If the project is in a region with no accessibility code, demonstrate three foundational physical accessibility elements:

- ramps for accessible routes and entrances
- clear width for human passage
- front approach wheelchair accessible section

In these cases, including fully accessible restrooms on all floors is highly encouraged.

RAMPS FOR ACCESSIBLE ROUTES AND ENTRANCES

Ramps are required in and around buildings to provide accessibility for individuals with mobility impairments, as mandated by the Americans with Disabilities Act (ADA). Unlike solutions that require separate or specialized accommodations, integrated ramps allow people with mobility disabilities, as well as people with strollers, to use the same paths as other occupants, thereby promoting dignity and inclusion through shared building circulation. Ramps must be installed at primary entrances where steps are present to ensure at least one accessible entry point. Inside buildings, ramps are necessary where level changes exceeding 0.5 inches occur, such as between floors or platforms, if elevators or lifts are not available. In parking areas, ramps should connect accessible parking spaces to building entrances via an accessible route. Emergency exits must also include ramps or accessible egress routes to support safe evacuation. All ramps must comply with ADA design standards, including proper slope, width, landings, handrails, and slip-resistant surfaces, to ensure they are safe and functional for all users.

CLEAR WIDTH FOR HUMAN PASSAGE

Projects must provide a minimum clear width of 32 inches (0.815 meters) at all doorways meant for human passage and 36 inches (0.915 meters) for circulation paths, with passing spaces of 60 inches (1.525 meters) provided at least every 200 feet (61 meters) along any path less than 60 inches (1.525 meters) wide. Teams are strongly encouraged to consider a minimum clear doorway width of 36 inches (0.91 meters) for enhanced accessibility for all occupants.

FRONT APPROACH WHEELCHAIR ACCESSIBLE SECTION

For all transaction surfaces, including reception desks and service counters, a front approach wheelchair accessible section is required to promote height inclusivity and ensure individuals using wheelchairs can access and navigate spaces safely, efficiently, and with dignity.

Projects must demonstrate that the accessible portions of counters are no higher than 36 inches (0.914 meters) above the floor and at least 36 inches (0.914 meters) wide.

ACCESSIBILITY AND INCLUSION STRATEGIES

Accessibility and inclusion strategies are organized into four categories: physical diversity, safety and aging, social health, and navigation. These strategies were selected based on research showing they provide the most significant benefits to the largest number of people beyond what is typically required by local accessibility codes, including ADA.

Projects must include at least 10 of the following strategies found in Tables 1 through 4. Findings from the *IPp2: Human Impact Assessment* must be used to understand unique occupant needs to inform strategy selection. Do not select strategies that are present in the existing local code, unless the team demonstrates increased stringency.

If fewer than 10 strategies are relevant to the project due to project type variations, teams may submit up to three alternative inclusive design strategies that meet the intent of credit for compliance.

TABLE 1. Accessibility for Physical Diversity Strategies

Accessibility for Physical Diversity	Examples, Specifications, and Explanation
Wave-to-open or vertical hand/foot press door operators at all regularly used building entrances.	No-touch door activation devices accommodate a wide range of users, including those with mobility impairments, people carrying objects or children, and individuals with temporary injuries.
Meeting spaces to accommodate mobility devices for at least 10% of occupants.	Accommodations include, but are not limited to, clear floor spaces and pathways of travel without obstruction. Users must be able to park mobility devices next to or within the seating area to avoid exclusion.
Accessible and inclusive equipment and activities in fitness facilities. And Open and accessible route to and around the equipment.	Projects with fitness equipment and activities must be designed to accommodate a range of physical abilities. This includes providing an inclusive range of strength and stretch training equipment. A minimum clearance of 36 inches (0.915 meters) is the appropriate threshold for accessible routes between exercise equipment, because this aligns with ADA requirements for accessible paths and allows adequate space for wheelchair users to navigate between equipment.
Alternate accessible route that starts and terminates at the same location as inaccessible routes.	Directness and proximity of the accessible route are essential to maintain a similar level of convenience to the inaccessible route. The alternative accessible route must be in the same general area as the circulation paths. Per *International Code Council* (ICC), the maximum allowable difference between the start and end points of an accessible route and its inaccessible counterpart is 200 feet (61 meters) in most indoor facilities, although it is recommended to minimize this distance for those with mobility challenges ideally to 100 feet (30 meters).[47]

TABLE 2. Accessibility for Safety and Aging

Accessibility for Safety and Aging	Examples and Explanation
Nonslip flooring	Acceptable nonslip flooring includes, but is not limited to, textured tiles, rubber flooring, and epoxy-coated flooring.
Fix area rugs to floor below, and provide transition strips over high traffic areas*	Solutions include nonslip backing, double-sided carpet tape, and a rug gripper to keep the rug in place. Transition strips to smooth the connection between two different types of flooring materials are required in high traffic areas.
Provide visual indication or railings at all full-height glazing, except in private residences.	Providing visual indicators, such as markers or patterns, will make it more noticeable for all users, including those with vision impairments. Adding railings at full-height glass also helps provide a physical barrier for people to easily detect and use for support, although not required.
Provide audible and visual alerts for emergency alerts.	Providing emergency alerts ensures that all users, including those with hearing or visual impairments, can respond to emergencies. This includes, but is not limited to, loud unique alarm sounds, voice announcements, flashing strobe lights, or LED beacons.

(*continued*)

47. International Code Council, "Accessible and Usable Buildings and Facilities," *2017 ICC A117.1*, 2017, https://codes.iccsafe.org/content/icca117-12017P4/chapter-4-accessible-routes.

TABLE 2. Accessibility for Safety and Aging (*continued*)

Accessibility for Safety and Aging	Examples and Explanation
Provide closed risers (visually and physically) on all stairs.	Closed risers eliminate gaps, reduce the risk of tripping or falling—particularly for small children or pets—and make stairs more secure. In addition, closed risers also offer privacy and reassurance through a more solid visual structure.
Use visual contrast between walls and floors, walls and doors, and walls and casework.	Incorporating distinct visual contrast between different architectural elements improves navigation and wayfinding, particularly for individuals with visual impairments. Projects must use contrasting colors, textures, and/or material for walls, floors, doors, casework (e.g., drawers and cabinet doors, or other permanently installed features).
Provide visual, tactile, contrasting, or photoluminescent warnings at floor level changes.	Incorporate bold color contrasts or reflective strips to mark transitions, use textured materials—such as ribbed or grooved strips—to provide tactile cues, and apply varied surface finishes to create both visual and tactile distinctions. Application of photoluminescent markings are recommended to ensure visibility in low-light conditions, particularly in staircases or emergency exit areas.

* Not required for Core and Shell projects.

TABLE 3. Accessibility for Social Health

Accessibility for Social Health	Examples and Explanation
Provide lactation rooms or lactation pods.	Dedicated lactation rooms, or breastfeeding support rooms, support higher rates of breastfeeding, improved work performance, and enhanced physical and emotional well-being.[48] It is recommended that these rooms include features such as an electrical outlet, seating, and a table, along with access to a sink and refrigerator either within the room or on the same floor, to better accommodate pumping and breastfeeding needs.
Provide at least one fully accessible all-gender single-use restroom OR one multi-use all-gender restroom on each floor of the building.	Providing all-gender restrooms creates a more inclusive and welcoming environment for people who may prefer gender-neutral options.[49]
At least one adult changing station or table in a designated, accessible restroom or family restroom.	Locate the changing station in a family or accessible restroom that complies with ADA guidelines for door width, turning space (60 inches or 1.52 meters diameter), and clear floor area (30 × 48 inches or 0.76 × 1.22 meters). Use a height-adjustable adult changing table with a weight capacity of at least 350 lbs (160 kg) to accommodate diverse user needs. Ensure the table is at least 72 inches (1.83 meters) long and 30 inches (0.76 meters) wide for adequate space.
Provide signage in all languages spoken by more than 5% of the local population.	Ensure services and information are accessible to diverse communities. Provide additional signage for people with limited English proficiency (LEP) when they constitute 5% of the population or 1,000 individuals, whichever is smaller.
Support neurodivergent users by achieving at least one point under *EQc2: Occupant Experience*, Option 1. Biophilic Environments, Path 1. Integrated Biophilic Design.	Biophilic design is particularly valuable for neurodivergent populations, which is approximately 15%–20% of the global population.[50] Achieve at least one point under *EQc2: Occupant Experience*, Option 1. Biophilic Environments, Path 1. Integrated Biophilic Design.

48. Carolina B. De Souza, Sonia I. Venancio, and Regina P. G. V. C. da Silva, "Breastfeeding Support Rooms and Their Contribution to Sustainable Development Goals: A Qualitative Study," *Frontiers in Public Health* 9 (2021): 732061, https://doi.org/10.3389/fpubh.2021.732061.
49. Markus Harwood-Jones, Kel Martin, and Lee Airton, "Research and Recommendations on Gender-Inclusive Washrooms and Changerooms," Kingston, Ontario, Canada: Queen's University, 2021, https://www.queensu.ca/hreo/sites/hreowww/files/uploaded_files/Washroom%20Report%20-%20Digital.pdf.
50. Nancy Doyle, "Neurodiversity at Work: A Biopsychosocial Model and the Impact on Working Adults," *British Medical Bulletin* 135, (October 14, 2020): 108–125, https://pubmed.ncbi.nlm.nih.gov/32996572/.

TABLE 4. Accessibility for Navigation

Accessibility for Navigation	Examples and Explanation
Provide wayfinding signage that clearly indicates exits, entrances, and major functions in the project.	Signage should be clear, concise, and easy to understand, using both text and universally recognized symbols to accommodate individuals with varying levels of literacy.
Provide non-text diagrams and symbols at signage.	Diagrams and symbols can convey instructions or warnings more quickly than text alone, making it easier for people with varying levels of literacy or visual impairments to understand. Provide alternative signage where applicable.
Provide Braille, visual and auditory cues, and/or continuous linear indicators on paths of travel.	Braille, visual and auditory cues, or continuous linear indicators enhance safety and usability for everyone, including those with vision impairments.
Use pattern and color blocking to identify key access spaces.	Contrasting colors and unique patterns help differentiate important spaces, such as entrances, exits, and pathways, and make them more recognizable and easier to navigate, particularly for the visually impaired. Provide visual contrast where applicable.
Provide haptic maps or tactile maps for wayfinding.	Wayfinding remains a significant barrier during daily life for people with visual impairments.[51] Haptic or tactile maps enable more users to form a clearer and more comprehensive understanding of an environment.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	All	Comply with local accessibility codes	Confirmation of whether there is a locally applicable accessibility code for the project.
			If the project has not met all locally applicable accessibility codes: contract documents highlighting the appropriate ramps, door widths, and front approach wheelchair accessible counter sections.
		Accessibility and inclusion strategies	Project strategies (check, from the standard list in Requirements, which strategies have been implemented).
			Narrative identifying locations in documents where project strategies are documented (file name and page number, at minimum).
		Alternative strategies	Evidence of up to three alternative inclusive design strategies that meet the intent of credit.

REFERENCED STANDARDS

- ADA (https://www.ada.gov/law-and-regs/design-standards/)
- ANSI A117.1 (https://codes.iccsafe.org/content/ICCA117.12017P7)
- OSHA (https://www.osha.gov/laws-regs)

51. Loes Ottink, et al., "Cognitive Map Formation Through Tactile Map Navigation in Visually Impaired and Sighted Persons," *Scientific Reports* 12 (July 7, 2022): 11499, https://www.nature.com/articles/s41598-022-15858-4.

FURTHER EXPLANATION

COMPLYING WITH LOCAL ACCESSIBILITY CODES

- Countries that adopt *ANSI A117.1-2017* accessibility standards referenced within the International Building Code will typically achieve compliance with most physical accessibility provisions outlined in the 2010 Americans with Disabilities Act (ADA) standards during new construction projects. These projects are not required to demonstrate compliance with the three foundational strategies for projects with no accessibility code in place.
- Many countries have their own national disability rights laws and accessibility guidelines in place. Projects in compliance with these relevant codes are also not required to demonstrate compliance with the three strategies outlined in the first section of this credit.

ADDITIONAL GUIDANCE FOR PROJECTS WITHOUT ACCESSIBILITY CODE

- Accessible routes are a continuous, unobstructed path, with a running slope not steeper than 1:20 that can be independently negotiated by all users, including those with mobility aids. These include routes to and through a building or site with a consistently clear width of at least 36 inches (92 centimeters); a cross slope not steeper than 1:48; and stable, firm, and slip-resistant ground/floor surfaces and turning spaces. This includes the use of accessible slopes, ramps, and/or curb cuts as necessary to eliminate abrupt changes in level (greater than 1/4-inch (6.35 millimeters)); refer to the 2010 ADA Standards for Accessible Design for any scenarios, definitions, or exceptions not addressed here.
- Accessible slopes are part of an accessible route used to address changes in level with an inclined ground or floor surface not greater than one unit vertical to 20 units horizontal (5% slope); 1/4 to 1/2-inch (6.35 to 12.70 millimeters) level changes may be beveled with a slope no greater than one unit vertical to two units horizontal (50% slope).

ACCESSIBILITY FOR PHYSICAL DIVERSITY

- Accessible and inclusive fitness equipment include a range of strength training and stretching equipment designed to accommodate differing physical abilities with an accessible route to and around each piece of equipment.

ACCESSIBILITY FOR SAFETY AND AGING

- For guidance on slip-resistant and non-slip flooring, refer to *ANSI A326.3-2021 Dynamic Coefficient of Friction (DCOF) of Hard Surface Flooring Materials*.
- For product use classification, see Table 1, adapted from *ANSI A326.3-2021*, Table 5.

TABLE 5. Product Use Classifications

Classification	Reference Category	Criteria
Interior, Dry	ID	≥ 0.42 dry DCOF (per Section 10.1)
Interior, Wet	IW	≥ 0.42 wet DCOF (per Section 9.1) or manufacturer declared
Interior, Wet Plus	IW+	Manufacturer declared
Exterior, Wet	EW	Manufacturer declared
Oils/Greases	O/G	Manufacturer declared

NOTE: Note: DCOF = dynamic coefficient of friction.

Accessibility for Social Health

- Accessible restrooms are a gender-specific or unisex restroom that can be independently navigated by all users, including those with mobility aids due to the following provisions aligned with ADA: a minimum 30 × 48-inch clear space next to the toilet, a minimum 60-inch diameter clear turning space, a toilet seat height between 17–19 inches, securely anchored grab bars on both sides and behind the toilet, a door width of at least 32 inches with lever or U-shaped door handles (or push mechanisms) placed 34-48 inches above the floor that do not require more than 5 pounds of force to operate, minimum knee clearance under the sink of 30W × 48L × 29H inches, toilet paper dispensers within reach of the toilet, and signage outside the door indicating the restroom type.
- Adult changing stations (also known as a universal changing table) are a securely anchored, height-adjustable table at least 72 inches (1.83 meters) long by 30 inches (0.76 meters) wide with a weight capacity of at least 350 pounds (160 kilograms) designed to enable use by caregivers and those in their care.

Accessibility for Navigation

- Key access spaces are interior transition zones, circulation hubs, or entry points that serve as important navigational points for major functions of a project. Major functions, or the primary purpose or intended use of the building, must be available to all occupants and therefore clearly communicated at key access spaces.

Resilient Spaces

EQc4
New Construction (1–2 points)
Core and Shell (1–2 points)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To support design features that increase the capacity for occupants to adapt to changing climate conditions and be protected from events that may compromise the quality of the indoor environment and subsequently occupant health and well-being.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–2**
Option 1. Management Mode for Episodic Outdoor Ambient Conditions—New Construction Only AND/OR	1
Option 2. Management Mode for Respiratory Diseases—New Construction Only AND/OR	1
Option 3. Design for Occupant Thermal Safety During Power Outages	1–2
Path 1. Consider Extreme Heat AND/OR	1
Path 2. Consider Extreme Cold AND/OR	1
Option 4. Operable Windows	1–2

Comply with any of the following options for up to 2 points:

OPTION 1. MANAGEMENT MODE FOR EPISODIC OUTDOOR AMBIENT CONDITIONS—NEW CONSTRUCTION ONLY (1 POINT)

This option applies to LEED BD+C: New Construction projects only.

Design systems with the capability to operate an episodic outdoor event management mode as described in *ASHRAE Guideline 44*. The mode should address varying outdoor conditions or events that could negatively influence IAQ, such as wildland fire smoke. Include the management mode in the design and commissioning documents. Verify proper implementation of the mode during commissioning.

AND/OR

OPTION 2. MANAGEMENT MODE FOR RESPIRATORY DISEASES—NEW CONSTRUCTION ONLY (1 POINT)

Design occupied spaces with the capability to operate an infection risk management mode that provides the minimum equivalent clean airflow rates outlined in *ASHRAE 241-2023*, Section 5.1. Include the management mode in the design and commissioning documents as outlined in *ASHRAE 241-2023*, Section B10.2 Design Documentation. Verify proper implementation of the mode during commissioning.

AND/OR

OPTION 3. DESIGN FOR OCCUPANT THERMAL SAFETY DURING POWER OUTAGES (1–2 POINTS)

PATH 1. CONSIDER EXTREME HEAT (1 POINT)

Demonstrate through thermal modeling that a building will passively maintain thermally habitable conditions during a power outage that lasts two days during peak summertime conditions of a TMY. Designate specific thermal safety zones where habitable conditions will be maintained during a power outage.

AND/OR

PATH 2. CONSIDER EXTREME COLD (1 POINT)

Demonstrate through thermal modeling or Passive House certification that a building will passively maintain thermally habitable conditions during a power outage that lasts two days during peak wintertime conditions of a TMY. Designate specific thermal safety zones where habitable conditions will be maintained during a power outage.

AND/OR

OPTION 4. OPERABLE WINDOWS (1–2 POINTS)

Design 50% for 1 point or 75% for 2 points of the regularly occupied spaces to have operable windows with the capability to provide access to outdoor air during heat waves or localized power outages. The windows must meet the opening size and location requirements of *ASHRAE 62.1-2022*, Section 6.4.

REQUIREMENTS EXPLAINED

Incorporating resilient design solutions into our buildings increases the adaptive capacity of our communities, strengthening their capacity to respond to climate change and natural disasters.

This credit addresses a building's ability to remain functional, maintain the quality of the indoor environment, and protect occupant health and well-being during major, episodic, disruptive events such as extreme weather conditions, wildfires, pandemics, or power outages. Although this credit addresses the design of events separately, teams are encouraged to consider and design for the possibility of multiple events occurring at the same time (such as a wildfire and extreme heat event).

Leveraging information from *IPp1: Climate Resilience Assessment* and *IPp2: Human Impact Assessment*, select two strategies for up to 2 points. Teams can select any two options or paths, even those not identified as a high priority under *IPp1: Climate Resilience Assessment*.

Building readiness plans may be developed and communicated to facilitate operating in specialized modes only when necessary, and to educate building occupants on the transition from normal operations to management mode for the duration of the condition or event.

OPTION 1. MANAGEMENT MODE FOR EPISODIC OUTDOOR AMBIENT CONDITIONS

Episodic outdoor ambient conditions can range from incidents such as the release of toxic chemicals outside a building to the widespread presence of wildfire smoke. Having an episodic outdoor event management mode facilitates the protection of building occupants from these and other outdoor pollution events.

System design for event management mode

ASHRAE Guideline 44, "Protecting Building Occupants from Smoke During Wildfire and Prescribed Burn Events," specifies enhanced modes of operation to preserve IAQ during periods of heightened outdoor air pollution.

Refer to *ASHRAE Guideline 44* to design HVAC systems capable of operating in a smoke-ready mode or other event management mode. Teams can leverage guidance from *Guideline 44* to design and apply similar modes of operation for any events that impact outdoor air quality including increases in nearby construction activity or chemical gas releases.

Commissioning requirements

ASHRAE Guideline 44 prescribes testing HVAC systems in smoke-ready conditions. Include the requisite sequences of operation in design documents and ensure that event management mode is included in the commissioning scope of work to verify that all equipment responds as intended.

OPTION 2. MANAGEMENT MODE FOR RESPIRATORY DISEASES

Following the COVID-19 pandemic, industry experts developed strategies to reduce airborne infectious disease transmission in buildings for the protection of public health and to facilitate keeping buildings operational during periods of heightened risk. *ASHRAE Standard 241-2023*, "Control of Infectious Aerosols," specifies an infection risk management mode with ventilation and filtration strategies for reducing occupant exposure to airborne pathogens that cause significant personal and economic damage each year.

Projects pursuing this option must design occupied spaces with the capability to operate in an Infection Risk Management Mode. This mode provides minimum equivalent clean airflow rates, calculated as the equivalent clean airflow rate per person multiplied by the anticipated number of people in a space. The building owner and facility manager must determine when to apply this mode of operation.

Commissioning requirements

Include the requisite sequences of operation in design documents and ensure that Infection Risk Management Mode is included in the commissioning scope of work to verify that all equipment responds as intended.

OPTION 3. DESIGN FOR OCCUPANT THERMAL SAFETY DURING POWER OUTAGES

During power outages when backup power is unavailable, mechanical cooling, heating, and ventilation become inaccessible. Designing spaces to sustain thermal habitability passively or through manual occupant controls allows the building to remain livable until power is restored.

Projects that pursue Option 3 have two paths to consider: one for extreme cold and one for extreme heat. Based on the project's location, teams should determine which option (or both) is appropriate.

Thermal safety zones

Designate thermal safety zones where thermal habitability can be maintained during a loss of power. Analyze conditions on the assumption that building occupants will congregate in the safety zones during an outage. This may increase the expected occupant density above normal assumptions for the space. Include enough thermal safety zones so the occupant density does not exceed one person per 20 square feet (1.9 square meters).

> **Example**
>
> An office building has 400 employees. If 20,000 square feet (1,858 square meters) of space is identified as being thermally safe, teams must analyze the space, assuming 400 people will be in that 20,000 square feet area (1,858 square meters).
>
> A 20,000 square feet (1,858 square meters) zone can accommodate up to 1,000 people. Therefore, the project would meet the sizing requirements.

Thermal habitability

Define habitable conditions as applicable to the project type. Thermally safe conditions may differ from a healthcare facility to a typical office building. For example, the heat stress index for an office building will be different than a nursing home. Consider the project type and the population when performing the initial analysis. Thermal habitability is not thermal comfort and will therefore be different than the comfort zone prescribed in *ASHRAE Standard 55*.

Natural ventilation

Thermal safety zones must have access to natural ventilation. This is achievable through operable windows, doors, operable panels, or louvers.

Thermal models

Thermal models analyze heat transfer within a building and account for climate, insulation, glazing specifications, solar gains, envelope leakage rates, and ventilation. Use computer simulation software to perform the thermal modeling for each path, based on project-specific inputs. Consider using modeling tools that are approved for Passive House compliance.

The analysis uses a two-day period. This was selected as an entry-level duration for LEED projects for design purposes. A four-day period has been used previously in the LEED v4 pilot credit Passive Survivability and Back-up Power During Disruptions. A time frame of 72 hours (3 days) is often used for general emergency preparedness planning (such as disaster-ready kits). For example, extreme heat or cold periods can last longer than two days. According to the U.S. EPA using heat wave tracking data by the National Oceanic and Atmospheric Administration (NOAA), the average heat wave in major U.S. urban areas has been about four days long.[52]

PATH 1. CONSIDER EXTREME HEAT

Demonstrate with a thermal model that the building will passively maintain habitable conditions for at least two days during a power outage in hot, summer months. The two-day period must represent the peak summertime conditions of a TMY.

AND/OR

PATH 2. CONSIDER EXTREME COLD

Demonstrate with a thermal model or through Passive House certification that the building will passively maintain habitable conditions for at least two days during a power outage in cold winter months. The two-day period must represent the peak wintertime conditions of a TMY.

OPTION 4. OPERABLE WINDOWS

During a power outage, ventilation may rely on operable windows, doors, or other openings to maintain airflow. Operable windows can also be used for thermal comfort, depending on outside weather conditions, and may minimize reliance on building systems and support adaptability for future changes in building use.

For this option, design operable windows to support ventilation in at least 50% of the regularly occupied floor area. Specify and size the windows to meet minimum window opening area and locations. The natural ventilation procedure in *ASHRAE 62.1-2022* Section 6.4 includes calculations and minimum openable area tables for determining these minimums. The opening sizes and locations will depend on the designer's approach to opening placement. For example, openings may be placed on one side of a zone, on opposite sides of a zone, or in the corner of a zone. The information provided in the tables is based solely on buoyancy-driven flow and does not address thermal comfort.

52. United States Environmental Protection Agency, *Climate Change Indicators: Heat Waves*, U.S. Environmental Protection Agency, June 2024, https://www.epa.gov/climate-indicators/climate-change-indicators-heat-waves#ref6.

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Management Mode for Episodic Outdoor Ambient Conditions	All	Design documents confirming management mode design and sequence of options.
			Final commissioning report confirming that management mode was tested during Commissioning.
	Option 2. Management Mode for Respiratory Diseases	All	Design documents confirming management mode design and sequence of options.
			Final commissioning report confirming that management mode was tested during Commissioning.
	Option 3. Design for Occupant Thermal Safety during Power Outages	Path 1 and Path 2	Thermal model report and results and identify thermal safety zones.
	Option 4. Operable Windows	All	*ASHRAE Standard 62.1* calculations for opening areas and distances for all regularly occupied spaces.
			Percentage of spaces with operable windows.

REFERENCED STANDARDS

- ASHRAE Guideline 44 (https://store.accuristech.com/ashrae/standards/guideline-44-2024-protecting-building-occupants-from-smoke-during-wildfire-and-prescribed-burn-events?product_id=2923808)
- ASHRAE 241-2023 (https://store.accuristech.com/ashrae/standards/ashrae-241-2023?product_id=2567398)
- FEMA P-361 (https://www.fema.gov/sites/default/files/documents/fema_p-361_safe-rooms-for-tornadoes-and-hurricanes_122024.pdf)
- FEMA E-74 (2012) (https://www.fema.gov/sites/default/files/2020-07/fema_earthquakes_reducing-the-risks-of-nonstructural-earthquake-damage-a-practical-guide-fema-e-74.pdf)
- REDi Version 1.0 (https://www.redi.arup.com)

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FURTHER EXPLANATION

SUMMERTIME/WINTERTIME CONDITIONS

- Extreme cold, or a cold wave, is a period of marked and unusual cold weather characterized by a sharp and significant drop in air temperatures near the surface (maximum, minimum, and daily average) over a large area and persisting below certain thresholds for at least two consecutive days during the cold season.[53]

53. World Meteorological Organization, "Guidelines on the Definition and Monitoring of Extreme Weather and Climate Events," accessed November 18, 2019, https://library.wmo.int/records/item/58396-guidelines-onthe-definition-and-characterization-of-extreme-weather-and-climate-events.

LEVERAGING *IPP1: CLIMATE RESILIENCE ASSESSMENT* AND *IPP2: HUMAN IMPACT ASSESSMENT* TO SELECT APPLICABLE OPTIONS(S) AND DEVELOP IMPLEMENTATION STRATEGIES

- For Option 1. Management Mode for Episodic Outdoor Ambient Conditions use information from *IPp1: Climate Risk Assessment* to identify the likelihood of an episodic event occurring during the project design service life. Consider the local jurisdictional risk rating for episodic events. Also use information from *IPp2: Human Impact Assessment*, specifically the "Occupant Experience" section, to determine project exposure to an episodic event. Consider outdoor air conditions, including, but not limited to, air quality index, ground level ozone, and ambient PM2.5.
- For Option 2. Management Mode for Respiratory Diseases use information from *IPp2: Human Impact Assessment*, specifically the "Occupant Experience" section, to determine the current prevalence of environmental features that may cause respiratory stress and disease.

Air Quality Testing and Monitoring

EQc5
New Construction (1–2 points)

IMPACT AREA ALIGNMENT:
☐ Decarbonization
☑ Quality of Life
☐ Ecological Conservation and Restoration

INTENT

To support better management of indoor air quality and identify opportunities for health-based approaches to building operations.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction	**1–2**
Option 1. Preoccupancy Air Testing	1–2
Path 1. Particulate Matter and Inorganic Gases	1
AND/OR	
Path 2. Volatile Organic Compounds	1
AND/OR	
Option 2. Continuous Indoor Air Monitoring	1

OPTION 1. PREOCCUPANCY AIR TESTING (1–2 POINTS)

After construction ends and before occupancy, but under ventilation conditions typical for occupancy, conduct baseline IAQ testing. Retail projects may conduct the testing within 14 days of occupancy. The number of measurements should be specified according to Table 1 and taken in representative locations of the building.

TABLE 1. Number of Measurements Required for Preoccupancy Air Testing

Total Occupied Floor Area, sq. ft. (sq. m.)	Number of Measurements
≤ 5,000 (500)	1
> 5,000 (500) and ≤ 15,000 (1,500)	2
> 15,000 (1,500) and ≤ 25,000 (2,500)	3
> 25,000 (2,500) and ≤ 200,000 (20,000)	4 plus 1 additional measurement per each 25,000 sq. ft. (2,500 sq. m.) above 25,000 sq. ft.
> 200,000	10 plus 1 additional measurement per each 50,000 sq. ft. (4,600 sq. m.) above 200,000 sq. ft.

PATH 1. PARTICULATE MATTER AND INORGANIC GASES (1 POINT)

Test for the particulate matter (PM) and inorganic gases listed in Table 2 using an allowed test method and demonstrate that the contaminants do not exceed the concentration limits listed in the table. Measure for a four-hour period and calculate peak concentration for carbon monoxide and average concentration for ozone, PM2.5, and PM10.

TABLE 2. Limits for PM and Inorganic Gases

Contaminant (CAS#)	Concentration Limit (μg/m^3)	Allowed Test Methods (Laboratory-based)	Direct Reading Instrument Minimum Specifications
Carbon monoxide (CO)	9 ppm; no more than 2 ppm above outdoor levels	*ISO 4224* *EPA Compendium Method IP-3* *GB/T 18883-2002* for projects in China	Direct calibrated electrochemical instrument with accuracy of ± 3% of reading and resolution of 0.1 ppm NDIR CO sensors with accuracy of 1% of 10 ppm full scale and display resolution of less than 0.1 ppm
Particulates (for projects in attainment areas)	ISO class 8 or lower per *ISO 14644-1:2015* OR meet PM 10: 50 μg/m^3 PM 2.5: 12 μg/m^3	n/a *IP-10A*	Accuracy (±): Greater of 5 μg/m^3 or 20% of reading Resolution (±): 5 μg/m^3
Particulates (for projects in nonattainment areas)	ISO class 8 or lower per *ISO 14644-1:2015* OR meet PM 10: 50 μg/m^3 PM 2.5: 35 μg/m^3	n/a *IP-10A*	Accuracy (±): Greater of 5 μg/m^3 or 20% of reading Resolution (±): 5 μg/m^3
Ozone	0.07 ppm OR 0.01 ppm for projects pursuing *EQc1: Enhanced Air Quality*, Option 1, Path 2	*ISO 13964* *ASTM D5149-02* EPA designated methods for ozone	Monitoring device with accuracy greater of 5 ppb or 20% of reading and resolution (5 min. average data) ± 5 ppb

NOTE: ppm = parts per million; ppb = parts per billion.

AND/OR

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PATH 2. VOLATILE ORGANIC COMPOUNDS (1 POINT)

Perform a screening test for TVOC. Use *ISO 16000-6*, *EPA TO-17*, or *EPA TO-15* to collect and analyze the air sample. Calculate the TVOC value per *EN 16516:2017*; *CDPH Standard Method v1.2 2017*, Section 3.9.4; or an alternative calculation method if full method description is included in test report.

If the TVOC levels exceed 500 μg/m^3, investigate for potential issues by comparing the individual VOC levels from the gas chromatography-mass spectrometry (GC/MS) results to associated cognizant authority health-based limits. Correct any identified issues and retest if necessary.

Test for the individual VOCs listed in Table 3 using an allowed test method and demonstrate that the contaminants do not exceed the concentration limits listed in the table. Laboratories that conduct the tests must be accredited under *ISO/IEC 17025* for the test methods they use.

TABLE 3. Volatile Organic Compound Limits

Contaminant (CAS#)	Concentration Limit (μg/m^3)	Allowed Test Methods
Formaldehyde 50-00-0	20 μg/m^3 (16 ppb)	*ISO 16000-3, 4* *EPA TO-11a* *EPA comp. IP-6A* *ASTM D5197-16*
Acetaldehyde 75-07-0	140 μg/m^3	
Benzene 71-43-2	3 μg/m^3	*ISO 16000-6* *EPA IP-1* *EPA TO-17* *EPA TO-15* *ISO 16017-1, 2* *ASTM D6196-15*
Hexane (n-) 110-54-3	7,000 μg/m^3	
Naphthalene 91-20-3	9 μg/m^3	
Phenol 108-95-2	200 μg/m^3	
Styrene 100-42-5	900 μg/m^3	
Tetrachloroethylene 127-18-4	35 μg/m^3	
Toluene 108-88-3	300 μg/m^3	
Vinyl acetate 108-05-4	200 μg/m^3	
Dichlorobenzene (1,4-) 106-46-7	800 μg/m^3	
Xylenes–total 108-38-3, 95-47-6, and 106-42-3	700 μg/m^3	

AND/OR

OPTION 2. CONTINUOUS INDOOR AIR MONITORING (1 POINT)

Provide indoor air monitors for all the following parameters:

- Carbon dioxide (CO_2)
- Particulate matter (PM2.5)
- Total volatile organic compounds
- Temperature
- Relative humidity

Monitors must be building grade or better and located 3–6 feet (1–2 meters) above the floor.

REQUIREMENTS EXPLAINED

This credit helps the project gain a better understanding of the indoor air. Testing before the building is occupied prevents exposing occupants to unsatisfactory air. Continuous indoor air monitoring during operation makes it possible to track contaminant levels over time and to proactively identify any issues and faults.

Teams may use both options for a total of 2 points and may find it beneficial to perform air testing prior to occupancy at the same time as setting up the continuous monitoring systems.

OPTION 1. PRE-OCCUPANCY AIR TESTING

Construction activities and building materials can introduce contaminants that may negatively affect a building's IAQ. Testing after construction incentivizes the contractors to follow construction management practices in accordance with *EQp1: Construction Management* and follow low-emitting material specifications. It also verifies that the indoor environment is acceptable for human occupancy and ensures that ventilation systems are effectively maintaining adequate IAQ.

Number of measurements

The number of measurement points required per floor area (square feet or square meters) is outlined in Table 1 of the rating system, which helps with planning for testing and the associated costs. The floor area in Table 1 reflects the total occupied floor area for the project, including all regularly and irregularly occupied areas. For example, corridors are irregularly occupied and must be included in the total area for this calculation. Unoccupied areas, such as mechanical and electrical rooms, are excluded.

Projects may choose to take more measurements beyond the minimum if desired. Exceeding the minimum number of measurements does not earn additional points but will provide a more comprehensive assessment of the IAQ.

Measurement locations

Measurement locations must be selected to best represent the project occupancy and function(s). Use the following criteria to determine representative locations for the project:

- **Regularly occupied spaces:** Prioritize regularly occupied spaces. Irregularly occupied spaces must be included in the total area for determining the occupied floor area but need not be tested.
- **Multiple space types:** If more than one measurement is necessary per Table 1, test multiple space types. For example, in an office building, test open-office spaces but also consider closed offices, conference rooms, quiet space, and other occupied space types. In a school building, test classroom spaces but also consider the auditorium, administrative offices, student assembly areas, and lab spaces.
- **Different ventilation systems:** If the project has multiple ventilation systems, identify a measurement location in areas served by each ventilation system, up to the required number of measurement points.
- **Multiple floors:** For projects with multiple floors, select locations on different floors.

- **Spaces where the highest concentrations of contaminants are likely to occur:** Areas of greater concentrations of contaminants could be due to the construction or fit-out of the space, or a lower ventilation rate. For example, private offices may have a higher concentration of contaminants compared to open offices because of a higher density of furniture and finish materials in an enclosed space.

Failed testing

If a test fails, take corrective action (e.g., clean and flush out the space) and retest. All test locations must meet the concentration limits in Table 2 for Path 1 compliance and/or Table 3 for Path 2 compliance.

Timing for air quality testing

Air quality measurements must be conducted after construction is complete and before occupancy. For the purposes of this credit, construction is complete once all furniture and finishes are installed, construction punch-list items that would generate VOCs or other contaminants are complete, and testing and balancing of the HVAC system has been conducted.

Testing must be done under normal operating ventilation conditions. If there are unoccupied setbacks in the ventilation system, test during normal occupied hours to achieve the typical ventilation conditions.

All testing and retesting must be completed before occupancy.

Retail Projects

- Retail projects may perform testing within 14 days of occupancy. This is to accommodate the unique compressed construction timeline for typical retail projects.

PATH 1. PARTICULATE MATTER AND INORGANIC GASES

Each location must be tested for all contaminants in Table 2, which outlines the approved test methods for each contaminant. Teams can use laboratory-based testing or take measurements using direct reading instruments. If using direct readings, all instruments must meet the minimum specifications of Table 2. Alternative methods may be used for Path 1 contaminants if the project team documents that the accuracy and resolution specifications in Table 2 are met.

PATH 2. VOLATILE ORGANIC COMPOUNDS

A screening test for TVOC and a test for each individual VOC in Table 3 must be performed at each measurement location. Because VOC testing and analysis is complex, it must be performed using specific methods by a laboratory that is accredited under *ISO/IEC 17025* for the test method used.

TVOC screening is intended to serve as a general indicator of the VOC levels in the building and is used to capture situations where investigation of individual VOCs beyond those targeted in Table 3 may be needed. Although projects are not required to meet a specific TVOC threshold, they are required to report TVOC results. If the TVOC concentration exceeds 500 $\mu g/m^3$, the team must work with the laboratory to compare the individual VOC levels from the GC/MS results to associated cognizant health-based limits and perform corrective actions as necessary.

OPTION 2. CONTINUOUS INDOOR AIR MONITORING

During occupancy, airborne contaminants can enter a building from outdoors or be introduced by indoor sources and activities such as cleaning, cooking, candles, 3D printers, or improperly vented heating appliances. Permanently installed continuous indoor air monitors enable the identification of potential issues and timely corrective actions to ensure systems designed to maintain IAQ in the building continue to work as intended through the project's operational phase.

Number of monitors

A successful monitoring strategy must consider the data collection purpose and dedicated resources for ongoing data management. Fewer, well-managed monitors are usually more beneficial than numerous neglected monitors. Include at least one monitor per 25,000 square feet (2,500 square meters) of total occupied floor area. This density is a good entry point for getting started with IAQ monitoring. Additional monitors can be added as desired up to the best-practice density of one monitor per 5,000 square feet (500 square meters) of total occupied floor area.

Monitor locations

Monitors must be placed to best represent the project occupancy and function(s). This will vary depending on the purpose of the monitoring. Use the following criteria to determine representative locations for the project:

- **Multiple space types:** Consider including monitors in multiple space types. For example, in an office building, monitor the open-office spaces but also consider closed offices, conference rooms, quiet spaces, and other occupied space types. In a school building, monitor classroom spaces but also consider the auditorium, administrative offices, student assembly areas, and lab spaces.
- **Different ventilation systems:** If the project has multiple ventilation systems, consider placing monitors in areas served by each ventilation system.
- **Multiple floors:** For projects with multiple floors, consider placing monitors on different floors.
- **Spaces where the highest concentrations of contaminants are likely to occur:** Areas of greater concentrations of contaminants could be due to the construction or fit-out of the space, a lower ventilation rate or air filtration level, the presence of combustion or operable windows, or occupant activities. For example, cafeterias may have a higher concentration of contaminants compared to classrooms because of cooking.
- **Spaces occupied by at-risk populations or spaces designated for cleaner air:** Consider placing monitors in areas where people who are more susceptible to poor IAQ congregate. For example, this may include spaces with infants, children, pregnant women, acute care facilities, and assisted living facilities.

If monitoring to support IAQ management during wildfires and prescribed burn events, review *ASHRAE Guideline 44*, Section 5.5.1.2 for considerations for monitor placement.

Monitors must be permanently installed at a height corresponding to the breathing zone of a typical occupant. In most situations, the breathing zone is 3–6 feet (0.9–1.8 meters) above finished floor height, based on the location of an occupant's head when seated or standing. Alternative mounting heights based on the anticipated occupant position in a space may be considered.

Where possible, place monitors at least three feet (0.9 meters) away from doors, windows, air filters, air supply outlets, exhaust intakes, stoves, printers, and other potential airborne contaminant sources or sinks. In areas where this is not possible, locate monitors closer to air returns than air diffusers.

Monitors located in ducts do not meet the requirements.

Monitor specifications

Select indoor air monitoring devices that measure carbon dioxide (CO_2), fine particulate matter (PM2.5), TVOC, temperature, and relative humidity. Monitors must meet the building grade requirements in Table 4 (based on *RESET Grade B*[54]), be *RESET Grade B* accredited, or validated *UL 2095 Grade B*.

Hourly reporting

Monitors must report hourly (or higher frequency, including 15-minute data for CO_2) data to a remote location that logs pollutant levels over time. A digital display, or integration with the building management system, is not required to achieve the credit.

TABLE 4. *RESET Grade B* Monitor Specifications

***RESET Grade B* Monitor Specifications**	**CO_2**	**PM2.5**	**TVOC**	**Temperature**	**Relative Humidity**
Data loss	10%				
Operating range for temperature	0–40°C				
Operating range for relative humidity	10–80% RH noncondensing				
Data output interval	5 min				
Sampling type		Active airflow			
Sensor output resolution	5 ppm	1 µg/m^3	4.4 ppb	0.1°C	1% RH
Measuring range	400–5,000 ppm	0–500 µg/m^3	65–870 ppb	0–40°C	10–80% RH
Accuracy*	400–2000 ppm: ±50 && 3% 2,000–5,000: ±50 && 5%	0–150: ±5 && 15% 150–500: ±5 && 20%	65–260 ppb: ±8.7 && 15% 260–870: ±8.7 && 20%	±1°C	±8% RH
Performance check and recalibration	Required	Required	Required	Required	Required

*EXAMPLE: If a reference monitor is reading 900 ppm, a Grade B monitor's reading must read within 50 + (0.03 × 900) = ±77. The Grade B monitor's reading must be between 823 and 977 ppm.

54. RESET, GIGAbase Canada, "Indoor Air Quality Monitors," 2025, accessed April 5, 2025, https://www.reset.build/directory/monitors/type/indoor.

EQ

DOCUMENTATION

Project Types	Options	Paths	Documentation
All	Option 1. Pre-Occupancy Air Testing	Path 1 and Path 2	Completed air quality testing report, including time, date, testing methods complying with credit requirements, results and limits of the tested contaminants in all locations, and lab accreditation scope for Path 2 VOCs if applicable.
			Evidence of testing locations.
			Document confirming substantial completion of construction, highlighting the date. The intent is to confirm that construction was complete prior to the time/date of the air quality testing.
	Option 2. Continuous Indoor Air Monitoring	All	Evidence of monitoring locations and description of monitoring approach.
			Specifications of building grade air monitors.

REFERENCED STANDARDS

- ISO 4224 (https://www.iso.org/standard/32229.html)
- EPA Compendium Method IP-3, May 1990 (https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=30003ULE.txt)
- ISO 13964 (https://www.iso.org/standard/23528.html)
- ASTM D5149-24 (https://www.astm.org/d5149-24.html)
- EPA-designated methods for Ozone (https://www.epa.gov/system/files/documents/2025-06/amtic-list-june-2025_final-508-compliant.pdf)
- ISO IEC 17025 (https://www.iso.org/ISO-IEC-17025-testing-and-calibration-laboratories.html)
- CDPH Standard Method v1.2-2017 (https://www.cdph.ca.gov/Programs/cls/dehl/ehl/Pages/AQS/VOCs.aspx)
- Reset Air Accredited Monitors (https://www.reset.build/directory/monitors)
- UL 2905 (https://www.shopulstandards.com/ProductDetail.aspx?productId=ULE2905_2_S_20230110)

FURTHER EXPLANATION

OPTION 1. PREOCCUPANCY AIR TESTING, PATH 1. PARTICULATE MATTER AND INORGANIC GASES

- The regional air quality, including attainment and nonattainment status, is identified in two other prerequisites: *IPp2: Human Impact Assessment* and *EQp2: Fundamental Air Quality*.
- If the project is identified as being in a nonattainment area for PM2.5, a higher indoor concentration limit is applicable (35 micrograms per cubic meter vs. 12 micrograms per cubic meter). This higher limit acknowledges it can be challenging to significantly reduce indoor PM2.5 levels when outdoor PM2.5 levels are high.

- Attainment areas are geographic areas that meet or exceed particular National Ambient Air Quality Standards regarding particulate matter, ozone, carbon monoxide, sulfur dioxide, nitrogen dioxide, and/or lead as set by the U.S. Environmental Protection Agency.
- Nonattainment areas are geographic areas that do not meet (or that contribute to ambient air quality in a nearby area that does not meet) the standard.

Project Priorities

OVERVIEW

The historical Innovation credit category has evolved in LEED v5 to become the Project Priorities (PR) credit category. The goal is greater flexibility for projects to address their unique context and priorities, including typology, culture, location, areas of innovation, and individual performance objectives. Credits can be added to the library as they are developed, which enables an adaptive and agile response to rapidly evolving industry knowledge, developing technologies, and emerging innovative solutions. In addition, projects are empowered to pursue improvements that are most meaningful to their specific goals and circumstances.

For example, the evolution of the building industry over the last 15 years has fostered a need for more sector-specific sustainability metrics. Additionally, greater adoption of reporting has prompted real estate organizations to establish targets in areas such as decarbonization, occupant health, and biodiversity. The PR credit category aims to provide recognition for projects pursuing these goals outside of the established credits in LEED v5.

New metrics and strategies can be continually applied to LEED without waiting for the next version to debut, thereby allowing for a more nimble and dynamic development of credits and compliance paths in between releases of new rating system versions.

By embracing flexibility and encouraging continuous innovation, the PR credit category ensures that LEED remains a dynamic tool for advancing sustainability. It empowers project teams to align their efforts with evolving best practices, sector-specific goals, and emerging global challenges and ensures that buildings remain resilient, forward-thinking, and impactful over time.

PROJECT PRIORITIES CREDIT

Project Priorities

PRc1

New Construction (1–9 points)

Core and Shell (1–9 points)

INTENT

To promote achievement of credits that address geographically sensitive or adaptation-specific environmental, social equity, and public health priorities. To encourage projects to think creatively to test and accelerate new sustainable building practices and strategies.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1–9**
Regional Priority	1–9
Project-Type Credits	
Exemplary Performance	
Pilot Credits	
Innovation Strategies	

Achieve any combination of the following for a maximum of 9 points:

REGIONAL PRIORITY

Achieve a regional priority credit from USGBC's Project Priority Library. These credits have been identified by USGBC as having additional regional importance for the project's region.

PROJECT-TYPE CREDITS

Achieve a project-type credit from USGBC's Project Priority Library. These credits have been identified by USGBC as addressing unique needs for the given adaptation or building application.

EXEMPLARY PERFORMANCE

Achieve an exemplary performance credit from USGBC's Project Priority Library. These credits have been identified by USGBC as going above and beyond an existing LEED v5 prerequisite or credit in the LEED v5 priority areas of scale, decarbonization, resilience, health, equity, and/or ecosystems.

PILOT CREDITS

Achieve a pilot credit from USGBC's Project Priority Library.

INNOVATIVE STRATEGIES

Achieve significant, measurable, environmental performance using a strategy not addressed in the LEED v5 green building rating system.

Identify all the following criteria:

- the intent of the proposed innovation strategy
- proposed requirements for compliance
- proposed submittals to demonstrate compliance
- the design approach or strategies used to meet the requirements

REQUIREMENTS EXPLAINED

Teams earn recognition for implementing innovative measures addressing distinct focus areas in their projects through the *PRc1: Project Priorities*. This credit offers multiple pathways for projects to address their respective priorities and go beyond the requirements listed in other LEED credits. This flexibility enables teams to effectively address the distinct needs of their projects and fosters innovation and adaptability. Each project can chart its path forward based on its own goals.[1]

Projects prioritize efforts based on their unique contexts. Teams can choose the best credits for addressing their project's goals and targets. Some projects may concentrate most of their effort toward a single priority area, including project-type-specific priorities or exemplary performance. Other projects might choose to address different priority areas more uniformly.

For example, an office building in a coastal city prone to hurricanes and flooding might prioritize enhancing resiliency to regional climate challenges with applicable credit pathways focused on flood mitigation, building safety, and reinforced construction materials and design. Similarly, an urban mixed-use development comprising residential and commercial spaces might have a variety of sustainable priorities to address, such as incorporating renewable energy efficiency, providing indoor environmental quality to building tenants, and promoting methods for active or cleaner forms of transportation.

1. Pacific Northwest National Laboratory, "Green Buildings," n.d., https://www.pnnl.gov/explainer-articles/green-buildings.

To achieve the maximum nine points available, project teams should incorporate as many credits under each pathway as they prefer by using any combination of project type credits, exemplary performance credits, regional priorities, innovation strategies, and pilot credit pathways.

Project type

Achieve a project-type credit from the USGBC's Project Priority Library. USGBC has identified these credits as addressing unique needs for the given adaptation or building application.

> **Example strategies: Project type**
>
> A data center project might focus on project-type credits specific to data centers that address energy efficiency, advanced cooling technologies, and renewable energy integration.

Exemplary performance

Achieve exemplary performance requirements of an existing LEED v5 credit eligible for exemplary performance as specified in USGBC's Project Priority Library. Exemplary performance earns points by exceeding the credit requirements or achieving the next incremental percentage threshold for the credit.

Regional priority

Identify the environmental, social equity, or public health priorities for the project's location and achieve LEED credits that address those regional priorities. Regional priority credits address geographically specific environmental and/or social priorities for the project's region.

Innovation strategies

Achieve innovation credits from the USGBC's Project Priority Library. Alternatively, achieve innovation credits by adopting new strategies not addressed in the LEED rating system that demonstrate reduced environmental impacts, increased decarbonization, and improved social impacts. Projects must submit documentation identifying the intent of the proposed innovation credit, proposed requirements for compliance, proposed submittals to demonstrate compliance, and the design approach or strategies used to meet the requirements.

Pilot credits

Achieve pilot credits from the USGBC's Project Priority Library. USGBC has identified these credits to explore new aspects of sustainable design, building, and construction and to potentially include in future additions of the LEED rating system.

DOCUMENTATION

Project Types	Options/ Paths	Documentation
All	All	Provide the supporting documentation for the identified strategy. This may consist of contract documents (plans, specifications, elevations, sections, construction details, etc.), calculations, reports, manufacturer product information, photographs, and/or descriptive narratives.
		For Regional Priority strategies, also include a description of regional/local priority, with evidence and description of strategies adopted by the project to address the aforementioned priority.
		For Innovative strategies not listed in the Project Priority Library, also include the intent of the strategy and a summary of the significant, measurable, environmental performance improvement proposed (i.e., the requirements for compliance).
		For Pilot credits, also include the evidence of Pilot Credit registration and the completed Pilot Credit survey.

REFERENCED STANDARDS

- None

LEED AP

PRc2
New Construction (1 point)
Core and Shell (1 point)

IMPACT AREA ALIGNMENT:
☑ Decarbonization
☑ Quality of Life
☑ Ecological Conservation and Restoration

INTENT

To encourage team integration required by a LEED AP and to streamline the application and certification process.

REQUIREMENTS

ACHIEVEMENT PATHWAYS	POINTS
New Construction and Core and Shell	**1**
LEED AP	1

At least one principal participant of the project team must be a LEED AP with a specialty appropriate for the project.

REQUIREMENTS EXPLAINED

The credit rewards projects that include a LEED AP with an active credential on the project team at the time of certification review.

A key design team member must have a LEED AP with a Building Design and Construction specialty. While all LEED AP credentials provide an understanding of the green building community and certification requirements, team members with the Building Design and Construction specialty have extensive knowledge and experience with prerequisites and credits for a New Construction or Core and Shell project.

LEED APs without specialty do not qualify for this credit.

DOCUMENTATION

Project Types	Options/ Paths	Documentation
All	All	Identify the LEED AP with specialty.
		Confirm that the LEED AP has a BD+C Specialty, and that the GBCI Credential Certificate, current at the time of review, is uploaded to the AP's GBCI Profile.

REFERENCED STANDARDS

- None

Appendix I

LEED PLATINUM REQUIREMENTS

NEW CONSTRUCTION

EAc1: Electrification

Five points are required.

EAc3: Enhanced Energy Efficiency

Eight points are required.

EAc4: Renewable Energy

100% of site energy use from any combination of Tier 1, Tier 2, and Tier 3 renewable energy.

MRc2: Reduce Embodied Carbon

20% reduction in embodied carbon.

CORE AND SHELL

EAc1: Electrification

Four points are required.

EAc3: Enhanced Energy Efficiency

Five points are required.

EAc4: Renewable Energy

100% of base building energy use is from any combination of Tier 1, Tier 2, and Tier 3 renewable energy.

MRc2: Reduce Embodied Carbon

20% reduction in embodied carbon.

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